Fixed Income Markets

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Fixed Income Markets

Management, Trading, Hedging

Second Edition

MOORAD CHOUDHRY
DAVID MOSKOVIC
MAX WONG

With contributions from Suleman Baig, Zhuoshi Liu, Michele Lizzio and Alexandru Voicu

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For Lindsay . . .

YOU are the best thing that ever happened to me.

—Moorad Choudhry

Contents

Foreword

I have been writing book chapters and forewords covering the fixed income markets anecdotally for over a decade. And like the markets that underlie the various themes I have commented on, the focus of my musings has evolved over this period as well. A consistent theme throughout my annotations, however, has been a basic assertion that the "science" of fixed income investing has been uniquely influenced by mathematics and computing technology. By most accounts, fixed income analytics as a formal study only began to take shape as a proper science during the 1970–1980s; and as with most developing sciences, analytical models rooted in mathematics became the means by which practitioners looked to enumerate the developing fixed income concepts. Interestingly, this was also the period that most would suggest when financial globalisation started. Whilst one can debate whether finance's coming of age was coincidental with globalisation or perhaps its key driver, it would be hard to argue that the electronic connectivity between borrower and saver has been anything short of revolutionary. Irrespective of the cause or effect, though, the world's escalating financial interconnectivity certainly warranted, if not necessitated, new thinking around fixed income valuation and risk management.

Consequently, analytical frameworks ranging from portfolio theory (i.e., optimal portfolio construction) to option theory (i.e., human choice) had their origins during this period. While each concept was unique in its own right, the collective insights helped practitioners appreciate that investment framing and decisioning are multivariate problems; financial markets are complex and dynamic by nature. In other words, fixed income value is not static by definition. Thus, rather than continuing to limit investors to simple "point in time" measures of value, valuation theory evolved to capture the effects that second, third, . . . , and *N*th order factors have on investment performance over time. In fact, many of the concepts in this book (i.e., duration, convexity, correlation, . . . , optionality) had their genesis during those early days of fixed income science. Although the new theories were helpful in better framing value sensitivity, it was the advent of computing technology that provided for its practical application and mass acceptance. Risk managers could for the first time capture large quantums of diverse information and feed them through the developing mathematical tools in a scalable way. In other words, scenario analysis for fixed income performance became feasible; risk managers had the practical ability to vary inputs (or "conditions") to build a probabilistic picture of all the potential returns one could possibly expect. The ability to understand the upside and downside sensitivity of an investment to a series of variables (i.e., credit spreads, interest rates, . . . default rates) categorically redefined the fundamental nature of fixed income analysis. Because they could be put into practice, these new theories and capabilities gained broad acceptance, and the creation of a professional industry ensued.

As a statement of fact, there is nothing particularly enlightening or controversial about the above observations. Undoubtedly, it is hard to dispute that the introduction of new valuation theories further empowered through computing would not be accretive to the knowledge base of any subject matter. Surely, research, scientific method, and technology are at the core of any industry's advancement? If so, why then is it that the advances in the field of finance have, by contrast, been so uniquely targeted and discredited since the financial crisis? Keeping in mind, we are already a half a decade on from its commencement and yet still no reprieve from the sceptics. The often accepted notion that finance's technological advancements were the sole catalyst for the crisis should be duly troubling to any impartial observer. Furthermore, the critics openly blame the complexity of derivatives, securitisation, and hybrid capital (collectively "financial engineering") specifically. While these types of oversimplifications make for popular journalism, I would suggest that these criticisms are misdirected and have never been well reasoned in logic or supported in fact.

Certainly, the targeting of the advances in finance seems misplaced when, ironically, many of these instruments have been at the cornerstone of the crisis's prescribed remedies. Governments, for example, have been fulfilling their quantitative easing programs through the purchasing of securitised assets; while troubled banks have been recapitalising themselves through the issuance of new forms of hybrid capital instruments, to name another.

With that irony in mind, it seems odd that the world is yet to move on from the fruitless task of looking to assign culpability for the challenges faced by the global financial system. In my opinion, efforts by critics would be better spent trying to understand the scope of knowledge afforded by the industry's work; the faculty of its tools are powerful and should be embraced. It is easier to identify and take lessons from the crisis when the observations are spanned and compared against the currently accepted analytic frameworks. One should never forget that analytical frameworks are theories that need to evolve; they are not absolute truths. Knowledge and technology are symbiotic creatures that feed off of one another; and like the financial markets mentioned above, share a dynamic relationship.

As such, this is why books like this make for worthwhile reading; these types of broad references help contextualise the aggregate body of knowledge that has accumulated around fixed income. A solid understanding of the current thinking is the necessary foundation from which current practitioners and future students can then hope to build better tools. Hence, I challenge the next generation of thought leaders in the field to identify what the key lessons learned from the crisis are; and consequently, help set the direction of where the study of finance should go (i.e., what are the next generation tools needed?).

As the story is still playing out, I don't think it is a cop-out to leave those questions to the next generation; realistically, both those questions are books in themselves and beyond the limits of a foreword. Equally, the debate would be long and fraught with controversy; and for the many that have tried, they invariably take a biased side and create drama out of what should be more of a philosophical message. I have, though, in different forums openly suggested that "financial engineering" in itself was not the root cause of the financial crisis; instead, it was the simple means by which these new tools and technology were employed and understood. Securitisation technology, for example, was not some inherently destructive technology that created the U.S. subprime crisis; it was bad judgement underpinning poorly underwritten mortgages. Securitisation technology was simply the practical delivery mechanism by which investors could take exposure to the mortgage asset class; the technology in itself didn't change the behaviour of the underlying asset class. So, I would argue that the technology was not flawed; but rather the choice of the investable asset class (or "inputs") were flawed. Practitioners simply lost perspective on the natural limitation of tools; again, their outputs are not absolutes. Perhaps this is where one key message from the crisis lies; or at least the key philosophical one.

So, despite all the advancements in financial thinking; their utility and usefulness are still dependent on human judgment. Judgment in terms of the inputs used as well as judgment as to how the outputs are interpreted. As we look forward, I think it is particularly important that we heed those cautions on the power of financial tools; we are in a period of unprecedented financial experimentation (i.e., global quantitative easing, structural bank reform, automated trading/exchanges). I appreciate that this all sounds like common sense; but, overreliance and underappreciation of the capabilities and consequences of developing technology are not a new challenge for society. In the extreme, atomic energy for instance, has had its moments of glory and shame in history. In spite of those moments, though, we as a society continue to utilise, research, and improve upon its application. Clearly, this is a dramatic comparison; albeit the messaging is the same. Discrediting efforts in science simply because we feel our current capabilities and knowledge are insufficient or incomplete is a societal failure—not one ascribable to the science itself. In that regard, I hope the sceptics will reflect on some of these observations and reorient some of their criticisms toward a more construct tone. Removing such distractions will allow the industry to more appropriately place all its efforts on developing the new tools needed rather than defending itself. It should not be lost on critics that the study of finance is an important science, which has yielded many notable achievements. In the end, advancing our understanding of the world's financial system is in everyone's interest; it is as vital to a healthy society as is energy above.

OLDRICH MASEK
Managing Director, Global Securitised Products
JPMorgan Chase
21 November 2013

The statements, views, and opinions expressed in this foreword are the writer's own and do not necessarily reflect those of his employer, JPMorgan Chase & Co., its affiliates, other employees, or clients.

Preface

The first edition of this book was published precisely 10 years ago. A lot has happened in the debt capital markets since then. The little matter of the global bank crash (in reality, the U.S. and Western European bank crash) had considerable impact, for starters. However fixed income instruments and their analysis have not materially changed from what readers of the first edition would have been familiar with. That said, there are always new things to talk about.

To begin at the beginning, we reprise this text from the Preface of the first edition:

> *The market in bond market securities, also known as the fixed income market, is large and diverse, and one that plays an important part in global economic development. The vast majority of securities in the world today are debt instruments, with outstanding volume estimated at over $23 trillion. In this book we provide a concise and accessible description of the main elements of the markets, concentrating on the instruments used and their applications.*

Notwithstanding our second sentence, we have had to update a considerable segment of the original text. The book remains broken into four parts, covering introduction to bonds, selected market instruments, derivatives, and trading strategy. However, we have made the book more practical, even more practitioner based, and completely up to the minute. That's why we changed the subtitle of the book from the original "Instruments, Applications, Mathematics" to the still more relevant "Management, Trading, Hedging". And for this I owe much to the co-authors I brought in to assist me with this second edition, namely David Moskovic, Max Wong, Suleman Baig, Zhuoshi Liu, Michele Lizzio, and Alex Voicu. David has a chapter entirely to himself, the excellent value-added that is the final chapter of the book, and also contributes in the chapters on the yield curve, credit derivatives, and securitisation. The others have all fed in their insight and expertise throughout the book chapters, while Messrs. Liu and Lizzio have also contributed to the book's associated website.

The highlights of this brand new Second Edition include:

- A chapter on convertibles, including the new post-crash-inspired instrument that is the contingent convertible bond (CoCo);
- A section on credit-linked notes (CLNs), and how they are hedged by the issuer; also includes worked examples of CLN repackaged securities and treatment on default;
- A chapter on the hedging, collateral, and correlation issues associated with valuing and managing a portfolio of derivatives;
- An updated chapter on value-at-risk and its shortcomings during the period of the crash;
- Latest developments in debt markets trading and hedging, including multi-currency yield curves, OIS discounting, CSA curves, and collateral management.

The Second Edition also features a companion website with Excel spreadsheet models that the reader should hopefully find very useful, including:

- Two different yield curve models, one incorporating cubic spline methodology and the other Svensson 94 methodology;
- A credit default swap pricing model;
- A convertible bond pricing model;
- An option-embedded bond pricing model, for both callable and puttable bonds.

At the same time we have thinned out, or removed entirely, text and chapters that time has passed by, including those on structured finance and synthetic investment products. As always, readers are encouraged to send in comments and feedback, please feel free to e-mail me directly at mooradchoudhry@gmail.com.

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All the best. . . .

Moorad Choudhry
Surrey, England
31st December 2013

About the Authors

Moorad Choudhry was a UK government bond trader and gilt repo trader at ABN Amro Hoare Govett during 1992–1997 and a gilts proprietary trader at Hambros Bank during 1997. He later worked in structured finance at JPMorgan and KBC Financial Products, and led the deal team that closed Picaros Funding (winner of the *Euromoney* Structured Finance Deal of the Year Award for 2005) and Red Sea Master Trust, the world's first in-house multi-SPV securitisation transaction. He is currently IPO Treasurer, Group Treasury at the Royal Bank of Scotland.

David Moskovic has been trading hybrid derivatives at The Royal Bank of Scotland since December 2009. Prior to that he worked in market risk and as a quantitative analyst. He qualified as a Chartered Accountant at Ernst & Young before moving to RBS. David obtained a Masters in Physics at Queens' College, Cambridge, achieving first class honours. He is ranked amongst the top 100 chess players in the country.

Max Wong is a risk professional with 18 years of experience in financial services, and author of *Bubble Value at Risk: A Countercyclical Risk Management Approach*. He was an open outcry local trader at Simex futures exchange during the Asian crisis (1998) and a quant risk manager during the global financial crisis (2008). He is currently Head of Risk Model Testing at Royal Bank of Scotland in Singapore. He holds a BSc in Physics and MSc in Financial Engineering.

If you want to know the reasons for
The things I feel inside
You can run away and hide if you want to
But if you need me
Show me a way to get through
Cause I've not see anyone like you
Since I was a man

If you want to know for certain
What is written in my soul
Where my wildest dreams unfold
Like an endless stream
If you need me
Show me a way to walk tall
Cause I'd not seen anything at all
Until I met you

—*If You Need Me, Gordon Lightfoot © 1980*

Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

PART One

Introduction to Bonds

In Part One, we describe the key concepts in fixed-income market analysis, which cover the basics of the bond instrument. The building blocks described here are generic and are applicable in any market. The analysis is simplest when restricted to plain-vanilla *default-free* bonds; as the instruments become more complex, we are required to introduce additional techniques and assumptions. Part One comprises five chapters. We begin with bond pricing and yield, followed by traditional interest-rate risk measures such as modified duration and convexity. This is followed by a look at spot and forward rates, the derivation of such rates from market yields, and the concept of the yield curve. Yield-curve analysis and the modelling of the term structure of interest rates is one of the most heavily researched areas of financial economics. The treatment here is kept as concise as possible, which sacrifices some detail, but bibliographies at the end of each chapter will direct interested readers to what are the most accessible and readable references in this area.

While we do not describe specifics of particular markets, it is important to remember that the general concepts discussed here are pertinent to debt markets in every jurisdiction.

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CHAPTER 1

The Bond Instrument

Bonds are debt-capital market instruments that represent a cash flow payable during a specified time period heading into the future. This cash flow represents the interest payable on the loan and the loan redemption. So, essentially, a bond is a loan, albeit one that is tradable in a secondary market. This differentiates bond-market securities from commercial bank loans.

In the analysis that follows, bonds are assumed to be default-free, which means that there is no possibility that the interest payments and principal repayment will not be made. Such an assumption is reasonable when one is referring to government bonds such as U.S. Treasuries, UK gilts, Japanese JGBs, and so on. However, it is unreasonable when applied to bonds issued by corporates or lower-rated sovereign borrowers. Nevertheless, it is still relevant to understand the valuation and analysis of bonds that are default-free, as the pricing of bonds that carry default risk is based on the price of risk-free securities. Essentially, the price investors charge borrowers that are not of risk-free credit standing is the price of government securities plus some credit risk premium.

BOND-MARKET BASICS

All bonds are described in terms of their issuer, maturity date, and coupon. For a default-free conventional, or plain-vanilla, bond, this will be the essential information required. Nonvanilla bonds are defined by further characteristics such as their interest basis, flexibilities in their maturity date, credit risk, and so on.

Figure 1.1 shows screen DES from the Bloomberg system. This page describes the key characteristics of a bond. From Figure 1.1, we see a description of a bond issued by the Singapore government, the 4.625% of 2010. This tells us the following bond characteristics:

Issue date	July 2000
Coupon	4.625%
Maturity date	1 July 2010
Issue currency	Singapore dollars
Issue size	SGD 3.4 million
Credit rating	AAA/Aaa

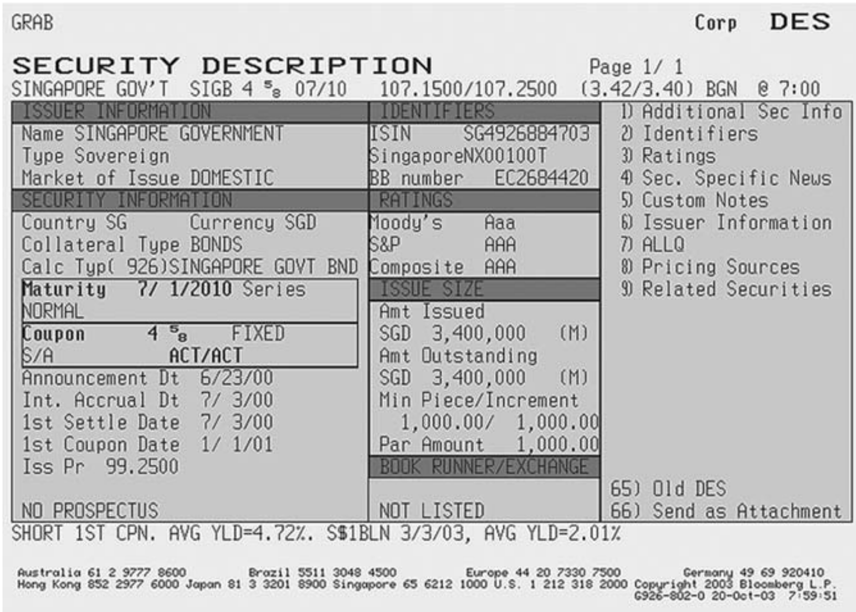

FIGURE 1.1 Bloomberg Screen DES Showing Details of $4\frac{5}{8}$ % 2010 Issued by Republic of Singapore as of 20 October 2003

Calling up screen DES for any bond, provided it is supported by Bloomberg, will provide us with its key details. Later on, we will see how nonvanilla bonds include special features that investors take into consideration in their analysis.

We will consider the essential characteristics of bonds later in this chapter. First, we review the capital market, and an essential principle of finance, the time value of money.

CAPITAL MARKET PARTICIPANTS

The debt capital markets exist because of the financing requirements of governments and corporates. The source of capital is varied, but the total supply of funds in a market is made up of personal or household savings, business savings, and increases in the overall money supply. Growth in the money supply is a function of the overall state of the economy, and interested readers may wish to consult the references at the end of this chapter, which include several standard economic texts. Individuals save out of their current income for future consumption, while business savings represent retained earnings. The entire savings stock represents the capital available in a market. The requirements of savers and borrowers differ significantly, in that savers have a short-term investment horizon while borrowers prefer to take a longer-term view. The constitutional weakness of what would otherwise be *unintermediated* financial markets led, from an early stage, to the development of financial intermediaries.

Financial Intermediaries

In its simplest form a financial intermediary is a *broker* or *agent*. Today we would classify the broker as someone who acts on behalf of the borrower or lender, buying or selling a bond as instructed. However, intermediaries originally acted between borrowers and lenders in placing funds as required. A broker would not simply on-lend funds that have been placed with it, but would accept deposits and make loans as required by its customers. This resulted in the first banks. A *retail bank* deals mainly with the personal financial sector and small businesses, and in addition to loans and deposits also provides cash transmission services. A retail bank is required to maintain a minimum cash reserve, to meet potential withdrawals, but the remainder of its deposit base can be used to make loans. This does not mean that the total size of its loan book is restricted to what it has taken in deposits: loans can also be funded in the wholesale market. An *investment bank* will deal with governments, corporates, and institutional investors. Investment banks perform an agency role for their customers, and are the primary vehicle through which a corporate will borrow funds in the bond markets. This is part of the bank's corporate finance function; it will also act as wholesaler in the bond markets, a function known as *market making*. The bond-issuing function of an investment bank, by which the bank will issue bonds on behalf of a customer and pass the funds raised to this customer, is known as *origination*. Investment banks will also carry out a range of other functions for institutional customers, including export finance, corporate advisory, and fund management.

Other financial intermediaries will trade not on behalf of clients but for their own *book*. These include *arbitrageurs* and speculators. Usually such market participants form part of investment banks.

Investors

There is a large variety of players in the bond markets, each trading some or all of the different instruments available to suit their own purposes. We can group the main types of investors according to the time horizon of their investment activity.

Short-term institutional investors. These include banks and building societies, money-market fund managers, central banks, and the treasury desks of some types of corporates. Such bodies are driven by short-term investment views, often subject to close guidelines, and will be driven by the total return available on their investments. Banks will have an additional requirement to maintain *liquidity,* often in fulfilment of regulatory authority rules, by holding a proportion of their assets in the form of easily tradable short-term instruments.

Long-term institutional investors. Typically these types of investors include pension funds and life assurance companies. Their investment horizon is long term, reflecting the nature of their liabilities; often they will seek to match these liabilities by holding long-dated bonds.

Mixed horizon institutional investors. This is possibly the largest category of investors and will include general insurance companies, most corporate bodies, and sovereign wealth funds. Like banks and financial-sector companies, they are also very active in the primary market, issuing bonds to finance their operations.

Market professionals. This category includes the banks and specialist financial intermediaries mentioned earlier, firms that one would not automatically classify as "investors" although they will also have an investment objective. Their time horizon will range from one day to the very long term. Proprietary traders will actively position themselves in the market in order to gain trading profit, for example in response to their view on where they think interest rate levels are headed. These participants will trade directly with other market professionals and investors, or via brokers. Market makers or *traders* (called *dealers* in the United States) are wholesalers in the bond markets; they make two-way prices in selected bonds. Firms will not necessarily be active market makers in all types of bonds; smaller firms often specialise in certain sectors. In a two-way quote the *bid price* is the price at which the market maker will buy stock, so it is the price the investor will receive when selling stock. The *offer price* or *ask price* is the price at which investors can buy stock from the market maker. As one might expect, the bid price is always higher than the offer price, and it is this *spread* that represents the theoretical profit to the market maker. The bid-offer spread set by the market maker is determined by several factors, including supply and demand, and liquidity considerations for that particular stock, the trader's view on market direction and *volatility* as well as that of the stock itself and the presence of any market intelligence. A large bid-offer spread reflects low liquidity in the stock, as well as low demand.

As mentioned earlier, *brokers* are firms that act as intermediaries between buyers and sellers and between market makers and buyers/sellers. Floor-based stock exchanges such as the New York Stock Exchange (NYSE) also feature *specialists,* members of the exchange who are responsible for maintaining an orderly market in one or more securities. These are known as *locals* on the London International Financial Futures and Options Exchange (LIFFE). Locals trade securities for their own account to counteract a temporary imbalance in supply and demand in a particular security; they are an important source of *liquidity* in the market. Locals earn income from brokerage fees and also from pure trading, when they sell securities at a higher price than the original purchase price.

Markets

Markets are that part of the financial system where capital market transactions, including the buying and selling of securities, takes place. A market can describe a traditional stock exchange; that is, a physical trading floor where securities trading occurs. Many financial instruments are traded over the telephone or electronically; these markets are known as *over-the-counter* (OTC) markets. A distinction is made between financial instruments of up to one year's maturity and instruments of over one year's maturity. Short-term instruments make up the *money market* while all other instruments are deemed to be part of the *capital market*. There is also a distinction made between the *primary market* and the *secondary market*. A new issue of bonds made by an investment bank on behalf of its client is made in the primary market. Such an issue can be a *public* offer, in which anyone can apply to buy the bonds, or a *private* offer where the customers of the investment bank are offered the stock. The secondary market is the market in which existing bonds and shares are subsequently traded.

WORLD BOND MARKETS

The origin of the spectacular increase in the size of global financial markets was the rise in oil prices in the early 1970s. Higher oil prices stimulated the development of a sophisticated international banking system, as they resulted in large capital inflows to developed country banks from the oil-producing countries. A significant proportion of these capital flows were placed in *eurodollar* deposits in major banks. The growing trade deficit and level of public borrowing in the United States also contributed. The past 20 years has seen tremendous growth in capital markets volumes and trading. As capital controls were eased and exchange rates moved from fixed to floating, domestic capital markets became internationalised. Growth was assisted by the rapid advance in information technology and the widespread use of financial engineering techniques. Today we would think nothing of dealing in virtually any liquid currency bond in financial centres around the world, often at the touch of a button. Global bond issues, underwritten by the subsidiaries of the same banks, are commonplace. The ease with which transactions can be undertaken has also contributed to a very competitive market in liquid currency assets.

The world bond market has increased in size more than 15 times in the past 30 years. As at the end of 2013, outstanding volume stood at over $25 trillion.

The market in U.S. Treasury securities is the largest bond market in the world. Like the government bond markets in the United Kingdom, Germany, France, and other developed economies, it also very liquid and transparent. Of the major government bond markets in the world, the U.S. market makes up nearly half of the total. The Japanese market is second in size, followed by the German market. A large part of the government bond market is concentrated therefore in just a few countries. Government bonds are traded on major exchanges as well as *over-the-counter* (OTC). Generally OTC refers to trades that are not carried out on an exchange but directly between the counterparties. Bonds are also listed on exchanges.

Companies finance their operations in a number of ways, from equity to short-term debt such as bank overdrafts. It is often advantageous for companies to fix longer-term finance, which is why bonds are so popular. Bonds are also attractive as a means of raising finance because the interest payable on them to investors is tax deductible for the company. Dividends on equity are not tax deductible. A corporate needs to get a reasonable mix of debt versus equity in its funding however, as a high level of interest payments will be difficult to service in times of recession or general market downturn. For this reason the market views unfavourably companies that have a high level of debt. Corporate bonds are also traded on exchanges and OTC. One of the most liquid corporate bond types is the *Eurobond,* which is an international bond issued and traded across national boundaries. Sovereign governments have also issued Eurobonds.

OVERVIEW OF THE MAIN BOND MARKETS

So far we have established that bonds are debt capital market instruments, which means that they represent loans taken out by governments and corporations. The duration of any particular loan will vary from 2 years to 30 years or longer. In this chapter we introduce just a small proportion of the different bond instruments that trade in the market, together with a few words on different country markets. This will set the scene for later chapters, where we look at instruments and markets in greater detail.

Domestic and International Bonds

In any market there is a primary distinction between *domestic* bonds and other bonds. Domestic bonds are issued by borrowers domiciled in the country of issue, and in the currency of the country of issue. Generally they trade only in their original market. A *Eurobond* is issued across national boundaries and can be in any currency, which is why they are also sometimes called *international* bonds. It is now more common for Eurobonds to be referred to as international bonds, to avoid confusion with "euro bonds", which are bonds denominated in *euros,* the currency of 17 countries of the European Union (EU). As an issue of Eurobonds is not restricted in terms of currency or country, the borrower is not restricted as to its nationality either. There are also *foreign* bonds, which are domestic bonds issued by foreign borrowers. An example of a foreign bond is a *Bulldog,* which is a sterling bond issued for trading in the United Kingdom (UK) market by a foreign borrower. The equivalent foreign bonds in other countries include *Yankee* bonds (United States), *Samurai* bonds (Japan), *Alpine* bonds (Switzerland) and *Matador* bonds (Spain).

There are detail differences between these bonds, for example in the frequency of interest payments that each one makes and the way the interest payment is calculated. Some bonds such as domestic bonds pay their interest *net,* which means net of a withholding tax such as income tax. Other bonds including Eurobonds make *gross* interest payments.

Government Bonds As their name suggests, government bonds are issued by a government or *sovereign.* Government bonds in any country form the foundation for the entire domestic debt market. This is because the government market will be the largest in relation to the market as a whole. Government bonds also represent the best *credit risk* in any market as people generally do not expect the government to go bankrupt. As we see in a later chapter, professional institutions that analyse borrowers in terms of their credit risk always rate the government in any market as the highest credit available. While this may sometimes not be the case, it is usually a good rule of thumb.[1] The government bond market is usually also the most *liquid* in the domestic market due to its size and will form the benchmark against which other borrowers are rated. Generally, but not always, the yield offered on government debt will be the lowest in that market.

United States Government bonds in the United States are known as *Treasuries.* Bonds issued with an original maturity of between 2 and 10 years are known as *notes* (as in "Treasury note") while those issued with an original maturity of over

[1] Occasionally one may come across a corporate entity that one may view as better rated in terms of credit risk compared to the government of the country in which the company is domiciled. However, the main credit rating agencies will not rate a corporate entity at a level higher than its country of domicile.

10 years are known as *bonds*. There is no difference between notes and bonds, and they trade the same way in the market. Treasuries pay semiannual coupons. The U.S. Treasury market is the largest single bond market anywhere and trades on a 24-hour basis all around the world. A large proportion of Treasuries are held by foreign governments and corporations. It is a very liquid and *transparent* market.

United Kingdom The UK government issues bonds known as *gilt-edged securities* or *gilts*.[2] The gilt market is another very liquid and transparent market, with prices being very competitive. Many of the more esoteric features of gilts such as "tick" pricing (where prices are quoted in 32nds and not decimals) and *special ex-dividend* trading have been removed in order to harmonise the market with other European Union sovereign bonds. Gilts still pay coupon on a semiannual basis though, unlike eurozone paper. The UK government also issues bonds known as *index-linked* gilts, whose interest and redemption payments are linked to the rate of inflation. There are also older gilts with peculiar features such as no redemption date and quarterly paid coupons.

Germany Government bonds in Germany are known as *bunds,* BOBLs, or *Schatze*. These terms refer to the original maturity of the paper and have little effect on trading patterns. Bunds pay coupon on an annual basis and are of course now denominated in euros.

Table 1.1 summarises the main characteristics of a selected sample of sovereign bond markets. A set of sovereign yield curve data as of December 2013 is given in Table 1.2.

Nonconventional Bonds

The definition of bonds given earlier in this chapter referred to conventional or *plain-vanilla* bonds. There are many variations on vanilla bonds and we can introduce a few of them here.

Floating rate notes. The bond market is often referred to as the *fixed income* market, or the *fixed interest* market in the United Kingdom. Floating rate notes (FRNs) do not have a fixed coupon but instead link their interest payments to an external reference, such as the three-month bank lending rate. Bank interest rates will fluctuate constantly during the life of the bond and so an FRN's cash flows are not known with certainty. Usually FRNs pay a fixed margin or *spread* over the specified reference rate; occasionally the spread is not fixed and such a bond is known as a *variable rate note*.

Because FRNs pay coupons based on the three-month or six-month bank rate, they are essentially money-market instruments and are treated by bank dealing desks as such.

Index-linked bonds. An index-linked bond has its coupon and redemption payment, or possibly just either one of these, linked to a specified index. When governments issue index-linked bonds the cash flows are linked to a price index such as consumer or commodity prices. Corporates have issued index-linked bonds that are connected to inflation or a stock market index.

[2] This is because early gilt issues are said to have been represented by certificates that were edged with gold leaf, hence the term gilt-edged. In fact the story is almost certainly apocryphal, and it is unlikely that gilt certificates were ever edged with gold.

TABLE 1.1 Selected Government Bond Market Characteristics

	Credit Rating	Maturity Range	Dealing Mechanism	Benchmark Bonds	Issuance	Coupon and Day-Count Basis
Australia	AAA	2–15 years	OTC Dealer network	5, 10 years	Auction	Semiannual, act/act
Canada	AAA	2–30 years	OTC Dealer network	3, 5, 10 years	Auction, subscription	Semiannual, act/act
France	AAA	BTAN: 1–7 years OAT: 10–30 years	OTC Dealer network Bonds listed on Paris Stock Exchange	BTAN: 2, 5 years OAT: 10, 30 years	Dutch auction	BTAN: Semiannual, act/act OAT: Annual, act/act
Germany	AAA	OBL: 2, 5 years BUND: 10, 30 years	OTC Dealer network Listed on Stock Exchange	The most recent issue	Combination of Dutch auction and proportion of each issue allocated on fixed basis to institutions	Annual, act/act
South Africa	A	2–30 years	OTC Dealer network Listed on Johannesburg SE	2, 7, 10, 20 years	Auction	Semiannual, act/365
Singapore	AAA	2–15 years	OTC Dealer network	1, 5, 10, 15 years	Auction	Semiannual, act/act
Taiwan	AA–	2–30 years	OTC Dealer network	2, 5, 10, 20, 30 years	Auction	Annual, act/act
United Kingdom	AAA	2–50 years	OTC Dealer network	5, 10, 30 years	Auction, subsequent issue by “tap” subscription	Semiannual, act/act
United States	AAA	2–20 years	OTC Dealer network	2, 5, 10 years	Auction	Semiannual, act/act

TABLE 1.2 Selected Government Bond Markets, Yield Curves as at 2 December 2013

Term (years)	Australia	Germany	Japan	United Kingdom	United States
1				0.43	0.110
2	2.74	0.12	0.08	0.48	0.250
3					
4					
5	3.49	0.66	0.18	1.58	1.251
7					
10	4.32	1.7	0.61	2.83	2.753
15	4.63			3.17	
20				3.36	
30		2.62	1.65	3.64	3.756

Zero-coupon bonds. Certain bonds do not make any coupon payments at all, and these are known as *zero-coupon bonds*. A zero-coupon bond or *strip* has only one cash flow, the redemption payment at maturity. If we assume that the maturity payment is 100% or *par*, the issue price will be at a discount to par. Such bonds are also known therefore as *discounted* bonds. The difference between the price paid on issue and the redemption payment is the interest realised by the bondholder. As we will discover when we look at strips, this has certain advantages for investors, the main one being that there are no coupon payments to be invested during the bond's life. Both governments and corporates issue zero-coupon bonds. Conventional coupon-bearing bonds can be *stripped* into a series of individual cash flows, which would then trade as separate zero-coupon bonds. This is a common practice in government bond markets where the borrowing authority does not actually issue strips, and they have to be created via the stripping process.

Amortised bonds. A conventional bond will repay on maturity the entire nominal sum initially borrowed on issue. This is known as a *bullet* repayment (which is why vanilla bonds are sometimes known as bullet bonds). A bond that repays portions of the borrowing in stages during its life is known as an *amortised* bond.

Bonds with embedded options. Some bonds include a provision in their offer particulars that gives either the bondholder and/or the issuer an option to enforce early redemption of the bond. The most common type of option embedded in a bond is a *call feature*. A call provision grants the issuer the right to redeem all or part of the debt before the specified maturity date. An issuing company may wish to include such a feature as it allows it to replace an old bond issue with a lower coupon rate issue if interest rates in the market have declined. As a call feature allows the issuer to change the maturity date of a bond, it is considered harmful to the bondholder's interests; therefore the market price of the bond at any time will reflect this. A bond issue may also include a provision that allows the investor to change the maturity of the bond. This is known as a *put feature* and gives the bondholder the right to sell the bond back to the issuer at par on specified dates. The advantage to the bondholder is that if interest rates rise after the issue date, thus depressing the bond's value, the investor can realise par value by *putting* the bond back to the issuer. A *convertible* bond is an issue giving the bondholder the right to exchange the bond for a specified amount of shares (equity) in the issuing company. This feature allows the investor to take advantage of favorable movements in the price of the issuer's shares. The presence of embedded options in a bond makes valuation more complex compared to plain-vanilla bonds, and will be considered separately.

Bond warrants. A bond may be issued with a *warrant* attached to it, which entitles the bondholder to buy more of the bond (or a different bond issued by the same borrower) under specified terms and conditions at a later date. An issuer may include a warrant in order to make the bond more attractive to investors. Warrants are often detached from their host bond and traded separately.

Finally there is a large class of bonds known as *asset-backed securities*. These are bonds formed from pooling together a set of loans such as mortgages or car loans and issuing bonds against them. The interest payments on the original loans serve to back the interest payable on the asset-backed bond. We will look at these instruments in some detail in a later chapter.

TIME VALUE OF MONEY

The principles of pricing in the bond market are exactly the same as those in other financial markets, which state that the price of any financial instrument is equal to the present (today's) value of all the future cash flows from the instrument. Bond prices are expressed as per 100 nominal of the bond, or "percent". So for example, if the price of a U.S. dollar–denominated bond is quoted as "98.00", this means that for every \$100 nominal of the bond a buyer would pay \$98.[3] The interest rate or discount rate used as part of the present value (price) calculation is key, as it reflects where the bond is trading in the market and how it is perceived by the market. All the determining factors that identify the bond, including the nature of the issuer, the maturity, the coupon and the currency, influence the interest rate at which a bond's cash flows are discounted, which will be similar to the rate used for comparable bonds. First, we consider the traditional approach to bond pricing for a plain-vanilla instrument, making certain assumptions to keep the analysis simple, and then we present the more formal analysis commonly encountered in academic texts.

Present Value 101

Bonds or *fixed-income*[4] instruments are debt-capital market securities and therefore have maturities longer than one year. This differentiates them from money-market

[3] The convention in certain markets is to quote a price per 1,000 nominal, but this is less common.

[4] The term "fixed income" originated at a time when bonds were essentially plain-vanilla instruments paying a fixed coupon per year. In the United Kingdom, the term "fixed interest" was used. These days, many bonds do not necessarily pay a fixed coupon each year, for instance, asset-backed bond issues were invariably issued in a number of tranches, with each tranche paying a different fixed or floating coupon. The market is still commonly referred to as the fixed-income market, however.

securities. Bonds have more intricate cash flow patterns than money-market securities, which usually have just one cash flow at maturity. This makes bonds more involved to price than money-market instruments, and their prices more responsive to changes in the general level of interest rates. There is a large variety of bonds. The most common type is the plain-vanilla (or *straight, conventional,* or *bullet*) bond. This is a bond paying a regular (annual or semiannual) fixed interest payment or coupon over a fixed period to maturity or redemption, with the return of principal (the par or nominal value of the bond) on the maturity date. All other bonds are variations on this.

The key identifying feature of a bond is its issuer, the entity that is borrowing funds by issuing the bond into the market. Issuers are generally categorised as one of four types: governments (and their agencies), local governments (or municipal authorities), supranational bodies such as the World Bank, and corporates. Within the municipal and corporate markets there is a wide range of issuers, each assessed as having differing abilities to maintain the interest payments on their debt and repay the full loan on maturity. This ability is identified by a *credit rating* for each issuer. The *term to maturity* of a bond is the number of years over which the issuer has promised to meet the conditions of the debt obligation. The maturity of a bond refers to the date that the debt will cease to exist, at which time the issuer will redeem the bond by paying the principal. The practice in the bond market is to refer to the "term to maturity" of a bond as simply its "maturity" or "term". Some bonds contain provisions that allow either the issuer or the bondholder to alter a bond's term. The term to maturity of a bond is its other key feature. First it indicates the time period over which the bondholder can expect to receive coupon payments and the number of years before the principal is paid back. Secondly, it influences the yield of a bond. Finally, the price of a bond will fluctuate over its life as yields in the market change. The volatility of a bond's price is dependent on its maturity. All else being equal, the longer the maturity of a bond, the greater its price volatility resulting from a change in market interest rates.

The principal of a bond is the amount that the issuer agrees to repay the bondholder on maturity. This amount is also referred to as the *redemption value, maturity value, par value,* or *face value.* The coupon rate, or nominal rate, is the interest rate that the issuer agrees to pay each year during the life of the bond. The annual amount of interest payment made to bondholders is the coupon. The cash amount of the coupon is the coupon rate multiplied by the principal of the bond. For example, a bond with a coupon rate of 8% and a principal of $1,000 will pay annual interest of $80. In the United States, the usual practice is for the issuer to pay the coupon in two semiannual installments. All bonds make periodic coupon payments, except for zero-coupon bonds. Such bonds are issued at a discount and redeemed at par. The holder of a zero-coupon bond realises interest by buying the bond at this discounted value, below its principal value. Interest is therefore paid on maturity, with the exact amount being the difference between the principal value and the discounted value paid on purchase.

Present Value and Discounting

As fixed-income instruments are essentially a collection of cash flows, we begin by reviewing the key concept in cash-flow analysis, that of discounting and *present* *value*. It is essential to have a firm understanding of the main principles of this before moving on to other areas. When reviewing the concept of the time value of money, assume that the interest rates used are the market-determined rates of interest.

Financial arithmetic has long been used to illustrate that £1 received today is not the same as £1 received at a point in the future. Faced with a choice between receiving £1 today or £1 in one year's time, we would be indifferent given a rate of interest of, say, 10% that was equal to our required nominal rate of interest. Our choice would be between £1 today or £1 plus 10p—the interest on £1 for one year at 10% per annum. The notion that money has a time value is a basic concept in the analysis of financial instruments. Money has time value because of the opportunity to invest it at a rate of interest. A loan that has one interest payment on maturity is accruing simple interest. On short-term instruments, there is usually only the one interest payment on maturity, hence simple interest is received when the instrument expires. The terminal value of an investment with simple interest is given by (1.1).

$$F = P(1 + r) \tag{1.1}$$

where F is the terminal value or future value
P is the initial investment or present value
r is the interest rate.

The market convention is to quote interest rates as annualised interest rates, which is the interest that is earned if the investment term is one year. Consider a three-month deposit of \$100 in a bank, placed at a rate of interest of 6%. In such an example, the bank deposit will earn 6% interest for a period of 90 days. As the annual interest gain would be \$6, the investor will expect to receive a proportion of this, which is calculated as follows:

$$\$6.00 \times 90/360$$

Therefore, the investor will receive \$1.50 interest at the end of the term. The total proceeds after the three months is therefore \$100 plus \$1.50. Note, we use 90/360 as that is the convention in the U.S. markets. For a small number of currencies, including Hong Kong dollars and Sterling, a 365-day denominator is used. If we wish to calculate the terminal value of a short-term investment that is accruing simple interest, we use the following expression:

$$F = P(1 + r \times \text{days/year}) \tag{1.2}$$

The fraction days/year refers to the numerator, which is the number of days the investment runs, divided by the denominator which is the number of days in the year. The convention in most markets (including the dollar and euro markets) is to have a 360-day year. In the sterling markets, the number of days in the year is taken to be 365. For this reason, we simply quote the expression as "days" divided by "year" to allow for either convention.

Let us now consider an investment of \$100 made for three years, again at a rate of 6%, but this time fixed for three years. At the end of the first year, the investor will be credited with interest of \$6. Therefore for the second year the interest rate of 6% will be accruing on a principal sum of \$106, which means that at the end of year 2 the interest credited will be \$6.36. This illustrates how compounding works, which is the principle of earning interest on interest. The outcome of the process of compounding is the future value of the initial amount. The expression is given in (1.3).

$$FV = PV(1+r)^n \tag{1.3}$$

where FV is the future value
PV is initial outlay or *present value*
r is the periodic rate of interest (expressed as a decimal)
n is the number of periods for which the sum is invested.

When we compound interest, we have to assume that the reinvestment of interest payments during the investment term is at the same rate as the first year's interest. That is why we stated that the 6% rate in our example was fixed for three years. We can see, however, that compounding increases our returns compared to investments that accrue only on a simple-interest basis.

Now let us consider a deposit of \$100 for one year, at a rate of 6% but with quarterly interest payments. Such a deposit would accrue interest of \$6 in the normal way, but \$1.50 would be credited to the account every quarter, and this would then benefit from compounding. Again assuming that we can reinvest at the same rate of 6%, the total return at the end of the year will be:

$$100\times[(1+0.015)\times(1+0.015)\times(1+0.015)\times(1+0.015)] = 100\times[(1+0.015)^4]$$

which gives us 100 × 1.06136, a terminal value of \$106.136. This is some 13 cents more than the terminal value using annual compounded interest.

In general, if compounding takes place m times per year, then at the end of n years mn interest payments will have been made and the future value of the principal is given by (1.4).

$$FV = PV(1+r/m)^{mn} \tag{1.4}$$

As we showed in our example, the effect of more frequent compounding is to increase the value of the total return when compared to annual compounding. The effect of more frequent compounding is shown in the following table, where we consider the annualised interest-rate factors, for an annualised rate of 6%.

$$\text{Interest-rate factor} = (1 + r/m)^{m}$$

Compounding Frequency		*Interest-Rate Factor*
Annual	$(1 + r)$	= 1.060000
Semiannual	$(1 + r/2)^2$	= 1.060900
Quarterly	$(1 + r/4)^4$	= 1.061364
Monthly	$(1 + r/12)^{12}$	= 1.061678
Daily	$(1 + r/365)^{365}$	= 1.061831

This shows us that the more frequent the compounding, the higher the interest-rate factor. The last case also illustrates how a limit occurs when interest is compounded continuously. Equation 1.4 can be rewritten as follows:

$$\begin{aligned} FV &= PV\left[\left(1+\frac{r}{m}\right)^{m/r}\right]^{rn} \\ &= PV\left[\left(1+\frac{1}{m/r}\right)^{m/r}\right]^{rn} \\ &= PV\left[\left(1+\frac{1}{k}\right)^{k}\right]^{rn} \end{aligned} \tag{1.5}$$

where $k = m/r$. As compounding becomes continuous and m and hence n approach infinity, the expression in the square brackets in (1.5) approaches a value known as e, which is shown in the following equation.

$$e = \lim_{K\to\infty}\left(1+\frac{1}{k}\right)^{k} = 2.718281\ldots$$

If we substitute this into (1.5) this gives us:

$$FV = PVe^{rn} \tag{1.6}$$

where we have continuous compounding. In (1.6), e^{rn} is known as the exponential function of rn, and it tells us the continuously compounded interest rate factor. If $r = 6\%$ and $n = 1$ year then:

$$e^{r} = (2.718281)^{0.06} = 1.061837$$

This is the limit reached with continuous compounding.

The convention in both wholesale and personal (retail) markets is to quote an annual interest rate. A lender who wishes to earn the interest at the rate quoted has to place her funds on deposit for one year. Annual rates are quoted irrespective of the maturity of a deposit, from overnight to 50 years. For example if one opens a bank account that pays interest at a rate of 3.5% but then closes it after six months, the actual interest earned will be equal to 1.75% of the sum deposited.

The actual return on a three-year building society bond (fixed deposit) that pays 6.75% fixed for three years is 21.65% after three years. The quoted rate is the annual one-year equivalent. An overnight deposit in the wholesale or interbank market is still quoted as an annual rate, even though interest is earned for only one day.

The convention of quoting annualised rates is to allow deposits and loans of different maturities and different instruments to be compared on the basis of the interest rate applicable. We must be careful when comparing interest rates for products that have different payment frequencies. As we have seen from the foregoing paragraphs, the actual interest earned will be greater for a deposit earning 6% on a semiannual basis than one earning 6% on an annual basis. The convention in the money markets is to quote the equivalent interest rate applicable when taking into account an instrument's payment frequency.

We saw how a future value could be calculated given a known present value and rate of interest. For example, \$100 invested today for one year at an interest rate of 6% will generate $100 \times (1 + 0.06) = \$106$ at the end of the year. The future value of \$100 in this case is \$106. We can also say that \$100 is the present value of \$106 in this case.

In (1.3) we established the following future-value relationship:

$$FV = PV\,(1 + r)^n$$

By reversing this expression we arrive at the present-value calculation given in (1.7).

$$PV = \frac{FV}{(1+r)^n} \tag{1.7}$$

where the symbols represent the same terms as before. Equation 1.7 applies in the case of annual interest payments and enables us to calculate the present value of a known future sum.

To calculate the present value for a short-term investment of less than one year, we will need to adjust what would have been the interest earned for a whole year by the proportion of days of the investment period. Rearranging the basic equation, we can say that the present value of a known future value is:

$$PV = \frac{FV}{\left(1 + r \times \dfrac{\text{days}}{\text{year}}\right)} \tag{1.8}$$

Given a present value and a future value at the end of an investment period, what then is the interest rate earned? We can rearrange the basic equation again to solve for the *yield*.

When interest is compounded more than once a year, the formula for calculating present value is modified, in shown (1.9).

$$PV = \frac{FV}{\left(1 + \dfrac{r}{m}\right)^{mn}} \tag{1.9}$$

where, as before, FV is the cash flow at the end of year n, m is the number of times a year interest is compounded, and r is the rate of interest or discount rate. Illustrating this, therefore, the present value of \$100 that is received at the end of five years at a rate of interest rate of 5%, with quarterly compounding is:

$$
\begin{aligned}
PV &= \frac{100}{\left(1+\frac{0.05}{4}\right)^{(4)(5)}} \\
&= \$78.00
\end{aligned}
$$

Interest rates in the money markets are always quoted for standard maturities—for example, overnight, tom next (the overnight interest rate starting tomorrow, or "tomorrow to the next"), spot next (the overnight rate starting two days forward), one week, one month, two months, and so on, up to one year. If a bank or corporate customer wishes to deal for nonstandard periods, an interbank desk will calculate the rate chargeable for such an "odd date" by interpolating between two standard-period interest rates. If we assume that the rate for all dates in between two periods increases at the same steady state, we can calculate the required rate using the formula for straight-line interpolation, shown in (1.10).

$$r = r_1 + (r_2 - r_1) \times \frac{n - n_1}{n_2 - n_1} \tag{1.10}$$

where r is the required odd-date rate for n days
r_1 is the quoted rate for n_1 days
r_2 is the quoted rate for n_2 days

Let us imagine that the one-month (30-day) offered interest rate is 5.25% and that the two-month (60-date) offered rate is 5.75%. If a customer wishes to borrow money for a 40-day period, what rate should the bank charge? We can calculate the required 40-day rate using the straight-line interpolation process. The increase in interest rates from 30 to 40 days is assumed to be 10/30 of the total increase in rates from 30 to 60 days. The 40-day offered rate would therefore be:

$$5.25\% + (5.75\% - 5.25\%) \times 10/30 = 5.4167\%$$

What about the case of an interest rate for a period that lies just before or just after two known rates and not roughly in between them? When this happens we extrapolate between the two known rates, again assuming a straight-line relationship between the two rates and for a period after (or before) the two rates. So if the one-month offered rate is 5.25% while the two-month rate is 5.75%, the 64-day rate is:

$$5.25 + (5.75 - 5.25) \times 34/30 = 5.8167\%$$

Discount Factors

An *n*-period discount factor is the present value of one unit of currency (£1 or $1) that is payable at the end of period *n*. Essentially, it is the present-value relationship expressed in terms of $1. If *d(n)* is the *n*-year discount factor, then the five-year discount factor at a discount rate of 6% is given by

$$d(5) = \frac{1}{(1+0.06)^5} = 0.747258$$

The set of discount factors for every time period from one day to 30 years or longer is termed the *discount function*. Discount factors may be used to price any financial instrument that is made up of a future cash flow. For example, what would be the value of $103.50 receivable at the end of six months if the six-month discount factor is 0.98756? The answer is given by:

$$0.98756 \times 103.50 = 102.212$$

In addition, discount factors may be used to calculate the future value of any present investment. From the earlier example, $0.98756 would be worth $1 in six months' time, so by the same principle a present sum of $1 would be worth

$$1/d(0.5) = 1/0.98756 = 1.0126$$

at the end of six months.

It is possible to obtain discount factors from current bond prices. Assume a hypothetical set of bonds and bond prices at the date of 7 Dec 2000 as given in Table 1.3, and assume further that the first bond in the table matures in precisely six months' time (these are semiannual coupon bonds).

Taking the first bond, this matures in precisely six months' time, and its final cash flow will be 103.0, comprising the $3.50 final coupon payment and the $100 redemption payment. The price or present value of this bond is 101.65, which allows us to calculate the six-month discount factor as:

$$d(0.5) \times 103.50 = 101.65$$

which gives $d(0.5)$ equal to 0.98213.

TABLE 1.3 Hypothetical Set of Bonds and Bond Prices at the Date of 7 Dec 2000

Coupon	Maturity Date	Price
7%	7-Jun-01	101.65
8%	7-Dec-01	101.89
6%	7-Jun-02	100.75
6.50%	7-Dec-02	100.37

TABLE 1.4 Discount Factors Calculated Using Bootstrapping Technique

Coupon	Maturity Date	Term (years)	Price	*d(n)*
7%	7-Jun-01	0.5	101.65	0.98213
8%	7-Dec-01	1.0	101.89	0.94194
6%	7-Jun-02	1.5	100.75	0.92211
6.50%	7-Dec-02	2.0	100.37	0.88252

From this first step we can calculate the discount factors for the following six-month periods. The second bond in Table 1.3, the 8% 2001, has the following cash flows:

- \$4 in six months' time
- \$104 in one year's time

The price of this bond is 101.89, which again is the bond's present value, and this comprises the sum of the present values of the bond's total cash flows. So we are able to set the following:

$$101.89 = 4 \times d(0.5) + 104 \times d(1)$$

However, we already know $d(0.5)$ to be 0.98213, which leaves only one unknown in the preceding expression. Therefore we may solve for $d(1)$, and this is shown to be 0.94194.

If we carry on with this procedure for the remaining two bonds, using successive discount factors, we obtain the complete set of discount factors as shown in Table 1.4. The continuous function for the two-year period from today is known as the discount function, shown in Figure 1.2.

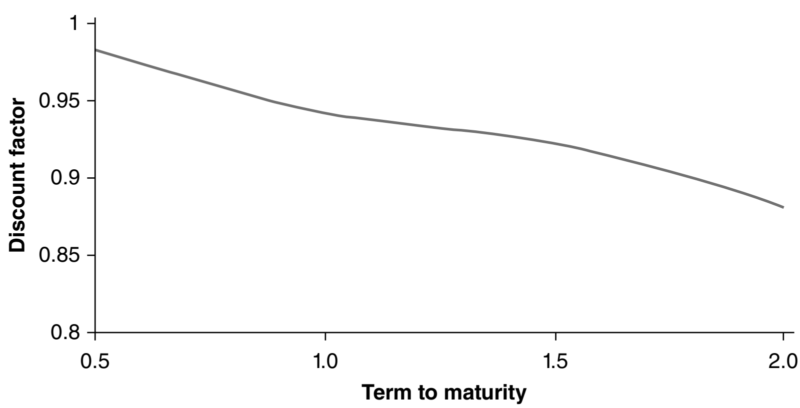

FIGURE 1.2 Hypothetical Discount Function

This technique, which is known as *bootstrapping,* is conceptually neat but presents problems when we do not have a set of bonds that mature at precise six-month intervals. In addition, liquidity issues connected with specific individual bonds can also cause complications. However, it is still worth being familiar with this approach.

Note from Figure 1.2 how discount factors decrease with increasing maturity: this is intuitively obvious, since the present value of something to be received in the future diminishes the further into the future we go.

BOND PRICING AND YIELD: THE TRADITIONAL APPROACH

Bond Pricing

The interest rate that is used to discount a bond's cash flows (and therefore called the discount rate) is the rate required by the bondholder. This is therefore known as the bond's yield. The yield on the bond will be determined by the market and is the price demanded by investors for buying it, which is why it is sometimes called the bond's return. The required yield for any bond will depend on a number of political and economic factors, including what yield is being earned by other bonds of the same class. Yield is always quoted as an annualised interest rate, so that for a bond paying semiannually exactly half of the annual rate is used to discount the cash flows.

The fair price of a bond is the present value of all its cash flows. Therefore, when pricing a bond, we need to calculate the present value of all the coupon interest payments and the present value of the redemption payment, and sum these. The price of a conventional bond that pays annual coupons can therefore be given by (1.11).

$$\begin{aligned} P &= \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \ldots\ldots \frac{C}{(1+r)^N} + \frac{M}{(1+r)^N} \\ &= \sum_{n=1}^{N} \frac{C}{(1+r)^n} + \frac{M}{(1+r)^N} \end{aligned} \tag{1.11}$$

where P is the price
C is the annual coupon payment
r is the discount rate (therefore, the required yield)
N is the number of years to maturity (therefore, the number of interest periods in an annually paying bond)
M is the maturity payment or par value (usually 100% of currency)

Note that (1.11) applies only for fixed coupon bonds where the "recovery rate" (RR) on default of the issuer is zero. In other words, we can only assume it for default-risk free bonds. The RR term will be explained and considered in later chapters.

For long-hand calculation purposes, the first half of (1.11) is usually simplified and is sometimes encountered in one of the two ways shown in (1.12).

$$\sum_{n=1}^{N} \frac{C}{(1+r)^n} = C\left[\frac{1-\left[\frac{1}{(1+r)^N}\right]}{r}\right]$$

or

$$\sum_{n=1}^{N} \frac{C}{(1+r)^n} = \frac{C}{r}\left[1-\frac{1}{(1+r)^N}\right] \tag{1.12}$$

The price of a bond that pays semiannual coupons is given by the expression in (1.13), which is our earlier expression modified to allow for the twice-yearly discounting:

$$\begin{aligned} P &= \frac{C/2}{\left(1+\frac{1}{2}r\right)} + \frac{C/2}{\left(1+\frac{1}{2}r\right)^2} + \frac{C/2}{\left(1+\frac{1}{2}r\right)^3} + \ldots\ldots \frac{C/2}{\left(1+\frac{1}{2}r\right)^{2N}} + \frac{M}{\left(1+\frac{1}{2}r\right)^{2N}} \\ &= \sum_{n=1}^{2N} \frac{C/2}{\left(1+\frac{1}{2}r\right)^n} + \frac{M}{\left(1+\frac{1}{2}r\right)^{2N}} \\ &= \frac{C}{r}\left[1-\frac{1}{\left(1+\frac{1}{2}r\right)^{2N}}\right] + \frac{M}{\left(1+\frac{1}{2}r\right)^{2N}} \end{aligned} \tag{1.13}$$

Note how we set $2N$ as the power to which to raise the discount factor, as there are two interest payments every year for a bond that pays semiannually. Therefore, a more convenient function to use might be the number of interest periods in the life of the bond, as opposed to the number of years to maturity, which we could set as *n*, allowing us to alter the equation for a semiannually paying bond as:

$$P = \frac{C}{r}\left[1-\frac{1}{(1+\frac{1}{2}r)^{2n}}\right] + \frac{M}{(1+\frac{1}{2}r)^{2n}} \tag{1.14}$$

The formula in (1.14) calculates the fair price on a coupon-payment date, so that there is no accrued interest incorporated into the price. It also assumes that there is an even number of coupon-payment dates remaining before maturity. The concept of accrued interest is an accounting convention, and treats coupon interest as accruing every day that the bond is held; this amount is added to the discounted present value of the bond (the *clean* price) to obtain the market value of the bond, known as the *dirty* price.

The date used as the point for calculation is the settlement date for the bond, the date on which a bond will change hands after it is traded. For a new issue of bonds, the settlement date is the day when the stock is delivered to investors and payment is received by the bond issuer. The settlement date for a bond traded in the secondary market is the day when the buyer transfers payment to the seller of the bond and when the seller transfers the bond to the buyer. Different markets will have different settlement conventions. For example, Australian government bonds normally settle two business days after the trade date (the notation used in bond markets is "T + 2"), whereas Eurobonds settle on T + 3. The term *value date* is sometimes used in place of settlement date. However, the two terms are not strictly synonymous. A settlement date can only fall on a business date, so that an Australian government bond traded on a Friday will settle on a Tuesday. However, a value date can sometimes fall on a non-business day; for example, when accrued interest is being calculated.

The standard formula also assumes that the bond is traded for a settlement on a day that is precisely one interest period before the next coupon payment. The price formula is adjusted if dealing takes place in between coupon dates. If we take the value date for any transaction, we then need to calculate the number of calendar days from this day to the next coupon date. We then use the following ratio i when adjusting the exponent for the discount factor:

$$i = \frac{\text{Days from value date to next coupon date}}{\text{Days in the interest period}}$$

The number of days in the interest period is the number of calendar days between the last coupon date and the next one, and it will depend on the day-count basis used for that specific bond. The price formula is then modified as shown in (1.15).

$$P = \frac{C}{(1+r)^{i}} + \frac{C}{(1+r)^{1+i}} + \frac{C}{(1+r)^{2+i}} + \ldots\ldots \frac{C}{(1+r)^{n-1+i}} + \frac{M}{(1+r)^{n-1+i}} \tag{1.15}$$

where the variables C, M, n and r are as before. Note that (1.15) assumes r for an annually paying bond and is adjusted to $r/2$ for a semiannually paying bond.

EXAMPLE 1.1

In these examples we illustrate the long-hand price calculation, using both expressions for the calculation of the present value of the annuity stream of a bond's cash flows.

1.1 (A)

Calculate the fair pricing of a U.S. Treasury, the 4% of February 2014, which pays semiannual coupons, with the following terms:

C = \$4.00 per \$100 nominal

M = \$100

N = 10 years (that is, the calculation is for value the 17th February 2004)

r = 4.048%

(*continued*)

EXAMPLE 1.1 (*Continued*)

$$
\begin{aligned}
P &= \frac{\$4.00}{0.04048}\left\{1-\frac{1}{[1+\tfrac{1}{2}(0.04048)]^{20}}\right\}+\frac{\$100}{[1+\tfrac{1}{2}(0.04048)]^{20}} \\
&= \$32.628+\$66.981 \\
&= \$99.609 \text{ or } 99-19+
\end{aligned}
$$

The fair price of the Treasury is \$99 – 19+, which is composed of the present value of the stream of coupon payments (\$32.628) and the present value of the return of the principal (\$66.981).

This yield calculation is shown at Figure 1.3, the Bloomberg YA page for this security. We show the price shown as 99 – 19+ for settlement on 17 Feb 2004, the date it was issued.

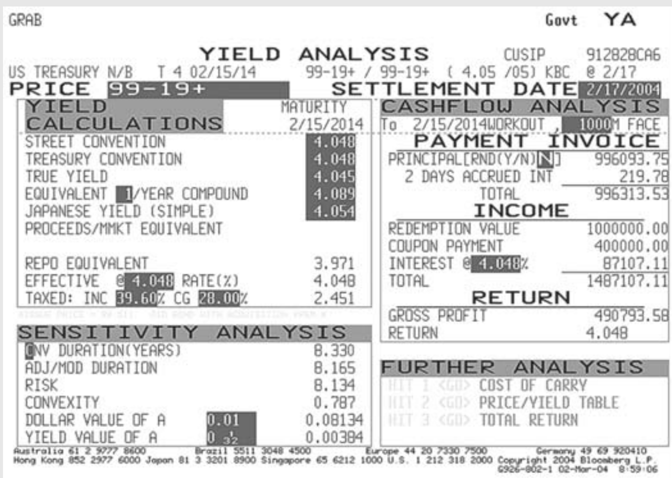

FIGURE 1.3 Bloomberg YA Page for Yield Analysis

1.1(B)

What is the price of a 5% coupon sterling bond with precisely five years to maturity, with semiannual coupon payments, if the yield required is 5.40%?

As the cash flows for this bond are 10 semiannual coupons of £2.50 and a redemption payment of £100 in 10 six-month periods from now, the price of the bond can be obtained by solving the following expression, where we substitute $C = 2.5$, $n = 10$, and $r = 0.027$ into the price equation (the values for C and r reflect the adjustments necessary for a semiannual paying bond).

EXAMPLE 1.1 (*Continued*)

$$P = 2.5\left[\frac{1-\left[\dfrac{1}{(1.027)^{10}}\right]}{0.027}\right]+\frac{100}{(1.027)^{10}}$$

$$= 21.65574 + 76.61178$$

$$= \$98.26752$$

The price of the bond is $98.2675 per $100 nominal.

1.1(C)

What is the price of a 5% coupon euro bond with five years to maturity paying annual coupons, again with a required yield of 5.4%?

In this case there are five periods of interest, so we may set $C = 5$, $n = 5$, with $r = 0.05$.

$$P = 5\left[\frac{1-\left[\dfrac{1}{(1.054)^{5}}\right]}{0.054}\right]+\frac{100}{(1.054)^{5}}$$

$$= 21.410121 + 76.877092$$

$$= £98.287213$$

Note how the annual-paying bond has a slightly higher price for the same required annualised yield. This is because the semiannual paying sterling bond has a higher effective yield than the euro bond, resulting in a lower price.

1.1(D)

Consider our 5% sterling bond again, but this time the required yield has risen and is now 6%. This makes $C = 2.5$, $n = 10$, and $r = 0.03$.

$$P = 2.5\left[\frac{1-\left[\dfrac{1}{(1.03)^{10}}\right]}{0.03}\right]+\frac{100}{(1.03)^{10}}$$

$$= 21.325507 + 74.409391$$

$$= £95.7349$$

(*continued*)

EXAMPLE 1.1 (*Continued*)

As the required yield has risen, the discount rate used in the price calculation is now higher, and the result of the higher discount is a lower present value (price).

1.1(E)

Calculate the price of our sterling bond, still with five years to maturity but offering a yield of 5.1%.

$$P = 2.5\left[\frac{1-\left[\frac{1}{(1.0255)^5}\right]}{0.0255}\right] + \frac{100}{(1.0255)^5}$$

$$= 21.823737 + 77.739788$$

$$= £99.563523$$

To satisfy the lower required yield of 5.1%, the price of the bond has fallen to £99.56 per £100.

1.1(F)

Calculate the price of the 5% sterling bond one year later, with precisely four years left to maturity and with the required yield still at the original 5.40%. This sets the terms in 1.1(b) unchanged, except now $n = 8$.

$$P = 2.5\left[\frac{1-\left[\frac{1}{(1.027)^8}\right]}{0.027}\right] + \frac{100}{(1.027)^8}$$

$$= 17.773458 + 80.804668$$

$$= £98.578126$$

The price of the bond is £98.58. Compared to 1.1(B) this illustrates how, other things being equal, the price of a bond will approach par (£100 percent) as it approaches maturity.

There also exist *perpetual* or *irredeemable* bonds which have no redemption date, so that interest on them is paid indefinitely. They are also known as undated bonds. An example of an undated bond is the 3½% War Loan, a UK gilt originally issued in 1916 to help pay for the 1914–1918 war effort. Most undated bonds date from a long time in the past, and it is unusual to see them issued today. In structure, the cash flow from an undated bond can be viewed as a continuous annuity. The fair price of such a bond is given from (1.11) by setting $N = \infty$, such that:

$$P = \frac{C}{r} \tag{1.16}$$

In most markets, bond prices are quoted in decimals, in minimum increments of 1/100ths. This is the case with Eurobonds, euro-denominated bonds, and gilts, for example. Certain markets—including the U.S. Treasury market and South African and Indian government bonds, for example—quote prices in ticks, where the minimum increment is 1/32nd. One tick is therefore equal to 0.03125. A U.S. Treasury might be priced at "98-05" which means "98 and five ticks". This is equal to 98 and 5/32nds which is 98.15625.

Bonds that do not pay a coupon during their life are known as zero-coupon bonds or strips, and the price for these bonds is determined by modifying (1.11) to allow for the fact that $C = 0$. We know that the only cash flow is the maturity payment, so we may set the price as:

$$P = \frac{M}{(1+r)^N} \tag{1.17}$$

where M and r are as before and N is the number of years to maturity. The important factor is to allow for the same number of interest periods as coupon bonds of the same currency. That is, even though there are no actual coupons, we calculate prices and yields on the basis of a quasi-coupon period. For a U.S. dollar or a

EXAMPLE 1.2

What is the total consideration for £5 million nominal of a gilt, where the price is 114.50?

The price of the gilt is £114.50 per £100, so the consideration is:

$$1.145 \times 5{,}000{,}000 = £5{,}725{,}000$$

What consideration is payable for \$5 million nominal of a U.S. Treasury, quoted at an all-in price of 99-16?

The U.S. Treasury price is 99-16, which is equal to 99 and 16/32, or 99.50 per \$100. The consideration is therefore:

$$0.9950 \times 5{,}000{,}000 = \$4{,}975{,}000$$

If the price of a bond is below par, the total consideration is below the nominal amount; whereas if it is priced above par, the consideration will be above the nominal amount.

EXAMPLE 1.3

1.3(A)

Calculate the price of a gilt strip with a maturity of precisely five years, where the required yield is 5.40%.

These terms allow us to set $N = 5$ so that $n = 10$, $r = 0.054$ (so that $r/2 = 0.027$), with $M = 100$ as usual.

$$\begin{aligned} P &= \frac{100}{(1.027)^{10}} \\ &= £76.611782 \end{aligned}$$

1.3(B)

Calculate the price of a French government zero-coupon bond with precisely five years to maturity, with the same required yield of 5.40%.

$$\begin{aligned} P &= \frac{100}{(1.054)^{5}} \\ &= £76.877092 \end{aligned}$$

sterling zero-coupon bond, a five-year zero coupon bond would be assumed to cover 10 quasi-coupon periods, which would set the price equation as:

$$P = \frac{M}{(1 + \tfrac{1}{2}r)^n} \tag{1.18}$$

We have to note carefully the quasi-coupon periods in order to maintain consistency with conventional bond pricing.

An examination of the bond price formula tells us that the yield and price for a bond are related. A key aspect of this relationship is that the price changes in the opposite direction to the yield. This is because the price of the bond is the net present value of its cash flows; if the discount rate used in the present value calculation increases, the present values of the cash flows will decrease. This occurs whenever the yield level required by bondholders increases. In the same way, if the required yield decreases, the price of the bond will rise. This property was observed in Example 1.2. As the required yield decreased, the price of the bond increased, and we observed the same relationship when the required yield was raised.

The relationship between any bond's price and yield at any required yield level is illustrated in a stylised manner in Figure 1.4, which is obtained if we plot the yield against the corresponding price; this shows a convex curve. In practice the curve is not quite as perfectly convex as illustrated in Figure 1.4, but the diagram is representative.

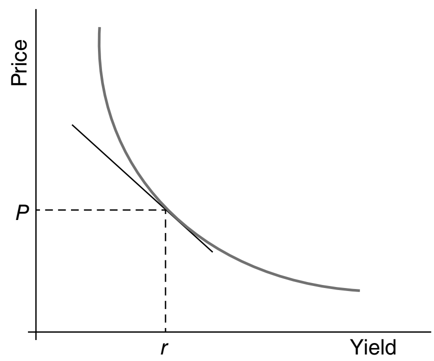

FIGURE 1.4 The Price/Yield Relationship

SUMMARY OF THE PRICE/YIELD RELATIONSHIP

At issue, if a bond is priced at par, its coupon will equal the yield that the market requires from the bond.

If the required yield rises above the coupon rate, the bond price will decrease.

If the required yield goes below the coupon rate, the bond price will increase.

BOND YIELD

We have observed how to calculate the price of a bond using an appropriate discount rate known as the bond's yield. We can reverse this procedure to find the yield of a bond where the price is known, which would be equivalent to calculating the bond's internal rate of return (IRR). The IRR calculation is taken to be a bond's yield to maturity or redemption yield and is one of various yield measures used in the markets to estimate the return generated from holding a bond. In most markets, bonds are generally traded on the basis of their prices, but because of the complicated patterns of cash flows that different bonds can have they are generally compared in terms of their yields. This means that a market maker will usually quote a two-way price at which she will buy or sell a particular bond, but it is the yield at which the bond is trading that is important to the market maker's customer. This is because a bond's price does not actually tell us anything useful about what we are getting. Remember, that in any market there will be a number of bonds with different issuers, coupons, and terms to maturity. Even in a homogenous market such as the Treasury market, different bonds and notes will trade according to their own specific characteristics. To compare bonds in the market, therefore, we need the yield on any bond, and it is yields that we compare, not prices.

The yield on any investment is the interest rate that will make the present value of the cash flows from the investment equal to the initial cost (price) of the investment. Mathematically, the yield on any investment, represented by r, is the interest rate that satisfies (1.19), which is simply the bond price equation we've already reviewed.

$$P = \sum_{n=1}^{N} \frac{C_n}{(1+r)^n} \tag{1.19}$$

But as we have noted there are other types of yield measure used in the market for different purposes. The simplest measure of the yield on a bond is the current yield, also known as the flat yield, interest yield or running yield. The running yield is given by (1.20).

$$rc = \frac{C}{P} \times 100 \tag{1.20}$$

where rc is the current yield.

In (1.20) C is not expressed as a decimal. Current yield ignores any capital gain or loss that might arise from holding and trading a bond and does not consider the time value of money. It essentially calculates the bond coupon income as a proportion of the price paid for the bond, and to be accurate would have to assume that the bond was more like an annuity rather than a fixed-term instrument.

The current yield is useful as a rough-and-ready interest-rate calculation; it is often used to estimate the cost of or profit from a short-term holding of a bond. For example, if other short-term interest rates such as the one-week or three-month rates are higher than the current yield, holding the bond is said to involve a running cost. This is also known as *negative carry* or *negative funding*. The term is used by bond traders and market makers and leveraged investors. The carry on a bond is a useful measure for all market practitioners as it illustrates the cost of holding or funding a bond. The funding rate is the bondholder's short-term cost of funds. A private investor could also apply this to a short-term holding of bonds.

The yield to maturity or gross redemption yield is the most frequently used measure of return from holding a bond.[5] Yield to maturity (YTM) takes into account the pattern of coupon payments, the bond's term to maturity, and the capital gain (or loss) arising over the remaining life of the bond. We saw from our bond price formula in the previous section that these elements were all related and were important components determining a bond's price. If we set the IRR for a set of cash flows to be the rate that applies from a start-date to an end-date we can assume the IRR to be the YTM for those cash flows. The YTM therefore is equivalent to the internal rate of return on the bond, the rate that equates the value of the discounted cash flows on the bond to its current price. The calculation

[5] In this book the terms *yield to maturity* and *gross redemption yield* are used synonymously. The latter term is encountered in sterling markets.

assumes that the bond is held until maturity, and therefore it is the cash flows to maturity that are discounted in the calculation. It also employs the concept of the time value of money.

As we would expect, the formula for YTM is essentially that for calculating the price of a bond. For a bond paying annual coupons, the YTM is calculated by solving (1.11). Note that the expression in (1.11) has two variable parameters, the price P and yield r. It cannot be rearranged to solve for yield r explicitly, and, in fact, the only way to solve for the yield is to use the process of numerical iteration. The process involves estimating a value for r and calculating the price associated with the estimated yield. If the calculated price is higher than the price of the bond at the time, the yield estimate is lower than the actual yield, and so it must be adjusted until it converges to the level that corresponds with the bond price.[6] For the YTM of a semiannual coupon bond, we have to adjust the formula to allow for the semiannual payments, shown in (1.13).

EXAMPLE 1.4 YIELD TO MATURITY FOR SEMIANNUAL COUPON BOND

A semiannual paying bond has a dirty price of \$98.50, an annual coupon of 6%, and there is exactly one year before maturity. The bond therefore has three remaining cash flows, comprising two coupon payments of \$3 each and a redemption payment of \$100. Equation 1.12 can be used with the following inputs:

$$98.50 = \frac{3.00}{\left(1+\frac{1}{2}rm\right)} + \frac{103.00}{\left(1+\frac{1}{2}rm\right)^2}$$

Note that we use half of the YTM value rm because this is a semiannual paying bond. The preceding expression is a quadratic equation, which is solved using the standard solution for quadratic equations, which is noted in the following equations.

$$ax^2 + bx + c = 0$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

In our expression, if we let $x = (1 + rm/2)$, we can rearrange the expression as follows:

$$98.50x^2 - 3.0x - 103.00 = 0$$

(*continued*)

[6] Bloomberg also uses the term *yield-to-workout*, where *workout* refers to the maturity date for the bond.

EXAMPLE 1.4 (*Continued*)

We then solve for a standard quadratic equation, and there will be two solutions, only one of which gives a positive redemption yield. The positive solution is $rm/2 = 0.037929$ so that $rm = 7.5859\%$.

As an example of the iterative solution method, suppose that we start with a trial value for rm of $r_1 = 7\%$ and plug this into the right-hand side of (1.12). This gives a value for the right-hand side of:

$$\text{RHS}_1 = 99.050$$

which is higher than the left-hand side (LHS = 98.50); the trial value for rm was therefore too low. Suppose then that we try next $r_2 = 8\%$ and use this as the right-hand side of the equation. This gives:

$$\text{RHS}_2 = 98.114$$

$$rm = r_1 + (r_2 - r_1)\frac{RHS_1 - LHS}{RHS_1 - RHS_2}$$

our linear approximation for the redemption yield is $rm = 7.587\%$, which is near the exact solution.

To differentiate redemption yield from other yield and interest-rate measures described in this book, we henceforth refer to it as *rm*.

Note that the redemption yield, as discussed earlier in this section, is the gross redemption yield, the yield that results from payment of coupons without deduction of any withholding tax. The net redemption yield is obtained by multiplying the coupon rate C by (1 – marginal tax rate). The net yield is what will be received if the bond is traded in a market where bonds pay coupon net, which means net of a withholding tax. The net redemption yield is always lower than the gross redemption yield.

We have already alluded to the key assumption behind the YTM calculation, namely that the rate *rm* remains stable for the entire period of the life of the bond. By assuming the same yield, we can say that all coupons are reinvested at the same yield *rm*. For the bond in Example 1.4, this means that if all the cash flows are discounted at 7.59% they will have a total net present value of 98.50. This is patently unrealistic since we can predict with virtual certainty that interest rates for instruments of similar maturity to the bond at each coupon date will not remain at this rate for the life of the bond. In practice, however, investors require a rate of return that is equivalent to the price that they are paying for a bond, and the redemption yield is, to put it simply, as good a measurement as any. A more accurate measurement might be to calculate present values of future cash flows using the discount rate that is equal to the forward interest rates at that point, known as the forward interest rate. However, forward rates are simply interest rates today for execution at a future date, and so a YTM measurement calculated using forward rates can be as speculative as one calculated using the conventional formula. So a YTM calculation made using forward rates would not be realised in practice either. We shall see later how the zero-coupon interest rate is the true interest rate for any term to maturity. However, despite the limitations presented by its assumptions, the YTM is the main measure of return used in the markets.

We have noted the difference between calculating redemption yield on the basis of both annual and semiannual coupon bonds. Analysis of bonds that pay semiannual coupons incorporates semiannual discounting of semiannual coupon payments. This is appropriate for most UK and U.S. bonds. However, government bonds in most of continental Europe and most Eurobonds pay annual coupon payments, and the appropriate method of calculating the redemption yield is to use annual discounting. The two yields measures are not therefore directly comparable. We could make a Eurobond directly comparable with a UK gilt by using semiannual discounting of the Eurobond's annual coupon payments. Alternatively we could make the gilt comparable with the Eurobond by using annual discounting of its semiannual coupon payments. The price/yield formulae for different discounting possibilities we encounter in the markets are listed in the following equations (as usual we assume that the calculation takes place on a coupon payment date so that accrued interest is zero).

Semiannual discounting of annual payments:

$$\mathrm{P}_d = \frac{C}{(1+\frac{1}{2}rm)^2} + \frac{C}{(1+\frac{1}{2}rm)^4} + \frac{C}{(1+\frac{1}{2}rm)^6} + \ldots \frac{C}{(1+\frac{1}{2}rm)^{2N}} + \frac{M}{(1+rm)^{2N}} \tag{1.21}$$

Annual discounting of semiannual payments:

$$\mathrm{P}_d = \frac{C/2}{(1+rm)^{\frac{1}{2}}} + \frac{C/2}{(1+rm)} + \frac{C/2}{(1+rm)^{\frac{3}{2}}} + \ldots \frac{C/2}{(1+rm)^N} + \frac{M}{(1+rm)^N} \tag{1.22}$$

Consider a bond with a dirty price of 97.89, a coupon of 6%, and five years to maturity. This bond would have the following gross redemption yields under the different yield-calculation conventions:

Discounting	Payments	Yield to Maturity (%)
Semiannual	Semiannual	6.500
Annual	Annual	6.508
Semiannual	Annual	6.428
Annual	Semiannual	6.605

This proves what we have already observed: namely, that the coupon and discounting frequency will affect the redemption yield calculation for a bond. We can see that increasing the frequency of discounting will lower the yield, while increasing the frequency of payments will raise the yield. When comparing yields for bonds that trade in markets with different conventions, it is important to convert all the yields to the same calculation basis. Intuitively we might think that doubling a semiannual yield figure will give us the annualised equivalent; in fact, this will result in an inaccurate figure due to the multiplicative effects of discounting and one that is an underestimate of the true annualised yield. The correct procedure for producing an annualised yields from semiannual and quarterly yields is given by the following expressions.

The general conversion expression is given by (1.23):

$$rm_a = (1 + \text{interest rate})^m - 1 \tag{1.23}$$

where m is the number of coupon payments per year.

Specifically we can convert between yields using the expressions given in (1.24) and (1.25).

$$\begin{aligned} rm_a &= [(1 + \tfrac{1}{2} rm_s)^2 - 1] \\ rm_s &= [(1 + rm_a)^{\frac{1}{2}} - 1] \times 2 \end{aligned} \tag{1.24}$$

$$\begin{aligned} rm_a &= [(1 + \tfrac{1}{4} rm_q)^4 - 1] \\ rm_q &= [(1 + rm_a)^{\frac{1}{4}} - 1] \times 4 \end{aligned} \tag{1.25}$$

where rm_q, rm_s, and rm_a are, respectively, the quarterly, semiannually, and annually compounded yields to maturity.

The market convention is sometimes simply to double the semiannual yield to obtain the annualised yields, despite the fact that this produces an inaccurate result. It is only acceptable to do this for rough calculations. An annualised yield obtained by multiplying the semiannual yield by two is known as a bond equivalent yield.

While YTM is the most commonly used measure of yield, it has one major disadvantage. The disadvantage is that implicit in the calculation of the YTM is the assumption that each coupon payment as it becomes due is reinvested at the rate rm.

EXAMPLE 1.5

A UK gilt paying semiannual coupons and a maturity of 10 years has a quoted yield of 4.89%. A European government bond of similar maturity is quoted at a yield of 4.96%. Which bond has the higher effective yield?

The effective annual yield of the gilt is:

$$rm = (1 + {}^1/_2 \times 0.0489)^2 - 1 = 4.9498\%$$

Therefore, the gilt does indeed have the lower yield.

This is clearly unlikely, due to the fluctuations in interest rates over time and as the bond approaches maturity. In practice, the measure itself will not equal the actual return from holding the bond, even if it is held to maturity. That said, the market standard is to quote bond returns as yields to maturity, bearing the key assumptions behind the calculation in mind.

Another disadvantage of this measure of return arises where investors do not hold bonds to maturity. The redemption yield measure will not be of great value where the bond is not being held to redemption. Investors might then be interested in other measures of return, which we can look at later.

To reiterate then, the redemption yield measure assumes that:

1. The bond is held to maturity;
2. All coupons during the bond's life are reinvested at the same (redemption yield) rate.

Therefore the YTM can be viewed as a prospective yield if the bond is purchased on issue and held to maturity. Even then the actual realised yield on maturity would be different from expected or anticipated yield and is closest to reality only where an investor buys a bond on first issue and holds the YTM figure because of the inapplicability of the second condition in the preceding list.

In addition, as coupons are discounted at the yield specific for each bond, it actually becomes inaccurate to compare bonds using this yield measure. For instance, the coupon cash flows that occur in two years time from both a two-year and five-year bond will be discounted at different rates (assuming we do not have a flat yield curve). This would occur because the YTM for a five-year bond is invariably different from the YTM for a two-year bond. However, it would clearly not be correct to discount a two-year cash flow at different rates, because we can see that the present value calculated today of a cash flow in two years' time should be the same whether it is sourced from a short- or long-dated bond. Even if the first condition noted earlier for the YTM calculation is satisfied, it is clearly unlikely for any but the shortest maturity bond that all coupons will be reinvested at the same rate. Market interest rates are in a state of constant flux and would thus affect money reinvestment rates. Therefore, although yield to maturity is the main market measure of bond levels, it is not a true interest rate. This is an important result, and we shall explore the concept of a true interest rate in Chapter 2.

Market Yield Measures

In Figure 1.3 we saw the Bloomberg page YA, used for yield analysis of fixed income instruments. Readers will notice from this figure that there are a number of different measures of yield shown on the page, all related to the standard yield-to-maturity calculation. This is because different markets and instruments use slightly different conventions when calculating yield-to-maturity.

All market yield measures define a bond's maturity date as the date when the final principal amount becomes due. This is almost invariably the stated maturity date of the bond, but in certain cases this date may fall on a non–business day, so the next working date is used instead.

Otherwise there are subtle differences in the way the different measures are calculated, although they may still give the same result. Here we explain these differences:

- **Street convention:** the standard yield-to-maturity calculation;
- **True yield:** this is the standard yield-to-maturity calculated with coupon dates moved whenever they fall on a non–business day, to the next valid business day. Moving this date is pertinent to the yield measure because it affects the number of days in an interest period;
- **Treasury convention:** the yield calculated using simple interest for the first coupon period, and compounded interest for subsequent interest periods. Assume an actual/actual accrued interest basis (this term is explained in the next section);
- **Consortium yield:** this is used for bonds that use an actual/365 day–count basis; it assumes 182.5 days in each (semiannual) interest period;
- **DMO yield:** a yield associated with United Kingdom gilts, the street convention equivalent as defined by the UK Debt Management Office;
- **Equivalent/year compound:** this is the street convention method adjusted for actual cash flow and compounding frequency; it uses the actual date of cash flows as they would be received;
- **Japanese yield:** this is a simple yield calculation using the annualised cash flow, expressed as a percentage of the original clean price used at purchase;
- **Money-market equivalent:** this is the yield of the bond but adjusted to make it equate to money-market yield convention. It would only be used for a bond that had less than 365 days left to maturity. This calculation compounds the remaining coupon payments to the maturity date, and this total amount is used to calculate a yield, based on simple interest, quoted using the present value of the total cash flows. Assumes an actual/360-day count basis;
- **Repo equivalent:** this is the yield calculated with interest accumulated on an overnight basis, with bond prices fixed, and assuming actual/360-day basis;
- **Effective rate:** the bond yield realised by investing coupon income at a specified reinvestment rate from now until maturity.

Investors will use the yield measure appropriate to the market and instrument they are analysing.

Accrued Interest, Clean and Dirty Bond Prices

Our discussion of bond pricing up to now has ignored coupon interest. All bonds accrue interest on a daily basis, and this is then paid out on the coupon date. The calculation of bond prices using present-value analysis does not account for coupon interest or *accrued interest*. In all major bond markets, the convention is to quote price as a *clean price*. This is the price of the bond as given by the net present value of its cash flows, but excluding coupon interest that has accrued on the bond since the last dividend payment. As all bonds accrue interest on a daily basis, even if a bond is held for only one day, interest will have been earned by the bondholder. However, we have referred already to a bond's *all-in* price, which is the price that is actually paid for the bond in the market. This is also known as the *dirty price* (or *gross price),* which is the clean price of a bond plus accrued interest.

In other words, the accrued interest must be added to the quoted price to get the total consideration for the bond.

Accruing interest compensates the seller of the bond for giving up all of the next coupon payment even though she will have held the bond for part of the period since the last coupon payment. The clean price for a bond will move with changes in market interest rates; assuming that this is constant in a coupon period, the clean price will be constant for this period. However, the dirty price for the same bond will increase steadily from one interest payment date until the next. On the coupon date, the clean and dirty prices are the same and the accrued interest is zero. Between the coupon payment date and the next ex-dividend date the bond is traded cum dividend, so that the buyer gets the next coupon payment. The seller is compensated for not receiving the next coupon payment by receiving accrued interest instead. This is positive and increases up to the next ex-dividend date, at which point the dirty price falls by the present value of the amount of the coupon payment. The dirty price at this point is below the clean price, reflecting the fact that accrued interest is now negative. This is because after the ex-dividend date the bond is traded "ex-dividend"; the seller not the buyer receives the next coupon, and the buyer has to be compensated for not receiving the next coupon by means of a lower price for holding the bond.

The net interest accrued since the last ex-dividend date is determined as follows:

$$AI = C \times \left[\frac{N_{xt} - N_{xc}}{Day\,Base} \right] \tag{1.26}$$

where
- AI is the next accrued interest
- C is the bond coupon
- N_{xc} is the number of days between the ex-dividend date and the coupon payment date (seven business days for UK gilts)
- N_{xt} is the number of days between the ex-dividend date and the date for the calculation
- *Day Base* is the day-count base (365 or 360)

Certain bonds do not have an ex-dividend period (for example, Eurobonds) and accrue interest right up to the coupon date.

Interest accrues on a bond from and including the last coupon date up to and excluding what is called the value date. The value date is almost always the settlement date for the bond, or the date when a bond is passed to the buyer and the seller receives payment. Interest does not accrue on bonds whose issuer has subsequently gone into default. Bonds that trade without accrued interest are said to be trading flat or clean. By definition therefore,

$$\text{Clean price of a bond} = \text{Dirty price} - \text{Accrued interest}$$

For bonds that are trading ex-dividend, the accrued coupon is negative and would be subtracted from the clean price. The calculation is given by (1.27).

$$AI = -C \times \left[\frac{\text{Days to next coupon}}{Day\,Base} \right] \tag{1.27}$$

As we noted, certain classes of bonds—for example, U.S. Treasuries and Eurobonds—do not have an ex-dividend period and therefore trade cum dividend right up to the coupon date.

The accrued-interest calculation for a bond is dependent on the day-count basis specified for the bond in question. When bonds are traded in the market, the actual consideration that changes hands is made up of the clean price of the bond together with the accrued that has accumulated on the bond since the last coupon payment; these two components make up the dirty price of the bond. When calculating the accrued interest, the market will use the appropriate day-count convention for that bond. A particular market will apply one of five different methods to calculate accrued interest:

Actual/365	Accrued = Coupon × Days/365
Actual/360	Accrued = Coupon × Days/360
Actual/actual	Accrued = Coupon × Days/actual number of days in the interest period
30/360	See following text
30E/360	See following text

When determining the number of days in between two dates, include the first date but not the second; thus, under the actual/365 convention, there are 37 days between 4th August and 10th September. The last two conventions assume 30 days in each month; so, for example, there are "30 days" between 10th February and 10th March. Under the 30/360 convention, if the first date falls on the 31st, it is changed to the 30th of the month, and if the second date falls on the 31st *and* the first date is on the 30th or 31st, the second date is changed to the 30th. The difference under the 30E/360 method is that if the second date falls on the 31st of the month, it is automatically changed to the 30th.

The accrued interest day-count basis for selected country bond markets is given in Table 1.5.

Van Deventer (1997) presents an effective critique of the accrued interest concept, believing essentially that it is an arbitrary construct that has little basis in economic reality. He states:

> The amount of accrued interest bears no relationship to the current level of interest rates.
>
> *Van Deventer, 1997, p. 11*

This is quite true; the accrued interest on a bond that is traded in the secondary market at any time is not related to the current level of interest rates, and is the same irrespective of where current rates are. As Example 1.6 makes clear, the accrued interest on a bond is a function of its coupon, which reflects the level of interest rates at the time the bond was issued. Accrued interest is therefore an accounting concept only, but at least it serves to recompense the holder for interest earned during the period the bond was held. It is conceivable that the calculation could be adjusted for present value, but, at the moment, accrued interest is the convention that is followed in the market.

TABLE 1.5 Selected Country Market Accrued Interest Day-Count Basis

Market	Coupon Frequency	Day-Count Basis	Ex-Dividend Period
Australia	Semiannual	Actual/actual	Yes
Austria	Annual	Actual/actual	No
Belgium	Annual	Actual/actual	No
Canada	Semiannual	Actual/actual	No
Denmark	Annual	30E/360	Yes
Eurobonds	Annual	30/360	No
France	Annual	Actual/actual	No
Germany	Annual	Actual/actual	No
Eire	Annual	Actual/actual	No
Italy	Annual	Actual/actual	No
New Zealand	Semiannual	Actual/actual	Yes
Norway	Annual	Actual/365	Yes
Spain	Annual	Actual/actual	No
Sweden	Annual	30E/360	Yes
Switzerland	Annual	30E/360	No
United Kingdom	Semiannual	Actual/actual	Yes
United States	Semiannual	Actual/actual	No

EXAMPLE 1.6

1.6(A): ACCRUAL CALCULATION FOR 7% TREASURY 2002

This gilt has coupon dates of 7th June and 7th December each year. £100 nominal of the bond is traded for value 27th August 1998. What is accrued interest on the value date?

On the value date, 81 days have passed since the last coupon date. Under the old system for gilts, act/365, the calculation was:

$$7 \times 81/365 = 1.55342$$

Under the current system of act/act, which came into effect for gilts in November 1998, the accrued calculation uses the actual number of days between the two coupon dates, giving us:

$$7 \times 81/183 \times 0.5 = 1.54918$$

(*continued*)

EXAMPLE 1.6 (*Continued*)

1.6(B)

Mansur buys £25,000 nominal of the 7% 2002 gilt for value on 27th August 1998, at a price of 102.4375. How much does he actually pay for the bond?

The clean price of the bond is 102.4375. The dirty price of the bond is 102.4375 + 1.55342 = 103.99092.

The total consideration is therefore

$$1.0399092 \times 25{,}000 = £25{,}997.73$$

EXAMPLE 1.6(C)

A Norwegian government bond with a coupon of 8% is purchased for settlement on 30th July 1999 at a price of 99.50. Assume that this is seven days before the coupon date and therefore the bond trades ex-dividend. What is the all-in price?

$$\text{The accrued interest } = -8 \times 7 / 365 = -0.153424$$

$$\text{The all-in price is therefore } 99.50 - 0.1534 = 99.3466$$

EXAMPLE 1.6(D)

A bond has coupon payments on 1st June and 1st December each year. What is the day-base count if the bond is traded for value date on 30th October, 31st October and 1st November 1999, respectively? There are 183 days in the interest period.

	30th October	31st October	1st November
Act/365	151	152	153
Act/360	151	152	153
Act/Act	151	152	153
30/360	149	150	151
30E/360	149	150	150

SELECTED BIBLIOGRAPHY AND REFERENCES

The following readings are recommended; Sundaresan (1997) presents a high-quality overview of the debt markets as a whole, and there is much practical application as well as theoretical treatment. Higson (1995) is a very accessible treatment that places fixed-income instruments in the context of the capital markets, and is also very good for an introduction to financial arithmetic.

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CHAPTER 2

Bond Instruments and Interest-Rate Risk

In the introductory chapter, we described traditional concepts in bond pricing. This chapter is a continuation of the first, and we discuss here the sensitivity of bond prices to changes in market interest rates, the key concepts of duration and convexity. The analysis is again the traditional approach; for present-day purposes when one speaks of "duration" or modified duration the interest risk exposure measure used is generally a per basis point measure.

DURATION, MODIFIED DURATION, AND CONVEXITY

Bonds pay a part of their total return during their lifetime, in the form of coupon interest, so that the term to maturity does not reflect the true period over which the bond's return is earned. Additionally, if we wish to gain an idea of the trading characteristics of a bond and compare this to other bonds of, say, similar maturity, term to maturity is insufficient, and so we need a more accurate measure. A plain-vanilla coupon bond pays out a proportion of its return during the course of its life, in the form of coupon interest. If we were to analyse the properties of a bond, we should conclude quite quickly that its maturity gives us little indication of how much of its return is paid out during its life, nor any idea of the timing or size of its cash flows, and hence its sensitivity to moves in market interest rates. For example, if comparing two bonds with the same maturity date but different coupons, the higher-coupon bond provides a larger proportion of its return in the form of coupon income than does the lower-coupon bond. The higher-coupon bond provides its return at a faster rate; its value is theoretically therefore less subject to subsequent fluctuations in interest rates.

We may wish to calculate an average of the time to receipt of a bond's cash flows, and use this measure as a more realistic indication of maturity. However, cash flows during the life of a bond are not all equal in value, so a more accurate measure would be to take the average time to receipt of a bond's cash flows, but weighted in the form of the cash flows' present value. This is, in effect, duration. We can measure the speed of payment of a bond, and hence its price risk relative to other bonds of the same maturity, by measuring the average maturity of the bond's cash-flow stream. Bond analysts use duration to measure this property (it is sometimes known as Macaulay's duration, after its inventor, who first introduced it in 1938).[1] Duration is the weighted average time until the receipt of cash flows from a bond, where the weights are the present values of the cash flows, measured in years. When he introduced the concept, Macaulay used the duration measure as an alternative for the length of time that a bond investment had remaining to maturity.

Duration

The original "duration" calculation has no real application, and hence value, in the fixed income markets. However, it's worth understanding because we use it to then arrive at the basis point risk exposure measure we consider later.

Recall that the price/yield formula for a plain-vanilla bond is as given in (2.1), assuming complete years to maturity paying annual coupons, and with no accrued interest at the calculation date. The yield to maturity reverts to the symbol r in this section.

$$P = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \ldots\ldots \frac{C}{(1+r)^N} + \frac{M}{(1+r)^N} \tag{2.1}$$

If we take the first derivative of this expression we obtain (2.2).

$$\frac{dP}{dr} = \frac{(-1)C}{(1+r)^2} + \frac{(-2)C}{(1+r)^3} + \ldots\ldots + \frac{(-n)C}{(1+r)^{n+1}} + \frac{(-n)M}{(1+r)^{n+1}} \tag{2.2}$$

If we rearrange (2.2), we will obtain the expression in (2.3), which is our equation to calculate the approximate change in price for a small change in yield.

$$\frac{dP}{dr} = \frac{-1}{(1+r)}\left[\frac{1C}{(1+r)} + \frac{2C}{(1+r)^2} + \ldots + \frac{nC}{(1+r)^n} + \frac{nM}{(1+r)^n}\right] \tag{2.3}$$

Readers may feel a sense of familiarity regarding the expression in brackets in equation (2.3) as this is the weighted average time to maturity of the cash flows from a bond, where the weights are, as in our previous example, the present values of each cash flow. The expression in (2.3) gives us the approximate measure of the change in price for a small change in yield. If we divide both sides of (2.3) by P we obtain the expression for the approximate percentage price change, given in (2.4).

$$\frac{dP}{dr}\frac{I}{P} = -\frac{1}{(1+r)}\left[\frac{1C}{(1+r)} + \frac{2C}{(1+r)^2} + \ldots + \frac{nC}{(1+r)^n} + \frac{nM}{(1+r)^n}\right]\frac{1}{P} \tag{2.4}$$

[1] Macaulay, F., *Some Theoretical Problems Suggested by the Movements of Interest Rates, Bond Yields and Stock Prices in the United States since 1865* (New York: National Bureau of Economic Research, 1938). Although it is frequently quoted, it's rare to meet someone who has actually read this work. However, it remains a fascinating treatise and is well worth reading; it is available from Risk Classics Library, under the title *The Movements of Interest Rates, Bond Yields and Stock Prices in the United States since 1856*.

If we divide the bracketed expression in (2.4) by the current price of the bond P, we obtain the definition of Macaulay duration, given in (2.5).

$$D = \frac{\frac{1C}{(1+r)} + \frac{2C}{(1+r)^2} + \ldots + \frac{nC}{(1+r)^n} + \frac{nM}{(1+r)^n}}{P} \tag{2.5}$$

Equation (2.5) is simplified using $\sum$ as shown by (2.6).

$$D = \frac{\sum_{n=1}^{N} \frac{nC_n}{(1+r)^n}}{P} \tag{2.6}$$

where C represents the bond cash flow at time n.

Example 2.1 calculates the Macaulay duration for a hypothetical bond, the 8% 2019 annual coupon bond, with the cash flows shown at Table 2.1.

EXAMPLE 2.1

Calculating the Macaulay duration for the 8% 2019 annual coupon bond:

Issued	30th September 2009
Maturity	30th September 2019
Price	102.497
Yield	7.634%

TABLE 2.1 Duration Calculation for the 8% 2009 Bond

Period (n)	Cash Flow	PV at Current Yield*	$n \times$ PV
1	8	7.43260	7.4326
2	8	6.90543	13.81086
3	8	6.41566	19.24698
4	8	5.96063	23.84252
5	8	5.53787	27.68935
6	8	5.14509	30.87054
7	8	4.78017	33.46119
8	8	4.44114	35.529096
9	8	4.12615	37.13535
10	108	51.75222	517.5222
Total		102.49696	746.540686

*Calculated as $C/(1+r)^n$

Macaulay Duration = 746.540686/102.497 = 7.283539998 years

Modified Duration = 7.28354/1.07634 = 6.76695

The Macaulay duration value given by (2.6) is measured in years. An interesting observation by Galen Burghardt in *The Treasury Bond Basis* is that, "measured in years, Macaulay's duration is of no particular use to anyone" (Burghardt 1994, p. 90). This is correct. However, as a risk measure and hedge calculation measure, duration transformed into modified duration was the primary measure of interest-rate risk used in the markets, and is still widely used alongside the value-at-risk measure for market risk.

If we substitute the expression for Macaulay duration (2.5) into (2.4) for the approximate percentage change in price we obtain (2.7).

$$\frac{dP}{dr}\frac{1}{P} = -\frac{1}{(1+r)}D \tag{2.7}$$

This is the definition of modified duration, given as (2.8).

$$MD = \frac{D}{(1+r)} \tag{2.8}$$

Modified duration is clearly related to duration, then. In fact we can use it to indicate that, for small changes in yield, a given change in yield results in an inverse change in bond price. We can illustrate this by substituting (2.8) into (2.7), giving us (2.9).

$$\frac{dP}{dr}\frac{1}{P} = -\mathrm{M}D \tag{2.9}$$

If we are determining duration long-hand, there is another arrangement we can use to shorten the procedure. Instead of equation (2.1), we use (2.10) as the bond price formula, which calculates price based on a bond being comprised of an annuity stream and a redemption payment, and summing the present values of these two elements. Again we assume an annual coupon bond priced on a date that leaves a complete number of years to maturity and with no interest accrued.

$$P = C\left[\frac{1-\dfrac{1}{(1+r)^n}}{r}\right] + \frac{M}{(1+r)^n} \tag{2.10}$$

This expression calculates the price of a bond as the present value of the stream of coupon payments and the present value of the redemption payment. If we take the first derivative of (2.10) and then divide this by the current price of the bond P, the result is another expression for the modified duration formula, given in (2.11).

$$MD = \frac{\dfrac{C}{r^2}\left[1-\dfrac{1}{(1+r)^n}\right] + \dfrac{n\left(M-\dfrac{C}{r}\right)}{(1+r)^{n+1}}}{P} \tag{2.11}$$

We have already shown that modified duration and duration are related; to obtain the expression for Macaulay duration from (2.11) we multiply it by $(1 + r)$. This

EXAMPLE 2.2

The 8% 2019 bond: using equation (2.11) for the modified duration calculation.

Coupon	8%, annual basis
Yield	7.634%
n	10
Price	102.497

Substituting the above terms into the equation, we obtain:

$$MD = \frac{\frac{8}{(0.07634^2)}\left[1 - \frac{1}{(1.07634)^{10}}\right] + \frac{10\left(100 - \frac{8}{0.07634}\right)}{(1.07634)^{11}}}{102.497}$$

$$MD = 6.76695$$

To obtain the Macaulay duration, we multiply the modified duration by $(1 = r)$, in this case 1.07634, which gives us a value of 7.28354 years.

shorthand formula is demonstrated in Example 2.2 for the same hypothetical bond, the annual coupon 8% 2019.

For an irredeemable bond duration is given by:

$$D = \frac{1}{rc} \tag{2.12}$$

where $rc = (C/P_d)$ is the *running yield* (or *current yield*) of the bond. This follows from equation (2.6) as $N \to \infty$, recognising that for an irredeemable bond $r = rc$. Equation (2.12) provides the limiting value to duration. For bonds trading at or above par, duration increases with maturity and approaches this limit from below. For bonds trading at a discount to par, duration increases to a maximum at around 20 years and then declines towards the limit given by (2.12). So, in general, duration increases with maturity, with an upper bound given by (2.12).

Properties of Macaulay Duration

A bond's duration is always less than its maturity. This is because some weight is given to the cash flows in the early years of the bond's life, which brings forward the average time at which cash flows are received. In the case of a zero-coupon bond, there is no present value weighting of the cash flows, for the simple reason that there are no cash flows, and duration for a zero coupon bond is equal to its term to maturity. Duration varies with coupon, yield, and maturity. The following three factors imply higher duration for a bond:

1. The lower the coupon,
2. The lower the yield,
3. Broadly, the longer the maturity.

Duration increases as coupon and yield decrease. As the coupon falls, more of the relative weight of the cash flows is transferred to the maturity date, and this causes duration to rise. Because the coupon on index-linked bonds is generally much lower than on vanilla bonds, this means that the duration of index-linked bonds will be much higher than for vanilla bonds of the same maturity. As yield increases, the present values of all future cash flows fall, but the present values of the more distant cash flows fall relatively more than those of the nearer cash flows. This has the effect of increasing the relative weight given to nearer cash flows and hence of reducing duration.

The effect of the coupon frequency. As we have already stated, certain bonds such as most Eurobonds pay coupon annually compared to, say, gilts, which pay semiannual coupons. If we imagine that every coupon is divided into two parts, with one part paid a half-period earlier than the other, this will represent a shift in weight to the left, as part of the coupon is paid earlier. Thus, increasing the coupon frequency shortens duration and, of course, decreasing coupon frequency has the effect of lengthening duration.

Duration as maturity approaches. Using our definition of duration, we can see that initially it will decline slowly and then at a more rapid pace as a bond approaches maturity.

Duration of a portfolio. Portfolio duration is a weighted average of the duration of the individual bonds. The weights are the present values of the bonds divided by the full price of the entire portfolio, and the resulting duration calculation is often referred to as a *market-weighted* duration. This approach is, in effect, similar to the duration calculation for a single bond. Portfolio duration has the same application as duration for an individual bond, and can be used to structure an immunised portfolio.

Modified Duration

Although it is common for newcomers to the market to think intuitively of duration, much as Macaulay originally did, as a proxy measure for the time to maturity of a bond, such an interpretation is to miss the main point of duration, which is a measure of price volatility or interest-rate risk.

Using the first term of a Taylor's expansion of the bond price function,[2] we can show the following relationship between price volatility and the duration measure, which is expressed as (2.13).

$$\Delta P = -\left[\frac{1}{(1+r)}\right] \times \text{Macaulay duration} \times \text{Change in yield} \tag{2.13}$$

[2] For an accessible explanation of the Taylor expansion, see C. Butler, *Mastering Value-at-Risk* (Upper Saddle River, NJ: FT Prentice Hall, 1998), 112–114.

where r is the yield to maturity for an annual-paying bond (for a semiannual coupon bond, we use $r/2$). If we combine the first two components of the right-hand side, we obtain the definition of modified duration. Equation (2.13) expresses the approximate percentage change in price as being equal to the modified duration multiplied by the change in yield. We saw in the previous section how the formula for Macaulay duration could be modified to obtain the modified duration for a bond. There is a clear relationship between the two measures. From the Macaulay duration of a bond can be derived its modified duration, which gives a measure of the sensitivity of a bond's price to small changes in yield. As we have seen, the relationship between modified duration and duration is given by (2.14).

$$MD = \frac{D}{1+r} \tag{2.14}$$

where MD is the modified duration in years. However, it also measures the approximate change in bond price for a 1% change in bond yield. For a bond that pays semiannual coupons, the equation becomes:

$$MD = \frac{D}{1+\frac{1}{2}r} \tag{2.15}$$

This means that the following relationship holds between modified duration and bond prices:

$$\Delta P = -MD \times \Delta r \times P \tag{2.16}$$

In the UK markets, the term *volatility* was sometimes used to refer to modified duration (see Example 2.3), but this is now rare, to avoid confusion with option markets' use of the same term, which refers to implied volatility and is something different.

EXAMPLE 2.3 USING MODIFIED DURATION

An 8% annual coupon bond is trading at par with a duration of 2.74 years. If yields rise from 8% to 8.50%, then the price of the bond will fall by:

$$\begin{aligned}\Delta P &= -D \times \frac{\Delta(r)}{1+r} \times P \\ &= -(2.74) \times \left(\frac{0.005}{1.080}\right) \times 100 \\ &= -£1.2685\end{aligned}$$

That is, the price of the bond will now be £98.7315.

(*continued*)

EXAMPLE 2.3 (*Continued*)

The modified duration of a bond with a duration of 2.74 years and yield of 8% is obviously:

$$MD = \frac{2.74}{1.08}$$

which gives us *MD* equal to 2.537 years.

In the earlier example of the five-year bond with a duration of 4.31 years, the modified duration can be calculated to be 3.99. This tells us that for a 1% move in the yield to maturity, the price of the bond will move (in the opposite direction) by 3.99%.

We can use modified duration to approximate bond prices for a given yield change. This is illustrated with the following expression:

$$\Delta P = -MD \times (\Delta r) \times P \tag{2.17}$$

For a bond with a modified duration of 3.99, priced at par, an increase in yield of 1 basis point (100 basis = 1%) leads to a fall in the bond's price of:

$$\Delta P = (-3.24/100) \times (+0.01) \times 100.00$$
$$\Delta P = \$0.0399, \text{or } 3.99 \text{ cents}$$

In this case, 3.99 cents is the basis-point value of the bond, which is the change in the bond price given a one-basis-point change in the bond's yield. The basis-point value of a bond can be calculated using (2.18).

$$BPV = \frac{MD}{100}\frac{P}{100} \tag{2.18}$$

Basis-point values (BPVs) are used in hedging. To hedge a bond position requires an opposite position to be taken in the hedging instrument. So, if we are long a 10-year bond, we may wish to sell short a similar 10-year bond as a hedge against it. Similarly a short position in a bond will be hedged through a purchase of an equivalent amount of the hedging instrument. In fact, there are a variety of hedging instruments available, both on- and off-balance-sheet. Once the hedge is put on, any loss in the primary position should, in theory, be offset by a gain in the hedge position, and vice versa. The objective of a hedge is to ensure that the price change in the primary instrument is equal to the price change in the hedging instrument. If we are hedging a position with another bond, we use the BPVs of each bond to calculate the amount of the hedging instrument required. This is important because each bond will have different BPVs, so that to hedge a long position in, say, £1 million nominal of a 30-year bond does not mean we simply sell £1 million of another 30-year bond. This is because the BPVs of the two bonds will almost certainly be different. Also, there may not be another 30-year bond in that particular bond. What if we have to hedge with a 10-year bond? How much nominal of this bond would be required?

We need to know the ratio given in (2.19) to calculate the nominal hedge position.

$$\frac{BPV_p}{BPV_h} \tag{2.19}$$

where

BPV_p is the basis-point value of the primary bond (the position to be hedged)
BPV_h is the basis-point value of the hedging instrument.

The *hedge ratio* is used to calculate the size of the hedge position and is given in (2.20).

$$\frac{BPV_p}{BPV_h} \times \frac{\text{Change in yield for primary bond position}}{\text{Change in yield for hedge instrument}} \tag{2.20}$$

The second ratio in (2.20) is known as the *yield beta*.

Example 2.4 illustrates use of the hedge ratio. In Example 2.5 we illustrate further the approximate nature of the modified duration calculation.

EXAMPLE 2.4 CALCULATING HEDGE SIZE USING BASIS-POINT VALUE

A trader holds a long position of £1 million of the 8% 2019 bond. The modified duration of the bond is 11.14692, and its price is 129.87596. The basis-point value of this bond is therefore 0.14477. The trader decides, to protect against a rise in interest rates, to hedge the position using the 0% 2009 bond, which has a BPV of 0.05549. If we assume that the yield beta is 1, what nominal value of the zero-coupon bond must be sold in order to hedge the position?

The hedge ratio is:

$$(0.14477 / 0.05549) \times 1 = 2.60894$$

Therefore, to hedge £1 million of the 20-year bond, the trader shorts £2,608,940 of the zero-coupon bond. If we use the respective BPVs to see the net effect of a 1-basis-point rise in yield, the loss on the long position is approximately equal to the gain in the hedge position.

EXAMPLE 2.5 THE NATURE OF THE MODIFIED DURATION APPROXIMATION

TABLE 2.2 Nature of the Modified Duration Approximation

Bond	8% 2009
Maturity (years)	10
Modified duration	6.76695
Price duration of basis point	0.06936
Yield: 6.00%	114.72017
6.50%	110.78325
7.00%	107.02358
7.50%	103.43204
7.99%	100.0671311
8.00%	100.00000
8.01%	99.932929
8.50%	96.71933
9.00%	93.58234
10.00%	87.71087

Yield Change	*Price Change*	*Estimate using Price Duration*
down 1 bp	0.06713	0.06936
up 1 bp	0.06707	0.06936

Table 2.2 in Example 2.5 shows the change in price for one of our hypothetical bonds, the 8% 2009, for a selection of yields. We see that for a one-basis-point change in yield, the change in price given by the dollar duration figure, while not completely accurate, is a reasonable estimation of the actual change in price. For a large move, however, say 200 basis points, the approximation is significantly in error, and analysts would not use it. Notice also for our hypothetical bond how the dollar duration value, calculated from the modified-duration measurement, underestimates the change in price resulting from a fall in yields but overestimates the price change for a rise in yields. This is a reflection of the price/yield relationship for this bond. Some bonds will have a more pronounced convex relationship between price and yield, and the modified-duration calculation will underestimate the price change resulting from both a fall or a rise in yields.

Convexity

Modified duration is a first-order measure of interest-rate risk: it measures the slope of the present value/yield profile. It is, however, only an approximation of the actual change in bond price given a small change in yield to maturity. Similarly for modified

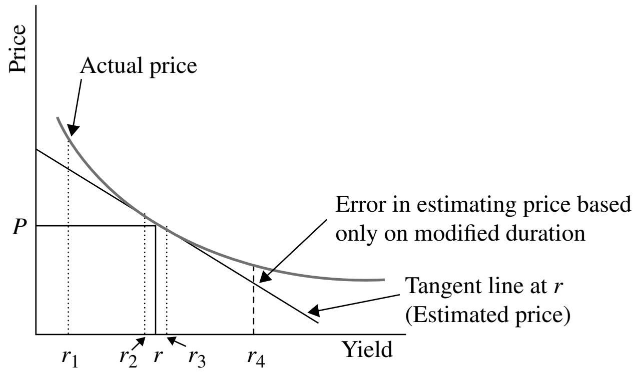

FIGURE 2.1 Approximation of the Bond Price Change Using Modified Duration
Reproduced with permission from F. Fabozzi, *Fixed Income Mathematics* (McGraw-Hill, 1997).

duration, which describes the price sensitivity of a bond to small changes in yield. However, as Figure 2.1 illustrates, the approximation is an underestimate of the actual price at the new yield. This is the weakness of the modified duration measure.

Convexity is a second-order measure of interest-rate risk; it measures the curvature of the present value/yield profile. Convexity can be regarded as an indication of the error one makes when using modified duration, as it measures the degree to which the curvature of a bond's price/yield relationship diverges from the straight-line estimation. The convexity of a bond is positively related to the dispersion of its cash flows; thus, other things being equal, if one bond's cash flows are more spread out in time than another's, then it will have a higher dispersion and hence a higher convexity. Convexity is also positively related to duration.

The second-order differential of the bond price equation with respect to the redemption yield r is:

$$\begin{aligned}\frac{\Delta P}{P} &= \frac{1}{P}\frac{\Delta P}{P}(\Delta r) + \frac{1}{2P}\frac{\Delta^2 P}{\Delta r^2}(\Delta r)^2 \\ &= -MD(\Delta r) + \frac{CV}{2}(\Delta r)^2\end{aligned} \tag{2.21}$$

where CV is the convexity.

From equation (2.21), convexity is the rate at which price variation to yield changes with respect to yield. That is, it describes a bond's modified duration changes with respect to changes in yield. It can be approximated by expression (2.22).

$$CV = 10^8\left(\frac{\Delta P'}{P} + \frac{\Delta P''}{P}\right) \tag{2.22}$$

where

$\Delta P'$ is the change in bond price if yield increases by 1 basis point (0.01)
$\Delta P''$ is the change in bond price if yield decreases by 1 basis point (0.01)

EXAMPLE 2.6

A 5% annual coupon is trading at par with three years to maturity. If the yield increases from 5.00 to 5.01%, the price of the bond will fall (using the bond price equation) to:

$$\begin{aligned} P'_d &= \frac{5}{(0.0501)}\left[1-\frac{1}{(1.0501)^3}\right]+\frac{100}{(1.0501)^3} \\ &= 99.97277262 \end{aligned}$$

or by $\Delta P'_d = -0.02722737$. If the yield falls to 4.99 per cent, the price of the bond will rise to:

$$\begin{aligned} P''_d &= \frac{5}{(0.0499)}\left[1-\frac{1}{(1.0499)^3}\right]+\frac{100}{(1.0499)^3} \\ &= 100.0272376 \end{aligned}$$

or by $P''_d = 0.027237584$. Therefore

$$\begin{aligned} CV &= 10^8\left(\frac{-0.02722737}{100}+\frac{0.027237584}{100}\right) \\ &= 10.2 \end{aligned}$$

Examples 2.6 and 2.7 illustrate the convexity calculation.

Appendix 2.2 provides the mathematical derivation of the formula.

The unit of measurement for convexity using (2.22) is the number of interest periods. For annual coupon bonds, this is equal to the number of years; for bonds paying coupon on a different frequency, we use (2.23) to convert the convexity measure to years.

$$CV_{years} = \frac{CV}{C^2} \tag{2.23}$$

The convexity measure for a zero-coupon bond is given by (2.24).

$$CV = \frac{n(n+1)}{(1+r)^2} \tag{2.24}$$

Convexity is a second-order approximation of the change in price resulting from a change in yield. This is given by:

$$\Delta P = \tfrac{1}{2} \times CV \times (\Delta r)^2 \tag{2.25}$$

The reason we multiply the convexity by half to obtain the convexity adjustment is because the second term in the Taylor expansion contains the coefficient ½. The convexity approximation is obtained from a Taylor expansion of the bond price formula. An illustration of Taylor expansion of the bond price/yield equation is given in Appendix 2.3.

The formula is the same for a semiannual coupon bond.

Note that the value for convexity given by the expressions above will always be positive; that is, the approximate price change due to convexity is positive for both yield increases and decreases.

In Example 2.7(B) we saw that the price change estimated using modified duration will be quite inaccurate, and that the convexity measure is the approximation of the size of the inaccuracy. The magnitude of the price change as estimated by both

EXAMPLE 2.7 SECOND-ORDER INTEREST-RATE RISK

2.7(A)

A 5% annual coupon bond is trading at par with a modified duration of 2.639 and convexity of 9.57. If we assume a significant market correction and yields rise from 5% to 7%, the price of the bond will fall by:

$$\begin{aligned}
\Delta P_d &= -MD \times (\Delta r) \times P_d + \frac{CV}{2} \times (\Delta r)^2 \times P_d \\
&= -(2.639) \times (0.02) \times 100 + \frac{9.57}{2} \times (0.02)^2 \times 100 \\
&= -5.278 + 0.1914 \\
&= -\$5.0866
\end{aligned}$$

to \$94.9134. The first-order approximation, using the modified-duration value of 2.639, is –\$5.278, which is an overestimation of the fall in price by \$0.1914.

2.7(B)

The 5% 2009 bond is trading at a price of £96.23119 (a yield of 5.50%) and has precisely 10 years to maturity, and its convexity parameter *CV* equals 100. If the yield rises to 7.50%, a change of 200 basis points, the percentage price change due to the convexity effect is given by:

$$(0.5) \times 96.23119 \times (0.02)^2 \times 100 = 1.92462\%$$

If we use an HP calculator to find the price of the bond at the new yield of 7.50%, we see that it is £82.83980, a change in price of 13.92%. The convexity measure of 1.92462% is an approximation of the error we would make when using the modified-duration value to estimate the price of the bond following the 200-basis-point rise in yield.

If the yield of the bond were to fall by 200 basis points, the convexity effect would be the same, as given by the expression in equation (2.25).

duration and convexity is obtained by summing the two values. However, it only makes any significant difference if the change in yield is very large. If we take our hypothetical bond again, the 5% 2009 bond, its modified duration is 7.64498. If the yield rises by 200 basis points, the approximation of the price change given by modified duration and convexity is:

$$\begin{aligned}&\text{Modified duration} = 7.64498 \times 2 = -15.28996\\&\text{Convexity} = 1.92462\end{aligned}$$

Note that the modified duration is given as a negative value, because a rise in yields results in a fall in price. This gives us a net percentage price change of 13.36534. As we saw in Example 2.7B, the actual percentage price change is 13.92%. So, in fact, using the convexity adjustment has given us a noticeably more accurate estimation. Let us examine the percentage price change resulting from a fall in yields of 1.50% from the same starting yield of 5.50%. This is a decrease in yield of 150 basis points, so our convexity measurement needs to be recalculated. The convexity value is:

$$(0.5) \times 96.23119 \times (0.0150)^2 \times 100 = 1.0826\%$$

So the price change is based on:

$$\begin{aligned}&\text{Modified duration} = 7.64498 \times 1.5 = 11.46747\\&\text{Convexity} = 1.0826\end{aligned}$$

This gives us a percentage price change of 12.55007. The actual price change was 10.98843%, so here the modified duration estimate is actually closer! This illustrates that the convexity measure is effective for larger yield changes only. Example 2.8 shows us that for very large changes, a closer approximation for bond price volatility is given by combining the modified duration and convexity measures.

EXAMPLE 2.8 CONVEXITY

The hypothetical bond is the 5% 2009, again trading at a yield of 5.50% and priced at 96.23119 (again assuming the convexity coefficient *CV* equals 100). If the yield rises to 8.50%, a change of 300 basis points, the percentage price change due to the convexity effect is given by:

$$(0.5) \times 96.23119 \times (0.03)^2 \times 100 = 4.3304\%$$

Meanwhile, as before, the modified duration of the bond at the initial yield is 7.64498. At the new yield of 8.50% the price of the bond is 77.03528 (check using an HP calculator).

EXAMPLE 2.8 (*Continued*)

The price change can be approximated using:

$$\text{Modified duration} = 7.64498 \times 3.0 = -22.93494$$
$$\text{Convexity} = 4.3304$$

This gives a percentage price change of 18.60454%. The actual percentage price change was 19.9477%, but our estimate is still closer than that obtained using only the modified-duration measure. The continuing error reflects the fact that convexity is also a dynamic measure and changes with yield changes; the effect of a large yield movement compounds the inaccuracy given by convexity.

EXAMPLE 2.9 USING MICROSOFT EXCEL®

The Excel® spreadsheet package has two duration functions, *Duration* and *Mduration,* which can be used for the two main measures. Not all installations of the software may include these functions, which have to be installed using the "Analysis ToolPak" add-in macro. The syntax required for using these functions is the same in both cases, and for Macaulay duration is: *Duration* (settlement, maturity, coupon, yield, frequency, basis).

As noted in the Excel 2010 help file, "dates should be entered by using the DATE function, or as results of other formulas or functions. For example, use DATE(2008,5,23) for the 23rd day of May, 2008. Problems can occur if dates are entered as text".

The other parameters are:

- Settlement: the settlement date
- Maturity: the maturity date
- Coupon: the coupon level
- Yield: the yield to maturity
- Frequency: annual or semiannual
- Basis: the day-count basis.

Once the parameters have been set up, Excel® will calculate the duration and modified duration for you.

Convexity is an attractive property for a bond to have. What level of premium will be attached to a bond's higher convexity? This is a function of the current yield levels in the market as well as market volatility. Remember that modified duration and convexity are functions of yield level and that the effect of both is magnified at lower yield levels. As well as the relative level, investors will value convexity higher if the current market conditions are volatile. Remember that the cash effect of convexity is noticeable only for large moves in yield. If an investor expects market yields to move only by relatively small amounts, she will attach a lower value to convexity and vice-versa for large movements in yield. Therefore the yield premium attached to a bond with higher convexity will vary according to market expectations of the future size of interest-rate changes.

The convexity measure increases with the square of maturity, and it decreases with both coupon and yield. As the measure is a function of modified duration, index-linked bonds have greater convexity than conventional bonds. We discussed how the price/yield profile will be more convex for a bond of higher convexity, and that such a bond will outperform a bond of lower convexity whatever happens to market interest rates. High convexity is therefore a desirable property for bonds to have. In principle, a more convex bond should fall in price less than a less convex one when yields rise, and rise in price more when yields fall. That is, convexity can be equated with the potential to outperform. Thus, other things being equal, the higher the convexity of a bond the more desirable it should in principle be to investors. In some cases, investors may be prepared to accept a bond with a lower yield in order to gain convexity. We noted also that convexity is in principle of more value if uncertainty, and hence expected market volatility, is high, because the convexity effect of a bond is amplified for large changes in yield. The value of convexity is therefore greater in volatile market conditions.

For a conventional vanilla bond, convexity is almost always positive. Negative convexity resulting from a bond with a concave price/yield profile would not be an attractive property for a bondholder. The most common occurrence of negative convexity in the cash markets is with callable bonds. We illustrated that for most bonds, and certainly when the convexity measure is high, the modified-duration measurement for interest-rate risk becomes more inaccurate for large changes in yield. In such situations it becomes necessary to use the approximation given by our convexity equation, to measure the error we have made in estimating the price change based on modified duration only. The expression was given earlier in this chapter.

The following points highlight the main convexity properties for conventional vanilla bonds.

A fall in yields leads to an increase in convexity. A decrease in bond yield leads to an increase in the bond's convexity; this is a property of positive convexity. Equally a rise in yields leads to a fall in convexity.

For a given term to maturity, higher coupon results in lower convexity. For any given redemption yield and term to maturity, the higher a bond's coupon, the lower its convexity. Therefore, among bonds of the same maturity, zero-coupon bonds have the highest convexity.

For a given modified duration, higher coupon results in higher convexity. For any given redemption yield and modified duration, a higher coupon results in a higher convexity. Contrast this with the earlier property; in this case, for bonds of the same modified duration, zero-coupon bonds have the lowest convexity.

APPENDICES

APPENDIX 2.1 FORMAL DERIVATION OF MODIFIED-DURATION MEASURE

Given that duration is defined as:

$$D = \frac{\sum_{n=1}^{N} \frac{nC_n}{(1+r)^n}}{P} \tag{A2.1.1}$$

if we differentiate P with respect to r we obtain:

$$\frac{dP}{dr} = -\sum_{n=1}^{N} nC_n (1+r)^{-n-1} \tag{A2.1.2}$$

Multiplying A2.1.2 by $(1 + r)$ we obtain:

$$(1+r)\frac{dP}{dr} = -\sum_{n=1}^{N} nC_n (1+r)^{-n} \tag{A2.1.3}$$

We then divide the expression by P giving us:

$$\frac{dP}{dr}\frac{1+r}{P} = -\sum_{n=1}^{N} \frac{nC_n}{(1+r)^n P} = -D \tag{A2.1.4}$$

If we then define modified duration as $D/(1 + r)$ then:

$$-\frac{dP}{dr}\frac{1}{P} = MD \tag{A2.1.5}$$

Thus, modified duration measures the proportionate impact on the price of a bond resulting from a change in its yield. The sign in (A2.1.5) is negative because of the inverse relationship between bond prices and yields (that is, rising yields result in falling prices). So if a bond has a modified duration of 6.767, then a rise in yield of 1% means that the price of the bond will fall by 6.767%. As we discuss in the main text, however, this is an approximation only and is progressively more inaccurate for greater changes in yield.

APPENDIX 2.2 MEASURING CONVEXITY

The modified duration of a plain-vanilla bond is:

$$MD = \frac{D}{(1+r)} \tag{A2.2.1}$$

We know that:

$$\frac{dP}{dr}\frac{1}{P} = -MD \tag{A2.2.2}$$

This shows that for a percentage change in the yield, we have an inverse change in the price by the amount of the modified-duration value. If we multiply both sides of (A2.2.2) by any particular change in the bond yield, given by *dr,* we obtain expression (A2.2.3)

$$\frac{dP}{P} = -MD \times dr \tag{A2.2.3}$$

Using the first two terms of a Taylor expansion, we obtain an approximation of the bond price change, given by (A2.2.4).

$$dP = \frac{dP}{dr}dr + \frac{1}{2}\frac{d^2P}{dr^2}(dr)^2 + \text{approximation error} \tag{A2.2.4}$$

If we divide both sides of (A2.2.4) by P to obtain the percentage price change, the result is the expression in (A2.2.5).

$$\frac{dP}{P} = \frac{dP}{dr}\frac{1}{2}dr + \frac{1}{2}\frac{d^2P}{dr^2}(dr)^2 + \frac{\text{approximation error}}{P} \tag{A2.2.5}$$

The first component of the right-hand side of (A2.2.4) is the expression in (A2.2.3), which is the cash price change given by the duration value. Therefore, equation (A2.2.4) is the approximation of the price change. Equation (A2.2.5) is the approximation of the price change as given by the modified-duration value. The second component in both expressions is the second derivative of the bond price equation. This second derivative captures the convexity value of the price/yield relationship and is the cash value given by convexity. As such, it is referred to as dollar convexity in the US markets. The dollar convexity is stated as (A2.2.6).

$$CV_{dollar} = \frac{d^2P}{dr^2} \tag{A2.2.6}$$

If we multiply the dollar convexity value by the square of a bond's yield change, we obtain the approximate cash value change in price resulting from the convexity effect. This is shown by (A2.2.7).

$$dP = (CV_{dollar})(dr)^2 \tag{A2.2.7}$$

If we then divide the second derivative of the price equation by the bond price, we obtain a measure of the percentage change in bond price as a result of the convexity effect. This is the measure known as convexity and is the convention used in virtually all bond markets. This is given by the expression in (A2.2.8).

$$CV = \frac{d^2P}{dr^2}\frac{1}{2} \tag{A2.2.8}$$

To measure the amount of the percentage change in bond price as a result of the convex nature of the price/yield relationship we can use (A2.2.9).

$$\frac{dP}{P} = \frac{1}{2}CV(dr)^2 \tag{A2.2.9}$$

For long-hand calculations, note that for the second derivative of the bond assumptions apply to the expressions; namely that the bond pays annual coupon and that the price equation is in (A2.2.10), which can be simplified to (A2.2.11). Also that the bond has a precise number of interest periods to maturity. If the bond is a semiannual paying one, the yield value r is replaced by $r/2$.

$$\frac{d^2P}{dr^2} = \sum_{n=1}^{N} \frac{n(n+1)C}{(1+r)^{n+2}} + \frac{n(n+1)M}{(1+r)^{n+2}} \tag{A2.2.10}$$

Alternatively, we differentiate to the second order the bond price equation as given by (A2.2.11), giving us the alternative expression (A2.2.12).

$$P = \frac{C}{r}\left[1 - \frac{1}{(1+r)^n}\right] + \frac{100}{(1+r)^n} \tag{A2.2.11}$$

$$\frac{d^2P}{dr^2} = \frac{2C}{r^3}\left[1 - \frac{1}{(1+r)^n}\right] - \frac{2C}{r^2(1+r)^{n+1}} + \frac{n(n+1)\left(100 - \frac{C}{r}\right)}{(1+r)^{n+2}} \tag{A2.2.12}$$

APPENDIX 2.3 TAYLOR EXPANSION OF THE PRICE/YIELD FUNCTION

Let us summarise the bond price formula as (A2.3.1), where C represents all the cash flows from the bond, including the redemption payment.

$$P = \sum_{n=1}^{N} \frac{C_n}{(1+r)^n} \tag{A2.3.1}$$

We therefore derive the following:

$$\frac{dP}{dr} = -\sum_{n=1}^{N} \frac{C_n n}{(1+r)^{n+1}} \tag{A2.3.2}$$

$$\frac{d^2P}{dr^2} = \sum_{n=1}^{N} \frac{C_n n(n+1)}{(1+r)^{n+2}} \tag{A2.3.3}$$

This then gives us:

$$\Delta P = \left[\frac{dP}{dr}\Delta r\right] + \left[\frac{1}{2!}\frac{d^2P}{dr^2}(\Delta r)^2\right] + \left[\frac{1}{3!}\frac{d^3P}{dr^3}(\Delta r)^3\right] + \ldots\ldots \tag{A2.3.4}$$

The first expression in (A2.3.4) is the modified duration measure, while the second expression measures convexity. The more powerful the changes in yield, the more expansion is required to approximate the change to greater accuracy. Expression (A2.3.4) therefore gives us the equations for modified duration and convexity, shown by (A2.3.5) and (A2.3.6), respectively.

$$MD = -\frac{dP/dr}{P} \tag{A2.3.5}$$

$$CV = -\frac{d^2P/dr^2}{P} \tag{A2.3.6}$$

We can therefore state the following:

$$\frac{\Delta P}{P} = \left[-(MD)\Delta r\right] + \left[\frac{1}{2}(CV)(\Delta r)^2\right] + \text{residual error} \tag{A2.3.7}$$

$$\Delta P = -\left[P(MD)\Delta r\right] + \left[\frac{P}{2}(CV)(\Delta r)^2\right] + \text{residual error} \tag{A2.3.8}$$

Example A2.3.1 illustrates use of this approach for a modified duration and convexity calculation.

EXAMPLE A2.3.1

Consider a three-year bond with (annual) coupon of 5% and yield of 5%. At a price of par we have:

$$\frac{dP}{dr} = -\left[\frac{5}{(1.05)^2} + \frac{5(2)}{(1.05)^3} + \frac{105(3)}{(1.05)^4}\right] = 263.9048$$

$$D = \frac{dP}{dr}\left[\frac{1+r}{P}\right] = 263.9048\left[\frac{1.05}{100}\right] = 2.771$$

$$MD = \frac{2.771}{1.05} = 2.639$$

EXAMPLE A2.3.1 (*Continued*)

$$\frac{d^2P}{dr^2} = \left[\frac{5(1)(2)}{(1.05)^3} + \frac{5(2)(3)}{(1.05)^4} + \frac{105(3)(4)}{(1.05)^5}\right] = 957.3179$$

$$CV = \frac{d^2P/dr^2}{P} = \frac{957.3179}{100} = 9.573$$

SELECTED BIBLIOGRAPHY AND REFERENCES

The references given in Chapter 1 are also useful for duration, modified duration, and convexity. Readers nevertheless may wish to have a look at the following texts. Burghardt's book, although a specialised subject not at all concentrating on interest-rate risk, nevertheless contains many useful insights and will help assist the reader to develop a deeper understanding of bond instruments. It is an excellent text and highly recommended. The same can be applied to Garbade (1996), another very high-quality work that contains many valuable insights on (among other things) duration, convexity, and the practicalities of risk and hedging from a trader's perspective. It is well worth purchasing.

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CHAPTER 3

Bond Pricing, Spot, and Forward Rates

In this chapter, we present basic concepts of yield and interest rates, and then follow this with a discussion of yield-curve analysis and the term structure of interest rates.

BASIC CONCEPTS

We are familiar with two types of fixed-income security, zero-coupon bonds, also known as discount bonds or strips, and coupon bonds. A zero-coupon bond makes a single payment on its maturity date, while a coupon bond makes regular interest payments at regular dates up to and including its maturity date. A coupon bond may be regarded as a set of strips, with each coupon payment and the redemption payment on maturity being equivalent to a zero-coupon bond maturing on that date. This is not a purely academic concept—witness events before the advent of the formal market in U.S. Treasury strips, when a number of investment banks had traded the cash flows of Treasury securities as separate zero-coupon securities.[1] The literature we review in this section is set in a market of default-free bonds, whether they are zero-coupon bonds or coupon bonds. The market is assumed to be liquid, so that bonds may be freely bought and sold. Prices of bonds are determined by the economy-wide supply and demand for the bonds at any time, so they are macroeconomic and not set by individual bond issuers or traders.

[1] The term *strips* comes from Separate Trading of Registered Interest and Principal of Securities, the name given when the official market was introduced by the Treasury. The banks would purchase Treasuries, which would then be deposited in a safe-custody account. Receipts were issued against each cash flow from each Treasury, and these receipts traded as individual zero-coupon securities. The market-making banks earned profit arising from the arbitrage difference in the price of the original coupon bond and the price at which the individual strips were sold. The U.S. Treasury formalised trading in strips after 1985, after legislation had been introduced that altered the tax treatment of such instruments. The market in UK gilt strips trading began in December 1997. Strips are also traded in France, Germany, the Netherlands, and New Zealand among other countries.

Zero-Coupon Bonds

A zero-coupon bond is the simplest fixed-income security. It is an issue of debt, the issuer promising to pay the face value of the debt to the bondholder on the date the bond matures. There are no coupon payments during the life of the bond, so it is a discount instrument, issued at a price that is below the face or principal amount. We denote as $P(t,T)$ the price of a discount bond at time t that matures at time T, with $T \geq t$. The term to maturity of the bond is denoted with n, where $n = T - t$. The price increases over time until the maturity date when it reaches the maturity or par value. If the par value of the bond is £1, then the yield to maturity of the bond at time t is denoted by $r(t,T)$, where r is actually "one plus the percentage yield"; that is, the rate of return earned by holding the bond from t to T. We have

$$P(t,T) = \frac{1}{[1 + r(t,T)]^n} \tag{3.1}$$

The yield may be obtained from the bond price and is given by

$$r(t,T) = \left[\frac{1}{P(t,T)}\right]^{1/n} - 1 \tag{3.2}$$

which is sometimes written as

$$r(t,T) = P(t,T)^{-(1/n)} - 1 \tag{3.3}$$

Analysts and researchers frequently work in terms of logarithms of yields and prices, or continuously compounded rates. One advantage of this is that it converts the nonlinear relationship in (3.2) into a linear relationship.[2]

The bond price at time t_2 where $t \leq t_2 \leq T$ is given by

$$P(t_2, T) = P(t,T)e^{(t_2 - t)r(t,T)} \tag{3.4a}$$

which is natural given that the bond price equation in continuous time is

$$P(t,T) = e^{-r(t,T)(T-t)} \tag{3.4b}$$

so that the yield is given by

$$r(t,T) = -\log[P(t,T)]^{1/n} \tag{3.5}$$

[2] A linear relationship in X would be a function $Y = f(X)$ in which the X values change via a power or index of 1 only and are not multiplied or divided by another variable or variables. So, for example, terms such as X^2, $\div X$ and other similar functions are not linear in X, nor are terms such as XZ or X/Z where Z is another variable. In econometric analysis, if the value of Y is solely dependent on the value of X, then its rate of change with respect to X, or the derivative of Y with respect to X, denoted dY/dX, is independent of X. Therefore, if $Y = 5X$, then $dY/dX = 5$, which is independent of the value of X. However, if $Y = 5X^2$, then $dY/dX = 10X$, which is not independent of the value of X. Hence this function is not linear in X. The classic regression function $E(Y \mid X_i) = a + bX_i$ is a linear function with slope b and the regression "curve" is represented geometrically by a straight line.

where $n = T - t$, which is sometimes written as

$$r(t, T) = -\left[\frac{1}{n}\right] \log P(t, T) \tag{3.6}$$

The expression in (3.4) includes the exponential function, hence the use of the term “continuously compounded”.

The term structure of interest rates is the set of zero-coupon yields at time t for all bonds ranging in maturity from (t, t + 1) to (t, $t + m$) where the bonds have maturities of {0, 1, 2, . . . , m}. A good definition of the term structure of interest rates is given by Sundaresan, who states that it “. . . refers to the relationship between the yield to maturity of default-free zero-coupon securities and their maturities” (Sundaresan 1997, p. 176).

The yield curve is a plot of the set of yields for $r(t, t + 1)$ to $r(t, t +$ m) against m at time t. For example, Figures 3.1–3.3 show the zero-coupon yield curve for U.S. Treasury strips and United Kingdom gilt strips, and the redemption yield curve for Chinese government bonds, on 2nd December 2013. Each of the curves exhibits peculiarities in its shape, although the most common type of curve is gently upward-sloping, as is the UK curve. The U.S. and UK curves are compared to their yield-to-maturity curve as well. We explore further the shape of the yield curve later in this chapter.

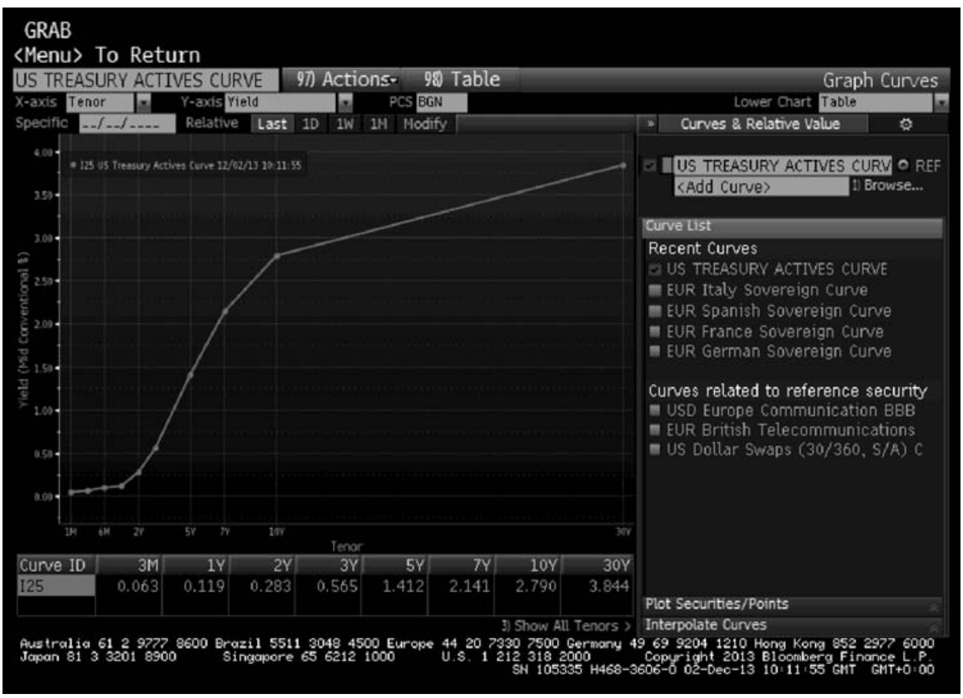

FIGURE 3.1 U.S. Treasury Zero-Coupon Yield Curve, December 2013

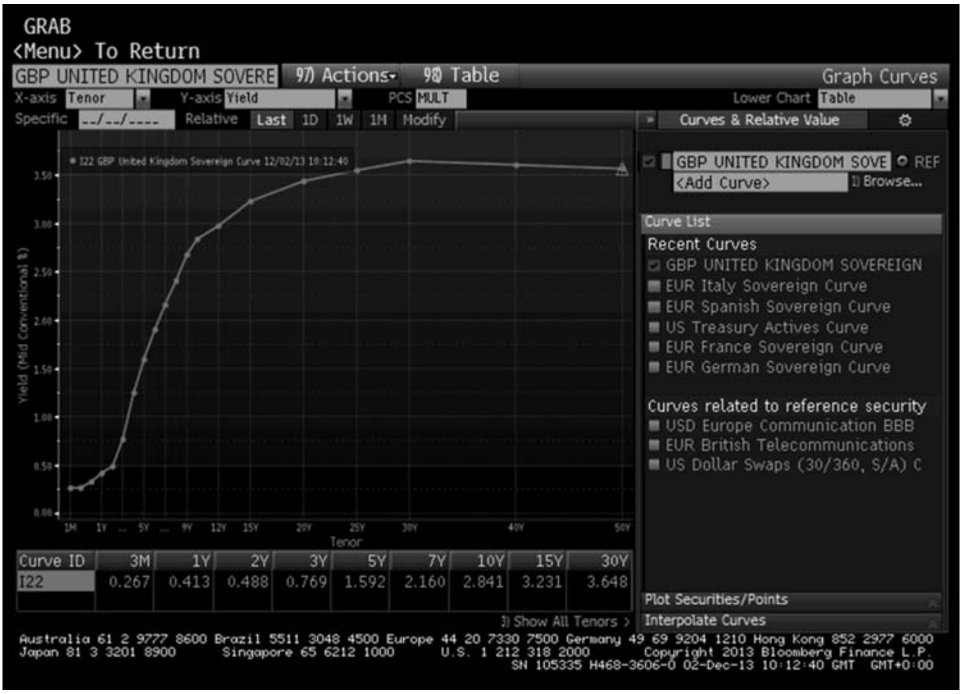

FIGURE 3.2 UK Gilt Zero-Coupon Yield Curve, December 2013

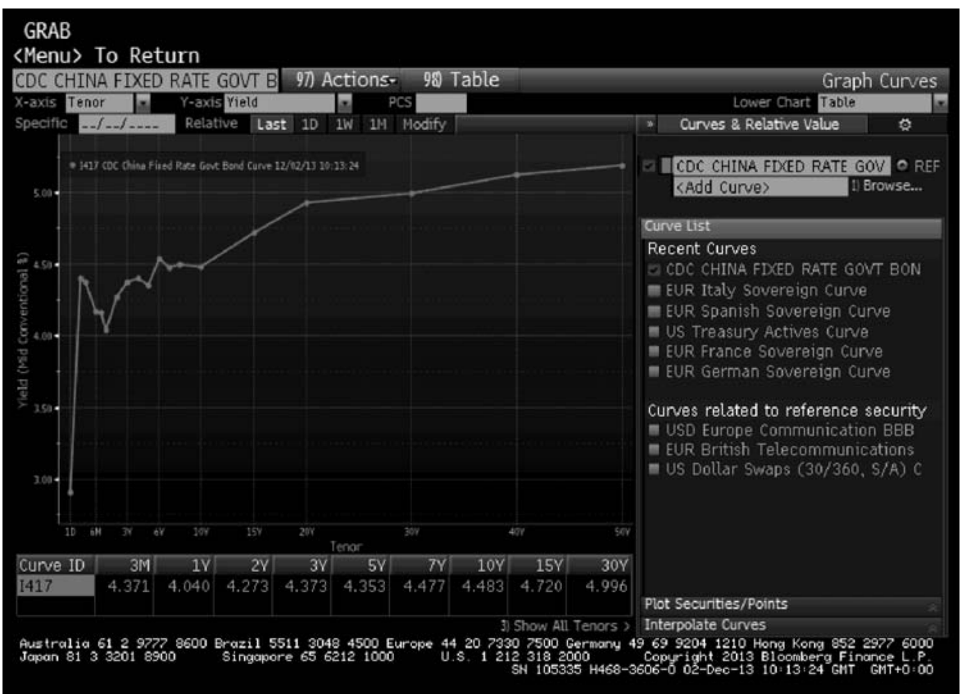

FIGURE 3.3 Chinese Government Yield Curve, December 2013

COUPON BONDS

The majority of bonds pay periodic interest or coupon payments during their life, and are known as coupon bonds. We have already noted that such bonds may be viewed as a package of individual zero-coupon bonds. The coupons have a nominal value that is a percentage of the nominal value of the bond itself, with steadily longer maturity dates, while the final redemption payment has the nominal value of the bond itself and is redeemed on the maturity date. We denote a bond issued at time i and maturing at time T as having an w-element vector of payment dates $(t_1, t_2, \ldots, t_{w-1}, T)$ and matching date payments $C_1, C_2, \ldots, C_{w-1}, Cw$.

In the academic literature these coupon payments are assumed to be made in continuous time, so that the stream of coupon payments is given by a positive function of time $C(t)$, $i < t < T$. The rate that discounts the bond's cash-flow stream C_w to its price $P(t, T)$ is r. The price is given by

$$P(t,T)\sum_{t_i>t} C_i e^{-(t_i-t)r(t,T)} \tag{3.7}$$

for a bond at time t that matures at time T. In other words the bond pays $P(t, T)$ and an investor that purchases the bond will receive the coupon payments as long as she continues to hold the bond.[3]

The yield to maturity at time t of a bond that matures at T is the interest rate that relates the price of the bond to the future returns on the bond which says that the bond price is given by the present value of the cash-flow stream of the bond, discounted at the rate $r(t, T)$. For a zero-coupon, (3.7) reduces to (3.5). In the academic literature, where coupon payments are assumed to be made in continuous time, the $\sum$ summation in (3.7) is replaced by the $\int$ integral. We will look at this in a moment.

In some texts, the plot of the yield to maturity at time t for the term of the bonds m is described as the term structure of interest rates but this is technically incorrect. The "term structure" is the plot of zero-coupon bond rates only. Plotting yields to maturity is generally described as graphically depicting the yield curve, rather than the term structure. Of course, given the law of one price, there is a relationship between the yield-to-maturity yield curve and the zero-coupon term structure, and given the first one can derive the second.

The expression in (3.7) obtains the continuously compounded yield to maturity $r(t, T)$. It is the use of the exponential function that enables us to describe the yield as continuously compounded.

The market frequently uses the measure known as current yield which is

$$rc = \frac{C}{Pd} \times 100 \tag{3.8}$$

where P_d is the dirty price of the bond. The measure is also known as the running yield or flat yield. Current yield is not used to indicate the interest rate or discount rate and therefore should not be mistaken for the yield to maturity.

[3] In theoretical treatment, this is the discounted clean price of the bond. For coupon bonds in practice, unless the bond is purchased for value on a coupon date, it will be traded with interest accrued. The interest that has accrued on a pro-rata basis from the last coupon date is added to the clean price of the bond, to give the market "dirty" price that is actually paid by the purchaser.

BOND PRICE IN CONTINUOUS TIME

Fundamental Concepts

In this section we present an introduction to the bond price equation in continuous time.[4] The necessary background on price processes is introduced in the next chapter; readers will see the logic in this as we introduce term structure modelling there.

Consider a trading environment where bond prices evolve in a w-dimensional process

$$X(t)[X_1(t), X_2(t), X_3(t), \ldots, X_w(t)],\ t > 0 \tag{3.9}$$

where the random variables are termed state variables that reflect the state of the economy at any point in time. The markets assume that the state variables evolve through a process described as geometric Brownian motion or a Weiner process. It is therefore possible to model the evolution of these variables, in the form of a stochastic differential equation.

The market assumes that the cash-flow stream of assets such as bonds and (for equities) dividends is a function of the state variables. A bond is characterised by its coupon process

$$C(t) = C[X_1(t), X_2(t), X_3(t), \ldots, X_w(t), t] \tag{3.10}$$

The coupon process represents the cash flow that the investor receives during the time that she holds the bond. Over a small incremental increase in time of dt from the time t the investor can purchase $1 + C(t)dt$ units of the bond at the end of the period $t + dt$. Assume that there is a very short-term discount security such as a Treasury bill that matures at $t + dt$, and during this period the investor receives a return of $r(t)$. This rate is the annualised short-term interest rate or short rate, which in the mathematical analysis is defined as the rate of interest charged on a loan that is taken out at time t and which matures almost immediately. For this reason, the rate is also known as the instantaneous rate.

The short rate is given by

$$r(t) = r(t, t) \tag{3.11}$$

and if we assume the short rate is deterministic,

$$r(t) = -\frac{\partial \log P(t,t)}{\partial t} \tag{3.12}$$

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[4] See also Avellaneda and Laurence (2000); Baxter and Rennie (1996); Campbell, Lo, and MacKinlay (1997); Neftci (2000); Ross (1999), and Shiller (1990). These are all excellent texts, and well recommended. For an accessible and highly readable introduction, Ross's book is worth buying for Chapter 4 alone, as is Avellaneda's for his Chapter 12. For a general introduction to the main pricing concepts, see Campbell et al. (1997), Chapter 10. Chapter 3 in Jarrow (1996) is an accessible introduction for discrete-time bond pricing.

If we continuously reinvest the short-term security such as the T-bill at this short rate, we obtain a cumulative amount that is the original investment multiplied by (3.13).[5]

$$M(t) = \exp\left[\int_0^t r(s)ds\right] \tag{3.13}$$

where M is a money-market account that offers a return of the short rate $r(t)$.

If we say that the short rate is constant, making $r(t) = r$, then the price of a risk-free bond that pays £1 on maturity at time T is given by

$$P(t,T) = e^{-r(T-t)} \tag{3.13a}$$

What (3.13a) states is that the bond price is simply a function of the continuously compounded interest rate, with the right-hand side of (3.13a) being the discount factor at time t. At $t = T$ the discount factor will be 1, which is the redemption value of the bond and hence the price of the bond at this time.

Consider the following scenario. A market participant may undertake the following:

- It can invest $e^{-r(T-t)}$ units cash in a money-market account today, which will have grown to a sum of £1 at time T;
- It can purchase the risk-free zero-coupon bond today, which has a maturity value of £1 at time T.

The market participant can invest in either instrument, both of which we know beforehand to be risk-free, and both of which have identical payouts at time T and have no cash flow between now and time T. As interest rates are constant, a bond that paid out £1 at T must have the same value as the initial investment in the money-market account, which is $e_t^{-r(T-t)}$. Therefore equation (3.13a) must apply. This is a restriction placed on the zero-coupon bond price by the requirement for markets to be arbitrage-free.

If the bond was not priced at this level, arbitrage opportunities would present themselves. Consider if the bond was priced higher than $e^{-r(T-t)}$. In this case, an investor could sell short the bond and invest the sale proceeds in the money-market account. On maturity at time T, the short position will have a value of –£1 (negative, because the investor is short the bond) while the money market will have accumulated £1, which the investor can use to pay the proceeds on the zero-coupon bond. However, the investor will have surplus funds because at time t

$$P(t,T) - e^{-r(T-t)} > 0$$

and so will have profited from the transaction at no risk to himself.

[5] This expression uses the integral operator. The integral is the tool used in mathematics to calculate sums of an infinite number of objects, that is where the objects are uncountable. This is different to the $\sum$ operator, which is used for a countable number of objects. For a readable and accessible review of the integral and its use in quantitative finance, see Neftci (2000, pp. 59–66), a summary of which is given at Appendix 3.1.

The same applies if the bond is priced below $e^{-r(T-t)}$. In this case, the investor borrows $e^{-r(T-t)}$ and buys the bond at its price $P(t, T)$. On maturity the bond pays £1, which is used to repay the loan amount. However, the investor will gain because

$$e^{-r(T-t)} - P(t,T) > 0$$

Therefore the only price at which no arbitrage profit can be made is if

$$P(t,T) = e^{-r(T-t)} \tag{3.13b}$$

In the academic literature the price of a zero-coupon bond is given in terms of the evolution of the short-term interest rate, in what is termed the risk-neutral measure.[6] The short rate $r(t)$ is the interest rate earned on a money-market account or short-dated risk-free security such as the T-bill suggested above, and it is assumed to be continuously compounded. This makes the mathematical treatment simpler. With a zero-coupon bond we assume a payment on maturity of 1 (say \$1 or £1), a one-off cash flow payable on maturity at time T. The value of the zero-coupon bond at time t is therefore given by

$$P(t,T) = \exp\left[-\int_t^T r(s)ds\right] \tag{3.14}$$

which is the redemption value of 1 divided by the value of the money-market account, given by (3.13).

The bond price for a coupon bond is given in terms of its yield as

$$P(t,T) = \exp[-(T-t)r(T-t)] \tag{3.15}$$

Expression (3.14) is very commonly encountered in the academic literature. Its derivation does not occur so frequently, however. We present it in Appendix 3.2, which is a summary of the description given in Ross (1999). This reference is highly recommended reading.

The expression (3.14) represents the zero-coupon bond pricing formula when the spot rate is continuous or stochastic, rather than constant. The rate $r(s)$ is the risk-free return earned during the very short or infinitesimal time interval $(t, t + dt)$. The rate is used in the expressions for the value of a money-market account (3.13) and the price of a risk-free zero-coupon bond (3.15).

Stochastic Rates in Continuous Time

In the academic literature the bond price given by (3.14) evolves as a martingale process under the risk-neutral probability measure $\tilde{P}$. This is an advanced branch of fixed-income mathematics, and is outside the scope of this book. However, it will be

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[6] This is part of arbitrage pricing theory. For details on this, see Cox, Ingersoll, and Ross (1985), while Duffie (1992) is a fuller treatment for those with a strong grounding in mathematics.

introduced in introductory fashion in the next chapter.[7] However, under this analysis the bond price is given as

$$P(t,T) = E_t^{\tilde{P}}\left[e^{-\int_t^T r(s)ds}\right] \tag{3.16}$$

where the right-hand side of (3.16) is viewed as the randomly evolved discount factor used to obtain the present value of the £1 maturity amount. Expression (3.16) also states that bond prices are dependent on the entire spectrum of short-term interest rates $r(s)$ in the future during the period $t < s < T$. This also implies that the term structure at time t contains all the information available on short rates in the future.[8]

From (3.16) we say that the function f $P(t,T)$ is the discount curve (or discount function) at time t. Avellaneda and Laurence (2000) note that the markets usually replace the term $(T - t)$ with a term meaning time to maturity, so the function becomes

$$\tau \to P_t^{t+\tau}, \tau > 0 \text{ where } \tau = (T - t).$$

Under a constant spot rate, the zero-coupon bond price is given by

$$P(t,T) = e^{-r(t,T)(T-t)} \tag{3.17}$$

From (3.16) and (3.17) we can derive a relationship between the yield $r(t, T)$ of the zero-coupon bond and the short rate $r(t)$, if we equate the two right-hand sides, namely

$$e^{-r(t,T)(T-t)} = E_t^{\tilde{P}}\left[e^{-\int_t^T r(s)ds}\right] \tag{3.18}$$

Taking the logarithm of both sides we obtain

$$r(t,T) = \frac{-\log E_t^{\tilde{P}}\left[e^{-\int_t^T r(s)ds}\right]}{T-t} \tag{3.19}$$

This describes the yield on a bond as the average of the spot rates that apply during the life of the bond, and under a constant spot rate the yield is equal to the spot rate.

With a zero-coupon bond and assuming that interest rates are positive, $P(t, T)$ is less than or equal to 1. The yield of the bond is, as we have noted, the continuously compounded interest rate that equates the bond price to the discounted present value of the bond at time t. This is given by

$$r(t,T) = -\frac{\log(P(t,T))}{T-t} \tag{3.20}$$

[7] Interested readers should consult Neftci (2000), Chapters 2, 17, and 18, which are highly recommended. Another accessible text is Baxter and Rennie (1996).

[8] This is related to the view of the short rate evolving as a martingale process. For a derivation of (3.16), see Neftci (2000, p. 417).

so we obtain

$$P(t,T) = e^{-(T-t)r(T-t)} \tag{3.21}$$

In practice, this means that an investor will earn $r(t, T)$ if she purchases the bond at t and holds it to maturity.

Coupon Bonds

Using the same principles as in the previous section, we can derive an expression for the price of a coupon bond in the same terms of a risk-neutral probability measure of the evolution of interest rates. Under this analysis, the bond price is given by

$$P_c = 100.E_t^{\tilde{P}}\left[e^{-\int_t^T r(s)ds}\right] + \sum_{n:t_n>t}^{N} \frac{C}{w} E_t^{\tilde{P}}\left[e^{-\int_{t_n}^{T_r} r(s)ds}\right] \tag{3.22}$$

where P_c is the price of a coupon bond
C is the bond coupon
t_n is the coupon date, with $n < N$, and $t = 0$ at the time of valuation
w is the coupon frequency[9]

and where 100 is used as the convention for principal or bond nominal value (that is, prices are quoted percent, or per 100 nominal).

Expression (3.22) is written in some texts as

$$P_c = 100e^{-rN} + \int_n^N Ce^{-rn}dt \tag{3.23}$$

We can simplify (3.22) by substituting Df to denote the discount factor part of the expression and assuming an annual coupon, which gives us

$$P = 100 \cdot Df_N + \sum_{n:t_n>t}^{N} C \cdot Df_n \tag{3.24}$$

which states that the market value of a risk-free bond on any date is determined by the discount function on that date.

We know from Chapter 2 that the actual price paid in the market for a bond includes accrued interest from the last coupon date, so that price given by (3.24) is known as the clean price, and the traded price, which includes accrued interest, is known as the dirty price.

[9] Conventional or plain-vanilla bonds pay coupon on an annual or semiannual basis. Other bonds, notably certain floating-rate notes and mortgage- and other asset-backed securities also pay coupon on a monthly basis, depending on the structuring of the transaction.

FORWARD RATES

An investor can combine positions in bonds of differing maturities to guarantee a rate of return that begins at a point in the future. That is, the trade ticket would be written at time t but would cover the period T to $T + 1$ where $t < T$ (sometimes written as beginning at T_1 and ending at T_2, with $t < T_1 < T_2$). The interest rate earned during this period is known as the forward rate.[10] The mechanism by which this forward rate can be guaranteed is described in the box, which essentially summarises Jarrow (1996) and Campbell, Lo, and MacKinlay (1997). The latter text is especially excellent.

THE FORWARD RATE

An investor buys one unit of a zero-coupon bond maturing at time T, priced at $P(t, T)$ and simultaneously sells $P(t, T)/P(t, T + 1)$ bonds that mature at $T + 1$. From Table 3.1 we see that the net result of these transactions is a zero cash flow. At time T there is a cash inflow of 1, and then at time $T + 1$ there is a cash outflow of $P(t, T)/P(t, T + 1)$. These cash flows are identical to a loan of funds made during the period T to $T + 1$, contracted at time t. The interest rate on this loan is given by $P(t, T)/P(t, T + 1)$, which is therefore the forward rate. That is,

$$f(t,T) = \frac{P(t,T)}{P(t,T+1)} - 1 \tag{3.25}$$

Together with our earlier relationships on bond price and yield, from (3.25) we can define the forward rate in terms of yield, with the return earned during the period $(T, T + 1)$ being

$$f(t,T,T+1) = \frac{1}{(P(t,T+1)/P(t,T))} - 1 = \frac{[1+r(t,T+1)]^{(T+1)}}{[1+r(t,T)]^{T}} - 1 \tag{3.26}$$

TABLE 3.1 No-Arbitrage Illustration of Derivation of Forward Rate Expression

	Time		
Transactions	t	T	T+1
Buy 1 unit of T-period bond	—$P(t,T)$	+1	
Sell $P(t,T)/P(t,T+1)$ of T+1 period bonds	+$[(P(t,T)/P(t,T+1)]P(t,T+1)$		—$P(t,T)/P(t,T+1)$
Net cash flows	0	+ 1	–P(t,T)/P(t, T+1)

(*continued*)

[10] See the footnote on p. 639 of Shiller (1990) for a fascinating insight on the origin of the term "forward rate", which Mr. Shiller ascribes to John Hicks in his book *Value and Capital*, 2nd ed. (New York: Oxford University Press, 1946).

(Continued)

From (3.25) we can obtain a bond price equation in terms of the forward rates that hold from t to T,

$$P(t,T) = \frac{1}{\prod_{k=t}^{T-1} f(t,k)} \tag{3.27}$$

A derivation of this expression can be found in Jarrow (1996), Chapter 3. Equation (3.27) states that the price of a zero-coupon bond is equal to the nominal value, here assumed to be 1, receivable at time T after it has been discounted at the set of forward rates that apply from t to T.[11]

When calculating a forward rate, it is as if we are writing an interest rate contract today that is applicable at the forward start date; in other words, we trade a forward contract. The law of one price, or no-arbitrage, is used to calculate the rate. For a loan that begins at T and matures at $T + 1$, similar to the way we described in the box above, consider a purchase of a $T + 1$ period bond and a sale of p amount of the T-period bond. The net cash position at t must be zero, so p is given by

$$P = \frac{P(t,T+1)}{P(t,T)}$$

and to avoid arbitrage the value of p must be the price of the $T + 1$–period bond at time T. Therefore, the forward yield is given by

$$f(t,T+1) = -\frac{\log P(t,T+1) - \log P(t,T)}{(T+1) - T} \tag{3.28}$$

If the period between T and the maturity of the later-dated bond is reduced, we now have bonds that mature at T and T_2, and $T_2 = T + \Delta t$, then as the incremental change in time Δt becomes progressively smaller we obtain an instantaneous forward rate, which is given by

$$f(t,T) = -\frac{\partial}{\partial T} \log \mathrm{P}(t,T) \tag{3.29}$$

[11] The symbol $\prod$ means "take the product of", and is defined as $\prod_{i=1}^{n} x_i = x_1 * x_2 * \ldots\ldots * x_n$, so that $\prod_{k=t}^{T-1} f(t,k) = f(t,t) * f(t,t+1) * \ldots\ldots * f(t,T-1)$, which is the result of multiplying the rates that obtain when the index k runs from t to T–1.

This rate is defined as the forward rate and is the price today of forward borrowing at time T. The forward rate for borrowing today where $T = t$ is equal to the instantaneous short rate $r(t)$. At time t, the spot and forward rates for the period (t, t) will be identical; at other maturity terms they will differ. For all points other than at (t, t) the forward-rate yield curve will lie above the spot-rate curve if the spot curve is positively sloping. The opposite applies if the spot-rate curve is downward sloping.

Campbell et al. (1997, pp. 400–401) observe that this property is a standard one for marginal and average cost curves. That is, when the cost of a marginal unit (say, of production) is above that of an average unit, then the average cost will increase with the addition of a marginal unit. This results in the average cost rising when the marginal cost is above the average cost. Equally the average cost per unit will decrease when the marginal cost lies below the average cost. We illustrate this further in Example 3.1.

EXAMPLE 3.1 THE SPOT AND FORWARD YIELD CURVES

From the discussion in this section, we see that it is possible to calculate bond prices, spot, and forward rates provided that one has a set of only one of these parameters. Therefore, given the following set of zero-coupon rates, observed in the market, given in Table 3.2, we calculate the corresponding forward rates and zero-coupon bond prices as shown. The initial term structure is upward sloping. The two curves are illustrated in Figure 3.4A.

There are technical reasons why the theoretical forward rate has a severe kink at the later maturity, but we shall not go into an explanation of this as it is outside the scope of this book.

Essentially, the relationship between the spot- and forward-rate curves is as stated by Campbell, Lo, and MacKinlay (1997). The forward-rate curve will lie above the spot-rate curve if the latter is increasing, and will lie below it if the spot-rate curve is decreasing. This relationship can be shown mathematically; the forward rate or marginal rate of return is equal to the spot rate or average rate of return plus the rate of increase of the spot rate, multiplied by the sum of the increases between t and T. If the spot rate is constant (a flat curve), the forward-rate curve will be equal to it.

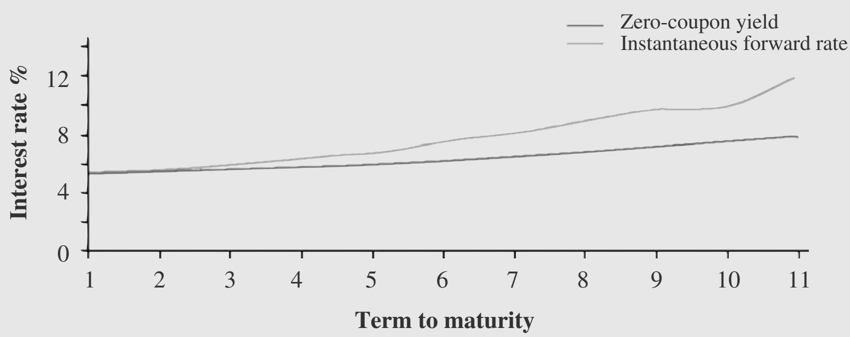

FIGURE 3.4A Hypothetical Zero-Coupon and Forward Yield Curves

(continued)

EXAMPLE 3.1 (*Continued*)

TABLE 3.2 Hypothetical Zero-Coupon Yield and Forward Rates

Term to maturity (0, T)	Spot Rate $r(0, T)$*	Forward Rate $f(0, T)$*	Bond Price $P(0, T)$
0			1
1	1.054	1.054	0.94877
2	1.055	1.056	0.89845
3	1.0563	1.059	0.8484
4	1.0582	1.064	0.79737
5	1.0602	1.068	0.7466
6	1.0628	1.076	0.69386
7	1.06553	1.082	0.64128
8	1.06856	1.0901	0.58833
9	1.07168	1.0972	0.53631
10	1.07526	1.1001	0.48403
11	1.07929	1.1205	0.43198

*Interest rates are given as $(1 + r)$ so to obtain the rate r itself we subtract 1.

However, an increasing spot-rate curve does not always result in an increasing forward curve; only one that lies above it will produce this result. It is possible for the forward curve to be increasing or decreasing while the spot rate is increasing. If the spot rate reaches a maximum level and then stays constant, or falls below this high point, the forward curve will begin to decrease at a maturity point earlier than the spot-curve high point. In the example in Figure 3.4A, the rate of increase in the spot rate in the last period is magnified when converted to the equivalent forward rate; if the last spot rate had been below the previous-period rate, the forward-rate curve would look like that in Figure 3.4B.

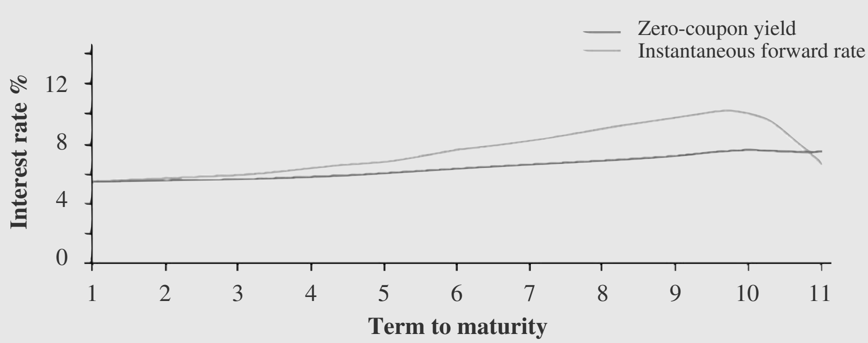

FIGURE 3.4B Hypothetical Spot and Forward Yield Curves

Calculating Spot Rates in Practice

Researchers have applied econometric techniques to the problem of extracting a zero-coupon term structure from coupon bond prices. The most well-known approaches are described in McCulloch (1971, 1975), Schaefer (1981), Nelson and Siegel (1987), Deacon and Derry (1994), Adams and Van Deventer (1994), and Waggoner (1997), to name but a few. The most accessible article is probably the one by Deacon and Derry.[12] In addition, a good overview of all the main approaches is contained in James and Webber (2000), and Chapters 15–18 of their book provide an excellent summary.

We have noted that a coupon bond may be regarded as a portfolio of zero-coupon bonds. By treating a set of coupon bonds as a larger set of zero-coupon bonds, we can extract an (implied) zero-coupon interest-rate structure from the yields on the coupon bonds.

If the actual term structure is observable, so that we know the prices of zero-coupon bonds of £1 nominal value $P_1, P_2, \ldots\ldots, P_N$, then the price P_C of a coupon bond of nominal value £1 and coupon C is given by

$$P_c = P_1 C + P_2 C + \ldots\ldots + P_n(1+C) \tag{3.30}$$

Conversely if we can observe the coupon bond yield curve, so that we know the prices $P_{C1}, P_{C2}, \ldots\ldots, P_{CN}$, then we may use (3.30) to extract the implied zero-coupon term structure. We begin with the one-period coupon bond, for which the price is

$$P_{c1} = P_1(1 + C)$$

so that

$$P_1 = \frac{P_{c1}}{(1+C)} \tag{3.31}$$

This process is repeated. Once we have the set of zero-coupon bond prices $P_1, P_2, \ldots\ldots, P_{N-1}$ we obtain P_N using

$$P_N = \frac{P_{CN} - P_{N-1}C - \ldots\ldots - P_1 C}{1+C} \tag{3.32}$$

At this point, we apply a regression technique known as ordinary least squares (OLS) to fit the term structure. The next chapter discusses this area in greater detail (we have segregated this so that readers who do not require an extensive familiarity with this subject may skip the next chapter). Interested readers should also consult the references at the end of Chapter 4.

Expression (3.30) restricts the prices of coupon bonds to be precise functions of the other coupon bond prices. In fact, this is unlikely in practice because specific bonds will be treated differently according to liquidity, tax effects, and so on. For this

[12] This is in the author's opinion. Those with a good grounding in econometrics will find all these references both readable and accessible. Further recommended references are given in the bibliography.

reason, we add an error term to (3.30) and estimate the value using cross-sectional regression against all the other bonds in the market. If we say that these bonds are numbered $i = 1, 2, \ldots, I$, then the regression is given by

$$P_{C_i N_i} = P_1 C_i + P_2 C_i + \ldots\ldots + P_{N_i}(1 + C_i) + u_i \tag{3.33}$$

for $i = 1, 2, \ldots, I$ and where Ci is the coupon on the ith bond and N_i is the maturity of the ith bond. In (3.33), the regressor parameters are the coupon payments at each interest period date, and the coefficients are the prices of the zero-coupon bonds P_1 to P_N where $j = 1, 2, \ldots, N$. The values are obtained using OLS, as long as we have a complete term structure and that $I \geq N$.

In practice, we will not have complete term structure of coupon bonds, and so we are not able to identify the coefficients in (3.33). McCulloch (1971, 1975) described a spline estimation method, which assumes that zero-coupon bond prices vary smoothly with term to maturity. In this approach, we define P_N, a function of maturity $P(N)$, as a discount function given by

$$P(N) = 1 + \sum_{j=1}^{J} a_j f_j(N) \tag{3.34}$$

The function $f_j(N)$ is a known function of maturity N, and the coefficients a_j must be estimated. We arrive at a regression equation by substituting (3.34) into (3.33) to give us (3.35), which can be estimated using OLS.

$$\prod_i = \sum_{j=1}^{J} a_j X_j + u_i,\ i = 1, 2 \ldots\ldots, I \tag{3.35}$$

where

$$\prod_i \equiv P_{C_i N_i} - 1 - C_i N_i$$

$$X_{ij} \equiv f_j(N_i) + C_i \sum_{l=1}^{N_i} f_j(l).$$

The function $f_j(N)$ is usually specified by setting the discount function as a polynomial. In certain texts, including McCulloch, this is carried out by applying what is known as a spline function. Considerable academic research has gone into the use of spline functions as a yield curve fitting technique; however, we are not able to go into the required level of detail here, which is left to the next chapter. Please refer to the bibliography for further information. For a specific discussion on using regression techniques for spline curve fitting methods, see Suits, Mason, and Chan (1978).

TERM STRUCTURE HYPOTHESES

As befits a subject that has been the target of extensive research, a number of hypotheses have been put forward that seek to explain the term structure of interest rates. These hypotheses describe why yield curves assume certain shapes, and relate maturity terms with spot and forward rates. In this section, we briefly review these hypotheses.

The Expectations Hypothesis

Simply put, the expectations hypothesis states that the slope of the yield curve reflects the market's expectations about future interest rates. There are, in fact, four main versions of the hypothesis, each distinct from and not compatible with the others. The expectations hypothesis has a long history, first being described in 1896 by Fisher and later developed by Hicks (1946) among others.[13] As Shiller (1990) describes, the thinking behind it probably stems from the way market participants discuss their view on future interest rates when assessing whether to purchase long-dated or short-dated bonds. For instance, if interest rates are expected to fall, investors will purchase long-dated bonds in order to lock in the current high long-dated yield. If all investors act in the same way, the yield on long-dated bonds will, of course, decline as prices rise in response to demand, and this yield will remain low as long as short-dated rates are expected to fall, and will revert to a higher level only once the demand for long-term rates is reduced. Therefore, downward-sloping yield curves are an indication that interest rates are expected to fall, while an upward-sloping curve reflects market expectations of a rise in short-term interest rates.

Let us briefly consider the main elements of the discussion. The unbiased expectations hypothesis states that current forward rates are unbiased predictors of future spot rates. Let $f_t(T,T + 1)$ be the forward rate at time t for the period from T to $T + 1$. If the one-period spot rate at time T is r_T, then according to the unbiased expectations hypothesis

$$f_t(T,T+1) = E_t[r_T] \tag{3.36}$$

which states that the forward rate $f_t(T, T + 1)$ is the expected value of the future one-period spot rate given by r_T at time T.

The return-to-maturity expectations hypothesis states that the return generated from an investment of term t to T by holding a $(T - t)$–period bond will be equal to the expected return generated by holding a series of one-period bonds and continually rolling them over on maturity. More formally we write

$$\frac{1}{P(t,T)} = E_t[(1+r_t)(1+r_{t+1})\ldots\ldots(1+r_{T-1})] \tag{3.37}$$

The left-hand side of (3.37) represents the return received by an investor holding a zero-coupon bond to maturity, which is equal to the expected return associated with rolling over £1 from time t to time T by continually reinvesting one-period maturity bonds, each of which has a yield of the future spot rate r_t. A good argument for this hypothesis is contained in Jarrow (1996, p. 52), which states that essentially in an environment of economic equilibrium the returns on zero-coupon bonds of similar maturity cannot be significantly different, otherwise investors would not hold the bonds with the lower return. A similar argument can be put forward with relation to coupon bonds of differing maturities. Any difference in yield would not therefore disappear as equilibrium was reestablished. However, there are a number of reasons

[13] See the footnote on p. 644 of Shiller (1990) for a fascinating historical note on the origins of the expectations hypothesis. An excellent overview of the hypothesis itself is contained in Ingersoll (1987), 389–392.

why investors will hold shorter-dated bonds, irrespective of the yield available on them, so it is possible for the return-to-maturity version of the hypothesis not to apply. In essence, this version represents an equilibrium condition in which expected holding-period returns are equal, although it does not state that this return is the same from different bond-holding strategies.

From (3.36) and (3.37) we can determine that the unbiased expectations hypothesis and the return-to-maturity hypothesis are not compatible with each other, unless there is no correlation between future interest rates. As Ingersoll (1987) notes, although it would be both possible and interesting to model such an economic environment, it is not related to reality, as interest rates are highly correlated. Given positive correlation between rates over a period of time, bonds with maturity terms longer than two periods will have a higher price under the unbiased expectations hypothesis than under the return-to-maturity version. Bonds of exactly two-period maturity will have the same price.

The yield-to-maturity expectations hypothesis is described in terms of yields. It is given by

$$\left[\frac{1}{P(t,T)}\right]^{\frac{1}{T-t}} = E_t\left[\{(1+r_t)(1+r_{t+1})\ldots\ldots(1+r_{T-1})\}^{\frac{1}{T-t}}\right] \tag{3.38}$$

where the left-hand side specifies the yield to maturity of the zero-coupon bond at time t. In this version, the expected holding period yield on continually rolling over a series of one-period bonds will be equal to the yield that is guaranteed by holding a long-dated bond until maturity. The local expectations hypothesis states that all bonds will generate the same expected rate of return if held over a small term. It is given by

$$\frac{E_t[P(t+1,T)]}{P(t,T)} = 1 + r_t \tag{3.39}$$

This version of the hypothesis is the only one that is consistent with no-arbitrage, because the expected rates of return on all bonds are equal to the risk-free interest rate. For this reason, the local expectations hypothesis is sometimes referred to as the risk-neutral expectations hypothesis.

Liquidity Premium Hypothesis

The liquidity premium hypothesis arises from the natural desire for borrowers to borrow long while lenders prefer to lend short. It states that current forward rates differ from future spot rates by an amount that is known as the liquidity premium. It is expressed as

$$f_t(T,T+1) = E_t[r_T] + \pi_t(T,T+1) \tag{3.40}$$

Equation (3.40) states that the forward rate $f_t(T,T+1)$ is the expected value of the future one-period spot rate given by r_T at time T plus the liquidity premium, which is a function of the maturity of the bond (or term of loan). This premium reflects the conflicting requirements of borrowing and lenders, while traders and speculators will borrow short and lend long, in an effort to earn the premium. The liquidity premium hypothesis has been described in Hicks (1946).

Segmented-Markets Hypothesis

The segmented-markets hypothesis seeks to explain the shape of the yield curve by stating that different types of market participants invest in different sectors of the term structure, according to their requirements. So, for instance, the banking sector has a requirement for short-dated bonds, while pension funds will invest in the long end of the market. This was first described in Culbertson (1957). There may also be regulatory reasons why different investors have preferences for particular maturity investments. *A preferred-habitat* theory was described in Modigliani and Sutch (1967), which states not only that investors have a preferred maturity but also that they may move outside this sector if they receive a premium for so doing. This would explain humped shapes in yield curves. The preferred-habitat theory may be viewed as a version of the liquidity-preference hypothesis, where the preferred habitat is the short end of the yield curve, so that longer-dated bonds must offer a premium in order to entice investors to hold them. This is described in Cox, Ingersoll, and Ross (1981).

EVIDENCE ON THE EXPECTATIONS HYPOTHESIS

The actions of market investors would seem to suggest that the expectations hypothesis does not apply in practice. This is not surprising, because the increased risk associated with longer-dated securities, both interest-rate risk and credit risk, suggest that returns should increase monotonically along the term structure. This is not a new idea. Hicks (1946) first proposed the idea of a liquidity premium embedded in the term structure. Fama (1986) and Longstaff (1990) showed that short-term returns reflected a liquidity premium.

Dhillon and Lasser (1998) presented evidence of a monotonically increasing term premium for U.S. Treasury strips. This is, of course, inconsistent with the expectations hypothesis. Nevertheless the result is not surprising, and reflects investor beliefs on the market. The existence of a liquidity premium, and an increasing relationship between liquidity premium and the maturity structure, reflects market belief on how the risk-return profile should behave.

APPENDICES

APPENDIX 3.1 THE INTEGRAL

The approach used to define integrals begins with an approximation involving a countable number of objects, which is then gradually transformed into an uncountable number of objects. A common form of integral is the Riemann integral.

Given a calculable or deterministic function that has been graphed for a period of time, let us say we require the area represented by this graph. The function is $f(t)$ and is graphed over the period $[0, T]$. The area of the graph is given by the integral

$$\int_0^T f(s)ds \tag{A3.1.1}$$

which is the area represented by the graph. This can be calculated using the Riemann integral, for which the area represented is shown in Figure A3.1.

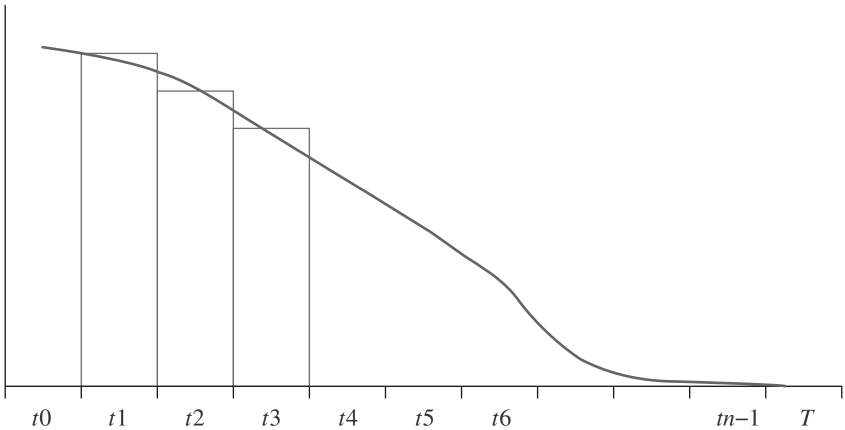

FIGURE A3.1 Calculating Area Using the Integral

The General Definition

To make the calculation, the time interval is separated into a series n of intervals, given by $t_0 = 0 < t_1 < t_2 < \ldots < t_{n-1} < t_n < T$.

The approximate area under the graph is given by the sum of the area of each of the rectangles, for which we assume each segment outside the graph is compensated by the area under the line that is not captured by any of the rectangles. Therefore, we can say that an approximating measure is described by

$$\sum_{i=0}^{n} f\left[\frac{t_i + t_{i-1}}{2}\right](t_i - t_{i-1}) \tag{A3.1.2}$$

This states that the area under the graph can be approximated by taking the sum of the n rectangles, which are created from the base x-axis that begins from t_0 through to T_n and the y-axis as height, described as

$$f((t_i + t_{i-1})/2)$$

This approximation only works if a sufficiently small base has been used for each interval, and if the function $f(t)$ is a smooth function; that is, it does not experience sudden swings or kinks.

The definition of the Riemann integral is, given that

$$\max_i |t_i - t_{i-1}| \mapsto 0$$

defined by the limit

$$\sum_{i=0}^{n} f\left[\frac{t_i + t_{i-1}}{2}\right](t_i - t_{i-1}) \to \int_0^T f(s)ds \tag{A3.1.3}$$

If the approximation is not sufficiently accurate, we can adjust it by making the intervals smaller still, which will make the approximation closer. This approach cannot be used if the function is not smooth; in mathematics this requirement is stated by saying that the function must be integrable or Riemann integrable.

Other integral forms are also used. A good introduction to these is given in Neftci (2000), Chapter 4.

APPENDIX 3.2 THE DERIVATION OF THE BOND PRICE EQUATION IN CONTINUOUS TIME

This section summarises the approach described in Ross (1999), on pages 54–56. This is an excellent reference, very readable and accessible, and is highly recommended. We replace Ross's use of investment at time 0 for maturity at time t with the terms t and T respectively, which is consistent with the price equations given in the main text. We also use the symbol M for the maturity value of the money-market account, again to maintain consistency with the expressions used in Chapters 2–4 of this book.

Assume a continuously compounded deterministic interest rate $r(s)$ that is payable on a money-market account at time s. This is the instantaneous interest rate at time s. Assume further that an investor places x in this account at time s; after a very short time period of time h, the account would contain

$$x_h \approx x(1 + r(s)h) \tag{A3.2.1}$$

Say that $M(T)$ is the amount that will be in the account at time T if an investor deposits £1 at time t. To calculate $M(T)$ in terms of the spot rate $r(s)$, where $t \leq s \leq T$, for an incremental change in time of h, we have

$$M(s + h) \approx M(s)(1 + r(s)h) \tag{A3.2.2}$$

which leads us to

$$M(s + h) - M(s) \approx M(s)r(s)h \tag{A3.2.3}$$

and

$$\frac{M(s + h) - M(s)}{h} \approx M(s)r(s) \tag{A3.2.4}$$

The approximation given by (A3.2.4) turns into an equality as the time represented by h becomes progressively smaller. At the limit given as h approaches zero, we say

$$M'(s) = M(s)r(s) \tag{A3.2.5}$$

which can be rearranged to give

$$\frac{M'(s)}{M(s)} = r(s) \tag{A3.2.6}$$

From expression (A3.2.6) we imply that in a continuous time process

$$\int_t^T \frac{M'(s)}{M(s)} ds = \int_t^T r(s)\, ds \tag{A3.2.7}$$

and that

$$\log(M(T)) - \log(M(r)) = \int_t^T r(s)\, ds \tag{A3.2.8}$$

However, we deposited £1 at time t, that is $M(t) = 1$, so from (A3.2.8) we obtain the value of the money-market account at T to be

$$M(T) = \exp\left[\int_t^T r(s)\, ds\right] \tag{A3.2.9}$$

which was our basic equation shown as (A3.2.1).

Let us now introduce a risk-free zero-coupon bond that has a maturity value of £1 when it is redeemed at time T. If the spot rate is constant, then the price at time t of this bond is given by

$$P(t, T) = e^{-r(T-t)} \tag{A3.2.10}$$

where r is the continuously compounded instantaneous interest rate. The right-hand side of (A3.2.10) is the expression for the present value of £1, payable at time T, discounted at time t at the continuously compounded constant interest rate r.

So we say that $P(t, T)$ is the present value at time t of £1 to be received at time T. Since a deposit of $1/M(T)$ at time t will have a value of 1 at time T, we are able to say that

$$P(t, T) = \frac{1}{M(T)} = \exp\left[-\int_t^T r(s)ds\right] \tag{A3.2.11}$$

which is our bond price equation in continuous time.

If we say that the *average* of the spot interest rates from t to T is denoted by $rf(T)$, so we have $rf(T) = \frac{1}{T}\int_t^T r(s)ds$, then the function $rf(T)$ is the term structure of interest rates.

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CHAPTER 4

Interest-Rate Modelling

In Chapter 3 we introduced basic concepts of bond interest rates and analysis. We build on these concepts in this and the next chapter, which are a review of the initial and subsequent work conducted in this field. Term-structure modelling has been extensively researched in the financial economics literature; it is possibly the most heavily covered subject in financial economics! It is not possible to deliver a comprehensive summary in just two chapters, but we aim to cover the main topics in depth. As ever, interested readers are directed to the bibliography listing, which contains the more accessible titles in this area.

In this chapter we review the more commonly used interest-rate models. In the next chapter we discuss some of the techniques used to fit a smooth yield curve to market-observed bond yields.

BACKGROUND

Term-structure modelling is based on theory describing the behaviour of interest rates. A model would seek to identify the elements or factors that are believed to explain the dynamics of interest rates. These factors are *stochastic,* a subset of random, in nature, so that we cannot predict with certainty the future level of any particular factor.[1] An interest-rate model must therefore specify a statistical process that describes the stochastic property of these factors, in order to arrive at a reasonably accurate representation of the behaviour of interest rates.

The first term-structure models covered in the academic literature described the interest-rate process as one where the short rate[2] follows a statistical process and where all other interest rates are a function of the short rate. So the dynamics of the short rate drive all other term interest rates. These models are known as one-factor models. A one-factor model assumes that all term rates follow on from when the short rate is specified; that is, they are not randomly determined. Two-factor interest-rate models have also been described. For instance, the model described by Brennan and Schwartz (1979) specified the factors as the short rate and a long-term rate, while a model described by Fong and Vasicek (1991) specified the factors as the short rate and short-rate volatility.

[1] The word *stochastic* is derived from a term in ancient Greek, the word *stokhos* which means "a bull's eye". The connection? Throwing darts at a board and aiming for the bull's eye is a stochastic process, as it will contain a number of misses.

[2] We came across this expression in Chapter 3.

Basic Concepts

The foundation of interest-rate models is grounded in probability theory, so readers may wish to familiarise themselves with this subject. An excellent introduction to this is given in Ross (1999), which we referred to in Chapter 3, while a fuller treatment is given in the same author's better-known book, *Probability Models* (2000).

In a one-factor model of interest rates, the short rate is assumed to be a random or stochastic variable, with the dynamics of its behaviour being uncertain and acting in an unpredictable manner. A random variable such as the short rate is defined as a variable whose future outcome can assume more than one possible value. Random variables are either discrete or continuous. A discrete variable moves in identifiable breaks or jumps. So, for example, while time is continuous, the trading hours of an exchange-traded future are not continuous, as the exchange will be shut outside business hours. Interest rates are treated in academic literature as being continuous, whereas, in fact, rates such as central-bank base rates move in discrete steps. A continuous variable moves in a manner that has no breaks or jumps. So, if an interest rate can move in a range from 5% to 10%, if it is continuous it can assume any value between this range; for instance, a value of 5.671291%. Although this does not reflect market reality, assuming that interest rates and the processes they follow are continuous allows us to use calculus to derive useful results in our analysis.

The short rate is said to follow a stochastic process, so although the rate itself cannot be predicted with certainty—since it can assume a range of possible values in the future—the process by which it changes from value to value can be assumed, and hence modelled. The dynamics of the short rate, therefore, are a stochastic process or probability distribution. A one-factor model of the interest rate actually specifies the stochastic process that describes the movement of the short rate.

The analysis of stochastic processes employs mathematical techniques originally used in physics. An instantaneous change in value of a random variable x is written as dx. The changes in the random variable are assumed to be normally distributed. The shock to this random variable that generates it to change value, also referred to as *noise,* follows a randomly generated process known as a Weiner process or geometric Brownian motion. This is described in Appendix 4.1. A variable following a Weiner process is a random variable, termed x or z, whose value alters instantaneously, but whose patterns of change follow a normal distribution with mean 0 and standard deviation 1. If we assume that the yield r of a zero-coupon bond follows a continuous Weiner process with mean 0 and standard deviation 1, this would be written $dr = dz$.

Changes or "jumps" in the yield that follow a Weiner process are scaled by the volatility of the stochastic process that drives interest rates, which is given by σ. So the stochastic process for the change in yields is given by $dr = \sigma dz$.

The value of this volatility parameter is user-specified; that is, it is set at a value that the user feels most accurately describes the current interest-rate environment. Users often use the volatility implied by the market price of interest-rate derivatives such as caps and floors.

So far, we've said that the zero-coupon bond yield is a stochastic process following a geometric Brownian motion that drifts with no discernible trend. However, under this scenario, over time the yield would continuously rise to a level of infinity or fall to infinity, which is not an accurate representation of reality. We need to add to the model a term that describes the observed trend of interest rates moving up and down in a cycle. This expected direction of the change in the short rate is the second parameter in an interest-rate model, which in some texts is referred to by a letter such as *a* or *b* and in other texts is referred to as μ.

The short-rate process can therefore be described in the functional form given by (4.1)

$$dr = adt + \sigma dz \tag{4.1}$$

where dr is the change in the short rate
a is the expected direction of change of the short rate or *drift*
dt is the incremental change in time
σ is the standard deviation of changes in the short rate
dz is the random process.

Equation (4.1) is sometimes seen with dW or dx in place of dz. It assumes that, on average, the instantaneous change in interest rates is given by the function adt, with random shocks specified by σdz. It is similar to a number of models, such as those first described by Vasicek (1977), Ho and Lee (1986), Hull and White (1990) and others.

To reiterate then, (4.1) states that the change in the short rate r over an infinitesimal period of time dt, termed dr, is a function of:

- The drift rate or expected direction of change in the short rate a
- A random process dz

The two significant properties of the geometric Brownian motion are:

- The drift rate is equal to the expected value of the change in the short rate. Under a zero drift rate, the expected value of the change is also zero, and the expected value of the short rate is given by its current value.
- The variance of the change in the short rate over a period of time T is equal to T, while its standard deviation is given by $\sqrt{T}$.

The model given by (4.1) describes a stochastic short-rate process, modified with a drift rate to influence the direction of change. However, a more realistic specification would also build in a term that describes the long-run behaviour of interest rates to drift back to a long-run level. This process is known as mean reversion, and is perhaps is best known by the Hull-White model. A general specification of mean reversion would be a modification given by (4.2).

$$dr = a(b - r)dt + \sigma dz \tag{4.2}$$

where b is the long-run mean level of interest rates and where a now describes the speed of mean reversion.

Equation (4.2) is known as an Ornstein-Uhlenbeck process. When r is greater than b, it will be pulled back towards b, although random shocks generated by dz will delay this process. When r is below b the short rate will be pulled up towards b.

Ito's Lemma

Having specified a term-structure model, for market practitioners it becomes necessary to determine how security prices related to interest rates fluctuate. The main instance of this is where we wish to determine how the price P of a bond moves over time and as the short rate r varies. The formula used to do this is known as Ito's lemma. For the background on the application of Ito's lemma, see Hull (1997) or Baxter and Rennie (1996). Ito's lemma transforms the dynamics of the bond price P into a stochastic process in the following form:

$$dP = P_r dr + \tfrac{1}{2} P_{rr} (dr)^2 + P_t \tag{4.3}$$

The subscripts indicate partial derivatives.[3] The terms dr and $(dr)^2$ are dependent on the stochastic process that is selected for the short rate r. If this process is the Ornstein-Uhlenbeck process that was described in (4.2), then the dynamics of P can be specified as (4.4).

$$\begin{aligned} dP &= P_r[a(b-r)dt + \sigma dz] + \tfrac{1}{2} \mathrm{P}_{rr}\sigma^2 dt + P_t dt \\ &= [P_r a(b-r) + \tfrac{1}{2} P_{rr}\sigma^2 + P_t]dt + P_r \sigma dz \\ &= a(r,t)dt + \sigma(r,t)dz \end{aligned} \tag{4.4}$$

What we have done is to transform the dynamics of the bond price in terms of the drift and volatility of the short rate. Equation (4.4) states that the bond price depends on the drift of the short rate, and the volatility.

Ito's lemma is used as part of the process of building a term-structure model. The generic process this follows involves the following:

- Specifying the random or stochastic process followed by the short rate, for which we must make certain assumptions about the short rate itself.
- Using Ito's lemma to transform the dynamics of the zero-coupon bond price in terms of the short rate.
- Imposing no-arbitrage conditions, based on the principle of hedging a position in one bond with one in another bond of a different maturity (for a one-factor model), in order to derive the partial differential equation of the zero-coupon bond price. We note that this is for a one-factor model: for a two-factor model we would require two bonds as hedging instruments.
- Solving the partial differential equation for the bond price, which is subject to the condition that the price of a zero-coupon bond on maturity is 1.

In the next section we review some of the models that are used in this process.

[3] This is the great value of Ito's lemma, a mechanism by which we can transform a partial differential equation.

ONE-FACTOR TERM-STRUCTURE MODELS

In this section, we discuss briefly a number of popular term-structure models and attempt to summarise the advantages and disadvantages of each which render them useful or otherwise under certain conditions and user requirements.

The Vasicek Model

The Vasicek model (1977) was the first term-structure model described in the academic literature, and is a yield-based one-factor equilibrium model. It assumes that the short-rate process follows a normal distribution. The model incorporates mean reversion and is popular with certain practitioners as well as academics because it is analytically tractable.[4] Although it has a constant volatility element, the mean reversion feature means that the model removes the certainty of a negative interest rate over the long term. However, other practitioners do not favour the model because it is not necessarily arbitrage-free with respect to the prices of actual bonds in the market.

So the instantaneous short rate is described in the Vasicek model as

$$dr = a(b - r)dt + \sigma dz \tag{4.5}$$

where a is the speed of the mean reversion and b is the mean reversion level or the long-run value of r.

z is the standard Weiner process or Brownian motion with a 0 mean and 1 standard deviation. In Vasicek's model, the price at time t of a zero-coupon bond that matures at time T is given by

$$P(t,T) = A(t,T)e^{-B(t,T)r(t)} \tag{4.6}$$

where $r(t)$ is the short rate at time t and

$$B(t,T) = \frac{1 - e^{-a(T-t)}}{a}$$

and

$$A(t,T) = \exp\left[\frac{\big(B(t,T) - T + t\big)\big(a^2 b - \sigma^2/2\big)}{a^2} - \frac{\sigma^2 B(t,T)^2}{4a}\right]$$

The derivation of (4.6) is given in a number of texts (not least the original article!). We recommend section 5.3 in Van Deventer and Imai (1997) for its accessibility.

[4] Tractability is much prized in a yield-curve model, and refers to the ease with which a model can be implemented; that is, with which yield curves can be computed.

Note that in certain texts the model is written as

$$dr = \kappa(\theta - r)dt + \sigma dz$$

or

$$dr = \alpha(\mu - r)dt + \sigma dZ \quad dr = \alpha(\mu - r)dt + \sigma dz$$

but it just depends on which symbol the particular text is using. We use the form shown in (4.5) because it is consistent with our introductory discussion in the previous section.

In Vasicek's model, the short rate *r* is normally distributed so it can therefore be negative with positive probability. The occurrence of negative rates is dependent on the initial interest-rate level and the parameters chosen for the model, and is an extreme possibility. For instance, a very low initial rate, such as that observed in the Japanese economy for some time now, and volatility levels set with the market, have led to negative rates when using the Vasicek model. This possibility, which also applies to a number of other interest-rate models, is inconsistent with a no-arbitrage market because investors will hold cash rather than opt to invest at a negative interest rate.[5] However, for most applications the model is robust, and its tractability makes it popular with practitioners.

The Hull and White Model

The model described by Hull and White (1990) is another well-known model that fits the theoretical yield curve that one would obtain using Vasicek's model extracted from the actual observed market yield curve. As such, it is sometimes referred to as the "extended Vasicek model", with time-dependent drift.[6] The model is popular with practitioners precisely because it enables them to calculate a theoretical yield curve that is identical to yields observed in the market, which can then be used to price bonds and bond derivatives and also to calculate hedges.

The model is given in (4.7).

$$dr = a\left[\frac{b(t)}{a} - r\right]dt + \sigma dz \tag{4.7}$$

where *a* is the rate of mean reversion and $\frac{b(t)}{a}$ is a time-dependent mean reversion. The price at time *t* of a zero-coupon bond with maturity *T* is

$$P(t,T) = A(t,T)e^{-B(t,T)r(t)}$$

[5] This is stated in Black (1995).

[6] Haug (1998) states that the Hull-White model is essentially the Ho and Lee model with mean reversion.

where $r(t)$ is the short rate at time t and

$$B(t,T)=\frac{1-e^{-a(t,T)}}{a}$$

$$\text{In } A(t,T)=\text{In}\left[\frac{P(0,T)}{P(0,t)}\right]-B(t,T)\frac{\partial P(0,t)}{\partial t}-\frac{v(t,T)^2}{2}$$

and

$$v(t,T)=\frac{1}{2a^3}\sigma^2(e-^{aT}-e^{at})^2(e^{2at}-1)$$

The Hull-White model is commonly used by derivatives traders.

Further One-Factor Term-Structure Models

The academic literature and market application have thrown up a large number of term-structure models as alternatives to the Vasicek model and models based on it such as the Hull-White model. As with these two models, each possesses a number of advantages and disadvantages. As we noted in the previous section, the main advantage of Vasicek-type models is their analytic tractability, with the assumption of the dynamics of the interest-rate allowing the analytical solution of bonds and bond instruments. The main weakness of these models is that they permit the possibility of negative interest rates. While negative interest rates are not a market impossibility,[7] the thinking would appear to be that they are a function of more than one factor, and that therefore modelling them using Vasicek-type models is not tenable. This aspect of the models does not necessarily preclude their use in practice, and will depend on the state of the economy at the time. To consider an example, during 1997–2000 Japanese money-market interest rates were frequently below ½ of 1%, and at this level even low levels of volatility below 5% will imply negative interest rates with high probability if using Vasicek's model. In this environment, practitioners may wish to use models that do not admit the possibility of negative interest rates; perhaps those that model more than the short rate alone, so-called two-factor and multifactor models. We look briefly at these in the next section. First we consider, again briefly, a number of other one-factor models. As usual, readers are encouraged to review the bibliography articles for the necessary background and further detail on application.

[7] Negative interest rates manifest themselves most obviously in the market for specific bonds in repo, which have gone excessively special. However, academic researchers often prefer to work with interest-rate environments that do not consider negative rates a possibility (for example, see Black (1995)).

The Cox, Ingersoll, and Ross Model

Although published officially in 1985, the Cox-Ingersoll-Ross model was apparently described in academic circles in 1977 or perhaps earlier, which would make it the first interest-rate model. Like the Vasicek model, it is a one-factor model that defines interest-rate movements in terms of the dynamics of the short rate. However, it incorporates an additional feature whereby the variance of the short rate is related to the level of interest rates, and this feature has the effect of not allowing negative interest rates. It also reflects a higher interest-rate volatility in periods of relatively high interest rates, and correspondingly lower volatility when interest rates are lower.

The model is given in (4.8).

$$dr = \kappa(b - r)dt + \sigma\sqrt{rdz} \tag{4.8}$$

The derivation of the zero-coupon bond price equation given in (4.9) is contained in Ingersoll (1987), Chapter 18. The symbol τ represents the term to maturity of the bond or $(T - t)$.

$$P(r,\tau) = A(\tau)e^{-B(\tau)r} \tag{4.9}$$

where

$$A(\tau) = \left[\frac{2\gamma e^{(\gamma+k)\frac{\tau}{2}}}{g(\tau)}\right]^{\frac{2kb}{\sigma^2}}$$

$$B(\tau) = \frac{-2(1 - e^{\gamma\tau})}{g(\tau)}$$

$$\gamma = \sqrt{(k)^2 + 2\sigma^2}$$

$$g(\tau) = 2\gamma + (k + \gamma)(e^{\gamma\tau} - 1)$$

Some researchers[8] have stated that the difficulties in determining parameters for the CIR model have limited its use among market practitioners.

The Black, Derman, and Toy Model

The Black-Derman-Toy model (1990) also removes the possibility of negative interest rates and is commonly encountered in the markets. The parameters specified in

[8] For instance, see Van Deventer and Imai (1997) citing Fleseker (1993) on p. 336, although the authors go on to state that the CIR model is deserving of further empirical analysis and remains worthwhile for practical application.

the model are time-dependent, and the dynamics of the short-rate process incorporate changes in the level of the rate. The model is given in (4.10).

$$d[\ln(r) = [\vartheta(t) - \phi(t)\ln(r)]dt + (t)dz \tag{4.10}$$

The popularity of the model among market practitioners reflects the following:

- It fits the market-observed yield curve, similar to the Hull-White model;
- It makes no allowance for negative interest rates;
- It models the volatility levels of interest rates in the market.

Against this, the model is not considered particularly tractable or capable of being programmed for rapid calculation. Nevertheless, it is important in the market, particularly for interest-rate derivative market makers. An excellent and accessible description of the BDT model is contained in Sundaresan (1997, pp. 240–244); Tuckman (1996, pp. 102–106) is also recommended.

THE HEATH, JARROW, AND MORTON MODEL

We have devoted a separate section to the approach described by Heath, Jarrow, and Morton (1992) because it is a radical departure from the earlier family of interest-rate models. It is rarely encountered in practice though.

The Heath-Jarrow-Morton (HJM) approach to the specification of stochastic state variables is different from that used in earlier models. The previous models describe interest-rate dynamics in terms of the short rate as the single or (in two- and multifactor models) key state variable. With multifactor models, the specification of the state variables is the fundamental issue in practical application of the models themselves. In the HJM model, the entire term structure, and not just the short rate, is taken to be the state variable. In Chapter 3, it was shown how the term structure can be defined in terms of default-free zero-coupon bond prices, yields, spot rates, or forward rates. The HJM approach uses forward rates. So, in the single-factor HJM model the change in forward rates at current time t, with a maturity at time u, is captured by:

- A volatility function
- A drift function
- A geometric Brownian or Weiner process that describes the shocks or noise experienced by the term structure

We present here a brief introduction to the HJM model. A recommended fuller treatment is presented in Chapter 8 of James and Webber (2000) and Chapter 5 of Baxter and Rennie (1996). The author suggests reading James and Webber first; both accounts, while presented as an introduction, nevertheless require a good grounding in financial calculus.

The Single-Factor HJM Model

To establish the HJM framework, we present our familiar simple interest-rate model. If we have a forward-rate term structure $f(0, T)$ that is T-integrable, the dynamics of the forward term structure may be described by:

$$df(t,T) = a(t,T)dt + \sigma dz \tag{4.11}$$

where a is the drift rate and σ the constant volatility level. The expression dz represents the geometric Brownian motion or Weiner process and is written as dW in some texts, or dW_t to represent the expression as applicable to the current time t. This is the stochastic differential equation for the forward rate. To transform this expression into the equation for the price of an asset we would apply Ito calculus, but this is something that we shall leave to specialised texts on financial mathematics. The forward rate expressed as an integral equivalent of (4.11) is

$$f(t,T) = f(0,T) + \int_0^t s(s,T)\,ds + sdz \tag{4.12}$$

which assumes that the forward rate is normally distributed. Crucially, the different forward rates of maturity $f(0,1)$, $f(0,2)$, . . . , $f(0, T)$ are assumed to be perfectly correlated. The random element is the Brownian motion dz, and the impact of this process is felt over time, rather than over different maturities.

As we noted, the term structure can be defined in terms of bond prices, yields, spot rates, and forward rates. The HJM model is based on the instantaneous forward rate $f(t, T)$, and the single-factor version states that, given an initial forward-rate term structure $f(0, T)$ at time t, the forward rate for each maturity T is given by

$$df(t,T) = a(t,T)dt + \sigma(t,T)dz \tag{4.13}$$

in the differential form, but which is usually seen in integral form:

$$f(t,T) = f(0,T) + \int_0^t a(s,T)\,ds + \int_0^t \sigma(s,T)dz_{(s)} \tag{4.14}$$

where s is an incremental move forward in time (so that $0 \le t \le T$ and $t \le s \le T$). Note that the expressions in (4.13) and (4.14) are simplified versions of the formal model.

Under (4.13) and (4.14), the forward rate for any maturity period T will develop as described by the drift and volatility parameters $a(t,T)$ and $\sigma(t,T)$. In the single-factor HJM model, the random character of the forward-rate process is captured by the Brownian motion dz.[9] Under HJM, the primary assumption is that for each T, the drift and volatility processes are dependent only on the history of the Brownian-motion process up to the current time t, and on the forward rates themselves up to time t.

[9] Note that certain texts use a for the drift term and W_t for the random term.

The Multifactor HJM Model

Under the single-factor HJM model, the movement in forward rates of all maturities is perfectly correlated. This can be too much of a restriction for market application; for example, when pricing an interest-rate instrument that is dependent on the yield spread between two points on the yield curve. In the multifactor model, each of the state variables is described by its own Brownian-motion process.[10] So, for example, in an *m*-factor model there would be *m* Brownian motions in the model, dz_1, dz_2, ... , dz_m. This allows each *T*-maturity forward rate to be described by its own volatility level $\sigma_i(t,T)$ and Brownian-motion process dz_i. Under this approach, the different forward rates given by the different maturity bonds that describe the current term structure evolve under more appropriate random processes, and different correlations between forward rates of differing maturities can be accommodated.

The multifactor HJM model is given in equation (4.15).

$$f(t,T) = f(0,T) + \int_0^t a(s,T)\,ds + \sum_{i=1}^{m} \int_0^t \sigma(s,T)\,dz_i(s) \tag{4.15}$$

Equation (4.15) states that the dynamics of the forward-rate process, beginning with the initial rate $f(0, T)$, are specified by the set of Brownian-motion processes and the drift parameter.

For practical application, the evolution of the forward-rate term structure is usually carried out as a binomial-type path-dependent process. However path-independent processes have also been used. The HJM approach has become popular in the market, both for yield-curve modelling and for pricing derivative instruments, due to the realistic effect of matching yield-curve maturities to different volatility levels, and it is reasonably tractable when applied using the binomial-tree approach. Simulation modelling based on Monte Carlo techniques are also used. For further detail on the former approach, see Jarrow (1996).

CHOOSING A TERM-STRUCTURE MODEL

Selection of an appropriate term-structure model is more of an art than a science. The different types of model available, and the different applications and user requirements, mean that it is not necessarily clear-cut which approach should be selected. In essence, a practitioner's requirements will determine whether a single-factor model or a two- or multifactor model is more appropriate. The Ho-Lee and BDT models, for example, are *arbitrage* models, which means that they are designed to match the current term structure. With arbitrage (or arbitrage-free) models, assuming that the specification of the evolution of the short rate is correct, the law of no-arbitrage can be used to determine the price of interest-rate derivatives. There is also a class of interest-rate models known as *equilibrium* models, which make an assumption of the dynamics of the short rate in the same way as arbitrage models do, but are not designed to match the current term structure. With equilibrium models, therefore, the price of zero-coupon bonds given by the model-derived term structure is not re-

[10] For a good introduction, see Chapter 6 in Baxter and Rennie (1996).

quired to (and does not) match prices seen in the market. This means that the prices of bonds and interest-rate derivatives are not given purely by the short-rate process. Overall then, arbitrage models take the current yield curve as described by the market prices of default-free bonds as given, whereas equilibrium models do not.

What considerations must be taken into account when deciding which term-structure model to use? Some of the key factors include:

Ease of application. The key input to arbitrage models is the current spot-rate term structure, which is straightforward to determine using the market price of bonds currently trading in the market. This is an advantage over equilibrium models, whose inputs are more difficult to obtain.

Capturing market imperfections. The term structure generated by an arbitrage model will reflect the current market term structure, which may include pricing irregularities due to liquidity and other considerations. If this is not desired, it is a weakness of the arbitrage approach. Equilibrium models would not reflect pricing imperfections.

Pricing bonds and interest-rate derivatives. Traditional seat-of-the-pants market-making often employs a combination of the trader's nous, the range of prices observed in the market (often from interdealer broker screens), and gut feeling to price bonds. For a more scientific approach or for relative value trading,[11] a yield-curve model may well be desirable. In this case, an equilibrium model is clearly the preferred model, as the trader will want to compare the theoretical price given by the model compared to the actual price observed in the market. An arbitrage model would not be appropriate because it would take the observed yield curve, and hence the market bond price, as given, and so would assume that the market bond prices were correct. Put another way, using an arbitrage model for relative-value trading would suggest to the trader that there was no gain to be made from entering into, say, a yield-curve spread trade. Pricing derivative instruments, such as interest-rate options or swaptions, require a different emphasis. This is because the primary consideration of the derivative market maker is the technique and price of hedging the derivative. That is, upon writing a derivative contract the market maker will simultaneously hedge the exposure using either the underlying asset or a combination of this and other derivatives such as exchange-traded futures. The derivative market maker generates profit through extracting premium and from the difference in price over time between the price of the derivative and the underlying hedge position. For this reason, only an arbitrage model is appropriate, as it would price the derivative relative to the market, which is important for a market maker. An equilibrium model would price the derivative relative to the theoretical market, which would not be appropriate since it is a market instrument that is being used as the hedge.

Use of models over time. At initial use, the parameters used in an interest-rate model, most notably the drift, volatility and (if applicable) mean reversion

[11]For example, yield-curve trades where bonds of different maturities are spread against each other, with the trader betting on the change in spread as opposed to the direction of interest rates, are a form of relative-value trade.

rate reflect the state of the economy up to that point. This state is not constant, and so consequently, over time, any model must be continually recalibrated to reflect the current state of the market. That is, the drift rate used today when calculating the term structure may well be a different value tomorrow. This puts arbitrage models at a disadvantage, as their parameters will be changed continuously in this way. Put another way, use of arbitrage models is not consistent over time. Equilibrium-model parameters are calculated from historic data or from intuitive logic, and so may not be changed as frequently. However, their accuracy over time may suffer. It is up to users to decide whether they prefer the continual tweaking of the arbitrage model over the more consistent use of the equilibrium model.

This is just the beginning; there are a range of issues that must be considered by users when selecting an interest-rate model. For example, in practice it has been observed that models incorporating mean reversion work more accurately than those that do not feature this. Another factor is the computer processing power available to the user, and it is often the case that single-factor models are preferred precisely because processing is more straightforward. A good account of the different factors to be considered when assessing which model to use is given in Chapter 15 of James and Webber (2000).

APPENDIX

APPENDIX 4.1 GEOMETRIC BROWNIAN MOTION

Brownian motion was described in 1827 by the English scientist Robert Brown, and defined mathematically by the American mathematician Norbert Weiner in 1918. As applied to the price of a security, consider the change in price of a security as it alters over time. The time now is denoted as 0, with $P(t)$ as the price of the security at time t from now. The collection of prices $P(t)$, $0 \leq t \leq \infty$ is said to follow a Brownian motion with *drift* parameter μ and variance parameter σ^2 if for all non-negative values of t and T the random variable

$$P(t+T)-P(t)$$

is independent of all the prices P that have been recorded up to time t. That is, the historic prices do not influence the value of the random variable. Also the random variable is normally distributed with a mean uT and variance $\sigma_2 T$.

Standard Brownian motion has two drawbacks when applied to model security prices. The first and most significant is that, as the security price is a normally distributed random variable, it can assume negative values with non-negative probability, a property of the normal distribution. This cannot happen with equity prices and only rarely, under special conditions, with interest rates. The second drawback of standard Brownian motion is that the difference between prices over an interval is assumed to follow a normal distribution irrespective of the price of the security at the start of the interval. This is not realistic, as the probabilities are affected by the initial price of the security.

For this reason the geometric Brownian motion model is used in quantitative finance. Let us consider this now. Again, the time now is 0, and the security price at time t from now is given by $P(t)$. The collection of prices P(t), $0 \le t \le \infty$ follows a geometric Brownian motion with drift μ and standard deviation or volatility σ if for non-negative values of t and T the random variable $P(t + T)/P(t)$ is independent of all prices up to time t. In addition, the value

$$\log\left[\frac{P(t+T)}{P(t)}\right]$$

is a normally distributed random variable with mean uT and variance $\sigma^2 T$.

What is the significance of this? It is this: Once the parameters μ and σ have been ascertained, the present price of the security, and the present price only, determines the probabilities of future prices. The history of past prices has no impact. Also the probabilities of the ratio of the price at future time T to the price now are not dependent on the present price. The practical impact of this is that the probability that the price of a security doubles at some specified point in the future is identical whether the price now is 5 or 50.

For our purposes, we need only be aware that at an initial price of $P(0)$, the expected price at time t is a function of the two parameters of geometric Brownian motion. The expected price, given an initial price $P(0)$, is given by

$$E[P(t)] = P_0 e^{t(\mu+\sigma^2/2)} \tag{A4.1.1}$$

(A4.1) states that under geometric Brownian motion the expected price of a security is the present price increasing at the rate of $\mu + \sigma^2/2$. The evolution of a price process, including an interest rate, under varying parameters is shown in Figure A4.1, with an initial price level at 100.

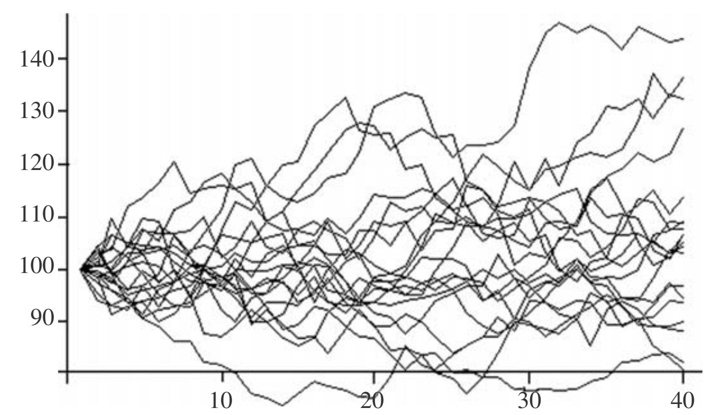

FIGURE A4.1 Evolution of Brownian or Weiner Process

SELECTED BIBLIOGRAPHY AND REFERENCES

Readers are first directed to the specific references cited in the text. If your interest is a general treatment of term-structure models, we suggest James and Webber (2000), while Baxter and Rennie (1996) is an excellent treatment of the mathematics involved. For a general exposition of the theory of the yield curve, Chapter 18 of Ingersoll (1987) is strongly recommended. Rebonato (1996) looks at the practical issues involved in implementing interest-rate models, and is a technical treatment. Finally, Van Deventer and Imai (1997) is an excellent, accessible account of, among other things, term-structure modelling and yield-curve smoothing techniques.

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CHAPTER 5

Fitting the Yield Curve

In this chapter we present some of the techniques used to actually fit the term structure. The term-structure models described in the previous chapter defined the interest-rate process under various assumptions about the nature of the stochastic process that drives these rates. However, the zero-coupon curve derived by models such as those described by Vasicek (1977), Brennan, and Schwartz (1979), and Cox, Ingersoll, and Ross (1985) do not fit the observed market rates or spot rates implied by market yields, and generally market yield curves are found to contain more variable shapes than those derived using term-structure models. Hence, the interest-rate models described in Chapter 4 are required to be calibrated to the market and, in practice, they are calibrated to the market yield curve. This is carried out in two ways; the model is either calibrated to market instruments such as money-market products and interest-rate swaps, which are used to construct the yield curve, or the yield curve is constructed from market-instrument rates, and the model is calibrated to this constructed curve. If the latter approach is preferred, there are a number of non-parametric methods that may be used. We will consider these later.

The academic literature contains a good deal of research into the empirical estimation of the term structure, the object of which is to fit a zero-coupon curve[1] that is a reasonably accurate fit to the market prices *and* is a smooth function. There is an element of trade-off between these two objectives. The second objective is as important as the first, however, in order to derive a curve that makes economic sense. (It would be possible to fit the curve perfectly at the expense of smoothness, but this would have little practical value).

In this chapter, we review some of the methods used to fit the yield curve. An excellent account of the approaches we discuss in this chapter is given in Anderson et al. (1996). Unfortunately, this book is out of print, but an excellent working paper that formed part of the input to this book is still available, which is Deacon and Derry (1994). A selection of other useful references is given as usual in the bibliography.

YIELD-CURVE SMOOTHING

An approach that has been used to estimate the term structure was described by Carleton and Cooper (1976) and which assumed that default-free bond cash flows

[1] The zero-coupon or spot curve, or equivalently the forward-rate curve or the discount function: all would be describing the same thing.

are payable on specified discrete dates, with a set of unrelated discount factors that apply to each cash flow. These discount factors were then estimated as regression coefficients, with each bond cash flow acting as the independent variables, and the bond price for that date acting as the dependent variable.[2] Using simple linear regression in this way produces a discrete discount function, not a continuous one, and forward rates that are estimated from this function are very jagged. An approach more readily accepted by the market was described by McCulloch (1971), who fitted the discount function using polynomial splines. This method produces a continuous function, and one that is linear so that the ordinary least-squares regression technique can be employed. In a later study, Langetieg and Smoot (1981)[3] use an extended McCulloch method, fitting cubic splines to zero-coupon rates instead of the discount function, and using nonlinear methods of estimation.

That is the historical summary of early efforts. We know that the term structure can be described as the complete set of discount factors, the discount function, which can be extracted from the price of default-free bonds trading in the market. The bootstrapping technique described in an earlier chapter may be used to extract the relevant discount factors. However, there are a number of reasons why this approach is problematic in practice. First, it is unlikely that the complete set of bonds in the market will pay cash flows at precise six-month intervals every six months from today to 30 years or longer. An adjustment is made for cash flows received at irregular intervals and for the lack of cash flows available at longer maturities. Another issue is the fact that the technique presented earlier allowed practitioners to calculate the discount factor for six-month maturities, whereas it may be necessary to determine the discount factor for nonstandard periods, such as four-month or 14.2-year maturities.

A third issue concerns the market price of bonds. These often reflect specific investor considerations, which include:

- The liquidity or lack thereof of certain bonds, caused by issue sizes, market-maker support, investor demand, nonstandard maturity, and a host of other factors.
- The fact that bonds do not trade continuously, so that some bond prices will be "newer" than others.
- The tax treatment of bond cash flows and the effect that this has on bond prices.
- The effect of the bid-offer spread on the market prices used.

The statistical term used for bond prices subject to these considerations is *error*. It is also common to come across the statement that these effects introduce noise into market prices.

To construct a fit to the yield curve that better handles the above considerations, smoothing techniques are used to derive the complete set of discount factors—known as the discount function—from market bond prices. Using the simple technique presented in Chapter 1, we graph the discount function for the UK gilt prices as at 12th June 2000. This is shown in Figure 5.2. The yield curve plotted from gilt redemption yields is shown in Figure 5.1. Figure 5.3 shows the zero-coupon yield curve and forward-rate curve that correspond to the discount function from the date.

From Figure 5.2 we see that the discount function is quite smooth, while the zero-coupon curve is also relatively smooth, although not as smooth as the discount

[2] The basics of regression are summarised briefly in Appendix 5.1.

[3] Reference in Vasicek and Fong (1982).

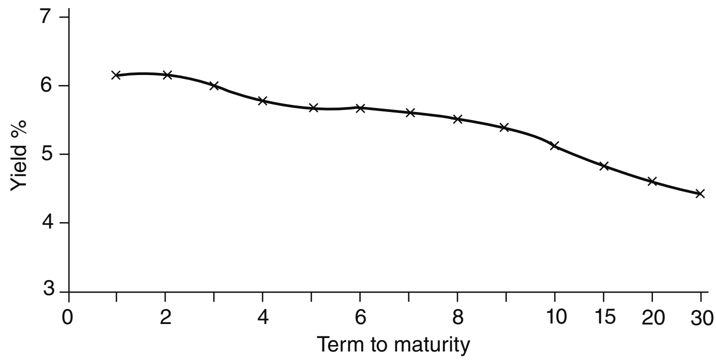

FIGURE 5.1 Gilt Gross Redemption Yields, 12th June 2000

function. The forward rate curve is distinctly unsmooth, if the reader will permit such as an expression, and there is obviously something wrong. In fact, the jagged nature of implied forward rates is one of the main concerns of the fixed-income analyst, and indicates in the first instance that the discount function and zero-coupon curve are not as smooth as they appear. Using the naive estimation method here, the main reason why the forward rates oscillate wildly[4] is that minor errors at the discount-factor stage are magnified many times over when translated into the forward rate. That is, any errors in the discount factors (which errors may stem from any of the reasons given above) are compounded when spot rates are calculated from them, and these are compounded into larger errors when calculating forward rates.

Smoothing Techniques

A common technique that may be used, but which is not accurate and so not recommended, is linear interpolation.[5] In this approach, the set of bond prices are used to graph a redemption yield curve (as in the previous section), and where bonds are not available for the required maturity term, the yield is interpolated from actual yields. Using gilt yields for 26th June 1997, we plot this as shown at Figure 5.4. The interpolated yields are those that are not marked by a cross. Figure 5.4 looks reasonable for any practitioner's purposes. However, spot and forward yields that are obtained from this curve are apt to behave in an unrealistic fashion, as shown in Figure 5.5. The forward curve is very bumpy, and each bump will correspond to a bond used in the

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[4] I've been looking for an excuse to use this expression, for the most obscure reason which we can finally reveal: see Goddard (2002, p. 140).

[5] At a conference in London in 2013, I was intrigued to hear a senior Treasury person from a UK High Street bank state, almost with pride, that his firm used the "SLI model" when constructing the bank's internal Funds Transfer Price curve. On enquiry from me what this model was (as I'd never heard of it), this person replied, to much mirth from the audience, that it was the "straight line interpolation" model. This surprised me. The bank in question has a balance sheet size of hundreds of billions. If one is using an "SLI model" curve to value assets or liabilities of such magnitude, even a 1 basis point error can have severe negative consequences. I do hope this lack of respect for more scientific or accurate methods of curve interpolation is not common in the UK banking market.

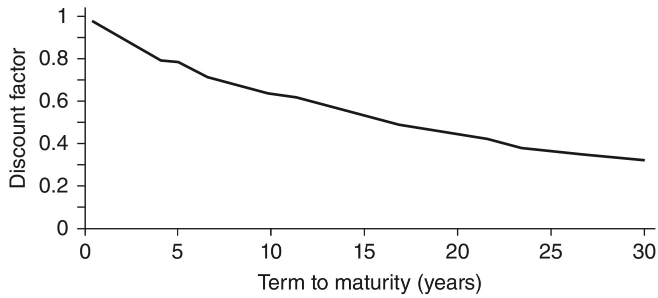

FIGURE 5.2 Discount Factors from Gilt Prices, 12th June 2000

original set. The spot rate has a kink at 21.5 years, and so the forward curve jumps significantly at this point. This curve would appear to be particularly unrealistic.

For this reason, bond traders and analysts do not bother with linear interpolation and instead use multiple regression or spline-based methods. One approach might be to assume a functional form for the discount function and estimate parameters of this form from the prices of bonds in the market. We consider these approaches next.

Using a Cubic Polynomial

A simple functional form for the discount function is a cubic polynomial. This approach consists of approximating the set of discount factors using a cubic function of time. If we say that $d(t)$ is the discount factor for maturity t, we approximate the set of discount factors using the following cubic function.

$$\hat{d}(t) = a_0 + a_1(t) + a_2(t)^2 + a_3(t)^3 \tag{5.1}$$

In some texts, the coefficients sometimes are written as *a, b,* and *c* rather than a_1 and so on.

The discount factor for $t = 0$, that is at time now, is 1. Therefore $a_0 = 1$, and (5.1) can then be rewritten as:

$$\hat{d}(t) - 1 = a_1(t) + a_2(t)^2 + a_3(t)^3 \tag{5.2}$$

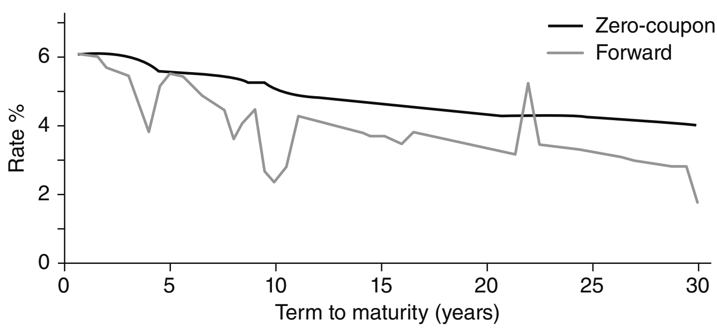

FIGURE 5.3 Zero-Coupon (spot) and Forward Rates Obtained from Gilt Yields, 12th June 2000

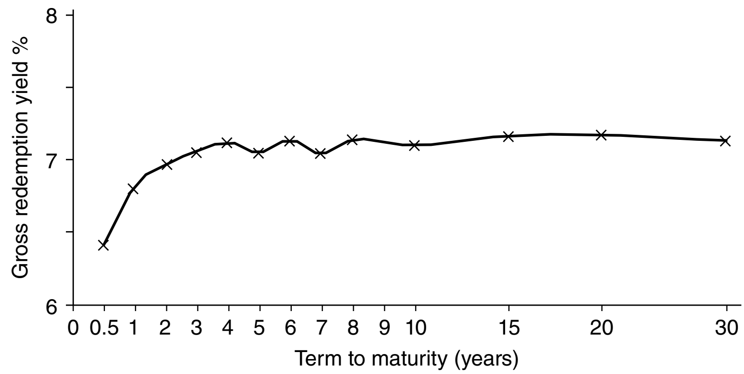

FIGURE 5.4 Linear Interpolation of Bond Yields, 26th June 1997

The market price of a traded coupon bond can be expressed in terms of discount factors. So in (5.3) we show the expression for the price of an N-maturity bond paying identical coupons C at regular intervals and redeemed at maturity at M.

$$P = d(t_1)C + d(t_2)C + \ldots\ldots + d(t_N)(C + M) \tag{5.3}$$

Using the cubic polynomial equation (5.2), expression (5.3) is transformed into:

$$P = C[1 + a_1(t_1) + a_2(t_1)^2 + a_3(t_1)^3] + \ldots + (C + M)[1 + a_1(t_N) + a_2(t_N)^2 + a_3(t_N)^3] \tag{5.4}$$

We require the coefficients of the cubic function in order to start describing the yield curve, so we rearrange (5.4) in order to express it in terms of these coefficients. This is shown in (5.5).

$$\begin{aligned} P = M + \sum C + a_1[C(t_1) + \ldots + (C + M)(t_N)] + a_2[C(t_1)^2 + \ldots + (C + M)(t_N)^2] \\ a_3[C(t_1)^3 + \ldots + (C + M)(t_N)^3] \end{aligned} \tag{5.5}$$

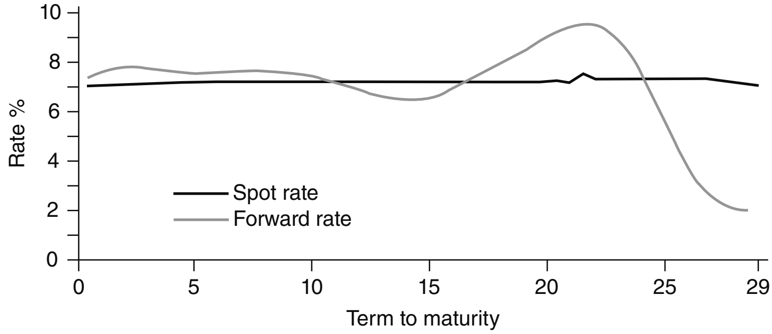

FIGURE 5.5 Spot and Forward Rates Implied from Figure 5.4

In the same way, we can express the pricing equation for each bond in our data set in terms of the unknown parameters of the cubic function. From (5.5) we may write

$$P-(M+\sum C)=a_1X_1+a_2X_2+a_3X_3 \tag{5.6}$$

where X_i is the appropriate expression in square brackets in (5.5); this is the form in which the expression is encountered commonly in textbooks.

We illustrate this interpolation technique in Example 5.1.

EXAMPLE 5.1

A benchmark semiannual coupon four-year bond with a coupon of 8% is trading at a price of 101.25. Assume the first coupon is precisely six months from now, so that $t_1 = 0.5$ and so $t_N = 4$. Set up the cubic function expression.

We have $C = 4$ and $M = 100$ so therefore:

$$100+\sum C=100+(8\times 4)=132$$

$$P-(100+\sum C)=101.25-132=-30.75$$

$$X_1=(4\times 0.5)+(4\times 1)+(4\times 1.5)+\ldots+(104\times 4)=472$$

$$X_2=[4\times(0.5)^2]+[4\times(1)^2]+[4\times(1.5)^2]+\ldots+[104\times(4)^2]=1{,}796$$

$$X_3=[4\times(0.5)^3]+[4\times(1)^3]+[4\times(1.5)^3]+\ldots+[104\times(4)^3]=7{,}528$$

This means that we now have an expression for the three coefficients, which is:

$$472a_1+1796a_2+7528a_3=-30.75$$

The prices for all other bonds are expressed in terms of the unknown parameters. To calculate the coefficient values, we use a statistical technique such as linear regression, for example ordinary least squares (OLS) to find the line of best fit values of the cubic equation. An introduction to this technique is given in Appendix 5.1.

In practice, the cubic polynomial approach is sometimes too limited a technique, requiring one equation per bond, and does not have the required flexibility to fit market data satisfactorily. The resulting curve is not really a curve but rather a set of independent discount factors that have been joined with a line of best fit. In addition, the impact of small changes in the data can be significant at the nonlocal level;

so, for example, a change in a single data point at the early maturities can result in badly behaved longer maturities.

Alternatively, a piecewise cubic polynomial approach is used, whereby $d(t)$ is assumed to be a different cubic polynomial over each maturity range. This means that the parameters a_1, a_2, and a_3 will be different over each maturity range. We will look at a special case of this use, the cubic spline, a little later.

NON-PARAMETRIC METHODS

Outside of the cubic-polynomial approach described in the previous section there are two main approaches to fitting the term structure. These are usually grouped into *parametric* and *non-parametric* curves. Parametric curves are based on term-structure models such as the Vasicek model or the Longstaff and Schwartz model. Non-parametric curves are not derived from an interest-rate model and are general approaches, described using a set of parameters. They include spline-based methods.

Spline-Based Methods

A spline is a statistical technique and a form of linear-interpolation method. There is more than one way of applying them, and the most straightforward method to understand the process is the spline function fitted using regression techniques. For the purposes of yield-curve construction, this method can cause curves to jump wildly and is oversensitive to changes in parameters.[5] However, we feel it is the most accessible method to understand and an introduction to the basic technique, as described in Suits, Mason, and Chan (1978), is given in Appendix 5.2.[6]

An nth order spline is a piecewise-polynomial approximation with n-degree polynomials that are differentiable $n - 1$ times. Piecewise means that the different polynomials are connected at arbitrarily selected points known as knot points (see Appendix 5.2). A cubic spline is a three-order spline, and is a piecewise cubic polynomial that is differentiable twice along all its points.

The x-axis in the regression is divided into segments at arbitrary points known as knot points. At each knot point the slopes of adjoining curves are required to match, as must the curvature. Figure 5.6 is a cubic spline. The knot points are selected at 0, 2, 5, 10, and 25 years. At each of these points, the curve is a cubic polynomial, and with this function we could accommodate a high and low in each space bounded by the knot points.

Cubic-spline interpolation assumes that there is a cubic polynomial that can estimate the yield curve at each maturity gap. One can think of a spline as a number of separate polynomials of $y = f(X)$, where X is the complete range, divided into user-specified segments, which are joined smoothly at the knot points. If we have a set of bond yields $r_0, r_1, r_2, \ldots\ldots r_n$ at maturity points $t_{0,} t_1, t_2, \ldots\ldots t_n$, we can estimate the cubic-spline function in the following way:

- The yield on bond i at time t is expressed as a cubic polynomial of the form $r\,(\mathrm{t}) = a_i + b_i t + c_i t^2 + d_i t^3$ for the interval over t_i and t_{i-1}.

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[5] For instance, see James and Webber (2000), section 15.3.

[6] The original article by Suits, Mason, and Chan (1978) is excellent and highly recommended.

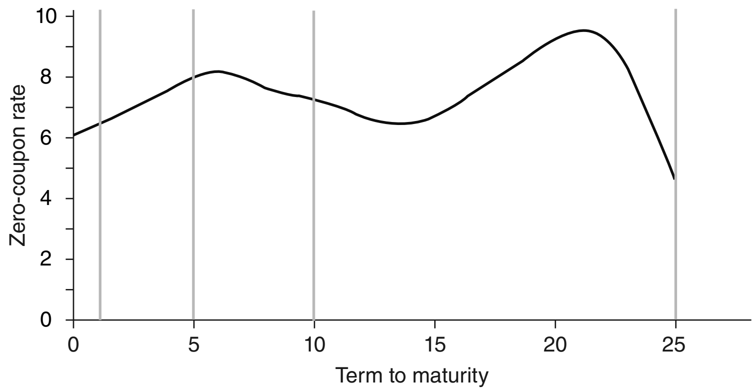

FIGURE 5.6 Cubic Spline with Knot Points at 0, 2, 5, 10, and 25 Years

- The coefficients of the cubic polynomial are calculated for all n intervals between the $n + 1$ data points, which results in $4n$ unknown coefficients that must be computed.
- These equations can be solved because they are made to fit the observed data. They are twice differentiable at the knot points, and these derivatives are equal at these points.
- The constraints specified are that the curve is instantaneously straight at the start of the curve (the shortest maturity) and instantaneously straight at the end of the curve, the longest maturity, that is $r(0) = 0$.

An accessible and readable account of this technique can be found in Van Deventer and Imai (1997).

The general formula for a cubic spline is:

$$s(\tau) = \sum_{i=0}^{3} a_i \tau^i + \frac{1}{3!}\sum_{p=0}^{n-1} b_p(\tau - X_p)^3 \tag{5.7}$$

where τ is the time of receipt of cash flows and where X_p refers to the points being joined and adjacent polynomials which are known as knot points, with $\{X_0, \ldots\ldots X_n\}, X_p < X_p + 1, p = 0, \ldots\ldots, n - 1$. In addition $(\tau - X_p) = \max(\tau - X_p, 0)$. The cubic spline is twice differentiable at the knot points. In practice, the spline is written down as a set of basis functions with the general spline being made up of a combination of these. One way to do this is by using what are known as B-splines. For a specified number of knot points $\{X_0, \ldots\ldots, X_n\}$ this is given by (5.8),

$$B_p(\tau) = \sum_{j=p}^{p+4}\left[\prod_{j=p, i\neq i}^{p+4} \frac{1}{X_i - X_j}\right](\tau - X_p)^3 \tag{5.8}$$

where $B_p(\tau)$ are cubic splines that are approximated on $\{X_0, \ldots\ldots, X_n\}$ with the following function:

$$\delta(\tau) = \delta(\tau | \lambda_{-3}, \ldots\ldots, \lambda_{n-1}) = \sum_{p=-3}^{n-1} \lambda_p B_p(\tau) \tag{5.9}$$

with $\lambda = (\lambda_{-3},\ldots\ldots,\lambda_{n-1})$ the required coefficients. The maturity periods $\tau_1,\ldots\ldots,\tau_n$ specify the B-splines so that $B = \left\{B_p(\tau_j)\right\}_{p=-3,\ldots,n-1,j-1,\ldots,m}$ and $\hat{\delta} = (\delta(\tau_1),\ldots\ldots,\delta(\tau_m))$ This allows us to set

$$\hat{\delta} = B\lambda \tag{5.10}$$

and therefore the regression equation

$$\lambda^* = \arg\frac{\min}{\lambda}\left\{\varepsilon'\varepsilon \middle| \varepsilon = P - D\lambda\right\} \tag{5.11}$$

with $D = C(t)\ B'$

where $C(t)$ is the cash flow of a bond at time t.

$\varepsilon'\varepsilon$ are the minimum errors. The regression in (5.11) is computed using ordinary least-squares regression.

An illustration of the use of B-splines is given in Steeley (1991) and, with a complete methodology, by Didier Joannas in Choudhry, Joannas, Landuyt, Pereira, and Pienaar (2010).

Appendix 5.2 provides background on splines fitted using regression methods.

Nelson and Siegel Curves

The curve-fitting technique first described by Nelson and Siegel (1985) has since been applied and modified by other authors, which is why they are sometimes described as a family of curves. These curves provide a satisfactory rough fit of the complete term structure, with some loss of accuracy at the very short and very long end. In the original curve, the authors specify four parameters. The approach is not a bootstrapping technique but, rather, a method for estimating the zero-coupon rate function from the yields observed on T-bills, under an assumed function for forward rates.

The Nelson and Spiegel curve states that the implied forward-rate yield curve may be modelled along the entire term structure using the following function:

$$rf(m,\beta) = \beta_0 + \beta_1 \exp\left[\frac{-m}{t_1}\right] + \beta_2\left[\frac{m}{t_1}\right]\exp\left[\frac{-m}{t_1}\right] \tag{5.12}$$

where $\beta = (\beta_0, \beta_1, \beta_2, t_1)'$ is the vector of parameters describing the yield curve, and m is the maturity at which the forward rate is calculated. There are three components; the constant term, a decay term, and a term reflecting the humped nature of the curve. The shape of the curve will gradually lead into an asymptote at the long end, the value of which is given by β_0, with a value of $\beta_0 + \beta_1$ at the short end.

A version of the Nelson and Siegel curve is the Svensson model (1994) with an adjustment to allow for the humped characteristic of the yield curve. This is fitted by adding an extension, as shown by (5.13).

$$rf(m,\beta) = \beta_0 + \beta_1 \exp\left[\frac{-m}{t_1}\right] + \beta_2\left[\frac{m}{t_1}\right]\exp\left[\frac{-m}{t_1}\right] + \beta_3\left[\frac{m}{t_2}\right]\exp\left[\frac{-m}{t_2}\right] \tag{5.13}$$

The Svensson curve is modelled, therefore, using six parameters, with additional input of β_3 and t_2.

Nelson and Siegel curves are popular in the market because they are straightforward to calculate. Jordan and Mansi (2000) state that one of the advantages of these curves is that they force the long-date forward curve into a horizontal asymptote. Another is that the user is not required to specify knot points, the choice of which determines the effectiveness or otherwise of cubic spline curves. The disadvantage they note is that these curves are less flexible than spline-based curves and there is therefore a chance that they do not fit the observed data as accurately as spline models.[7] James and Webber (2000, pp. 444–445) also suggest that Nelson and Siegel curves are slightly inflexible due to the limited number of parameters, and are accurate for yield curves that have only one hump, but are unsatisfactory for curves that possess both a hump and trough. Nevertheless, they produce reasonably accurate and smooth forward curves.

COMPARING CURVES

Whichever curve is chosen will depend on the user's requirements and the purpose for which the model is required. The choice of modelling methodology is usually a trade-off between simplicity and ease of computation and accuracy. Essentially, the curve chosen must fulfill the qualities of

- *Accuracy.* Is the curve a reasonable fit of the market curve? Is it flexible enough to accommodate a variety of yield-curve shapes?
- *Model consistency.* Is the curve-fitting method consistent with a theoretical yield-curve model such as Vasicek or Cox-Ingersoll-Ross?
- *Simplicity.* Is the curve reasonably straightforward to compute; that is, is it *tractable?*

The different methodologies all fit these requirements to greater or lesser extent. A good summary of the advantages and disadvantages of some popular modelling methods can be found in James and Webber (2000, Chapter 15).

YIELD CURVE CONSTRUCTION: A PRACTICAL GUIDE

In this section we demonstrate how to construct a family of yield curves to reprice vanilla derivative instruments, and how credit risk can be added on top of risk-free curves. We show that there are further practical difficulties associated with this exercise.

Background

The first thing required to price almost any trade is a family of yield curves that reprices liquid collateralised interest-rate swaps and FX forwards. This is because,

[7] This is an excellent article, strongly recommended. A good overview introduction to curve fitting is given in the introduction, and the main body of the article gives a good insight into the type of research that is currently being undertaken in yield-curve analysis.

without correct discount rates, we cannot evaluate the present value of future cash flows. Many trades contain a leg that either pays Libor or references it in another way, and thus we also need to be able to reprice vanilla swaps. Previously, there were no material differences between swaps referencing three-month (3m) and six-month (6m) Libor, or between Libor at the risk-free rate. Thus, people would use a single yield curve to project and discount all swaps in U.S. dollars (USD). Even when there were material spreads on cross-currency (XCCY) interest-rate swaps, traders who dealt only with swaps in a single currency, such as Japanese yen (JPY), would ignore this effect and discount all their JPY flows at the (JPY) Libor rate. They knew that this would not reprice a XCCY swap, but as this was not an instrument they traded, they did not care. These traders were in serious danger of losing money due to their approximations. Once credit risk issues worsened, material spreads started to develop between the Libor projection rates, and the risk-free rates paid on collateral the overnight-index swap (OIS) rate, even in USD. Furthermore, there could be material spreads between the risk-free rates in different collateral agreements (known as Credit Support Annexes or CSAs), as they are an annexe to the standard ISDA derivative legal agreement. For example, sterling (GBP) overnight-index swaps or SONIA represents transactions that have more credit risk than the transactions at OIS, and thus OIS would generally be at a lower rate than SONIA (where SONIA is cross-currency-swapped into USD, to allow for a fair comparison).

We reference here also the discussion on CSAs (see Chapter 21) and how they define the relevant risk-free rate, in order to show how to construct our family of discount and projection curves.

Method to Create a Family of Libor Projection and Risk-Free Discounting Curves

Step 1: OIS Discounting Curve We use the quoted prices of OIS swaps in USD, collateralised at the OIS rate, to create our first risk-free discounting curve. In an OIS swap, one party will pay the overnight OIS rate every night, while the other party pays a fixed-rate leg. For example, if a four-year OIS swap is quoted at 4% then we know that the four-year OIS rate is 4% (for the given payment frequency and basis on the fixed leg of the swap). We use OIS swaps of different tenors to build an OIS curve that starts with the overnight rate, and continues out to the longest tenor of the quoted swaps. This allows us to create a *single-bootstrapped* curve containing the OIS discount rates.

For consistency, we should use OIS swaps that are collateralised at OIS. If the swap was collateralised using a different interest rate, then the relative weightings of the front-end and back-end payments would be affected, thus potentially changing the quoted levels. For example, if the collateral rate used was much higher than OIS, back-end cash flows would be reduced in value, increasing the relative weighting of the front-end flows. If the OIS curve were upward sloping, this would result in a reduction in the longer-dated par OIS rates.

After this step, we now know the correct discount rate to apply to a future USD-denominated cash flow that is priced under a CSA referencing OIS.

Step 2: Three-Month USD Libor Projection Curve We now seek out prices for term three-month USD Libor swaps. As explained above, these should be collateralised by OIS, as the discounting rate can affect the longer-dated par projection rates. So far, OIS is the only discount rate whose level we know.

These Libor swaps exchange a fixed rate (the market convention being semi-annual bond basis for USD) for the floating three-month Libor rate. The prices of these swaps tell us the projection rates for the Libor-indexed swaps, starting at three-months (if we include the rate itself), out to the longest tenor swap available.

The Libor rates themselves, though used for the fixings, do not trade very liquid themselves. To get increased granularity at the front-end of the curve, we can include index-traded Libor futures. However, we shall not dwell on this point, as it does not affect the general principle.

We create a pair of *dual-bootstrapped* curves. The projection curve contains the swap levels. The discount curve contains the swap levels plus the OIS–Libor spread required to reprice the OIS discount factors from Step One. Note that, in practice, this spread is almost certainly negative, as Libor contains three-month credit risk to banks, whereas OIS is the closest thing we have to the "true" risk-free rate. (Because it refers to overnight interest-rate exposure, and to fully collateralised derivatives; hence "risk-free".)

Note that, by definition, this curve pair always contains the par swap rates themselves, and prices them to zero. Thus, par swaps cannot have any risk to the discounting curve. However, off-market swaps (swaps that are priced away from the current market rate) will have discounting delta, as they can be viewed as a par swap rate plus a set of independent fixed cash flows. The discounting of the fixed cash flows creates the OIS delta. As we have represented OIS as a spread to Libor, a single fixed cash flow will have identical risk to both the swap rate and the OIS-Libor spread.

We now know the correct projection rate for a three-month USD Libor swap, discounted at OIS.

Step 3: Six-Month USD Libor Projection Curve Six-month USD Libor swaps also trade in the market. As they are typically less liquid than the three-month swaps, they usually trade in the form of "3's vs. 6's" basis swaps. One party receives the three-month USD Libor rate plus a spread, while the other party receives the six-month USD Libor rate. The spread is usually positive, as credit spreads are usually upward sloping, and longer-dated deposits result in reduced liquidity.

As before, we should try to ensure the quoted swaps are collateralised at OIS, as we do not yet have another discount rate to bootstrap with our projection rates.

Our curve family now becomes a triple bootstrap. We add an additional projection curve, representing six-month Libor swaps. This curve contains the three-month Libor swap rates plus the quoted spreads. A par swap of six-month Libor against three-month Libor will have risk only to the 3's–6's spread. However, a par swap of six-month Libor vs. fixed rate will have identical risk to the spread and the three-month Libor swap rate. An off-market swap will also have OIS risk.

We could continue this process to add in projection curves to represent any other swaps that trade, such as 1m or 12m.

We now know the correct projection rate for a six-month USD Libor swap, discounted at OIS.

Step 4: Construction of Curve Families in Other Currencies with Their Own Collateral Types We can repeat steps 1–3 for any other currency we are interested in. For example, in euros (EUR), we start with euro overnight-index (EONIA) swaps, and then move on to three-month Euribor, and six-month Euribor.

We now know the Libor projection and EONIA discount rates in EUR, and we can repeat for other currencies.

Step 5: Construction of XCCY Collateral Curves We have prices of XCCY swaps, usually collateralised with USD cash, at OIS. For example, imagine a USD/EUR XCCY swap. One part receives the USD notional on day one, pays 3m USD Libor, and returns the same notional amount on the swap maturity date. The other party pays 3m EURIBOR + X, receives the EUR notional on day one, and returns it at the end. The notional exchange is fixed at the spot FX rate. We can use the price of these swaps to solve for the discount rate on the EUR leg required to price to par. This rate is the equivalent of OIS, but XCCY-swapped into EUR. We shall call it EUR-OIS.

We should add this new discount rate to our curve family by expressing it as a spread to EURIBOR + USD-OIS, remembering that USD-OIS itself is a spread to USD 3m Libor.

It is unlikely that these XCCY swaps trade liquid on more than one collateral type. Thus, we can get USD-EONIA by assuming that the FX forwards using EONIA as collateral will be the same as the FX forwards using OIS as collateral. The FX forwards using OIS as collateral can be derived directly from EURUSD_spot *df_USD-OIS / df_EUR-OIS. Note that, although our assumption seems sensible, it is not guaranteed to be true if no liquid instrument trades to define the price.

If we take all the currencies from steps 1–4 and follow this procedure for each of them, we now know the discount rates, for all these collateral types, in all these currencies.

Step 6: Construction of Curves in Other Currencies, Which Use OIS Collateral For some currencies, the collateral used for the vanilla swaps is USD cash, paying OIS (or GBP paying SONIA, etc.), rather than collateral in the domestic currency. This is likely to be the case outside the major currencies, such as for Polish zloty (PLN) or South African rand (ZAR).

So for PLN, we shall have prices for PLN swaps and XCCY swaps (typically 3m USD Libor vs. fixed in PLN), all collateralised in OIS. The XCCY swap gives us PLN-OIS, and then the Warsaw rate fixing (Wibor) swap gives us the PLN projection rates. Assuming the FX forwards are again independent of collateral type, we can now deduce PLN-EONIA etc.

In practice, there is an extra complication on XCCY swaps, as they usually trade with mark-to-market resets on the non-USD notional, every period. This complicates the pricing somewhat, but we shall ignore this for current purposes.

We now possess a full family of projection and discount curves, for all currencies and collateral types.

Step 7: Extension to Risky Curves in USD Now let us try to use these risk-free curves as a basis to reprice a credit-risky instrument, such as a term bank deposit. As previously explained, we do this by discounting all the cash flows at the risk-free rate, and then adding our credit curve on top to price the recovery leg and the probability of receiving all the coupons and the final notional.

The bank's Treasury desk will publish a term structure of deposits that trade at a fixed spread above Libor, in USD. Alternatively, we could back the deposit rates out from secondary-market bond prices, but let us assume we are using the former method, for now.

We start by taking all the coupons and notional amounts, and discounting them at OIS, as that is the closest thing we have to the "true" risk-free rate. Now we include the recovery leg and default probabilities, to solve for the credit spread that reprices the bond to the value of the initial notional. The credit spread we have solved for is the rate, above OIS, at which the bank can fund itself.

We can now reprice unsecured USD-denominated deposits.

Step 8: Extension to Risky Curves in Other Currencies If we have published deposit rates/secondary-market bond prices in the given currency, we can solve for the credit spread in the same way as we did above. If no deposit exists in the given currency, then we have to make some assumptions to determine the risky spread.

If we assume that the uncollateralised FX forwards are identical to the collateralised ones, then we can solve for the credit spread that maintains the forward rates. This will mean that the credit spreads will be always wider in currencies that have higher collateral discounting. In reality, this assumption is likely to be false. For example, a currency such as PLN is likely to strongly depreciate against USD when credit spreads widen, as USD is seen as a safe haven currency compared to PLN. To include the effects of these correlations, we should reprice our PLN deposit using a stochastic credit, FX and rates model which calibrates to the risk-free curves, USD-credit curve, and the FX/IR/credit volatilities and correlations.

We can now reprice unsecured deposits in all currencies.

Why Do XCCY Bases Appear? In 2011, the Basis on the PLN leg of a PLN/USD XCCY swap was very negative. Why might that be? There are several possible reasons, and we explain the major one now.

Economically, there are many more investment opportunities in USD than in PLN. Thus, holders of PLN will want to borrow USD today, and repay it in the future. Being PLN-rich, they can fund this by depositing PLN today, and receiving the PLN notional back in the future. Thus there is a large supply of PLN-denominated deposits, and a large demand for USD-denominated ones. This depresses the interest rates on PLN compared to USD, thus generating a large negative XCCY basis on the PLN leg of XCCY basis swaps. Note that this does not mean that Wibor rates fall. These can remain high due a higher PLN base rate and a higher degree of credit risk associated with making short-term deposits at Warsaw-based Polish banks, compared to London-based ones.

How Incorrect Curve Construction Leads to Mispricing There are many ways that using incorrect curves can lead to material misprices, we shall look at some of the more common ones.

A GBP-swaps trader might think that he can simply discount at GBP-Libor, ignoring both the XCCY basis and the collateral type, arguing that, if all he trades are

GBP swaps, then the correct integration of these swap rates into a full multi-currency world is irrelevant to him. By construction, he will still get all at-the-money (ATM) GBP swap rates correct, as he has the same swap rates, as model inputs, as everyone else. Indeed, if all he was ever pricing were ATM swaps, then he would not need a model at all, as the sole purpose of the model is interpolate the prices of off-market instruments from the liquid instruments used as inputs.

In reality, he will always end up with some off-market swaps, if he has net risk to Libor, and rates move. In this circumstance, he may wish to rehedge. Using his incorrect model, he will misrepresent the present values (PV) of the off-market swaps, and thus also their risks. For example, imagine swap rates move against him, and he has to post collateral at the GBP overnight index swap (SONIA) rate. He will be completely unaware that he now has risk to SONIA. He will not hedge, and thus if SONIA falls, he will experience an unexplained loss, as the return he receives on the collateral he posted falls.

More serious is a case where he has to price an off-market swap: for example, to unwind a transaction. As he has the wrong discount rate, he will misvalue the present value of the future net cash flows, and thus trade at a wrong price with his counterparty. The counterparty could synthetically re-create the risk on the off-market swap from a combination of on-market Libor and SONIA swaps, and thus lock in an immediate profit, thus arbitraging the trader who is using the incorrect model.

Real World Example

In 2011, the cross-currency basis on one-year EURs vs. USD reached lows of –100 to –150 bps. This seemed very strange at first sight, not only given its previous history (see Figure 5.7). As explained above, the cross-currency basis represents supply/demand imbalances for a named currency, against USD. However, we also know that Libor, theoretically, is an actual rate around where banks can borrow money short-

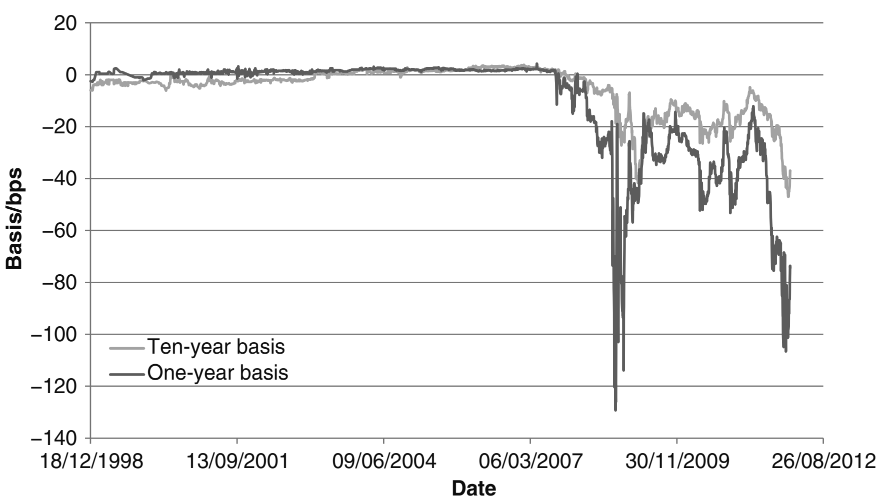

FIGURE 5.7 EUR-USD Cross-Currency Basis History 1998–2012

term. As Libor rates are published out to 12 months, the deposit rates out to this duration should be close to Libor, and thus have almost zero cross-currency basis. That does not mean that the cross-currency basis must be exactly zero. We know that Libor rates are not representative of real trades, and thus are only approximations of actual borrowing rates. We also know that different banks have different levels of riskiness, and can thus be expected to borrow above or below Libor (generally above), which represents the average of a specific panel of banks. However, these differences should be relatively immaterial at short tenors, where liquidity is usually high, and credit risk is relatively low. We expect much larger variations in deposit and loan rates in longer tenors.

In fact, we can show that there appears to be an almost perfect arbitrage opportunity. If the cross-currency basis is –150 bps, then (assume) a bank could borrow USD at Libor flat, and then execute a collateralised FX swap to exchange the USD for EUR today and back to EUR in a year's time. The forward FX rate will be at the level dictated by Libor + the cross-currency basis. Thus the amount of EUR that has to be repaid will be reduced by the negative cross-currency basis. The EUR received today will be deposited at EURIBOR flat. At the end of the trade, the USD amounts on the FX-swap and the physical loan will match, but the EUR notional on the EUR deposit will exceed the notional on the EUR leg of the cross-currency swap.

By executing these two trades, a profit can be locked in. This is not a perfect arbitrage, as the FX swap is collateralised but the deposit and loan are not. However, over a one-year horizon (or even less, e.g., one month), the covariance between the Libor rates and the FX rate will be far too low to account for the size of the cross-currency basis.

How is it that this opportunity could exist? Due to the economic crisis in 2008–09, many European banks found it hard to borrow money in USD, and only have access to EUR funds, using liquidity lines provided by the European Central Bank (ECB). This meant that, in order to gain access to USD, these banks had to borrow in EUR, and then cross-currency-swap (or FX-swap) the EUR into USD. In these swaps, the bank is paying EUR (receiving USD) on day one, and receiving EUR (repaying USD) at maturity. This creates a demand for future EUR compared to future USD, and thus generates the cross-currency basis.

There are other banks that can more readily borrow USD: for example, through liquidity lines created by the Federal Reserve Bank. However, if the banks who cannot borrow USD represent a significant share of the market, then the mismatch between the FX swap and the loan/deposit will continue to exist. The banks that can borrow in USD have access to a complete market, and can arbitrage the banks that have access only to an incomplete market. Hence the large cross-currency basis in EUR versus USD during and after the financial crash.

If there was globally a demand for EUR, rather than USD, then it would be the banks that can borrow in USD that would get arbitraged by those that can borrow in EUR; however, at time of writing that is not the case. Some large global or universal banks, such as HSBC, have liquidity lines to both the ECB and the Fed. In fact, there exist mutual liquidity lines between the ECB and the Fed, whereby the Fed provides USD lines to the ECB, and the ECB provides EUR lines to the Fed. This means that banks who have lines to the ECB can borrow some USD, but in a much smaller volume than banks that have direct access to the Fed. In December 2011, the size of the

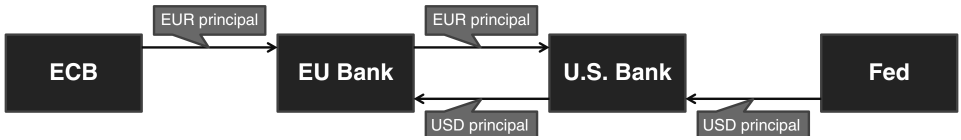

FIGURE 5.8 Cross-Currency Swap: Initial Payment of Principal

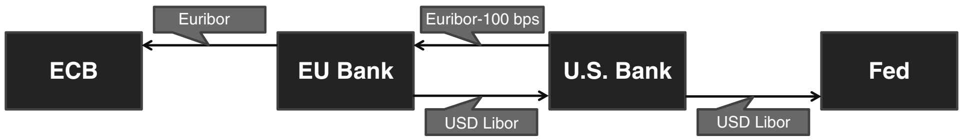

FIGURE 5.9 Interest Payments

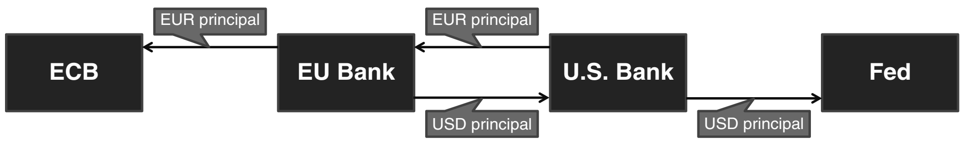

FIGURE 5.10 Final Payment of Principal

USD lines offered by the ECB was increased, which led to the XCCY basis tightening, but not disappearing.

Figures 5.8 to 5.10 show the cash flow patterns at the start, during, and at the end of a vanilla XCCY swap.

Multi-Currency Yield Curve

The Credit Support Annexe (CSA) side agreement to the standard ISDA derivatives agreement often allows counterparties to a derivative to post collateral in the form of cash in a different denomination to the currency of the trade. As a result, banks now construct multi-currency yield curves to assist them with the funding decision when paying or receiving collateral.

The concept of the single-currency curve is straightforward and is covered in detail elsewhere in this chapter. Here we consider the concept of the multi-currency curve. As this is most relevant to collateralised interest-rate swaps (IRS), we place the discussion in this context.

IRS Rates and Discounting Levels Prior to the crash of 2008, the market consensus had been to use Libor levels to discount swaps. Post-crash, the convention is to use the relevant CSA rate for collateralised derivatives, or the bank's actual cost of funds (COF) rate for uncollateralised derivatives. In a conventional positive-sloping yield curve environment, forward Libor rates will be increasing along the term structure. In this environment a vanilla IRS, paying floating and receiving fixed, will have net positive cash flows at the start of its life (when the floating rate payment, typically three-month or six-month Libor, is lower than the fixed rate), and net negative flows towards the end of its life (when the floating Libor fix is higher than the fixed rate). If discounting rates rise, then this will affect cash flows at the end of the swap's life by more than it affects flows at the start of the swap.

In other words, if market interest rates rise, the mark-to-market (mtm) value of a receive-fixed swap will be increasing as discounting rates rise. In turn, this means that the break-even rate of the swap moves lower; hence, a market-making swap bank will require a lower fixed rate if it is to price the swap correctly as discounting rates rise.

On the other hand, consider a receive-fixed IRS where the fixed payment is on an annual basis, while the floating leg pays quarterly three-month Libor. If discounting rates rise, then this will affect the fixed-leg payments by more than the floating-leg payments, because fixed payments are received on a lower frequency and at a later time, so there is more time for the impact of discounting to take effect. In this case, the value of the receive-fixed swap is decreasing as discounting rates rise. Here, the break-even rate of the swap moves higher as rates rise.

The break-even rate to use when pricing an IRS is a function of the current discounting level. This is separate to the current interbank swap rate. The question then is what discount level to use when constructing the forward Libor curve from current swap rates. This is where the funding impact of swaps traded under a CSA agreement becomes pertinent. The party that is negative mtm on a collateralised swap will, under London Clearing House (LCH) and SwapClear rules, pass collateral in the local currency, which attracts the overnight-index swap (OIS) interest rate.

This therefore dictates that a dealer bank should use the local currency OIS curve as inputs to construct a curve of projected Libor rates from current interbank swap rates. Swaps traded in euro currency would be priced off the EONIA curve, sterling swaps off the SONIA curve, and so on.

Constructing Multi-Currency Yield Curves

Where a swap is being priced in the local currency, it uses the projected three-month or six-month Libor rate and OIS curve for valuation. A different issue arises when a bank posts collateral to cover a negative mtm swap in a currency other than that of the swap. This means a dealer bank needs to consider using a discount curve to value a swap across different currencies. The best way to look at this is to view it from the angle of a cross-currency swap, and the spread for such products. These swaps, which exchange currencies at inception and then return the same amount, pay floating-fixed (or floating-floating) interest in two different currencies. The principle for discounting them is identical to that of vanilla IRS.

We illustrate with an example. Consider a floating-floating cross-currency swap between EUR and GBP. The latter currency has higher forward rates. If the discount rate falls in both currencies, the EUR leg value will decrease by more than the GBP side. This is because a higher value of the EUR cash flows than the GBP cash flows is paid at maturity on the re-exchange of principal. To allow for this, the price of the GBP leg needs to be higher, to cover against this impact.

The question here is what discount level to use when pricing the cross-currency swap. In general the market uses the OIS rate. Therefore this rate should be used when building the multi-currency curve.

The starting point is that we set discount curves in all the main currencies, which are the relevant OIS curve. We can extract the discount factors for each currency from these curves, which we call Df_{CCY} for a general discount factor and Df_{OIS} for the relevant discount factor for the OIS in that currency. If we assume that FX rates are not correlated to interest rates (a big assumption but necessary in this analysis), this implies that forward FX rates—which are a deposit product, as forward FX rates are simply spot FX rate adjusted for the deposit interest rate in each currency—are not a function of the discounting level in each currency. This further implies that the ratio of forward discount factors is constant.

This enables us to set the following relationship for any two currencies and any two rates:

$$\frac{Df_{CCY1}^{RefB}(t)}{Df_{CCY2}^{RefB}(t)} = \frac{DF_{CCY1}^{RefA}(t)}{DF_{CCY2}^{RefA}(t)}$$

We can then use this relationship to obtain the discount factor for any currency pair. Taking the base currency of euro, we set the currency to euro and the reference level to EONIA, as follows:

$$\frac{Df_{CCY}^{Ref}(t)}{Df_{EUR}^{Ref}(t)} = \frac{Df_{CCY}^{EONIA}(t)}{DF_{EUR}^{EONIA}(t)}$$

Setting one reference level to EONIA, we use these EONIA levels to obtain the reference level between any two currencies CCY1, CCY2, shown as follows:

$$\frac{Df_{CCY2}^{REF}(t)}{Df_{CCY1}^{REF}(t)} = \left(\frac{Df_{CCY2}^{REF}(t)}{Df_{EUR}^{REF}(t)} \Big/ \frac{Df_{CCY1}^{REF}(t)}{Df_{EUR}^{REF}(t)} \right) = \left(\frac{Df_{CCY2}^{EONIA}(t)}{Df_{EUR}^{EONIA}(t)} \Big/ \frac{Df_{CCY1}^{EONIA}(t)}{Df_{EUR}^{EONIA}(t)} \right) = \frac{Df_{CCY2}^{EONIA}(t)}{Df_{CCY1}^{EONIA}(t)}$$

Using this relationship, we build the yield curve by obtaining the relevant reference rate for regular intervals on the term structure, and then obtaining the discount factors for each point along the term structure. This set of discount factors is then used to extract the yield curve.

Where the CSA in place allows a choice of two or more currencies out of EUR, GBP, or USD for cash collateral, we will need to account for more than one reference level. This is done by constructing curves for each separate reference level as before, and then building a new curve by looking at each daily forward rate from the three curves, and taking the highest point for each point along the curve as the data point for the curve. This process has "equalised" the curves of the three different currencies. (Note that this process does not take into account the optionality value inherent in the fact that the poster of collateral has a choice as to which currency it posts). A stylised illustration of such a "hybrid" curve is shown in Figure 5.11.

The graph in Figure 5.11 shows the cross currency basis adjusted discount curves for IRS traded under four CSA scenarios: USD cash only, EUR cash only, GBP cash only, and a choice of either of the three. A hybrid curve is not like a conventional yield curve constructed using Nelson-Siegel or cubic spline, because it is not a smooth curve. As the underlying forward curve changes from one currency to another, there is a kink at the changeover point. Hence as we show in Figure 5.11, at each point in time the orange hybrid curve overlaps with exactly one of the three single currency curves and only three curves can be seen at any given time. Under conventional analysis, the bank that is negative mtm will post the currency of highest yield, or conversely of lowest funding cost.

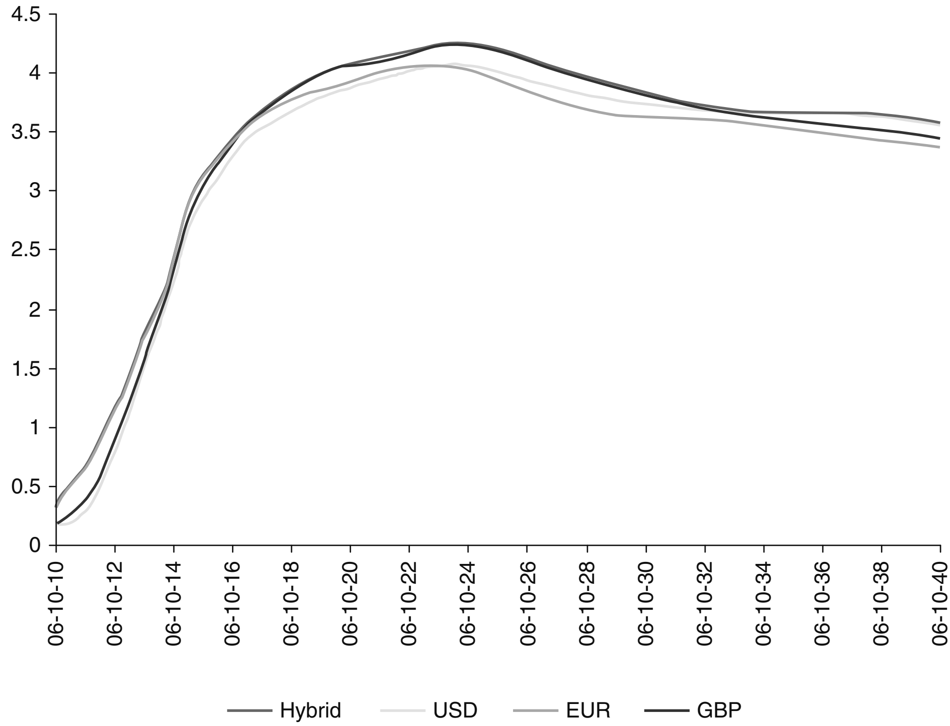

FIGURE 5.11 Hybrid Discount Yield Curve

The assumption of forward FX rates being uncorrelated to funding rates is perhaps the biggest issue for discussion. Certainly one would be right to state that FX spot rates do have positive correlation with changes in interest rates. The impact is greater where one of the currencies is a core currency such as USD, EUR, GBP, and possibly CHF, which are held as reserve deposits by other countries' central banks. However, we make the assumption with respect to forward FX rates to enable us to construct a multi-currency curve. The issue of which currency we wish to receive as collateral is a separate discussion.

APPENDICES

APPENDIX 5.1 LINEAR REGRESSION: ORDINARY LEAST SQUARES

The main purpose of regression analysis is to estimate or predict the average value of a dependent variable given the known values of an independent or explanatory variable. In one way, it measures the relationship between two sets of data. This data might be the relationship between family income and expenditure; here the income would be the independent variable and expenditure the dependent variable. More relevant for our purposes, it could also be the relationship between the change in price of a corporate bond, priced off the benchmark yield curve, and changes in the price of a short-dated benchmark bond and the price of a long-term bond. In this case, the price of the corporate bond is the dependent variable, while the prices of the short- and long-term government bonds are independent variables. If there is a linear relationship between the dependent and independent variables, we may use a regression function to determine the relationship between them. In this appendix, we provide a basic overview and introduction to regression and ordinary least squares.

Regression analysis is a key part of financial market econometrics, and advanced econometric analysis is extensively used in fixed-income analysis. For a background in basic econometrics, we recommend the book of the same name by Damodar Gujarati (1995), which is excellent. Gujarati's book is very accessible and readable, and provides a good grounding in the topic. Another excellent introduction to econometrics is Brooks (2002). For more advanced applications, we recommend *The Econometrics of Financial Markets* by Campbell, Lo, and MacKinlay (1997).

If our data set consists of the entire population, rather than a statistical sample of the population, we estimate the relationship between two data sets using the population-regression function. Given an independent variable X and dependent variable Y it can be shown that the conditional mean $E(Y \mid X_i)$ is a function of X_i. This would be written

$$E(Y \mid X_i) = f(X_i) \tag{A5.1.1}$$

where $f(X_i)$ is a function of the independent variable X_i. Equation (A5.1.1) is termed the two-variable population-regression function and states that average value of Y given X_i is a linear function of X_i. It can further be shown that

$$E(Y \mid X_i) = \alpha + \beta X_i \tag{A5.1.2}$$

where α and β are the regression coefficients; they are unknown but fixed parameters. The α term is the intercept coefficient and β is the slope coefficient. These are shown in Figure A5.1.1. The objective of regression analysis is to estimate the values of the regression coefficients using observations of the values of X and Y.

Equation (A5.1.2) is a two-variable regression; it is sometimes written as

$$Y = \beta_1 + \beta_2 X_i \tag{A5.1.3}$$

Where there are more than two variables, we use a multi-variable regression.

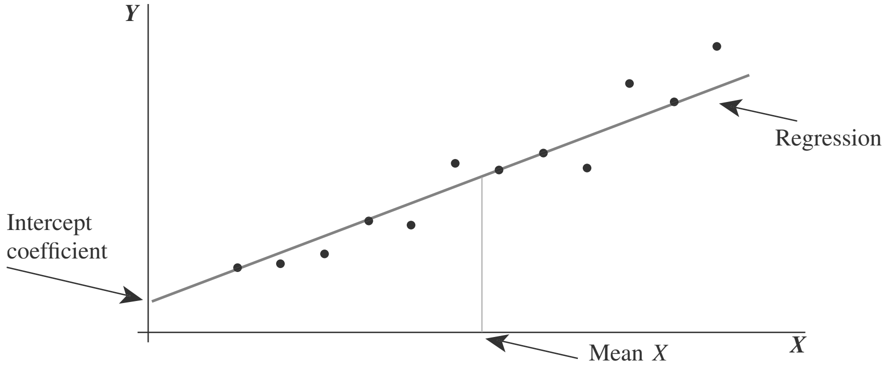

FIGURE A5.1.1 Regression Line Passing through Values of X and Y

It can further be shown that given a value for X_i, the independent variable value will be clustered around the average value for Y at that X_i; in other words, around the conditional expectation. The deviation of an individual Y_i around its expected value is defined as

$$u_i = Y_i - E(Y \mid X_i) \tag{A5.1.4}$$

This is rearranged to give

$$Y_i = E(Y \mid X_i) + u_i \tag{A5.1.5}$$

where the term u_i is an unknown random variable and is known as the stochastic disturbance or stochastic error term or simply error term. It is written as ε in some textbooks.

The value of the dependent variable is the sum of two elements: the systematic or deterministic element, which is given by the regression function above, and a random or nonsystematic element, which is the error term. Essentially, the error term captures all those elements that have been missed out or left out of the regression model.

In practice it is most unlikely that we will have data sets available for the entire population, so we use statistical sample data instead. When regression is carried out on sample data we use the sample regression function (SRF), which is (A5.1.6).

$$Y_i = \hat{\alpha} + \hat{\beta} X_i + \hat{u}_i \tag{A5.1.6}$$

where

$\hat{\alpha}$ is the estimator of α
$\hat{\beta}$ is the estimator of β

The SRF is determined using a statistical technique known as ordinary least squares or OLS. This approach is covered in any number of statistics and econometrics text books. We suggest Chapter 3 in Gujarati (1995). A very brief description is given here.

Let us expand our regression model. Assume we have N observations and m independent variables. Say that Y_i is the ith observation on the dependent variable and X_{it} is the tth observation on the tth independent variable. The regression function of the relationship between the dependent and independent variables is given by

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots\ldots + \beta_m X_{mi} + \varepsilon_i \tag{A5.1.7}$$

and where we must estimate the $\beta_1, \beta_2, \ldots\ldots \beta_m$ regression coefficients. This is done using OLS. From (A5.1.7) ε_i is the error in the model, the random element that is left out in predicting the ith value of the dependent variable. From our earlier description we know that

$$\varepsilon_i = Y_i - \beta_1 X_{1i} - \beta_2 X_{2i} - \ldots\ldots - \beta_m X_{mi} \tag{A5.1.8}$$

We require the sum of squared errors, which is given by $\varepsilon_1^2 + \varepsilon_2^2 + \ldots\ldots + \varepsilon_i^2$, and OLS is determining the coefficients that minimise this sum of squared errors.

Earlier in the chapter we illustrated a bond pricing equation, which was $472a_1 + 1796a_2 + 7528a_3 = -30.75$.

What this expression tells us is that –30.75 is the value of the dependent variable. There are three independent variables with values of 472, 1,796, and 7,528. As there are three unknowns, we require only four such bond price equations and we can solve for a_1, a_2, and a_3. This may appear slightly daunting if it is to be carried out by hand, and in fact software applications are used to speed the process. Using such a package, we can calculate the values that minimise the sum of squared errors on the model.

If we say that the values are $\hat{\alpha}_1$, $\hat{\alpha}_2$, and $\hat{\alpha}_3$, then the OLS estimate of the discount function, given the market bond prices used to derive the coefficient equations, is given by

$$\hat{d}(t) = 1 + \hat{a}_1 t + \hat{a}_2 t^2 + \hat{a}_3 t^3 \tag{A5.1.9}$$

APPENDIX 5.2 REGRESSION SPLINES

This appendix summarises Suits et al. (1978), an excellent account of how a spline function may be fitted using regression methods. The article is very accessible and strongly recommended.

A standard econometric approach is that of piecewise linear regression. This method is not suitable for fitting a relationship that is not purely linear, however (such as a term structure), as illustrated by Figure A5.2.1.

To get around this problem, one approach would be to join a series of linear regressions, at arbitrary points specified by the user. This is described by

$$Y = [a_1 + b_1(X - X_0)]D_1 + [a_2 + b_2(X - X_1)]D_2 + [a_3 + b_3(X - X_2)]D_3 + u \tag{A5.2.1}$$

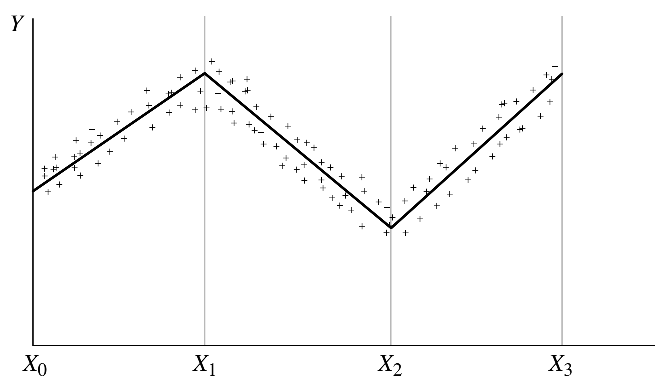

FIGURE A5.2.1 Piecewise Linear Regression: Unsuitable for Yield Curve Construction

In (A5.2.1) D_i is what is known as a dummy variable, with a value of 1 for all observations whenever $X_{i-1} \le X \le X_i$ and a value of 0 at other times. As it stands, (A5.2.1) is discontinuous at X_1 and X_2, but this can be removed by imposing the following constraints:

$$\begin{aligned} a_2 &= a_1 + b_1(X_1 - X_0) \\ a_3 &= a_2 + b_2(X_2 - X_1) \end{aligned} \tag{A5.2.2}$$

If (A5.2.2) is substituted into (A5.2.1), the following is obtained:

$$\begin{aligned} Y = a_1 &+ b_1[(X - X_0)D_1 + (X_1 - X_0)D_2 + (X_2 - X_1)D_3] \\ &+ b_2[(X - X_1)D_2 + (X_2 - X_1)D_3] + b_3[(X - X_2)D_3] + u \end{aligned} \tag{A5.2.3}$$

The expression in (A5.2.3) has converted a piecewise linear regression into a multiple regression. *Y* is the dependent variable and is regressed on three composite variables, the values for which are obtained from:

- The *X* data sets
- The values for X_i at the points at which the curve is required to "bend"
- The widths of the selected intervals
- The three dummy variables

As Suits et al. state, it would be possible to calculate the coefficients by hand (!), but there are a number of standard software packages that the user can employ to solve the regression.

The disadvantages of piecewise linear regression if used for a number of applications, including yield-curve fitting, are twofold; first, the derivatives of the function are not continuous, and this discontinuity can seriously distort the curve at the derivative points, which would make the curve meaningless at these points. Secondly, and more crucial for yield-curve applications, it may not be obvious where the linear segments should be placed as the scatter diagram of observations may indicate several possibilities. This situation may make it desirable to specify X_i at user-specified (arbitrary) points, and this makes linear regression unsuitable.

To get around these problems we use a spline function. For this the linear function described by equation (A5.1.1) is replaced with a set of piecewise polynomial functions. It would be possible to use polynomials of any degree, but the most common approach is to use cubic polynomials. The *x*-axis is divided into three intervals, at the points X_0, X_1, X_2, and X_3. These points are known as knot points and are illustrated in Figure A5.2.2.

In Figure A5.2.2, the segments chosen, following Suits, Mason, and Chan (1978) are at equal intervals. This is not essential to the procedure, however, and for applications, including yield-curve modelling, is not undertaken. Instead, the knots are placed at points where the user thinks the relationship changes most. For example, for the term structure the knots may be placed at 0, 2-, 5-, and 10-year maturities. If more than four knots are required—for instance, to go beyond to 20- and 30-year maturities—then the analyst will require a greater number of composite variables, as discussed later. The downside associated with a greater number of intervals is that, as more composite variables are required to fit the curve, additional degrees of freedom are lost.

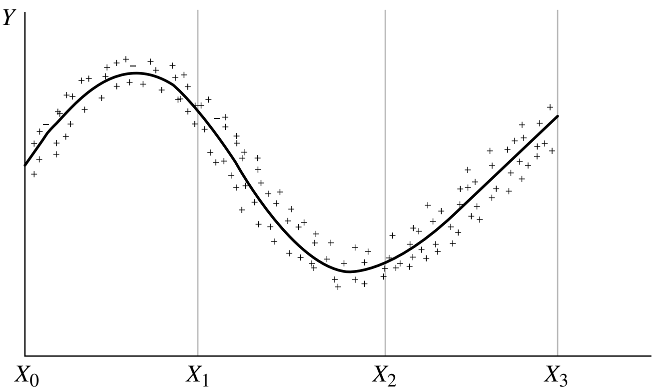

FIGURE A5.2.2 Knot Points within a Spline Function

The regression relationship now becomes

$$\begin{aligned} Y = & [a_1 + b_1(X - X_0) + C_1(X - X_0)^2 + d_1(X - X_0)^3]D_1] \\ & + [a_2 + b_2(X - X_1) + c_2(X - X_1)^2 + d_2(X - X_1)^3]D_2 \\ & + [a_3 + b_3(X - X_2) + c_3(X - X_2)^2 + d_3(X - X_2)^3]D_3 + u \end{aligned} \tag{A5.2.4}$$

where D_i is a dummy variable specified by the interval i.

This time, both the function described by (A5.2.4) and its derivatives are discontinuous at the knot points, but this feature can be removed by applying constraints to the coefficients. The constraints ensure the following:

- For a_i the values of the very short end and the very long end are equal at the start and end knot points.
- For b_i the first derivatives, the slope of the left- and right-hand sides of each knot point, are equal.
- For c_i the second derivatives are equal.

The constraints are given by (A5.2.5).

$$\begin{aligned} a_2 &= a_1 + b_1(X - X_0) + c_1(X - X_0)^2 + d_1(X - X_0)^3 \\ a_3 &= a_2 + b_2(X - X_1) + c_2(X_2 - X_1)^2 + d_2(X_2 - X_1)^3 \\ b_2 &= b_1 + 2c_1(X_1 - X_0) + 3d_1(X_1 - X_0)^2 \\ b_3 &= b_2 + 2c_2(X_2 - X_1) + 3d_2(X_2 - X_1)^2 \\ c_2 &= c_1 + 3d_1(X_1 - X_0) \\ c_3 &= c_2 + 3d_2(X_2 - X_1) \end{aligned} \tag{A5.2.5}$$

Using the expression above, a spline function becomes a multiple regression of the dependent variable Y on five composite variables and Suits et al. show this to be

$$Y = a_1 + b_1(X - X_0) + c_1(X - X_0)^2 + d_1(X - X_0)^3 + (d_2 - d_1)(X - X_1)^3 D_1^* + (d_3 - d_2)(X - X_2)^3 D_2^* \quad \text{(A5.2.6)}$$

where D_1^* and D_2^* are dummy variables where $D_i^* = 1$ if $X \geq X_i$, at other times the dummy variable is equal to 0. To compute the coefficients of the regression, we can use the least-squares procedure, available on standard software packages.

Finally, if we wish to select more than three intervals, that is more than four knot points, which is common in yield-curve applications, we can use (A5.2.7), derived in Suits et al. (1978). This is a multiple regression, a fitted spline function with $n + 1$ intervals, with knot points at $X_0, X_1, X_2, \ldots\ldots, X_{n+1}$ and corresponding dummy variables $D_1^*, D_2^*, \ldots\ldots D_n^*$.

$$Y = a_1 + b_1(X - X_0) + c_1(X - X_0)^2 + d_1 X - X_0)^3 + \sum_{i=1}^{n}(d_{i+1} - d_i)(X - X_i)^3 D_1^* \quad \text{(A5.2.7)}$$

With each extra interval, an extra composite variable is required, and this results in the loss of one more degree of freedom.

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PART Two

Selected Market Instruments

The second part of the book reviews selected instruments traded in the debt capital markets. The products have been chosen to give the reader an idea of the depth of variety in the market. So we consider money-market instruments, followed by hybrid securities and callable bonds. We also consider index-linked bonds and structured finance securities. Some of the techniques used to analyse these more complex products are also described and explained.

Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

CHAPTER 6

The Money Markets

Money-market securities are securities with maturities of up to 12 months, so they are short-term debt obligations. Money-market debt is an important part of the global financial markets, and facilitates the smooth running of the banking industry as well as providing working capital for industrial and commercial corporate institutions. The market allows issuers, who are financial organisations as well as corporates, to raise funds for short-term periods at relatively low interest rates. These issuers include sovereign governments, who issue Treasury bills, corporates issuing commercial paper, and banks issuing bills and certificates of deposit. At the same time, investors are attracted to the market because the instruments are often liquid and may carry relatively low credit risk. Investors in the money market include banks, local authorities, corporations, money-market investment funds, and individuals; however, the money market essentially is a wholesale market, and the denominations of individual instruments are relatively large.

Although the money market has traditionally been defined as the market for instruments maturing in one year or less, frequently the money-market desks of banks trade instruments with maturities of up to two years, both cash and off-balance-sheet.[1] In addition to the cash instruments that go to make up the market, the money markets also consist of a wide range of over-the-counter off-balance-sheet derivative instruments. These instruments are used mainly to establish future borrowing and lending rates, and to hedge or change existing interest-rate exposure. This activity is carried out by banks, central banks, and corporates. The main derivatives are short-term interest-rate futures, forward-rate agreements, and short-dated interest-rate swaps.

In this chapter, we review the cash instruments traded in the money market as well as the two main money-market derivatives: interest-rate futures and forward-rate agreements.

OVERVIEW

The cash instruments traded in the money market include the following:

- Treasury bill
- Time deposit

[1] The author has personal experience in market-making on a desk that combined cash and derivative instruments of up to two years maturity as well as government bonds of up to three years maturity.

- Certificate of deposit
- Commercial paper
- Bankers acceptance
- Bill of exchange

We can also add the market in repurchase agreements or repo, which are essentially secured cash loans, to this list.

A Treasury bill is used by sovereign governments to raise short-term funds, while certificates of deposit (CDs) are used by banks. The other instruments are used by corporates and, occasionally, banks. Each instrument represents an obligation on the borrower to repay the amount borrowed on the maturity date together with interest if this applies. The instruments above fall into one of two main classes of money-market securities: those quoted on a yield basis and those quoted on a discount basis. These two terms are discussed below.

The calculation of interest in the money markets often differs from the calculation of accrued interest in the corresponding bond market. Generally, the day-count convention in the money market is the exact number of days that the instrument is held over the number of days in the year. In the sterling market, the year base is 365 days, so the interest calculation for sterling money-market instruments is given by (6.1).

$$i = n/365 \tag{6.1}$$

Money markets that calculate interest based on a 365-day year are listed in Appendix 6.1. However, the majority of currencies, including the U.S. dollar and the euro, calculate interest based on a 360-day base.

Settlement of money-market instruments can be for value today (generally only when traded in before mid-day), tomorrow, or two days forward, known as spot.

SECURITIES QUOTED ON A YIELD BASIS

Two of the instruments in the list above are yield-based instruments.

Money-Market Deposits

These are fixed-interest term deposits of up to one year with banks and securities houses. They are also known as time deposits or clean deposits. They are not negotiable, and so they cannot be liquidated before maturity. The interest rate on the deposit is fixed for the term and related to the London Interbank Offer Rate (Libor) of the same term. Interest and capital are paid on maturity.

The effective rate on a money-market deposit is the annual equivalent interest rate for an instrument with a maturity of less than one year.

Certificates of Deposit

Certificates of deposit (CDs) are receipts from banks for deposits that have been placed with them. They were first introduced in the sterling market in 1958. The deposits themselves carry a fixed rate of interest related to Libor and have a fixed term to maturity, and so they cannot be withdrawn before maturity. However, the certificates themselves can be traded in a secondary market; that is, they are negotiable.[2] CDs are therefore very similar to negotiable money-market deposits, although the yields are below the equivalent deposit rates because of the added benefit of secondary market liquidity. Most CDs issued are of between one and three months maturity, although they do trade in maturities of one to five years. Interest is paid on maturity except for CDs lasting longer than one year, where interest is paid annually or, occasionally, semiannually.

Banks issue CDs to raise funds to finance their business activities. This is usually where they have a funding shortfall due to a surplus of customer assets over customer liabilities (deposits), but it may be for operational reasons and funding diversity as well. A CD will have a stated interest rate and fixed maturity date and can be issued in any denomination. On issue, a CD is sold for face value, so the settlement proceeds of a CD on issue always equal its nominal value. The interest is paid, together with the face amount, on maturity. The interest rate is sometimes called the coupon, but unless the CD is held to maturity this will not equal the yield, which is of course the current rate available in the market and varies over time. In the United States, CDs are available in smaller denomination amounts to retail investors. The largest group of CD investors, however, are banks themselves, money-market funds, corporates, and local-authority treasurers. Unlike coupons on bonds, which are paid in rounded amounts, CD coupon is calculated to the exact day.

CD yields. The coupon quoted on a CD is a function of the credit quality of the issuing bank, and its expected liquidity level in the market and, of course, the maturity of the CD, as this will be considered relative to the money-market yield curve. As CDs are issued by banks as part of their short-term funding and liquidity requirement, issue volumes are driven by the demand for bank loans and the availability of alternative sources of funds for bank customers. The credit quality of the issuing bank is the primary consideration, however.

If the current market price of the CD including accrued interest is P and the current quoted yield is r, the yield can be calculated given the price, using (6.2).

$$r = \left\{ \frac{M}{P} \times \left[1 + C\left(\frac{N_{im}}{B} \right) \right] - 1 \right\} \times \left(\frac{B}{N_{sm}} \right) \tag{6.2}$$

The price can be calculated given the yield using (6.3).

$$P = M \times \left[1 + C\left(\frac{N_{im}}{B} \right) \right] / \left[1 + r\left(\frac{N_{sm}}{B} \right) \right] \tag{6.3}$$

where C is the quoted coupon on the CD
M is the face value of the CD
B is the year day-basis (365 or 360)

[2] A small number of CDs are nonnegotiable.

F is the maturity value of the CD
N_{im} is the number of days between issue and maturity
N_{sm} is the number of days between settlement and maturity
N_{is} is the number of days between issue and settlement

After issue, a CD can be traded in the secondary market. The secondary market in CDs in the United Kingdom is generally liquid, and CDs will trade at the rate prevalent at the time, which will invariably be different from the coupon rate on the CD at issue. When a CD is traded in the secondary market, the settlement proceeds will need to take into account interest that has accrued on the paper and the different rate at which the CD has now been dealt. The formula for calculating the settlement figure is given in (6.4), which applies to the sterling market and its 365-day count basis.

$$\text{Proceeds} = \frac{M \times \text{Tenor} \times C \times 100 + 36500}{\text{Days remaining} \times r \times 100 + 36500} \tag{6.4}$$

The tenor of a CD is the life of the CD in days, while days remaining is the number of days left to maturity from the time of trade. The return on holding a CD is given by (6.5).

$$\text{Return} = \left[\frac{\left(1 + \text{Purchase yield} \times \dfrac{\text{Days from purchase to maturity}}{B}\right)}{1 + \text{Sale yield} \times \dfrac{\text{Days from sale to maturity}}{B}} - 1 \right] \times \frac{B}{\text{Days held}} \tag{6.5}$$

Securities Quoted on a Discount Basis

The remaining money-market instruments are all quoted on a discount basis, and so are known as discount instruments. This means that they are issued at a discount to face value, and are redeemed on maturity at face value. Treasury bills, bills of exchange, bankers acceptances, and commercial paper are examples of money-market securities that are quoted on a discount basis; that is, they are sold on the basis of a discount to par. The difference between the price paid at the time of purchase and the redemption value (par) is the interest earned by the holder of the paper. Explicit interest is not paid on discount instruments. Rather, interest is reflected implicitly in the difference between the discounted issue price and the par value received at maturity. Note that in some markets CP is quoted on a yield basis, but not in the United Kingdom or in the United States, where they are discount instruments.

Treasury Bills

Treasury bills or T-bills are government IOUs of short duration, often three-month maturity. For example, if a bill is issued on 10th January, it will mature on 10th April. Bills of one-month and six-month maturity are also issued, but only rarely in the UK market. On maturity, the holder of a T-bill receives the par value of the bill.

In a developed economy capital market, T-bill yields are regarded as the nearest proxy to the "risk-free" yield, as they represent the yield from short-term government debt. In emerging markets, they are often the most liquid instruments available for investors.

Pricing T-bills is at the pure discount. If the three-month yield at the time of issue is 5.25%, the price of the bill at issue is:

$$P = \frac{10\text{m}}{\left(1 + 0.0525 \times \dfrac{91}{365}\right)} = £9,870,800.69$$

In the UK and U.S. markets, the interest rate on discount instruments is quoted as a discount rate rather than a yield. This is the amount of discount expressed as an annualised percentage of the face value, and not as a percentage of the original amount paid. By definition, the discount rate is always lower than the corresponding yield. If the discount rate on a bill is *d*, then the amount of discount is given by (6.6) below.

$$d_{value} = M \times d \times \frac{n}{B} \tag{6.6}$$

The price *P* paid for the bill is the face value minus the discount amount, given by equation (6.7).

$$P = 100 \times \left[1 - \frac{\mathrm{d} \bullet \left(\dfrac{N_{sm}}{365} \right)}{100} \right] \tag{6.7}$$

If we know the yield on the bill, then we can calculate its price at issue by using the simple present-value formula, as shown in (6.8).

$$P = M / \left[1 + r \left(\frac{N_{sm}}{365} \right) \right] \tag{6.8}$$

The discount rate *d* for T-bills is calculated using (6.9).

$$d = (1 - P) \times \frac{B}{n} \tag{6.9}$$

The relationship between discount rate and true yield is given by (6.10).

$$\begin{aligned} d &= \frac{r}{\left(1 + r \times \dfrac{n}{B}\right)} \\ r &= \frac{d}{\left(1 - d \times \dfrac{n}{B}\right)} \end{aligned} \tag{6.10}$$

If a T-bill is traded in the secondary market, the settlement proceeds from the trade are calculated using (6.11).

$$\text{Proceeds} = \text{M} - \left(\frac{M \times \text{days remaining} \times d}{B \times 100} \right) \tag{6.11}$$

Bond equivalent yield. In certain markets, including the UK and U.S. markets, the yields on government bonds that have a maturity of less than one year are compared to the yields of Treasury bills. However, before the comparison can be made, the yield on a bill must be converted to a "bond equivalent" yield. Therefore, the bond-equivalent yield of a U.S. Treasury bill is the coupon of a theoretical Treasury bond trading at par that has an identical maturity date. If the bill has 182 days or less until maturity, the calculation required is the conventional conversion from discount rate to yield, with the exception that it is quoted on a 365-day basis (in the UK market, the quote basis is essentially the same unless it is a leap year; so the conversion element in (6.12) is not necessary). The calculation for the U.S. market is given by (6.12).

$$r = \frac{d}{\left(1 - d \times \frac{days}{360}\right)} \times \frac{365}{360} \tag{6.12}$$

where r is the bond-equivalent yield that is being calculated.

Note that if there is a bill and a bond that mature on the same day in a period under 182 days, the bond-equivalent yield will not be precisely the same as the yield quoted for the bond in its final coupon period, although it is a very close approximation. This is because the bond is quoted on actual/actual basis, so its yield is actually made up of two-times the actual number of days in the interest period.

Bankers Acceptances

A bankers acceptance is a written promise issued by a borrower to a bank to repay borrowed funds. The lending bank lends funds and in return accepts the bankers acceptance. The acceptance is negotiable and can be sold in the secondary market. The investor who buys the acceptance can collect the loan on the day that repayment is due. If the borrower defaults, the investor has legal recourse to the bank that made the first acceptance. Bankers acceptances are also known as bills of exchange, bank bills, trade bills, or commercial bills.

Essentially, bankers acceptances are instruments created to facilitate commercial trade transactions. The instrument is called a bankers acceptance because a bank accepts the ultimate responsibility to repay the loan to its holder. The use of bankers acceptances to finance commercial transactions is known as acceptance financing. The transactions for which acceptances are created include the import and export of goods, the storage and shipping of goods between two overseas countries, where neither the importer nor the exporter is based in the home country,[3] and the storage and shipping of goods between two entities based as home. Acceptances are discount instruments and are purchased by banks, local authorities, and money-market investment funds.

[3] A bankers acceptance created to finance such a transaction is known as a third-party acceptance.

The rate that a bank charges a customer for issuing a bankers acceptance is a function of the rate at which the bank thinks it will be able to sell it in the secondary market. A commission is added to this rate. For ineligible bankers acceptances (see below) the issuing bank will add an amount to offset the cost of the additional reserve requirements.

Commercial Paper

Commercial paper (CP) is a short-term money-market funding instrument issued by corporates. In the United Kingdom and United States, it is a discount instrument, with sterling paper being dealt with on a 365-day basis. They trade essentially as T-bills but with higher yields as they are unsecured corporate obligations. CP is an important part of the U.S. money market and began as a U.S. instrument before being introduced in other money centres around the world. The instrument ranges in maturity from 30 to 270 days (although typical maturities are 30 to 90 days) and is usually issued in response to investor demand or for short-term working-capital considerations. As the paper is unsecured, investor sentiment usually requires that any issue be rated by a rating agency such as Moody's Investors Services or Standard & Poor's, although high-yield CP also exists.

Another significant market exists in euro-commercial paper (ECP), which is similar in concept to CP but is not restricted to the 270-day maturity.[4] The market in ECP exists in money centres globally. Standard settlement of ECP is for spot value (which is two business days forward), whereas standard settlement of U.S. and UK settlement is on a same-day basis. ECP can be issued both as a discount instrument or a yield-bearing instrument, although the latter is rarer.

For yield-bearing ECP, the calculation of settlement proceeds in the secondary market is given by (6.13).

$$\text{Proceeds} = M \times \left[\frac{1 + \dfrac{C \times T}{36{,}000}}{1 + \dfrac{r \times N}{36{,}000}} \right] \tag{6.13}$$

where M is the face amount
C is the coupon
r is the yield
T is the paper's original maturity or tenor
N is the time from settlement to maturity

For paper issued on a discount basis, the proceeds are given by

$$\text{Proceeds} = \left[\frac{M}{1 + \dfrac{r \times N}{36{,}000}} \right] \tag{6.14}$$

The majority of ECP is issued in U.S. dollars, although euro, sterling, and Japanese yen are also popular currencies.

[4] In the U.S. market this is a Securities and Exchange Commission requirement.

U.S. TREASURY BILLS

The Treasury-bill market in the United States is one of the most liquid and transparent debt markets in the world. Consequently, the bid-offer spread on these bills is very narrow. The Treasury issues bills at a weekly auction each Monday, made up of 91-day and 182-day bills. Every fourth week, the Treasury also issues 52-week bills as well. As a result there are large numbers of Treasury bills outstanding at any one time. The interest earned on Treasury bills is not liable to state and local income taxes.

FEDERAL FUNDS

Commercial banks in the United States are required to keep reserves on deposit at the Federal Reserve. Banks with reserves in excess of required reserves can lend these funds to other banks, and these interbank loans are called federal funds or fed funds and are usually overnight loans. Through this market, commercial banks with excess funds are able to lend to banks that are short of reserves, thus facilitating liquidity. The transactions are very large denominations, and are lent at the fed funds rate, which is a very volatile interest rate because it fluctuates with market shortages.

PRIME RATE

The prime interest rate in the United States is often said to represent the rate at which commercial banks lend to their most creditworthy customers. In practice, many loans are made at rates below the prime rate, so the prime rate is not the best rate at which highly rated firms may borrow. Nevertheless, the prime rate is a benchmark indicator of the level of U.S. money-market rates, and is often used as a reference rate for floating-rate instruments. As the market for bank loans is highly competitive, all commercial banks quote a single prime rate, and the rate for all banks changes simultaneously.

MARKET RATES AND PRICES

Money-market yields and prices can be determined from a number of sources. Figures 6.1 through 6.5 show some key Bloomberg screens, which are MMR, MMCV, and BTMM, as of 20th October 2003.

Figure 6.1 shows U.S. money-market deposit rates. These are cash rates for overnight to 11-month terms. Figure 6.2 shows the same page for the Philippines money market, in this case prime bank-offered rates.

The money-market equivalent of the Bloomberg "IYC" yield-curve menu is MMCV, which shows money-market yield curves. We show in Figure 6.3 the money-market curves for three countries: Australia, Singapore, and Thailand. This screen is also available in table form, with the rates that are used to draw the curve listed in a table.

A useful summary screen for money-market rates on Bloomberg is BTMM. Figure 6.4 shows this page for the markets in Malaysia. This is a composite page that

GRAB Govt MMR

8:23 **USD Money Market Rates** Page 1 / 4

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SECURITY	TIME	BID	ASK	CHANGE	HIGH	LOW	PRV CLS
Fed Funds							
2)FDFD	10/17	.9375	1.0000	--	1.0000	1.0000	1.0000
Depo (Non-Japanese							
4)O/N	8:05	.9400	1.0600	+.0400	1.0650	.9500	.9900
5)USD Depo 2 Day	8:05	.9300	1.0500	+.0150	1.0250	.9950	1.0100
6)USD Depo 3 Day	8:05	.9300	1.0500	+.0150	1.0250	1.0050	1.0100
7)USD Depo 1 Wk	8:05	.9600	1.0800	+.0200	1.0500	1.0250	1.0300
8)USD Depo 2 Wk	8:05	.9800	1.1000	+.0200	1.0700	1.0500	1.0500
9)USD DEPO 3 Wk	8:05	.9800	1.1000	+.0200	1.0750	1.0600	1.0550
10)USD Depo 1 Mo	8:05	.9800	1.1000	+.0150	1.0750	1.0100	1.0600
11)USD Depo 2 Mo	8:05	.9900	1.1100	+.0150	1.0850	1.0200	1.0700
12)USD Depo 3 Mo	8:05	1.0400	1.1600	+.0200	1.1400	1.0450	1.1200
13)USD Depo 4 Mo	8:05	1.0700	1.1900	+.0575	1.1650	1.1050	1.1075
14)USD Depo 5 Mo	8:05	1.0900	1.2100	+.0625	1.1850	1.1275	1.1225
15)USD Depo 6 Mo	8:05	1.1300	1.2500	+.0725	1.2200	1.1475	1.1475
16)USD Depo 7 Mo	8:05	1.1600	1.2800	+.0350	1.2550	1.2050	1.2150
17)USD Depo 8 Mo	8:05	1.2000	1.3200	+.0200	1.2950	1.2550	1.2650
18)USD Depo 9 Mo	8:05	1.2400	1.3600	+.0150	1.3400	1.2850	1.3150
19)USD Depo 10 Mo	8:03	1.3500	1.3800	--	1.3950	1.3550	1.3650
20)USD Depo 11 Mo	8:03	1.4000	1.4300	+.0100	1.4350	1.3950	1.4050

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2003 Bloomberg L.P.
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FIGURE 6.1 U.S. Money Market Rates

GRAB Govt BIL

8:24 **PRIME RATES BY BANKS** Page 1 / 3

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SECURITY	TICKER	CUR VAL	DATE	PREVIOUS	DATE	CHANGE	PCT CHN
1)GENERAL AVE	PLNDAVE	8.6134	10/17	8.6134	10/16	--	--
Local Banks							
3)ASIA UNITED	PLNDAUB	9.6290	10/17	9.6290	10/16	--	--
4)BNK OF COMMERCE	PLNDCOM	8.6290	10/17	8.6290	10/16	--	--
5)EAST WEST	PLNDEWB	10.5060	10/17	10.5060	10/16	--	--
6)EXPORT & INDS	PLNDEXIN	8.6290	10/17	8.6290	10/16	--	--
7)I-BANK	PLNDIBAN	10.6290	10/17	10.6290	10/16	--	--
8)PBCOM	PLNDPCOM	9.5000	10/17	9.5000	10/16	--	--
9)PHILTRUST	PLNDPTC	8.6290	10/17	8.6290	10/16	--	--
10)VETERANS BANK	PLNDPVB	9.0000	10/17	9.0000	10/16	--	--
11)TA BANK	PLNDTAB	8.9120	10/17	8.9120	10/16	--	--
Subsidiary FB's							
13)UOB	PLNDUOB	10.3750	10/17	10.3750	10/16	--	--
14)SANTANDER	PLNDBSAN	7.7500	10/17	7.7500	10/16	--	--
15)CHINATRUST	PLNDCHIB	8.8790	10/17	8.8790	10/16	--	--
16)MAY BANK	PLNDMAYB	10.1290	10/17	10.1290	10/16	--	--
17)LOCAL BKS AVE	PLNDAVLS	9.3228	10/17	9.3228	10/16	--	--
Foreign Banks							
19)ANZ BANK	PLNDANZ	7.4000	10/17	7.4000	10/16	--	--
20)BANGKOK BANK	PLNDBKB	6.6290	10/17	6.6290	10/16	--	--

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2003 Bloomberg L.P.
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FIGURE 6.2 Philippines Money Market Prime Rates

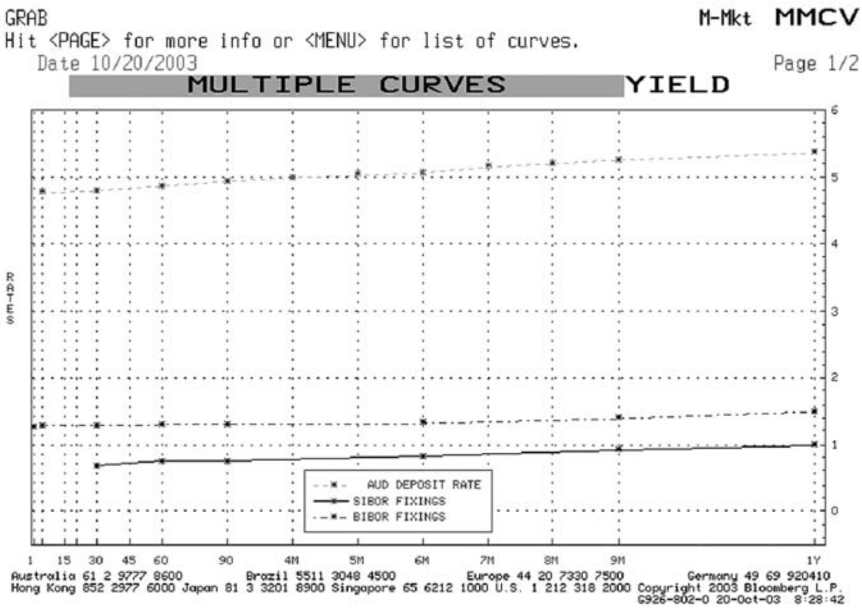

FIGURE 6.3 Money Market Yield Curves on Screen MMCV

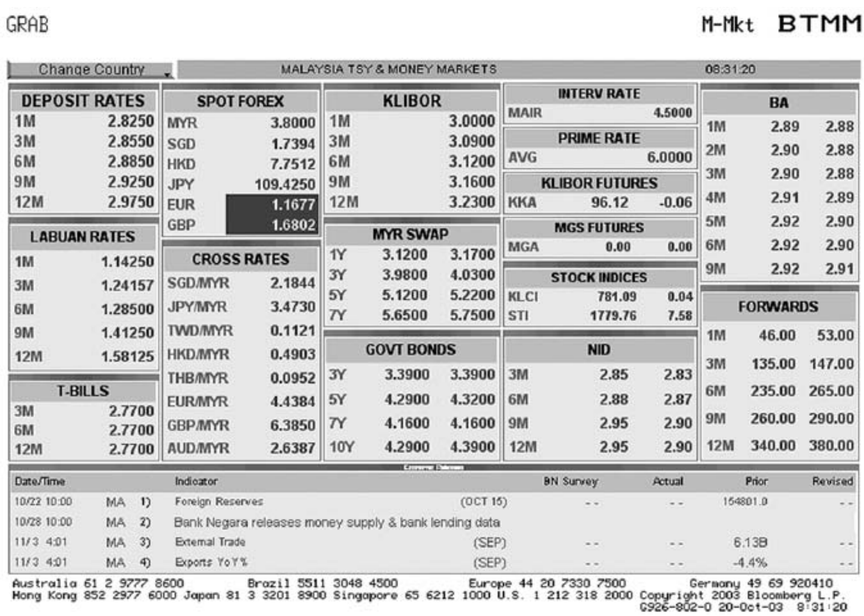

FIGURE 6.4 Malaysian Market Rates on BTMM

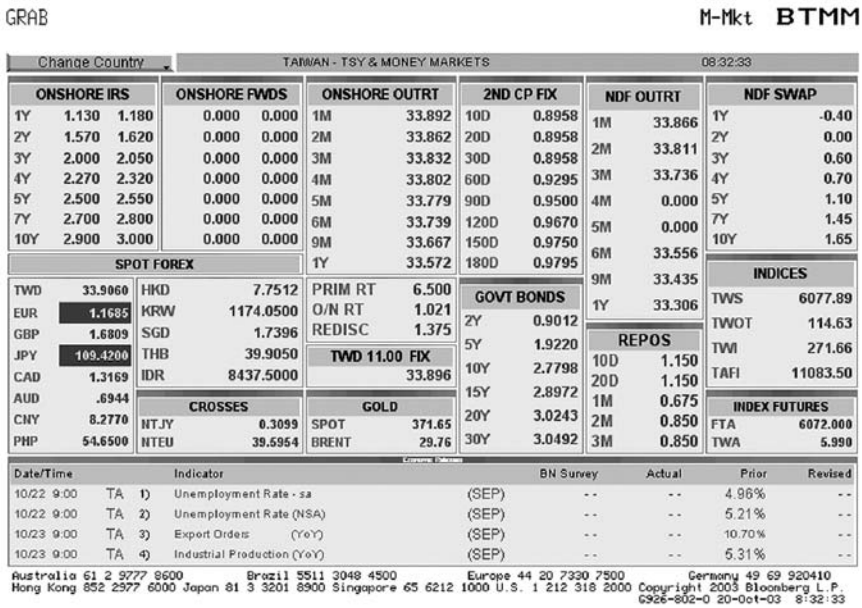

FIGURE 6.5 Taiwan Money Market Rates on BTMM

shows a range of sectors; for instance, government-bond yields, T-bills, cash-deposit rates, forward rates, and bankers acceptances. Figure 6.5 shows the same page for the markets in Taiwan.

THE REPO INSTRUMENT

Repo is a short-term secured cash instrument that should always be labelled as part of the money markets. There is a wide range of uses to which repo might be put. Structured transactions that are very similar to repo include total return swaps, and other structured repo trades include floating-rate repo, which contains an option to switch to a fixed rate at a later date. In the equity market, repo is often conducted in a basket of stocks, which might be constituent stocks in an index such as the FTSE100 or CAC40 or user-specified baskets. Market makers borrow and lend equities with differing terms to maturity, and generally the credit rating of the institution involved in the repo transaction is of more importance than the quality of the collateral. Central banks' use of repo also reflects its importance; it is a key instrument in the implementation of monetary policy in many countries. Essentially then, repo markets have vital links and relationships with global money markets, bond markets, futures markets, swap markets, and over-the-counter (OTC) interest rate derivatives.

In this section we discuss key features of the repo instrument.

Key Features

Repo is essentially a secured loan. The term comes from *sale and repurchase agreement;* however, this is not necessarily the best way to look at it. Although in a *classic* *repo* transaction legal title of an asset is transferred from the "seller" to the "buyer" during the term of the repo, in the author's opinion this detracts from the essence of the instrument: a secured loan of cash. It is therefore a money-market instrument.

There are a number of benefits in using repo, which concurrently have been behind its rapid growth. These include the following:

- Market makers generally are able to finance their long bond and equity positions at a lower interest cost if they repo out the assets; equally they are able to cover short positions;
- There is greater liquidity in specific individual bond issues;
- Greater market liquidity lowers the cost of raising funds for capital market borrowers;
- Central banks are able to use repo in their open market operations;
- Repo reduces *counterparty risk* in money-market borrowing and lending, because of the security offered by the collateral given in the loan;
- Investors have an added investment option when placing funds;
- Institutional investors and other long-term holders of securities are able to enhance their returns by making their inventories available for repo trading.

The maturity of the majority of repo transactions is between overnight and three months, although trades of six months and one year are not uncommon. It is possible to transact in longer-term repo as well. Because of this, repo is best seen as a money-market product.[5] However, because of the nature of the collateral, repo market participants must keep a close eye on the market of the asset collateral, whether this is the government bond market, Eurobonds, equity, or other assets.[6] The counterparties to a repo transaction will have different requirements, for instance to "borrow" a particular asset against an interest in lending cash. For this reason it is common to hear participants talk of trades being *stock-driven* or *cash-driven.* A corporate treasurer who invests cash while receiving some form of security is driving a cash-driven repo, whereas a market maker that wishes to cover a short position in a particular stock, against which she lends cash, is entering into a stock-driven trade.

There is a close relationship between repo and both the bond and money markets. The use of repo has contributed greatly to the liquidity of government, Eurobond and emerging market bond markets. Although it is a separate and individual market itself, operationally repo is straightforward to handle, in that it settles through clearing mechanisms used for bonds.

Financial institutions will engage in both repo and reverse repo trades. Investors also, despite their generic name, will be involved in both repo and reverse repo. Their money-market funds will be cash-rich and engage in investment trades; at the same

[5] The textbook definition of a "money-market" instrument is of a debt product issued with between one day and one year to maturity, while debt instruments of greater than one year maturity are known as capital market instruments. In practice the money-market desks of most banks will trade the yield curve to up to two years maturity, so it makes sense to view a money-market instrument as being of up to two years maturity.

[6] This carries on to bank organisation structure. In most banks, the repo desk for bonds is situated in the money markets area, while in others it will be part of the bond division (the author has experience of banks employing each system). Equity repo is often situated as part of the back office settlement or Treasury function.

time they will run large fixed-interest portfolios, the returns for which can be enhanced through trading in repo. Central banks are major players in repo markets and use repo as part of daily liquidity or *open market* operations and as a tool of monetary policy.

Repo itself is an over-the-counter market conducted over the telephone, with rates displayed on screens. These screens are supplied by both brokers and market makers themselves. Increasingly, electronic dealing systems are being used, with live dealing rates displayed on screen and trades being conducted at the click of a mouse button.

There are three main basic types of repo: the *classic* repo, the *sell/buy-back,* and *tri-party repo.* A sell/buy-back, referred to in some markets as a *buy-sell,* is a spot sale and repurchase of assets transacted simultaneously. It does not require a dealing and settlement system that can handle the concept of a classic repo and is often found in emerging markets. A classic repo is economically identical, but the repo rate is explicit, and the transaction is conducted under a legal agreement that defines the legal transfer of ownership of the asset during the term of the trade. Classic repo, the type of transaction that originated in the United States, is a sale and repurchase of an asset where the repurchase price is unchanged from the "sale" price. Hence the transaction is better viewed as a loan and borrow of cash. In a tri-party repo a third party acts as an agent on behalf of both seller and buyer of the asset, but otherwise the instrument is identical to classic repo.

Repo Products

A repo agreement is a transaction in which one party sells securities to another, and at the same time and as part of the same transaction commits to repurchase identical securities on a specified date at a specified price. The seller delivers securities and receives cash from the buyer. The cash is supplied at a predetermined rate of interest—*the repo rate*—that remains constant during the term of the trade. On maturity, the original seller receives back collateral of equivalent type and quality, and returns the cash plus repo interest. One party to the repo requires either the cash or the securities and provides *collateral* to the other party, as well as some form of compensation for the temporary use of the desired asset. Although legal title to the securities is transferred, the seller retains both the economic benefits and the market risk of owning them. This means that the "seller" will suffer loss if the market value of the collateral drops during the term of the repo, as they still retain beneficial ownership of the collateral. The "buyer" in a repo is not affected in profit/loss account terms if the value of the collateral drops, although as we shall see later, there will be other concerns for the buyer if this happens.

We have given here the legal definition of repo. However, the purpose of the transaction as we have described above is to borrow or lend cash, which is why we have used quotation marks when referring to sellers and buyers. The "seller" of stock is really interested in borrowing cash, on which they will pay interest at a specified interest rate. The "buyer" requires security or *collateral* against the loan they have advanced, and/or the specific security to borrow for a period of time. The first and most important thing to state is that repo is a secured loan of cash, and would be categorised as a money-market yield instrument.[7]

We now look at the main repo instruments in turn.

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[7] That is, a money-market product quoted as a yield instrument, similar to a bank deposit or a Certificate of Deposit. The other class of money-market products is *discount* instruments such as a Treasury bill or commercial paper.

The Classic Repo

The *classic repo* is the instrument encountered in the United States, United Kingdom, and other markets. In a classic repo one party will enter into a contract to sell securities, simultaneously agreeing to purchase them back at a specified future date and price. The securities can be bonds or equities but also money-market instruments such as T-bills. The buyer of the securities is handing over cash, which on the termination of the trade will be returned to them, and on which they will receive interest.

The seller in a classic repo is selling or *offering* stock, and therefore receiving cash, whereas the buyer is buying or *bidding* for stock, and consequently paying cash. So if the one-week repo interest rate is quoted by a market-making bank as "5½ – 5¼", this means that the market maker will bid for stock, that is, lend the cash, at 5.50% and offers stock or pays interest on borrowed cash at 5.25%. In some markets the quote is reversed.

Illustration of Classic Repo

There will be two parties to a repo trade, let us say Bank A (the seller of securities) and Bank B (the buyer of securities). On the trade date the two banks enter into an agreement whereby on a set date, the *value* or *settlement* date Bank A will sell to Bank B a nominal amount of securities in exchange for cash.[8] The price received for the securities is the market price of the stock on the value date. The agreement also demands that on the termination date Bank B will sell identical stock back to Bank A at the previously agreed price; consequently, Bank B will have its cash returned with interest at the agreed repo rate.

In essence a repo agreement is a secured loan (or *collateralised* loan) in which the repo rate reflects the interest charged on the cash being lent.

On the value date, stock and cash change hands. This is known as the start date, *on-side* date, *first leg*, or *opening leg,* while the termination date is known as the *second leg, off-side leg*, or *closing leg.* When the cash is returned to Bank B, it is accompanied by the interest charged on the cash during the term of the trade. This interest is calculated at a specified rate known as the *repo rate.* It is important to remember that although in legal terms the stock is initially "sold" to Bank B, the economic effects of ownership are retained with Bank A. This means that if the stock falls in price it is Bank A that will suffer a capital loss. Similarly, if the stock involved is a bond and there is a coupon payment during the term of the trade, this coupon is to the benefit of Bank A, and although Bank B will have received it on the coupon date, it must be handed over on the same day or immediately after to Bank A. This reflects the fact that although legal title to the collateral passes to the repo buyer, economic costs and benefits of the collateral remain with the seller.

A classic repo transaction is subject to a legal contract signed in advance by both parties. A standard document will suffice; it is not necessary to sign a legal agreement prior to each transaction.

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[8] The two terms are not necessarily synonymous. The value date in a trade is the date on which the transaction acquires value; for example, the date from which accrued interest is calculated. As such it may fall on a nonbusiness day such as a weekend or public holiday. The settlement date is the day on which the transaction settles or *clears,* and so can only fall on a business day.

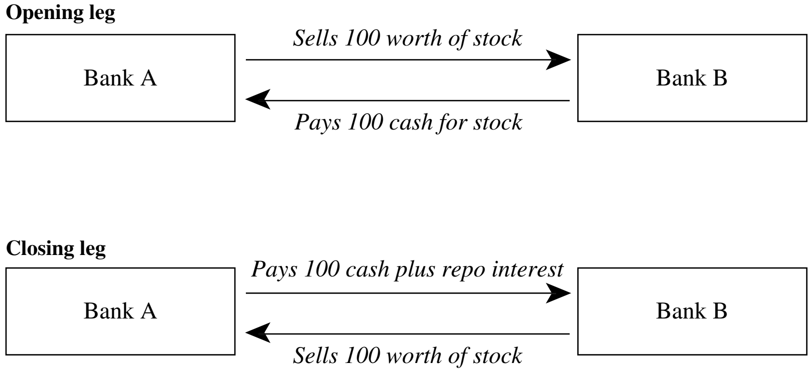

FIGURE 6.6 Classic Repo Transaction for 100-Worth of Collateral Stock

Note that although we have called the two parties in this case "Bank A" and "Bank B", it is not only banks that get involved in repo transactions, and we have used these terms for the purposes of illustration only.

The basic mechanism is illustrated in Figure 6.6.

A seller in a repo transaction is entering into a repo, whereas a buyer is entering into a *reverse repo*. In Figure 6.6 the repo counterparty is Bank A, while Bank B is entering into a reverse repo. That is, a reverse repo is a purchase of securities that are sold back on termination. As is evident from Figure 6.6 every repo is a reverse repo, and the name given to a deal is dependent on whose viewpoint one is looking at the transaction.

Example of Classic Repo The basic principle is illustrated with the following example. This considers a *specific* repo; that is, one in which the collateral supplied is specified as a particular stock, as opposed to a *general collateral* (GC) trade in which a basket of collateral can be supplied, of any particular issue, as long as it is of the required type and credit quality.

We consider first a classic repo in the UK gilt market between two market counterparties, in the 5.75% Treasury 2009 gilt stock. The terms of the trade are given in Table 6.1 and illustrated in Figure 6.7. Note that the terms of a classic repo trade are identical, irrespective of which market the deal is taking place in. So the basic trade, illustrated in Table 6.1, would be recognisable for bond repo in European and Asian markets.

The repo counterparty delivers to the reverse repo counterparty £10 million nominal of the stock, and in return receives the purchase proceeds. The clean market price of the stock is £104.60. In this example no *margin* (called haircut) has been taken so the start proceeds are equal to the market value of the stock, which is £10,505,560.11. It is common for a rounded sum to be transferred on the opening leg. The repo interest is 5.75%, so the repo interest charged for the trade is

$$10{,}505{,}560 \times 5.75\% \times 7/365$$

or £11,584.01. The sterling market day-count basis is actual/365, and the repo interest is based on a seven-day repo rate of 5.75%. Repo rates are agreed at the time of the trade and are quoted, like all interest rates, on an annualised basis. The settlement price

TABLE 6.1 Terms of Classic Repo Trade

Trade date	5 July 2000
Value date	6 July 2000
Repo term	1 week
Termination date	13 July 2000
Collateral (stock)	UKT 5.75% 2009
Nominal amount	£10,000,000
Price	104.60
Accrued interest (29 days)	0.4556011
Dirty price	105.055601
Settlement proceeds (*wired amount*)	£10,505,560.11
Repo rate	5.75%
Repo interest	£11,584.90
Termination proceeds	£10,517,145.01

(dirty price) is used because it is the market value of the bonds on the particular trade date and therefore indicates the cash value of the gilts. By doing this the cash investor minimises credit exposure by equating the value of the cash and the collateral.

On termination the repo counterparty receives back its stock, for which it hands over the original proceeds plus the repo interest calculated above.

Market participants who are familiar with the Bloomberg trading system will use screen RRRA for a classic repo transaction. For this example the relevant screen entries are shown in Figure 6.8. This screen is used in conjunction with a specific stock, so in this case it would be called up by entering

UKT 5.75 09 <GOVT> RRRA <GO>

where "UKT" is the ticker for UK gilts. Note that the date format for Bloomberg screens is the U.S. style, which is mm/dd/yy. The screen inputs are relatively

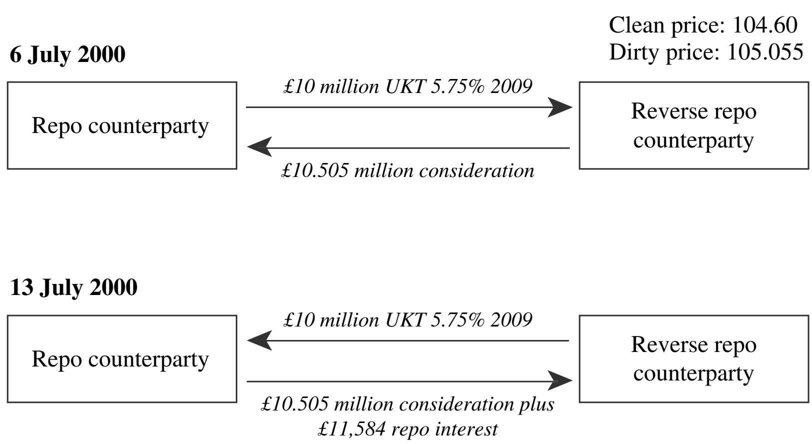

FIGURE 6.7 Diagram of Classic Repo Trade

self-explanatory, with the user entering the terms of the trade that are detailed in Table 6.1. There is also a field for calculating margin, labelled "collateral" on the screen. As no margin is involved in this example, it is left at its default value of 100.00%. The bottom of the screen shows the opening leg cash proceeds or "wired amount", the repo interest and the termination proceeds.

If we wanted to use screen RRRA for other securities, we would enter the relevant bond ticker, for example "T" for U.S. Treasuries, "B" for German Bunds, and so on. The principles are the same for classic repo trades whatever the jurisdiction, and the screen may be used for all markets that undertake classic repo.

What if a counterparty is interested in investing £10 million against gilt collateral? Let us assume that a corporate treasury function with surplus cash wishes to invest this amount in repo for a one-week term. It invests this cash with a bank that deals in gilt repo. We can use Bloomberg screen RRRA to calculate the nominal amount of collateral required. Figure 6.9 shows the screen for this trade, again against the 5.75% Treasury 2009 stock as collateral. We see from Figure 6.9 that the terms of the trade are identical to that in Table 6.1, including the bond price and the repo rate; however, the opening leg wired amount is entered as £10 million, which is the cash being invested. Therefore, the nominal value of the gilt collateral required will be different, as we now require a market value of this stock of £10 million. From the screen we see that this is £9,518,769. The cash amount is different from the example in Figure 6.8, so of course the repo interest charged is different, and is £11,027 for the seven-day term. The diagram in Figure 6.10 illustrates the transaction details.

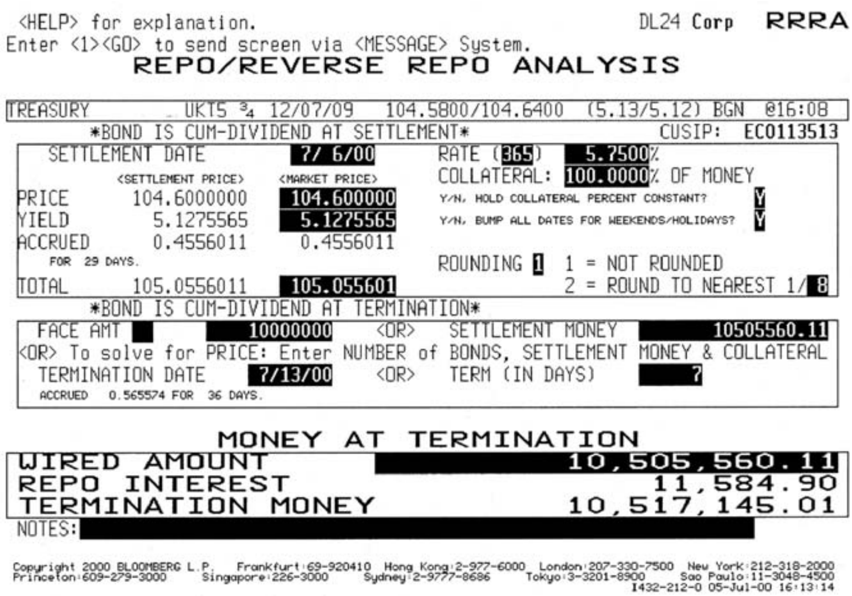

FIGURE 6.8 Bloomberg Screen RRRA for Classic Repo Transaction, Trade Date 5 July 2000

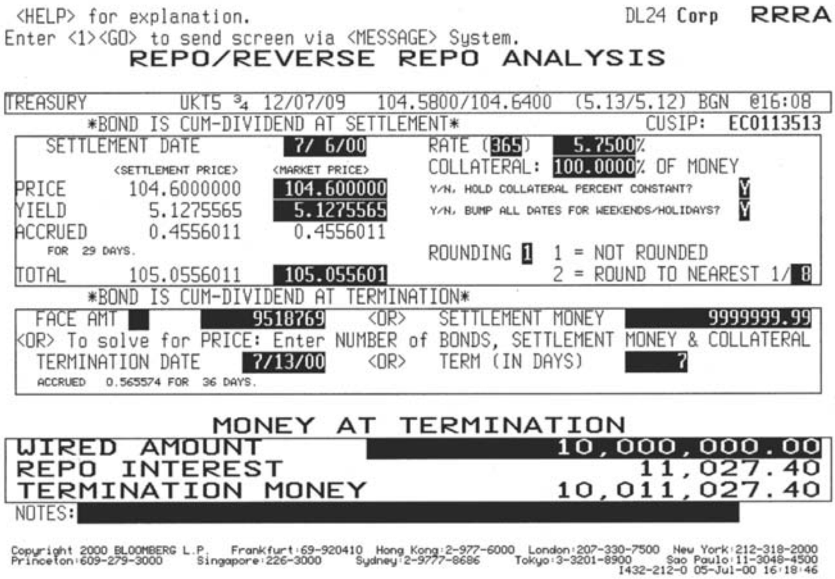

FIGURE 6.9 Bloomberg Screen for the Classic Repo Trade Illustrated in Figure 6.10

The Sell/Buy-Back

We next consider the sell/buy-back, which is economically identical to the classic repo but is described under different cash flow terms.

Definition In addition to classic repo, there exists *sell/buy-back*. A sell/buy-back is defined as an outright sale of a bond on the value date, and an outright repurchase of that bond for value on a *forward* date. The cash flows therefore become a sale of the bond at a *spot* price, followed by repurchase of the bond at the *forward* price. The

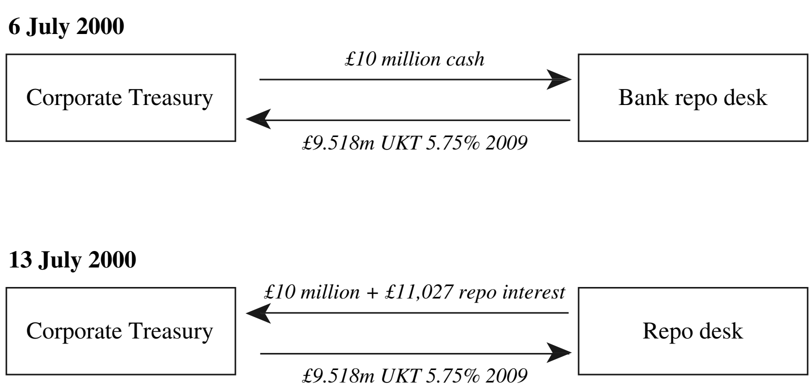

FIGURE 6.10 Corporate Treasury Classic Repo

forward price calculated includes the interest on the repo, and is therefore a different price to the spot price.[9] That is, repo interest is realised as the difference between the spot price and forward price of the collateral at the start and termination of the trade. The sell/buy-back is entered into for the same reasons as a classic repo, but was developed initially in markets where no legal agreement existed to cover repo transactions, and where the settlement and IT systems of individual counterparties were not equipped to deal with repo. Over time, sell/buy-backs have become the convention in certain markets, most notably Italy, and so the mechanism is still used. In many markets therefore, sell/buy-backs are not covered by a legal agreement, although the standard legal agreement used in classic repo now includes a section that describes them.[10]

A sell/buy-back is a spot sale and forward repurchase of bonds transacted simultaneously, and the repo rate is not explicit, but is implied in the forward price. Any coupon payments during the term are paid to the seller; however, this is done through incorporation into the forward price, so the seller will not receive it immediately, but on termination. This is a disadvantage when compared to classic repo. However, there will be compensation payable if a coupon is not handed over straightaway, usually at the repo rate used in the sell/buy-back. As sell/buy-backs are not subject to a legal agreement in most cases, in effect the seller has no legal right to any coupon, and there is no provision for marking-to-market and *variation margin*. This makes the sell/buy-back a higher-risk transaction when compared to classic repo, even more so in volatile markets.

Note that in some markets the term *repo* is used to describe what are in fact sell/buy-backs. The Italian market is a good example of where this convention is followed.

A general diagram for the sell/buy-back is given in Figure 6.11.

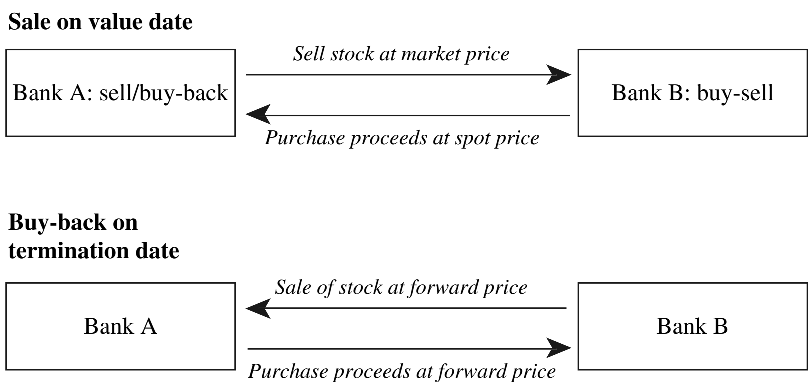

FIGURE 6.11 Sell/Buy-Back Transaction

[9] The "forward price" is calculated only for the purpose of incorporating repo interest; it should not be confused with a forward interest rate, which is the interest rate for a term starting in the future and which is calculated from a spot interest rate. Nor should it be taken to be an indication of what the market price of the bond might be at the time of trade termination, the price of which could differ greatly from the sell/buy-back forward price.

[10] This is the ICMA Global Master Repurchase Agreement.

Example of Sell/Buy-Back

We use the same terms of trade given in Figure 6.8 earlier, but this time the trade is a sell/buy-back.[11] In a sell/buy-back we require the forward price on termination, and the difference between the spot and forward price incorporates the effects of repo interest. It is important to note that this forward price has nothing to do with the actual market price of the collateral at the time of forward trade. It is simply a way of allowing for the repo interest that is the key factor in the trade. Thus in sell/buy-back, the repo rate is not explicit (although it is the key consideration in the trade); rather, it is implicit in the forward price.

In this example, one counterparty sells £10 million nominal of the UKT 5.75% 2009 at the spot price of 104.60, this being the market price of the bond at the time. The consideration for this trade is the market value of the stock, which is £10,505,560 as before. Repo interest is calculated on this amount at the rate of 5.75% for one week, and from this the termination proceeds are calculated. The termination proceeds are divided by the nominal amount of stock to obtain the forward dirty price of the bond on the termination date. For various reasons, the main one being that settlement systems deal in clean prices, we require the forward clean price, which is obtained by subtracting from the forward dirty price the accrued interest on the bond on the termination date. At the start of the trade the 5.75% 2009 had 29 days' accrued interest, therefore on termination this figure will be 29 + 7 or 36 days.

Bloomberg users access a different screen for sell/buy-backs, which is BSR. This is shown in Figure 6.12. Entering in the terms of the trade, we see from Figure 6.12 that the forward price is 104.605876. However, the fundamental nature of this transaction is evident from the bottom part of the screen: the settlement amount ("wired amount"), repo interest, and termination amount are identical for the classic repo trade described earlier. This is not surprising; the sell/buy-back is a loan of £10.505 million for one week at an interest rate of 5.75%. The mechanics of the trade do not differ on this key point.

Screen BSR on Bloomberg has a second page, which is shown in Figure 6.13. This screen summarises the cash proceeds of the trade at start and termination. Note how the repo interest is termed "funding cost". This is because the trade is deemed to have been entered into by a bond trader who is funding his book. This will be considered later, but we can see from the screen details that during the one week of the trade the bond position has accrued interest of £10,997. This compares unfavourably with the repo funding cost of £11,584.

If there is a coupon payment during a sell/buy-back trade and it is not paid over to the seller until termination, a compensating amount is also payable on the coupon amount, usually at the trade's repo rate. When calculating the forward price on a sell/buy-back where a coupon will be paid during the trade, we must subtract the coupon amount from the forward price. Note also that sell/buy-backs are not possible on an open basis, as no forward price can be calculated unless a termination date is known.

[11] The Bank of England discourages sell/buy-backs in gilt repo, and it is unusual, if not unheard of, to observe them in this market. However, we use these terms of trade for comparison purposes with the classic repo example given in the previous section. The procedure and the terms of the trade would be identical in other markets such as Italy and Portugal where sell/buy-back trades are the norm.

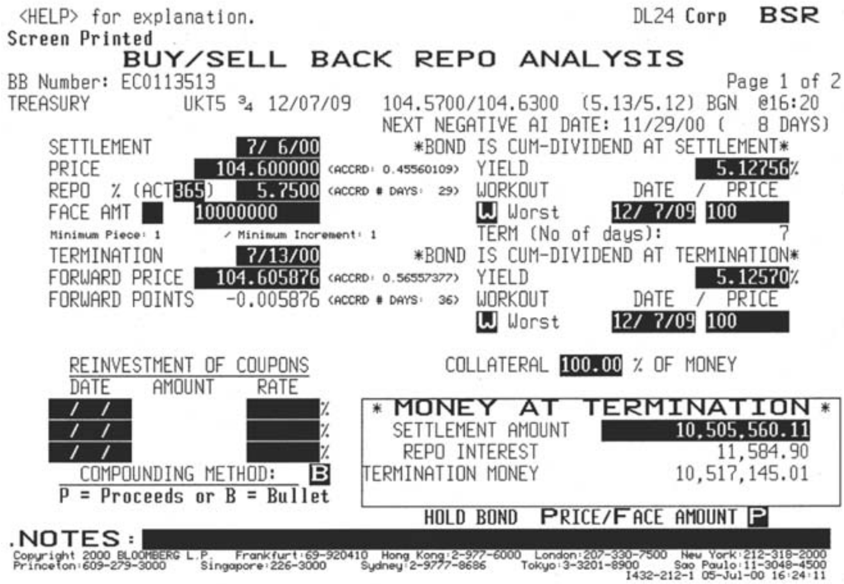

FIGURE 6.12 Bloomberg Screen BSR for Sell/Buy-Back Trade in 5.75% 2009, Trade Date 5 July 2000

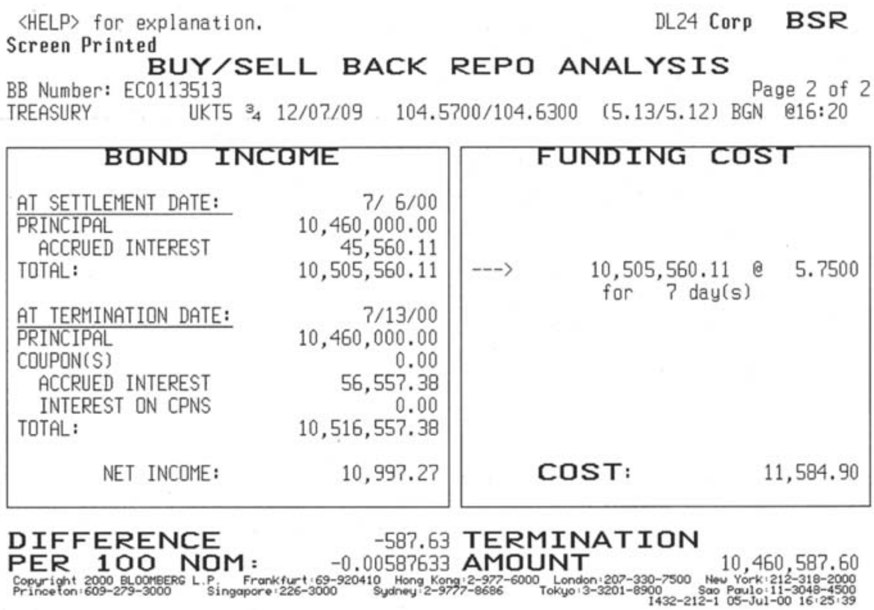

FIGURE 6.13 Bloomberg Screen BSR Page 2 for Sell/Buy-Back Trade in 5.75% 2009 Gilt Shown in Figure 6.12

REPO VARIATIONS

In the earlier section we described the standard classic repo trade, which has a fixed term to maturity and a fixed repo rate. Generally, classic repo trades will range in maturity from overnight to one year; however it is possible to transact longer maturities than this if required. The overwhelming majority of repo trades are between overnight and three months in maturity, although longer-term trades are not uncommon. A fixed-maturity repo is sometimes called a *term repo*. One could call this the plain-vanilla repo. It is usually possible to terminate a vanilla repo before its stated maturity date if this is required by one or both of the counterparties.[12]

A repo that does not have a specified fixed maturity date is known as an *open repo*. In an open repo the borrower of cash will confirm each morning that the repo is required for a further overnight term. The interest rate is also fixed at this point. If the borrower no longer requires the cash or requires the return of his collateral, the trade will be terminated at one day's notice.

In the remainder of this section we present an overview of the most common variations on the vanilla repo transaction that are traded in the markets.

Tri-Party Repo

The tri-party repo mechanism is designed to make the repo arrangement accessible to a wider range of market counterparties. Essentially it introduces a third-party agent in between the two repo counterparties, who can fulfil a number of roles from security custodian to cash account manager. The tri-party mechanism allows bond and equity dealers full control over their inventory, and incurs minimal settlement cost to the cash investor, but gives the investor independent confirmation that their cash is fully collateralised. Under a tri-party agreement, the securities dealer delivers collateral to an independent third-party custodian, such as Euroclear or Clearstream,[13] who will place it into a segregated tri-party account. The securities dealer maintains control over which precise securities are in this account (multiple substitutions are permitted), but the custodian undertakes to confirm each day to the investor that their cash remains fully collateralised by securities of suitable quality. A tri-party agreement needs to be in place with all three parties before trading can commence. This arrangement reduces the administrative burden for the cash-investor, but is not, in theory, as secure as a conventional delivery-versus-payment structure. Consequently the yield on the investor's cash (assuming collateral of identical credit quality) should be slightly higher. The structure is shown in Figure 6.14.

[12] The term *delivery repo* is sometimes used to refer to a vanilla classic repo transaction where the supplier of cash takes delivery of the collateral, whether in physical form or as a book-entry transfer to his account in the clearing system (or his agent's account).

[13] Clearstream was previously known as Cedel Bank. Other tri-party providers include Deutsche Bank and Bank of New York.

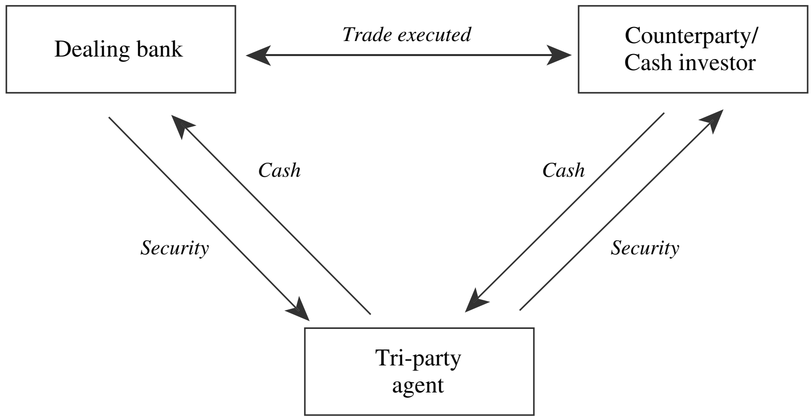

FIGURE 6.14 Tri-Party Repo Structure

The first tri-party repo deal took place in 1993 between the European Bank for Reconstruction and Development (EBRD) and Swiss Bank Corporation.[14]

A tri-party arrangement is, in theory, more attractive to smaller market participants as it removes the expense of setting up in-house administration facilities that would be required for conventional repo. This is mainly because the delivery and collection of collateral is handled by the tri-party agent. Additional benefits to cash-rich investors include:

- No requirement to install repo settlement and monitoring systems;
- No requirement to take delivery of collateral or to maintain an account at the clearing agency;
- Independent monitoring of market movements and margin requirements;
- In the event of default, a third-party agent that can implement default measures.

Set against the benefits is of course the cost of tri-party repo, essentially the fee payable to the third-party agent. This fee will include a charge for setting up accounts and arrangements at the tri-party agent, and a custodian charge for holding securities in the clearing system.

As well as being attractive to smaller banks and cash-rich investors, the larger banks will also use tri-party repo, in order to be able to offer it as a service to their smaller-size clients. The usual arrangement is that both dealer and cash investor will pay a fee to the tri-party agent based on the range of services that are required, and this will be detailed in the legal agreement in place between the market counterparty and the agent. This agreement will also specify, among other details, the specific types of security that are acceptable as collateral to the cash lender; the repo rate that is earned by the lender will reflect the nature of collateral that is supplied. In every other respect, however, the tri-party mechanism offers the same flexibility of

[14] Stated in D. Corrigan et al., *Repo: The Ultimate Guide* (London: Pearls of Wisdom Publishing, 1999), 27.

conventional repo and may be transacted from maturities ranging from overnight to one year.

The tri-party agent is an agent to both parties in the repo transaction. It provides a collateral management service overseeing the exchange of securities and cash, and managing collateral during the life of the repo. It also carries out daily marking-to-market, and substitution of collateral as required. The responsibilities of the agent can include:

- The setting up of the repo account;
- Monitoring of cash against purchased securities, both at inception and at maturity;
- Initial and ongoing testing of *concentration* limits;
- The safekeeping of securities handed over as collateral;
- Managing the substitution of securities, where this is required;
- Monitoring the market value of the securities against the cash lent out in the repo;
- Issuing margin calls to the borrower of cash.

The tri-party agent will issue close-of-business reports to both parties. The contents of the report can include some or all of the following:

- Tri-party repo cash and securities valuation;
- Corporate actions;
- Pre-advice of expected income;
- Exchange rates;
- Collateral substitution.

The extent of the duties performed by the tri-party agent is dependent on the sophistication of an individual party's operation. Smaller market participants who do not wish to invest in extensive infrastructure may outsource all repo-related functions to the tri-party agent.

Tri-party repo was originally conceived as a mechanism through which repo would become accessible to smaller banks and nonbank counterparties. It is primarily targeted at cash-rich investors. However, users of the instrument range across the spectrum of market participants and include, on the investing side, cash-rich financial institutions such as banks, fund managers including life companies and pension funds, and savings institutions such as UK building societies. On the borrowing side, users include bond and equity market makers, and banks with inventories of high-quality assets such as government bonds and highly-rated corporate bonds.

Hold-in-Custody Repo

This is part of the general collateral (GC) market, and is more common in the United States than elsewhere. Consider the case of a cash-rich institution investing in GC as an alternative to deposits or commercial paper. The better the quality of collateral, the lower the yield the institution can expect, while the mechanics of settlement may also affect the repo rate. The most secure procedure is to take physical possession of the collateral. However, if the dealer needs one or more substitutions during the

TRI-PARTY REPO: DEAL MECHANICS

The process of cash and collateral flow in a tri-party repo trade is illustrated in Figures 6.15 and 6.16.

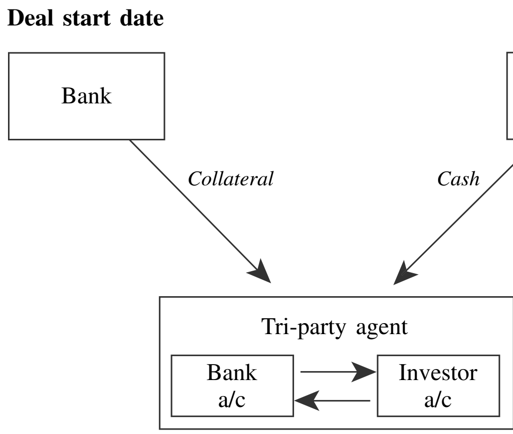

FIGURE 6.15 Tri-Party Repo Flow

Cash is transferred to the Bank's account and securities to the Investor's account after the tri-party agent confirms the collateral quality and adequacy.

The tri-party agent effects a simultaneous transfer of cash (original capital plus repo interest) versus securities.

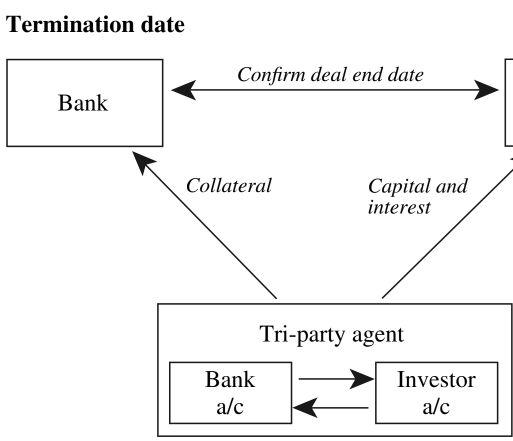

FIGURE 6.16 Tri-Party Repo Flow

(*continued*)

(Continued)

Table 6.2 shows the acceptable collateral types as advised by the institutional trust arm of a U.S. investment bank.

TABLE 6.2 Tri-Party Acceptable Collateral: U.S. Investment Bank Counterparty

Government bonds	Cash
Government guaranteed/local authority bonds	Certificate of Deposits Delivery by Value (DBV)
Supranational bonds	Letters of credit
Eurobonds	Equities
Corporate bonds	American depositary receipts
ABS/MBS	Warrants
Convertible bonds	

term of the trade, the settlement costs involved may make the trade unworkable for one or both parties. Therefore, the dealer may offer to hold the securities in his own custody against the investor's cash. This is known as a *hold-in-custody* (HIC) repo. The advantage of this trade is that since securities do not physically move, no settlement charges are incurred. However, this carries some risk for the investor because they only have the dealer's word that their cash is indeed fully collateralised in the event of default. Thus this type of trade is sometime referred to as a "Trust Me" repo; it is also referred to as a *due-bill repo* or a *letter repo*.

In the U.S. market there have been cases in which securities houses that went into bankruptcy and defaulted on loans were found to have pledged the same collateral for multiple HIC repo trades. Investors dealing in HIC repo must ensure:

- They only invest with dealers of good credit quality, since an HIC repo may be perceived as an unsecured transaction;
- They receive a higher yield on their cash in order to compensate them for the higher credit risk involved.

A *safekeeping repo* is identical to an HIC repo whereby the collateral from the repo seller is not delivered to the cash lender but held in "safe keeping" by the seller. This has advantages in that there is no administration and cost associated with the movement of stock. The risk is that the cash lender must entrust the safekeeping of collateral to the counterparty and has no means of confirming that the security is indeed segregated, and only being used for one transaction.

Due to the counterparty risk inherent in an HIC repo, it is rare to see it transacted either in the U.S. market or elsewhere. Certain securities are not suitable for delivery; for example, the class of mortgage securities known as *whole loans* in the United States, and these are often funded using HIC repo (termed *whole-loan repo)*.

Cross-Currency Repo

All of the examples of repo trades discussed so far have used cash and securities denominated in the same currency: for example, gilts trading versus sterling cash, and so on. In fact, there is no requirement to limit oneself to single-currency transactions. It is possible to trade say, gilts versus U.S. dollar cash (or any other currency), or pledge Spanish government bonds against borrowing Japanese government bonds. A cross-currency repo is essentially a plain-vanilla transaction, but where collateral that is handed over is denominated in a different currency to that of the cash lent against it. Other features of cross-currency repo include:

- Possible significant daylight credit exposure on the transaction if securities cannot settle versus payment;
- A requirement for the transaction to be covered by appropriate legal documentation;
- Fluctuating foreign exchange rates, which mean that it is likely that the transaction will need to be marked-to-market frequently in order to ensure that cash or securities remain fully collateralised.

It is also necessary to take into account the fluctuations in the relevant exchange rate when marking securities used as collateral, which are obviously handed over against cash that is denominated in a different currency.

We illustrate the instrument mechanics in Example 6.1.

The repo market has allowed the hedge fund to borrow in sterling at a rate below the cost of unsecured borrowing in the money market (4.95%). The repo market maker is "overcollateralised" by the difference between the value of the bonds (in £) and the loan proceeds (2%). A rise in USD yields or a fall in the USD exchange rate value will adversely affect the value of the bonds, causing the market maker to be undercollateralised.

EXAMPLE 6.1 CROSS-CURRENCY REPO

On 4 January 2000 a hedge fund manager funds a long position in U.S. Treasury securities against sterling, for value the following day. It is offered a bid of 4.90% in the one-week, and the market maker also requires a 2% margin. The one-week Libor rate is 4.95%, and the exchange rate at the time of trade is £1/$1.63. The terms of the trade are given below.

Trade date	January 2000
Settlement date	January 2000
Stock (collateral)	U.S. Treasury 6.125% 2001
Nominal amount	$100 million
Repo rate	4.90% (sterling)
Term	7 days

(*continued*)

EXAMPLE 6.1 (*Continued*)

Maturity date	12 January 2001
Clean price	99-19
Accrued interest	5 days (0.0841346)
Dirty price	99.6778846
Gross settlement amount	$99,677,884.62
Net settlement amount (after 2% haircut)	$97,723,416.29
Net wired settlement amount in sterling	£59,953,016.13
Repo interest	£56,339.41
Sterling termination money	£60,009,355.54

Repo-to-Maturity

A *repo-to-maturity* is a classic repo where the termination date on the repo matches the maturity date of the bond in the repo. We can discuss this trade by considering the Bloomberg screen used to analyse repo-to-maturity, which is REM. The screen used to analyse a reverse repo-to-maturity is RRM.

Screen REM is used to analyse the effect of borrowing funds in repo to purchase a bond, where the bond is the collateral security. This is conventional, and we considered this earlier. In essence, the screen will compare the financing costs on the borrowed funds to the coupons received on the bond up to and including maturity. The key determining factor is the repo rate used to finance the borrowing. From Figure 6.17 we see that the screen calculates the break-even rate, which is the rate at which the financing cost equals the bond return. The screen also works out cash flows at start and termination, and the borrowed amount is labelled as the "repo principal". This is the bond total consideration. Under "outflows" we see the repo interest at the selected repo rate, labelled as "Int. Exp". Gross profit is the total inflow minus total outflow, which in our example is zero because the repo rate entered is the break-even rate. The user will enter the actual repo rate payable to calculate the total profit.

A reverse repo-to-maturity is a reverse repo with matching repo termination and bond expiry dates. This shown in Figure 6.18.

Repo-to-maturity is a low-risk trade as the financing profit on the bond position is known with certainty to the bond's maturity. For financial institutions that operate on an accruals basis rather than a mark-to-market basis, the trade can guarantee a profit and not suffer any losses in the interim while they hold the bond.

REPO MARGIN

To reduce the level of risk exposure in a repo transaction, it is common for the lender of cash to ask for a margin, which is where the market value of collateral is higher than the cash value of cash lent out in the repo. This is a form of protection should the

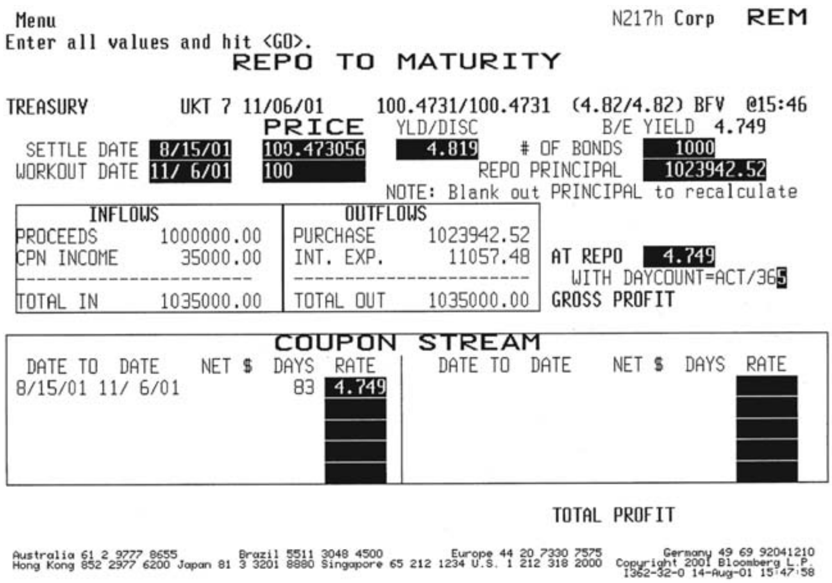

FIGURE 6.17 Bloomberg Screen REM; Used for Repo-to-Maturity Analysis, for UK Treasury 7% 2001 on 14 August 2001

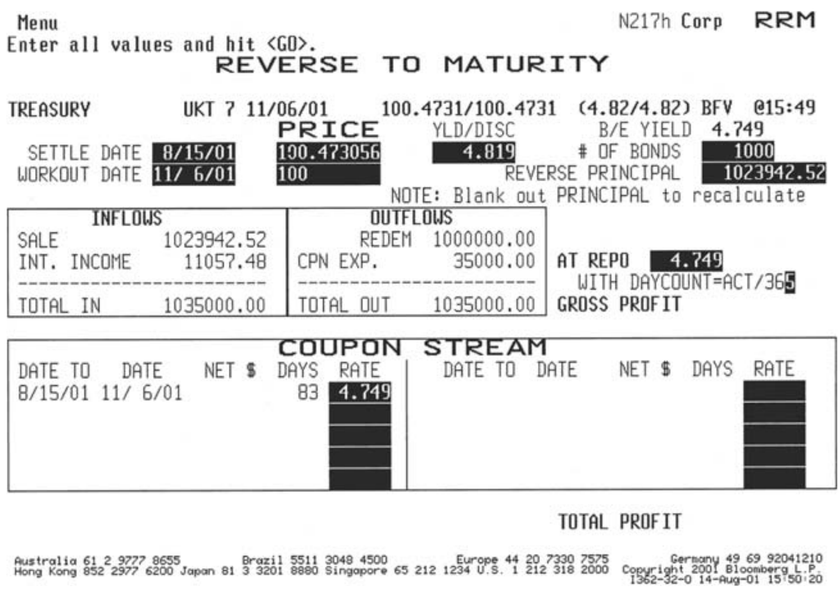

FIGURE 6.18 Bloomberg Screen RRM, Used for Reverse Repo-to-Maturity Analysis, for UK Treasury 7% 2001 on 14 August 2001

cash-borrowing counterparty default on the loan. Another term for margin is *overcollateralisation* or *haircut*. There are two types of margin: an *initial margin* taken at the start of the trade, and *variation margin*, which is called if required during the term of the trade.

Initial Margin

The cash proceeds in a repo are typically no more than the market value of the collateral. This minimises credit exposure by equating the value of the cash to that of the collateral. The market value of the collateral is calculated at its *dirty* price, not clean price—that is, including accrued interest. This is referred to as *accrual pricing*. To calculate the accrued interest on the (bond) collateral, we require the day-count basis for the particular bond.

The start proceeds of a repo can be less than the market value of the collateral by an agreed amount or percentage. This is known as the *initial margin* or *haircut*. The initial margin protects the buyer against:

- A sudden fall in the market value of the collateral;
- Illiquidity of collateral;
- Other sources of volatility of value (for example, approaching maturity);
- Counterparty risk.

The margin level of repo varies from 0–2% for collateral such as U.S. Treasuries or German Bonds, to 5% for cross-currency and equity repo, to 10–35% for emerging market debt repo.

In both classic repo and sell/buy-back, any initial margin is given to the supplier of cash in the transaction. This remains true in the case of specific repo. For initial margin the market value of the bond collateral is reduced (or given a *haircut*) by the percentage of the initial margin and the nominal value determined from this reduced amount. In a stock loan transaction, the lender of stock will ask for margin.

There are two methods for calculating the margin; for a 2% margin this could be one of the following:

1. (dirty price of the bonds) × 0.98
2. ((dirty price of the bonds)/1.02)

The two methods do not give the same value! The RRRA repo page on Bloomberg uses the second method for its calculations, and this method is turning into something of a convention.

For a 2% margin level the GMRA defines a "margin ratio" as:

$$\text{Collateral value/Cash} = 102\%$$

The size of margin required in any particular transaction is a function of the following:

- The credit quality of the counterparty supplying the collateral; for example, a central bank counterparty, interbank counterparty, and corporate will all suggest different margin levels;

- The term of the repo; an overnight repo is inherently lower risk than a one-year risk;
- The duration (price volatility) of the collateral; for example, a T-bill compared to the long bond;
- The existence or absence of a legal agreement; repo traded under a standard agreement is considered lower risk.

Certain market practitioners, including a chap who worked at JPMorgan in the United States and who once interviewed the author in 2001, and particularly those that work on bond research desks, believe that the level of margin is a function of the volatility of the collateral stock. This may be either, say, one-year historical volatility or the implied volatility given by option prices. Given a volatility level of, say, 10%, suggesting a maximum expected price movement of –10% to +10%, the margin level may be set at, say, 5% to cover expected movement in the market value of the collateral. This approach to setting initial margin is regarded as onerous by most repo traders, given the differing volatility levels of stocks within GC bands. The counterparty credit risk and terms of trade remain the most influential elements in setting margin, followed by quality of collateral.[15]

In the final analysis, margin is required to guard against market risk—the risk that the value of collateral will drop during the course of the repo. Therefore the margin call must reflect the risks prevalent in the market at the time; extremely volatile market conditions may call for large increases in initial margin.

Variation Margin

The market value of the collateral is maintained through the use of *variation margin*. So if the market value of the collateral falls, the buyer calls for extra cash or collateral. If the market value of the collateral rises, the seller calls for extra cash or collateral. In order to reduce the administrative burden, margin calls can be limited to changes in the market value of the collateral in excess of an agreed amount or percentage, which is called a *margin maintenance limit*.

The standard market documentation that exists for the three structures covered so far includes clauses that allow parties to a transaction to call for variation margin during the term of a repo. This can be in the form of extra collateral (if the value of the collateral has dropped in relation to the asset exchanged) or a return of collateral, if the value has risen. If the cash-borrowing counterparty is unable to supply more collateral where required, they will have to return a portion of the cash loan. Both parties have an interest in making and meeting margin calls, although there is no obligation. The level at which variation margin is triggered is often agreed beforehand in the legal agreement put in place between individual counterparties. Although primarily viewed as an instrument used by the supplier of cash against a fall in the value of the collateral, variation margin can of course also be called by the repo seller if the value of the collateral has risen in value.

An illustration of variation margin being applied during the term of a trade is given in Example 6.2.

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[15] In his years dealing repo, during 1992–1997 and again in 2003–2008, and managing a repo desk as part of a bank treasury function during 2008–2010, the author never once came across a repo market maker who set margin levels in line with collateral volatility levels. So much for certain bank research desks . . .

EXAMPLE 6.2 VARIATION MARGIN

Figure 6.19 shows a 60-day repo in the 5% Treasury 2004, a UK gilt, where a margin of 2% is taken. The repo rate is 5%. The start of the trade is 5 January 2000. The clean price of the gilt is 95.25.

Nominal amount	1,000,000
Principal	£952,500.00
Accrued interest (29 days)	£3,961.75
Total consideration	**£956,461.75**

The consideration is divided by 1.02, the amount of margin, to give £937,707.60. Assume that this is rounded up to the nearest pound.

Loan amount	£937,708.00
Repo interest at 5%	£8,477.91
Termination proceeds	£946,185.91

Assume that one month later there has been a catastrophic fall in the bond market and the 5% 2004 gilt is trading down at 92.75. Following this drop, the market value of the collateral is now:

Principal	£927,500
Accrued interest (59 days)	£8,082.19
Market value	£935,582.19

However, the repo desk has lent £937,708 against this security, which exceeds its market value. Under a variation margin arrangement, it can call margin from the counterparty in the form of general collateral securities or cash.

The formula used to calculate the amount required to restore the original margin of 2% is given by:

$$\text{Margin adjustment} = ((\text{original consideration} + \text{repo interest charged to date}) \times (1 + \text{initial margin})) - (\text{new all-in price} \times \text{nominal amount})$$

This therefore becomes:

$$((937{,}708 + 4{,}238.96) \times (1 + 0.02)) - (0.93558219 \times 1{,}000{,}000) = £25{,}203.71$$

The margin requirement can be taken as additional stock or cash. In practice, margin calls are made on what is known as a portfolio basis, based on the net position resulting from all repos and reverse repos in place between the two

EXAMPLE 6.2 (*Continued*)

counterparties, so that a margin delivery may be made in a general collateral stock rather than more of the original repo stock. The diagrams in Figure 6.19 show the relevant cash flows at the various dates.

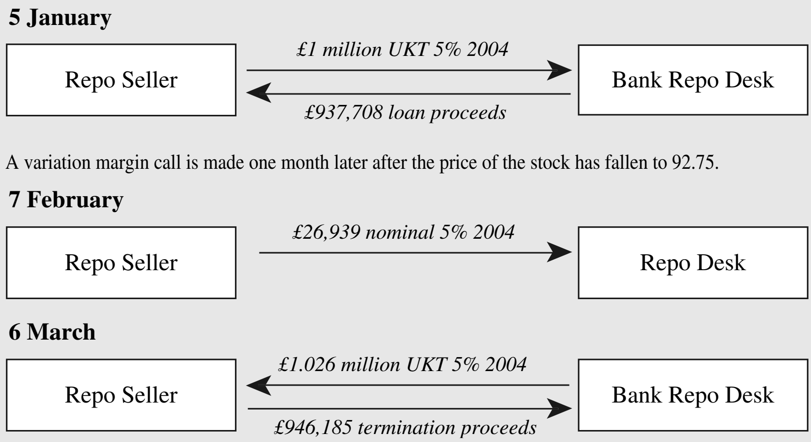

FIGURE 6.19 Margin Call

MONEY-MARKET DERIVATIVES

The market in short-term interest-rate derivatives is a large and liquid one, and the instruments involved are used for a variety of purposes. Here we review the two main contracts used in money-markets trading; the short-term interest-rate future and the forward-rate agreement. In Chapter 1, we introduced the concept of the forward rate. Money-market derivatives are priced on the basis of the forward rate and are flexible instruments for hedging against or speculating on forward interest rates. The FRA and the exchange-traded interest-rate future both date from around the same time, and although initially developed to hedge forward interest-rate exposure, they now have a range of uses. The instruments are introduced and analysed, and there is a review of the main uses that they are put to. Readers interested in the concept of convexity bias in swap and futures pricing may wish to refer to the chapter listed in the bibliography, which is by the author and is an accessible introduction.

Forward-Rate Agreements

A forward-rate agreement (FRA) is an OTC derivative instrument that trades as part of the money markets. It is essentially a forward-starting loan, but with no exchange of principal, so that only the difference in interest rates is traded—only the interest applicable on the notional amount between the rate dealt at and the actual rate prevailing at the time of settlement changes hands. That is, FRAs are off-balance-sheet (OBS) instruments. By trading today at an interest rate that is effective at some point in the future, FRAs enable banks and corporates to hedge interest-rate exposure. They are also used to speculate on the level of future interest rates.

An FRA is an agreement to borrow or lend a notional cash sum for a period of time lasting up to 12 months, starting at any point over the next 12 months, at an agreed rate of interest (the FRA rate). The "buyer" of a FRA is borrowing a notional sum of money while the "seller" is lending this cash sum. Note how this differs from all other money-market instruments. In the cash market, the party buying a CD or bill, or bidding for stock in the repo market, is the lender of funds. In the FRA market, to "buy" is to "borrow". We use the term "notional" because with an FRA no borrowing or lending of cash actually takes place, as it is an OBS product. The notional sum is simply the amount on which interest payment is calculated.

So when a FRA is traded, the buyer is borrowing (and the seller is lending) a specified notional sum at a fixed rate of interest for a specified period, the "loan" to commence at an agreed date in the future. The *buyer* is the notional borrower, and so, if there is a rise in interest rates between the date that the FRA is traded and the date that the FRA comes into effect, she will be protected. If there is a fall in interest rates, the buyer must pay the difference between the rate at which the FRA was traded and the actual rate, as a percentage of the notional sum. The buyer may be using the FRA to hedge an actual exposure, that is an actual borrowing of money, or simply speculating on a rise in interest rates. The counterparty to the transaction, the *seller* of the FRA, is the notional lender of funds, and has fixed the rate for lending funds. If there is a fall in interest rates, the seller will gain, and if there is a rise in rates, the seller will pay. Again, the seller may have an actual loan of cash to hedge or be a speculator.

In FRA trading, only the payment that arises as a result of the difference in interest rates changes hands. There is no exchange of cash at the time of the trade. The cash payment that does arise is the difference in interest rates between that at which the FRA was traded and the actual rate prevailing when the FRA matures, as a percentage of the notional amount. FRAs are traded by both banks and corporates and between banks. The FRA market is very liquid in all major currencies, and rates are readily quoted on screens by both banks and brokers. Dealing is over the telephone or over a dealing system such as Reuters.

The terminology quoting FRAs refers to the borrowing time period and the time at which the FRA comes into effect (or matures). Hence, if a buyer of an FRA wished to hedge against a rise in rates to cover a three-month loan starting in three months' time, she would transact a "three-against-six month" FRA, or more usually a 3 × 6 or 3-vs-6 FRA. This is referred to in the market as a "threes-sixes" FRA, and means a three-month loan in three months' time. So a "ones-fours" FRA (1 v 4) is a three-month loan in one month's time, and a "threes-nines" FRA (3 v 9) is six-month money in three months' time.

We illustrate the use of FRAs in hedging in Example 6.3.

EXAMPLE 6.3

A company knows that it will need to borrow £1 million in three months' time for a 12-month period. It can borrow funds today at Libor +50 basis points. Libor rates today are at 5%, but the company's treasurer expects rates to go up to about 6% over the next few weeks. So the company will be forced to borrow at higher rates unless some sort of hedge is transacted to protect the borrowing requirement. The treasurer decides to buy a 3 × 15 ("threes-fifteens") FRA to cover the 12-month period beginning three months from now. A bank quotes 5% for the FRA, which the company buys for a notional £1 million. Three months from now rates have indeed gone up to 6%, so the treasurer must borrow funds at 6½% (the Libor rate plus spread). However, she will receive a settlement amount, which will be the difference between the rate at which the FRA was bought and today's 12-month Libor rate (6%) as a percentage of £1 million, which will compensate for some of the increased borrowing costs.

In virtually every market, FRAs trade under a set of terms and conventions that are identical. The British Bankers' Association (BBA) drew up standard legal documentation to cover FRA trading. The following standard terms are used in the market.

Notional sum: The amount for which the FRA is traded.

Trade date: The date on which the FRA is dealt.

Settlement date: The date on which the notional loan or deposit of funds becomes effective; that is, is said to begin. This date is used, in conjunction with the notional sum, for calculation purposes only as no actual loan or deposit takes place.

Fixing date: This is the date on which the reference rate is determined; that is, the rate to which the FRA dealing rate is compared.

Maturity date: The date on which the notional loan or deposit expires.

Contract period: The time between the settlement date and maturity date.

FRA rate: The interest rate at which the FRA is traded.

Reference rate: This is the rate used as part of the calculation of the settlement amount, usually the Libor rate on the fixing date for the contract period in question.

Settlement sum: The amount calculated as the difference between the FRA rate and the reference rate as a percentage of the notional sum, paid by one party to the other on the settlement date.

These terms are illustrated in Figure 6.20.

The spot date is usually two business days after the trade date; however, it can by agreement be sooner or later than this. The settlement date will be the time period after the spot date referred to by the FRA terms. For example, a 1 × 4 FRA will have

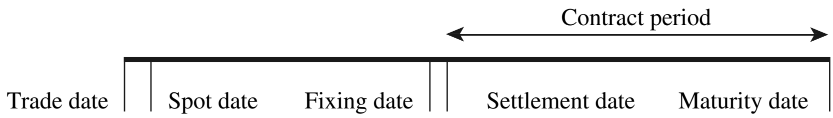

FIGURE 6.20 Key Dates in an FRA Trade

a settlement date one calendar month after the spot date. The fixing date is usually two business days before the settlement date. The settlement sum is paid on the settlement date, and as it refers to an amount over a period of time that is paid up front, at the start of the contract period, the calculated sum is discounted. This is because a normal payment of interest on a loan/deposit is paid at the end of the time period to which it relates; because an FRA makes this payment at the *start* of the relevant period, the settlement amount is a discounted figure.

With most FRA trades the reference rate is the Libor fixing on the fixing date.

The settlement sum is calculated after the fixing date, for payment on the settlement date. We may illustrate this with a hypothetical example. Consider a case where a corporate has bought £1 million notional of a 1 v 4 FRA, and dealt at 5.75%, and that the market rate is 6.50% on the fixing date. The contract period is 90 days. In the cash market, the extra interest charge that the corporate would pay is a simple interest calculation, and is:

$$\frac{6.50-5.75}{100}\times 1{,}000{,}000\times\frac{91}{365}=£1{,}869.86$$

This extra interest that the corporate is facing would be payable with the interest payment for the loan, which (as it is a money-market loan) is when the loan matures. Under an FRA, then, the settlement sum payable should, if it was paid on the same day as the cash-market interest charge, be exactly equal to this. This would make it a perfect hedge. As we noted above, though, FRA settlement value is paid at the start of the contract period; that is, the beginning of the underlying loan and not the end. Therefore, the settlement sum has to be adjusted to account for this, and the amount of the adjustment is the value of the interest that would be earned if the unadjusted cash value was invested for the contract period in the money market. The amount of the settlement value is given by (6.15).

$$\text{Settlement}=\frac{(r_{ref}-r_{FRA})\times M\times\dfrac{n}{B}}{1+(r_{ref}\times\dfrac{n}{B})} \tag{6.15}$$

where r_{ref} is the reference interest fixing rate
r_{FRA} is the FRA rate or contract rate
M is the notional value
n is the number of days in the contract period
B is the day-count base (360 or 365)

The expression in equation (6.15) simply calculates the extra interest payable in the cash market, resulting from the difference between the two interest rates, and then discounts the amount because it is payable at the start of the period and not, as would happen in the cash market, at the end of the period.

In our hypothetical illustration, as the fixing rate is higher than the dealt rate, the corporate buyer of the FRA receives the settlement sum from the seller. This then compensates the corporate for the higher borrowing costs that he would have to pay in the cash market. If the fixing rate had been lower than 5.75%, the buyer would pay the difference to the seller, because the cash market rates will mean that he is subject to a lower interest rate in the cash market. What the FRA has done is hedge the corporate, so that whatever happens in the market, it will pay 5.75% on its borrowing.

A market maker in FRAs is trading short-term interest rates. The settlement sum is the value of the FRA. The concept is exactly as with trading short-term interest-rate futures; a trader who buys an FRA is running a long position, so that if on the fixing date $r_{ref} > r_{FRA}$, the settlement sum is positive and the trader realises a profit. What has happened is that the trader, by buying the FRA, "borrowed" money at an interest rate, which subsequently rose. This is a gain, exactly like a short position in an interest-rate future, where if the price goes down (that is, interest rates go up), the trader realises a gain. Equally a "short" position in an FRA, put on by selling an FRA, realises a gain if on the fixing date $r_{ref} < r_{FRA}$.

FRA Pricing

As their name implies, FRAs are forward-rate instruments and are priced using the forward-rate principles we established earlier. Consider an investor who has two alternatives; either a six-month investment at 5% or a one-year investment at 6%. If the investor wishes to invest for six months and then roll the investment over for a further six months, what rate is required for the roll-over period such that the final return equals the 6% available from the one-year investment? If we view an FRA rate as the break-even forward rate between the two periods, we simply solve for this forward rate, and that is our approximate FRA rate. This rate is sometimes referred to as the interest rate "gap" in the money markets (not to be confused with an interbank desk's gap risk, the interest-rate exposure arising from the net maturity position of its assets and liabilities).

We can use the standard forward-rate break-even formula to solve for the required FRA rate. We established this relationship earlier when discussing the calculation of forward rates that are arbitrage-free. The relationship given in (6.16) connects simple (bullet) interest rates for periods of time of up to one year, where no compounding of interest is required. As FRAs are money-market instruments we are not required to calculate rates for periods in excess of one year,[16] where compounding would need to be built into the equation.

$$(1 + r_2 t_2) = (1 + r_1 t_1)(1 + r_f t_f) \qquad (6.16)$$

[16] Although it is, of course, possible to trade FRAs with contract periods greater than one year, for which a different pricing formula must be used.

where r_2 is the cash-market interest rate for the long period
r_1 is the cash-market interest rate for the short period
r_f is the forward rate for the gap period
t_2 is the time period from today to the end of the long period
t_1 is the time period from today to the end of the short period
t_f is the forward-gap time period, or the contract period for the FRA

This is illustrated diagrammatically in Figure 6.21.

The time period t_1 is the time from the dealing date to the FRA settlement date, while t_2 is the time from the dealing date to the FRA maturity date. The time period for the FRA (contract period) is t_2 minus t_1. We can replace the symbol "t" for time period with "n" for the actual number of days in the time periods themselves. If we do this and then rearrange the equation to solve for r_{fra} the FRA rate, we obtain (6.17).

$$r_{FRA} = \frac{r_2 n_2 - r_1 n_1}{n_{fra}\left[1 + r_1 \frac{n_1}{365}\right]} \tag{6.17}$$

where n_1 is the number of days from the dealing date or spot date to the settlement date
n_2 is the number of days from dealing date or spot date to the maturity date
r_1 is the spot rate to the settlement date
r_2 is the spot rate from the spot date to the maturity date
n_{fra} is the number of days in the FRA contract period
r_{FRA} is the FRA rate

If the formula is applied to, say, the U.S. money markets, the 365 in the equation is replaced by 360, the day-count base for that market.

In practice, FRAs are priced off the exchange-traded short-term interest-rate future for that currency, so that sterling FRAs are priced off the London International Financial Futures Exchange (LIFFE) short sterling futures. Traders normally use a spreadsheet pricing model that has futures prices directly fed into it.

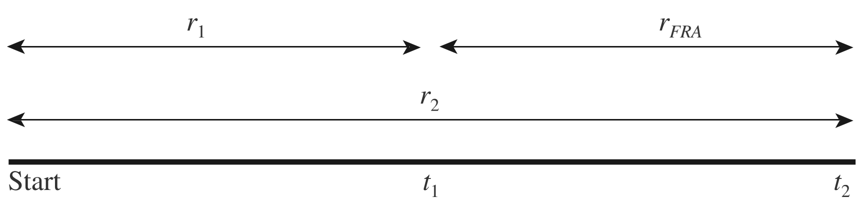

FIGURE 6.21 Rates Used in FRA Pricing

FRA positions are also usually hedged with other FRAs or short-term interest-rate futures.

FRA Prices in Practice

The dealing rates for FRAs are possibly the most liquid and transparent of any non-exchange-traded derivative instrument. This is because they are calculated directly from exchange-traded interest-rate contracts. The key consideration for FRA market makers, however, is how the rates behave in relation to other market interest rates. The forward rate calculated from two period spot rates must, as we have seen, be set such that it is arbitrage-free. If, for example, the six-month spot rate was 8.00% and the nine-month spot rate was 9.00%, the 6 v 9 FRA would have an approximate rate of 11%. What would be the effect of a change in one or both of the spot rates? The same arbitrage-free principle must apply. If there is an increase in the short-rate period, the FRA rate must decrease, to make the total return unchanged. The extent of the change in the FRA rate is a function of the ratio of the contract period to the long period. If the rate for the long period increases, the FRA rate will increase, by an amount related to the ratio between the total period to the contract period. The FRA rate for any term is generally a function of the three-month Libor rate generally, the rate traded under an interest-rate future. A general rise in this rate will see a rise in FRA rates.

SHORT-TERM INTEREST-RATE FUTURES

Description

A futures contract is a transaction that fixes the price today for a commodity that will be delivered at some point in the future. Financial futures fix the price for interest rates, bonds, equities, and so on, but trade in the same manner as commodity futures. Contracts for futures are standardised and traded on exchanges. In London, the main futures exchange is LIFFE, although commodity futures are also traded on, for example, the International Petroleum Exchange and the London Metal Exchange. The money markets trade short-term interest-rate futures, which fix the rate of interest on a notional fixed-term deposit of money (usually for 90 days or three months) for a specified period in the future. The sum is notional because no actual sum of money is deposited when buying or selling futures; the instrument is off-balance-sheet. Buying such a contract is equivalent to making a notional deposit, while selling a contract is equivalent to borrowing a notional sum.

The three-month interest-rate future is the most widely used instrument for hedging interest-rate risk.

The LIFFE exchange in London trades short-term interest-rate futures for major currencies including sterling, euros, yen, and Swiss francs. Table 6.3 summarises the terms for the short sterling contract as traded on LIFFE.

Figure 6.22 shows Bloomberg screen DES for the Eurodollar contract, traded on CBOT but actually available for trading 24 hours a day electronically across various global exchanges.

TABLE 6.3 Description of LIFFE Short Sterling Future Contract

Name	90-day sterling Libor future
Contract size	£500,000
Delivery months	March, June, September
Delivery date	First business day after last trading day
Last trading day	Third Wednesday of delivery month
Price	100 minus yield
Tick size	0.005
Tick value	£6.25
Trading hours	0825 – 1830 LIFFE Connect (electronic screen trading)

The original futures contracts related to physical commodities, which is why we speak of *delivery* when referring to the expiry of financial futures contracts. Exchange-traded futures such as those on LIFFE are set to expire every quarter during the year. The short sterling contract is a deposit of cash; so, as its price refers to the rate of interest on this deposit, the price of the contract is set as:

$$P = 100 - r$$

where P is the price of the contract and r is the rate of interest at the time of expiry implied by the futures contract. This means that if the price of the contract rises, the

FIGURE 6.22 Bloomberg Screen DES Showing Eurodollar Contract Terms

rate of interest implied goes down, and vice versa. For example, the price of the June 2004 short sterling future (written as Jun04 or M04, from the futures identity letters of H, M, U, and Z for contracts expiring in March, June, September, and December, respectively) at the start of trading when this contract become the front month on 1st March 2004 was 95.52, which implied a three-month Libor rate of 4.48% on expiry of the contract in June. If a trader bought 20 contracts at this price and then sold them just before the close of trading that day, when the price had risen to 95.55, an implied rate of 4.45%, she would have made 6 ticks profit or £1,500. That is, a 6-tick upward price movement in a long position of 20 contracts is equal to £1,500. This is calculated as follows:

$$\text{Profit} = \text{ticks gained} \times \text{tick value} \times \text{number of contracts}$$
$$\text{Loss} = \text{ticks lost} \times \text{tick value} \times \text{number of contracts}$$

The tick value for the short sterling contract is straightforward to calculate, since we know that the contract size is £500,000, there is a minimum price movement (tick movement) of 0.005% and the contract has a three-month "maturity".

$$\begin{aligned}\text{Tick value} &= 0.005\% \times £500{,}000 \times 3/12 \\ &= £6.25\end{aligned}$$

The profit made by the trader in our example is logical because if we buy short sterling futures we are depositing (notional) funds; if the price of the futures rises, it means the interest rate has fallen. We profit because we have "deposited" funds at a higher rate beforehand. If we expected sterling interest rates to rise, we would sell short sterling futures, which is equivalent to borrowing funds and locking in the loan rate at a lower level.

Note how the concept of buying and selling interest-rate futures differs from FRAs: if we buy an FRA we are borrowing notional funds, whereas if we buy a futures contract we are depositing notional funds. If a position in an interest-rate futures contract is held to expiry, cash settlement will take place on the delivery day for that contract.

Short-term interest-rate contracts in other currencies are similar to the short sterling contract and trade on exchanges such as Eurex in Frankfurt and CBOT in Chicago.

Bloomberg users can calculate the futures strip required to hedge a holding of a short-dated bond using screen TED. Figure 6.23 shows the strip required to hedge a holding on 24 March 2004 of £10 million nominal of a UK gilt, the $6^3/_4$% 2004 gilt, that matures on 26 November 2004. A total of 54 contracts are required, spread between the June, September, and December 2004 short sterling contracts.

Pricing Interest-Rate Futures

The price of a three-month interest-rate futures contract is the implied interest rate for that currency's three-month rate at the time of expiry of the contract. Therefore, there is always a close relationship and correlation between futures prices, FRA rates (which are derived from futures prices) and cash-market rates. On the day of expiry, the price of the future will be equal to the Libor rate as fixed that day. This is known

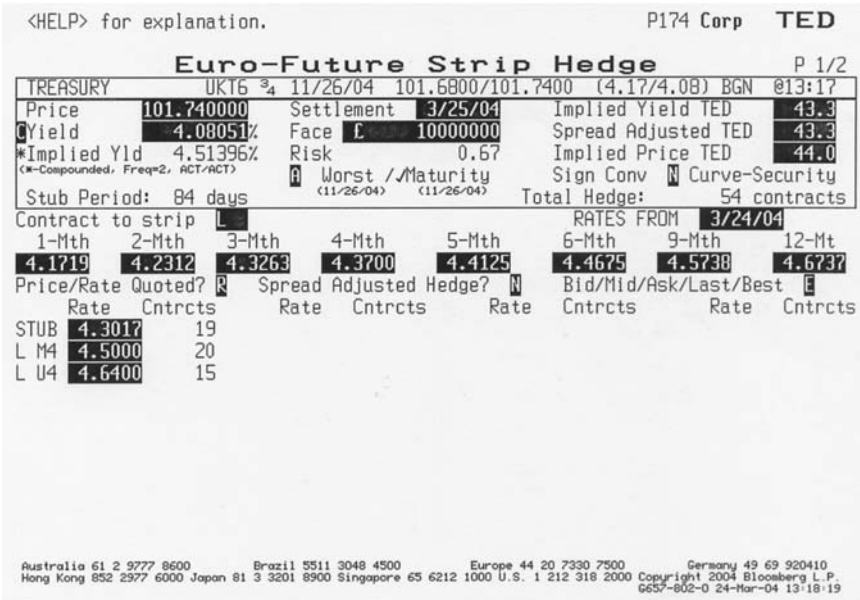

FIGURE 6.23 Bloomberg Page TED, Showing Futures Strip Hedge for £10 Million Position in UK Gilt 6.75% 2004, as of 24 March 2004

as the Exchange Delivery Settlement Price (EDSP) and is used in the calculation of the delivery amount. During the life of the contract its price will be less closely related to the actual three-month Libor rate *today,* but closely related to the *forward rate* for the time of expiry.

Equation (3.25) was our basic forward-rate formula for money-market maturity forward rates, which we adapted to use as our FRA price equation. If we incorporate some extra terminology to cover the dealing dates involved it can also be used as our futures price formula. Let us say that:

T_0 is the trade date
T_M is the contract expiry date
T_{CASH} is the value date for cash market deposits traded on T_0
T_1 is the value date for cash market deposits traded on T_M
T_2 is the maturity date for a three-month cash market deposit traded on T_M

We can then use equation (3.25) as our futures price formula to obtain P_{fut}, the futures price for a contract up to the expiry date.

$$P_{fut} = 100 - \left[\frac{r_2 n_2 - r_1 n_1}{n_f \left(1 + r_1 \frac{n_1}{365} \right)} \right] \tag{6.18}$$

where P_{fut} is the futures price
r_1 is the cash-market interest rate to T_1
r_2 is the cash-market interest rate to T_2
n_1 is the number of days from T_{CASH} to T_1
n_2 is the number of days from T_{CASH} to T_2
n_f is the number of days from T_1 to T_2

The formula uses a 365 day-count convention, which is applies in the sterling money markets; where a market uses a 360-day base this must be used in the equation instead.

In practice, the price of a contract at any one time will be close to the theoretical price that would be established by (6.18). Discrepancies will arise for supply-and-demand reasons in the market, as well as because Libor rates are often quoted only to the nearest sixteenth or 0.0625. The price between FRAs and futures are correlated very closely. In fact, banks will often price FRAs using futures, and use futures to hedge their FRA books. When hedging an FRA book with futures, the hedge is quite close to being exact, because the two prices track each other almost tick for tick. However, the tick value of a futures contract is fixed, and uses (as we saw above) a 3/12 basis, while FRA settlement values use a 360- or 365-day base. The FRA trader will be aware of this when putting on her hedge.

In our discussion of forward rates in Chapter 3, we emphasised that they were the market's view on future rates using all information available today. Of course, a futures price today is very unlikely to be in line with the actual three-month interest rate that is prevailing at the time of the contract's expiry. This explains why prices for futures and actual cash rates will differ on any particular day. Up until expiry, the futures price is the implied forward rate; of course, there is always a discrepancy between this forward rate and the cash market rate *today*. The gap between the cash price and the futures price is known as the basis. This is defined as:

$$\text{Basis} = \text{Cash price} - \text{Futures price}$$

At any point during the life of a futures contract prior to final settlement—at which point futures and cash rates converge—there is usually a difference between current cash-market rates and the rates implied by the futures price. This is the difference we've just explained. In fact, the difference between the price implied by the current three-month interbank deposit and the futures price is known as simple basis, but it is what most market participants refer to as the basis. Simple basis consists of two separate components, theoretical basis and value basis. Theoretical basis is the difference between the price implied by the current three-month interbank deposit rate and that implied by the theoretical fair futures price based on cash-market forward rates, given by (6.18). This basis may be either positive or negative, depending on the shape of the yield curve.

Futures contracts do not, in practice, provide a precise tool for locking into cash-market rates today for a transaction that takes place in the future, although this is what they are, in theory, designed to do. Futures do allow a bank to lock in a rate for a transaction to take place in the future, and this rate is the forward rate. The basis is the difference between today's cash-market rate and the forward rate on a particular date in the future. As a futures contract approaches expiry, its price and the rate in the cash market will converge (the process is given the name convergence). As we noted earlier, this is given by the EDSP, and the two prices (rates) will be exactly in line at the exact moment of expiry.

Hedging Using Interest-Rate Futures

Banks use interest-rate futures to hedge interest-rate risk exposure in cash and derivative instruments. Bond-trading desks also often use futures to hedge positions in bonds of up to two or three years' maturity, as contracts are traded up to three years' maturity. The liquidity of such "far month" contracts is considerably lower than for near-month contracts and the "front month" contract (the current contract, for the next maturity month). When hedging a bond with a maturity of, say, two years' maturity, the trader will put on a strip of futures contracts that matches as nearly as possible the expiry date of the bond. The purpose of a hedge is to protect the value of a current or anticipated cash market or OBS position from adverse changes in interest rates. The hedger will try to offset the effect of the change in interest rate on the value of his cash position with the change in value of her hedging instrument. If the hedge is an exact one, the loss on the main position should be compensated by a profit on the hedge position. If the trader is expecting a fall in interest rates and wishes to protect against such a fall, she will buy futures, known as a long hedge, and will sell futures (a short hedge) if wishing to protect against a rise in rates.

Bond traders also use three-month interest-rate contracts to hedge positions in short-dated bonds. For instance, a market maker running a short-dated bond book would find it more appropriate to hedge his book using short-dated futures rather than the longer-dated bond futures contract. When this happens, it is important to accurately calculate the correct number of contracts to use for the hedge. To construct a bond hedge, it will be necessary to use a strip of contracts, thus ensuring that the maturity date of the bond is covered by the longest-dated futures contract. The hedge is calculated by finding the sensitivity of each cash flow to changes in each of the relevant forward rates. Each cash flow is considered individually, and the hedge values are then aggregated and rounded to the nearest whole number of contracts.

Example 6.4 demonstrates the calculation of the forward rate for a money-market (sub one-year) tenor.

EXAMPLE 6.4 FORWARD RATE CALCULATION FOR MONEY-MARKET TERM

Consider two positions:

- A repo of £250 million gilts GC from 2nd January 2000 for 30 days at 6.500%,
- A reverse repo of £250 million gilts GC from 2nd January 2000 for 60 days at 6.625%.

EXAMPLE 6.4 (*Continued*)

- The two positions generate a 30-day forward 30-day interest-rate exposure (a 30- versus 60-day forward rate). This exposure can be hedged by buying an FRA that matches the gap, or by selling a strip of futures. What forward rate must be used if the trader wished to hedge this exposure, assuming no bid-offer spreads and a 365 day–count base?

The 30-day by 60-day forward rate can be calculated using the forward-rate formula in 6.19.

$$rf = \left[\left(\frac{1+(rs_2 \times \frac{L}{M})}{1+(rs_1 \times \frac{S}{M})}\right) - 1\right] \times \frac{M}{L-S} \tag{6.19}$$

where rf is the forward rate
rs_2 is the long-period rate
rs_1 is the short-period rate
L is the long-period days
S is the short-period days
M is the day-count base

Using this formula, we obtain a 30 v 60 day forward rate of 6.713560%. This assumes no bid-offer spread, which will of course apply.

This interest-rate exposure can be hedged using interest-rate futures or forward-rate agreements (FRAs). Either method is an effective hedging mechanism, although the trader must be aware of:

- Basis risk that exists between repo rates and the forward rates implied by futures and FRAs
- Date mismatch between expiry of futures contracts and the maturity dates of the repo transactions.

Therefore, an FRA will probably be preferred.

APPENDIX

APPENDIX 6.1

Currencies using money-market year base of 365 days:

- Sterling
- Hong Kong dollar

- Malaysian ringgit
- Singapore dollar
- South African rand
- Taiwan dollar
- Thai baht

In addition, the domestic markets, but not the international markets, of the following currencies also use a 365-day base:

- Australian dollar
- Canadian dollar
- Japanese yen
- New Zealand dollar

To convert an interest rate i quoted on a 365-day basis to one quoted on a 360-day basis (i^*) use the expressions given in (A6.1.1).

$$\begin{aligned} i &= i^* \times \frac{365}{360} \\ i^* &= i \times \frac{360}{365} \end{aligned} \tag{A6.1.1}$$

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CHAPTER 7

Hybrid Securities and Structured Securities

In this chapter, we describe certain exotic or structured notes that have been introduced into the fixed-income market. The motivations behind the development and use of these products are varied, but include the desire for increased yield without additional credit risk, as well as the need to alter, transform, or transfer risk exposure and risk-return profiles. Certain structured notes were also developed as hedging instruments. The instruments themselves have been issued by banks, corporate institutions, and sovereign authorities. By using certain types of notes, investors can gain access to different markets, sometimes synthetically, that were previously not available to them. For instance, by purchasing a structured note an investor can take a position that reflects her views on a particular exchange rate and anticipated changes in yield curve but in a different market. The investment instrument can be tailored to suit the investor's particular risk profile.

Note that this is only the tip of the iceberg, and many different types of bonds and structured notes are available. Indeed, if any particular investor or issuer requirement has not been met, it is often a relatively straightforward process whereby an investment bank will structure a note that meets specific requirements.

FLOATING-RATE NOTES

Floating-rate notes (FRNs) are not structured notes as such. However, we describe them here as a prelude to a discussion about other bond types.

FRNs are bonds that have variable rates of interest; the coupon rate is linked to a specified index and changes periodically as the index changes. An FRN is usually issued with a coupon that pays a fixed spread over a reference index; for example, the coupon may be 50 basis points over the six-month interbank rate. An FRN whose spread over the reference rate is not fixed is known as a variable rate note. Since the value for the reference benchmark index is not known, it is not possible to calculate the redemption yield for an FRN. Additional features have been added to FRNs, including floors (the coupon cannot fall below a specified minimum rate), caps (the coupon cannot rise above a maximum rate), and callability. There also exist perpetual FRNs. As in other markets, borrowers frequently issue paper with specific or even esoteric terms in order to meet particular requirements or meet customer demand.

Generally the reference interest rate for FRNs is the London interbank reference rate known as Libor. An FRN will pay interest at Libor plus a quoted margin (or spread). The interest rate is fixed for a three-month or six-month period and is reset in line with the Libor fixing at the end of the interest period. Hence, at the coupon reset date for a sterling FRN paying six-month Libor +0.50%, if the Libor fix is 7.6875%, then the FRN will pay a coupon of 8.1875%. Interest therefore will accrue at a daily rate of £0.0224315.

On the coupon reset date, an FRN will be priced precisely at par. Between reset dates, it will trade very close to par because of the way in which the coupon is reset. If market rates rise between reset dates, an FRN will trade slightly below par; similarly, if rates fall, the paper will trade slightly above. This makes FRNs very similar in behaviour to money-market instruments traded on a yield basis, although FRNs have much longer maturities, of course. Investors can opt to view FRNs as essentially money-market instruments or as alternatives to conventional bonds. For this reason, one can use two approaches in analysing FRNs. The first approach is known as the margin method. This calculates the difference between the return on an FRN and that on an equivalent money-market security. There are two variations on this: simple margin and discounted margin.

The simple-margin method is sometimes preferred because it does not require the forecasting of future interest rates and coupon values. Simple margin is defined as the average return on an FRN throughout its life compared with the reference interest rate. It has two components: a quoted margin either above or below the reference rate, and a capital gain or loss element, which is calculated under the assumption that the difference between the current price of the FRN and the maturity value is spread evenly over the remaining life of the bond. Simple margin is given by (7.1).

$$\text{Simple margin} = \frac{(M - P_d)}{(100 \times T) + M_q} \tag{7.1}$$

where P_d is $P + AI$, the dirty price
M is the par value
T is the number of years from settlement date to maturity
M is the quoted margin

A quoted margin that is positive reflects yield for an FRN that is offering a higher yield than the comparable money-market security.

At certain times, the simple-margin formula is adjusted to take account of any change in the reference rate since the last coupon reset date. This is done by defining an adjusted price, which is either:

$$AP_d = P_d + (re + M) \times \frac{N_{sc}}{365} \times 100 - \frac{C}{2} \times 100$$

or (7.2)

$$AP_d = P_d + (re + M) \times \frac{N_{sc}}{365} \times P_d - \frac{C}{2} \times 100$$

where AP_d is the adjusted dirty price
re is the current value of the reference interest rate (such as Libor)
$C/2$ is the next coupon payment (that is, C is the reference interest rate on the last coupon reset date plus M_q)
N_{sc} is the number of days between settlement and the next coupon date

The upper equation in (7.2) ignores the current yield effect: all payments are assumed to be received on the basis of par, and this understates the value of the coupon for FRNs trading below par and overstates the value when they are trading above par. The lower equation in (7.2) takes account of the current yield effect.

The adjusted price AP_d replaces the current price P_d in (7.1) to give an adjusted simple margin. The simple-margin method has the disadvantage of amortising the discount or premium on the FRN in a straight line over the remaining life of the bond rather than at a constantly compounded rate. The discounted-margin method uses the latter approach. The distinction between simple margin and discounted margin is exactly the same as that between simple yield to maturity and yield to maturity. The discounted-margin method does have a disadvantage in that it requires a forecast of the reference interest rate over the remaining life of the bond.

The discounted margin is the solution to (7.3), given for an FRN that pays semi-annual coupons.

$$P_d = \left\{ \frac{1}{1 + \frac{1}{2}(re + DM)^{days/year}} \right\} \bullet \left\{ \frac{C}{2} + \sum_{t=1}^{N-1} \frac{(re^* + M) \times 100/2}{[1 + \frac{1}{2}(re^* + DM)]^t} + \frac{M}{[1 + \frac{1}{2}(re^* + DM)]^{N-1}} \right\} \tag{7.3}$$

where DM is the discounted margin
re is the current value of the reference interest rate
re^* is the assumed (or forecast) value of the reference rate over the remaining life of the bond
M is the quoted margin
N is the number of coupon payments before redemption

Equation (7.3) may be stated in terms of discount factors instead of the reference rate. The yield-to-maturity spread method of evaluating FRNs is designed to allow direct comparison between FRNs and fixed-rate bonds. The yield to maturity on the FRN (*rmf*) is calculated using (7.3) with both ($re + DM$) and ($re^* + DM$) replaced with *rmf*. The yield to maturity on a reference bond (*rmb*) is calculated using (1.11). The yield-to-maturity spread is defined as:

$$\text{Yield-to-maturity spread} = rmf - rmb$$

If this is positive, the FRN offers a higher yield than the reference bond.

In addition to plain-vanilla FRNs, some of the other types of floating-rate bonds that have traded in the market include:

Collared FRNs: These offer caps and floors on an instrument, thus establishing a maximum and minimum coupon on the deal. Effectively, these securities contain two embedded options—the issuer buying a cap and selling a floor to the investor.

Step-up recovery FRNs: Where coupons are fixed against comparable longer maturity bonds, thus providing investors with the opportunity to maintain exposure to short-term assets while capitalising on a positive sloping yield curve.

INDEXED AMORTISING NOTE

Description

A type of hybrid note is the indexed amortising note or IAN. They were introduced in the U.S. domestic market in the 1990s at the demand of investors in asset-backed notes known as collateralised mortgage obligations or CMOs. IANs are fixed-coupon unsecured notes issued with a nominal value that is not fixed. That is, the nominal amount may reduce in value ahead of the legal maturity according to the levels recorded by a specified reference index such as six-month Libor. If the reference remains static or its level decreases, the IAN value will amortise in nominal value. The legal maturity of IANs is short- to medium-term, with the five-year maturity being common. The notes have been issued by banks and corporates, although a large volume has been issued by U.S. government agencies. The yield payable on IANs is typically at a premium above that of similar credit-quality conventional debt. The amortisation schedule on an IAN is linked to the movement of the specified reference index, which is easily understood. This is considered an advantage to certain mortgage-backed notes, which amortise in accordance with less clearly defined patterns such as a prepayment schedule.

An issuer of IANs will arrange a hedge that makes the funding obtained more attractive; for example, a straight Libor-type exposure. This is most commonly arranged through a swap arrangement that mirrors the note structure. A diagrammatic representation is shown in Figure 7.1. In fact, it is more common for the swap arrangement to involve a series of options on swaps. The coupon available on an IAN might be attractive to investors when the volatility on swaptions is high and there is a steep, positively sloping yield curve. Under such an environment the option-element of an IAN would confer greatest value.

The terms of a hypothetical IAN issue are given in Table 7.1.

Under the terms of issue of the note summarised in Table 7.1, the coupon payable is the current two-year government benchmark plus a fixed spread of

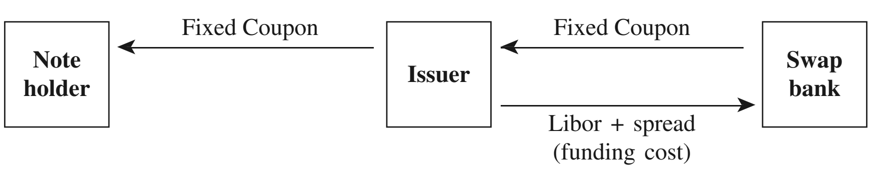

FIGURE 7.1 IAN Hedge Arrangement

TABLE 7.1 Hypothetical IAN Issue

Issuer	Mortgage Agency
Nominal value	$250,000,000
Legal maturity	6 years
Coupon	2-year Treasury plus 100 bps
Interest basis	Monthly
"Lock-out period"	Three years
Reference index	6-month Libor
6m Libor fixing on issue	5.15%
Minimum level of note	20%

Average Life Sensitivity

Libor Rate	Amortisation Rate	Average Life (years)
5.15%	100%	3
6.00%	100%	3
7.00%	21%	4.1
8.00%	7%	5.6
9.00%	0.00%	6

1%. The note has a legal maturity of six years. However, it will mature in three years if the six-month Libor rate is at a level of 6.00% or below two years from the date of issue. If the rate is above 6.00%, the maturity of the note will be extended. The lock-out of three years means that the note has a minimum life of three years, irrespective of what happens to Libor rates. Amortisation takes place if on subsequent rate-fixing dates after the lock-out period the Libor rate rises. The maximum maturity of the note is six years. If at any time there is less than 20% of the nominal value in issue, the note is cancelled in full.

Potential Benefits for Investors

The IAN structure offers advantages to investors under certain conditions. If the credit quality is acceptable, the notes offer a high yield over a relatively low term to maturity. The amortisation structure is easier to understand than that on mortgage-backed securities, which contain prepayment schedules that are based on assumptions that may not apply. This means that investors will know with certainty how the amortisation of the note will proceed, given the level of the reference index at any given time. The lock-out period of a note is usually set at a period that offers investor comfort, such as three years; during this time no amortisation can take place.

As with the other instruments described here, IANs can be tailored to meet individual investor requirements. The legal maturity and lock-out period are features that are most frequently subject to variation, with the yield premium decreasing as the lock-out period becomes closer to the formal maturity. The reference index can be a government benchmark or an interbank rate such as the swap rate. However, the most common reference is the Libor rate.

SYNTHETIC CONVERTIBLE NOTE

Description

Synthetic convertible notes are fixed-coupon securities whose total return is linked to an external source such as the level of an equity index, or the price of a specific security. The fixed-coupon element is typically at a low level, and the investor has greater exposure to the performance of the external index. A common arrangement has the note redeeming at par, but redeemable at a greater amount if the performance of the reference index exceeds a stated minimum. However, the investor has the safety net of redemption at par. Another typical structure is a zero-coupon note, issued at par and redeemable at par, but redeemable at a higher level if a specified equity index performs above a prespecified level.

Table 7.2 lists the terms of a hypothetical synthetic convertible note issue that is linked to the FTSE-100 equity index. This note will pay par on maturity, but if the level of the FTSE-100 has increased by more than 10% from the level on note issue, the note will be redeemed at par plus this amount. Note, however, that this is an investment suitable only for someone who is very bullish on the prospects for the FTSE-100. If this index does not rise by the minimum level, the investor will have received a coupon of 0.5%, which was roughly five percent age points below the level for two-year sterling at this time.

TABLE 7.2 Terms of a Synthetic Convertible Note Issue

Nominal value	£50,000,000
Term to maturity	Two years
Issue date	17-Jun-99
Maturity date	17-Jun-01
Issue price	£100
Coupon	0.50%
Interest basis	Semiannual
Redemption proceeds	Min [100, Formula level]
Formula level	100 + [100 × (R(I) – (1.1 × R(II)) / R(II)]
Index	FTSE-100
R(I)	Index level on maturity
R(II)	Index level on issue
Hedge terms	
Issuer pays	Libor
Swap bank pays	Redemption proceeds in accordance with formula

Potential Benefits for Investors

Similarly to a convertible bond, a synthetic convertible note provides investors with a fixed coupon together with additional market upside potential if the level of the reference index performs above a certain level. Unlike the convertible, however, the payoff is in the form of cash.

The reference can be virtually any publicly quoted source, and notes have been issued whose payout is linked to the exchange rate of two currencies, the days on which Libor falls within a specified range, the performance of a selected basket of stocks (say, "technology stocks"), and so on.

INTEREST-DIFFERENTIAL NOTES

Interest-differential notes (IDNs) are hybrid securities aimed at investors who wish to put on a position that reflects their view on the interest-rate differential between rates of two different currencies. Notes in the U.S. market are usually denominated in U.S. dollars, whereas Euromarket notes have been issued in a wide range of global currencies.

There are a number of variations of IDNs. Notes may pay a variable coupon and a fixed redemption amount, or a fixed coupon and a redemption amount that is determined by the level or performance of an external reference index. IDNs have also been issued with payoff profiles that are linked to the differentials in interest rates of two specified currencies, or between one currency across different maturities.

Example of IDN

Here, we discuss a five-year note that is linked to the differential between U.S. dollar-Libor and euro-libor (see Table 7.3).

The return on this note is a function of the spread between the U.S. dollar-Libor rate and euro-libor. An increase in the spread results in a higher coupon payable on the note, while a narrowing of the spread results in a lower coupon payable. Such a structure will appeal to an investor who has a particular view on the U.S.-dollar and EUR yield curves. For instance, assume that the U.S.-dollar curve is inverted and the euro curve is positively sloping. A position in an IDN (structured as above) on these two currencies allows an investor to avoid outright yield-curve plays in each currency and, instead, put on a trade that reflects a view on the relative level of interest rates in each currency. An IDN in this environment would allow an investor to earn a high yield while taking a view that is different from the market consensus.

When analysing an IDN, an investor must regard the note to be the equivalent to a fixed-coupon bond together with a double indexation of an interest-rate differential. The effect of this double indexation on the differential is to create two long positions in a five-year U.S. dollar fixed-rate note and two short positions in a EUR fixed-rate note. The short position in the EUR note means that the EUR exchange-rate risk is removed and the investor has an exposure only to the EUR interest-rate risk, which is the desired position.

The issuer of the note hedges the note with a swap structure as with other hybrid securities. The arrangement involves both U.S.-dollar and EUR interest-rate swaps. The swap bank takes the opposite position in the swaps.

TABLE 7.3 IDN Example

Term to Maturity	Five Years	
Coupon	[(2 × USD Libor) – (2 × EUR Libor) – .50%]	
Current USD Libor	6.15%	
Current EUR Libor	3.05%	
Rate differential	2.65%	
First coupon fix	5.70%	
Current five-year benchmark	4.75%	
Yield spread over benchmark	0.95%	
Change in Libor Spread at Rate Reset Spread over Benchmark (bps p.a.)		
75	4.78%	2.34%
50	3.90%	1.88%
25	3.15%	1.21%
0	2.65%	0.95%
–25	1.97%	0.56%
–50	1.32%	0.34%
–75	0.89%	0.12%
–100	0.32%	–0.28%

Table 7.3 also illustrates the return profiles possible under different interest-rate scenarios. One possibility shows that the IDN provides a 95-basis-point yield premium over the five-year government benchmark yield. However, this assumes, rather unrealistically, that the interest differential between the U.S. dollar and EUR interest rates remains constant through the final coupon-setting date. More significantly, though, we see that the yield premium available on the note increases as the spread differential between the two rates increases. In a spread-tightening environment, the note offers a premium over the government yield as long as the tightening does not exceed 100 basis points each year.

Potential Benefits for Investors

IDN-type instruments allow investors to put on positions that reflect their view on foreign interest-rate direction and/or levels, but without having to expose themselves to currency (exchange-rate) risk at the same time. The notes may also be structured in a way that allows investors to take a view on any maturity point of the yield curve. For instance, the coupon may be set in accordance with the differential between the 10-year government benchmark yields of two specified countries. As another approach, investors can arrange combinations of different maturities in the same currency, which is a straight yield curve or relative-value trade in a domestic or foreign currency.

The risk run by a noteholder is that the interest-rate differential moves in the opposite direction from the one that is sought, which reduces the coupon payable and may even result in a lower yield than what is available on the benchmark bond.

CONVERTIBLE BONDS

Convertible bonds have a long history in the capital markets, having been issued by utility companies in the United States in the nineteenth century. Today they are common in all global debt markets.

A convertible bond is a corporate debt security that gives the bondholder the right, without imposing an obligation, to convert the bond into another security under specified conditions, usually the ordinary shares of the issuing company. Thus a convertible bond provides an investor with an exposure to the underlying equity, but allied with a regular coupon payment and promise of capital repayment on maturity if no conversion takes place. From the investor's viewpoint a convertible usually presents higher value compared to the dividend stream of the equity, as well as an opportunity to share in any upside performance of the equity. The option represented by the convertibility feature carries value, for which investors are willing to pay a premium. This premium is in the form of a lower bond yield compared to a vanilla bond of equivalent liquidity and credit quality. Because of their structure, convertibles display the characteristics of both debt and equity instruments, and are often referred to as *hybrid* instruments.

In this section we provide a description of convertible bonds and an overview of issues in their valuation. We also describe a bond type that was introduced in the wake of the crash of 2008–09, the contingent convertible (CoCo) bond.

Basic Features

Convertible bonds are typically fixed-coupon securities that are issued with an option to be converted into the equity of the issuing company under specified terms and conditions. They have been issued as both senior and subordinated securities. The investors' view on the performance of the issuing company's shares is a major factor, because investors are buying into the right to subscribe for the shares at a later date and, if exercised, at a premium on the open market price prevailing at first issue. For this reason the price of a convertible bond at any time will reflect changes in the price of the underlying equity; it also reflects changes in interest rates. Convertibles are typically medium- to long-dated instruments with maturities of 5 to 10 years. The coupon on a convertible is below the level payable on the same issuer's nonconvertible bond of the same maturity, reflecting the additional value of the equity option element. This is a key reason why issuer companies favour convertibles, because through them they can raise funds at a below-market (to them) interest rate. The bonds are usually convertible into shares under a set ratio and at a specified price.

The option element in a convertible cannot be stripped out of the bond element, and so is termed an "embedded option". The valuation of the bond takes into account this embedded optionality. Note also that unlike a straight equity option, there is no additional payment to make on conversion: the holder simply exchanges the bond for the specified number of shares. One could view the price paid for exercising the option as being the loss of the "bond" element, which is the regular coupon and redemption proceeds on maturity, but this should be viewed as more of an "opportunity cost" rather than a payment. This bond element is often referred to as the "bond floor", which is the straight debt element of the convertible. The bond floor can be viewed as the level at which a vanilla bond issued by the same company would trade; that is, its yield and price. It generally accounts for between 50 and 80% of the total value.

Some convertibles are callable by the issuer, under prespecified conditions. These are known as *convertible calls* and remove one of the advantages of the straight convertible—that conversion is at the discretion of the bondholder—because by calling a bond the issuer is able to force conversion, on terms potentially unfavourable to the investor. There are two types of call option. *Hardcall* is nonconditional while *softcall* is conditional. If a bond is hardcall protected for any time after issue, then the issuer may not early redeem the bond. During softcall protection, early redemption is possible under certain conditions, normally that the underlying share price must trade above a certain level for a specific period. This level is usually around 130% of the conversion price.

Investors rarely convert voluntarily. They may do so during an event such as a call or a tender offer, or if the share price has risen by a considerable amount. The main reason why early redemption is not generally in the investor's interest is because it will erode the "time value" of the option element, as well as remove the yield advantage of holding the convertible. It also removes the downside protection afforded by the bond. That is why softcall options build in a significant equity upside element before the issuer can redeem early. Issuers will call a bond to effect conversion into shares, at which point the bondholder will convert or simply receive cash for par value. Generally, there will be value in conversion at this point so converting is not problematic. Issuers also call convertibles if there is an opportunity to reissue debt at a lower interest rate.

Investor Analysis

When evaluating convertible securities, the investor will consider the expected performance of the underlying shares, the future prospects of the company itself, and the relative attraction of the bond as a pure fixed income instrument in the event that the conversion feature proves to be worthless. He will also take into account the credit quality of the issuer, the yield give-up suffered as a result of purchasing the convertible over a conventional bond, the conversion premium ratio, and the fixed-income advantage gained over a purchase of the underlying shares in the first place.

Consider a convertible bond issued by hypothetical borrower ABC plc, which confers a right, but not the obligation, to the bondholder to convert into the underlying shares of ABC plc at a specified price during the next 10 years (see Table 7.4).

The terms and conditions under which a convertible is issued, and the terms under which it may be converted into the issuer's ordinary shares, are listed in the offer particulars or *prospectus*. The legal obligations of issuers and the rights of bondholders are stated in the *indenture* of the bond.

The ratio of exchange between the convertible bond and the ordinary shares can be stated either in terms of a *conversion price* or a *conversion ratio*. The conversion ratio is given by (7.4).

$$\text{Conversion ratio} = \text{Bond denomination}/\text{Conversion price} \tag{7.4}$$

TABLE 7.4 ABC plc 10% 2019 Convertible Hypothetical Bond Terms

Issuer	ABC plc
Coupon	10%
Maturity	December 2019
Issue size	£50,000,000
Face value	£1,000
Number of bonds	50,000
Issue price	£100
Current price	£103.50
Conversion price	£8.50
Dividend yield	3.50%

so that for the ABC plc bond it is

$$£1{,}000 / £8.50 = 117.64 \text{ shares}$$

Conversion terms for a convertible do not necessarily remain constant over time. In certain cases convertible issues will provide for increases or *step ups* in the conversion price at periodic intervals. A £1,000 denomination face value bond may be issued with a conversion price of, say, £8.50 a share for the first three years, £10 a share for the next three years, and £12 for the next five years, and so on. Under this arrangement, the bond will convert to fewer ordinary shares over time, which is a logical arrangement, given that the share price is expected to rise during this period. The conversion price is also adjusted for any corporate actions that occur after the convertibles have been issued, such as rights issues or stock dividends. For example, if there was a 2-for-1 rights issue, the conversion price would be halved. This provision protects the convertible bondholders and is known as an *antidilution* clause.

The *parity* or *intrinsic value* of a convertible refers to the value of the underlying equity, expressed as a percentage of the nominal value of the bond. Parity is given by (7.5).

$$\text{Parity} = \text{Share price} / \text{Conversion price} \tag{7.5}$$

or

$$(\text{Share price} \times \text{Conversion ratio}) / \text{Face value}$$

The bond itself may be analysed—in the first instance—as a conventional fixed-income security, so using its coupon and maturity date we may calculate a current yield (running yield) and yield-to-maturity. The *yield advantage* is the difference between the current yield and the *dividend yield* of the underlying share, given by (7.6).

$$\text{Yield advantage} = \text{Current yield} - \text{Dividend yield} \tag{7.6}$$

For the ABC plc bond the current yield of the bond is 9.66%, which results in a yield advantage of 6.16%. Equity investors also use another measure, the *break-even* value, which is given by (7.7).

$$\text{Break-even} = (\text{Bond price} - \text{Parity}) / \text{Yield advantage} \tag{7.7}$$

The conversion price is the price paid for the shares when conversion takes place.

$$\text{Conversion price} = \frac{\text{Par value of bond}}{\text{Conversion ratio}} \tag{7.8}$$

The *conversion premium* is the percentage by which the conversion price exceeds the current share price. The ABC plc bond has a conversion ratio of 117.64 (that is, 117.64 shares are received in return for the bond with a par value of £1,000) and therefore a conversion price of £8.50. If the current price of the share is £6.70, then we have:

$$\begin{aligned}\text{Percentage conversion premium} &= \frac{\text{Conversion price} - \text{Share price}}{\text{Share price}} \\ &= \frac{£8.50 - £6.70}{£6.70} \\ &= 26.87\%\end{aligned} \tag{7.9}$$

The *conversion value* of the bond is given by:

$$\text{Conversion value} = \text{Share price} \times \text{Conversion ratio} \tag{7.10}$$

This shows the current value of the shares received in exchange for the bond. As the current share price is £6.70, then the current conversion value is given by:

$$\text{Conversion value} = £6.70 \times 117.64 = £788.19$$

If the bond is trading at 103.50 (per 100), then the *percentage conversion price premium*, or the percentage by which the current bond price exceeds the current conversion value is given by (7.11).

$$\text{Percentage conversion price premium} = \frac{\text{Price of bond} - \text{Conversion value}}{\text{Conversion value}} \tag{7.11}$$

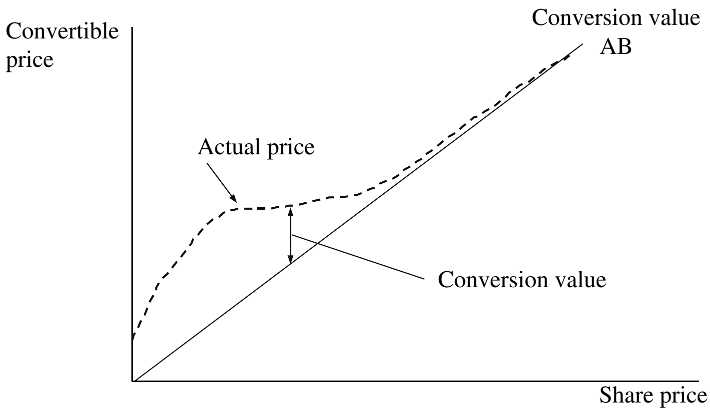

FIGURE 7.2 Convertible Bond and Conversion Premium

In our example the premium value is:

$$= \frac{1{,}035 - 788.19}{788.19} = 31.32\%$$

The premium value in a convertible is illustrated in Figure 7.2, which shows the value of the convertible bond minus the conversion feature (represented by the line AB). It is sometimes referred to as the *straight line* value and is the conventional redemption yield measure. The minimum value of a convertible bond is the higher of its straight line value and conversion value.

Investors are concerned with the point at which the ratio of the parity of the bond to the investment value moves far above the bond floor. At this point, the security trades more like equity than debt, the "equity exposure investment" we described earlier. The opposite to this is when the equity price falls to low levels, to the point at which it will need to appreciate by a very large amount before the conversion option has any value; at this point the convertible trades as a yield investment.

Fair Value of a Convertible Bond: The Binomial Model

The fair price of a convertible bond is the one that provides no opportunity for arbitrage profit, that is, it precludes a trading strategy of running simultaneous but opposite positions in the convertible and the underlying equity in order to realise a profit. Under this approach, we consider an application of the binomial model to value a convertible security. Following the usual conditions of an option-pricing model such as Black and Scholes (1973) or Cox, Ross, and Rubinstein (1979), we assume no dividend payments, no transaction costs, a risk-free interest rate, and no bid-offer spreads.

Application of the binomial model requires a binomial tree detailing the price outcomes from the start period, which is shown in Figure 7.3. In the case of a convertible bond this will refer to the prices for the underlying asset, which is the ordinary share of the issuing company.

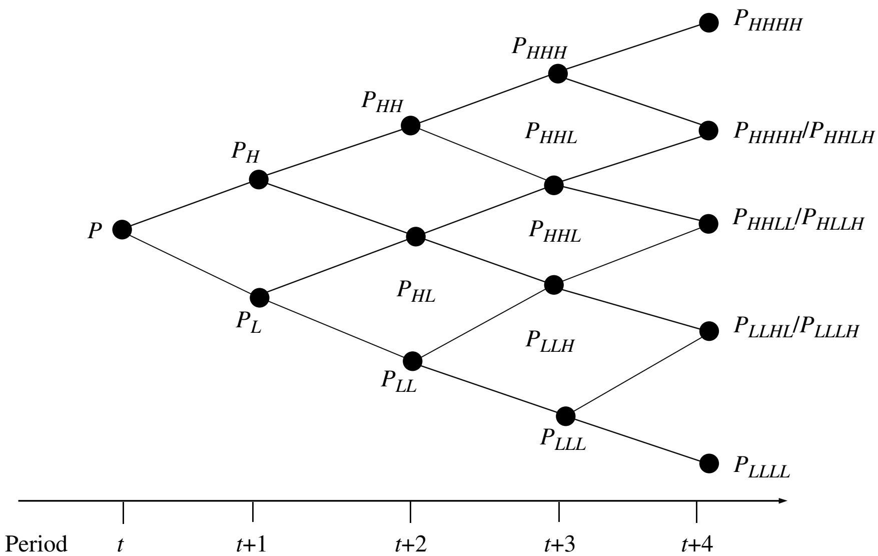

FIGURE 7.3 Underlying Equity Price Binomial Tree

If we accept that the price of the equity follows such a path, we assume that it follows a *multiplicative binomial process*. This is a geometric process that accelerates as the share price increases and decelerates as the share price falls. This assumption is key to the working of the model and has been the subject of some debate. Although it is not completely accurate, by assuming that market returns follow this pattern we are able to model a time series of share prices that is useful as a means of illustrating the principle behind binomial pricing of an embedded option.

In Figure 7.3 the current price of the underlying share is given as P, in period 1 at time t. In the next period, time $t + 1$, the share price can assume a price of P_H or P_L, with P_H higher than P and P_L lower than P. If the price is P_H in period 2, the price in period 3 can be P_{HH} or P_{HL}, and so on. The tree may be drawn for as many periods as required.

The value of a convertible bond is a function of a number of variables. For the purposes of this analysis we set parameters required as shown below.

P_{conv} is the price of the convertible bond
P_{share} is the price of the underlying equity
C is the bond coupon
r is the risk-free interest rate
N is the time to maturity
σ is the annualised share price volatility
c is the call option feature
rd is the dividend yield on the underlying share

In the first instance, we wish to calculate the value of a call option on the underlying shares of a convertible bond. For Figure 7.3 if we state that the probability of a price increase is 50%, this leaves the probability of a price decrease as $1 - p$ or 50%. If we were to construct a portfolio of δ shares, funded by borrowing X

pounds sterling, which mirrored the final payoff of the call option, we can state that the call option must be equal to the value of the portfolio, to remove any arbitrage possibilities. To solve for this, we set the following constraints:

$$\delta P_H + rX = P_H - E \tag{7.12}$$

$$\delta P_L + rX = 0 \tag{7.13}$$

where E is the strike price of the option. If we say that the exercise price is 100, then it will be higher than P_L and lower than P_H. This is illustrated by Figure 7.4.

From equation (7.13) we set:

$$rX = -\delta P_L \tag{7.14}$$

$$X = \frac{-(\delta P_L)}{r} \tag{7.15}$$

Substituting (7.15) into (7.12) gives us:

$$\begin{aligned} &\delta P_H - \delta P_L = P_H - E \\ &\Rightarrow \delta = \frac{P_H - E}{P(H-L)} \end{aligned} \tag{7.16}$$

The number of shares that would be held in the portfolio is δ, which is known as the *delta* or *hedge ratio*.

We use the relationships above to solve for the value of the borrowing component X, by substituting (7.16) into (7.13), giving us:

$$\frac{P_H - E}{P(H-L)} \cdot P_L + rX = 0$$

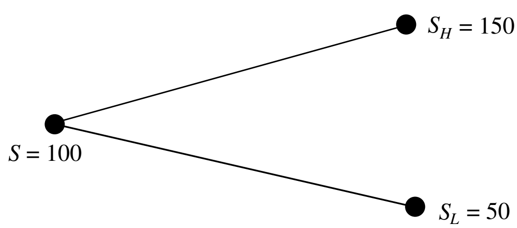

FIGURE 7.4 Binomial Price Outcome for Call Option

and rearranging for X we obtain:

$$X = \frac{-L(P_H - E)}{r(H - L)} \tag{7.17}$$

Given the relationships above, we may set the fair value of a call option as (7.18):

$$c = \delta P + r \tag{7.18}$$

Following Black-Scholes, an estimate of the extent of the increase and decrease in prices from period 1 is given by:

$$H = e^{r+\sigma} \tag{7.19}$$

$$L = e^{r-\sigma} \tag{7.20}$$

where r is the risk-free interest rate and σ is the volatility of the share price at each time period.

We are now in a position to apply this analysis to a convertible security. Table 7.5 sets the terms and parameters for a hypothetical convertible bond and underlying share price.

Using the parameters above, we may calculate the value of the underlying equity, which is priced at 100 in period 1. Note how we require a value for equity volatility and a current equity price. The volatility is assumed to be constant at 10%, and the maturity of the bond is five years or 1,825 days. The price is calculated over 10 periods, with each time period equal to 180 days. The risk-free interest rate is 4.00%, but under Black-Scholes we require the continuously compounded rate, which is calculated as:

$$r = e^{0.04} = 1.040811$$

or 4.08%.

TABLE 7.5 Hypothetical Convertible Bond Terms

Equity price	100
Conversion price	100
Coupon	5.00% semiannual
Time to maturity	5 years
Volatility	10.00%
Risk-free rate r	4.00%
Credit-risk rate	6.00%
Time step	6 months
H	Calculated below
L	Calculated below

The one time-period or 180-day equivalent rate is $(e^{0.02})-1$ or 2.0201%. This is therefore the risk-free interest rate to use. The volatility level of 10% is an annualised figure, following market convention. This may be broken down per time period as well, and this is calculated by multiplying the annual figure by the square root of the time period required. This is shown as follows:

$$\sigma \times \sqrt{t/N} = 0.10 \times \sqrt{0.5/1} = 0.070711$$

This allows the calculation of H and L, which is given as:

$$H = e^{0.020201+0.070711} = 1.0951726$$

$$L = e^{0.020201-0.070711} = 0.9507451$$

Using these parameters the price tree for the underlying equity, from time periods t_0 to t_{10} under the assumptions given, is shown in Figure 7.5. The last period is t_{10}, five years from now.

The valuation now proceeds to the conventional bond element of the convertible. The bond has a coupon of 5.00% payable semiannually and a maturity of five years. We assume a credit-risky spread of 200 basis points above the risk-free interest rate, so a discount rate of 6.00% is used to value the bond. This credit spread is a subjective measure based on the perceived credit risk of the bond issuer. On maturity the bond must be priced at par or 100, while the final coupon is worth 2.50. Therefore, the value of the bond on maturity in the final period (at t_{10}) must be 102.50. This is shown in Figure 7.6. The value of the bond at earlier nodes along the binomial tree is then calculated using straightforward discounting, using the 6% discount rate. For example, at t_9 the valuation is obtained by taking the maturity value of 102.50 and discounting at the credit-adjusted semiannual interest rate:

$$P_{conv} = (102.5/1.03) + 2.5 = 102.01$$

This process is continued all the way back to time t_0 to give a start price of 98.23 for the bond part of the convertible.

We then take the analysis further for a convertible bond plus its embedded option. Figure 7.7 shows the price tree for the conventional bond where the share price and conversion price is equal to 100 in the current time period. Note how the conventional bond element of the convertible provides a floor for its price in later periods.

The convertible price accounts for both the conventional bond element and the embedded option element. If we assume the share price in period t_9 is 97.01, then in period t_{10} the share can assume only one of two possible values, 106.25 or 92.24 (see Figure 7.5). In these cases the value of the call option c_H and c_L will be equal to the higher of the bond's conversion value or its redemption value, which is 106.25 if there is a rise in the price of the underlying or 102.50 if there is a fall in the price of the underlying. These are the range of possible final values for the bond; however, we require the current (present) value, so we discount this at the appropriate rate.

t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
100.00	109.51726	119.94031	131.35535	143.85679	157.54802	172.54228	188.96358	206.94775	226.64351	248.21377
	95.07451	104.12300	114.03266	124.88545	136.77113	149.78800	164.04372	179.65619	196.75455	215.48020
		90.39162	98.99442	108.41599	118.73422	130.03447	142.41020	155.96375	170.80723	187.06341
			85.93938	94.11846	103.07596	112.88597	123.62963	135.39579	148.28176	162.39413
				81.70644	89.48266	97.99896	107.32578	117.54026	128.72688	140.97815
					77.68199	85.07520	93.17203	102.03945	111.75082	122.38644
						73.85577	80.88482	88.58284	97.01351	106.24654
							70.21801	76.90084	84.21970	92.23511
								66.75942	73.11310	80.07146
									63.47119	69.51191
										60.34492

FIGURE 7.5 Underlying Share Price Tree

t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
98.23	98.61	98.99	99.38	99.79	100.21	100.64	101.09	101.54	102.01	102.50
	98.61	98.99	99.38	99.79	100.21	100.64	101.09	101.54	102.01	102.50
		98.99	99.38	99.79	100.21	100.64	101.09	101.54	102.01	102.50
			99.38	99.79	100.21	100.64	101.09	101.54	102.01	102.50
				99.79	100.21	100.64	101.09	101.54	102.01	102.50
					100.21	100.64	101.09	101.54	102.01	102.50
						100.64	101.09	101.54	102.01	102.50
							101.09	101.54	102.01	102.50
								101.54	102.01	102.50
									102.01	102.50
										102.50

FIGURE 7.6 Conventional Bond Price Tree

To determine the correct rate to use, consider the corresponding price of the conventional bond when the share price is 63.47 at period t_9. The price of the bond is calculated on the basis that on maturity the bond will be redeemed irrespective of what happens to the share price. Therefore, the appropriate interest rate to use when discounting a conventional bond is the credit-adjusted rate, as this is a corporate bond carrying credit risk—it is not default-risk free. However, this does not apply at a different share price; consider the corresponding conventional bond price when the underlying share price is 341.25, in the same time period. The position of a bondholder at this point is essentially long of underlying stock and also receiving a coupon. A position equivalent to a risk-free bond may be put on synthetically by holding the convertible bond

t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
110.91	115.54	125.70	145.87	152.08	160.84	173.59	188.96	206.95	226.64	248.21
	106.32	108.73	120.95	134.73	141.27	151.06	164.04	179.66	196.75	215.48
		104.50	106.12	116.40	125.40	133.28	142.41	155.96	170.81	187.06
			102.55	104.39	110.41	115.44	123.63	135.40	148.28	162.39
				101.35	103.29	105.08	107.33	117.54	128.73	140.98
					100.21	102.33	101.09	102.04	111.75	122.39
						100.64	101.09	101.75	102.01	106.25
							101.09	101.54	102.01	102.50
								101.54	102.01	102.50
									102.01	102.50
										102.50

FIGURE 7.7 Convertible Bond Price Tree

and selling short one unit of the underlying equity. In the event of default, the position is hedged; therefore in this case the correct discount rate to use is the risk-free interest rate. The rate to use at any one time period is dependent on the price of the underlying share at that time and how this affects the behaviour of the convertible. This rate will be either the risk-free rate or a credit-adjusted rate. The adjusted rate can be obtained using (7.21) and indeed all the convertible prices at period $t + 9$ are obtained using (7.21).

$$r_{adjusted} = \delta \cdot r + (1 - \delta) \cdot credit\ adjustment \tag{7.21}$$

The process is then carried out "backwards" to complete the entire price tree. At period t_0 with the share price at 100, the fair value of the convertible is seen to be 110.91.

Model Parameters

The binomial model reviewed above will calculate the fair value for a convertible where certain parameters have been specified. It is apparent that altering any of the inputs to the model will have an impact on the price calculation. We consider now the effect of changing one of these parameters.

Share Price The price of the underlying share is a key parameter of the model. A change in the value of the underlying share will result in a change in the value of the convertible; specifically, a rise in the underlying will result in a rise in the price of the convertible, and a fall in the price of the underlying will result in a fall in the price of the convertible.

The *delta* of an option instrument measures the extent of this change. A partial measure of the change in the price of an option with respect to a change in the price of the underlying equity is given by the delta, which is

$$\delta = \frac{\text{Change in note price}}{\text{Change in underlying price}} \tag{7.22}$$

For convertible bonds, delta is defined in terms of the sensitivity of the bond's price to changes in its parity. The parity is given from the value of the underlying equity price. The value of this delta can be gauged from Figure 7.8, which illustrates parity and bond floor. The delta is seen from the relationship of the parity and note price.

Figure 7.8 depicts a graph drawn from the values in Figure 7.5. It shows that when the share price is at low levels relative to the conversion price, the sensitivity of the convertible to movements in the share price is low. However, when the share price is high, this sensitivity (delta) approaches unity, so that a unit move in the price of the share is matched by a unit move in the price of the bond. In the former case, the option is said to be "out-of-the-money", while in the latter case the option is "in-the-money". Where the delta approaches unity the option is "deeply in-the-money".

The delta is defined as the first derivative of the price of an instrument with respect to the price of the underlying asset. Here we are concerned with the delta being the measure of the change in the price of the convertible bond with respect to change in the price of the underlying share. The value of delta in our illustration is given by the

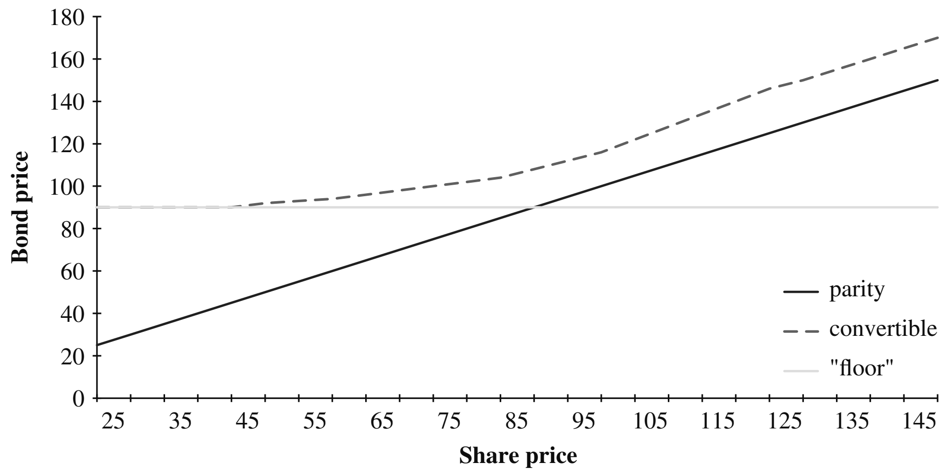

FIGURE 7.8 Convertible Bond Price Sensitivity

gradient of the convertible price line as shown in Figure 7.8. When the option feature of the convertible is deeply in-the-money, the convertible behaves more or less in the same way as the underlying share itself, but one that pays a coupon rather than a dividend.

Volatility A change in the volatility of the underlying share price will affect the price of the convertible, given that under the binomial model its value has been calculated based on a volatility level for the share. The measure of sensitivity of the convertible price to a change in the volatility of the underlying share is given by *vega*. Put simply, an increase in the volatility of the underlying has the effect that there is a greater probability of nodes on the binomial price tree being reached. This has the effect of increasing the value of the convertible.

Interest Rate A change in the risk-free interest rate will have an impact on the price of the convertible. The measure of sensitivity of a convertible bond's price to changes in the level of interest rates is given by *rho*. The key factor is that the price of deeply out-of-the-money convertibles is sensitive to changes in interest rates, because at this point the convertible behaves similarly to a conventional straight bond. Deep in-the-money convertibles behave more like the underlying equity and are little affected by changes in interest rates.

CONTINGENT CONVERTIBLE SECURITIES

Contingent convertible bonds (CoCos) are a type of convertible security that was introduced in the wake of the bank crash of 2008. They are debt instruments that will convert into the issuer's equity if a prespecified trigger point is reached in their life. This trigger point is usually a minimum level of regulatory capital. In some cases the trigger action will not convert the bond to equity, but instead write down a specified amount of the bond's face value. The trigger action takes place while the bank is still a going concern; the act of conversion is intended to restore the issuing bank's capital base to an acceptable level for regulatory purposes, and so maintain the bank as a going concern. In other words, CoCos are designed to be genuine loss-absorbing instruments and as such, CoCos are viewed by investors as essentially identical to equity, and so will price accordingly.

Basic Features

A CoCo is defined by three ingredients:

1. The loss-absorption mechanism.
2. The trigger event that effects the loss-absorption mechanism.
3. The issuer's host instrument within which the mechanism is embedded.

We consider these in turn.

Loss-Absorption Mechanism The kicking-in of the trigger at any time t will cause the CoCo to convert into the issuer's equity or otherwise have its notional amount N marked down by a specified "haircut". If converting, the bondholder will receive shares as per the *conversion ratio*. The conversion price P of a CoCo of face value N is given by

$$P = N / Conversion\ \ ratio$$

Upon conversion the loss to the bondholder L is a function of the conversion ratio and the share price S as at time t. Thus,

$$\begin{aligned} L &= N - CR \times S \\ &= N\left(1 - \frac{s}{p}\right) \\ &= N\left(1 - \prod CoCo\right) \end{aligned}$$

From the above we introduce a recovery rate value R for the CoCo (see Chapter 14). R is higher the closer the conversion price P is to the share price S at time of conversion.

Where a CoCo is written down as opposed to converted, the market has observed three different mechanisms:

1. *Full write-down:* At the trigger the bond value is wiped out completely, and the funds used to absorb the bank's losses.
2. *Partial write-down:* A specified notional amount, say 50%, is written down. In other words the investor takes a haircut on the original capital value. The bondholding is worth 50% of its original value, and the investor should, all being well, receive this amount back on maturity. The CoCo remains "performing", that is, coupons are still paid.
3. *Staggered write-down:* This is a flexible write-down mechanism, for example successive haircuts in multiples of 10%.

Trigger Event A CoCo with a capital ratio trigger is linked to the regulatory capital level of the issuing bank. Table 7.6 shows the levels for five banks that have issued CoCos, alongside their Core Equity Tier 1 (CET1) ratio as of March 2013.

TABLE 7.6 Accounting Trigger Examples for Existing CoCo Issues

	Trigger Level	Reported CET1
Barclays Bank	7.0	10.9
Credit Suisse	7.0	15.6
KBC Bank	7.0	13.8
Lloyds Bank	5.0	11.9
UBS	5.0	19.0

Source: JPMorgan Research.

For regulators, CET1 is the key indicator of a bank's economic robustness and ability to survive through the cycle. The principal guidance to the minimum level of capital required is given by Basel III, which states an absolute minimum of 4.5%. Given this, the lowest possible trigger point for a CoCo would have to be above this, and from our sample at Table 7.6 that is 5%.

Host Instrument To date, CoCos have been similar in structure to conventional convertible bonds that pay a fixed coupon. An issue from ZKB, which was callable, had a feature that converted the coupon to floating-rate should it not be called at the first call date. Another issue from the Bank of Cyprus also incorporated a conversion feature at the investor's option; in other words, in addition to the trigger event conversion, which one expects would be occurring while the equity price is falling, the investor has the option to convert to equity at set dates. This enables investors to benefit from a rising share price. The terms of this bond are summarised in Table 7.7.

TABLE 7.7 Bank of Cyprus CoCo Bond

Bank of Cyprus	
Issue date	18-May-2011
Issue size	EUR 890mln
Maturity	Perpetual
Yield at issue	6.50%
Coupon	6.50%
Call feature	Callable after five years. If not called, the bond coupon is set at 6-month Libor + 300 bps.
Coupon deferral	Yes (noncumulative)
Optional conversion price	3.3
Forced conversion price	Max[0.8 × S, 1]
Trigger	CET1 level
Trigger level	5%
Nonviability trigger	Yes

Source: Bloomberg LP.

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CHAPTER 8

Bonds with Embedded Options and Option-Adjusted Spread Analysis

In Chapter 1 we reviewed the yield-to-maturity calculation, the main measure of bond return used in the fixed-income markets. For conventional bonds, the yield calculation is relatively straightforward because the issue's redemption date is fixed. This means that the future cash flows that make up the total cash flows of the bond are known with certainty. As a result, data required to calculate the yield to maturity is known with certainty. Callable, put-able, and sinking-fund bonds, generally termed bonds with embedded options, are not as straightforward to analyse. This is because some aspect of their cash flows, such as the timing or the value of their future payments, is not certain. The term *embedded* is used because the option element cannot be separated from the bond itself. Since callable bonds have more than one possible redemption date, the collection of future cash flows contributing to their overall return is not clearly defined. If we wish to calculate the yield to maturity for such bonds, we must assume a particular redemption date and calculate the yield to this date. The market convention is to assume the first possible maturity date as the one to be used for yield calculation if the bond is priced above par, and the last possible date if the bond is priced below par. The term *yield-to-worst* is sometimes used to refer to a redemption-yield calculation made under this assumption; this is the Bloomberg term. If the actual redemption date of a bond is different from the assumed redemption date, the measurement of return will be meaningless and irrelevant.

The market therefore prefers to use other measures of bond return for callable bonds. The most common method of return calculation is something known as option-adjusted spread analysis, or OAS analysis, a good account of which can be found in Windas (1993). In this chapter we present one of the main methods by which callable bonds are priced. Although the discussion centres on callable bonds, the principles apply to all bonds with embedded-option elements in their structure.

UNDERSTANDING EMBEDDED-OPTION ELEMENTS IN A BOND

Consider a hypothetical sterling corporate bond issued by ABC plc[1] with a 6% coupon on 1st December 1999 and maturing on 1st December 2019. The bond is call-

[1] After writing this chapter, I found that there really was a company called "ABC plc"! Of course we refer here to a hypothetical corporate entity.

TABLE 8.1 Call Schedule for "ABC plc" 6% Bond Due December 2019

Date	Call Price
01-Dec-2004	103.00
01-Dec-2005	102.85
01-Dec-2006	102.65
01-Dec-2007	102.50
01-Dec-2008	102.00
01-Dec-2009	101.75
01-Dec-2010	101.25
01-Dec-2011	100.85
01-Dec-2012	100.45
01-Dec-2013	100.25
01-Dec-2014	100.00

able after five years, under the schedule shown in Table 8.1. We see that the bond is first callable at a price of 103.00, after which the call price falls progressively until December 2014, after which the bond is callable at par.

Although our example is hypothetical, this form of call provision is quite common in the corporate-debt market. In the rest of this chapter, we review the price behaviour of callable bonds. The basic case can be stated quite easily, though; in our example the ABC plc bond pays a fixed semiannual coupon of 6%. If the market level of interest rates rises after the bonds are issued, ABC plc effectively gains because it is paying below-market financing costs on its debt. If rates decline, however, investors gain from a rise in the capital value of their investment. But in this instance their upside is capped by the call provisions attached to the bond.

The difference between the value of a callable bond and that of an (otherwise identical) noncallable bond of similar credit quality is the value attached to the option element of the callable bond. This is an important relationship and one that we will consider. But first, a word on the basics of option instruments.

Basic Features of Options

An option is a contract between two parties. The buyer of an option has the right, but not the obligation, to buy or sell an underlying asset at a specified price during a specified period or at a specified time (usually the expiry date of the option contract). The price of an option is known as the premium, which is paid by the buyer to the seller or writer of the option. An option that grants the holder the right to buy the underlying asset is known as a call option; one that grants the right to sell the underlying asset is a put option. The option writer is short the contract; the buyer is long. If the owner of the option elects to exercise her option and enter into the underlying trade, the option writer is obliged to execute under the terms of the option contract. The price at which an option specifies that the underlying asset may be bought or sold is known as the exercise or strike price. The expiry date of an option is the last day on which it may be exercised. Options that can be exercised anytime from the time they are struck up to and including the expiry date are called American options. Those that can be exercised only on the expiry date are known as European options.[2]

The profit/loss payoff profiles for option buyers and sellers are quite different. The buyer of an option has her loss limited to the price of that option, while her profit can, in theory, be unlimited. The seller of an option has her profit limited to the option price, while her loss can, in theory, be unlimited, or at least potentially very substantial. The profit/loss profiles for the four main type of option positions are abundantly covered in existing literature.

The value or price of an option comprises two elements; its *intrinsic value* and its time value. The intrinsic value of an option is the value to the holder of an option if it were exercised immediately. That is, it is the difference between the strike price and the current price of the underlying asset. The holder of an option will only exercise it if there is underlying intrinsic value. For this reason, the intrinsic value is never less than zero. To illustrate, if a call option on a bond has a strike price of £100 and the underlying bond is currently trading at £103, the option has an intrinsic value of £3. An option with intrinsic value greater than zero is said to be *in-the-money.* An option where the strike price is equal to the price of the underlying is said to be *at-the-money,* while one whose strike price is above (call) or below (put) the underlying is said to be *out-of-the-money.*

The time value of an option is the difference between the intrinsic value of an option and its total value. An option with zero intrinsic value has value composed solely of time value. That is,

$$\text{Time value of an option} = \text{Option price} - \text{Intrinsic value}$$

The time value reflects the potential for an option to move into the money during its life, or move a higher level of being in-the-money, before expiry. Time value diminishes steadily for an option up to its expiry date, when it will be zero. The price of an option on expiry is composed solely of intrinsic value.

Later in this chapter, we will illustrate how the price of a bond with an embedded option is calculated by assessing the value of the "underlying" bond and the value of its associated option. The basic issues behind the price of the associated option are considered here.[3] The main factors behind the price of an option on an interest-rate instrument such as a bond are:

- The strike price of the option
- The current price of the underlying bond and its coupon rate
- The time to expiry
- The short-term risk-free rate of interest during the life of the option
- The expected volatility of interest rates during the life of the option

The effect of each of these factors will differ for call and put options and American and European options. There are a number of option-pricing models used in the market, the original of which is the Black-Scholes model. The fundamental

[2] There are also Bermudan options, and Asian options, but these needn't concern us here.

[3] For a technical review of option pricing, see the references cited in the bibliography.

principle behind the Black-Scholes model is that a synthetic option can be created and valued by taking a position in the underlying asset and borrowing or lending funds in the market at the risk-free rate of interest. It remains the basis for certain subsequent option models and is still used widely in the market.

The Call Provision

A bond with early redemption provisions essentially is a portfolio containing an underlying conventional bond, with the coupon and maturity date of the actual bond, and a put or call option on this underlying issue. Analysis therefore is interest-rate dependent; it must consider the possibility of the option being exercised when valuing the bond. The value of a bond with an option feature is the sum of the values of the individual elements; that is, the underlying bond and the option component. This is expressed in (8.1).

$$P_{bond} = P_{underlying} + P_{option} \tag{8.1}$$

Equation (8.1) states simply that the value of the actual bond is composed of the value of the underlying conventional bond together with the value of the embedded option(s). The relationship would hold for a true conventional bond, as the option-component value would be zero. For a put-able bond, an embedded put option is an attractive feature for investors as the put feature contributes to its value by acting as a floor on the bond's price. Thus the greater the value of the put, the greater the value of the actual bond. We can express this by rewriting (8.1) as (8.2).

$$P_{bond} = P_{underlying} + P_{put} \tag{8.2}$$

The equation in (8.2) states that the value of a put-able bond is equal to the sum of the values of the underlying conventional bond and the embedded put option. If any of the components of the total price were to increase in value, then so would the value of the put-able bond itself.

A callable bond is viewed as a conventional bond together with a short position in a call option, which acts as a cap on the actual bond's price. This "short" position in a call option reduces the total value of the actual bond, so we present the bond price in the form (8.3)

$$P_{bond} = P_{underlying} - P_{call} \tag{8.3}$$

Equation (8.3) states that the price of a callable bond is equal to the price of the underlying conventional bond less the price of the embedded call option. Therefore, if the value of the call option were to increase, the value of the callable bond would decrease. That is, when a bondholder of such a bond sells a call option, she receives the option price. The difference between the price of the option-free bond and the callable bond at any time is the price of the embedded-call option. The precise nature of the behaviour of the attached option element will depend on the terms of the callable bond issue. If the issuer of a callable bond is entitled to call the issue at any time after the first call date, the bondholder has effectively sold the issuer an American call option. However, the call-option price may vary with the date the option is exercised.

This occurs when the call schedule for a bond has different call prices according to which date the bond is called. The underlying bond at the time the call is exercised comprises the remaining coupon payments that would have been received by the bondholder had the issue not been called. However, for ease of explanation the market generally analyses a callable bond in terms of a long position in a conventional bond and a short position in a call option, as stated by (8.3). Note, of course, that the option is embedded; it does not trade in its own right. Nevertheless, it is clear that embedded options are important elements not only in the behaviour of a bond but in its valuation as well.

THE BINOMIAL TREE OF SHORT-TERM INTEREST RATES

In Chapter 3, we illustrated how a coupon-bond yield curve could be used to derive spot (zero-coupon) and implied forward rates. A forward rate is defined as the one-period interest rate for a term beginning at a forward date and maturing one period later. Forward rates form the basis upon which a binomial interest-rate tree is built. For an introduction to a binomial process, see Appendix 15.1 of Choudhry (2001).

An option model that used implied forward rates to generate a price for an option's underlying bond on a future date would implicitly assume that interest rates implied by the yield curve today for a date in the future would occur with certainty. Such an assumption would essentially repeat the errors associated with yield-to-worst analysis, which would be inaccurate because interest rates do not remain unchanged from a future today to a future pricing date. To avoid this inaccuracy, a binomial-tree model assumes that interest rates do not remain fixed but fluctuate over time. This is done by treating implied forward rates, sometimes referred to as short rates, as outcomes of a binomial process. In a binomial interest-rate process, we construct a binomial tree of possible short rates for each future time period.

AN INTRODUCTION TO ARBITRAGE-FREE PRICING

Consider a hypothetical situation. Assume that the short-term yield curve describes the following environment:

Six-month rate: 5.00%

One-year rate: 5.15%

Assume further that in six months' time the then six-month rate will be either 5.01% or 5.50%, and that the probability of either new rate is equal at 50% each. Our capital market is a semiannual one (the convention is for bonds to pay a semiannual coupon, as in the U.S. and UK domestic markets). We can illustrate this state as follows.

Figure 8.1 is a binomial interest-rate tree or lattice for the six-month interest rate. The tree is called "binomial" because there are precisely two possibilities for the future level of the interest rate. Using this lattice, we can calculate the tree for the prices of six-month and one-year zero-coupon bonds.

(*continued*)

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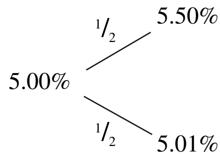

FIGURE 8.1 Two-Way Interest-Rate Projection

The six-month zero-coupon bond price today is given by 100/(1 + [0.05/2]) or 97.56098. The price tree is given in Figure 8.2.

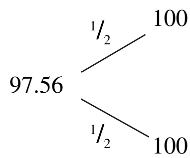

FIGURE 8.2 Initial Price Tree

Period 0 is today; period 1 is the point precisely six months from today. Given that we are dealing with a six-month zero-coupon bond, it is apparent that there is only one state of the world whatever the interest rate is in period 1; the maturity value of the bond, which is 100.

The binomial lattice for the one-year zero-coupon bond is given in Figure 8.3.

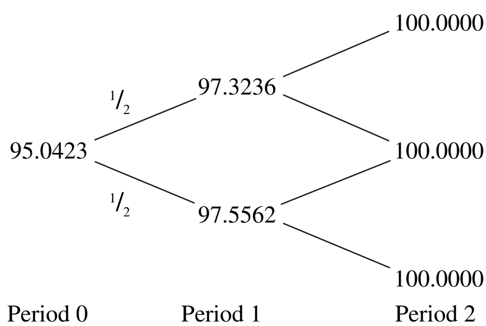

FIGURE 8.3 One-Year Binomial Lattice

At period 0, the price of the one-year zero-coupon bond is $100/(1 + [0.0515/2])^2$ or 95.0423. The price of the bond at period 1, at which point it is now a six-month piece of paper, is dependent on the six-month rate at the time, shown in the diagram. At period 2, the bond matures, and its price is 100. The model in Figure 8.3 demonstrates that the average or expected value of the price of the one-year bond at period 1 is $[(0.5 \times 97.3236) + (0.5 \times 97.5562)]$

(Continued)

or 97.4399. This is the expected price at period 1. Therefore, using this, the price at period 0 is:

$$97.4399/(1+[0.05/2])$$
$$\text{or } 95.06332$$

However, we know that the market price is 95.0423. This demonstrates a very important principle in financial economics, that markets do not price derivative instruments on the basis of their expected future value. At period 0, the one-year zero-coupon bond is a more risky investment than the shorter-dated bond; in the last six months of its life it will be worth either 97.32 or 97.55 depending on the direction of six-month rates. Investors' preference is for a bond that has a price of 97.4399 at period 1 with certainty. The price of such a bond at period 0 would be 97.4399/(1 + (0.05/2)) or 95.0633. In fact, the actual price of the one-year bond at that date, 95.0423, indicates the *risk premium* that the market places on the bond.

We can now consider the pricing of an option. What value should be given to a six-month call option maturing in six months' time (period 1) written on 100 nominal of the six-month zero-coupon bond "(zero-coupon bond)" at a strike price of 97.40? The binomial tree for this option is given in Figure 8.4. This shows that at period 1, if the six-month rate is 5.50%, the call option has no value, because the price of the bond is below the strike price. If, on the other hand, the six-month rate is at the lower level, the option has a value of 97.5562 – 97.40 or 0.1562.

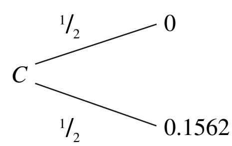

FIGURE 8.4 Binomial Path for Option Value

How do we calculate the price of the option? Option-pricing theory states that to do this one must construct a *replicating portfolio* and find the value of this portfolio. In our example, we must set up a portfolio of six-month and one-year zero-coupon bonds today that will have no value at period 1 if the six-month rate rises to 5.50%, but will have a value of 0.1562 if the rate at that time is 5.01%. If we let the value of the six-month and one-year bonds in the replicating portfolio be C_1 and C_2 respectively at period 1, we may set the following equations:

$$C_1 + 0.973236C_2 = 0 \tag{8.4}$$

$$C_1 + 0.975562C_2 = 0.1562 \tag{8.5}$$

(*continued*)

(*Continued*)

The value of the six-month zero-coupon bond in the replicating portfolio at period 1 is 100 when it matures. In the case of an interest-rate rise, the value of the one-year bond (now a six-month bond) at period 1 is 97.3236. The total value of the portfolio is given by the first expression above, which states that this value must also be equal to the value of the option. The second expression gives the value of the replicating portfolio in the event that rates decrease, when the option value is 0.1562.

Solving the expressions above gives us $C_1 = -65.3566$ and $C_2 = 67.1539$. What does this mean? Basically, to construct the replicating portfolio we purchase 67.15 of one-year zero-coupon bonds and sell short 65.36 of the six-month zero-coupon bond. However, the original intention behind the replicating portfolio was to price the option: the portfolio and the option have equal values. The value of the portfolio is a known quantity, as it is equal to the price of the six-month bond at period 0 multiplied by C_1 together with the price of the one-year bond multiplied by C_2. This is given by

$$(0.9756 \times -65.3566) + (0.950423 \times 67.1539) = 0.0627$$

That is, the price of the six-month call option is 0.06. This is the *arbitrage-free* price of the option; below this price a market participant could buy the option and simultaneously sell short the replicating portfolio and would be guaranteed a profit. If the option was quoted at a price above this, a trader could write the option and buy the portfolio. Note how the probability of the six-month rate increasing or decreasing was not part of the analysis. This reflects the arbitrage pricing logic. That is, the replicating portfolio must be equal in value to the option whatever direction interest rates move in. This means probabilities do not have an impact in the construction of the portfolio. This is not to say that probabilities do not have an impact on the option price; far from it. For example, if there is a very high probability that rates will increase (in our example), intuitively we can see that the value of an option to an investor will fall. However, this is accounted for by the market in the value of the option or callable bond at any one time. If probabilities change, the market price will change to reflect this.

Let us now turn to the concept of risk-neutral pricing. Notwithstanding what we have just noted about how the market does not price instruments using expected values, there exist risk-neutral probabilities for which the discounted expected value does give the actual price at period 0. If we let p be the risk-neutral probability of an interest-rate increase and $(1 - p)$ be the probability of a rate decrease, we may set p such that

$$\frac{97.3236p + 97.5562(1-p)}{1 + \tfrac{1}{2}0.05} = 95.0423$$

(Continued)

That is, we can calculate a value for p such that the discounted expected value, using the probability p rather than the actual probability of $^1/_2$, provides the true market price. The above expression solves to give $p = 0.5926$.

In our example from the option price tree in Figure 8.4, given the risk-neutral probability of 0.5926, we can calculate the option price to be

$$\frac{(0.5926 \times 0) + (0.4074 \times 0.1562)}{1 + \frac{1}{2}0.05} = 0.0621$$

This is virtually identical to the 0.062 option price calculated above. Put very simply, risk-neutral pricing works by first finding the probabilities that produce prices of the replicating or *underlying* security equal to the discounted expected value. An option on the security is valued by discounting this expected value under the risk-neutral probability.

We can now turn to binomial trees. In the description above we had a two-period tree. Moving to period 2, we might have Figure 8.5.

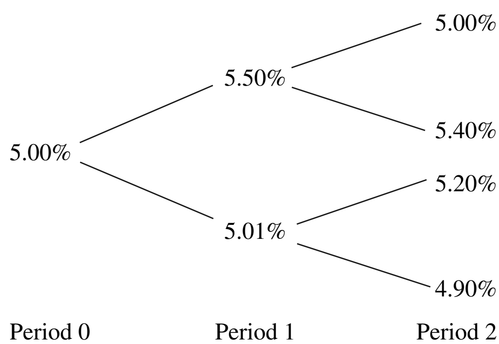

FIGURE 8.5 Two-Period Binomial Tree

This binomial tree is known as a nonrecombining tree, because each node branches out to two further nodes. This might seem more logical, and such trees are used in practice in the market. However, implementing it requires a considerable amount of computer processing power, and it is easy to see why. In period 1, there are two possible levels for the interest rate, and at period 2 there are four possible levels. After N interest periods, there will be 2^N possible values for the interest rate. If we wished to calculate the current price of a 10-year callable bond that paid semiannual coupons, we would have over one million possible values for the last period set of nodes. For a 20-year bond we would have over one trillion possible values. (Note also that in practice binomial models are not used with a six-month time step between nodes, but have much smaller time steps, further increasing the number of nodes.)

(*continued*)

(Continued)

For this reason certain market practitioners prefer to use a recombining binomial tree, where the upward-downward state has the same value as the downward-upward state. This is shown in Figure 8.6.

The number of nodes and possible values at the latest time step is much reduced in a recombining tree. For example, the number of nodes used to price a 20-year bond that was being priced with one-week time steps would be $52 \times 20 + 1$, or 1,041. Implementation is therefore more straightforward with a recombining tree.

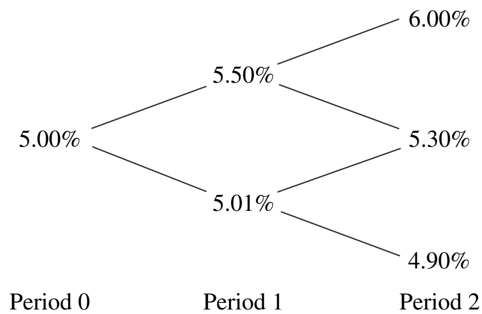

FIGURE 8.6 Recombining Binomial Tree

In the binomial tree, we model two interest rates as the possible outcomes of a previous time period, when the interest rate was known.

Pricing Callable Bonds

We can now consider a simple pricing method for callable bonds. We will assume a binomial term-structure model. It is well worth reading the box above beforehand, especially if one is not familiar with binomial models or the principle of the arbitrage-free pricing of financial instruments. Using the binomial model, we can derive a risk-neutral binomial lattice, where each lattice carries an equal probability

TABLE 8.2 Hypothetical Callable Bond Terms

Coupon Maturity	6% Three Years
Call Schedule	
Year 1	103.00
Year 1.5	102.00
Year 2	101.50
Year 2.5	101.00
Year 3	100.00

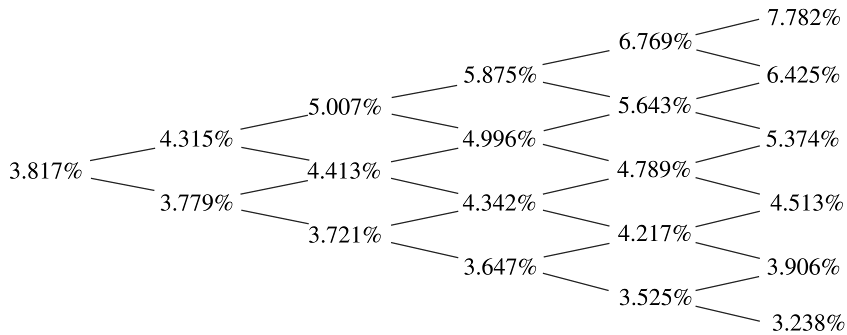

FIGURE 8.7 Interest-Rate Tree

of upward or downward moves, for the evolution of the six-month interest rate. The time step in the lattice is six months. This model is then used to price a hypothetical semiannual coupon bond as shown in Table 8.2.

The tree is shown at Figure 8.7.

In the first instance, we construct the binomial tree that describes the price process followed by the bond itself, if we ignore its call feature. This is shown in Figure 8.8. Note that the maturity value of the bond on the redemption date is given as 100.00; that is, we perform the analysis on the basis of the bond's ex-coupon value. The final cash flow would, of course, be 103.00.

We construct the tree from the final date backwards. At each of the nodes at year 3, the price of the bond will be 100.00, the (ex-coupon) par value. At year 2.5, the price of the bond at the highest yield will be that at which the yield of the bond is 7.782%. At this point, the price of the bond after six months will be 103.00 in both the "up" state and the "down" state. Following risk-neutral pricing, therefore, the price of the bond at this node is

$$P_{bond} = \frac{0.5 \times 103 + 0.5 \times 103}{1 + \dfrac{0.07782}{2}} = 99.14237$$

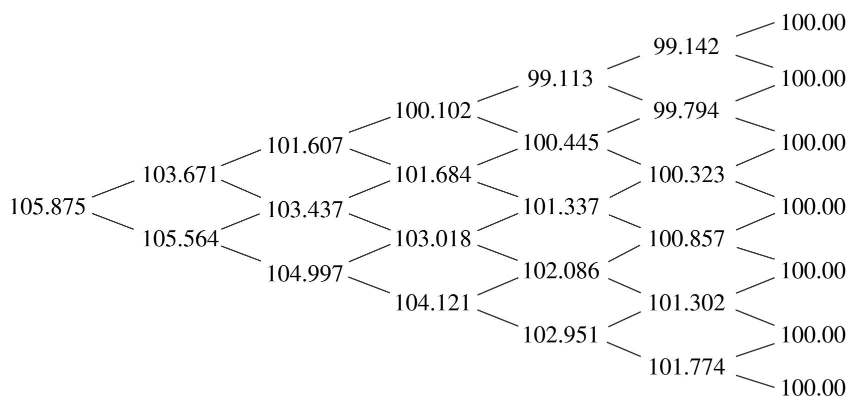

FIGURE 8.8 Bond Price Tree

The same process is used to obtain the prices for every node at year 2.5. Once all these prices have been calculated, we repeat the process for the prices at each node in year 2. At the highest yield, 6.769%, the two possible future values are

$$99.14237 + 3.0 = 102.14237 \ \ \textit{and} \ \ 99.79411 + 3.0 = 102.79411$$

Therefore, the price of the bond in this state is given by

$$P_{bond} = \frac{0.5 \times 102.14237 + 0.5 \times 102.79411}{1 + \dfrac{0.06769}{2}} = 99.14237$$

The same procedure is repeated until we have populated every node in the lattice. At each node the ex-coupon bond price is equal to the sum of the expected value and coupon, discounted at the appropriate six-month interest rate. The completed lattice is shown in Figure 8.8.

Once we have calculated the prices for the conventional element of the bond, we can calculate the value of the option element on the callable bond. This is shown in Figure 8.9. On the bond's maturity date, the option is worthless, because it is an option to call at 100, which is the price the bond is redeemed at in any case. At all other node points, a valuation analysis is called for.

The holder of the option in the case of a callable bond is the issuing company. At any time during the life of the bond, the holder will either exercise the option on the call date or elect to hold it to the next date. The option holder must consider:

- The value of holding the option for an extra period, denoted by P_{Ct}.
- The value of exercising the option straightaway, denoted P_C.

If the value of the former exceeds that of the latter, the holder will elect to not exercise, and if the value at the exercise date is higher, the holder will exercise immediately. At the year 2.5 call date, for example, there is no value in holding the option because it will be worthless at year 3. Therefore, at any point where the option is in-the-money the holder will exercise.

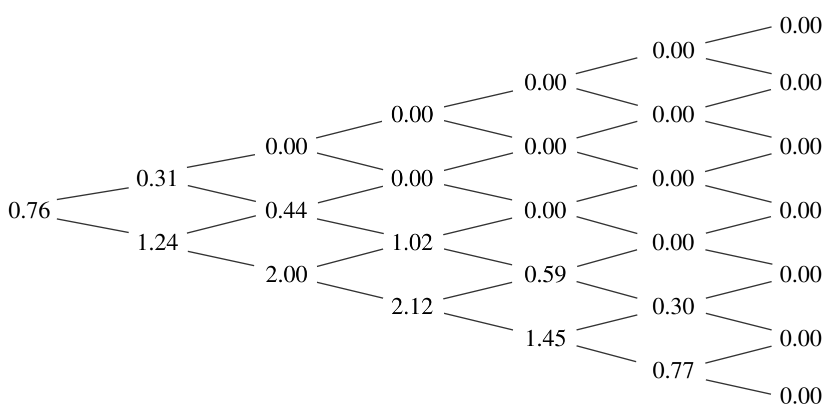

FIGURE 8.9 Embedded Option Price Tree

We can express the general valuation as follows. The value of the option for immediate exercise is V_t. The value if one is holding on to the option for a further period is V_T. Additionally, let *P* be the value of the bond at any particular node, *S* the call option price, and V_h and V_l the values of the option in the up-state and down-state, respectively. The value of the option at any specified node is *V*. The six-month interest rate at any specified node point is *r*. We have

$$V_T = \frac{0.5V_h + 0.5V_l}{1 + \frac{1}{2}r} \tag{8.6}$$

$$V_t = \max(0, P - S)$$

while the expression for *V* is $V = \max(V_t . V_t)$.

The rule is as demonstrated above; to work backwards in time and apply the expression at each node, which produces the option-value binomial lattice tree.

The general rule with an option is that they have more value "alive than dead". This means that sometimes it is optimal to run an in-the-money option rather than exercising straight away. The same is true for callable bonds. There are a number of factors that dictate whether an option should be exercised or not. The first is the asymmetric profile resulting when the price of the "underlying" asset rises; option holders gain if the price rises, but will only lose the value of their initial investment if the price falls. Therefore, it is optimal to run with the option position. There is also time value, which is lost if the option is exercised early. With callable bonds, it is often the case that the call price decreases as the bond approaches maturity. This is an incentive to delay exercise until a lower exercise price is available. The issue that may influence the decision to exercise sooner is coupon payments, as interest is earned sooner.

To return to our hypothetical example, we can now complete the price tree for the callable bond. Remember that the option in the case of a callable bond is held by the issuer, so the value of the option is subtracted from the price of the bond to obtain the actual value. We see from Figure 8.10 that the price of the callable bond today is 105.875 – 0.76 or 105.115. The price of the bond at each node in the lattice is also shown. By building a tree in this way, which can be programmed into a spreadsheet or as a front-end application, we are able to price a callable or put-able bond.

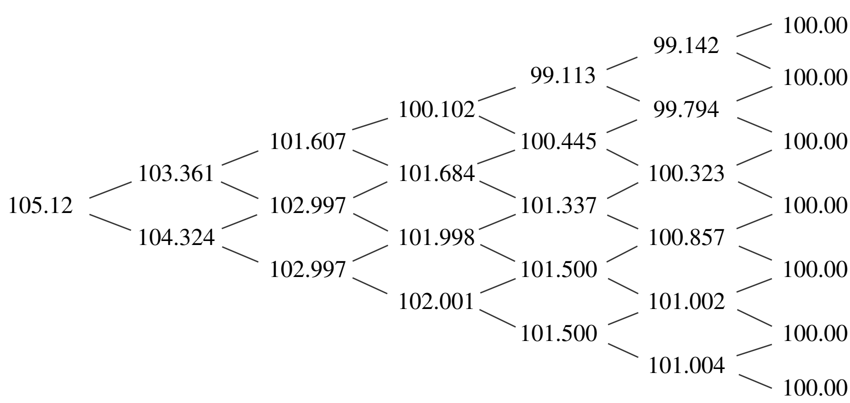

FIGURE 8.10 Final Price Tree

TABLE 8.3 Price/Yield Relationship

	Value of Price Change for	
Change in Yield	**Positive Convexity**	**Negative Convexity**
Fall of 100 bps	X%	Lower than Y%
Rise of 100 bps	Lower than X%	Y%

Price and Yield Sensitivity

As we saw in an earlier chapter, the price/yield relationship for a conventional vanilla bond is essentially convex in shape, while for a bond with an option feature attached this relationship changes as the price of the bond approaches par, at which the bond is said to exhibit negative convexity. This means that the rise in price will be lower than the fall in price for a large change in yield of a given number of basis points. We summarise the price/yield relationship for both conventional and option-feature bonds in Table 8.3.

The price/yield relationship for a callable bond exhibits negative convexity as interest rates fall. Option-adjusted spread analysis is used to highlight this relationship for changes in rates. This is done by effecting a parallel shift in the benchmark yield curve, holding the spread level constant, and then calculating the theoretical price along the nodes of the binomial price tree. The average present value then becomes the projected price for the bond. General results for a hypothetical callable bond compared to a conventional bond are shown in Figure 8.11. In our example, once the market rate falls below the 10% level, the bond exhibits negative convexity. This is because it then becomes callable at that point, which acts as an effective cap on the price of the bond.

The market analyses bonds with embedded options in terms of a yield spread, with a "cheap" bond trading at a higher yield spread and a "dear" bond trading at a lower yield spread. The usual convention is to quote yield spreads as the difference between the redemption yield of the bond being analysed and the equivalent-maturity government bond. This is not accurate because the redemption yield is, in effect, a meaningless number—there is not a single rate at which all the cash flows comprising

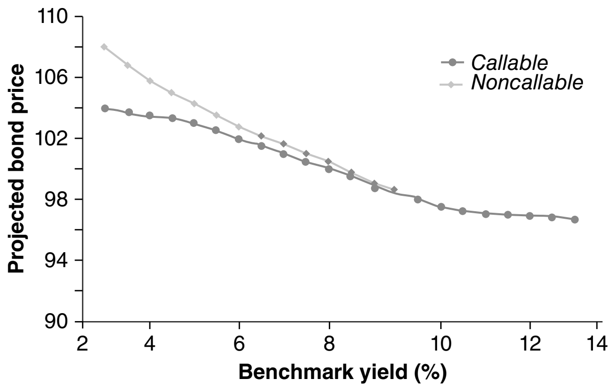

FIGURE 8.11 Projected Prices for Callable and Conventional Bonds with Identical Coupon and Final Maturity Dates

either bond should be discounted but a set of spot or forward rates that are used for each successive interest period. The correct procedure for discounting, therefore, is to determine the yield spread over the spot or forward-rate curve. With regard to the binomial tree, what we require then is the constant spread that, when added to all the short-rates on the binomial tree, makes the theoretical (model-derived) price equal to the observed market price. The constant spread that satisfies this requirement is known as the *option-adjusted spread* (OAS). The spread is referred to as an "option-adjusted" spread because it reflects the option feature attached to the bond. The OAS will depend on the volatility level assumed in running the model. For any given bond price, the higher the volatility level specified, the lower will be the OAS for a callable bond, and the higher for a put-able bond. Since the OAS is usually calculated relative to a government spot or forward-rate curve, it reflects the credit risk and any liquidity premium between the corporate bond and the government bond. Note that OAS analysis reflects the valuation model being used, and its accuracy is a reflection of the accuracy of the model itself.

Measuring Bond Yield Spreads

The binomial model evaluates the return of a bond by measuring the extent to which its return exceeds the returns determined by the risk-free short rates in the tree. The difference between these returns is expressed as a spread and may be considered the incremental return of a bond at a specified price. Determining the spread involves the following steps:

- The binomial tree is used to derive a theoretical price for the specified bond.
- The theoretical price is compared with the bond's observed market price.
- If the two prices differ, the rates in the binomial model are adjusted by a user-specified amount, which is the estimate of the spread.
- Using the adjusted rates, a new theoretical price is derived and compared with the observed price.
- The last two steps are repeated until the theoretical price matches the observed price.

This process is programmed into the normal binomial pricing process used at banks.

PRICE VOLATILITY OF BONDS WITH EMBEDDED OPTIONS

We have reviewed traditional duration and modified duration measures for bond interest-rate risk. Modified duration is essentially a predictive measure, used to describe the expected percentage change in bond price for a 1% change in yield. The measure is a snapshot in time, based on the current yield of the bond and the structure of its expected cash flows. In analysing a bond with an embedded option, the bondholder must assume a fixed maturity date, based on the current price of the bond, and calculate modified duration based on this assumed redemption date. However, under circumstances where it is not exactly certain what the final maturity is, modified duration may be calculated to the first call date and to the final maturity date. This would be of little use to bondholders in these circumstances, since it may be unclear which measure is appropriate. The problem is more acute for bonds that are continuously callable (or put-able) from the first call date up to maturity.

Effective Duration

To recap, the duration for any bond is calculated using (8.7), which assumes annualised yields.

$$D = \frac{\sum_{t=1}^{n} \frac{tC_t}{(1+rm)^t}}{P} \tag{8.7}$$

This measure can be approximated using (8.8):

$$D_{\text{approx}} = \frac{P_- - P_+}{2P_0(\Delta rm)} \tag{8.8}$$

where P_0 is the initial price of the bond
P_- is the estimated price of the bond if the yield falls by Δrm
P_+ is the estimated price of the bond if the yield rises by Δrm
Δrm is the change in the yield of the bond

The drawbacks of this measure are overcome to a certain extent when OAS analysis is used to measure the *effective duration* of a bond. Whereas traditional duration seeks to predict a bond's price changes based on a given price and assumed redemption date, effective duration is solved from actual price changes resulting from specified shifts in interest rates. Applying the analysis to a bond with an embedded option means that the new prices resulting from yield changes reflect changes in the cash flow. Effective duration may be thought of as a duration measure that recognises that yield changes may change the future cash flow of the bond. For bonds with embedded options, the difference between traditional duration and effective duration can be significant. For example, for a callable bond the effective duration is sometimes half that of its traditional duration measure. For mortgage-backed securities the difference is sometimes greater still.

To calculate effective duration using the binomial model and (8.8), we employ the following procedure:

- Calculate the OAS spread for the bond.
- Change the benchmark yield through a downward parallel shift.
- Construct an adjusted binomial tree using the new yield curve.
- Add the OAS adjustment to the short rates at each of the node points in the tree.
- Use the modified binomial tree constructed above to calculate the new value of the bond, which then becomes P_+ for use in equation (8.8).

To determine the lower price resulting from a rise in yields, we follow the same procedure but effect an upward parallel shift in the yield curve.

Effective duration for bonds that contain an embedded option is often referred to as option-adjusted spread duration. There are two advantages associated with using this measure. These are that, by incorporating the binomial tree into the analysis, the interest-rate-dependent nature of the cash flows is taken into account. This is done by holding the bond's OAS constant over the specified interest-rate shifts, in effect maintaining the credit spread demanded by the market at a constant level. This takes into account the behaviour of the embedded option as interest rates change. The second, and possibly more significant, advantage is that OAS duration is calculated on a parallel shift in the benchmark yield curve, which gives us an indication of the change in bond price with respect to changes in market interest rates rather with respect to changes in its own yield.

SINKING FUNDS

In some markets, corporate-bond issuers set up *sinking-fund* provisions. For example, consider the following hypothetical bond issue:

In the example of the ABC plc 8% 2019 bond, a proportion of the principal is paid out over a period of time. This is the formal provision. In practice, the actual payments made may differ from the formal requirements.

A sinking fund allows the bond issuer to redeem the nominal amount using one of two methods. The issuer may purchase the stipulated amount in the open market, and then deliver these bonds to the Trustee[4] for cancellation. Alternatively, the issuer may call the required amount of the bonds at par. This is in effect a partial call, similar to a callable bond for which only a fraction of the issue may be called. Generally the actual bonds called are selected randomly by certificate serial numbers. Readers will have noticed, however, that the second method by which a portion of the issue is redeemed is actually a call option, which carries value for the issuer. Therefore, the method by which the issuer chooses to fulfil its sinking-fund requirement is a function of the level of interest rates. If interest rates have risen since the bond was issued, so that the price of the bond has fallen, the issuer will meet its sinking-fund obligation by direct purchase in the open market. However, if interest rates have fallen, the issuing company will call the specified amount of bonds at par. In the hypothetical example given in Table 8.4, in effect ABC plc has 10 options embedded in the bond, each relating to £5 million nominal of the bonds. The options each have different maturities, so the first expires on 1st December 2009 and subsequent options maturing on 1st December each following year until 2018.[5] The decision to exercise the options as they fall due is made using the same binomial-tree method that we discussed earlier.

[4] Bond issuers appoint a Trustee that is responsible for looking after the interests of bondholders during the life of the issue. In some cases, the Trustee is appointed by the underwriting investment bank or the issuer's solicitors. Specialised arms of commercial and investment banks carry out the Trustee function: for example, Deutsche Bank, Bank of New York, Citibank, and others.

[5] The "options" are European options, in that they can only be exercised on the expiry date.

TABLE 8.4 Hypothetical Bond Sinking-Fund Terms

Issuer	ABC plc
Issue date	1-Dec-99
Maturity date	1-Dec-19
Nominal	£100 million
Coupon	8%
Sinking-fund provision	£5 million 1 December, 2009 to 2018

OPTION-ADJUSTED SPREAD ANALYSIS

The modified duration and convexity methods we described earlier are only suitable for use in the analysis of conventional fixed-income instruments with known fixed cash flows and maturity date. They are not satisfactory for use with bonds that contain embedded options, such as callable bonds, or instruments with unknown final redemption dates, such as mortgage-backed bonds.[6] For these and other bonds that exhibit uncertainties in their cash-flow pattern and redemption date, so-called option-adjusted measures are used. The most common of these are the option-adjusted spread (OAS) and option-adjusted duration (OAD). The techniques were developed to allow for the uncertain cash-flow structure of non-vanilla fixed income instruments, and to model the effect of the option element of such bonds.

A complete description of option-adjusted spread is outside the scope of this book; here, we present an overview of the basic concepts. An excellent introduction is Windas (1993).

Background

Option-adjusted spread analysis uses simulated interest rate paths as part of its calculation of bond yield and convexity. Therefore an OAS model is a *stochastic* model. The OAS refers to the yield spread between a callable or mortgage-backed bond and a government benchmark bond. The government bond chosen ideally will have similar coupon and duration values. Thus, the OAS is an indication of the value of the option element of the bond, as well as the premium required by investors in return for accepting the default risk of the corporate bond. When OAS is measured as a spread between two bonds of similar default risk, the yield difference between the bonds reflects the value of the option element only. This is rare, and the market convention is to measure OAS over the equivalent benchmark government bond. OAS is used in the analysis of corporate bonds that incorporate call or put provisions, as well as mortgage-backed securities with prepayment risk. For both applications the spread is calculated as the number of basis points over the yield of the government bond that would equate the price of both bonds.

[6] The term *embedded* is used because the option element of the bond cannot be stripped out and traded separately. An example is the call option inherent in a callable bond.

The essential components of the OAS technique are as follows:

- A simulation method such as Monte Carlo is used to generate sample interest rate paths, and a cash-flow pattern generated for each interest rate path;
- The value of the bond for each of the future possible rate paths is found, by discounting in the normal manner each of the bond's cash flows at the relevant interest rate (plus a spread) along the points of each path. This produces a range of values for the bond, and for a given price the OAS is the spread at which the average of the range of values is equal to the given price.

Thus OAS is a general stochastic model, with discount rates derived from the standard benchmark term structure of interest rates. This is an advantage over more traditional methods in which a single discount rate is used. The calculated spread is a spread over risk-free forward rates, accounting for both interest rate uncertainty and the price of default risk. As with any methodology, OAS has both strengths and weaknesses; however, it provides more realistic analysis than the traditional yield-to-maturity approach. It has been widely adopted by investors since its introduction in the late 1980s.

A Theoretical Framework

All bond instruments are characterised by the promise to pay a stream of future cash flows. The term structure of interest rates and associated discount function is crucial to the valuation of any debt security and underpins any valuation framework.[7] Armed with the term structure, we can value any bond, assuming it is liquid and default-free, by breaking it down into a set of cash flows and valuing each cash flow with the appropriate discount factor. Further characteristics of any bond, such as an element of default risk or embedded option, are valued incrementally over its discounted-cash-flow valuation.

Valuation under Known Interest Rate Environments We showed in Chapter 3 how forward rates can be calculated using the no-arbitrage argument. We use this basic premise to introduce the concept of OAS. Consider the spot interest rates for two interest periods:

Term (interest periods)	Spot rate
1	5%
2	6%

From Chapter 3, we can determine that the one-period interest rate starting one period from now is 7.009%. This is the implied one-period forward rate.

[7] The term structure of interest rates is the spot-rate yield curve; spot rates are viewed as identical to zero-coupon bond interest rates where there is a market of liquid zero-coupon bonds along regular maturity points. As such, a market does not exist anywhere, the spot-rate yield curve is considered a theoretical construct, which is most closely equated by the zero-coupon term structure derived from the prices of default-free liquid government bonds (see Chapter 4).

We may use the spot-rate term structure to value a default-free zero-coupon bond, so for example a two-period bond would be priced at \$89.[8] Using the forward rate, we obtain the same valuation, which is exactly what we expect.[9]

This framework can be used to value other types of bonds. Let us say we wish to calculate the price of a two-period bond that has the following cash flow stream:

Period 1	\$5
Period 2	\$105

Using the spot-rate structure above, the price of this bond is calculated to be \$98.21.[10] This would be the bond's fair value if it were liquid and default-free. Assume, however, that the bond is a corporate bond and carries an element of default risk, and is priced at \$97. What spread over the risk-free price does this indicate? We require the spread over the implied forward rate that would result in a discounted price of \$97. Using iteration, this is found to be 67.6 basis points. The calculation is:

$$\begin{aligned} P &= \frac{5}{1+(0.05+0.00676)} \\ &\quad + \frac{105}{[1+(0.05+0.00676] \times [1+(0.07009+0.00676)]} \\ &= 97.00 \end{aligned}$$

The spread of 67.6 basis points is implied by the observed market price of the bond, and is the spread over the expected path of interest rates. Another way of considering this is that it is the spread premium earned by holding the corporate bond instead of a risk-free bond with identical cash flows.

This framework can be used to evaluate relative value. For example, if the average sector spread of bonds with similar credit risk is observed to be 73 basis points, a fairer value for the example bond might be:

$$\begin{aligned} P &= \frac{5}{1+(0.05+0.0073)} \\ &\quad + \frac{105}{[1+(0.05+0.0073)] \times [1+(0.07009+0.0073)]} \\ &= 96.905 \end{aligned}$$

which would indicate that our bond is overvalued.

The approach just described is the OAS methodology in essence. However, it applies only in an environment in which the future path of interest rates is known with certainty. The spread calculated is the OAS spread under conditions of no uncertainty. We begin to appreciate, though, that this approach is preferable to the traditional one of comparing redemption yields; whereas the latter uses a single discount rate, the OAS approach uses the correct spot rate for each period's cash flow.

[8] $\$100/(1.062)^2 = \$88.9996$.

[9] $\$100/(1.05 \times 1.07009) = \$89$.

[10] $\$5/1.05 + \$105/[(1.05) \times 1.070091] = \$98.21$.

However, our interest lies with conditions of interest rate uncertainty. In practice, the future path of interest rates is not known with certainty. The range of possible values of future interest rates is a large one, although the probability of higher or lower rates that are very far away from current rates is low. For this reason the OAS calculation is based on the most likely future interest rate path among the universe of possible rate paths. This is less relevant for vanilla bonds, but for securities whose future cash flow is contingent on the level of future interest rates, such as mortgage-backed bonds, it is very important. The first step, then, is to describe the interest rate process in terms that capture the character of its dynamics.[11]

Ideally the analytical framework under conditions of uncertainty would retain the arbitrage-free character of our earlier discussion. But by definition the evolution of interest rates will not match the calculated forward rate, thus creating arbitrage conditions. This is not surprising. For instance, if the calculated one-period forward rate shown above actually *turns out to be* 7.50%, the maturity value of the bond at the end of period 2 would be \$100.459 rather than \$100. This means that there was an arbitrage opportunity at the original price of \$89. For the bond price to have been arbitrage-free at the start of period 1, it would have been priced at \$88.59.[12]

This is a meaningless argument, on a par with advocating the employment of a clairvoyant. However, it illustrates how different eventual interest rate paths correspond to different initial fair values. As interest rates can follow a large number of different paths of varying possibility, so bond prices can assume a number of fair values. The ultimate arbitrage-free price in real terms on maturity is as unknown as future interest rate levels! Let us look then at how the OAS methodology uses the most likely interest rate path when calculating fair value.

Valuation under Uncertain Interest Rate Environments We begin by assuming that the current spot-rate term structure is consistent with bond prices of zero option-adjusted spread, so that their price is the average expected for all possible evolutions of the future interest rate. By assuming this, we may state that the most likely future interest rate path, which lies in the centre of the range of all possible interest rate paths, will be a function of interest rate volatility. Here we use *volatility* to mean the average annual percentage deviation of interest rates around their mean value. So an environment of 0% volatility would be one of interest rate certainty and would generate only one possible arbitrage-free bond price. This is a worthwhile scenario, since it enables us to generalise the example in the previous section as the arbitrage-free model in times of uncertainty, but when volatility is 0%.

A rise in volatility generates a range of possible future paths around the expected path. The actual expected path that corresponds to a zero-coupon bond price incorporating zero OAS is a function of the dispersion of the range of alternative paths around it. This dispersion is the result of the dynamics of the interest rate process,

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[11] This is interest rate modelling as introduced in Chapter 4.

[12] The price of the bond at the start of period is \$89, shown earlier. This is worth (89 × 1.05) or \$93.45 at the end of period 1, and hence \$100 at the end of period 2 (93.45 × 1.07009). However if an investor at the time of rolling over the one-period bond can actually invest at 7.50%, this amount will mature to (83.45 × 1.075) or \$100.459. In this situation, *in hindsight* the arbitrage-free price of the bond at the start of period 1 would be 100/(1.05 × 1.075) or \$88.59.

so this process must be specified for the current term structure. We can illustrate this with a simple binomial model example. Consider again the spot-rate structure shown early on. Assume that there are only two possible future interest-rate scenarios, outcome 1 and outcome 2, both of equal probability. The dynamics of the short-term interest rate are described by a constant drift rate *a*, together with a volatility rate σ. These two parameters describe the evolution of the short-term interest rate. If outcome 1 occurs, the one-period interest rate one period from now will be:

$$5\% \times \exp(a + \sigma)$$

while if outcome 2 occurs, the period one rate will become:

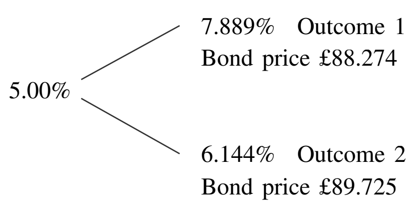
$$5\% \times \exp(a - \sigma)$$

In Figure 8.12 we present the possible interest rate paths under conditions of 0% and 25% volatility levels, and maintain our assumption that the current spot-rate structure price for a risk-free zero-coupon bond is identical to the price generated using the structure to obtain a zero option adjusted spread. To maintain the no-arbitrage condition, we know that the price of the bond at the start of period 1 must be \$89, so we calculate the implied drift rate by an iterative process. This is shown in Figure 8.12.

At 0% volatility, the prices generated by the up and down moves are equal, because the future interest rates are equal. Hence the forward rate is the same as before: 7.009%. When there is a multiple interest rate path scenario, the fair value of the bond is determined as the average of the discounted values for each rate path. Under conditions of certainty (0% volatility), the price of the bond is, not surprisingly, unchanged at both paths. The average of these is obviously \$89. Under 25% volatility, the up-move interest rate is 7.889%, and the down-move rate is 6.144%. The average of these rates is 7.017%. We can check the values by calculating the value of the bond at each outcome (or "node") and then obtaining the average of these values; this is shown to be \$89. Beginners can view the simple spreadsheet used to calculate the rates in Appendix 8.1, with the iterative process undertaken using the Microsoft Excel Goal Seek function.

25% volatility
Implied drift rate 20.61%

Expected future rate under zero volatility is 7.017%



Probability of up-move and down-move is identical at 50%.

FIGURE 8.12 Expected Interest Rate Paths under Conditions of Uncertainty

Term (interest periods)	Spot Rate
1	5.00%
2	6.00%
1v1 fwd rate	7.009%

0% *volatility*

Implied drift rate 33.776%

Expected future rate under zero volatility is 7.009%

$$5.00\% \begin{cases} \diagup \; 7.009\% \text{ Outcome 1} \\ \diagdown \; 7.009\% \text{ Outcome 2} \end{cases}$$

Probability of up-move and down-move is identical at 50%

25% *volatility*

Implied drift rate 20.61%

Expected future rate under zero volatility is 7.017%

$$5.00\% \begin{cases} \diagup \; 7.889\% \text{ Outcome 1, Bond price £88.274} \\ \diagdown \; 6.144\% \text{ Outcome 2, Bond price £89.725} \end{cases}$$

Probability of up-move and down-move is identical at 50%.

We now consider the corporate bond with \$5 and \$105 cash flows at the end of periods 1 and 2 respectively. In an environment of certainty, the bond price of \$97 implied an OAS of 67.6 basis points (bps). In the uncertain environment, we can use the same process as above to determine the spread implied by the same price. The process involves discounting the cash flows across each path with the spread added, to determine the price at each node. The price of the bond is the average of all the resulting prices; this is then compared to the observed market price (or required price). If the calculated price is lower than the market price, then a higher spread is required, and if the calculated price is higher than the market price, then the spread is too high and must be lowered.

Applying this approach to the model in Figure 8.12, under the 0% volatility the spread implied by the price of \$97 is, unsurprisingly, 67.6 bps. In the 25% volatility environment, however, this spread results in a price of \$97.296, which is higher than the observed price. This suggests the spread is too low. By iteration, we find that the spread that generates a price of \$97 is 89.76 bps, which is the bond's option-adjusted spread. This is shown below.

Outcome 1:

$$\begin{aligned} P &= \frac{5}{1+(0.05+0.00897)} \\ &+ \frac{105}{+[1+(0.05\ +0.00897)\times[1+(0.07887+0.00897)]} \\ &= £95.87 \end{aligned}$$

Outcome 2:

$$
\begin{aligned}
P &= \frac{5}{1+(0.05+0.00897)} \\
&\quad + \frac{105}{+[1+(0.05+0.00897)\times[1+(0.05244+0.00897)]} \\
&= £98.14
\end{aligned}
$$

The calculated price is the average of these two values and is [(95.87 + 98.13)/2] or $97 as required. The OAS of 89.76 bps in the binomial model is a measure of the value attached to the option element of the bond at 25% volatility.

The final part of this discussion introduces the value of the embedded option in a bond. Our example bond from earlier is now semiannually paying and carries a coupon value of 7%. It has a redemption value of $101.75. Assume that the bond is callable at the end of period 1, and that it is advantageous for the issuer to call the bond at this point if interest rates fall below 7%. We assume further that the bond is trading at the fair value implied by the discounting calculation earlier. With a principal nominal amount of $101.75, this suggests a market price of (101.75/97.00) or $104.89. We require the OAS implied by this price now that there is an embedded option element in the bond. Under conditions of 0% volatility the value of the call is zero, as the option is out-of-the-money when interest rates are above 7%. In these circumstances the bond behaves exactly as before, and the OAS remains 67.6 bps. However, in the 25% volatility environment it becomes advantageous to the issuer to call the bond in the down-state environment, as rates are below 7%. In fact, we can calculate that the spread over the interest rate paths that would produce an average price of 104.89 is 4 bps, which means that the option carries a cost to the bondholder of (67.6 – 4) or 63.6 bps. We illustrate this property in general terms in Figure 8.13. A conventional bond has a convex price/yield profile, but the introduction of a call feature limits the upside price performance of a bond, since there is a greater chance of its being called as market yields fall.

The Methodology in Practice

In practice, the forward-rate term structure is extracted using regression methods from the price of default-free government coupon bonds. Generally, OAS models used in the market are constructed so that they generate government prices that are identical to the prices observed in the market, because they assume that government bonds are fair value. This results in an implied forward-rate yield curve that is the "expected" path of future interest rates around which other rate possibilities are dispersed. Under this assumption then, an OAS value is a measure of the return over the government yield that an investor can expect to achieve by holding the option-embedded bond that is being analysed. Banks generally employ a simulation model such as Monte Carlo to generate paths of possible interest rate scenarios. This is a series of computer-generated random numbers that are used to derive interest rate

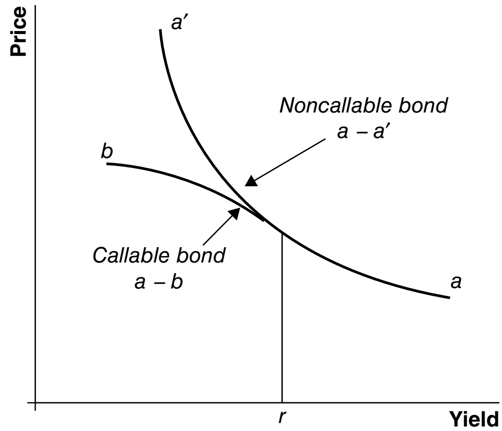


FIGURE 8.13 Impact of a Call Option on the Price/Yield (convexity) Profile of a Corporate Bond
Reproduced with permission from Frank J. Fabozzi, *Fixed Income Mathematics* (McGraw-Hill, 1997).

paths. To generate the paths, the simulation model runs using the two parameters introduced earlier: the deterministic drift term and the volatility term. Interest rates are assumed to be lognormally distributed.

When conducting relative value analysis, we strip a security down into the constituent cash flows relevant to each interest rate path. This is straightforward for vanilla bonds; however, for bonds such as mortgage-backed securities the cash flows are determined after assuming a level of prepayment. The results generated from the prepayment model are themselves based on interest rate scenarios. The OAS is then calculated by discounting the cash flows corresponding to each node on the interest-rate tree, and is the spread that equates the calculated average price to the observed market price. In practice, the model is run the other way: assuming a fair value spread over government bonds, the fair value price of a bond is the average price that is obtained by discounting all the cash flows at each relevant interest rate, together with the fair value spread.

OAS Analysis for a Hypothetical Corporate Callable Bond and Treasury Bond

We conclude this chapter with an illustration of the OAS technique. Consider a five-year semiannual corporate bond with a coupon of 8%. The bond incorporates a call feature that allows the issuer to call it after two years, and is currently priced at $104.25. This is equivalent to a yield to maturity of 6.979%. We wish to measure the value of the call feature to the issuer, and we can do this using the OAS technique. Assume that a five-year Treasury security also exists with a coupon of 8%, and is priced at $109.11, a yield of 5.797%. The higher yield of the corporate bond reflects the market-required premium, due to the corporate bond's default risk and call feature.

The valuation of both securities is shown in Table 8.5.

TABLE 8.5 OAS Analysis for Corporate Callable Bond and Treasury Bond

Bond	**Price**	**Yield**
Corporate bond	104.25	6.979%
Treasury 8% 2006	109.11	5.797%
OAS Spread	110.81 bps	

Period	YTM	Date	Spot Rate	Discount Factor	Cash Flow	Present Value	OAS-Adjusted Spot Rate	PV of OAS-Adjusted Cash Flows
0	5.000	25-Feb-2001	5.000	1				
1	5.000	27-Aug-2001	5.069	0.97521333	4	3.901125962	6.177	3.88016
2	5.150	25-Feb-2002	5.225	0.95034125	4	3.798913929	6.333	3.75822
3	5.200	26-Aug-2002	5.277	0.92582337	4	3.699381343	6.385	3.64011
4	5.250	25-Feb-2003	5.329	0.90137403	4	3.600632571	6.437	3.52394
5	5.374	25-Aug-2003	5.463	0.87567855	4	3.495761898	6.571	3.40300
6	5.500	25-Feb-2004	5.597	0.84928019	4	3.389528778	6.705	3.28196
7	5.624	25-Aug-2004	5.734	0.82277146	4	3.281916075	6.842	3.16080
8	5.750	25-Feb-2005	5.870	0.79585734	4	3.173623771	6.978	3.04022
9	5.775	25-Aug-2005	5.895	0.77285090	4	3.079766213	7.003	2.93453
10	5.800	27-Feb-2006	5.920	0.74972455	104	77.6869373	7.028	73.62754
						109.1075878		104.25049

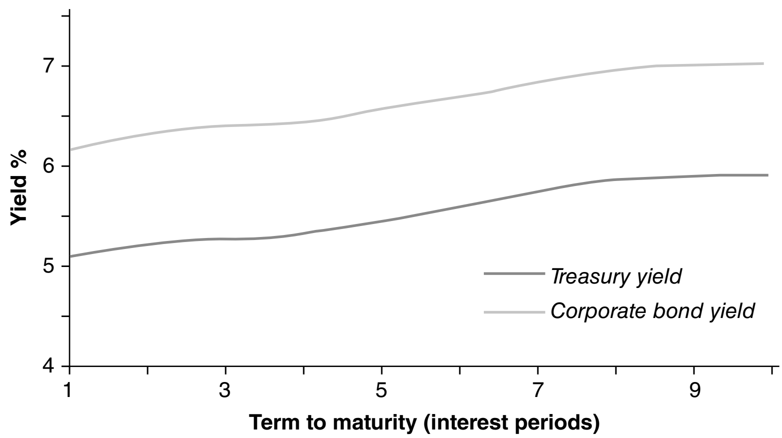


FIGURE 8.14 Yield Curves Illustrating OAS Yield Spread

Our starting point is the redemption yield curve, from which we calculate the current spot-rate term structure. This was done using RATE software and is shown in column 4. Using the spot-rate structure, we calculate the present value of the Treasury security's cash flows, which is shown in column 7. We wish to calculate the OAS that equates the price of the Treasury to that of the corporate bond. By iteration, this is found to be 110.81 bps. This is the semiannual OAS spread. The annualised OAS spread is double this. With the OAS spread added to the spot rates for each period, the price of the Treasury matches that of the corporate bond, as shown in column 9. The adjusted spot rates are shown in column 8.

Figure 8.14 illustrates the yield curve for the Treasury security and the corporate bond.

CORRECT WAY TO RISK-MANAGE A CALLABLE NOTE

Assume a callable note where the investor deposits 100 on day one. The note pays an annual coupon of 4% until maturity. After every coupon date, the bank has the right, but not the obligation, to call the note. If it does so, the bank returns the investor's deposit of 100, in addition to the coupon just paid. If the note is never called, then it redeems on its final maturity date, in 10 years' time.

How should the trading desk risk-manage this trade? The bank can issue new notes at a rate of Libor + funding spread, where the funding spread depends on its perceived creditworthiness. If the sum of Libor and the spread fall, then the bank will be able to issue new debt at a cheaper rate than before. If Libor and/or the funding-spread fall materially, the bank will decide to call the existing note, and issue a new note at a lower rate. Bear in mind that, by calling the note, the bank loses the option of calling it on future coupon dates. This future optionality has some value attributed to it because the funding rate of the bank may increase again in the future. Thus the bank will only consider calling the note if the funding levels have fallen sufficiently to compensate for the loss of the future optionality that has been thrown away by calling early.

If the trading desk had issued a noncallable note paying Libor + spread, it would simply deposit the cash received on the note with the internal treasury, to the duration of the note. The treasury would pay the trading desk Libor + spread, and the desk would be perfectly hedged.

If the trading desk had issued a noncallable fixed-rate note, then they would have a mismatch between the floating coupons received from the treasury and the fixed coupons paid to the client. They could dynamically manage this risk by trading a collateralised swap, receiving fixed, and paying floating. If the funding spread increased, then they would have to rehedge by decreasing the notional of the collateralised swap, as the ratio of the discounting on the note and the swap would increase.

Now we can consider how to manage the callable note. First of all, we deposit the note proceeds at the bank's Treasury desk for a term shorter than 10 years, to reflect the possibility of early termination. For example, if the pricing model tells us that the risk-neutral probability of the note surviving for 10 years is only 20%, then we should only deposit 20 for the full 10-year term. By calculating the call probabilities in our model (or by looking at the risks to the term-structure of our funding spread curve), we can calculate an amortising deposit schedule to place the money with our treasury. We can trade interest rate swaps to hedge the Libor inflows from treasury against the fixed coupon outflows to the investor. We should model both the volatility of Libor and our funding spread in pricing the note. A high positive correlation between the funding spread and Libor is beneficial to us, as it increases the value of the call. We do not want scenarios to occur where the funding spread rises when Libor falls, as the offsetting effects will reduce the overall volatility of the note's PV. We can dynamically hedge the interest rate volatility exposure by selling collateralised swaptions. The Libor/funding-spread correlation and the funding-spread volatility are very difficult to hedge. The only ways to hedge these indirectly would be to try to sell other notes with naturally opposing risks, or maybe to proxy-hedge the funding-spread volatility exposure by selling options on other credit spreads: for example, iTRAXX. In reality, we probably just need to make conservative assumptions about the levels of correlation and volatility, and warehouse the risks. If the expected credit volatility increases, then the expected maturity of the note will decrease, as an early call becomes more likely. We could manage this change in risk by shortening the average duration of the deposit with treasury, and decreasing the maturity of the interest rate swap hedges.

APPENDIX

APPENDIX 8.1 CALCULATING INTEREST RATE PATHS USING MICROSOFT EXCEL

The following table shows the cell references when using Microsoft Excel to calculate interest rate paths.

Price	89
Period 1 rate	1.05
Volatility	0.25
Drift	0.206054017

Up state	0.078891779	1.078892	Bond price	88.2740022	Average 88.9996
Down state	0.061440979	1.061441		89.72528584 Fwd rate %	0.070166379

	F	G	H	I	J	K	L
17							
18	Price	89					
19	Period 1 rate	1.05					
20	Volatility	0.25					
21	Drift	0.206054017					
22							
23							
24	Up state	= 0.05*EXP(G21+G20)	= 1+G24	Bond price	= 100/(G19*H24)	Average = (J24+J25)/2	
25	Down state	= 0.05*EXP(G21-H19)	= 1+G25		= 100/ (G19*H25)		
26							
27					Fwd rate %	= (G24+G25)/2	

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CHAPTER 9

Inflation-Indexed Bonds and Derivatives

In some countries there is a market in bonds whose return, both coupon and final redemption payment, is linked to the consumer price index. Investors' experiences with inflation-indexed bonds differ across countries, as they were introduced at different times and, as a result, the exact design of index-linked bonds varies across the different markets. This, of course, makes the comparison of issues such as yield difficult and has in the past acted as a hindrance to arbitrageurs seeking to exploit real yield differentials. In this chapter, we will highlight the basic concepts behind the structure of indexed bonds and how this may differ from that employed in another market. Not all index-linked bonds link both coupon and maturity payments to a specified index; in some markets only the coupon payment is index linked. Generally the most liquid market available will be the government bond market in index-linked instruments.

In this chapter we present the structure and analysis of index-linked bonds, as well as related index-linked derivative products. Appendix 9.1 lists those countries that currently issue public-sector indexed securities.

BASIC CONCEPTS

There are a number of reasons why investors and issuers alike are interested in inflation-indexed bonds. Before considering these, we look at some of the factors involved in security design.

Choice of index: In principle, bonds can be indexed to any number of variables, including various price indices, earnings, output, specific commodities, or foreign currencies. Although ideally the chosen index would reflect the hedging requirements of both parties, these may not coincide. For instance, the overwhelming choice of retail investors is for indexation to consumer prices, whereas pension funds prefer linking to earnings levels, to offset earnings-linked pension liabilities. In practice, most bonds have been linked to an index of consumer prices such as the UK Retail Price Index, since this is usually widely circulated and well understood and issued on a regular basis.

Indexation lags: In order to provide practically precise protection against inflation, interest payments for a given period would need to be corrected for actual inflation over the same period. However, for two important reasons, there are unavoidable lags between the movements in the price index and the adjustment to the bond cash flows as they are paid. This reduces the inflation-proofing properties of indexed bonds. Deacon and Derry (1998) state two reasons why indexation lags are necessary. First, inflation statistics can only be calculated and published with a delay. The data for one month is usually known well into the next month, and there may be delays in publication. This calls for a lag of at least one month. Secondly, in some markets the size of the next coupon payment must be known before the start of the coupon period in order to calculate the accrued interest; this leads to a delay equal to the length of time between coupon payments.[1]

The indexation lag is illustrated in Figure 9.1.

Coupon frequency: Index-linked bonds often pay interest on a semiannual basis, but long-dated investors such as fund managers whose liabilities may well include inflation-indexed annuities are also, at least in theory, interested in indexed bonds that pay on a quarterly or monthly basis.

Indexing the cash flows: There are five basic methods of linking the cash flows from a bond to an inflation index. These are:

1. *Interest-indexed bonds:* These pay a fixed real coupon and an indexation of the fixed principal every period; the principal repayment at maturity is not adjusted. In this case, all the inflation adjustment is fully paid out as it occurs and does not accrue on the principal. This type of bond has been issued in Australia, for example.
2. *Capital-indexed bonds*: The coupon rate is specified in real terms. Interest payments equal the coupon rate multiplied by the inflation-adjusted principal amount. At maturity, the principal repayment is the product of the nominal value of the bond multiplied by the cumulative change in the index. Compared with interest-indexed bonds of similar maturity, these bonds have higher duration and lower reinvestment risk. This type of bond has been issued in Australia, Canada, New Zealand, the United Kingdom, and the United States.
3. *Zero-coupon indexed bonds:* As their name implies, these pay no coupons but the principal repayment is scaled for inflation. These have the highest duration

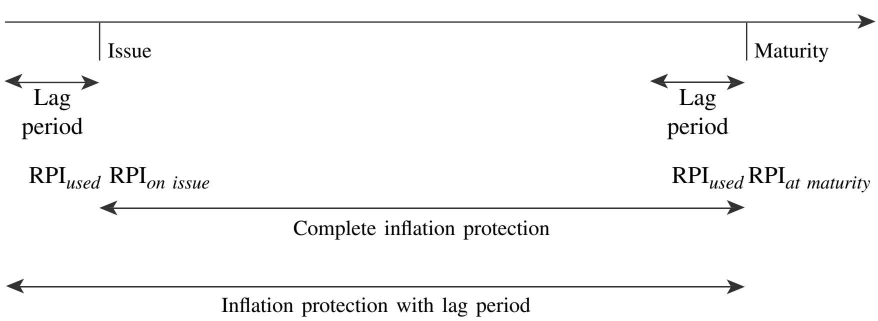

FIGURE 9.1 The Indexation Lag

[1] The same source cites various methods by which the period of the lag may be minimised; for example, the accrued-interest calculation for Canadian Real Return Bonds is based on cumulative movements in the consumer price index, which run from the last coupon date. This obviates the need to know with certainty the nominal value of the next coupon, unlike the arrangement for UK index-linked gilts. See Deacon and Derry (1998, pp. 30–31).

of all indexed bonds and have no reinvestment risk. This type of bond has been issued in Sweden.

4. *Indexed-annuity bonds:* The payments consist of a fixed annuity payment and a varying element to compensate for inflation. These bonds have the lowest duration and highest reinvestment risk of all index-linked bonds. They have been issued in Australia, although not by the central government.
5. *Current pay bond:* As with interest-indexed bonds, the principal cash flow on maturity is not adjusted for inflation. The difference with current pay bonds is that their term cash flows are a combination of an inflation-adjusted coupon and an indexed amount that is related to the principal. Thus, in effect, current pay bonds are an inflation-indexed floating-rate note. They have been issued in Turkey.

The choice of instrument will reflect the requirements of investors and issuers. Deacon and Derry (1998) cite duration, tax treatment, and reinvestment risk as the principal factors that influence instrument design. Although duration for an indexed bond measures something slightly different from duration for a conventional bond, being an indication of the bond price sensitivity due to changes in the real interest rate, as with a conventional bond it is higher for zero-coupon indexed bonds than for coupon bonds. Indexed annuities will have the shortest duration. Longer-duration instruments will (in theory) be demanded by investors that have long-dated hedging liabilities. Again, similarly to conventional bonds, investors holding indexed bonds are exposed to reinvestment risk, which means that the true yield earned by holding a bond to maturity cannot be determined when it is purchased, as the rate at which interim cash flows can be invested is not known. Hence, bonds that pay more of their return in the form of coupons are more exposed to this risk, which would be indexed annuities. Indexed zero-coupon bonds, like their conventional counterparts, do not expose investors to reinvestment risk. The tax regime in individual markets will also influence investor taste. For instance, some jurisdictions tax the capital gain on zero-coupon bonds as income, with a requirement that any tax liability be discharged as current income. This is unfavourable treatment as the capital is not available until maturity, which would reduce demand from institutional investors for zero-coupon instruments.

In three countries—Canada, New Zealand, and the United States—there exists a facility for investors to strip indexed bonds, thus enabling separate trading of coupon and principal cash flows.[2] Such an arrangement obviates the need for a specific issue of zero-coupon indexed securities, as the market can create them in response to investor demand.

Coupon stripping feature: Allowing market practitioners to strip indexed bonds enables them to create new inflation-linked products that are more specific to investors' needs, such as indexed annuities or deferred-payment indexed bonds. In markets that allow stripping of indexed government bonds, a strip is simply an individual uplifted cash flow. An exception to this is in New Zealand, where the cash flows are separated into three components; the principal, the principal-inflation adjustment, and the set of inflation-linked coupons (that is, an indexed annuity). The U.S. TIPS market is described in Example 9.1.

[2] In the United Kingdom, the facility of "stripping" exists for conventional gilts but not index-linked gilts. The term originates in the U.S. market, being an acronym for **S**eparate **T**rading of **R**egistered **I**nterest and **P**rincipal.

EXAMPLE 9.1 THE U.S. TIPS MARKET

TIPS is the term for what were first referred to as Treasury inflation-indexed securities. They were introduced in 1997 by the U.S. Treasury, to increase investment options for investors but also to diversify its funding sources and risk exposure. They are similar to UK index-linked gilts in that they have a fixed coupon on issue, but one that is then adjusted to account for changes in the level of inflation. At the time of writing, TIPS are issued only at 10-year maturities, through an auction held in January and July each year.[3]

Interest on TIPS securities can be viewed as a mix of the fixed coupon and a floating coupon. The fixed coupon is known on issue. The "floating" coupon is the issued nominal amount, which is made up on coupon payments rising as the amount of principal increases. The bond principal amount increases because it is linked to the U.S. Consumer Price Index (CPI).[4] The inflation-adjusted principal amount is paid out on maturity.

A general market convention is to measure TIPS performance on the basis of break-even yields, which is the nominal Treasury yield minus the TIPS yield. The lower the break-even yield, the cheaper TIPS are relative to conventional Treasuries of identical maturity.

INDEX-LINKED BOND YIELDS

Calculating Index-Linked Yields

Inflation-indexed bonds have either or both of their coupon and principal linked to a price index such as the retail price index (RPI), a commodity price index (for example, wheat), or a stock market index. In the United Kingdom, the reference is to the CPI, following practice in other markets where the price index is the consumer price index (CPI). If we wish to calculate the yield on such bonds, it is necessary to make forecasts of the relevant index, which are then used in the yield calculation. In the UK, both the principal and coupons on UK index-linked government bonds are linked to the CPI and are therefore designed to give a constant real yield. Most of the index-linked stocks that have been issued by the UK government have coupons of 2% or 2 1/2%. This is because the return from an index-linked bond represents, in theory, real return, as the bond's cash flows rise in line with inflation. Historically, real rate of return on UK market debt stock over the long term has been approximately 2%.

Indexed bonds differ in their make-up across markets. In some markets only the principal payment is linked, whereas other indexed bonds link only their coupon

[3] At the start of the TIPS programme the Treasury also issued 5-year and 30-year TIPS, but these have since been discontinued.

[4] Obviously this assumes that the economy experiences inflation. If it experienced deflation, the principal amount would of course decrease. In the event of deflation, TIPS are redeemed at a minimum of par. This is known as the deflation floor and is a safety net for investors.

payments and not the redemption payment. In the case of the former, each coupon and the final principal are scaled up by the ratio of two values of the CPI. The main CPI measure is the one reported for eight months before the issue of the gilt, and is known as the base CPI. The base CPI is the denominator of the index measure. The numerator is the CPI measure for eight months prior to the month coupon payment, or eight months before the bond maturity date.

The coupon payment of an index-linked gilt is given by (9.1).

$$\text{Coupon payment} = (C/2) \times \frac{CPI_{C-8}}{CPI_0} \tag{9.1}$$

Expression (9.1) shows the coupon divided by two before being scaled up, because index-linked gilts pay semiannual coupons. The formula for calculating the size of the coupon payment for an annual-paying indexed bond is modified accordingly.

The principal repayment is given by (9.2).

$$\text{Principal payment} = 100 \times \frac{CPI_{M-8}}{CPI_0} \tag{9.2}$$

where C is the annual coupon payment
CPI_0 is the CPI value eight months prior to the issue of the bond (the base CPI)
CPI_{C-8} is the CPI value eight months prior to the month in which the coupon is paid
CPI_{M-8} is the CPI value eight months prior to the bond redemption.

Price indices are occasionally "re-based", which means that the index is set to a base level again.

The coupon payout calculation is illustrated in Example 9.2.

EXAMPLE 9.2

An index-linked gilt with coupon of 4.625% was issued in April 1988 and matured in April 1998. The base measure required for this bond is the CPI for August 1987, which was 102.1. The CPI for August 1997 was 158.5. We can use these values to calculate the actual cash amount of the final coupon payment and principal repayment in April 1998, as shown below.

$$\text{Coupon payment} = (4.625/2) \times \frac{158.5}{102.1} = £3.58992$$

$$\text{Principal payment} = 100 \times \frac{158.5}{102.1} = 155.23996$$

(*continued*)

EXAMPLE 2.1 (*Continued*)

We can determine the accrued interest calculation for the last six-month coupon period (October 1987 to April 1998) by using the final coupon payment, given below.

$$3.58992 \times \frac{\text{No. of days accrued}}{\text{Actual days in period}}$$

The markets use two main yield measures for index-linked bonds, both of which are a form of yield to maturity. These are the *money* (or nominal) *yield,* and the *real yield.*

In order to calculate a money yield for an indexed bond, we require forecasts of all future cash flows from the bond. Since future cash flows from an index-linked bond are not known with certainty, we require a forecast of all the relevant future CPIs, which we then apply to all the cash flows. In fact, the market convention is to take the latest available CPI and assume a constant inflation rate thereafter, usually $2^1/_2$% or 5%. By assuming a constant inflation rate we can set future CPI levels, which in turn allow us to calculate future cash-flow values.

We obtain the forecast for the first relevant future CPI using (9.3).

$$CPI_1 = CPI_0 \times (1+\tau)^{m/12} \tag{9.3}$$

where CPI_1 is the forecast CPI level
CPI_0 is the latest available CPI
τ is the assumed future annual inflation rate
m is the number of months between CPI_0 and CPI_1

Consider an indexed bond that pays coupons every June and December. For analysis we require the CPI forecast value for eight months prior to June and December, which will be for October and April. If we are now in February, we require a forecast for the CPI for the next April. This sets $m = 2$ in equation (9.3). We can then use (9.4) to forecast each subsequent relevant CPI required to set the bond's cash flows.

$$CPI_{j+1} = CPI_1 \times (1+\tau)^{j/2} \tag{9.4}$$

where j is the number of semiannual forecasts after CPI_1 (which was our forecast CPI for April). For example, if the February CPI was 163.7 and we assume an annual inflation rate of 2.5%, then we calculate the forecast for the CPI for the following April to be:

$$\begin{aligned} CPI_1 &= 163.7 \times (1.025)^{2/12} \\ &= 164.4 \end{aligned}$$

and for the following October it would be:

$$\begin{aligned} CPI_3 &= 164.4 \times (1.025) \\ &= 168.5 \end{aligned}$$

Once we have determined the forecast CPIs, we can calculate the yield. Under the assumption that the analysis is carried out on a coupon date, so that accrued interest on the bond is zero, we can calculate the money yield *(ri)* by solving equation (9.5).

$$P_d = \frac{(C/2)(CPI_1/CPI_0)}{(1+\frac{1}{2}ri)} + \frac{(C/2)(CPI_2/CPI_0)}{(1+\frac{1}{2}ri)^2} + \cdots\cdots + \frac{([C/2]+M)(CPI_N/CPI_0)}{(1+\frac{1}{2}ri)^N} \tag{9.5}$$

where ri is the semiannualised money yield to maturity
N is the number of coupon payments (interest periods) up to maturity

Equation (9.5) is for semiannual paying indexed bonds such as index-linked gilts. The equation for annual coupon indexed bonds is given in equation (9.6).

$$P_d = \frac{(C)(CPI_1/CPI_0)}{(1+ri)} + \frac{C(CPI_2/CPI_0)}{(1+ri)^2} + \cdots\cdots + \frac{(C+M)(CPI_N/CPI_0)}{(1+ri)^N} \tag{9.6}$$

The real yield *ry* is related to the money yield through equation (9.7), as it applies to semiannual coupon bonds, which was first described by Fisher in *Theory of Interest* (1930).

$$(1+\tfrac{1}{2}ry) = (1+\tfrac{1}{2}ri)/(1+\tau)^{\frac{1}{2}} \tag{9.7}$$

To illustrate this, if the money yield is 5.5% and the forecast inflation rate is 2.5%, then the real yield is calculated using (9.7) as shown below.

$$\begin{aligned} ry &= \left\{ \frac{[1+\frac{1}{2}(0.055)]}{[1+(0.025)]^{\frac{1}{2}}} - 1 \right\} \times 2 \\ &= 0.0297 \text{ or } 2.97\% \end{aligned}$$

We can rearrange equation (9.5) and use (9.7) to solve for the real yield, shown in (9.8) and applicable to semiannual coupon bonds. Again, we use (9.8) where the calculation is performed on a coupon date.

$$\begin{aligned} P_d &= \frac{CPI_a}{CPI_0}\left[\frac{(C/2)(1+\tau)^{1/2}}{(1+\frac{1}{2}ri)} + \frac{(C/2)(1+\tau)}{(1+\frac{1}{2}ri)^2} + \ldots + \frac{([C/2]+M)(1+\tau)^{N/2}}{(1+\frac{1}{2}ri)^N}\right] \\ &= \frac{CPI_a}{CPI_0}\left[\frac{(C/2)}{(1+\frac{1}{2}ry)} + \ldots\ldots + \frac{(C/2)+M}{(1+\frac{1}{2}ry)^N}\right] \end{aligned} \tag{9.8}$$

where

$$CPI_a = \frac{CPI_1}{(1+\tau)^{1/2}}$$

CPI_0 is the base index level as initially described. CPI_a/CPI_0 is the rate of inflation between the bond's issue date and the date the yield calculation is carried out.

It is best to think of the equations for money yield and real yield by thinking of which discount rate to employ when calculating a redemption yield for an indexed bond. Equation (9.5) can be viewed as showing that the money yield is the appropriate discount rate for discounting money or nominal cash flows. We then rearrange this equation as given in (9.8) to show that the real yield is the appropriate discount rate to use when discounting real cash flows.

The yield calculation for a U.S. TIPS security is given in Appendix 9.2.

Assessing Yield for Index-Linked Bonds

Index-linked bonds do not offer *complete* protection against a fall in real value of an investment. That is, the return from index-linked bonds (including index-linked gilts) is not in reality a guaranteed real return, in spite of the cash flows being linked to a price index such as the CPI. The reason for this is the lag in indexation, which for index-linked gilts is eight months. The time lag means that an indexed bond is not protected against inflation for the last interest period of its life, which for gilts is the last six months. Any inflation occurring during this final interest period will not be reflected in the bond's cash flows and will reduce the real value of the redemption payment and hence the real yield. This can be a worry for investors in high-inflation environments. The only way to effectively eliminate inflation risk for bondholders is to reduce the time lag in indexation of payments to something like one or two months.

Bond analysts frequently compare the yields on index-linked bonds with those on conventional bonds, as this implies the market's expectation of inflation rates. To compare returns between index-linked bonds and conventional bonds, analysts calculate the break-even inflation rate. This is the inflation rate that makes the money yield on an index-linked bond equal to the redemption yield on a conventional bond of the same maturity. Roughly speaking, the difference between the yield on an indexed bond and a conventional bond of the same maturity is what the market expects inflation to be during the life of the bond. Part of the higher yield available on the conventional bond is therefore the inflation premium. In August 1999, the redemption yield on the $5^3/_4$% Treasury 2009, the 10-year benchmark gilt, was 5.17%. The real yield on the 2% index-linked 2009 gilt, assuming a constant inflation rate of 3%, was 2.23%. Using (9.5), this gives us an implied break-even inflation rate of:

$$\begin{aligned}\tau &= \left\{\frac{[1+\tfrac{1}{2}(0.0517)]}{[1+\tfrac{1}{2}(0.0223)]}\right\}^2 - 1\\ &= 0.029287 \text{ or } 2.9\%.\end{aligned}$$

If we accept that an advanced, highly developed and liquid market such as the gilt market is of at least semistrong form, if not strong form, then the inflation expectation in the market is built into these gilt yields. However, if this implied inflation rate understated what was expected by certain market participants, investors will start holding more of the index-linked bond rather than the conventional bond. This activity will then force the indexed yield down (or the conventional yield up). If investors had the opposite view and thought that the implied inflation rate overstated inflation expectations, they would hold the conventional bond. In our illustration above, the market is expecting long-term inflation to be at around 2.9% or less, and the higher yield of the $5^3/_4$% 2009 bond reflects this inflation expectation. A fund manager will take into account her view of inflation, among other factors, in deciding how much of the index-linked gilt to hold compared to the conventional gilt. It is often the case that investment managers hold indexed bonds in a portfolio against specific index-linked liabilities, such as pension contracts that increase their payouts in line with inflation each year.

The premium on the yield of the conventional bond over that of the index-linked bond is therefore compensation against inflation to investors holding it. Bondholders will choose to hold index-linked bonds instead of conventional bonds if they are worried by unexpected inflation. An individual's view on expected inflation will depend on several factors, including the current macroeconomic environment and the credibility of the monetary authorities, be they the central bank or the government. In certain countries such as the United Kingdom and New Zealand, the central bank has explicit inflation targets, and investors may feel that, over the long term, these targets will be met. If the track record of the monetary authorities is proven, investors may feel further that inflation is no longer a significant issue. In these situations the case for holding index-linked bonds is weakened.

The real-yield level on indexed bonds in other markets is also a factor. As capital markets around the world have become closely integrated in the last 20 years, global capital mobility means that high-inflation markets are shunned by investors. Therefore, over time, expected returns, certainly in developed and liquid markets, should be roughly equal, so that real yields are at similar levels around the world. If we accept this premise, we would then expect the real yield on index-linked bonds to be at approximately similar levels, whatever market they are traded in. For example, we would expect indexed bonds in the United Kingdom to be at a level near to that in, say, the U.S. market. In fact, in May 1999 long-dated index-linked gilts traded at just over 2% real yield, while long-dated indexed bonds in the U.S. were at the higher real-yield level of 3.8%. This was viewed by analysts as reflecting that international capital was not as mobile as had previously been thought, and that productivity gains and technological progress in the U.S. economy had boosted demand for capital there to such an extent that real yield had had to rise. However, there is no doubt that there is considerable information content in index-linked bonds, and analysts are always interested in the yield levels of these bonds compared to conventional bonds.

The Bloomberg YA page used for yield calculation shows the nominal and real yield analysis for inflation-linked bonds. Figure 9.2 shows this page for the U.S. TIPS security the 2% of 2014, which was issued in January 2004, as of 15 March 2004. We see that the nominal or money yield was 1.462%, against a real yield of 3.485%, assuming an inflation rate of 2.00%. We also see that the base CPI at issue was 184.77.

Further Views on Index-Linked Yields

The market analyses the trading patterns and yield levels of index-linked (I-L) gilts for their information content. The difference between the yield on I-L gilts and conventional gilts of the same maturity is an indication of the market's view on future inflation. Where this difference is historically low, it implies that the market considers that inflation prospects are benign. So the yield spread between index-linked gilts and the same maturity conventional gilt is roughly the market's view of expected inflation levels over the long term. For example, on 3rd November 1999 the 10-year benchmark, the $5^3/_4$% Treasury 2009, had a gross redemption yield of 5.280%. The 10-year index-linked bond, the $2^1/_2$% I-L

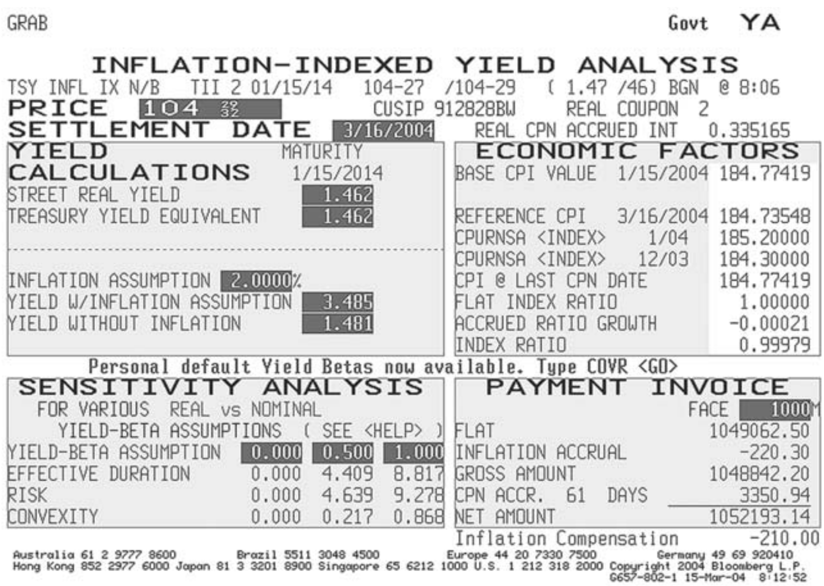

FIGURE 9.2 Bloomberg Page YA Showing Yield Calculation for U.S. 2% 2014 TIP Security, as of 15 March 2004

Treasury 2009, had a money yield of 5.046% and a real yield of 2.041%, the latter assuming an inflation rate of 3%. Roughly speaking, this reflects a market view on inflation of approximately 3.24% in the 10 years to maturity. Of course, other factors drive both conventional and I-L bond yields, including supply and demand, and liquidity. Generally, conventional bonds are more liquid than I-L bonds. An increased demand will depress yields for the conventional bond as well.[5] However, the inflation expectation will also be built into the conventional bond yield, and it is reasonable to assume the spread to be an approximation of the market's view on inflation over the life of the bond. A higher inflation expectation will result in a greater spread between the two bonds, this reflecting a premium to holders of the conventional issue as a compensation against the effects of inflation.

Table 9.1 shows the real yield of the $2^1/_2$% I-L 2009 bond at selected points since the beginning of 1997, alongside the gross redemption yield of the 10-year benchmark conventional bond at the time (we use the same I-L bond because there was no issue that matured in 2007 or 2008). Although the market's view on expected inflation rate over the 10-year period is, on the whole, stuck around the 3% level, there has been a downward trend in the period since the MPC became responsible for setting interest rates. As the MPC has an inflation target of 2.5%, the spread between the real yield on the 10-year linker and the 10-year conventional gilt implies that the market believes that the MPC will achieve its goal.[6]

The yield spread between I-L and conventional gilts fluctuates over time and is influenced by a number of factors and not solely by the market's view of future inflation (the implied forward inflation rate). As we discussed in the previous paragraph, however, the market uses this yield spread to gauge an idea of future inflation levels. The other term used to describe the yield spread is break-even inflation; that is, the level of inflation required that would equate nominal yields on I-L gilts with yields on conventional gilts. Figure 9.1 shows the implied forward inflation rate for the 15-year and 25-year terms to maturity as they fluctuated during 1998-1999. The data is from the Bank of England. For both maturity terms, the implied forward rate decreased significantly during the summer of 1998. Analysts, however, ascribed this to the rally in the conventional gilts, brought on by the flight to quality after the emerging-markets fallout beginning in July that year. This rally was not matched by I-L gilts' performance. The 25-year implied forward inflation rate touched 1.66% in September 1999, which was considered excessively optimistic given that that the

[5] For example, two days later, following a rally in the gilt market, the yield on the conventional gilt was 5.069%, against a real yield on the index-linked gilt of 1.973%, implying that the inflation premium had been reduced to 3.09%. This is significantly lower than the premium just two days later, which may reflect the fact that the MPC had just raised interest rates by 0.5% the day before, but the rally in gilts would have a greater impact on the conventional bond than on the linker.

[6] In fact, the MPC target centres on the RPIX measure of inflation, the "underlying" rate. This is the RPIX measure but with mortgage interest payments stripped out. During 2004 the MPC replaced its RPIX target with HICP, an EU harmonised measure of inflation.

TABLE 9.1 Real Yield on the $2^1/_2$% I-L 2009 versus the 10-Year Benchmark

	Yields (%)							
Bonds	**Feb-97**	**Jun-97**	**Feb-98**	**Jun-98**	**Feb-99**	**Jun-99**	**Sep-99**	**Nov-99**
2.5% I-L 2009*	3.451	3.707	3.024	2.880	1.937	1.904	2.352	2.041
7.25% 2007	7.211	7.021	-	-	-	-	-	-
9% 2008	-	-	5.959	5.911	-	-	-	-
5.75% 2009	-	-	-	-	4.379	4.907	5.494	5.280
Spread	3.760	3.314	2.935	3.031	2.442	3.003	3.142	3.239

* Real yield, assuming 3% rate of inflation.
Source: Bloomberg.

Bank of England (BoE) was working towards achieving a 2.5% rate of inflation over the long term! This suggested then that conventional gilts were significantly overvalued.[7] As we see in Figure 9.3, this implied forward rate for both maturity terms returned to more explainable levels later during the year, slightly above 3%. This is viewed as more consistent with the MPC's target, and can be expected to fall to just over 2.5% over the long term.

The fluctuation in implied forward inflation rates over time is illustrated well in Figure 9.3, which shows the pattern during 1998–99.

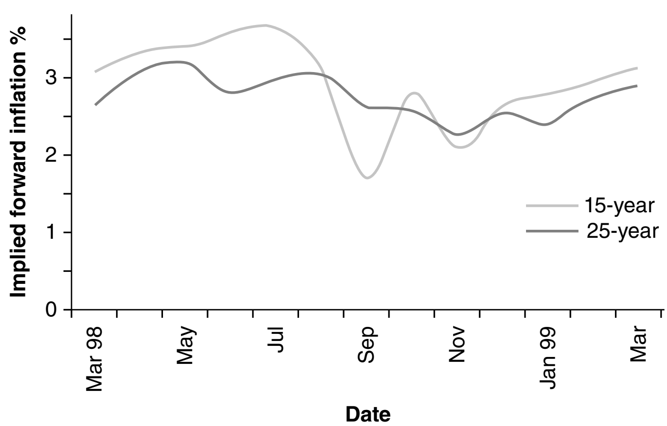

FIGURE 9.3 UK Implied Forward Inflation Rates during 1998–99

[7] As we have noted, the yield spread between I-L and conventional gilts reflects other considerations in addition to the forward inflation rate. As well as specific supply-and-demand issues, considerations include inflation risk premium in the yield of conventional gilts, and distortions created when modelling the yield curve. There is also a liquidity premium priced into I-L gilt yields that would not apply to benchmark conventional gilts. The effect of these is to generally overstate the implied forward inflation rate. Note also that the implied forward inflation rate applies to CPI, whereas the BoE's MPC inflation objective targets the RPIX measure of inflation, which is the headline inflation rate minus the impact of mortgage interest payments.

Analysis of Real Interest Rates Observing trading patterns in a liquid market in inflation-indexed bonds enables analysts to draw conclusions on nominal versus real interest indicators, and the concept of an inflation term structure. Such analysis is often problematic, however, as there is usually a significant difference between liquidity levels of conventional and indexed bonds. Nevertheless, as we discussed in the previous section, it is usually possible to infer market estimates of inflation expectations from the yields observed on indexed bonds, when compared to conventional yields.

Inflation Expectations

Where an indexed bond incorporates an indexation lag, there is an imperfect indexation, and the bond's return will not be completely inflation-proof. Deacon and Derry (1998) suggest that this means an indexed bond may be regarded as a combination of a true indexed instrument (with no lag) and an unindexed bond. Where the lag period is exactly one coupon period, the price/yield relationship is given by

$$P = \sum_{n:t_n>t}^{N} \frac{C\prod_{n=0}^{j-1}(1+ri_i)}{(1+rm_j)^j\prod_{i=1}^{j}(1+ri_i)} + \frac{M\prod_{i=0}^{n-1}(1+ri_i)}{(1+rm_n)^n\prod_{i=1}^{n}(1+ri_i)} \tag{9.9}$$

where ri is the rate of inflation between dates $i-1$ and i
rm is the redemption yield

and C and M are coupon and redemption payments, as usual. If the bond has just paid the last coupon ahead of its redemption date, (9.9) reduces to

$$P = \frac{C}{(1+rm)(1+ri)} + \frac{M}{(1+rm)(1+ri)} \tag{9.10}$$

In this situation, the final cash flows are not indexed and the price/yield relationship is identical to that of a conventional bond. This fact enables us to quantify the indexation element, as the yields observed on conventional bonds can be compared to that on the nonindexed element of the indexed bond. This implies a true real-yield measure for the indexed bond.

The Fisher identity is used to derive this estimate. Essentially, this describes the relationship between nominal and real interest rates, and in one form is given as

$$1 + y = (1+r)(1+i)(1+\rho) \tag{9.11}$$

where y is the nominal interest rate, r the real interest rate, and i the expected rate of inflation. ρ is a premium for the risk of future inflation. Using (9.11), assuming a value for the risk premium ρ, we can link the two bond price equations, which can, as a set of simultaneous equations, be used to obtain values for the real interest rate and the expected inflation rate.

If they exist, one approach is to use two bonds of identical maturity, one conventional and one indexed, and, ignoring lag effects, use the yields on both to determine the expected inflation rate, given by the difference between the redemption yields of each bond. In fact, as we noted in the previous section, this measures the average expected rate of inflation during the period from now to the maturity of the bonds. This is at best an approximation. It is a flawed measure because an assumption of the expected inflation rate has been made when calculating the redemption yield of the indexed bond in the first place. As Deacon and Derry (1998, p. 91) state, this problem is exacerbated if the maturity of both bonds is relatively short, because impact of the unindexed element of the indexed bond is greater the shorter its maturity. To overcome this flaw, a break-even rate of inflation is used. This is calculated by first calculating the yield on the conventional bond, followed by the yield on the indexed bond using an assumed initial inflation rate. The risk premium ρ is set to an assumed figure; say, 0. The Fisher identity is used to calculate a new estimate of the expected inflation rate i. This new estimate is then used to recalculate the yield on the indexed bond, which is then used to produce a new estimate of the expected inflation rate. The process is repeated iteratively until a consistent value for i is obtained.

The main drawback with this basic technique is that it is necessary for there to exist a conventional and an index-linked bond of identical maturity, so approximately similar maturities have to be used, further diluting the results. The yields on each bond will also be subject to liquidity, taxation, indexation, and other influences. There is also no equivalent benchmark (or on-the-run) indexed security. The bibliography cites some recent research that has investigated this approach.

An Inflation Term Structure

Where a liquid market in indexed bonds exists, across a reasonable-maturity term structure, it is possible to construct a term structure of inflation rates. In essence, this involves fitting the nominal and real interest-rate term structures, the two of which can then be used to infer an inflation term structure. This, in turn, can be used to calculate a forward expected inflation rate for any future term, or a forward inflation curve, in the same way that a forward interest-rate curve is constructed.

The Bank of England uses an iterative technique to construct a term structure of inflation rates.[8] First, the nominal interest rate term structure is fitted using a version of the Waggoner model (1997, also described in James and Webber (2000)). An initial assumed inflation term structure is then used to infer a term structure of real interest rates. This assumed inflation curve is usually set flat at 3 or 5%. The real interest rate curve is then used to calculate an implied real interest rate forward curve. Second, the Fisher identity is applied at each point along the nominal and real interest rate forward curves, which produces a new estimate of the inflation term structure. A new real interest rate curve is calculated from this curve. The process is repeated until a single consistent inflation term structure is produced.

[8] This is a term structure of *expected* inflation rates.

INFLATION DERIVATIVES: INTRODUCTION

Inflation-linked derivatives, or, simply, inflation derivatives, were introduced in 2001. They arose out of the desire of investors for real, inflation-linked returns and hedging rather than nominal returns. Although index-linked bonds are available for those wishing to have such returns, as we've observed in other asset classes, inflation derivatives can be tailor-made to suit specific requirements. Volume growth has been rapid since 2003, as shown in Figure 9.4 for the European market.

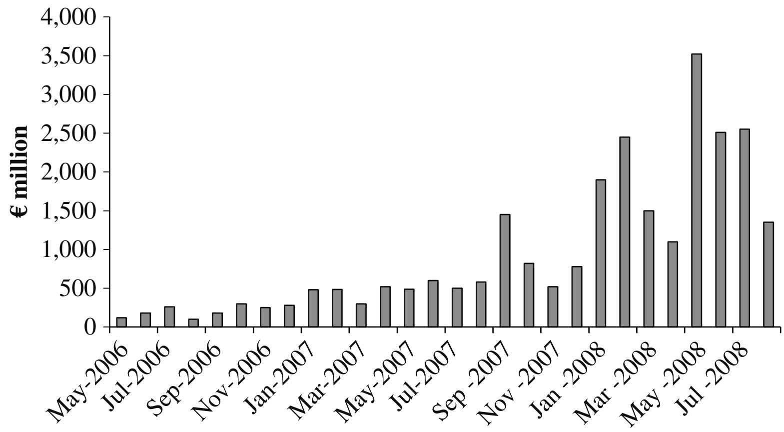

FIGURE 9.4 Inflation Derivatives Volumes, 2006–2009
Source: ICAP.

The UK market, which features a well-developed index-linked cash market, has seen the largest volume of business in inflation derivatives. They have been used by market-makers to hedge inflation-indexed bonds, as well as by corporates who wish to match future liabilities. For instance, the retail company Boots plc added to its portfolio of inflation-linked bonds when it wished to better match its future liabilities in employees' salaries, which were assumed to rise with inflation. Hence, it entered into a series of inflation derivatives with Barclays Capital, in which it received a floating-rate, inflation-linked interest rate and paid nominal fixed-rate interest rate. The swaps ranged in maturity from 18 to 28 years, with a total notional amount of £300 million.

While this is an example of a pension fund that wishes to *receive* inflation-linked returns, corporates that receive revenues linked to inflation often wish to hedge this by *paying* inflation. For instance, utility companies often have pricing structures dictated by regulatory authorities, who may cap price increases that can be passed to customers. Such companies are natural issuers of index-linked bonds, and may therefore be interested in paying inflation in exchange for a fixed-rate return. Another UK-market corporate deal involved

(*continued*)

(*Continued*)

National Air Traffic Services, which paid inflation in a £200 million swap (again with Barclays Capital) in 2002. For market-making banks who take on this exposure, the usual hedge is with an index-linked bond such as UK gilts or Treasury TIPS.

Inflation swaps may be priced in a number of ways. The most common method involves determining prices from index-linked bonds using the bootstrapping approach. Assuming a zero-coupon swap, the market maker calculates the inflation rate to use in the swap from the difference between the yields of index-linked and conventional bonds of the same maturity. This is the break-even inflation rate we discussed earlier in the chapter. Some banks use asset-correlation models to price the swaps, such as the type presented in Jarrow and Yildirim (2003), which uses three stochastic factors to price inflation-linked bonds and swaps.

INFLATION-INDEXED DERIVATIVES

Inflation-indexed derivatives, also known as inflation-linked derivatives or *inflation derivatives*, have become widely traded instruments in the capital markets in a relatively short space of time. They are traded generally by the same desks in investment banks that trade inflation-linked sovereign bonds, who use these instruments for hedging as well as to meet the requirements of clients such as hedge funds, pension funds, and corporates. They are a natural development of the inflation-linked bond market.

Inflation derivatives are an additional means by which market participants can have an exposure to inflation-linked cash flows. They can also improve market liquidity in inflation-linked products, as an earlier generation of derivatives did for interest rates and credit risk. As flexible OTS products, inflation derivatives offer advantages over cash products in certain circumstances. They provide:

- An ability to tailor cash flows to meet investors' requirements;
- A means by which inflation-linked exposures can be hedged;
- An instrument via which relative value positions can be put on across cash and synthetic markets;
- A building block for the structuring of more complex and hybrid products.

The inflation derivatives market in the UK was introduced after the introduction of the gilt repo market in 1996. In most gilt repo trades, index-linked (I-L) gilts could be used as collateral; this meant that both IL and conventional gilts could be used as hedging tools against positions in inflation derivatives. In the euro area, IL derivatives were introduced later but experienced significant growth during 2002–2003. The existence of a sovereign IL bond market can be thought of as a necessary precursor to the development of IL derivatives, and although there is no reason why this should be the case, up to now this has been the case. The reason for this is probably because such a cash market suggests that investors are aware of the attraction of IL products, and wish to invest in them. From a market in IL bonds then develops a market in IL swaps, which are the most common IL derivatives. The IL bond market also provides a ready reference point from which IL derivatives can be priced.

Market Instruments

We describe first some common inflation derivatives, before considering some uses for hedging and other purposes. We then consider IL derivatives pricing.

Inflation-Linked Bond Swap This is also known as a *synthetic index-linked bond*. It is a swap with the following two cash flow legs:

1. Pay (receive) the cash flows on a government IL bond;
2. Receive (pay) a fixed or floating cash flow.

This converts existing conventional fixed- or floating-rate investments into inflation-linked investments. An example of such a swap is given below.

IL Bond Swap

Nominal	100,000,000
Start date	15 March 2004
Maturity term	15 March 2009

Bank receives	Six-month Euribor flat [+ spread], semiannual, act/360
	or
	Fixed rate coupon x%, annual 30/360
Bank pays	Real coupon of y%
	y * [HICP (p – 3) / (HICP (s – 3))] * day count * notional
	Annual 30/360
	On maturity:
	Notional * max {0%, [HICP (m – 3) / HICP (s – 3) – 1]}

The symbols in the formulae above are

p	Payment date
s	Start date
m	Maturity date
HCIP	Harmonised Index of Consumer Prices

The “minus 3” in the formula for HCIP refers to a three-month lag for indexation, common in euro sovereign IL bond markets.

The swap is illustrated in Figure 9.5.

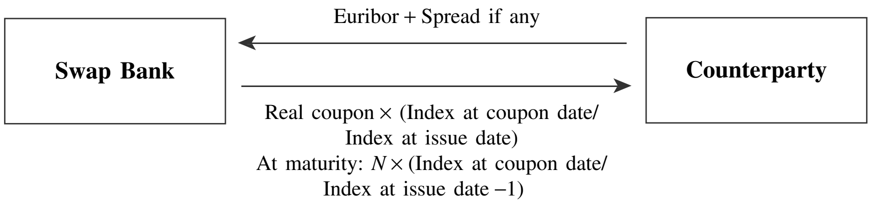

FIGURE 9.5 Synthetic Index-Linked Bond

Year-on-Year Inflation Swap This swap is commonly used to hedge issues of IL bonds. The swap is comprised of:

- Pay (receive) an index-linked coupon, which is a fixed-rate component plus the annual rate of change in the underlying index;
- Receive (pay) Euribor or Libor, plus a spread if necessary.

With these swaps, the IL leg is usually set at a floor of 0% for the annual change in the underlying index. This guarantees the investor a minimum return of the fixed-rate coupon.

This swap is also known as a *pay-as-you-go swap*. It is shown in Figure 9.6.

TIPS Swap The TIPS swap is based on the structure of U.S. TIPS securities. It pays a periodic fixed rate on an accreting notional amount, together with an additional one-off payment on maturity. This payout profile is identical to many government IL bonds. They are similar to synthetic IL bonds described above.

TIPS swaps are commonly purchased by pension funds and other long-dated investors. They may prefer the added flexibility of the IL swap market compared to the cash IL bond market. Figure 9.7 shows the TIPS swap.

Break-Even Swap This is also known as a zero-coupon inflation swap or simply a zero-coupon swap. It allows the investor to hedge away a break-even exposure. Compared to IL swaps such as the synthetic bond swap, which hedge a real yield exposure, the break-even swap has both cash flow legs paying out on maturity. The legs are:

- The total return on the inflation index
- A compounded fixed break-even rate

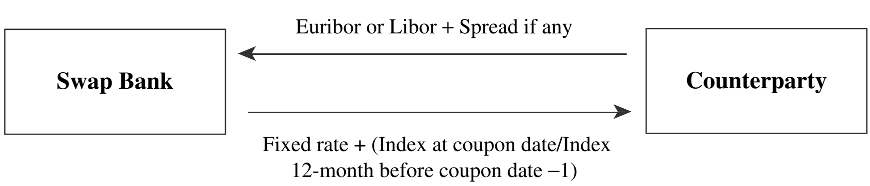

FIGURE 9.6 Year-on-Year Inflation Swap

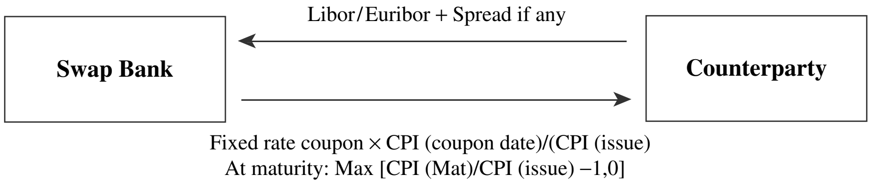

FIGURE 9.7 Illustration of a TIPS Swap

This structure enables IL derivative market makers to hedge their books. It is illustrated in Figure 9.8.

Real-Annuity Swap A real-annuity swap is used to hedge inflation-linked cash flows where this applies for payments such as rental streams, lease payments, project finance cash flows, and so on. It enables market participants who pay or receive such payments to replace the uncertainty of the future level of these cash flows with a fixed rate of growth. The swap is written on the same notional amount for both legs, but payout profiles differ as follows:

- The index-linked leg of the swap compounds its payments with the rate of change of the index;
- The fixed leg of the swap compounds its payments at a prespecified fixed rate.

These swaps are one of the most commonly traded. The fixed rate quoted for the swap provides a ready reference point against which to compare expected future rates of inflation. So for instance, if a bank is quoting for a swap with a fixed rate of 3.00%, and an investor believes that inflation rates will not rise above 3.00% for the life of the swap, then it will receive "fixed" (here meaning a fixed rate of growth) and pay inflation-linked on the swap.

The Inflation Term Structure and Pricing Inflation Derivatives

An inflation term structure is a necessary prerequisite to the pricing of inflation derivatives. It is constructed using the same principles we discussed in Chapters 3 and 4. Previously, to construct this curve we would have used IL bond prices as the set of market yields used as inputs to the curve. Now, however, we can also use the prices of IL derivatives. As with other markets, the derivative prices are often preferred to cash prices for two reasons; one, we can use a continuous set of prices rather than have to rely on available bond maturities, and secondly, there is usually greater liquidity in the OTC market.

In the case of IL products, the indexation element is not in fact a true picture, but rather a picture based on a lag of three, six, or seven months. This lag needs to be taken into account when constructing our curve.

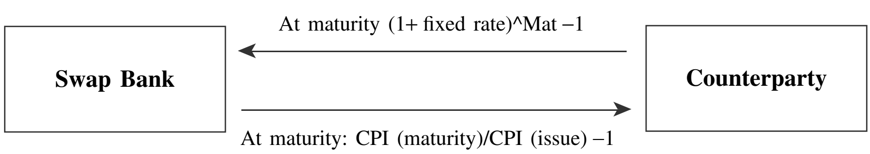

FIGURE 9.8 Break-Even Inflation Swap

The forward index value I at time T from time t ($t < T$) is given by

$$I(t,T) = \frac{I(t)P_r(t,T)}{P_r(t,T)} \tag{9.12}$$

where $I(t)$ is the index value at time t
$P_r(t,T)$ is the price at time t of a real zero-coupon bond of par value 1 maturing at T
$P_n(t,T)$ is the price at time t of a nominal zero-coupon bond of par value 1 maturing at T

Using (9.12) we can build a forward inflation curve provided we have the values of the index at present, as well as a set of zero-coupon bond prices of required credit quality. Following standard yield curve analysis, we may build the term structure from forward rates and therefore imply the real yield curve, or alternatively we may construct the real curve and project the forward rates. However, if we are using inflation swaps for the market price inputs, the former method is preferred because IL swaps are usually quoted in terms of a forward index value.

The curve can be constructed using standard bootstrapping techniques.

Inflation derivatives can be priced reasonably accurately once the inflation term structure is constructed. However some practitioners use stochastic models in pricing such products to account for the volatility surfaces. That is, they model the volatility of inflation as well. The recent literature describes such methods. For instance, Van Bezooyen, Exley, and Smith (1997), Hughston (1998) and Jarrow and Yildirim (2003) suggest an approach based on that described by Amin and Jarrow (1991). This assumes that suitable proxies for the real and nominal term structures are those of foreign and the domestic economies. In other words, the foreign exchange rate captures the information required to model the two curves. We describe this approach here.

We assume that the index follows a lognormal distribution, and we use normal models for the real and nominal forward rates. For the index we then have

$$\mathrm{d}I(t) = I(t)[\mu_1(t)\mathrm{d}t + \sigma(t)\mathrm{d}W(t)] \tag{9.13}$$

and we have

$$\mathrm{d}F_n(t,T) = a_n(t,T)\mathrm{d}t + \sigma_n(t,T)\mathrm{d}W(t) \tag{9.14}$$

$$\mathrm{d}F_r(t,T) = a_r(t,T)\mathrm{d}t + \sigma_r(t,T)\mathrm{d}W(t) \tag{9.15}$$

for the nominal and real forward-rate processes.

The dynamics of the zero-coupon bonds introduced earlier for (9.13) are given by

$$\begin{gathered}\mathrm{d}(\log P_k(t,T)) = [r_k(t) - \int_t^T \alpha_k(t,u)\mathrm{d}u]\mathrm{d}t + \sum\nolimits_k(t,T)\mathrm{d}W(t) \\ k = n, r\end{gathered} \tag{9.16}$$

and where

$$\sum\nolimits_k (t,T) = -\int \alpha_k(t,u)\mathrm{d}u$$
$$k = n, r$$

describes the zero-coupon bond volatilities. We use a one-factor model (see Chapter 4) for each of the term structures and one for the index. Therefore d*W(t)* is a combined three-dimensional vector of three correlated Brownian or Weiner processes, with a correlation of ρ. The volatility of each bond and the index is therefore also a three-element vector.

From the above, the price at time t of an option on the index struck at X and expiring at time T is given by

$$V_\phi^{Index}(t,T) = \phi[I(t)P_r(t,T)N(\phi h_1) - XP_n(t,T)N(\phi h_2)] \tag{9.17}$$

where

$\phi = 1$ for a call option and $\phi = -1$ for a put option, and where

$$h_1 = \left(\log\left(\frac{I(t)P_r(t,T)}{P_n(t,T)X}\right) + \frac{V(t,T)^2}{2}\right) / V(t,T)$$
$$h_2 = h_1 - V(t,T)$$

and where

$$V(t,T)^2 = \int_t^T \sum\nolimits_I (u,t) \cdot \rho \cdot \sum\nolimits_I (u,T)\mathrm{d}u$$

and where we define

$$\sum\nolimits_I (t,T) = \sum\nolimits_n (t,T) - \sum\nolimits_r (t,T) - \sigma_I(u)$$

The result above has been derived, in different forms, in all three references noted above. As with other options-pricing models, it needs to be calibrated to the market before it can be used. Generally this will involve using actual and projected forward inflation rates to fit the model to market prices and volatilities.

APPLICATIONS

We now describe some common applications of IL derivatives.

Hedging Pension Liabilities

This is perhaps the most obvious application. Assume a life assurance company or corporate pension fund wishes to hedge its long-dated pension liabilities, which are linked to the rate of inflation. It may invest in sovereign IL bonds such as IL gilts, or in IL corporate bonds that are hedged (for credit-risk purposes) with credit derivatives. However, the market in IL bonds is not always liquid, especially in IL corporate bonds. The alternative is to buy a synthetic IL bond. This is structured as a combination of a conventional government bond and an IL swap, in which the pension fund pays away the bond coupon and receives inflation-linked payments.

The net cash flow leaves the pension fund receiving a stream of cash flows that are linked to inflation. The fund is therefore hedged against its liabilities. In addition, because the swap structure can be tailor-made to the pension fund's requirements, the dates of cash flows can be set up exactly as needed. This is an added advantage over investing in the IL bonds directly.

Portfolio Restructuring Using Inflation Swaps

Assume that a bank or corporate has an income stream that is linked to inflation. Up to now, it has been funded by a mix of fixed- and floating-rate debt. Say that these are floating-rate bank loans and fixed-rate bonds. However, from an asset-liability management (ALM) point of view this is not optimal, because of the nature of a proportion of its income. It makes sense, therefore, to switch a part of its funding into an inflation-linked segment. This can be done using either of the following approaches:

- Issue an IL bond.
- Enter into an IL swap, with a notional value based on the optimum share of its total funding that should be inflation-linked, in which it pays inflation-linked cash flows and receives fixed-rate income.

The choice will depend on which approach provides cheapest funding and most flexibility.

Hedging a Bond Issue

Assume that a bank or corporate intends to issue an IL bond, and wishes to hedge against a possible fall in government IL bond prices, against which its issue will be priced. It can achieve this hedge using an IL gilt-linked derivative contract.

The bank or corporate enters into cash-settled contract for difference (CFD), which pays out in the event of a rise in government IL bond yields. The CFD has a term to maturity that ties in with the issue date of the IL bond. The CFD market maker has effectively shorted the government bond, from the CFD trade date until maturity. On the issue date, the market maker will provide a cash settlement if yields have risen. If yields have fallen, the IL bond issuer will pay the difference. However, this cost is netted out by the expected "profit" from the cheaper funding when the bond is issued. Meanwhile, if yields have risen and the bank or corporate issuer does have to fund at a higher rate, it will be compensated by the funds received by the CFD market maker.

APPENDICES

APPENDIX 9.1 CURRENT ISSUERS OF PUBLIC-SECTOR INDEXED SECURITIES

TABLE A9.1.1 Current Issuers of Public-Sector Inflation-Indexed Securities

Country	Date First Issued	Index Linking
Australia	1983	Consumer prices
	1991	Average weekly earnings
Austria	1953	Electricity prices

TABLE A9.1 (*Continued*)

Country	Date First Issued	Index Linking
Brazil	1964–1990	Wholesale prices
	1991	General prices
Canada	1991	Consumer prices
Chile	1966	Consumer prices
Colombia	1967	Wholesale prices
	1995	Consumer prices
Czech Republic	1997	Consumer prices
Denmark	1982	Consumer prices
France	1956	Average value of French securities
Greece	1997	Consumer prices
Hungary	1995	Consumer prices
Iceland	1964–1980	Cost of Building Index
	1980–1994	Credit Terms Index
	1995	Consumer prices
Ireland	1983	Consumer prices
Italy	1983	Deflator of GDP at factor cost
Mexico	1989	Consumer prices
New Zealand	1977–1984	Consumer prices
	1995	Consumer prices
Norway	1982	Consumer prices
Poland	1992	Consumer prices
Sweden	1952	Consumer prices
	1994	Consumer prices
Turkey	1994–1997	Wholesale prices
	1997	Consumer prices
United Kingdom	1981	Consumer prices
United States	1997	Consumer prices

Source: Deacon and Derry (1998). Used with permission of Prentice-Hall Europe.

APPENDIX 9.2 U.S. TREASURY INFLATION-INDEXED SECURITIES (TIPS)

Indexation Calculation

United States Treasury inflation-indexed securities link their coupon and principal to an Index Ratio of the Consumer Prices Index. The index ratio is given by

$$IR = \frac{CPI_{Settlement}}{CPI_{Issue}}$$

where "settlement" is the settlement date and "issue" is the issue date of the bond. The actual CPI used is that recorded for the calendar month three months earlier than the relevant date, this being the lag time. For the first day of any month, the reference CPI level is that recorded three months earlier; so, for example, on 1st May the relevant CPI measure would be that recorded on 1st February. For any other day in the month, linear interpolation is used to calculate the appropriate CPI level recorded in the reference month and the following month.

Cash Flow Calculation

The inflation adjustment for the security cash flows is given as the principal multiplied by the index ratio for the relevant date, minus the principal (p). This is termed the Inflation Compensation (ic), given as

$$\text{Inflation Compensation}_{\text{SetDate}} = (\text{Principal} \times \text{Index Ratio}_{\text{SetDate}}) - \text{Principal}$$

Coupon payments are given by

$$Interest_{DivDate} = C/2 \times (P + IC_{DivDate})$$

The redemption value of a TIPS is guaranteed by the Treasury to be a minimum of \$100, whatever value has been recorded by the CPI during the life of the bond.

Settlement price

The price/yield formula for a TIPS security is given by the following expressions:

$$\text{Price} = \text{Inflation} - \text{Adjusted price} + \text{Inflation} - \text{Adjuted accrued interest}$$
$$\text{Inflation} - \text{Adjusted price} = \text{Real price} \times \text{Index ratio}_{\text{SetDate}}$$

The real price is given by

$$\text{Real Price} = \left[\frac{1}{1 + \frac{f}{d}\frac{r}{2}}\right]\left(\frac{C}{2} + \frac{C}{2}\sum_{j=1}^{n}\phi_j + 100\phi^n\right) - RAI \qquad \text{(A9.2.1)}$$

where

$$\text{Inflation} - \text{Adjusted accrued interest} = RAI \times IR_{SetDate}$$

and where

$$\phi = \left[\frac{1}{1 + \frac{r}{2}}\right]$$

r is the annual real yield

RAI is the unadjusted accrued interest, which is $\frac{C}{2} \times \frac{(d-f)}{d}$

f is the number of days from the settlement date to the next coupon date

d is the number of days in the regular semiannual coupon period ending on the next coupon date

n is the number of full semiannual coupon periods between the next coupon date and the maturity date

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CHAPTER 10

Introduction to Securitisation and Asset-Backed Securities

Perhaps the best illustration of the innovation in the debt capital markets is the rise in the use and importance of *securitisation*. As defined in Sundaresan (1997, p. 359), securitisation is "a framework in which some illiquid assets of a corporation or a financial institution are transformed into a package of securities backed by these assets, through careful packaging, credit enhancements, liquidity enhancements, and structuring".

The flexibility of securitisation appeals to both issuers and investors. Financial engineering techniques employed by investment banks enable bonds to be created from any type of cash flow. The most typical such flows are those generated by high-volume loans such as residential mortgages and car and credit card loans, which are recorded as assets on bank or financial house balance sheets. In a securitisation, the loan assets are packaged together, and their interest payments are used to service the new bond issue.

In addition to the more traditional cash flows from mortgages and loan assets, investment banks underwrite bonds secured with flows received by leisure and recreational facilities, such as health clubs, and other entities, such as nursing homes. Bonds securitising mortgages are usually treated as a separate class, termed *mortgage-backed securities,* or MBSs. Those with other underlying assets are known as *asset-backed securities,* or ABSs. The type of asset class backing a securitised bond issue determines the method used to analyse and value it.

The asset-backed market represents a large and diverse group of securities, which are suited to a varied group of investors. Often these instruments are the only way for institutional investors to pick up yield while retaining assets with high credit ratings. They are popular with issuers because they represent a cost-effective means of removing assets from their balance sheets, thus freeing up lines of credit and enabling them to access lower-cost funding.

Instruments are available backed by a variety of assets covering the entire yield curve, with either fixed or floating coupons. In the United Kingdom, for example, it is common for mortgage-backed bonds to have floating coupons, mirroring the interest basis of the country's mortgages. To suit investor requirements, however, some of these structures have been modified, through swap arrangements, to pay fixed coupons.

The market in structured finance securities was hit hard in the wake of the 2007–2008 financial crisis. Investors shunned asset-backed securities in a mass flight to quality. As the global economy recovered from recession, interest in the securitisation resumed. We examine the fallout in the market later in this chapter. First we discuss the principal concepts that drive the desire to undertake securitisation.

Readers interested in more detail on ABS product, and on how to undertake a securitisation transaction, should consult the author's book *The Mechanics of Securitization*, also published by John Wiley & Sons.

THE CONCEPT OF SECURITISATION

Securitisation is a well-established practice in the global debt capital markets. It refers to the sale of assets, which generate cash flows, from the institution that owns them, to another company that has been specifically set up for the purpose, and the issuing of notes by this second company. These notes are backed by the cash flows from the original assets. The technique was introduced initially as a means of funding for mortgage banks in the United States, with the first such transaction generally recognised as having been undertaken by Salomon Brothers in 1979. Subsequently, the technique was applied to other assets such as credit card payments and leasing receivables, and has been employed worldwide. It has also been employed as part of asset-liability management, as a means of managing balance-sheet risk.

Reasons for Undertaking Securitisation

The driving force behind securitisation has been the need for banks to realise value from the assets on their balance sheet. Typically these assets are residential mortgages, corporate loans, and retail loans such as credit card debt. The following are factors that might lead a financial institution to securitise a part of its balance sheet:

- If revenues received from assets remain roughly unchanged but the size of assets has decreased, this will lead to an increase in the return on equity ratio.
- The level of capital required to support the balance sheet will be reduced, which again can lead to cost savings or allow the institution to allocate the capital to other, perhaps more profitable, business.
- The financial institution can obtain cheaper funding: Frequently the interest payable on ABS securities is considerably below the level payable on the underlying loans. This creates a cash surplus for the originating entity.

In other words, a bank will securitise part of its balance sheet for one or all of the following reasons:

- Funding the assets it owns
- Balance sheet capital management
- Risk management and credit risk transfer

We consider each of these in turn.

Funding Banks can use securitisation to (1) support rapid asset growth, (2) diversify their funding mix and reduce cost of funding, and (3) reduce maturity mismatches. All banks will not wish to be reliant on only a single or a few sources of funding, as this can be risky in times of difficult market liquidity. Banks aim to optimise their funding between a mix of retail, interbank, and wholesale sources. Securitisation has a key role to play in this mix. It also enables a bank to reduce its funding costs. This is because the securitisation process separates the credit rating of the originating institution from the credit rating of the issued notes. Typically most of the notes issued by special purpose vehicles (SPVs) will be more highly rated than the bonds issued directly by the originating bank itself. Although the liquidity of the secondary market in ABSs is frequently lower than that of the corporate bond market, and this adds to the yield payable by an ABS, it is frequently the case that the cost to the originating institution of issuing debt is still lower in the ABS market because of the latter's higher rating. Finally, there is the issue of maturity mismatches. The business of bank asset-liability management (ALM) is inherently one of maturity mismatch, because a bank often funds long-term assets, such as residential mortgages, with short-asset liabilities, such as bank account deposits or interbank funding. This funding gap can be mitigated via securitisation, as the originating bank receives funding from the sale of the assets, and the economic maturity of the issued notes frequently matches that of the assets.

Balance Sheet Capital Management Banks use securitisation to improve balance sheet capital management. This provides (1) regulatory capital relief, in some cases (depending on the form of the transaction), (2) "economic" capital relief, and (3) diversified sources of funding. As stipulated in the Bank for International Settlements (BIS) capital rules,[1] also known as the Basel rules, banks must maintain a minimum capital level for their assets, in relation to the risk of these assets. Under Basel I, for every \$100 of risk-weighted assets a bank must hold at least \$8 of capital; however, the designation of each asset's risk-weighting is restrictive. For example, with the exception of mortgages, customer loans are 100% risk weighted regardless of the underlying rating of the borrower or the quality of the security held. The anomalies that this raises, which need not concern us here, were partly addressed by the Basel II rules, which became effective from 2007. However, the Basel rules that have been in place since 1988 (and effective from 1992) were a key driver of securitisation. Because an SPV is not a bank, it is not subject to Basel rules, and needs only such capital as is economically required by the nature of the assets it contains. This is not a set amount, but is significantly below the 8% level required by banks in all cases. Although an originating bank does not obtain 100% regulatory capital relief when it sells assets off its balance sheet to an SPV because it will have retained a first-loss piece out of the issued notes, its regulatory capital charge will be significantly reduced after the securitisation.[2]

To the extent that securitisation provides regulatory capital relief, it can be thought of as an alternative to capital-raising, compared with the traditional sources of Tier 1 (equity), preferred shares, and perpetual loan notes with step-up coupon features. By reducing the amount of capital that has to be used to support the asset pool, a bank can also improve its return-on-equity (ROE) value. This will be received favourably by shareholders.

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[1] For further information on this, see Choudhry (2007).

[2] The "first loss" piece refers to the most junior tranche on the liabilities sides of the securitisation (the issued notes), and is the tranche that is exposed to the first of any default losses suffered by the underlying asset pool. In other words, it carries the most performance risk for the investor.

Risk Management Once assets have been securitised, the credit risk exposure on these assets for the originating bank is reduced considerably and, if the bank does not retain a first-loss capital piece (the most junior of the issued notes), it is removed entirely. This is because assets have been sold to the SPV. Securitisation can also be used to remove nonperforming assets from banks' balance sheets. This has the dual advantage of removing credit risk and removing a potentially negative sentiment from the balance sheet, as well as freeing up regulatory capital as before. Further, there is a potential upside from securitising such assets: If any of them start performing again, or there is a recovery value obtained from defaulted assets, the originator will receive any surplus profit made by the SPV.

Potential Benefits of Securitisation to Investors

In theory there are a number of benefits available to investors from investing in ABS notes, centred mainly on the alternative sectors that they allow investors to diversify into. The potential attractions include:

- Ability to diversify into sectors of exposure that might not be available in the regular bond markets (for example, residential mortgages or project finance loans).
- Access to different (and sometimes superior) risk-reward profiles.
- Access to sectors that are otherwise not open to them.

A key benefit of ABS notes is the ability to tailor risk-return profiles. For example, if there is a lack of assets of any specific credit rating, these can be created via securitisation. Securitised notes sometimes produce a better risk-reward performance than corporate bonds of the same rating and maturity. Although this might seem peculiar (why should one AA-rated bond perform better in terms of credit performance than another just because it is asset backed?), this occurs because the originator holds the first-loss piece in the structure.

A holding in an ABS also diversifies investor risk exposure. For example, rather than invest $100 million in an AA-rated corporate bond and be exposed to event risk associated with the issuer, investors can gain exposure to, for instance, 100 pooled assets. These pooled assets will, in theory, have lower concentration risk, although the experience of 2007–2008 showed that this theoretical diversification of concentration risk did not always occur in practice.

THE PROCESS OF SECURITISATION

We look now at the process of securitisation, the nature of the SPV structure, and issues such as credit enhancements and the cash flow waterfall.

Securitisation Process

The securitisation process involves a number of participants. First there is the *originator*, the firm whose assets are being securitised. The most common process involves an *issuer* acquiring the assets from the originator. The issuer is usually a company that has been specially set up for the purpose of the securitisation, which is the SPV and is usually domiciled offshore. The creation of an SPV ensures that the underlying asset pool is held separate from the other assets of the originator. This is done so that in the event that the originator is declared bankrupt or insolvent, the assets that have been transferred to the SPV will not be affected. This is known as being "bankruptcy remote". Conversely, if the underlying assets begin to deteriorate in quality and are subject to a ratings downgrade, investors have no recourse to the originator.

By holding the assets within an SPV framework, defined in formal legal terms, the financial status and credit rating of the originator becomes almost irrelevant to the bondholders. The process of securitisation often involves *credit enhancements*, in which a third-party guarantee of credit quality is obtained, so that notes issued under the securitisation are often rated at investment grade and up to AAA-grade.

The process of structuring a securitisation deal ensures that the liability side of the SPV—the issued notes—carries lower cost than the asset side of the SPV. This enables the originator to secure lower-cost funding that it would not otherwise be able to obtain in the unsecured market. This is a tremendous benefit for institutions with lower credit ratings.

Figure 10.1 illustrates the process of securitisation in simple fashion. Example 10.1 describes a typical securitisation process.

Mechanics of Securitisation Securitisation involves a true sale of the underlying assets from the balance sheet of the originator. This is why a separate legal entity, the SPV,

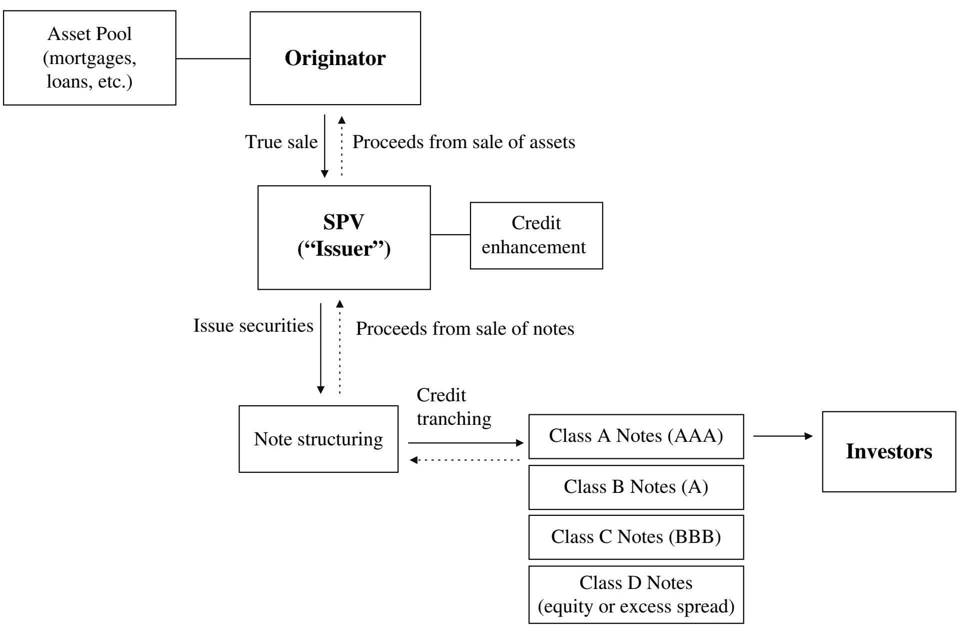

FIGURE 10.1 The Securitisation Process

is created to act as the issuer of the notes. The assets being securitised are sold onto the balance sheet of the SPV. The process involves:

- Undertaking due diligence on the quality and future prospects of the assets.
- Setting up the SPV and then effecting the transfer of assets to it.
- The underwriting of loans for credit quality and servicing.
- Determining the structure of the notes, including how many tranches are to be issued, in accordance to originator and investor requirements.
- The notes being rated by one or more credit rating agencies.
- The placing of notes in the capital markets.

The sale of assets to the SPV needs to be undertaken so that it is recognised as a true legal transfer. The originator will usually hire external legal counsel to advise it in such matters. The credit rating process will consider the character and quality of the assets, and also whether any enhancements have been made to the assets that will raise their credit quality. This can include *overcollateralisation,* which is when the principal value of notes issued is lower than the principal value of assets, and a liquidity facility is provided by a bank.

A key consideration for the originator is the choice of the underwriting bank that structures the deal and places the notes. The originator will award the mandate for its deal to an investment bank on the basis of fee levels, marketing ability, and track record with assets being securitised.

Securitisation Note Tranching As illustrated in Figure 10.1, in a securitisation the issued notes are structured to reflect specified risk areas of the asset pool, and thus are rated differently. The senior tranche is usually rated AAA. The lower-rated notes usually have an element of *overcollateralisation* and are thus capable of absorbing losses. The most junior note is the lowest rated or nonrated. It is often referred to as the *first-loss piece*, because it is impacted by losses in the underlying asset pool first. The first-loss piece is sometimes called the *equity piece* or equity note (even though it is a bond) and is usually held by the originator.

Financial Modelling The hypothetical XYZ Securities will construct a cash-flow model to estimate the size of the issued notes. The model will consider historical sales values, any seasonal factors in sales, credit card cash flows, and so on. Certain assumptions will be made when constructing the model: for example, growth projections, inflation levels, tax levels, and so on. The model will consider a number of different scenarios, and also calculate the minimum asset coverage levels required to service the issued debt. A key indicator in the model will be the debt service coverage ratio (DSCR). The more conservative the DSCR, the more comfort there will be for investors in the notes. For a residential mortgage deal, this ratio might be approximately 2.5 to 3.0; however, for an exotic asset class like an airline ticket receivables deal, the DSCR would be unlikely to be lower than 4.0. The model will therefore calculate the amount of notes that can be issued against the assets while maintaining the minimum DSCR.

Credit Rating It is common for securitisation deals to be rated by one or more of the formal credit ratings agencies such as Moody's, Fitch, or Standard & Poor's.

A formal credit rating will make it easier for XYZ Securities to place the notes with investors. The methodology employed by the ratings agencies takes into account both qualitative and quantitative factors, and will differ according to the asset class being securitised. The main issues in a typical ABS deal include:

- *Corporate credit quality:* These are risks associated with the originator, and are factors that affect its ability to continue operations, meet its financial obligations, and provide a stable foundation for generating future receivables. This might be analysed according to (1) the issuer's historical financial performance, including its liquidity and debt structure; (2) its status within its domicile country, for example whether it is state-owned; (3) the general economic conditions for industry and for airlines; and (4) the historical record and current state of the airline—for instance, its safety record and age of its airplanes.
- *The competition and industry trends:* the issuer's market share, the competition on its network.
- *Regulatory issues,* such as need for the issuer to comply with forthcoming legislation that would impact its cash flows.
- *Legal structure* of the SPV and transfer of assets.
- *Cash flow analysis*.

Based on the findings of the ratings agency, the arranger may redesign some aspect of the deal structure so that the issued notes are rated at the required level.

This is a selection of the key issues involved in the process of securitisation. Depending on investor sentiment, market conditions, and legal issues, the process from inception to closure of the deal may take anything from 3 to 12 months or more. After the notes have been issued, the arranging bank will no longer have anything to do with the issue; however, the bonds themselves require a number of agency services for their remaining life until they mature or are paid off. These agency services include paying agent, cash manager, and custodian.

Credit Enhancement

Credit enhancement refers to the group of measures that can be instituted as part of the securitisation process for ABS and MBS issues so that the credit rating of the issued notes meets investor requirements. The lower the quality of the assets being securitised, the greater the need for credit enhancement. This is often by one of the following methods:

Overcollateralisation: Where the nominal value of the assets in the pool is in excess of the nominal value of issued securities.

Pool insurance: An insurance policy provided by an insurance company to cover the risk of principal loss in the collateral pool. The claims-paying rating of the insurance company is important in determining the overall rating of the issue.

Senior/junior note classes: Credit enhancement is provided by subordinating a class of notes (class B notes) to the senior class notes (class A notes). The class B notes' right to their proportional share of cash flows is subordinated to the rights of the senior noteholders. Class B notes do not receive payments of principal until certain rating agency requirements have been met: specifically, satisfactory performance of the collateral pool over a predetermined period or, in many cases, until all of the senior note classes have been redeemed in full.

Margin step-up: A number of ABS issues incorporate a step-up feature in the coupon structure, which typically coincides with a call date. Although the issuer is usually under no obligation to redeem the notes at this point, the step-up feature was introduced as an added incentive for investors and serves to imply from the outset that the economic cost of paying a higher coupon is unacceptable, and so the issuer will seek to refinance by exercising its call option.

Excess spread: This is the difference between the return on the underlying assets and the interest rate payable on the issued notes (liabilities). The monthly excess spread is used to cover expenses and any losses. If any surplus is left over, it is held in a reserve account to cover against future losses or (if not required for that) as a benefit to the originator. In the meantime, the reserve account is a credit enhancement for investors.

All securitisation structures incorporate a *cash waterfall* process, whereby the cash that is generated by the asset pool is paid in order of payment priority. Only when senior obligations have been met can more junior obligations be paid. An independent third-party agent is usually employed to run tests on the vehicle to confirm that there is sufficient cash available to pay all obligations. If a test is failed, then the vehicle will start to pay off the notes, starting from the senior notes. The waterfall process is illustrated in Figure 10.2.

SECURITISING MORTGAGES

A mortgage is a long-term loan taken out to purchase residential or commercial property, which itself serves as security for the loan. The term of the loan is usually 20 to 25 years, but a shorter period is possible if the borrower, or *mortgagor,* wishes one. In exchange for the right to use the property during the term of the mortgage, the borrower provides the lender, or *mortgagee,* with a *lien,* or claim, against the property and agrees to make regular payments of both principal and interest. If the borrower defaults on the interest payments, the lender has the right to take over and sell the property, recovering the loan from the proceeds of the sale. The lien is removed when the debt is paid off.

A lending institution may have many hundreds of thousands of individual residential and commercial mortgages on its books. When these are pooled together and used as collateral for a bond issue, the result is an MBS. In the U.S. market, certain mortgage-backed securities were backed, either implicitly or explicitly, by the government. A government agency, the Government National Mortgage Association (GNMA, known as Ginnie Mae), and two government-sponsored agencies, the Federal Home Loan Mortgage Corporation and the Federal National Mortgage Association (Freddie Mac and Fannie Mae, respectively), purchased mortgages to

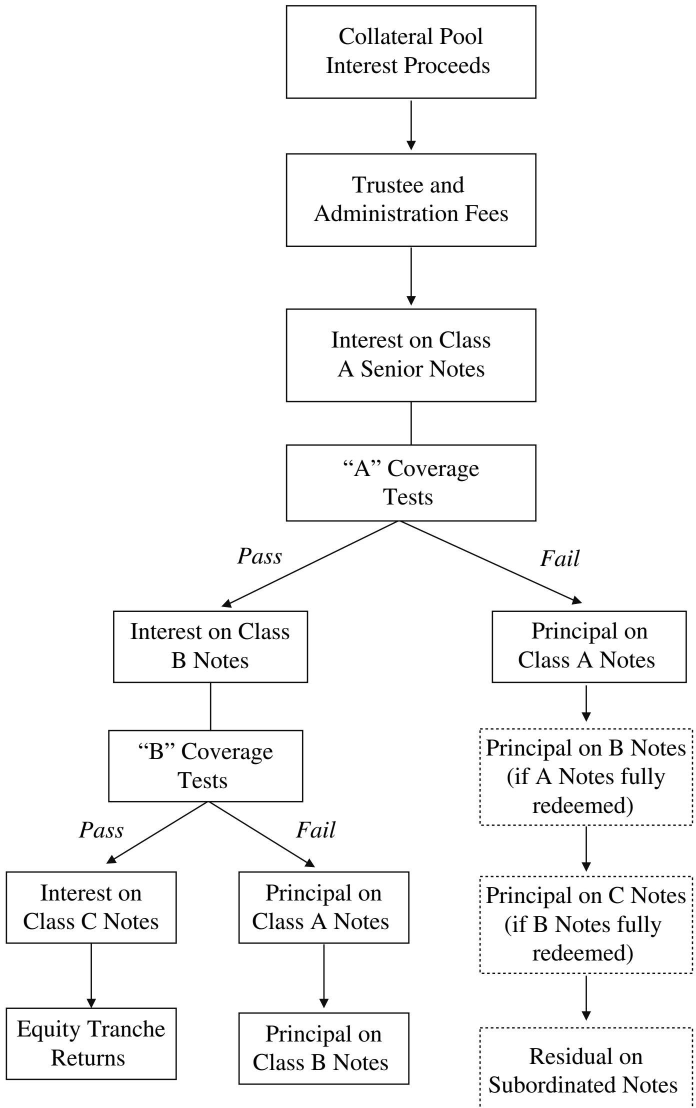

FIGURE 10.2 Cash Flow Waterfall (priority of payments)

pool and hold in their portfolios and, possibly, securitise. The MBSs created by these agencies traded essentially as risk-free instruments and were not rated by the credit agencies. Following the 2007–2008 financial crash, Fannie Mae and Freddie Mac were taken under explicit government control.

Mortgage-backed bonds not issued by government agencies are rated in the same way as other corporates. Some nongovernment agencies obtain mortgage insurance for their issues to boost their credit quality. The credit rating of the insurer then becomes an important factor in the bond's credit rating.

Growth of the Market

We list the following features of mortgage-backed bonds:

- Their yields were traditionally higher than those of corporate bonds with the same credit rating. In the mid-1990s, mortgage-backed bonds traded around 100 to 200 basis points above Treasury bonds; by comparison, corporates traded at a spread of around 80 to 150 for bonds of similar credit quality. This yield gap stems from the mortgage bonds' complexity and the uncertainty of mortgage cash flows. At the height of the structured finance market in 2007, MBS securities rated AAA paid a comfortable 20 to 30 bps over government security. That this was underpriced is reflected in spreads post-2008, which remain higher by some margin over equivalent-rated conventional securities.
- They offer investors a wider range of maturities, cash flows, and security collateral to choose from.
- The market is large and until the 2007 crash was very liquid; agency mortgage-backed bonds had the same liquidity as Treasury bonds. Post-2008 the liquidity reduced considerably.
- Unlike most other bonds, mortgage-backed securities pay monthly coupons, an advantage for investors who require frequent income payments.

EXAMPLE 10.1 SECURITISATION TRANSACTION

We illustrate the impact of securitising the balance sheet under original Basel I regulatory rules using a hypothetical example from ABC Bank PLC.

The bank has a mortgage book of £100 million, and the regulatory weight for this asset is 50%. The capital requirement is therefore £4 million (that is, 8% × 0.5% × £100 million). The capital is comprised of equity, estimated to cost 25%, and subordinated debt, which has a cost of 10.2%. The cost of straight debt is 10%. The ALM desk reviews a securitisation of 10% of the asset book, or £10 million. The loan book has a fixed duration of 20 years, but its effective duration is estimated at seven years, due to refinancings and early repayment. The net return from the loan book is 10.2%.

The ALM desk decides on a securitised structure that is made up of two classes of security, subordinated notes, and senior notes. The subordinated notes will be granted a single-A rating due to their higher risk, whereas the senior notes are rated triple-A. Given such ratings the required rate of return for the subordinated notes is 10.61%, and that of the senior notes is 9.80%. The senior notes have a lower cost than the current balance sheet debt, which has a cost of 10%. To obtain a single-A rating, the subordinated notes need to represent at least 10% of the securitised amount. The costs associated with the transaction are the initial cost of issue and the yearly servicing cost, estimated at 0.20% of the securitised amount (see the accompanying summary information).

EXAMPLE 10.1 (*Continued*)

ABC Bank PLC Mortgage Loan Book and Securitisation Proposal

Current funding	
Cost of equity	25%
Cost of subordinated debt	10.20%
Cost of debt	10%
Mortgage book	
Net yield	10.20%
Duration	7 years
Balance outstanding	£100 million
Proposed structure	
Securitised amount	£10 million
Senior securities:	
Cost	9.80%
Weighting	90%
Maturity	10 years
Subordinated notes:	
Cost	10.61%
Weighting	10%
Maturity	10 years
Servicing costs	0.20%

A bank's cost of funding is the average cost of all the funds it employed. The funding structure in our example is capital 4%, divided into 2% equity at 25%, 2% subordinated debt at 10.20%, and 96% debt at 10%. The weighted funding cost F therefore is:

$$F_{\text{balance sheet}} = (96\% \times 10\%) + [(8\% \times 50\%) \\ \times (25\% \times 50\%) + (10.20\% \times 50\%)] = 10.30\%$$

This average rate is consistent with the 25% before-tax return on equity given at the start. If the assets do not generate this return, the received return will change accordingly, because it is the end result of the bank's profitability. As currently the assets generate only 10.20%, they are performing below shareholder expectations. The return actually obtained by shareholders is such

(*continued*)

EXAMPLE 10.1 (*Continued*)

that the average cost of funds is identical to the 10.20% return on assets. We may calculate this return to be:

$$\text{Asset return} = 10.20\% = (96\% \times 10\%) + 8\% \times 50\% \times (\text{ROE} \times 50\% + 10.20\% \times 50\%)$$

Solving this relationship we obtain a return on equity of 19.80%, which is lower than shareholder expectations. In theory, the bank would find it impossible to raise new equity in the market because its performance would not compensate shareholders for the risk they are incurring by holding the bank's paper. Therefore, any asset that is originated by the bank would have to be securitised, which would also be expected to raise the shareholder return.

The ALM desk proceeds with the securitisation, issuing £9 million of the senior securities and £1 million of the subordinated notes. The bonds are placed by an investment bank with institutional investors. The outstanding balance of the loan book decreases from £100 million to £90 million. The weighted assets are therefore £45 million. Therefore, the capital requirement for the loan book is now £3.6 million, a reduction from the original capital requirement by £400,000, which can be used for expansion in another area, a possible route for which is given here.

Impact of Securitisation on Balance Sheet

Outstanding balances	Value (£m)	Capital required (£m)
Initial loan book	100	4
Securitised amount	10	0.4
Senior securities	9	Sold
Subordinated notes	1	Sold
New loan book	90	3.6
Total assets	90	
Total weighted assets	45	3.6

The benefit of the securitisation is the reduction in the cost of funding. The funding cost as a result of securitisation is the weighted cost of the senior notes and the subordinated notes, together with the annual servicing cost. The cost of the senior securities is 9.80%, whereas the subordinated notes have a cost of 10.61% (for simplicity here we ignore any differences in the duration and amortisation profiles of the two bonds). This is calculated as:

$$(90\% \times 9.80\%) + (10\% \times 10.61\%) + 0.20\% = 10.08\%$$

This overall cost is lower than the target funding cost obtained from the balance sheet, which was 10.30%. This is the quantified benefit of the securitisation process. Note that the funding cost obtained through securitisation is lower than the yield on the loan book. Therefore, the original loan can be sold to the structure issuing the securities for a gain.

ABS STRUCTURES: A PRIMER ON PERFORMANCE METRICS AND TEST MEASURES

This section is an introduction to the performance measures on the underlying collateral of the ABS and MBS product. These would be of most interest to potential investors in ABS notes, but would also be noted by (among others) ratings agencies.

Collateral Types

ABS performance is largely dependent on consumer credit performance, so typical ABS structures include trigger mechanisms (to accelerate amortisation) and reserve accounts (to cover interest shortfalls) to safeguard against poor portfolio performance. Though there is no basic difference in terms of the essential structure between CDO and ABS/MBS, some differences arise by the very nature of the collateral and the motives of the issuer. Interestingly, whereas a CDO portfolio may have 100–200 loans, ABS portfolios will often have thousands of obligors, thus providing the necessary diversity in the pool of consumers.

We discuss briefly some prominent asset classes.

Auto Loans Auto loan pools were some of the earliest to be securitised in the ABS market and still remain a major segment of the U.S. market. Investors traditionally have been attracted to the high asset quality involved and the fact that the vehicle offers an easily sellable, tangible asset in the case of obligor default. In addition, because a car is seen as an essential purchase and a short loan exposure (three to five years) provides a disincentive to refinance, no real prepayment culture exists. Prepayment speed is extremely stable, and losses are relatively low, particularly in the prime sector.

Performance analysis:

- **Loss curves** show expected cumulative loss through the life of a pool and so, when compared to actual losses, give a good measure of performance. In addition, the resulting loss forecasts can be useful to investors buying subordinate classes. Generally, prime obligors will have losses more evenly distributed, while nonprime and subprime lenders will have losses recognised earlier and so show a steeper curve. In both instances, losses typically decline in the latter years of the loan.
- The **absolute prepayment speed** (also abbreviated as **APS**) is a standard measure for prepayments, comparing actual period prepayments as a proportion to the whole pool balance. As with all prepayment metrics, this measure provides an indication of the expected maturity of the issued ABS and, essentially, the value of the call option on the issued ABS at any time.

Credit Cards For specialised credit card banks, particularly in the United States, the ABS market has become the primary vehicle to fund increases in the volume of unsecured credit loans to consumers. Credit card pools are different from other types of ABSs in that loans have no predetermined term. A single obligor's credit card debt is often no more than six months, so the structure has to differ from other ABSs in that repayment speed needs to be controlled either through scheduled amortisation or the inclusion of a revolving period (where principal collections are used to purchase additional receivables).

Since 1991, the stand-alone trust has been replaced with a master trust as the preferred structuring vehicle for credit card ABS. The master trust structure allows an issuer to sell multiple issues from a single trust and from a single, albeit changing, pool of receivables. Each series can draw on the cash flows from the entire pool of securitised assets with income allocated to each pro rata based on the invested amount in the master trust.

Consider the example structure represented by Figure 10.3. An important feature is excess spread, reflecting the high yield on credit card debt. In addition, a financial guarantee is included as a form of credit enhancement, given the low rate of recoveries and the absence of security on the collateral. Excess spread released from the trust can be shared with other series suffering interest shortfalls.

Performance analysis:

- The **delinquency ratio** is measured as the value of credit card receivables overdue for more than 90 days as a percentage of total credit card receivables. The ratio provides an early indication of the quality of the credit card portfolio.
- The **default ratio** refers to the total amount of credit card receivables written off during a period as a percentage of the total credit card receivables at the end of that period. Together, these two ratios provide an assessment of the credit loss on

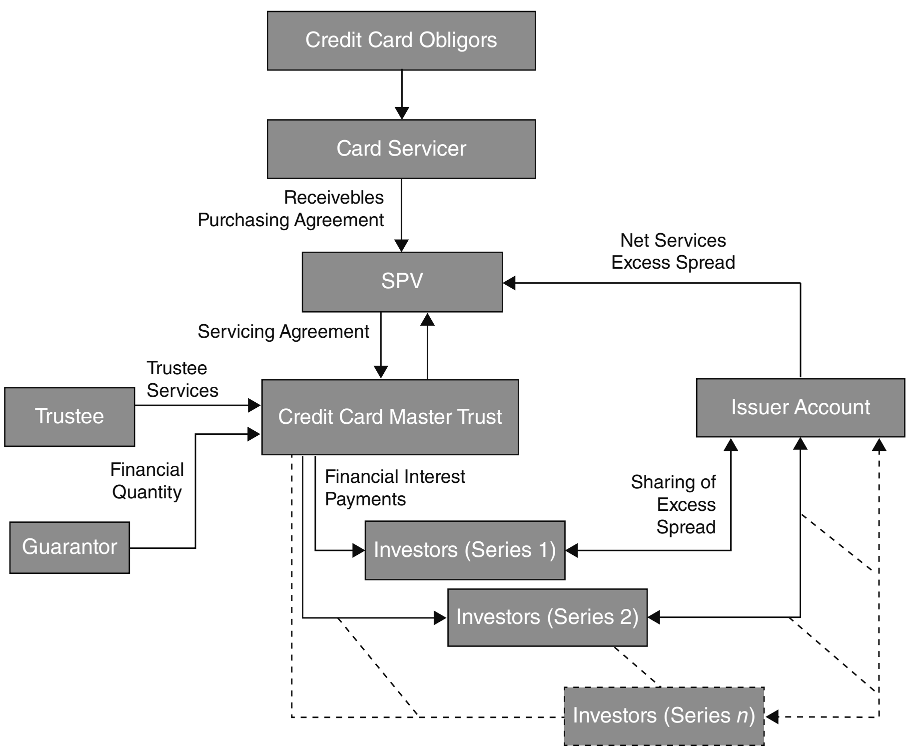

FIGURE 10.3 Master Trust Structure

the pool and are normally tied to triggers for early amortisation and so require reporting through the life of the transaction.

- The **monthly payment rate (MPR)**[3] reflects the proportion of the principal and interest on the pool that is repaid in a particular period. The ratings agencies require every nonamortising ABS to establish a minimum as an early amortisation trigger.

Mortgages The MBS sector is notable for the diversity of mortgage pools that are offered to investors. Portfolios can offer varying duration as well as both fixed and floating debt. The most common structure for agency MBS is pass-through, where investors are simply purchasing a share in the cash flow of the underlying loans. Conversely, nonagency MBS (including CMBS), have a senior and a tranched subordinated class with principal losses absorbed in reverse order.

The other notable difference between RMBS and CMBS is that the CMBS is a nonrecourse loan to the issuer as it is fully secured by the underlying property asset. Consequently, the debt service coverage ratio (DSCR) becomes crucial to evaluating credit risk.

Performance analysis:

- **Debt service coverage ratio (DSCR)** is given by Net operating income/Debt payments, and so indicates a borrower's ability to repay a loan. A DSCR of less than 1.0 means that there is insufficient cash flow generated by the property to cover required debt payments.
- The **weighted average coupon (WAC)** is the weighted coupon of the pool, which is obtained by multiplying the mortgage rate on each loan by its balance. The WAC will therefore change as loans are repaid, but at any point in time, when compared to the net coupon payable to investors, it gives us an indication of the pool's ability to pay.
- The **weighted average maturity (WAM)** is the average weighted (weighted by loan balance) of the remaining terms to maturity (expressed in months) of the underlying pool of mortgage loans in the MBS. Longer securities are by nature more volatile, so a WAM calculated on the stated maturity date avoids the subjective call of whether the MBS will mature and recognises the potential liquidity risk for each security in the portfolio. Conversely, a WAM calculated using the reset date will show the shortening effect of prepayments on the term of the loan.

The **weighted average life (WAL)** of the notes at any point in time is

$$s = \sum t \cdot \mathrm{PF}(s)$$

where $\mathrm{PF}(s)$ = Pool factor at s
t = Actual/365

We illustrate this measure using the example shown in Figure 10.4.

[3] This is not a prepayment measure because credit cards are nonamortising assets.

IPD	Dates	Actual Days (a)	PF(t)	Principal Paid	O/S	a/365	PF(t)*(a/365)
0	21/11/2003	66	1.00		89,529,500.00	0.18082192	0.1(8082192
1	26/01/2004	91	0.94	5,058,824.00	84,470,588.00	0.24931507	0.23522739
2	26/04/2004	91	0.89	4,941,176.00	79,529,412.00	0.24931507	0.22146757
3	26/07/2004	91	0.83	4,823,529.00	74,705,882.00	0.24931507	0.20803536
4	25/10/2004	91	0.78	4,705,882.00	70,000,000.00	0.24931507	0.19493077
5	24/01/2005	91	0.73	4,588,235.00	65,411,765.00	0.24931507	0.18215380
6	25/04/2005	91	0.68	4,470,588.00	60,941,176.00	0.24931507	0.16970444
7	25/07/2005	91	0.63	4,352,941.00	56,588,235.00	0.24931507	0.15758269
8	24/10/2005	92	0.58	4,235,294.00	52,352,941.00	0.25205479	0.14739063
9	24/01/2006	90	0.54	4,117,647.00	48,235,294.00	0.24657534	0.13284598
10	24/04/2006	91	0.49	4,000,000.00	44,235,294.00	0.24931507	0.12318314
11	24/07/2006	92	0.45	3,882,353.00	40,352,941.00	0.25205479	0.11360671
12	24/10/2006	92	0.41	3,764,706.00	36,588,235.00	0.25205479	0.10300784
13	24/01/2007	90	0.37	3,647,059.00	32,941,176.00	0.24657534	0.09072408
14	24/04/2007	91	0.33	3,529,412.00	29,411,765.00	0.24931507	0.08190369
15	24/07/2007	92	0.29	3,411,765.00	26,000,000.00	0.25205479	0.07319849
16	24/10/2007	92	0.25	3,294,118.00	22,705,882.00	0.25205479	0.06392448
17	24/01/2008	91	0.22	3,176,471.00	19,529,412.00	0.24931507	0.05438405
18	24/04/2008	91	0.18	3,058,824.00	16,470,588.00	0.24931507	0.04586606
19	24/07/2008		–	16,470,588.00	—	—	—
						WAL	2.57995911

FIGURE 10.4 Sample Weighted Average Life (WAL) Calculation

It is the time-weighted maturity of the cash flows that allows potential investors to compare the MBS with other investments with similar maturity. These tests apply uniquely to MBS because principal is returned through the life of the investment on such transactions.

Forecasting prepayments is crucial to computing the cash flows of MBS. Though the underlying payment remains unchanged, prepayments, for a given price, reduce the yield on the MBS. There are a number of methods used to estimate prepayment; two commonly used ones are the constant prepayment rate (CPR) and the PSA method.

The CPR approach is:

$$CPR = 1 - (1 - SMM)^{12}$$

where **single monthly mortality (SMM)** is the single-month proportional prepayment.

An SMM of 0.65% means that approximately 0.65% of the remaining mortgage balance at the beginning of the month, less the scheduled principal payment, will prepay that month.

The CPR is based on the characteristics of the pool and the current expected economic environment, as it measures prepayment during a given month in relation to the outstanding pool balance.

The **public securities association (PSA)** has a metric for projecting prepayment that incorporates the rise in prepayments as a pool seasons. A pool of mortgages is said to have 100% PSA if its CPR starts at 0 and increases by 0.2% each month until it reaches 6% in month 30. It is a constant 6% after that. Other prepayment scenarios can be specified as multiples of 100% PSA. This calculation helps derive an implied prepayment speed assuming mortgages prepay slower during their first 30 months of seasoning. That is,

$$PSA = [CPR/(0.2)\ (m)]*100$$

where m = number of months since origination

Summary of Performance Metrics

Table 10.1 lists the various performance measures we have introduced in this chapter and the asset classes to which they apply.

TABLE 10.1 Summary of ABS Analysis and Performance Metrics

Performance Measure	Calculation	Typical Asset Class
Public Securities Association (PSA)	PSA = [CPR/(0.2) (months)]*100	Mortgages, home equity, student loans
Constant prepayment rate (CPR)	$1 - (1 - SMM)^{12}$	Mortgages, home equity, student loans
Single monthly mortality (SMM)	Prepayment/Outstanding pool balance	Mortgages, home equity, student loans
Weighted average life (WAL)	$\sum$ (a/365).PF(s) where PF(s) =	Mortgages
Weighted average maturity (WAM)	Weighted maturity of the pool	Mortgages
Weighted average coupon (WAC)	Weighted coupon of the pool	Mortgages
Debt service coverage ratio (DSCR)	Net operating income/Debt payments	Commercial mortgages
Monthly payment rate (MPR)	Collections/Outstanding pool balance	All nonamortising asset classes
Default ratio	Defaults/Outstanding pool balance	Credit card
Delinquency ratio	Delinquents/Outstanding pool balance	Credit card
Absolute prepayment speed (ABS)	Prepayments/Outstanding pool balance	Auto loans, truck loans
Loss curves	Show expected cumulative loss	Auto loans, truck loans

THE SECURITISATION MARKET POST-2007

In this section we look briefly at the market in the immediate postcrash era and discuss salient features of interest to investors.

Market Observation

A standard feature during the recovery in an economic cycle is that risk aversion decreases once the worst of the recession is over. For example, during 2009, UK AAA-rated residential MBS spreads tightened over 300 bps and were trading at 180 bps over Libor. This is considerably below the yields seen only six months previously. Figure 10.5 shows UK residential mortgage-backed securitisation (RMBS) spreads over alternative bank funding sources (straight bank debt rated AA and bank-covered bonds) for the period 2005–2009.

For other asset classes, the spreads stayed wide into 2009. Figure 10.6 is the Bloomberg DES page for the AAA-rated senior tranche of a European collateralised loan obligation (CLO), Columbus Nova. We see that this was issued in August 2006 at par with a spread of 26 bps over three-month Libor. In September 2009, this same paper was offered at a price of $87.00, a spread of over 400 bps over Libor.[4]

The tranches of higher-quality underlying assets were trading at better spreads. On 3 September 2009, UK prime RMBS tranches were offered at prices still in the upper 90s, as shown in Table 10.2A Note, however, that the Granite issue, which was the vehicle of Northern Rock PLC, was being offered at a considerably lower price of €81.00.[5] All

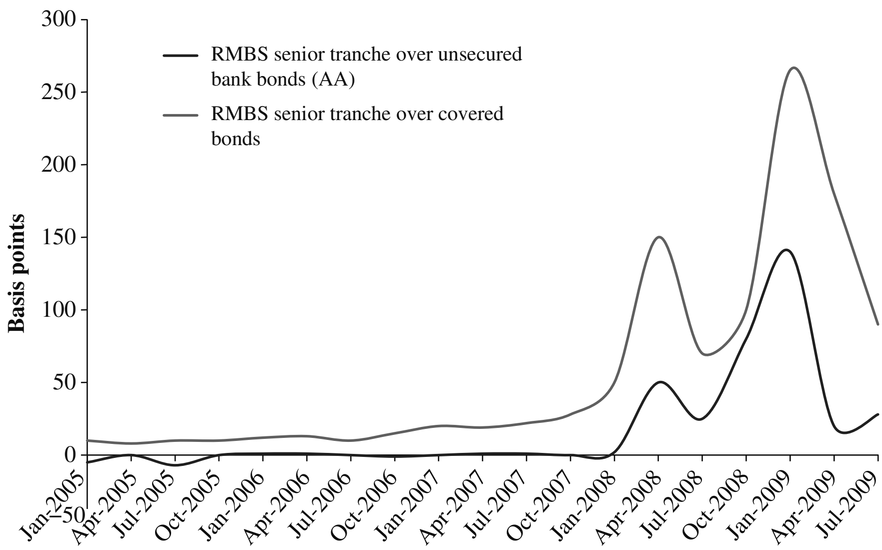

FIGURE 10.5 UK RMBS Spreads over Bank Debt and Covered Bonds
Source: Bloomberg L.P.

[4] Source: Communication from Deutsche Bank market-making desk.

[5] Northern Rock PLC, a commercial bank with its main business line in prime and subprime residential mortgages, suffered liquidity difficulties in August 2007 and was subsequently nationalised by the UK government.

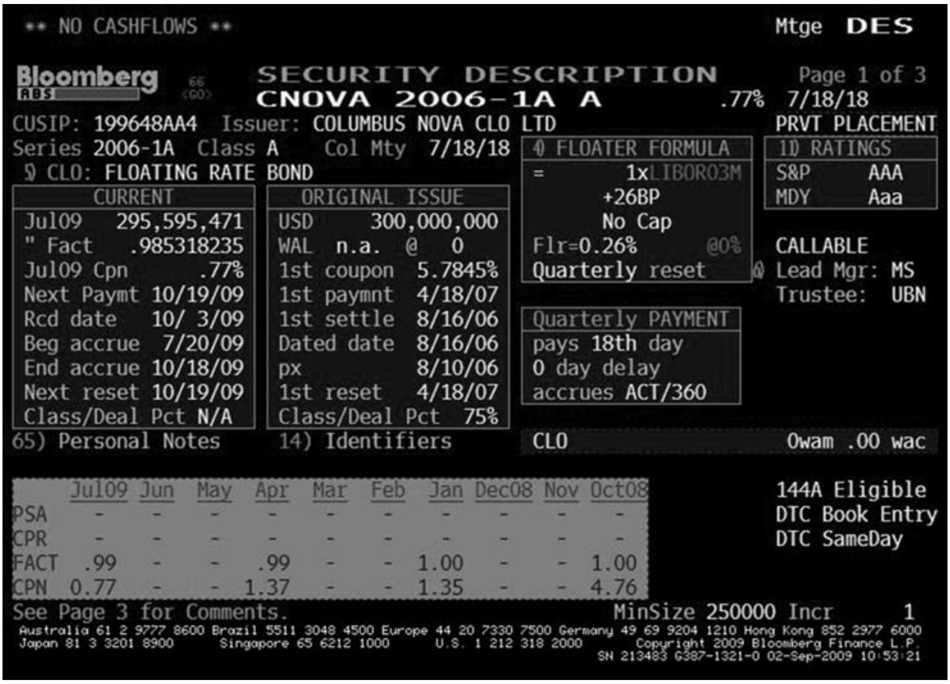

FIGURE 10.6 Bloomberg Page DES for European Bank CLO

these are EUR-denominated securities. Contrast this with Table 10.2B, which shows U.S. market commercial mortgage-backed securities (CMBS) yield spreads during August 2009. The commercial property sector in the United States was still suffering badly from the effects of the subprime fallout and the recession; consequently, even AAA-rated CMBS spreads remained high. In addition, USD-denominated securities generally paid a higher spread than EUR-denominated bonds of all classes.

In the postcrash market in the United States, the most common type of issuance was for auto-loan and credit card underlying. Figure 10.7 is a breakdown of ABS issuance in 2009. Note the virtual disappearance of the collateralised debt obligation

TABLE 10.2A Market-Maker's Offer Page, UK RMBS Senior Tranches, September 2009

Name	Rating	Currency	Spread (bps)	Price	WAL (years)	Factor	Issuer
PERM 2006-1	AAA/Aaa	EUR	285	94.73	2.0	1.00	HBOS PLC
LOTHM 2006-1X	AAA/Aaa	EUR	190	97.64	1.3	0.91	Standard Life
HMI 2007-1	AAA/Aaa	EUR	190	97.15	1.6	1.00	Abbey National
HFP 9	AAA/Aaa	EUR	185	99.18	0.5	1.00	Abbey National
GMFM 2007-1X	AAA/Aaa	EUR	190	95.90	2.3	1.00	Barclays Bank
ARKLE 2006-1X	AAA/Aaa	EUR	190	95.36	2.6	1.00	Lloyds TSB
GRANM 2005-2	AAA/Aaa	EUR		81.00		0.63	Northern Rock

Source: Market Makers page.

TABLE 10.2B U.S. CMBS Spreads to One-Month Libor, August 2009

Tranche	Spread
AAA	1,325
AA	3,250
A	4,250
BBB	5,400

[1-mo Libor 0.3%]
Source: Market Makers page.

(CDO) asset class. Figure 10.8 illustrates the impact on note spreads following the financial crisis.

Lower spreads provide a more feasible environment in which banks could start to reissue ABS again, as the cost of funding started to approach that of straight (unsecured) bank debt. As was the case prior to the crash, the economic rationale for banks to issue ABS over alternative sources of funding remains one of asset yield over liability yield. To make the case for ABS, banks must be able to place all tranches, and not just the AAA senior piece, into the market. In fact, under the postcrash regulatory regime, banks will need to do this if they are to secure any regulatory capital relief as a result of the deal (that is, they need to have transferred the entire credit risk to a third party).

During 2009, many bank assets (such as mortgages and student loans) did not yield sufficient margin to support a full capital structure ABS issue. Of course, a bank may wish to close a deal for other, more strategic reasons, such as encouraging investor interest and helping to reenergise the market, or to create collateral for posting at the central bank. A prime attraction, as was always the case, is that an ABS deal also

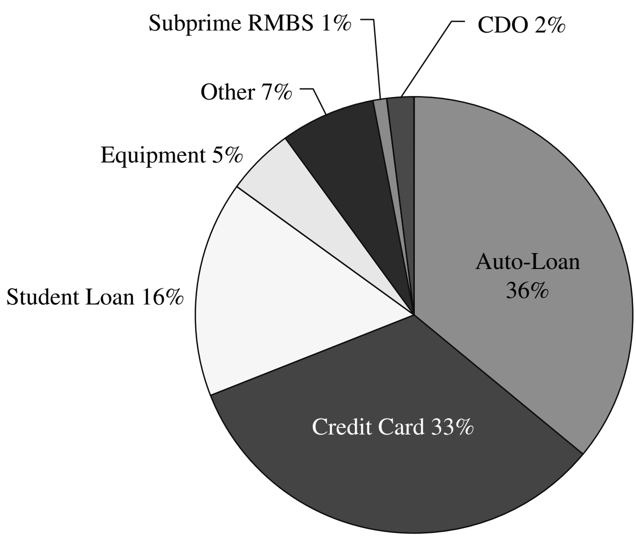

FIGURE 10.7 U.S. ABS Sector Breakdown, 2009
Source: Thomson Financial Securities Data.

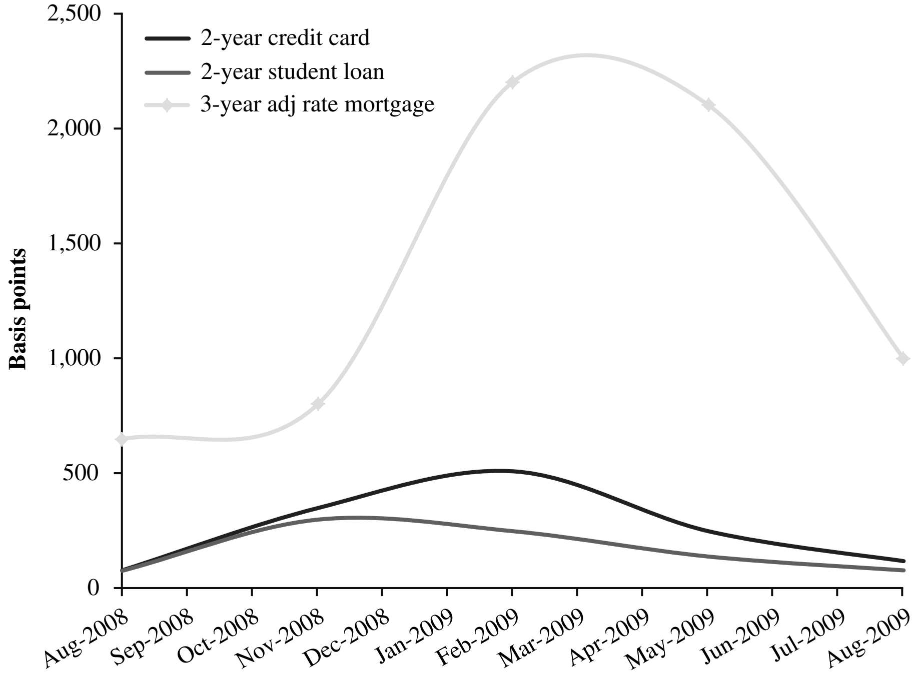

FIGURE 10.8 U.S. Floating-Rate Spreads, 2008–2009
Source: Bloomberg L.P. (yield).

diversifies funding sources for a bank, which is a key requirement of bank regulators such as the UK's Financial Services Authority (FSA) in the post-Lehman era.

Impact on Rating Agencies

The status and reputation of the three main credit rating agencies (CRAs) suffered grievously in the wake of the 2007–2008 financial crisis. This issue caused many former ABS investors to refrain from reentering the market. As a result, the agencies reviewed their rating methodologies and criteria, as well as their base case assumptions. Chapter 4 in Baig and Choudhry (2013) looks in detail at Moody's methodology for ABS and CDO products, and we also review their main ratings checklist process that ABS originators need to follow ahead of the ratings review. The immediate impact of the review process was that the CRAs revised their performance and loss assumptions to take into account actual losses that occurred in the recession. The CRAs also took steps to publish more transparent disclosures as part of their ratings process.

The stability of credit ratings was most heavily damaged in the mortgage market sector. Investors in subprime and prime RMBS, CMBS, and CDOs of ABS (in which the underlying included MBS) observed the greatest deterioration in ratings during 2008–2009. CRAs publish ratings transition tables detailing the percentage of securities that were still the same rating after one year (or longer). Moody's also publishes downgrade rates, which are defined as the number of securities downgraded divided by the total number of outstanding securities at the beginning of the observation period.

In a report from 2009, Moody's reported that 7.38% of all the structured finance securities that it rated were downgraded in the period 1999–2008. This is a surprisingly low figure. If CDOs of ABS, home equity ABS, and RMBS are excluded, the figure is 3.18%.[6] For 2008 only, however, the downgrade rate was 35.5%, and 12.1% when mortgage-related securities are excluded.

In response to the crisis and criticism of their methodology, the CRAs modified their approach. However, in essence their methodology remained in place, and changes were largely technical in nature. The definition of a rating itself did not change, but certain underlying assumptions were changed to reflect the observations from 2007 to 2008. In addition, all CRAs undertook to provide greater transparency and disclosures as part of the ratings process.

One significant change that was adopted was that the ratings process now considers the liquidity and funding scenario for the originating or sponsoring entity. This had not been reviewed prior to the crash. The impact of this change is material, because the CRAs now take into account the liquidity position of the originator. Sponsors that are overly reliant on the ABS market, or have few alternative sources for their funding, are required to provide a higher level of credit enhancements for the structure. In some cases, the top AAA/Aaa rating is not available to such sponsors.

Following is a summary comparison of the new approaches of each of the three main CRAs.

Moody's Investors Service According to its website, a Moody's rating is a qualitative description of a quantitative measure of relative creditworthiness of a security. It is a forward-looking, 12-month risk assessment. The analysis undertaken to arrive at a rating is both qualitative and quantitative, although for a structured finance security the qualitative review is limited in comparison. The rating itself represents a probability of the likelihood of a security making full and timely payment of its principal and interest liabilities.

Moody's rating approach is essentially unchanged from its precrisis model, but has been modified in certain aspects of its detail. Greater weight is now given to operational features of the structure, the relationship of the structure with the sponsor, and the sponsor's access to alternative sources of liquidity. Based on this approach, some asset-class ABS products are now no longer assigned an Aaa rating, irrespective of the amount of credit enhancement that is built into the vehicle; this has been observed, for example, with specific types of auto ABS securities, as well as certain credit card issuers and equipment lease deals.[7]

Included in the new Moody's approach is the provision of V-scores and parameter sensitivities in its ratings report. The V-score is a measure of the potential variability of the inputs to the ratings process, and is designed to highlight the risk

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[6] See Moody's Investors Service, *Structured Finance Rating Transitions: 1983–2008*, Special Report, March 2009.

[7] Some of the so-called credit card banks had relied excessively on the securitisation market to raise finances and grow their balance sheet. For such firms, the lack of alternative sources of finance means that their ABS product is now unlikely to be assigned an Aaa rating unless the sponsor arranges for funding diversity.

of such fluctuations to investors. Parameter sensitivities are a sensitivity analysis of these rating inputs.

Standard & Poor's An S&P rating is also a qualitative measure of relative creditworthiness of a security within the rated population; however, it is also specifically a measure of the probability of default. The S&P review process also includes payment priority, recovery rate, and credit stability. Since the 2007 crisis, S&P has emphasised that its ratings are comparable across security type and asset class, and that they represent an ability to withstand specific macroeconomic scenarios. An AAA-rated bond in theory would not default in an economic scenario similar to the 1930s depression, whereas a BBB-rated bond would survive an equity market fall of 50% and unemployment rate of 10% (statistics that are similar to those observed in the 2007–2008 crash).

Fitch Similar to the other CRAs, a Fitch rating is a measure of relative creditworthiness, with an emphasis (as with S&P) on probability of default. Fitch has also adopted a similar approach to Moody's in taking into account the liquidity and funding position of an originating entity when assigning the rating.

Alone among the CRAs, Fitch publishes an outlook for each rating, which is a medium-term view of rating stability alongside the 12-month formal rating. Two additional measures introduced by Fitch are loss severity ratings and recovery ratings. Loss severity ratings measure the adequacy of the size of the tranche under specified loss scenarios. The recovery rating is a measure of recovery value. Also unique to Fitch was the decision not to award the highest AAA rating to transactions that were deemed to be market value deals, which, broadly speaking, are those whose performance is linked to an external index or the trading value of underlying assets.

SPECIAL PURPOSE VEHICLES (SPVs)

What Is an SPV and How Is It Documented?

A special purpose vehicle (or special purpose entity) is a legal entity, usually a limited company, created to fulfil narrow, specific, or temporary objectives.

Usually, when a new company is set up, the sponsor (i.e., person/company who created the new company) is the same person as the owner. SPVs are explicitly designed so that this is not the case. In the cases below, the bank sets up the SPV, and is thus its sponsor. However, the shares in the SPV are owned by a third party, usually a charitable trust. Thus the directors of the SPV, who represent its shareholders, are independent to the bank and its directors.

Because the bank has no legal ownership over the SPV, all exchanges between the two can be treated as arm's-length transactions, and the SPV does not need to be consolidated in the bank's accounts. If the SPV purchases an asset and creates a new liability, these transactions are not included in the bank's balance sheet. If the bank defaults, then any assets owned by the SPV do not become pooled with the rest of the bank's assets to pay off its creditors. Thus, any investor in the SPV is largely insulated from any credit risk to the bank. There will be some residual credit risk to the bank only if there are uncollateralised transactions between the bank and the SPV, resulting in money being owed by the bank upon its default.

In practice, banks that sponsor transactions involving SPVs are required to "consolidate" the accounts of the SPV into their group financial statements. Only if 100% of the risk associated with the SPV (termed "significant risk transfer" by the UK regulatory authority, the PRA) is moved off the balance sheet of the sponsoring bank can a case be made for the accounts of the SPV not to be so consolidated.

Individual sets of transactions with the SPV can be compartmentalised under separate International Securities and Derivatives Association (ISDA) Master Agreements. Thus, each set of transactions with the SPV can be treated as being with a separate company, since a default on a payment on one set of transactions does not trigger a default on another set of transactions. Furthermore, a default on the payments in one transaction will not create a claim against another transaction.

The transactions that are done with the SPV are designed to be pass-through. Every cash flow into the SPV is matched by an identical flow out of the SPV. If the SPV purchases an asset to hold, then external funding must be provided to purchase the asset, and the proceeds from its future sale will be paid to an external legal entity. This means that, with the exception of any assets being held on the above basis, the SPV is cash-flat, with no net outstanding assets or liabilities.

The SPV, like any other company, defaults if it cannot pay its debts as they fall due. However, as each transaction is ring-fenced, a default on one transaction will only result in that transaction being wound up. All other transactions will continue as before. In the examples below, the transactions with the SPV are of a specific form. An investor provides the SPV with cash, which is used to purchase an asset (a bond) issued by a third party. The bank then enters into a collateralised or uncollateralised swap with the SPV, where the bank receives the coupons from the asset, and exchanges them for a different set of coupons. These new coupons are passed on to the investor. A transaction of this sort is called a "repack", because an existing asset is repackaged, by changing its coupons, to suit the risk appetite of the investor.

If the third-party bond issuer defaults, then the SPV will no longer receive the coupons on the bond. If the bank defaults, it will no longer receive the coupons on the repack swap. Sometimes, the repack swap will reference other entities (like a CDS), and thus a default on one of the reference entities will also result in the coupons on the swap ceasing (and usually a windfall cash payment to the protection buyer—either the bank or the SPV—on the swap). All these events constitute a default event that will result in the windup of the ring-fenced set of transactions. This default triggers the "waterfall".

The waterfall is a legal clause that determines the order of priority in the event of a default. It will state which debts get paid off first. In the above example, the SPV holds a bond (which may be in default and worth its recovery rate) as an asset. It owes money to the investor, and either owes money to, or is owed money by, the bank. After receiving any money it may be owed on the swap with the bank, the SPV liquidates its asset (the bond), and proceeds to pay off its creditors, in the order of priority specified by the waterfall. Usually, the waterfall states that any tax liabilities and legal costs must be paid off first. Then the swap counterparty (the bank) and the investor are paid off in some order. Finally, after all debts have been settled, if there are any residual funds remaining, these get paid to the shareholders, that is, the charitable trust. Usually there would be no funds remaining to be paid out under this clause.

In the following discussions, we look at all the risks on the transactions with the SPV from the point of view of the bank. The SPV itself, being a pass-through vehicle, is entirely risk-flat and can never be left with a net profit or loss. To work out the risks and returns from the investor's point of view, we can thus say:

$$\text{Investor's PV} = \text{Bond PV} - \text{Bank's swap PV}$$

Impact of the Waterfall

The wording of the waterfall is very important because it determines the cash flows to the bank (and investor) in the event of the collapse of the SPV. There are two main ways the waterfall can be structured; we shall consider the pricing impact of both.

1. The bank is senior to the investor, regardless of what triggered the collapse of the SPV.
2. The bank is senior to the investor, unless it was the default of the bank itself that triggered the collapse of the SPV. If it was the bank that defaulted first, then the investor becomes senior to the bank.

There is also the question of what would happen in the rare case where, following the SPV's collapse, the assets held by the SPV are worth more than the amount required to pay off the bank and the investor. Normally, in this instance, the surplus would pass to the investor. As this is an unlikely occurrence anyway, we shall not consider the impact of allowing the bank to keep the windfall.

Terms of the Transactions with the SPV

In the following sections, we give some examples of typical transactions that SPVs are used for, and analyse them in detail. All these trades share a common basis.

The investor puts upfront cash into the SPV. The SPV uses this cash to buy some bonds (either new or existing issues). It then enters into a swap with a counterparty (usually the bank that set up the SPV); the SPV passes the bond cash flows to the counterparty, and, in return, receives a different coupon. This SPV issues a bond to the investor, and passes on to him the swap coupons. At the final maturity of the transaction, the swap terminates, and the face value of the bond is received by the SPV and passed on to the investor.

The transaction may terminate early for a number of reasons:

- If the bond purchased by the SPV defaults.
- If the swap counterparty (the bank) defaults.
- If a predefined trigger event occurs (e.g., the SPV owes more collateral to the swap counterparty than the market value of the bond, or if a market event happens that was named as a termination event of the swap).

Any termination event would trigger the waterfall, described in the previous section.

The SPV-issued note is uncollateralised, although the investor may receive a non-zero recovery amount upon a termination event, depending on the mark-to-market of the bonds held by the SPV, and the swap with the bank.

The swap itself may be collateralised or uncollateralised. If it is uncollateralised, then the SPV, and thus the investor, is probably taking on some credit risk to the bank itself (except in rare cases where the swap MtM will always be in the favour of the bank, due to a large upfront payment). If it is collateralised, then the terms of the CSA between the SPV and the bank could take on various forms.

The bank could post collateral in a one-way agreement. This would be the most expensive option for the Investor, as this agreement is the least favourable for the bank, and thus the bank will pay a much lower swap coupon as a result. If the bank is senior in the waterfall, the investor would not gain anything from not having to post collateral to the bank, as the bank would have first claim upon the SPV's assets in the event of a windup. We could argue that, even under this one-way CSA, because the bank is senior in the waterfall, the trade is, after accounting for the tail risk that the assets of the SPV are lower in value than the swap MtM, effectively two-way collateralised. However, the bank still has to charge for the real-world funding cost of the uncollateralised swap. This real-world cost occurs because the bank almost certainly cannot borrow money at the risk-free rate, collateralised by the value of its trade with the SPV (i.e., it cannot "repo" the SPV!).

This funding cost can be mitigated by creating a two-way CSA, under which the SPV posts assets (i.e., bonds) to the Swap counterparty as collateral. If there is a repo market for these bonds, then the bank can use this to fund the trade, and should discount the swap cash flows at the repo rate.

In the case of a two-way CSA, a situation could arise where the MtM of the swap between the bank and the SPV is in the bank's favour by more than the MtM of the SPV's assets. Depending on the wording of the CSA documentation, this could count as either a windup event for the SPV, or be ignored. If it counts as a windup event, then the tail risk that the bank would have been exposed to upon the default of the bonds held by the SPV is removed (except for a small amount of gap risk). If it does not count as a windup event, then the SPV simply stops posting collateral, and the bank is exposed to the full tail-risk that the collateralised MtM of the swap exceeds the recovery value of the bonds upon their default.

In Chapter 21 we argue that we could ignore any tail-risk upon the bank's own default, by making the conservative assumption that the SPV would apply offset against the bank, even though this is not actually possible in general. However, if the swap is collateralised, then the bank's own credit risk has been ignored in pricing, and it has been implicitly assumed that the bank receives the full MtM of the swap upon its own default (as the collateral has already been posted). So in this case, there appears a tail risk that when the SPV is wound up by a default by the bank, the MtM of the SPV's bonds is insufficient to cover the MtM of the swap. This second tail-risk could only ever be a cost to the bondholders and not to the shareholders.

In general, a two-way CSA is likely to be advantageous to both the bank and the Investor: the Investor no longer has credit risk to the bank, and the bank no longer has a real-world funding cost on the swap. There will be, however, little or no value to the bank if the collateral has no rehypothecation value. In that case, the only potential value to the bank would be if the SPV was dissolved when the swap MtM exceeded the collateral value.

SPV Structures

Amortising Structures Amortising structures pay principal and interest to investors on a coupon-by-coupon basis throughout the life of the security, as illustrated in

Figure 10.9. They are priced and traded based on expected maturity and weighted-average life (WAL), which is the time-weighted period during which principal is outstanding. A WAL approach incorporates various prepayment assumptions, and any change in this prepayment speed will increase or decrease the rate at which principal is repaid to investors. Pass-through structures are commonly used in residential and commercial mortgage-backed deals (RMBS and CMBS) and consumer loan ABS.

Revolving Structures Revolving structures revolve the principal of the assets; that is, during the revolving period, principal collections are used to purchase new receivables that fulfill the necessary criteria. The structure is used for short-dated assets with a relatively high prepayment speed, such as credit card debt and auto loans. During the amortisation period, principal payments are paid to investors in a series of equal installments (*controlled amortisation*) or principal is "trapped" in a separate account until the expected maturity date and then paid in a single lump sum to investors (*soft bullet*).

Master Trust Frequent issuers under U.S. and UK law use *master trust* structures, which allow multiple securitisations to be issued from the same SPV. Under such schemes, the originator transfers assets to the master trust SPV. Notes are then issued out of the asset pool based on investor demand. Master trusts have been used by MBS and credit card ABS originators.

Repackaging Structures Involving an SPV

In this section, we consider a number of the most common repack structures that exist in the market, their risk, and their pricing. This list is by no means exhaustive. If the pricing of these examples is fully understood, however, then the reader should be able to apply the same principles to price most other realistic examples.

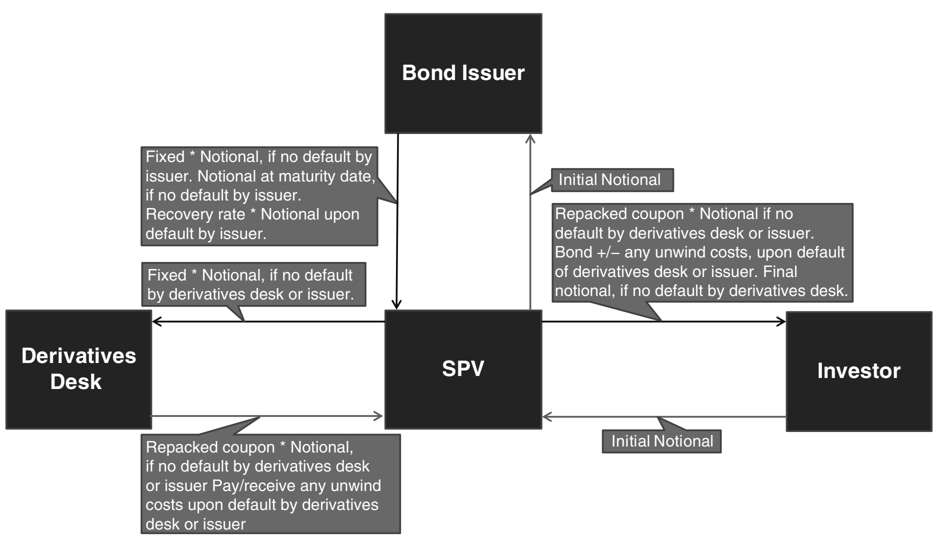

FIGURE 10.9 Vanilla Bond and Repackaging Structure

The principal motivation for the investor to do the transactions with the SPV instead of the bank is that, while the investor may like the risk profile of the repack swap offered by the bank, he may not be comfortable with taking credit risk to the bank on his entire investment. By transacting with the SPV, he gets access to the bank's financial structured products without being exposed to the full credit risk of the bank. Most/all of his credit exposure is to the issuer of the asset purchased by the SPV. The principal advantage for the bank is that it gets to offer its financial products to an investor, without bloating its balance sheet by creating additional assets and liabilities.

Vanilla Bond Repacked to Have a Different Coupon As illustrated in Figure 10.10 in the cash flow diagram, we have assumed that the bond purchased is a new issue, priced at 100. The same structure would work with a new or existing bond priced above or below 100. Either the investor would initially pay in the MtM of the bond, or he would still pay in 100, and the derivatives desk would pay/receive the difference.

The impact of a default depends on whether there is a CSA between the derivatives desk and the SPV or not. First of all, let us assume there is no CSA, and that the derivatives desk is senior in the Waterfall when the bond defaults.

On day one:

- The Investor pays the notional to the SPV.
- The SPV uses the notional to buy a third-party bond.

If there is no default by either the bond issuer or the derivatives desk, then:

- The SPV receives the fixed coupon on the bond.
- The SPV pays the fixed coupons to the derivatives desk.
- The derivatives desk pays the repack coupon to the SPV.
- The SPV passes on the repack coupon to the investor.

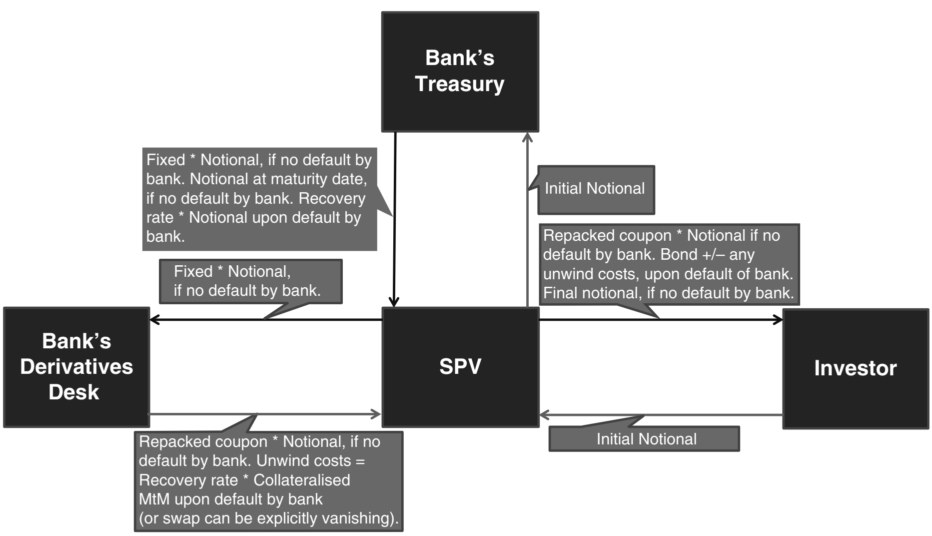

FIGURE 10.10 Repackaging Bank Own Debt

At the maturity of the bond, if there have been no default events, then:

- The SPV receives the bond's face value from the bond issuer.
- The SPV passes the face value to the Investor.
- The swap terminates.

If there is a default by the bond issuer, then:

- The SPV liquidates the bond, in return for its recovery value in cash.
- The uncollateralised MtM of the swap (i.e., including the pricing impact of the bank's funding curve) is exchanged between the SPV and the derivatives desk. Any payment to the derivatives desk is capped by the liquidation value of the bond.
- The SPV passes the remaining cash in the SPV to the investor, and the transaction ends.

If there is a default by the derivatives desk, then the collateralised mark-to-market of the credit linked swap between the Swap counterparty and the SPV is calculated.

If the collateralised swap MtM is in favour of the SPV then:

- The SPV claims the collateralised MtM from the derivatives desk, and receives their recovery rate multiplied by the collateralised MtM.
- The SPV pays the above sum to the Investor.
- The SPV passes ownership of the bond to the Investor, and the transaction ends.

If the collateralised swap MtM is in favour of the swap counterparty, then the payout depends on who is senior in the waterfall.

- If the swap counterparty is senior, then the bond is liquidated, the derivatives desk is paid in full, and any remaining cash is returned to the investor.
- If the investor is senior, then the Investor receives the bond, and the derivatives desk receives nothing.

If there is a CSA between the SPV and the derivatives desk, then there are some small changes to the above. A default by the derivatives desk is no longer an economic event (ignoring any gap risk), and the SPV would wind up with the investor getting the mark-to-market of the bond plus swap. If the fact that the bond mark-to-market was insufficient to cover the MtM of the swap was counted as a termination event, then the SPV would wind up at this moment, with the derivatives desk keeping the MtM of the swap, and the investor receiving the MtM owed to him; that is, nothing. A default by the bond would have the same consequences as above, except that the derivatives desk could not be exposed to any tail risk that the swap's MtM was not covered by the recovery on the bond (ignoring any gap risk).

If we can price the structure from the derivatives desk's point of view, then we know that, because the SPV is a pass-through vehicle that cannot end up with any assets upon termination, the value to the Investor is equal to the bond's value less the derivatives desk's value.

If the swap is collateralised, with a termination trigger when the collateral held by the SPV becomes insufficient to pay for the mark-to-market, then the swap has a regular collateralised value. Assuming that the CSA is symmetrical, and thus both sides post the same bond as held by the SPV as collateral, then the correct risk-free discount rate to use is the repo rate of that bond. In practice, the derivatives desk will be allowed to post cash instead of the bond, if it wants to, so that possible future illiquidity of the bond cannot present a problem. The desk will probably prefer to post the bond itself if possible, as the repo rate will usually be greater than OIS, which is beneficial to the party who has to post collateral, as they get to discount their future liability at a higher rate.

If the swap is uncollateralised, then the most practical approach is to discount it at the bank's (i.e., derivatives desk's) own funding rate. Ignoring any tail risk, the swap is, in theory, still collateralised by the fact that the Derivatives desk will have a claim upon the bond held by the SPV upon a default event. It is also not possible for the SPV to buy the bank's own debt, and thus offset does not apply upon a default by the bank. However, as the bank cannot rehypothecate the assets held by the SPV, it is still faced with a real-world funding cost to its shareholders. Thus, it is simpler to price the swap including the effect of its own credit spread.

Finally, the tail risk upon a default event has to be priced. Upon the bank's default, we have implicitly assumed, by using the bank's credit spread in pricing, that the bank only has a claim of collateralised MtM * recovery rate. This is conservative, as we said that offset is not possible. It is so unlikely that the MtM of the bond is less than this amount that any tail risk upon the bank's default can be ignored. Even if we are wrong, this tail risk is only a cost to the bondholders, not the shareholders.

Upon the bond's default, the bank is short the option:

$$Max(uncollateralisedswapMtM - RR_{bond}, 0)$$

This could be priced using the techniques described in the section on CLNs.

A typical use of this type of repack, at the time of writing would be Japanese investors, who like the high interest rates they can get from the FX-linked coupons in PRDs (Power Reverse Dual-currency notes), but who no longer want to take the credit risk of the issuing banks. Thus, to make the product attractive to the investor, the banks can take a Japanese government bond and add the PRD coupon onto it.

Repack of the Bank's Own Debt In all of the trades below, except the negative basis bond, the SPV collateral could be a bond issued by the bank acting as the swap counterparty. This is fairly rare, as one common reason why the bank uses an SPV to sell a bond to an Investor is because that investor does not want to be exposed to the full credit risk on the bank's debt. One case where such a transaction may appear attractive would be if the internal issuance curve shown to the derivative/ repack desks by the bank's treasury is at a level far below where the bank's own vanilla bonds trade in the secondary market. The derivative desk could then use the SPV structure to create a bond for an investor who wants both the bank's credit risk and a structured coupon.

Let us consider the vanilla bond repack described above, where the swap counterparty is also the bond issuer. This is illustrated in Figure 10.10.

There is now only one possible termination event leading to a windup of the SPV: a default by the bank. If the swap is in-the-money to the SPV, then the Investor will receive RR_Bank * (MtM of swap + Notional of bond). If the swap is in-the-money to the bank, the SPV will offset its collateralised value against the face value of the bank's bonds that it holds, and thus the bank will receive its own recovery rate on the MtM of the swap. The tail risk that the collateralised MtM of the swap exceeds the face value of the bond in the SPV can be ignored, for practical purposes. Thus the swap between the bank and the SPV can be treated as a pure uncollateralised one.

The end result to the investor is that he is in a situation almost identical to the one that would have existed had he bought a bond with the repacked coupon directly from the bank. There will be a small difference in his claim upon default (MtM of swap + Bond notional, rather than just the notional itself). However, the swap could be made to be explicitly vanishing upon the bank's default (i.e., its mark-to-market is deemed to be zero upon a default event), and thus we would have created a situation that is economically identical to the direct purchase of a bond with the repacked coupon.

In this example, it does not make any financial sense to make the swap itself collateralised. The only collateral that can be posted to the bank is its own debt, and no one is going to provide a repo market to the debt's issuer! Thus, being posted back its own bonds would not reduce the real-world funding cost faced by the bank.

Vanilla Bond Repacked into a Negative Basis Bond The purpose of this transaction is to remove the credit risk on a bond, resulting in a coupon that is greater than the risk-free rate. This trade is traditionally thought of as a way to take advantage when the implied risk premium in the CDS market is lower than the implied risk premium in the bond market. However, in the absence of a collateral agreement between the bank and the SPV, the structure is a little more complex.

First of all, let us assume there is no CSA between the bank and the SPV.

On Day one:

- The Investor pays the notional to the SPV
- The SPV uses the notional to buy a third-party bond

If there is no default by either the bond issuer or the swap counterparty, then:

- The SPV receives the fixed on the bond
- The SPV pays the fixed coupons to the swap counterparty
- The swap counterparty pays Libor + spread to the SPV
- The SPV passes on Libor + spread to the investor

At the maturity of the bond, if there have been no default events, then:

- The SPV receives the bond's face value from the bond issuer
- The SPV passes the face value to the Investor
- The swap terminates

If there is a default by the bond issuer, then:

- The SPV passes the defaulted bond to the swap counterparty
- The swap counterparty passes the cash face value of the bond to the SPV
- The SPV passes the cash face value on to the Investor and the transaction ends

If there is a default by the swap counterparty, then the collateralised mark-to-market of the credit-linked swap between the swap counterparty and the SPV is calculated (i.e., a trade where the swap counterparty receives fixed and pays Libor + spread until the issuer default, and the swap counterparty pays the face value in return for receiving the Bond (worth recovery) upon the issuer's default).

If the collateralised swap MtM is in favour of the SPV (which is likely), then:

- The SPV claims the collateralised MtM from the swap counterparty, and receives the counterparty's recovery rate multiplied by the collateralised MtM.
- The SPV pays the above sum to the Investor.
- The SPV passes ownership of the bond to the investor and the transaction ends.

If the collateralised swap MtM is in favour of the swap counterparty, then the payout depends on who is senior in the waterfall.

- If the swap counterparty is senior, then the bond is liquidated, the swap counterparty is paid in full, and any remaining cash is returned to the Investor.
- If the Investor is senior, then the investor receives the collateralised MtM of Libor + Spread and the final notional repayment, and any remaining balance is paid to the swap counterparty.

The transaction is illustrated in Figure 10.11.

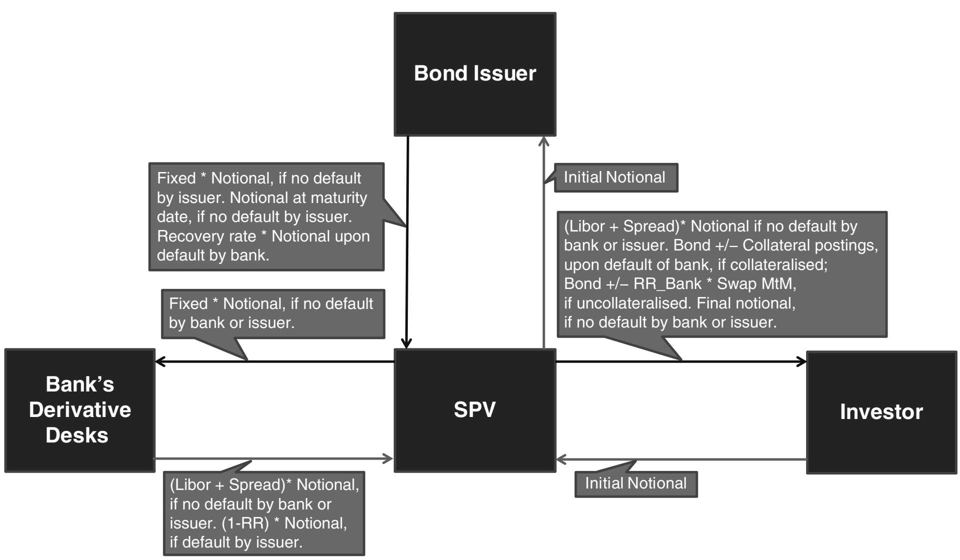

FIGURE 10.11 Vanilla Bond and Negative Basis Repack

Superficially, the Investor will have achieved his objective if Libor + Spread is greater than the risk-free rate. However, notice that the swap with the bank is not collateralised! As it is impossible for the CDS to have a positive value to the Swap counterparty upon a default by the third party, the only credit risk on the swap can be to the bank itself. Clearly the MtM of the credit protection will move in the SPV's favour as the bank's credit spread widens, due to their high correlation. It is practically impossible for the effect to be overridden by any correlation between the value of the fixed for floating swap and the credit spreads. Hence, the fact that the swap is uncollateralised is clearly in the favour of the bank.

Because the value of the uncollateralised CDS protection is much lower than the equivalent amount of collateralised protection, the bank can afford to receive a smaller spread on the swap compared to the collateralised version. Thus the value of Libor + X – Y may well exceed the risk-free rate, without fully compensating the Investor for the joint credit risk he faces to the bank and the third-party entity. It is also possible that the investor may not really be aware that he may still face significant credit risk in the transaction. Note that the dynamic hedging of the uncollateralised CDS will be identical to the situation discussed in Chapter 14, which covers CLNs where the Hybrids desk had sold uncollateralised protection to a theoretically risk-free counterparty: the SPV, in this special case is, from the point of view of the bank, a true risk-free counterparty!

These issues can be mitigated by collateralising the swap between the SPV and the bank. The value of the bonds in the SPV will always, in practice, be sufficient to cover the swap MtM. The investor now only faces some gap risk arising from a jump downwards in the value of the bonds (and upwards on the swap) upon a default by the bank. Ignoring that effect, we can now truly say that the investor has removed all the credit risk from the bond, and has achieved his goal of locking in a risk-free profit.

At first sight, this may appear to be a perfect trade for the investor. However, there are some downsides. There will be little or no possibility of his exiting the structure before the full maturity of the bond. Thus, he faces liquidity risk, as he may require his funds back before the termination of the trade. He may also miss out on possible future situations that could be even more profitable, after he has committed his funds for the full term. If the client is a natural risk-seeker, there may be other risky transactions that, in his personal risk measure (which is NOT the risk-neutral measure), are actually more profitable than this one. Indeed, some investors have a target rate of return on their funds that is likely to exceed the low rate of return on this investment. There are legal and transactional costs in setting up the structure that may reduce a theoretically profitable situation to an unprofitable one.

However, the fact that this structure can be set up means that there must be a limit on the basis that can exist between the risky rate implied from CDS and the risky rate implied from the underlying bond. If the risky rate on the bond significantly exceeds the cost of CDS protection, then arbitrageurs will enter into negative basis SPV transactions until the difference reduces to an unprofitable level. This transaction does not prevent the risky rate implied by the CDS from exceeding the rate implied by the bond. If that situation developed, however, then people holding the bond would sell it at a high price, and also sell CDS protection, thus getting a higher level of return while maintaining the same amount of risk to the survival of the entity. As before, the existence of this transaction does not mean the rate of the CDS can never exceed the rate implied by the bond. For example, some people who hold the bond may not be able to purchase the CDS (e.g., they don't have any ISDA agreements set up). In general, it is more common for the risky rates implied from bonds to exceed those implied from CDSs, as the bonds usually have greater liquidity risk.

One final practical point why the negative basis trade may not work is if it is possible for a "legal basis" to exist between the CDS protection and the bond; that is, if a situation can occur where the credit risk on the bond may not be precisely hedged by the credit risk on the CDS. One example at the time of writing is Greek Sovereign debt, because a "voluntary default" that many, but not all, debt holders agree to, would not trigger the CDS. Another example would be if the CDS does not reference the exact bond that is inside the SPV, and there is not perfect cross-default between the CDS reference obligation and the bond held.

For those people who are interested in the mathematical reasons why arbitrages can continue to exist in the markets, see Shleifer and Vishny, "The Limits of Arbitrage" (1997) and "Risk in Dynamic Arbitrage: The Price Effects of Convergence Trading" (Kondor, 2009).

Vanilla Bond Repacked into a Credit-Linked Note In Chapter 14 in the section on CLNs, we give some reasons why an investor would want to own a credit-linked note. If neither of the names he wishes to be exposed to is interested in issuing a CLN, then he could ask another bank whose credit risk he does not want to take, to synthesise his desired trade through an SPV structure. The transaction is illustrated in Figure 10.12.

As in the negative basis trade, there is the issue as to whether the CDS protection traded with the bank should be collateralised or uncollateralised. In this case, it is the bank buying the CDS protection, so the Investor will receive a larger coupon on the SPV note if the swap with the bank is collateralised. If there is also a mark-to-market trigger that collapses the structure when the value of the capital starts to fall

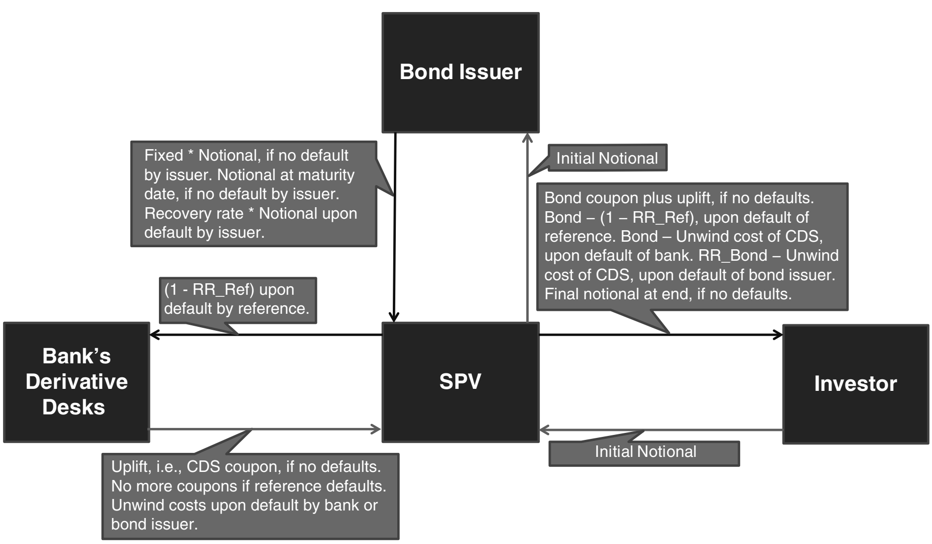

FIGURE 10.12 Vanilla Bond CLN Repack

compared to the swap mark-to-market, then the tail risk is avoided too. It is unlikely, however, that the Investor would want a collateral trigger. His view on the market is that the increased return paid to him, due to the credit exposure to the reference entity and to the issuer, is high compared to the real-world probability of either entity defaulting. Thus, when credit spreads widen, it is likely that he will want to weather the market move, rather than be forced to unwind at a loss. In this case, he will have to be prepared to accept a slightly reduced coupon to cover the bank's tail risk.

The tail risk will be far more material than in the preceding examples. First of all, the MtM of a CDS can typically reach a much higher level than the MtM of a vanilla swap. Secondly, an increase in the MtM to the bank is likely to coincide with a loss in value on the collateral, thus making the expected exposure more severe. As explained above, the swap is worth more to the bank if it is collateralised. However, this also means that the tail risk is greater if the swap is collateralised, so the Investor will not get the full benefit of collateralisation. From the bank's point of view, the hedging of the CDS and the tail risk is very similar to the case of the CLNs discussed in the Chapter 14.

SUMMARY AND CONCLUSIONS

Securitisation is a long-established technique that enables banks to manage their balance sheets with greater flexibility and precision. By employing it, a bank can work towards ensuring optimum treatment of its assets for funding, regulatory capital, and credit risk management purposes. In addition, in the postcrash era the securitisation tool continues to be used to create tradable notes that can then be used as collateral at the central bank.

The primary driver behind the decision to securitize part of the asset side of the balance sheet is one or more of the following:

- *Funding:* Assets that are unrepo-able may be securitised, which enables the bank to fund them via the capital market, or by creating notes that can be used as collateral at the central bank.
- *Regulatory capital:* This is a rarer reason driving securitisation in the postcrash, Basel III era, however it is still possible to structure a deal that reduces regulatory capital requirement, principally if "significant risk transfer" has taken place.
- *Risk management:* Assets on the balance sheet expose the originator to credit risk, and this can be managed (either reduced, removed, or hedged) via securitisation.

A secondary driver is client demand: the process of securitisation creates bonds that can be sold, and investor demand for a note of specified credit risk, liquidity, maturity, and underlying asset class may well cause a bond to be created, via securitisation, to meet this specific investor requirement. Whether a transaction is demand driven or issuer driven, it will always be created to meet at least one of the preceding requirements.

The mechanics of closing a securitisation deal, which we cover in detail in subsequent chapters, can take anything from a few months to up to a year or more. The most important parts of the process are the legal review and drafting of transaction documents, and the rating agency review. The involvement of third parties, such as lawyers, trustees, agency services providers, and the rating agencies, is the key driver behind the cost of closing a securitisation deal, and these costs are covered either by the deal itself or direct by the originating institution.

A wide range of asset classes can be securitised, with the most common being residential mortgages and corporate loans. Other asset classes include auto loans and credit card receivables. When assessing the risk exposure and performance of different types of ABS, investors will consider the behaviours and characteristics of the specific type of underlying asset. Some performance metrics are of course common to all types of assets, such as delinquency rate or the percentage of nonperforming loans.

The securitisation market remains a valuable tool for banks to use as part of their balance sheet management. The key focus for originators remains that of sourcing assets of sufficient credit quality and performance when transacting a deal, to ensure that investor requirements can be satisfied. That there remains demand, at lower levels, for ABS products speaks to the continuing needs of certain classes of investor for the risk-reward profiles of such notes.

The rating agencies' methodology for rating ABS notes takes into account lessons learned after the 2008 financial crash, and as a result certain of the more exotic underlying asset classes are less observed in the market. Also, market value–type transactions are no longer able to obtain the highest AAA rating.

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PART Three

Derivative Instruments

In the third part of the book, we consider the primary derivative instruments. Derivatives are *not* securities, and it is incorrect to refer to them as such. Hence, we have fixed-income securities in the cash markets and fixed-income derivatives (or interest-rate derivatives) in the synthetic markets. We look here at their application in the bond markets.

We concentrate on interest-rate swaps and how they are used by bond-market participants for hedging purposes, options, and credit derivatives. There is also a chapter on the theory of forward and futures pricing.

Readers who wish to learn more about derivative instruments have a large selection of titles to choose from. We recommend Blake (2000) as an introductory-level text. This book does a good job of placing all capital-market instruments within an integrated context, including the analysis and use of options. Marshal and Bansal (1992) and Levy (1999) are also very good introductory texts that cover a wide range of instruments, including options. Another excellent title is Jarrow and Turnbull (2000), while the author's personal favourite on options and other derivatives is Kolb (2007); this book is well worth purchasing. For an intermediate- to advanced-level treatment, the standard text is Hull (2011). Rubinstein (1999) is at the same level, and is very accessible and readable; it is highly recommended. An excellent advanced text on derivatives is Briys et al. (1998), which presents the concepts in a way that practitioners will appreciate, while remaining academically rigourous. Other advanced treatments that are written in clear and readable style are Pliska (1997) and Elliott and Kopp (1999). An excellent introduction to the mathematics contained in these two texts is given in Ross (1999).

Finally, two references cited elsewhere but worth mentioning again because of their high readability are Questa (1999) and Sundaresan (1997). These are introductory- and intermediate-level treatments, respectively, which discuss derivatives and price processes within the context of the fixed-income markets.

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CHAPTER 11

Forwards and Futures Valuation

Before our discussion of derivative instruments, we discuss the valuation and analysis of forward and futures contracts. A description of interest-rate futures was given in Chapter 6. Here we develop basic valuation concepts.

INTRODUCTION

A forward contract is an agreement between two parties in which the buyer contracts to purchase from the seller a specified asset, for delivery at a future date, at a price agreed today. The terms are set so that the present value of the contract is zero. For the forthcoming analysis, we use the following notation:

- P is the current price of the underlying asset, also known as the *spot* price
- P_T is the price of the underlying asset at the time of delivery
- X is the delivery price of the forward contract
- T is the term to maturity of the contract in years, also referred to as the time-to-delivery
- r is the risk-free interest rate
- R is the return of the payout or its *yield*
- F is the current price of the forward contract

The payoff of a forward contract is therefore given by

$$P_T - X \tag{11.1}$$

with X set at the start so that the present value of $(P_T - X)$ is zero. The payout yield is calculated by obtaining the percentage of the spot price that is paid out on expiry.

FORWARDS

When a forward contract is written, its delivery price is set so that the present value of the payout is zero. This means that the forward price F is then the price on delivery that would make the present value of the payout, on the delivery date, equal to zero. That is, at the start $F = X$. This is the case only on day 1 of the contract, however. From then until the contract expiry, the value of X is fixed, but the forward price F will

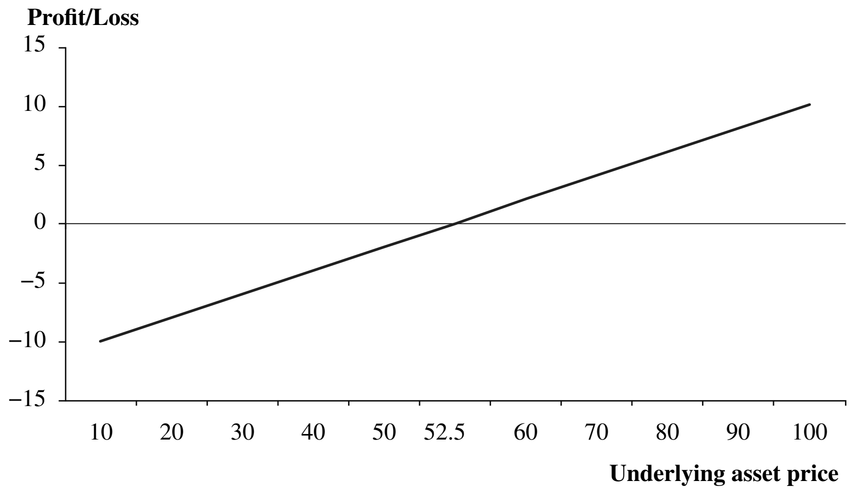

FIGURE 11.1 Forward Contract Profit/Loss Profile

fluctuate continuously until delivery. It is the behaviour of this forward price that we wish to examine. For instance, generally as the spot price of the underlying increases, so the price of a forward contract written on the asset also increases; and vice versa.

At this stage, it is important to remember that the forward price of a contract is not the same as the value of the contract, and the terms of the agreement are set so that at inception the value is zero. The relationship given above is used to show that an equation can be derived that relates F to P, T, r, and R.

Consider first the profit/loss profile for a forward contract. This is shown in Figure 11.1. The price of the forward can be shown to be related to the underlying variables as

$$F = S(r/R)^T \tag{11.2}$$

and for the one-year contract highlighted in Figure 11.1 is 52.5, where the parameters are

$$S = 50,\ r = 1.05,\ \text{and}\ R = 1.00$$

FUTURES

Forward contracts are tailor-made instruments designed to meet specific individual requirements. Futures contracts on the other hand are standardised contracts that are traded on recognised futures exchanges. Apart from this, the significant difference between them, and the feature that influences differences between forward and futures prices, is that profits or losses that are gained or suffered in futures trading are paid out at the end of the day. This does not occur with forwards. The majority of trading in futures contracts are always closed out; that is, the position is netted out to zero before the expiry of the contract. If a position is run into the delivery month, depending on the terms and conditions of the particular exchange, the long future may be delivered into. Settlement is by physical delivery in the case of commodity futures or in cash in the case of certain financial futures. Bond futures are financial futures where any bond that is in the *delivery basket* for that contract will be delivered to the long. With both physical and financial futures, only a very small percentage of contracts are actually delivered into because the majority of trading is undertaken for hedging and speculative purposes.

With futures contracts, as all previous trading profits and losses have been settled, on the day of expiry only the additional change from the previous day needs to be accounted for. With a forward contract all loss or gain is rolled up until the expiry day and handed over as a total amount on this day.[1]

FORWARDS AND FUTURES

Before looking closer at the relationship between forward and future price, we consider the two type of contract cash flow differences.

CASH FLOW DIFFERENCES

The cash flow treatment of the two contracts is illustrated in Table 11.1, which uses F to denote the price of the futures contract as well. The table shows the payoff schedule at the end of each trading day for the two instruments; assume that they have identical terms. With the forward there is no cash flow on intermediate dates, whereas with the futures contract there is. As with the forward contract, the price of the future fixes the present value of the futures contract at zero. Each day the change in price, which at the end of the day is *marked to market* at the *close* price, will have resulted in either a profit or gain,[2] which is handed over or received each day as appropriate. The process of daily settlement of price movements means that the nominal delivery price can be reset each day so that the present value of the contract is always zero. This means that the future and nominal delivery prices of a futures contract are the same at the end of each trading day.

We see in Table 11.1 that there are no cash flows changing hands between counterparties to a forward contract. The price of a futures contract is reset each day; after day 1 this means it is reset from F to F_1. The amount $(F_1 - F)$, if positive, is handed over by the short future to the long future. If this amount is negative, it is paid by the long future to the short. On the expiry day T of the contract, the long future will receive a settlement amount equal to $P_T - F_{T-1}$ which expresses the relationship between the price of the future and the price of the underlying asset. And as significant, the daily cash flows transferred when holding a futures contract cancel each other out, so that on expiry the value of the contract is (at this stage) identical to that for a forward, that is $(P_T - F)$.

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[1] We assume the parties have traded only one forward contract between them. If, as is more accurate to assume, a large number of contracts have been traded across a number of different maturity periods and perhaps instruments, as contracts expire only the net loss or gain is transferred between counterparties.

[2] Or no profit or gain if the closing price is unchanged from the previous day's closing price, a *doji* as technical traders call it.

TABLE 11.1 Cash Flow Process for Forwards and Futures Contracts

Time	Forward Contract	Futures Contract
0	0	0
1	0	$F_1 - F$
2	0	$F_2 - F_1$
3	0	$F_3 - F_2$
4	0	$F_4 - F_3$
5	0	$F_5 - F_4$
…	0	…
…	0	…
…	0	…
$T - 1$	0	$F_{T-1} - F_{T-2}$
T	$P_T - F$	$P_T - F_{T-1}$
Total	$P_T - F$	$P_T - F$

With exchange-traded contracts all market participants are deemed to conduct their trading with a central counterparty, the exchange's clearing house. This eliminates counterparty risk in all transactions, and the clearing house is able to guarantee each bargain because all participants are required to contribute to its clearing fund. This is by the process of *margin*, by which each participant deposits an *initial margin* and then, as its profits or losses are recorded, deposits further *variation margin* on a daily basis. The marking-to-market of futures contract is an essential part of this margin process. A good description of the exchange clearing process is contained in Kolb (2000).

This is the key difference between future and forward contracts. If holding a futures position that is recording a daily profit, the receipt of this profit on a daily basis is advantageous because the funds can be reinvested while the position is still maintained. This is not available with a forward. Equally, losses are suffered on a daily basis that are not suffered by the holder of a loss-making forward position.

RELATIONSHIP BETWEEN FORWARD AND FUTURE PRICE

In general, futures and forwards will trade at different prices. This is because the daily settlement of the profits and losses leads to what is known as a *convexity* effect on the price. Let us assume that the daily margins on the futures contract can be reinvested at the risk-free rate. Now let us consider the prices of a future and forward on an asset whose price moves are positively correlated to interest rates; that is, when the asset price rises, the risk-free rate is likely to also rise, and when the asset price falls, so does the risk-free rate.

In this case, would we rather be long the future or the forward contract, if we could enter into each at the same price?

Let us say that risk-free interest rates are currently at 1%, we can fund ourselves at the risk-free rate, and the contracts expire in one year's time.

First, consider a case where the price of the forward rises by 100 points and interest rates rise to 2%. The change in the present value of the forward contract is 100/1.02 = 98.04. However, the change in the price of the futures contract is 100, because the 100 is settled today, and thus no discount is required to calculate the present value.

Now, consider a case where the price of the forward falls by 100 points and interest rates fall to 0%. The change in the present value of both the forward and the future is –100.

Thus, the positive correlation between the asset price and interest rates has led to a situation where our profits are discounted compared to our losses, in the case of the future contract. Hence, we would rather enter into the forward than the future, and thus the future will trade at a lower price than the forward. The higher the correlation, and the greater the volatility of interest rates and the forward asset price, the greater the price differential between the forward and the future.

Conversely, a negative correlation between rates and the asset price would lead to profits on the future being discounted less than profits on the forward, and thus the future's price will be in excess of the forward's.

If there is no correlation, the future and forward will trade at the same price. In the normal situation of positive interest rate, however, the effective notional of one future contract will be greater than one forward contract, because price changes on the forward will be discounted and thus have a smaller present value than the price changes on the future.

The fact that forwards and forward-rate agreements (FRAs) are over-the-counter (OTC) instruments, whereas futures are exchange-traded, makes no difference in practice in terms of collateralisation. In the interbank market, OTC instruments are traded under the Credit Support Annexe (CSA) of the standard International Securities and Derivatives Association (ISDA) contract. This means they are cash-collateralised just like exchange-traded futures contracts. However, this does not do away with the need for the convexity adjustment. The difference in valuation arises because of the different amount of collateral posted, which is discounted in the case of the CSA forward and undiscounted in the case of the future.

One way of thinking about it is that both futures and forwards are prices in a way that is equivalent to discounting at the collateral funding interest rate, and hence in both cases correlation between the underlying and the collateral rate is important. For futures, the collateral rate is effectively 0%, whereas for forwards it is usually the overnight index (OIS) swap rate (see Chapter 13). Incidentally in an uncollateralised case, one would discount at the bank's collateral funding rate, which is strongly correlated to OIS but will lie above it. The other factor to bear in mind is that in the interbank market all derivative contracts are traded under an ISDA CSA, so that they are daily (usually cash) collateralised. This equates the cash flow treatment to that observed for an exchange-traded derivative contract.

THE FORWARD-SPOT PARITY

We can use the forward strategy to infer the forward price, provided we know the current price of the underlying and the money-market interest rate. A numerical example of the forward strategy is given in Figure 11.2, with the same parameters given earlier. We assume no-arbitrage and a perfect frictionless market.

	Cash Flows	
	Start Date	*Expiry*
Buy forward contract	0	$P_T - F$
Buy one unit of the underlying asset	–50	P_T
Borrow zero present-value of forward price	$F/1.05$	F
Total	$-50 + F/1.05$	$P_T - F$
Result		
Set $-50 + F/1.05$ equal to zero (no-arbitrage condition)		
Therefore $F = 52.5$		

FIGURE 11.2 Forward Strategy

What Figure 11.2 is saying is that it is possible to replicate the payoff profile we observed in Figure 11.1 by a portfolio composed of one unit of the underlying asset, which purchase is financed by borrowing a sum that is equal to the present value of the forward price. This borrowing is repaid on maturity and is equal to $(F/1.05)\times 1.05$ which is in fact F. In the absence of arbitrage opportunity, the cost of forming the portfolio will be identical to that of the forward itself. However, we have set the current cost of the forward contract at zero, which gives us

$$-50 + F/1.05 = 0$$

We solve this expression to obtain F, which is 52.50.

So to recap, the price of the forward contract is 52.50 although the present value of the forward contract when it is written is zero. Following Rubinstein we prove this in Figure 11.3.

What Figure 11.3 states is that the payoff profile for the forward can be replicated precisely by setting up a portfolio that holds R^{-T} units of the underlying asset, which is funded through borrowing a sum equal to the present value of the forward price. This borrowing is repaid at maturity, this amount being equal to

$$\left(Fr^{-T}\right)\times r^{T} = F$$

The portfolio has an identical payoff profile (by design) to the forward, this being $\left(P_T - F\right)$. In a no-arbitrage environment, the cost of setting up the portfolio must be equal to the current price of the forward, as they have identical payoffs, and if one was cheaper than the other, there would be a risk-free profit for a trader who bought the cheap instrument and shorted the dear one. However, we set the current cost of the forward (its present value) as zero, which means the cost of constructing the duplicating portfolio must therefore be zero as well. This gives us

	Start Date	*Expiry Date*
Buy forward contract	0	$P_T - F$
Buy R^{-T} units of the underlying asset	$-PR^{-T}$	P_T
Borrow zero present-value of forward price	Fr^{-T}	$-F$
Total	$-PR^{-T} + Fr^{-T}$	$P_T - F$

Set
$-PR^{-T} + Fr^{-T} = 0$
Therefore
$F = P(r / R)^T$

FIGURE 11.3 Algebraic Proof of Forward Price

$$-PR^{-T} + Fr^{-T} = 0$$

which allows us to solve for the forward price F.

The significant aspect for the buyer of a forward contract is that the payoff of the forward is identical to that of a portfolio containing an equivalent amount of the underlying asset, which has been constructed using borrowed funds. The portfolio is known as the *replicating portfolio*. The price of the forward contract is a function of the current underlying spot price, the risk-free or money-market interest rate, the payoff and the maturity of the contract.

To recap, the forward-spot parity states that

$$F = P(r / R)^T \tag{11.3}$$

It can be shown that neither of the possibilities $F > P(r/R)^T$ or $F < P(r/R)^T$ will hold unless arbitrage possibilities are admitted. The only possibility is equation (11.3), at which the futures price is *fair value*.

THE BASIS AND IMPLIED REPO RATE

For later analysis, we now introduce some terms used in the futures markets.

The difference between the price of a futures contract and the current underlying spot price is known as the *basis*. For bond futures contracts, which are written not on a specific bond but on a *notional* bond that can in fact be represented by any bond that fits within the contract terms, the size of the basis is given by equation (11.4).

$$Basis = P_{bond} - (P_{fut} \times CF) \tag{11.4}$$

where the basis is the *gross basis* and *CF* is the *conversion factor* for the bond in question. All delivery-eligible bonds are said to be in the *delivery basket*. The conversion factor equalises each deliverable bond to the futures price.[3] The size of the gross basis represents the cost of carry associated with the bond from today to the delivery date. The bond with the lowest basis associated with it is known as the *cheapest-to-deliver* bond.

The magnitude of the basis changes continuously, and this uncertainty is termed *basis risk*. Generally, the basis declines over time as the maturity of the contract approaches, and converges to zero on the expiry date. The significance of basis risk is greatest for market participants who use futures contracts for hedging positions held in the underlying asset. The basis is positive or negative according to the type of market in question, and is a function of issues such as *cost of carry*. When the basis is positive—that is $F > P$—the situation is described as a *contango*; a negative basis, $F < P$, is described as *backwardation*. Contango can occur in markets where the yield on the asset is less than the risk-free rate. For example, a precious metal such as gold will yield a low or negative interest yield to the owner, due to storage costs, and thus will commonly lead to contango. Equities usually (but not always) have dividend yields in excess of the risk-free rate, and thus commonly lead to backwardation. Commodities with seasonal demand or anticipated future changes in supply can lead to either contango or backwardation. Foreign exchange contracts will show either backwardation or contango, depending on the relative levels of the interest rates of the two currencies.

The hedging of futures and the underlying asset requires a keen observation of the basis. To hedge a position in a futures contract, one could run an opposite position in the underlying. However running such a position incurs the cost of carry referred to above, which, depending on the nature of the asset, may include storage costs, opportunity cost of interest foregone, funding costs of holding the asset, and so on. The futures price may be analysed in terms of the forward-spot parity relationship and the risk-free interest rate. If we say that the risk-free rate is

$$r - 1$$

and the forward-spot parity is

$$F = P(r/R)^T$$

we can set

$$r - 1 = R(F/P)^{1/T} - 1 \tag{11.5}$$

which must hold because of the no-arbitrage assumption.

This interest rate is known as the *implied repo rate*, because it is similar to a repurchase agreement carried out with the futures market. Generally, a relatively high

[3] For a description and analysis of bond futures contracts, the basis, implied repo, and the cheapest-to-deliver bond, see Burghardt et al. (1994), an excellent account of the analysis of the Treasury bond basis. Plona (1996) is also a readable treatment of the European government bond basis.

implied repo rate is indicative of high futures prices, and the same for low implied repo rates. The rates can be used to compare contracts with each other, when these have different terms to maturity and even underlying assets. The implied repo rate for the contract is more stable than the basis; as maturity draws near, the level of the rate becomes very sensitive to changes in the future's price, spot price, and (by definition) time to maturity.

The implied repo rate is constant during this time. If we start with

$$F = P(r/R)^T$$

we can derive

$$r = R(F/P)^{1/T}$$

where r is the implied repo rate. The cost of carry is defined as the net cost of holding the underlying asset from trade date to expiry date (or delivery date).

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CHAPTER 12

Bond Futures Contracts

The most widely used risk management instrument in the bond markets is the government-bond futures contract. This is usually an exchange-traded standardised contract that fixes the price today at which a specified quantity and quality of a bond will be delivered at a date during the expiry month of the futures contract. Unlike short-term interest-rate futures, which only require cash settlement, and which we encountered in the section on money markets, bond futures require the actual physical delivery of a bond when they are settled.

In this chapter we review bond futures contracts and their use for trading and hedging purposes.

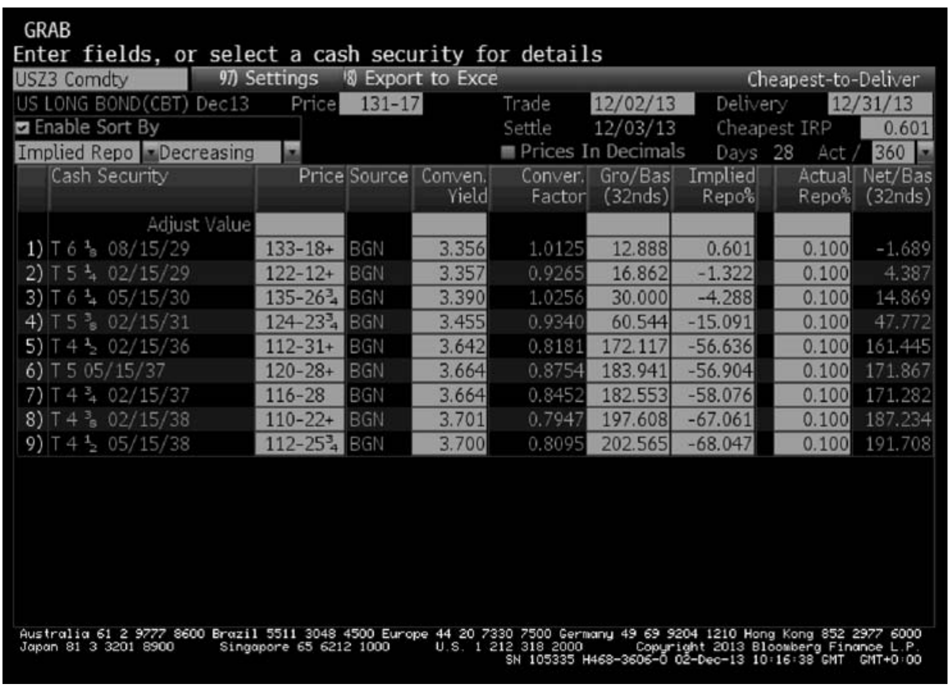
BACKGROUND

The concept of a bond futures contract is probably easier to grasp intuitively than a short-dated interest-rate future. This reflects the fact that a bond futures contract represents an underlying physical asset, the bond itself, and a bond must be delivered on expiry of the contract. In this way, bond futures are similar to commodity futures, which also require physical delivery of the underlying commodity.

A futures contract is an agreement between two counterparties that fixes the terms of an exchange that will take place between them at some future date. They are standardised agreements, as opposed to over-the-counter (OTC) ones, when traded on an exchange, so they are also referred to as exchange-traded futures. In the United Kingdom, financial futures are traded on London International Financial Futures Exchnage (LIFFE), which opened in 1982. There are four classes of contract traded on LIFFE: short-term interest-rate contracts; long-term interest-rate contracts (bond futures); currency contracts; and stock-index contracts. We discussed interest-rate futures contracts, which generally trade as part of the money markets, in an earlier chapter. In this section we will look at bond futures contracts, which are an important part of the bond markets; they are used for hedging and speculative purposes. Most futures contracts on exchanges around the world trade at three-month maturity intervals, with maturity dates fixed at March, June, September, and December each year. This includes the contracts traded on LIFFE. Therefore, at preset times during the year a contract for each of these months will expire, and a final settlement price is determined for it. The further out one goes, the less liquid the trading is in that contract. It is normal to see liquid trading only in the "front month" contract (the current contract, so that if we are trading in April 2013 the front month is the June 2013 future), and possibly one or two of the next contracts, for most bond futures contracts. The liquidity of contracts diminishes the further one trades out in the maturity range.

When a party establishes a position in a futures contract, it can either run this position to maturity or close out the position between the trade date and maturity. If a position is closed out, the party will have either a profit or loss to book. If a position is held until maturity, the party who is long the future will take delivery of the underlying asset (bond) at the settlement price; the party who is short futures will deliver the underlying asset. This is referred to as physical settlement or sometimes, confusingly, as cash settlement. Figure 12.1 shows the deliverable bonds into the US Treasury futures contract, and the prices of each cash bond at the time.

There is no counterparty risk associated with trading exchange-traded futures, because of the role of the clearing house, such as the London Clearing House (LCH). This is the body through which contracts are settled. A clearing house acts as the buyer to all contracts sold on the exchange, and the seller to all contracts that are bought. So in the London market, the LCH acts as the counterparty to all transactions, so that settlement is effectively guaranteed. The LCH requires all exchange participants to deposit margin with it, a cash sum that is the cost of conducting business (plus broker's commissions). The size of the margin depends on the size of a party's net open position in contracts (an open position is a position in a contract that is held overnight and not closed out). There are two types of margin, maintenance margin and variation margin. Maintenance margin is the minimum level required to be held at the clearing house; the level is set by the exchange. Variation


GRAB
Enter fields, or select a cash security for details
USZ3 Comdty 97) Settings 98) Export to Exce Cheapest-to-Deliver
US LONG BOND(CBT) Dec13 Price 131-17 Trade 12/02/13 Delivery 12/31/13
Enable Sort By Settle 12/03/13 Cheapest IRP 0.601
Implied Repo Decreasing Prices In Decimals Days 28 Act / 360

	Cash Security	Price	Source	Conven. Yield	Conver. Factor	Gro/Bas (32nds)	Implied Repo%	Actual Repo%	Net/Bas (32nds)
	Adjust Value								
1)	T 6 1/8 08/15/29	133-18+	BGN	3.356	1.0125	12.888	0.601	0.100	-1.689
2)	T 5 1/4 02/15/29	122-12+	BGN	3.357	0.9265	16.862	-1.322	0.100	4.387
3)	T 6 1/4 05/15/30	135-26 3/4	BGN	3.390	1.0256	30.000	-4.288	0.100	14.869
4)	T 5 3/8 02/15/31	124-23 3/4	BGN	3.455	0.9340	60.544	-15.091	0.100	47.772
5)	T 4 1/2 02/15/36	112-31+	BGN	3.642	0.8181	172.117	-56.636	0.100	161.445
6)	T 5 05/15/37	120-28+	BGN	3.664	0.8754	183.941	-56.904	0.100	171.867
7)	T 4 3/4 02/15/37	116-28	BGN	3.664	0.8452	182.553	-58.076	0.100	171.282
8)	T 4 3/8 02/15/38	110-22+	BGN	3.701	0.7947	197.608	-67.061	0.100	187.234
9)	T 4 1/2 05/15/38	112-25 3/4	BGN	3.700	0.8095	202.565	-68.047	0.100	191.708

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000
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FIGURE 12.1 Bond Futures Delivery Quotes, Bloomberg Page DLV, 2 December 2013 margin is the additional amount that must be deposited to cover any trading losses and as the size of the net open positions increases. Note that this is not like margin in, say, a repo transaction. Margin in repo is a safeguard against a drop in value of collateral that has been supplied against a loan of cash. The margin deposited at a futures exchange clearing house acts essentially as "good faith" funds, required to provide comfort to the exchange that the futures trader is able to satisfy the obligations of the futures contract that are being traded.

BOND FUTURES CONTRACTS

We have noted that futures contracts traded on an exchange are standardised. This means that each contract represents exactly the same commodity, and it cannot be tailored to meet individual customer requirements. In this section, we describe two very liquid and commonly traded contracts, starting with the U.S. T-Bond contract traded on the Chicago Board of Trade (CBOT). The details of this contract are given in Table 12.1.

The terms of this contract relate to a U.S. Treasury bond with a minimum maturity of 15 years and a notional coupon of 8%. We introduced the concept of the notional bond in the chapter on repo markets. A futures contract specifies a notional coupon to prevent delivery and liquidity problems that would arise if there was shortage of bonds with exactly the coupon required, or if one market participant purchased a large proportion of all the bonds in issue with the required coupon. For exchange-traded futures, a short future can deliver any bond that fits the maturity criteria specified in the contract terms. Of course, a long future would like to be delivered a high-coupon bond with significant accrued interest, while the short future would want to deliver a low-coupon bond with low interest accrued. In fact, this

TABLE 12.1 CBOT U.S. T-Bond Futures Contract Specifications

Unit of Trading	U.S. Treasury bond with notional value of $100,000 and a coupon of 8%
Deliverable grades	U.S. T-bonds with a minimum maturity of 15 years from first day of delivery month
Delivery months	March, June, September, December
Delivery date	Any business day during the delivery month
Last trading day	12:00 noon, seventh business day before last business day of delivery month
Quotation	Percent of par expressed as points and thirty-seconds of a point, e.g., 108–16 is 108 16/32 or 108.50
Minimum price movement	1/32
Tick value	$31.25
Trading hours	07.20–14.00 (trading pit) 17.20–20.05 22.3006.00 hours (screen trading)

issue does not arise because of the way the invoice amount (the amount paid by the long future to purchase the bond) is calculated. The invoice amount on the expiry date is given in equation (12.1).

$$Inv_{\text{amt}} = P_{fut} \times CF + AI \tag{12.1}$$

where Inv_{amt} is the invoice amount
P_{fut} is the price of the futures contract
CF is the conversion factor
AI is the bond accrued interest

Any bond that meets the maturity specifications of the futures contract is said to be in the delivery basket, the group of bonds that are eligible to be delivered into the futures contract. Every bond in the delivery basket will have its own *conversion factor,* which is used to equalise coupon and accrued interest differences of all the delivery bonds. The exchange will announce the conversion factor for each bond before trading in a contract begins; the conversion factor for a bond will change over time, but remains fixed for one individual contract. That is, if a bond has a conversion factor of 1.091252, this will remain fixed for the life of the contract. If a contract specifies a bond with a notional coupon of 7%, like the long gilt future on LIFFE, then the conversion factor will be less than 1.0 for bonds with a coupon lower than 7% and higher than 1.0 for bonds with a coupon higher than 7%. A formal definition of conversion factor is given below.

Although conversion factors equalise the yield on bonds, bonds in the delivery basket will trade at different yields, and, for this reason, they are not "equal" at the time of delivery. Certain bonds will be cheaper than others, and one bond will be the cheapest-to-deliver bond. The cheapest-to-deliver bond is the one that gives the greatest return from a strategy of buying a bond and simultaneously selling the futures contract, and then closing out positions on the expiry of the contract. This so-called cash-and-carry trading is actively pursued by proprietary trading desks in banks. If a contract is purchased and then held to maturity, the buyer will receive, via the exchange's clearing house, the cheapest-to-deliver gilt. Traders sometimes try to exploit arbitrage price differentials between the future and the cheapest-to-deliver gilt, known as basis trading. This is discussed in Choudhry (2003), where

CONVERSION FACTOR

The conversion factor (or price factor) gives the price of an individual cash bond such that its yield to maturity on the delivery day of the futures contract is equal to the notional coupon of the contract. The product of the conversion factor and the futures price is the forward price available in the futures market for that cash bond (plus the cost of funding, referred to as the gross basis).

the mathematical calculation of the conversion factor for the gilt future is given in Appendix 12.1.

We summarise the contract specification of the long gilt futures contract traded on LIFFE in Table 12.2. There is also a medium gilt contract on LIFFE, which was introduced in 1998 (having been discontinued in the early 1990s). This trades a notional five-year gilt, with eligible gilts being those of four to seven years maturity.

FUTURES PRICING

The Theoretical Principle

A futures exchange is probably the closest one gets to an example of the economist's perfect and efficient market. The immediacy and liquidity of the market will ensure that at virtually all times the price of any futures contract reflects fair value. In essence because a futures contract represents an underlying asset, albeit a synthetic one, its price cannot differ from the actual cash-market price of the asset itself. This is because the market sets futures prices such they are arbitrage-free. We can illustrate this with a hypothetical example.

Let us say that the benchmark 10-year bond, with a coupon of 8% is trading at par. This bond is the underlying asset represented by the long bond futures contract; the front month contract expires in precisely three months. If we also say that the three-month Libor rate (the repo rate) is 6%, what is fair value for the front month futures contract?

TABLE 12.2 LIFFE Long Gilt Future Contract Specifications

Unit of Trading	UK Gilt bond having a face value of £100,000, a notional coupon of 7% and a notional maturity of 10 years (changed from contract value of £50,000 from the September 1998 contract)
Deliverable grades	UK gilts with a maturity ranging from $8^3/_4$ to 13 years from the first day of the delivery month (changed from 10–15 years from the December 1998 contract)
Delivery months	March, June, September, December
Delivery date	Any business day during the delivery month
Last trading day	11:00 hours two business days before last business day of delivery month
Quotation	Percent of par expressed as points and hundredths of a point, for example 114.56 (changed from points and 1/32nds of a point, as in 114-17 meaning 114 17/32 or 114.53125, from the June 1998 contract)
Minimum price movement	0.01 of one point (one tick)
Tick value	£10
Trading hours	08:00–16:15 hours 16:22–18:00 hours All trading conducted electronically on LIFFE Connect

For the purpose of illustration, let us start by assuming the futures price to be 105. We could carry out the following arbitrage-type trade:

- Buy the bond for £100
- Simultaneously sell the future at £105
- Borrow £100 for three months at the repo rate of 6%

As this is a leveraged trade, we have borrowed the funds with which to buy the bond, and the loan is fixed at three months because we will hold the position to the futures contract expiry, which is in exactly three months' time. At expiry, as we are short futures, we will deliver the underlying bond to the futures clearing house and close out the loan. This strategy will result in cash flows for us as shown below.

Futures Settlement Cash Flows

Price received for bond	= 105.00
Bond accrued	= 2.00 (8% coupon for three months)
Total proceeds	= 107.00

Loan Cash Flows

Repayment of principal	= 100.00
Loan interest	= 1.500 (6% repo rate for three months)
Total outlay	= 101.50

The trade has resulted in a profit of £5.50, and this profit is guaranteed as we have traded the two positions simultaneously and held them both to maturity.We are not affected by subsequent market movements. The trade is an example of a pure arbitrage, which is risk-free. There is no cash outflow at the start of the trade because we borrowed the funds used to buy the bond. In essence, we have locked in the forward price of the bond by trading the future today, so that the final settlement price of the futures contract is irrelevant. If the situation described above were to occur in practice it would be very short-lived, precisely because arbitrageurs would buy the bond and sell the future to make this profit. This activity would force changes in the prices of both bond and future until the profit opportunity was removed.

So in our illustration, the price of the future was too high (and possibly the price of the bond was too low as well) and not reflecting fair value because the price of the synthetic asset was out of line with the cash asset. What if the price of the future was too low? Let us imagine that the futures contract is trading at 95.00. We could then carry out the following trade:

- Sell the bond at 100
- Simultaneously buy the future for 95
- Lend the proceeds of the short sale (100) for three months at 6%

This trade has the same procedure as the first one, with no initial cash outflow, except that we have to cover the short position in the repo market, through which we invest the sale proceeds at the repo rate of 6%. After three months, we are delivered a bond as part of the futures settlement, and this is used to close out our short position. How has our strategy performed?

Futures Settlement Cash Flows

Clean price of bond	=	95.00
Bond accrued	=	2.00
Total cash outflow	=	97.00

Loan Cash Flows

Principal on loan maturity	=	100.00
Interest from loan	=	1.500
Total cash inflow	=	101.500

The profit of £4.50 is again a risk-free arbitrage profit. Of course, our hypothetical world has ignored considerations such as bid-offer spreads for the bond, future, and repo rates, which would apply in the real world and impact any trading strategy. Yet again, however, the futures price is out of line with the cash market and has provided opportunity for arbitrage profit.

Given the terms and conditions that apply in our example, there is one price for the futures contract at which no arbitrage-profit opportunity is available. If we set the future price at 99.5, we would see that both trading strategies, buying the bond and selling the future or selling the bond and buying the future, yield a net cash flow of zero. There is no profit to be made from either strategy. So at 99.5 the futures price is in line with the cash market, and it will only move as the cash-market price moves; any other price will result in an arbitrage-profit opportunity.

Arbitrage-Free Futures Pricing

The previous section demonstrated how we can arrive at the fair value for a bond futures contract provided we have certain market information. The market mechanism and continuous trading will ensure that the fair price *is* achieved, as arbitrage-profit opportunities are eliminated. We can determine the bond future's price given:

- The coupon of the underlying bond, and its price in the cash market.
- The interest rate for borrowing or lending funds, from the trade date to the maturity date of the futures contract. This is known as the repo rate.

For the purpose of deriving this pricing model we can ignore bid-offer spreads and borrowing and lending spreads. If we set the following:

r	is the repo rate
rc	is the bond's running yield
P_{bond}	is the price of the cash bond
P_{fut}	is the price of the futures contract
t	is the time to the expiry of the futures contract

We can substitute these symbols into the cash-flow profile for our earlier trade strategy, that of buying the bond and selling the future. This gives us:

Futures Settlement Cash Flows

Clean price for bond $= P_{fut}$
Bond accrued $= rc.t.P_{bond}$
Total proceeds $= P_{fut} + (rc.t.P_{bond})$

Loan Cash Flows

Repayment of loan principal $= P_{bond}$
Loan interest $= r.t.P_{bond}$
Total outlay $= P_{bond} + (r.t.P_{bond})$

The profit from the trade would be the difference between the proceeds and outlay, which we can set as follows:

$$\text{Profit} = P_{fut} + rc.t.P_{bond} - (P_{bond} + r.t.P_{bond})$$

We have seen how the futures price is at fair value when there is no profit to be gained from carrying out this trade, so if we set profit at zero, we obtain the following:

$$0 = P_{fut} + rc.t.P_{bond} - (P_{bond} + r.t.P_{bond})$$

Solving this expression for the futures price P_{fut} gives us:

$$P_{fut} = P_{bond} + P_{bond}\,t(r - rc)$$

Rearranging this we get:

$$P_{fut} = P_{bond} + (1 + t[r - rc]) \tag{12.2}$$

If we repeat the procedure for the other strategy, that of selling the bond and simultaneously buying the future, and set the profit to zero, we will obtain the same expression for the futures price as given in equation (12.2).

It is the level of the repo rate in the market, compared to the running yield on the underlying bond, that sets the price for the futures contract. From the examples used at the start of this section we can see that it is the cost of funding compared to the repo rate that determines if the trade strategy results in a profit. The expression $[r - rc]$ from equation (12.2) is the net financing cost in the arbitrage trade, and is known as the *cost of carry.* If the running yield on the bond is higher than the funding cost (the repo rate), this is positive funding or *positive carry.* Negative funding (*negative carry*) is when the repo rate is higher than the running yield. The level of $[r - rc]$ will determine whether the futures price is trading above the cash-market price or below it. If we have positive carry (when $rc > r$) then the futures price will trade below the cash-market price, known as trading at a *discount*. Where $r > rc$ and we have negative carry, then the futures price will be at a premium over the cash-market price. If the net funding cost was zero, such that we had neither positive or negative carry, then the futures price would be equal to the underlying bond price.

The cost of carry related to a bond futures contract is a function of the yield curve. In a positive yield-curve environment, the three-month repo rate is likely to be lower than the running yield on a bond so that the cost of carry is likely to be positive. As there is generally only a liquid market in long bond futures out to contracts that mature up to one year from the trade date, with a positive yield curve it would be unusual to have a short-term repo rate higher than the running yield on the long bond. So, in such an environment, we would have the future trading at a discount to the underlying cash bond. If there is a negative-sloping yield curve, the futures price will trade at a premium to the cash price. It is in circumstances of changes in the shape of the yield curve that opportunities for relative value and arbitrage trading arise, especially as the bond that is cheapest-to-deliver for the futures contract may change with large changes in the curve.

A trading strategy that involved simultaneous and opposite positions in the cheapest-to-deliver bond (CTD) and the futures contract is known as cash-and-carry trading or basis trading. However, by the law of no-arbitrage pricing, the payoff from such a trading strategy should be zero. If we set the profit from such a trading strategy as zero, we can obtain a pricing formula for the fair value of a futures contract, which summarises the discussion above, and states that the fair-value futures price is a function of the cost of carry on the underlying bond. This is given in equation (12.3).

$$P_{fut} = \frac{(P_{bond} + AI_0) \times (1 + rt) - \sum_{i=1}^{N} \mathrm{Ci}(1 + rt_{i.del}) - AI_{del}}{CF} \tag{12.3}$$

where AI_0 is the accrued interest on the underlying bond today
AI_{del} is the accrued interest on the underlying bond on the expiry or delivery date (assuming the bond is delivered on the final day, which will be the case if the running yield on the bond is above the money-market rate)
C_i is the *i*th coupon
N is the number of coupons paid from today to the expiry or delivery date

HEDGING USING FUTURES

The Theoretical Position

Bond futures are used for a variety of purposes. Much of one day's trading in futures will be speculative; that is, a punt on the direction of the market. Another main use of futures is to hedge bond positions. In theory, when hedging a cash-bond position with a bond futures contract, if cash and futures prices move together, then any loss from one position will be offset by a gain from the other. When prices move exactly in lock-step with each other, the hedge is considered perfect. In practice, the price of even the cheapest-to-deliver bond (which one can view as being the bond being traded—implicitly—when one is trading the bond future) and the bond future will not move exactly in line with each other over a period of time. The difference between the cash price and the futures price is called the basis. The risk that the basis will change in an unpredictable way is known as *basis risk*.

Futures are a liquid and straightforward way of hedging a bond position. By hedging a bond position, the trader or fund manager is hoping to balance the loss on the cash position by the profit gained from the hedge. However, the hedge will not be exact for all bonds except the cheapest-to-deliver (CTD) bond, which we can assume is the futures contract underlying bond. The basis risk in a hedge position arises because the bond being hedged is not identical to the CTD bond. The basic principle is that if the trader is long (or net long, where the desk is running long and short positions in different bonds) in the cash market, an equivalent number of futures contracts will be sold to set up the hedge. If the cash position is short, the trader will buy futures. The hedging requirement can arise for different reasons. A market-maker will wish to hedge positions arising out of client business, when she is unsure when the resulting bond positions will be unwound. A fund manager may, for example, know that she needs to realise a cash sum at a specific time in the future to meet fund liabilities, and sell bonds at that time. The market maker will want to hedge against a drop in value of positions during the time the bonds are held. The fund manager will want to hedge against a rise in interest rates between now and the bond sale date, to protect the value of the portfolio.

When putting on the hedge position, the key is to trade the correct number of futures contracts. This is determined by using the hedge ratio of the bond and the future, which is a function of the volatilities of the two instruments. The number of contracts to trade is calculated using the hedge ratio, which is given by

$$\text{Hedge ratio} = \frac{\text{Volatility of bond to be hedged}}{\text{Volatility of hedging instrument}}$$

Therefore one needs to use the volatility values of each instrument. We can see from the calculation that if the bond is more volatile than the hedging instrument, then a greater amount of the hedging instrument will be required. Let us now look in greater detail at the hedge ratio.

THE FUTURES BASIS

The term "basis" is also used to describe the difference in price between the future and the deliverable cash bond. The basis is of considerable significance. It is often used to establish the fair value of a futures contract, as it is a function of the cost of carry. The gross basis is defined (for deliverable bonds only) as follows:

$$\text{Gross basis} = \text{Clean bond price} - (\text{futures price} \times \text{conversion factor}).$$

There are different methods available to calculate hedge ratios. The most common ones are the conversion-factor method, which can be used for deliverable bonds (also known as the price-factor method) and the modified-duration method (also known as the basis-point value method).

Where a hedge is put on against a bond that is in the futures delivery basket, it is common for the conversion factor to be used to calculate the hedge ratio. A conversion-factor hedge ratio is more useful, as it is transparent and remains constant, irrespective of any changes in the price of the cash bond or the futures contract. The number of futures contracts required to hedge a deliverable bond using the conversion-factor hedge ratio is determined using the following equation:

$$Number\ of\ contracts = \frac{M_{bond} \times CF}{M_{fut}} \tag{12.4}$$

where M is the nominal value of the bond or futures contract.

The conversion-factor method may only be used for bonds in the delivery basket. It is important to ensure that this method is only used for one bond. It is an erroneous procedure to use the ratio of conversion factors of two different bonds when calculating a hedge ratio. This will be considered again later.

Unlike the conversion-factor method, the modified-duration hedge ratio may be used for all bonds, both deliverable and nondeliverable. In calculating this hedge ratio the modified duration is multiplied by the dirty price of the cash bond to obtain the basis-point value (BPV). As we discovered in Chapter 2, the BPV represents the actual impact of a change in the yield on the price of a specific bond. The BPV allows the trader to calculate the hedge ratio to reflect the different price sensitivity of the chosen bond (compared to the CTD bond) to interest-rate movements. The hedge ratio calculated using BPVs must be constantly updated, because it will change if the price of the bond and/or the futures contract changes. This may necessitate periodic adjustments to the number of lots used in the hedge. The number of futures contracts required to hedge a bond using the BPV method is calculated using the following

$$Number\ of\ contracts = \frac{M_{bond}}{M_{fut}} \times \frac{BPV_{bond}}{BPV_{fut}} \tag{12.5}$$

where the BPV of a futures contract is defined with respect to the BPV of its CTD bond, as given by equation (12.6).

$$BPV_{fut} = \frac{BPV_{CTD\ bond}}{CF_{CTD\ bond}} \tag{12.6}$$

The simplest hedge procedure to undertake is one for a position consisting of only one bond, the cheapest-to-deliver bond. The relationship between the futures price and the price of the CTD given by equation (12.3) indicates that the price of the future will move for moves in the price of the CTD bond; so, therefore, we may set:

$$\Delta P_{fut} \cong \frac{\Delta P_{bond}}{CF} \tag{12.7}$$

where CF is the CTD conversion factor.

The price of the futures contract, over time, does not move tick for tick (although it may on an intraday basis) but rather by the amount of the change divided by the conversion factor. It is apparent, therefore, that to hedge a position in the CTD bond we must hold the number of futures contracts equivalent to the value of bonds held multiplied by the conversion factor. Obviously, if a conversion factor is less than one, the number of futures contracts will be less than the equivalent nominal value of the cash position; the opposite is true for bonds that have a conversion factor greater than one. However, the hedge is not as simple as dividing the nominal value of the bond position by the nominal value represented by one futures contract (!); this error is frequently made by graduate trainees and those new to the desk.

To measure the effectiveness of the hedge position, it is necessary to compare the performance of the futures position with that of the cash-bond position, and to see how much the hedge instrument mirrored the performance of the cash instrument. A simple calculation is made to measure the effectiveness of the hedge, given by equation (12.8), which is the percentage value of the hedge effectiveness.

$$Hedge\ effectiveness = -\left[\frac{Fut\ P/L}{Bond\ P/L}\right] \times 100 \tag{12.8}$$

Hedging a Bond Portfolio

The principles established above may be applied when hedging a portfolio containing a number of bonds. It is more realistic to consider a portfolio as holding not just bonds that are outside the delivery basket, but are also nongovernment bonds. In this case, we need to calculate the number of futures contracts to put on as a hedge based on the volatility of each bond in the portfolio compared to the volatility of the CTD bond. Note that, in practice, there is usually more than one futures contract that may be used as the hedge instrument. For example, in the sterling market it would be more sensible to use LIFFE's medium gilt contract, whose underlying bond has a notional maturity of four to seven years, if hedging a portfolio of short- to medium-dated bonds. However, for the purposes of illustration we will assume that only one contract, the long gilt, is available.

To calculate the number of futures contracts required to hold as a hedge against any specific bond, we use the expression in equation (12.9).

$$Hedge = \frac{M_{bond}}{M_{fut}} \times Vol_{bond/CTD} \times Vol_{CTD/fut} \tag{12.9}$$

where M is the nominal value of the bond or future
$Vol_{bond/CTD}$ is the relative volatility of the bond being hedged compared to that of the CTD bond
$Vol_{CTD/fut}$ is the relative volatility of the CTD bond compared to that of the future

It is not necessarily straightforward to determine the relative volatility of a bond vis-à-vis the CTD bond. If the bond being hedged is a government bond, we can calculate the relative volatility using the two bonds' modified duration. This is because the yields of both may be safely assumed to be strongly positively correlated. If, however, the bond being hedged is a corporate bond and/or a nonvanilla bond, we must obtain the relative volatility using regression analysis, as the yields between the two bonds may not be strongly positively correlated. This is apparent when one remembers that the yield spread of corporate bonds over government bonds is not constant and will fluctuate with changes in government-bond yields. To use regression analysis to determine relative volatilities, historical price data on the bond is required. The daily price moves in the target bond and the CTD bond are then analysed to assess the slope of the regression line. In this section, we will restrict the discussion to a portfolio of government bonds.

If we are hedging a portfolio of government bonds, we can use (12.10) to determine relative volatility values, which are based on the modified duration of each of the bonds in the portfolio.

$$Vol_{bond/CTD} = \frac{\Delta P_{bond}}{\Delta P_{CTD}} = \frac{MD_{bond} \times P_{bond}}{MD_{CTD} \times P_{CTD}} \tag{12.10}$$

where MD is the modified duration of the bond being hedged or the CTD bond, as appropriate. This preserves the terminology we introduced in Chapter 2.[1]

Once we have calculated the relative volatility of the bond being hedged, equation (12.11) (obtained by rearranging (12.7)) tells us that the relative volatility of the CTD bond to that of the futures contract is approximately the same as its conversion factor. We are then in a position to calculate the futures hedge for each bond in a portfolio.

$$Vol_{CTD/fut} = \frac{\Delta P_{CTD}}{\Delta P_{fut}} \approx CF_{CTD} \tag{12.11}$$

[1] In certain textbooks and research documents, it is suggested that the ratio of the conversion factors of the bond being hedged (if it is in the delivery basket) and the CTD bond can be used to determine the relative volatility of the target bond. This is a fallacious argument. The conversion factor of a deliverable bond is the price factor that will set the yield of the bond equal to the notional coupon of the futures contract on the delivery date, and it is a function mainly of the coupon of the deliverable bond. The price volatility of a bond, on the other hand, is a measure of its modified duration, which is a function of the bond's duration (that is, the term to maturity). Therefore, using conversion factors to measure volatility levels will produce erroneous results. It is important not to misuse conversion factors when arranging hedge ratios.

Table 12.3 shows a portfolio of five UK gilts on 20th October 1999. The nominal value of the bonds in the portfolio is £200 million, and the bonds have a market value, excluding accrued interest, of £206.84 million. Only one of the bonds is a deliverable bond, the $5^3/_4$% 2009 gilt, which is in fact the CTD bond. For the Dec99 futures contract, the bond had a conversion factor of 0.9124950. The fact that this bond is the CTD explains why it has a relative volatility of 1. We calculate the number of futures contracts required to hedge each position, using the equations listed above. For example, the hedge requirement for the position in the 7% 2002 gilt was calculated as follows:

$$\frac{5,000,000}{100,000} \times \frac{2.245 \times 101.50}{7.235 \times 99.84} \times 0.9124950 = 14.39$$

The volatility of all the bonds is calculated relative to the CTD bond, and the number of futures contracts determined using the conversion factor for the CTD bond. The bond with the highest volatility is, not surprisingly, the 6% 2028, which has the longest maturity of all the bonds and hence the highest modified duration. We note from Table 12.3 that the portfolio requires a hedge position of 2,091 futures contracts. This illustrates how a rough-and-ready estimate of the hedging requirement, based on nominal values, would be insufficient, as that would suggest a hedge position of only 2,000 contracts.

The effectiveness of the hedge must be monitored over time. No hedge will be completely perfect, however, and the calculation illustrated above, as it uses modified-duration value, does not take into account the convexity effect of the bonds. The reason why a futures hedge will not be perfect is because, in practice, the price of the futures contract will not move tick for tick with the CTD bond, at least not over a period of time. This is the basis risk that is inherent in hedging cash bonds with futures. In addition, the calculation of the hedge is only completely accurate for a parallel shift in yields, as it is based on modified duration, so as the yield curve changes around pivots, the hedge will move out of line. Finally, the long gilt future is not the appropriate contract to use to hedge three of the bonds in the portfolio, or over 25% of the portfolio by nominal value. This is because these bonds are short- or medium-dated, and so their price movements will not track the futures price as closely as longer-dated bonds. In this case, the more appropriate futures contract to use would have been the medium gilt contract, or (for the first bond, the 8% 2000) a strip of short sterling contracts. Using shorter-dated instruments would reduce some of the basis risk contained in the portfolio hedge.

THE MARGIN PROCESS

Institutions buying and selling futures on an exchange deal with only one counterparty at all times, the exchange clearing house. The clearing house is responsible for the settlement of all contracts, including managing the delivery process. A central clearing mechanism eliminates counterparty risk for anyone dealing on the exchange, because the clearing house guarantees the settlement of all transactions. The clearing house may be owned by the exchange itself, such as the one associated with the Chicago

TABLE 12.3 Bond Futures Hedge for Hypothetical Gilt Portfolio, 20 October 1999

		CTD			5.75% 2009		
		Modified duration			7.2345656		
		Conversion factor			0.9124950		
		Price			99.84		
Bond	**Nominal Amount (£m)**	**Price**	**Yield %**	**Duration**	**Modified Duration**	**Relative Volatility**	**Number of Contracts**
UKT 8%							
2000	12	102.17	5.972	1.072	1.01158797	0.143090242	15.67
UKT 7%							
2002	5	101.50	6.367	2.388	2.24505721	0.315483336	14.39
UKT 5%							
2004	38	94.74	6.327	4.104	3.85979102	0.50626761	175.55
UKT 5.75%							
2009	100	99.84	5.770	7.652	7.23456557	1	912.50
UKT 6%							
2028	45	119.25	4.770	15.031	14.3466641	2.368603078	972.60
Total	200						2,090.71

Mercantile Exchange (the CME Clearinghouse) or it may be a separate entity, such as the London Clearing House, which settles transactions on LIFFE. The LCH is also involved in running clearing systems for swaps and repo products in certain currencies.

One of the key benefits to the market of the clearing-house mechanism is that counterparty risk, as it is transferred to the clearing house, is virtually eliminated. The mechanism that enables the clearing house to accept the counterparty risk is the margining process that is employed at all futures exchanges. A bank or local trader must deposit margin before commencing dealing on the exchange; each day a further amount must be deposited or returned, depending on the results of the day's trading activity.

The exchange will specify the level of margin that must be deposited for each type of futures contract that a bank wishes to deal in. The initial margin will be a fixed sum per lot. So, for example, if the margin was £1,000 per lot, an opening position of 100 lots would require margin of £100,000. Once initial margin has been deposited, there is a mark-to-market of all positions at the close of business; exchange-traded instruments are the most transparent products in the market, and the closing price is not only known to everyone, it is also indisputable. The closing price is also known as the settlement price. Any losses suffered by a trading counterparty, whether closed out or run overnight, are entered as a debit on the party's account and must be paid the next day. Trading profits are credited and may be withdrawn from the margin account the next day. This daily process is known as variation margining. Thus, the margin account is updated on a daily basis, and the maximum loss that must be made up on any morning is the maximum price movement that occurred the previous day. It is a serious issue if a trading party is unable to meet a margin call. In such a case, the exchange will order it to cease trading, and will also liquidate all its open positions; any losses will be met out of the firm's margin account. If the level of funds in the margin account is insufficient, the losses will be made good from funds paid out of a general fund run by the clearing house, which is maintained by all members of the exchange.

Payment of margin is made by electronic funds transfer between the trading party's bank account and the clearing house. Initial margin is usually paid in cash, although clearing houses will also accept high-quality securities, such as T-bills or certain government bonds, to the value of the margin required. Variation margin is always cash. The advantage of depositing securities rather than cash is that the depositing firm earns interest on its margin. This is not available on a cash margin, and the interest forgone on a cash margin is effectively the cost of trading futures on the exchange. However, if securities are used, there is effectively no cost associated with trading on the exchange (we ignore, of course, infrastructure costs and staff salaries).

The daily settlement of exchange-traded futures contracts, as opposed to when the contract expires or the position is closed out, is the main reason why futures prices are not equal to forward prices for long-dated instruments. This is known as the convexity bias and is discussed in Chapter 15.

APPENDIX

APPENDIX 12.1 THE CONVERSION FACTOR FOR THE LONG GILT FUTURE

In this appendix, we describe the process used for the calculation of the conversion factor or price factor for deliverable bonds of the long gilt contract. The contract specifies a bond of maturity 8¾–13 years and a notional coupon of 7%.

For each bond that is eligible to be in the delivery basket, the conversion factor is given by the following expression:

$$\frac{P(7)}{100}$$

where the numerator $P(7)$ is equal to the price per £100 nominal of the deliverable gilt at which it has a gross redemption yield of 7%, calculated as at the first day of the delivery month, less the accrued interest on the bond on that day. This calculation uses the formula given in (A12.1.1) and the expression used to calculate accrued interest.

The numerator $P(7)$ is given by (A12.1.1).

$$P(7) = \frac{1}{1.035^{\frac{t}{s}}}\left[c_1 + \frac{c_2}{1.035} + \frac{C}{0.07}\left(\frac{1}{1.035} - \frac{1}{1.035^n}\right) + \frac{100}{1.035^n}\right] - AI \qquad \text{(A12.1.1)}$$

where
- c_1 is the cash flow due on the following quasi-coupon date, per £100 nominal of the gilt.
- c_1 will be zero if the first day of the delivery month occurs in the ex-dividend period or if the gilt has a long first coupon period and the first day of the delivery month occurs in the first full coupon period.
- c_1 will be less than $C/2$ if the first day of the delivery month falls in a short first coupon period. c_1 will be greater than $C/2$ if the first day of the delivery month falls in a long first coupon period and the first day of the delivery month occurs in the second full coupon period.
- c_2 is the cash flow due on the next-but-one quasi-coupon date, per £100 nominal of the gilt. c_2 will be greater than $C/2$ if the first day of the delivery month falls in a long first coupon period and in the first full coupon period. In all other cases, $c_2 = C/2$.
- C is the annual coupon of the gilt, per £100 nominal.
- t is the number of calendar days from and including the first day of the delivery month up to but excluding the next quasi-coupon date.
- s is the number of calendar days in the full coupon period in which the first day of the delivery month occurs.
- n is the number of full coupon periods between the following quasi-coupon date and the redemption date.
- AI is the accrued interest per £100 nominal of the gilt.

The accrued interest used in the formula above is given according to the following procedures.

If the first day of the delivery month occurs in a standard coupon period, and the first day of the delivery month occurs on or before the ex-dividend date, then:

$$AI = \frac{t}{s} \times \frac{C}{2} \qquad \text{(A12.1.2)}$$

If the first day of the delivery month occurs in a standard coupon period, and the first day of the delivery month occurs after the ex-dividend date, then:

$$AI = \left(\frac{t}{s} - 1\right) \times \frac{C}{2} \tag{A12.1.3}$$

where t is the number of calendar days from and including the last coupon date up to but excluding the first day of the delivery month

s is the number of calendar days in the full coupon period in which the first day of the delivery month occurs

If the first day of the delivery month occurs in a short first coupon period, and the first day of the delivery month occurs on or before the ex-dividend date, then:

$$AI = \frac{t^*}{s} \times \frac{C}{2} \tag{A12.1.4}$$

If the first day of the delivery month occurs in a short first coupon period, and the first day of the delivery month occurs after the ex-dividend date, then:

$$AI = \left(\frac{t^* - n}{s}\right) \times \frac{C}{2} \tag{A12.1.5}$$

where t^* is the number of calendar days from and including the issue date up to but excluding the first day of the delivery month

n is the number of calendar days from and including the issue date up to but excluding the next quasi-coupon date

If the first day of the delivery month occurs in a long first coupon period, and during the first full coupon period, then:

$$AI = \frac{u}{s_1} \times \frac{C}{2} \tag{A12.1.6}$$

If the first day of the delivery month occurs in a long first coupon period, and during the second full coupon period and on or before the ex-dividend date, then:

$$AI = \left(\frac{p_1}{s_1} + \frac{p_2}{s_2}\right) \times \frac{C}{2} \tag{A12.1.7}$$

If the first day of the delivery month occurs in a long first coupon period, and during the second full coupon period and after the ex-dividend date, then:

$$AI = \left(\frac{p_2}{s_2} - 1\right) \times \frac{C}{2} \tag{A12.1.8}$$

where u is the number of calendar days from and including the issue date up to but excluding the first day of the delivery month.

s_1 is the number of calendar days in the full coupon period in which the issue date occurs.

s_2 is the number of days in the next full coupon period after the full coupon period in which the issue date occurs.

p_1 is the number of calendar days from and including the issues date up to but excluding the next quasi-coupon date.

p_2 is the number of calendar days from and including the quasi-coupon date after the issue date up to but excluding the first day of the delivery month which falls in the next full coupon period after the full coupon period in which the issue date occurs.

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CHAPTER 13

Swaps

Swaps are one of the most important and useful instruments in the debt-capital markets, indeed the global economy. The main types of swap are interest-rate swaps, asset swaps, basis swaps, cross-currency swaps, and currency-coupon swaps. The market for swaps is organised by the International Swaps and Derivatives Association (ISDA). They are used by a wide range of institutions, including banks, mortgage banks and building societies, corporates, and local authorities. As the market has matured, the instrument has gained wider acceptance, and it is regarded as a plain-vanilla product in the debt-capital markets. Virtually all commercial and investment banks will quote swap prices for their customers, and as they are over-the counter (OTC) instruments, dealt over the telephone, it is possible for banks to tailor swaps to match the precise requirements of individual customers. There is also a close relationship between the bond market and the swap market, and corporate finance teams and underwriting banks keep a close eye on the government yield curve and the swap yield curve, looking out for possibilities regarding new issue of debt.

In this chapter, we review the use of interest-rate swaps from the point of view of the bond-market participant; this includes pricing and valuation and its use as a hedging tool. The bibliography lists further reading on important topics such as pricing, valuation, and credit risk.

INTEREST-RATE SWAPS

Background

Interest-rate swaps are the most important type of swap in terms of volume of transactions. They are used to manage and hedge interest-rate risk and exposure, while market makers will also take positions in swaps that reflect their view on the direction of interest rates. An interest-rate swap is an agreement between two counterparties to make periodic interest payments to one another during the life of the swap, on a predetermined set of dates, based on a notional principal amount. One party is the fixed-rate payer, and this rate is agreed at the time of trade of the swap; the other party is the floating-rate payer, the floating rate being determined during the life of the swap by reference to a specific market index. The principal or notional amount is not physically exchanged; hence, the term "off-balance-sheet", but is used merely to calculate the interest payments.[1] The fixed-rate payer receives floating-rate interest and is said to be "long" or to have "bought" the swap. The long side has conceptually purchased a floating-rate note (because it receives floating-rate interest) and issued a fixed-coupon bond (because it pays out fixed interest at intervals); that is, it has in principle borrowed funds. The floating-rate payer is said to be "short" or to have "sold" the swap. The short side has conceptually purchased a coupon bond (because it receives fixed-rate interest) and issued a floating-rate note (because it pays floating-rate interest). So an interest-rate swap is:

- An agreement between two parties
- To exchange a stream of cash flows
- Calculated as a percentage of a *notional* sum
- Calculated on different interest bases

For example, in a trade between Bank A and Bank B, Bank A may agree to pay fixed semiannual coupons of 10% on a notional principal sum of £1 million, in return for receiving from Bank B the prevailing six-month sterling Libor rate on the same amount. The known cash flow is the fixed payment of £50,000 every six months by Bank A to Bank B.

Interest-rate swaps trade in a secondary market, so their value moves in line with market interest rates, in exactly the same way as bonds. If a five-year interest-rate swap is transacted today at a rate of 5%, and five-year interest rates subsequently fall to 4.75%, the swap will have decreased in value to the fixed-rate payer, and correspondingly increased in value to the floating-rate payer, who has now seen the level of interest payments fall. The opposite would be true if five-year rates moved to 5.25%. Why is this? Consider the fixed-rate payer in an IR swap to be a borrower of funds; if she fixes the interest rate payable on a loan for five years, and then this interest rate decreases shortly afterwards, is she better off? No, because she is now paying above the market rate for the funds borrowed. For this reason, a swap contract decreases in value to the fixed-rate payer if there is a fall in rates. Equally a floating-rate payer gains if there is a fall in rates, as he can take advantage of the new rates and pay a lower level of interest; hence, the value of a swap increases to the floating-rate payer if there is a fall in rates.

A bank swaps desk will have an overall net interest-rate position arising from all the swaps it has traded that are currently on the book. This position is an interest-rate exposure at all points along the term structure, out to the maturity of the longest-dated swap. At the close of business each day, all the swaps on the book will be *marked-to-market* at the relevant tenor interest rate quoted for that day.

A swap can be viewed in two ways; either as a bundle of forward or futures contracts, or as a bundle of cash flows arising from the "sale" and "purchase" of cash-market instruments. If we imagine a strip of futures contracts, maturing every

[1] The author himself eschews the term "off-balance-sheet" because it is misleading; all derivatives have a very definite on-balance-sheet impact on any person transacting them, whether it is the contractual cash flows and/or the collateral cash flows. From an understanding of the balance sheet perspective, forget the accounting terminology; derivatives are on your balance sheet because their associated cash flows are on your balance sheet.

three or six months out to three years, we can see how this is conceptually similar to a three-year interest-rate swap. However, in the author's view it is better to visualise a swap as being a bundle of cash flows arising from cash instruments.

Let us imagine we have only two positions on our book:

1. A long position in £100 million of a three-year floating-rate note (FRN) that pays six-month Libor semiannually, and is trading at par
2. A short position in £100 million of a three-year gilt with coupon of 6% that is also trading at par

Being short a bond is the equivalent to being a borrower of funds. Assuming this position is kept to maturity, the resulting cash flows are shown in Table 13.1.

There is no net outflow or inflow at the start of these trades, as the £100 million purchase of the FRN is netted with receipt of £100 million from the sale of the gilt. The resulting cash flows over the three-year period are shown in the last column of Table 13.1. This net position is exactly the same as that of a fixed-rate payer in an (interest-rate) IR swap. As we had at the start of the trade, there is no cash inflow or outflow on maturity. For a floating-rate payer, the cash flow would mirror exactly a long position in a fixed-rate bond and a short position in an FRN. Therefore, the fixed-rate payer in a swap is said to be short in the bond market; that is, a borrower of funds. The floating-rate payer in a swap is said to be long the bond market.

Market Terminology

Virtually all swaps are traded under the legal terms and conditions stipulated in the ISDA standard documentation. The trade date for a swap is, not surprisingly, the date on which the swap is transacted. The terms of the trade include the fixed interest rate, the maturity and notional amount of the swap, and the payment bases of both legs of the swap. The date from which floating interest payments are determined is the setting date, which may also be the trade date. Most swaps fix the floating-rate payments to Libor, although other reference rates that are used include the U.S. Prime

TABLE 13.1 Three-Year Cash Flows

Cash Flows Resulting from Long Position in FRN and Short Position in Gilt

Period (6 mo)	FRN	Gilt	Net Cash Flow
0	–£100m	+£100m	£0
1	+(Libor × 100)/2	–3	+(Libor × 100)/2 – 3.0
2	+(Libor × 100)/2	–3	+(Libor × 100)/2 – 3.0
3	+(Libor × 100)/2	–3	+(Libor × 100)/2 – 3.0
4	+(Libor × 100)/2	–3	+(Libor × 100)/2 – 3.0
5	+(Libor × 100)/2	–3	+(Libor × 100)/2 – 3.0
6	+[(Libor × 100)/2] + 100	–103	+(Libor × 100)/2 – 3.0

The Libor rate is the six-month rate prevailing at the time of the setting; for instance, the Libor rate at period 4 will be the rate actually prevailing at period 4.

rate, the Fed Funds rate, euribor, the Treasury-bill rate, and the commercial-paper rate. In the same way as for a forward-rate agreement (FRA) and for eurocurrency deposits, the rate is fixed two business days before the interest period begins. The second (and subsequent) setting date will be two business days before the beginning of the second (and subsequent) swap periods. The effective date is the date from which interest on the swap is calculated, and this is typically two business days after the trade date. In a forward-start swap, the effective date will be at some point in the future, specified in the swap terms. The floating interest rate for each period is fixed at the start of the period, so that the interest payment amount is known in advance by both parties (the fixed rate is known, of course, throughout the swap by both parties).

Although for the purposes of explaining swap structures, both parties are said to pay interest payments (and receive them), in practice only the net difference between both payments changes hands at the end of each interest period. This eases the administration associated with swaps and reduces the number of cash flows for each swap. The counterparty that is the net payer at the end of each period will make a payment to the other counterparty. The first payment date will occur at the end of the first interest period, and subsequent payment dates will fall at the end of successive interest periods. The final payment date falls on the maturity date of the swap. The calculation of interest is given by equation (13.1).

$$I = M \times r \times \frac{n}{B} \tag{13.1}$$

where I is the interest amount, M is the nominal amount of the swap, and B is the interest day-base for the swap. Dollar- and euro-denominated swaps use an actual/360 day-count, similar to other money-market instruments in those currencies, while sterling swaps use an actual/365 day-count basis.

The cash flows in a vanilla interest-rate swap are illustrated in Figure 13.1.The counterparties in a swap transaction only pay across net cash flows, however, so at each interest payment date only one actual cash transfer will be made, by the net payer. This is shown as Figure 13.1(iii).

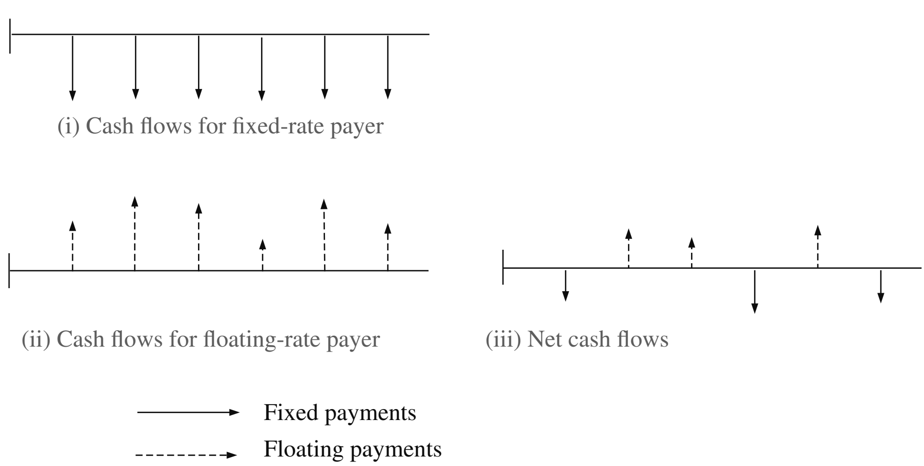

FIGURE 13.1 Cash Flows for Vanilla Interest-Rate Swap

Swap Spreads and the Swap Yield Curve

In the market, banks will quote two-way swap rates on screens or via a dealing system such as Reuters. Brokers will also be active in relaying prices in the market. The convention in the market is for the swap market maker to set the floating leg at Libor and then quote the fixed rate that is payable for that maturity. So for a five-year swap, a bank's swap desk might be willing to quote the following:

Floating-rate payer:	Pay 6-month-Libor	
		Receive fixed rate of 5.19%
Fixed-rate payer:	Pay fixed rate of 5.25%	
		Receive 6-month Libor

In this case, the bank is quoting an offer rate of 5.25%, which the fixed-rate payer will pay, in return for receiving Libor flat. The bid price quote is 5.19%, which is what a floating-rate payer will receive fixed. The bid-offer spread in this case is therefore 6 basis points. The fixed-rate quotes are always at a spread above the government bond yield curve. Let us assume that the five-year gilt is yielding 4.88%. In this case, then, the five-year swap bid rate is 31 basis points above this yield. So the bank's swap trader could quote the swap rates as a spread above the benchmark-bond yield curve, say 37-31, which is her swap spread quote. This means that the bank is happy to enter into a swap paying fixed 31 basis points above the benchmark yield and receiving Libor, and receiving fixed 37 basis points above the yield curve and paying Libor. The bank's screen on, say, Bloomberg or Reuters might look something like Table 13.2, which quotes the swap rates as well as the current spread over the government-bond benchmark.

The swap spread is a function of the same factors that influence the spread over government bonds for other instruments. For shorter-duration swaps of, say, up to three years, there are other yield curves that can be used in comparison, such as the cash-market curve or a curve derived from futures prices. For longer-dated swaps, the spread is determined mainly by the credit spreads that prevail in the corporate-bond market. Because a swap is viewed as a package of long and short positions in fixed- and floating-rate bonds, it is the credit spreads in these two markets that will determine the swap spread. This is logical; essentially, it is the premium for greater credit risk involved in lending to corporates that dictates that a swap rate will be higher than the same maturity government-bond yield. Technical factors will be responsible for day-to-day fluctuations in swap rates, such as the supply of corporate bonds and the level of demand for swaps, plus the cost to swap traders of hedging their swap positions.

TABLE 13.2 Swap Quotes

1-yr	4.50	4.45	+17
2-yr	4.69	4.62	+25
3-yr	4.88	4.80	+23
4-yr	5.15	5.05	+29
5-yr	5.25	5.19	+31
10-yr	5.50	5.40	+35

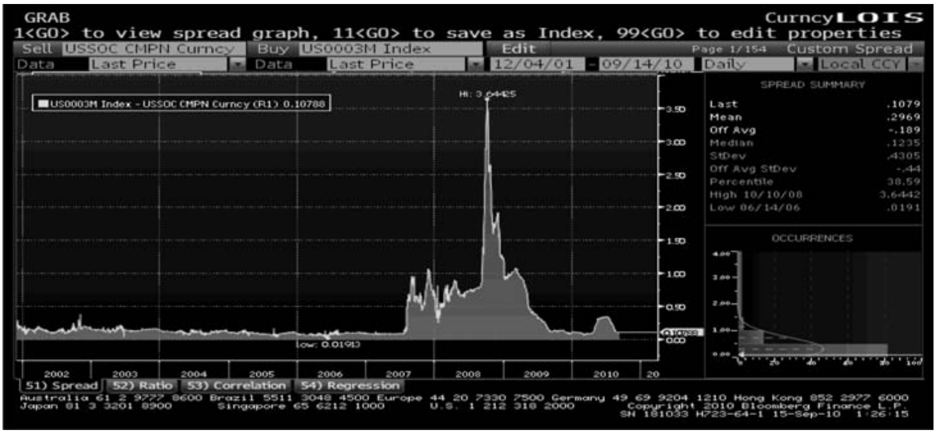

FIGURE 13.2 Libor-OIS Spread, 2002–2008

We can summarise by saying that swap spreads over government bonds reflect the supply and demand conditions of both swaps and government bonds, as well as the market's view on the credit quality of swap counterparties. There is considerable information content in the swap yield curve, much like that in the government-bond yield curve. During times of credit concerns in the market, such as the corrections in Asian and Latin American markets in the summer of 1998, and the 2008 bank crash, the swap spread will increase, more so at higher maturities. After the Lehman default in September 2008, the overnight index swap (OIS) spread over Libor widened considerably. The change in swap spreads is shown in Figure 13.2.

GENERIC SWAP VALUATION

Banks generally use par-swap (zero-coupon) swap pricing. We will look at this method in the next section. First, however, we will introduce an intuitive swap valuation method.

Intuitive Swap Pricing

Assume we have a vanilla interest-rate swap with a notional principal of N that pays n payments during its life, to a maturity date of T. The date of each payment is on t_i with $i = 1, \ldots n$. The present value today of a future payment is denoted by $PV(0, t)$. If the swap rate is r, the value of the fixed-leg payments is given by (13.2).

$$PV_{fixed} = N\sum_{i=1}^{n} PV(0,t_i)\times\left[r\times\left(\frac{t_i - t_{i-1}}{B}\right)\right] \tag{13.2}$$

where B is the money-market day base. The term $(t_i - t_{i-1})$ is simply the number of days between the ith and the i −1th payments.

The value of the floating-leg payments at the date t_1 for an existing swap is given by,

$$PV_{float} = N \times \left[rl \times \frac{t_1}{B} \right] + N - [N \times PV(t_1, t_n)] \tag{13.3}$$

where rl is the Libor rate that has been set for the next interest payment. We set the present value of the floating-rate payment at time 0 as follows:

$$PV(0, t_1) = \frac{1}{1 + rl(t_1)\left(\frac{t_1}{B}\right)} \tag{13.4}$$

For a new swap, the value of the floating payments is given by

$$PV_{float} = N\left\{ \left[rl \times \frac{t_1}{B} + 1 \right] \times PV(0, t_1) - PV(0, t_n) \right\} \tag{13.5}$$

The swap valuation is then given by $PV_{fixed} - PV_{float}$. The swap rate quoted by a market-making bank is that which sets $PV_{fixed} = PV_{float}$ and is known as the par or zero-coupon swap rate. We consider this next.

Zero-Coupon Swap Pricing

So far, we have discussed how vanilla swap prices are often quoted as a spread over the benchmark government-bond yield in that currency, and how this swap spread is mainly a function of the credit spread required by the market over the government (risk-free) rate. This method is convenient and also logical because banks use government bonds as the main instrument when hedging their swap books. However, because much bank swap trading is now conducted in nonstandard, tailor-made swaps, this method can sometimes be unwieldy, as each swap needs to have its spread calculated to suit its particular characteristics. Therefore, banks use a standard pricing method for all swaps known as zero-coupon swap pricing.

In Chapter 3, we referred to zero-coupon bonds and zero-coupon interest rates. Zero-coupon or spot rates, are true interest rates for their particular term to maturity. In zero-coupon swap pricing, a bank will view all swaps, even the most complex, as a series of cash flows. The zero-coupon rates that apply now for each of the cash flows in a swap can be used to value these cash flows. Therefore, to value and price a swap, each of the swap's cash flows are present-valued using known spot rates; the sum of these present values is the value of the swap.

In a swap, the fixed-rate payments are known in advance, and so it is straightforward to present-value them. The present value of the floating rate payments is usually estimated in two stages. First, the implied forward rates can be calculated using (13.6). We are quite familiar with this relationship from our reading of Chapter 3

$$rf_i = \left(\frac{df_i}{df_{i+1}} - 1 \right) N \tag{13.6}$$

where rf_i is the one-period forward rate starting at time i
df_i is the discount factor for the maturity period i
df_{i+1} is the discount factor for the period $i + 1$
N is the number of times per year that coupons are paid

By definition, the floating-payment interest rates are not known in advance, so the swap bank will predict what these will be, using the forward rates applicable to each payment date. The forward rates are those that are currently implied from spot rates. Once the size of the floating-rate payments has been estimated, these can also be valued by using the spot rates. The total value of the fixed and floating legs is the sum of all the present values, so the value of the total swap is the net of the present values of the fixed and floating legs.

While the term "zero-coupon" refers to an interest rate that applies to a discount instrument that pays no coupon and has one cash flow (at maturity), it is not necessary to have a functioning zero-coupon bond market in order to construct a zero-coupon yield curve. In practice, most financial pricing models use a combination of the following instruments to construct zero-coupon yield curves:

- Money-market deposits
- Interest-rate futures
- FRAs
- Government bonds.

Frequently an overlap in the maturity period of all instruments is used. FRA rates are usually calculated from interest-rate futures so it is only necessary to use one of either FRA or futures rates.

Once a zero-coupon yield curve (term structure) is derived, this may be used to value a future cash flow maturing at any time along the term structure. This includes swaps: to price an interest-rate swap, we calculate the present value of each of the cash flows using the zero-coupon rates and then sum all the cash flows. As we noted above, while the fixed-rate payments are known in advance, the floating-rate payments must be estimated, using the forward rates implied by the zero-coupon yield curve. The net present value of the swap is the net difference between the present values of the fixed- and floating-rate legs.

Calculating the Forward Rate from Spot-Rate Discount Factors

Remember that one way to view a swap is as a long position in a fixed-coupon bond that was funded at Libor, or against a short position in a floating-rate bond. The cash flows from such an arrangement would be paying floating-rate and receiving fixed-rate. In the former arrangement, where a long position in a fixed-rate bond is funded with a floating-rate loan, the cash flows from the principals will cancel out, as they are equal and opposite (assuming the price of the bond on purchase was par), leaving a collection of cash flows that mirror an interest-rate swap that pays floating and receives fixed. Therefore, as the fixed-rate on an interest-rate swap is the same as the coupon (and yield) on a bond priced at par, calculating the fixed-rate on an interest-rate swap is the same as calculating the coupon for a bond that we wish to issue at par.

The price of a bond paying semiannual coupons is given by (13.7), which may be rearranged for the coupon rate *r* to provide an equation that enables us to determine the par yield, and hence the swap rate *r*, given by (13.8).

$$\mathrm{P} = \frac{r_n}{2} df_1 + \frac{r_n}{2} df_2 + \ldots\ldots + \frac{r_n}{2} df_n + M df_n \tag{13.7}$$

where r_n is the coupon on an *n*-period bond with *n* coupons and *M* is the maturity payment. Assuming $P = 1$ and $M = 1$, it can be shown then that

$$\begin{aligned} r_n &= \frac{1 - df_n}{\frac{df_1}{2} + \frac{df_2}{2} + \ldots\ldots + \frac{df_n}{2}} \\ &= \frac{1 - df_n}{\sum_{i=1}^{n} \frac{df_i}{2}} \end{aligned} \tag{13.8}$$

For annual coupon bonds, there is no denominator for the discount factor, while for bonds paying coupons on a frequency of *N* we replace the denominator 2 with *N*.[2] The expression in (13.8) may be rearranged again, using *F* for the coupon frequency, to obtain an equation that may be used to calculate the *n*th discount factor for an *n*-period swap rate, given in (13.9).

$$df_n = \frac{1 - r_n \sum_{i=1}^{n-1} \frac{df_i}{N}}{1 + \frac{r_n}{N}} \tag{13.9}$$

The expression in (13.9) is the general expression for the *bootstrapping* process that we first encountered in Chapter 1. Essentially, to calculate the *n*-year discount factor we use the discount factors for the years 1 to $n - 1$, and the *n*-year swap rate or zero-coupon rate. If we have the discount factor for any period, we may use (13.9) to determine the same period zero-coupon rate, after rearranging it, shown in (13.10).

$$rs_n = \sqrt[t_n]{\frac{1}{df_n}} - 1 \tag{13.10}$$

Discount factors for spot rates may also be used to calculate forward rates. We know that

$$df_1 = \frac{1}{\left(1 + \frac{rs_1}{N}\right)} \tag{13.11}$$

[2] The expression also assumes an actual/365 day-count basis. If any other day-count convention is used, the $1/N$ factor must be replaced by a fraction made up of the actual number of days as the numerator and the appropriate year base as the denominator.

where rs is the zero-coupon rate. If we know the forward rate we may use this to calculate a second discount rate, shown by (13.12).

$$df_2 = \frac{df_1}{\left(1 + \frac{rf_1}{N}\right)} \tag{13.12}$$

where rf_1 is the forward rate. This is of no use in itself; however, we may derive from it an expression to enable us to calculate the discount factor at any point in time between the previous discount rate and the given forward rate for the period n to $n + 1$, shown in (13.13), which may then be rearranged to give us the general expression to calculate a forward rate, given in (13.14).

$$df_{n+1} = \frac{df_n}{\left(1 + \frac{rf_n}{N}\right)} \tag{13.13}$$

$$rf_n = \left(\frac{df_n}{df_{n+1}} - 1\right)N \tag{13.14}$$

The general expression for an n-period discount rate at time n from the previous period forward rates is given by (13.15).

$$\begin{aligned} df_n &= \frac{1}{\left(1 + \frac{rf_{n-1}}{N}\right)} \times \frac{1}{\left(1 + \frac{rf_{n-2}}{N}\right)} \times \ldots\ldots \times \frac{1}{\left(1 + \frac{rf_n}{N}\right)} \\ df_n &= \prod_{i-0}^{n-1} \left[\frac{1}{\left(1 + \frac{rf_i}{N}\right)}\right] \end{aligned} \tag{13.15}$$

From the above (13.7 to 13.15), we may combine equations (13.8) and (13.14) to obtain the general expression for an n-period swap rate and zero-coupon rate, given by (13.16) and (13.17), respectively.

$$r_n = \frac{\sum_{i=1}^{n} \frac{rf_{i-1} df_i}{N}}{\sum_{i=1}^{n} \frac{df_i}{F}} \tag{13.16}$$

$$1 + rs_n = \sqrt[t_n]{\prod_{i=0}^{n-1} \left(1 + \frac{rf_i}{N}\right)} \tag{13.17}$$

The two expressions do not tell us anything new, as we have already encountered their results in Chapter 3. The swap rate, which we have denoted as r_n is shown by (13.16) to be the weighted average of the forward rates. If we consider that a strip of FRAs constitutes an interest-rate swap, then a swap rate for a continuous period could be covered by a strip of FRAs. Therefore, an average of the FRA rates would be the correct swap rate. As FRA rates are forward rates, we may be comfortable with (13.16), which states that the *n*-period swap rate is the average of the forward rates from rf_0 to rf_n. To be accurate, we must weight the forward rates, and these are weighted by the discount factors for each period. Note that although swap rates are derived from forward rates, interest payments under a swap are paid in the normal way at the end of an interest period, while payments for an FRA are made at the beginning of the period and must be discounted.

Equation (13.17) states that the zero-coupon rate is calculated from the geometric average of (one plus) the forward rates. The *n*-period forward rate is obtained using the discount factors for periods n and $n - 1$. The discount factor for the complete period is obtained by multiplying the individual discount factors together, and exactly the same result would be obtained by using the zero-coupon interest-rate for the whole period to obtain the discount factor.

Illustrating Interest-Rate Swap Pricing

The rate charged on a newly transacted interest-rate swap is the one that gives its net present value as zero. The term *valuation* of a swap is used to denote the process of calculating the net present value of an existing swap, when marking-to-market the swap against current market interest rates. Therefore, when we price a swap, we set its net present value to zero; while, when we value a swap, we set its fixed rate at the market rate and calculate the net present value.

To illustrate the basic principle, we price a plain-vanilla interest-rate swap with the terms set out below; for simplicity we assume that the annual fixed-rate payments are the same amount each year, although in practice there would be slight differences. Also assume that we already have our zero-coupon yields as shown in Table 13.3.

We use the zero-coupon rates to calculate the discount factors, and then use the discount factors to calculate the forward rates. This is done using equation (13.14). These forward rates are then used to predict what the floating-rate payments will be at each interest period. Both fixed-rate and floating-rate payments are then present-valued at the appropriate zero-coupon rate, which enables us to calculate the net present value.

TABLE 13.3 Generic Interest-Rate Swap

Period	Zero-Coupon Rate %	Discount Factor	Forward Rate %	Fixed Payment	Floating Payment	PV Fixed Payment	PV Floating Payment
1	5.50	0.94	5.50	689,625	550,000.00	653,672.98	521,327.01
2	6.00	0.88	6.50	689,625	650,236.96	613,763.79	578,708.58
3	6.25	0.83	6.75	689,625	675,177.02	574,944.84	562,899.47
4	6.50	0.77	7.25	689,625	725,353.49	536,061.43	563,834.02
5	7.00	0.71	9.02	689,625	902,358.47	491,693.09	643,369.11
		4.16				2,870,137.00	2,870,137.00

The fixed-rate for the swap is calculated using equation (13.8) to give us:

$$\frac{1-0.71298618}{4.16187950}$$

or 6.8963%.

The swap terms are:

Nominal principal	£10 million
Fixed rate	6.8963%
Day count fixed	Actual/365
Day count floating	Actual/365
Payment frequency fixed	Annual
Payment frequency floating	Annual
Trade date	31st January 2000
Effective date	2nd February 2000
Maturity date	2nd February 2005
Term	Five years

For reference, the Microsoft Excel formulae are shown in Table 13.4. It is not surprising that the net present value is zero, because the zero-coupon curve is used to derive the discount factors, which are then used to derive the forward rates, which are used to value the swap. As with any financial instrument, the fair value is its break-even price or hedge cost, and in this case the bank that is pricing the five-year swap shown in Table 13.3 could hedge the swap with a series of FRAs transacted at the forward rates shown. If the bank is paying fixed and receiving floating, value of the swap to it will rise if there is a rise in market rates, and fall if there is a fall in market rates. Conversely, if the bank was receiving fixed and paying floating, the swap value to it would fall if there was a rise in rates, and vice versa.

This method is used to price any interest-rate swap, even an exotic one.

Valuation Using Final-Maturity Discount Factor

A short-cut to valuing the floating-leg payments of an interest-rate swap involves using the discount factor for the final maturity period. This is possible because, for the purposes of valuation, an exchange of principal at the beginning and end of the swap is conceptually the same as the floating-leg interest payments. This holds because, in an exchange of principal, the interest payments earned on investing the initial principal would be uncertain, as they are floating rate, while on maturity the original principal would be returned. The net result is a floating-rate level of receipts, exactly similar to the floating-leg payments in a swap. To value the principals, then, we need only the final maturity discount rate.

To illustrate, consider Table 13.3, where the present value of both legs was found to be £2,870,137. The same result is obtained if we use the five-year discount factor, as shown below.

$$PV_{\text{floating}} = (10,000,000 \times 1) - (10,000,000 \times 0.71298618) = 2,870,137$$

TABLE 13.4 Generic Interest-Rate Swap (Excel formulae)

CELL	C	D	E	F	G	H	I	J
21			10000000					
22								
23	Period	Zero-Coupon Rate %	Discount Factor	Forward Rate %	Fixed Payment	Floating Payment	PV Fixed Payment	PV Floating Payment
24	1	5.50	0.94	5.50	689,625	"(F24*10000000)/100	"G24/1.055	"H24/(1.055)
25	2	6.00	0.88	"((E24/E25)-1)*100	689,625	"(F25*10000000)/100	"G24/(1.06)^2	"H25/(1.06)^2
26	3	6.25	0.83	"((E25/E26)-1)*100	689,625	"(F26*10000000)/100	"G24/(1.0625)^3	"H26/(1.0625^3)
27	4	6.50	0.77	"((E26/E27)-1)*100	689,625	"(F27*10000000)/100	"G24/(1.065)^4	"H27/(1.065)^4
28	5	7.00	0.71	"((E27/E28)-1)*100	689,625	"(F28*10000000)/100	"G24/(1.07)^5	"H28/(1.07)^5
			"SUM(E24:E28)				2,870,137.00	2,870,137.00

The first term is the principal multiplied by the discount factor 1; this is because the present value of an amount valued immediately is unchanged (or rather, it is multiplied by the immediate-payment discount factor, which is 1.0000).

Therefore, we may use the principal amount of a swap if we wish to value the swap. This is, of course, for valuation only, as there is no actual exchange of principal in a swap.

Summary of IR Swap

Let us summarise the chief characteristics of swaps. A plain-vanilla swap has the following characteristics:

- One leg of the swap is fixed-rate interest, while the other will be floating-rate, usually linked to a standard index such as Libor.
- The fixed rate is fixed through the entire life of the swap.
- The floating rate is set in advance of each period (quarterly, semiannually, or annually) and paid in arrears.
- Both legs have the same payment frequency.
- The maturity can be standard whole years up to 30 years, or set to match the customer's requirements.
- The notional principal remains constant during the life of the swap.

Of course, to meet customer demand banks can set up swaps that have variations on any or all of the above standard points. Some of the more common variations are discussed in a following section.

ASSET SWAPS

Asset swaps combine an interest-rate swap with a bond and are seen as both cash market instruments and also as credit derivatives. They are used to alter the cash flow profile of a bond. The asset swap market is an important segment of the credit derivatives market since it explicitly sets out the price of credit as a spread over Libor. Pricing a bond by reference to Libor is commonly used, and the spread over Libor is a measure of credit risk in the cash flow of the underlying bond. Asset swaps can be used to transform the cash flow characteristics of reference assets, so that investors can hedge the currency, credit, and interest-rate risks of cash bonds they have purchased to create synthetic investments with more suitable cash-flow characteristics. An asset swap package involves transactions in which the investor acquires a bond position and then enters into an interest rate swap with the bank that sold him the bond. The investor pays fixed and receives floating. This transforms the fixed coupon of the bond into a Libor-based floating coupon.

As an example, consider an entity that wishes to insure against loss due to credit events such as default or bankruptcy of the issuer of a bond it is holding. For risk management purposes, the buyer holding this risky bond wishes to hedge the credit risk of this position. By means of an asset swap the protection seller will agree to pay the protection buyer Libor +/– a spread in return for the cash flows of the risky bond (there is no exchange of notional at any point). In the event of default the protection

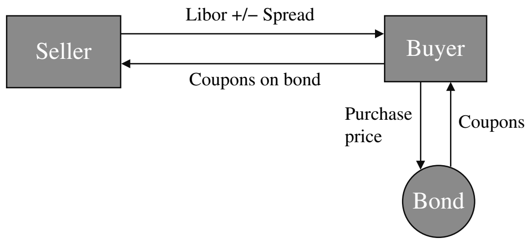

FIGURE 13.3 Asset Swap

buyer will continue to receive the Libor +/– spread from the protection seller. In this way the protection buyer has transformed its original risk profile by changing both its interest rate and credit risk exposure.

The generic structure of an asset swap is shown in Figure 13.3.

Asset Swaps Example

Assume that an investor holds a bond and enters into an asset swap with a bank. Then the value of an asset swap is the spread the bank pays over or under Libor. This is based on the following components:

1. Value of the coupons of the underlying asset compared to the market swap rate.
2. The accrued interest and the clean price premium or discount compared to par value. Thus when pricing the asset swap it is necessary to compare the par value and to the underlying bond price.

The spread above or below Libor reflects the credit spread difference between the bond and the swap rate.

The Bloomberg asset swap calculator pricing screen ASW shown in Figure 13.4 shows these components in the analysis of the swapped spread details.

Illustration of Asset Swap Terms

Let us assume that we have a credit-risky bond with the following details:

Currency: EUR
Issue date: 31 March 2000
Maturity: 31 March 2007
Coupon: 5.5% per annum
Price (Dirty): 105.3%
Price (Clean): 101.2%
Yield: 5%
Accrued interest: 4.1%
Credit rating: A1

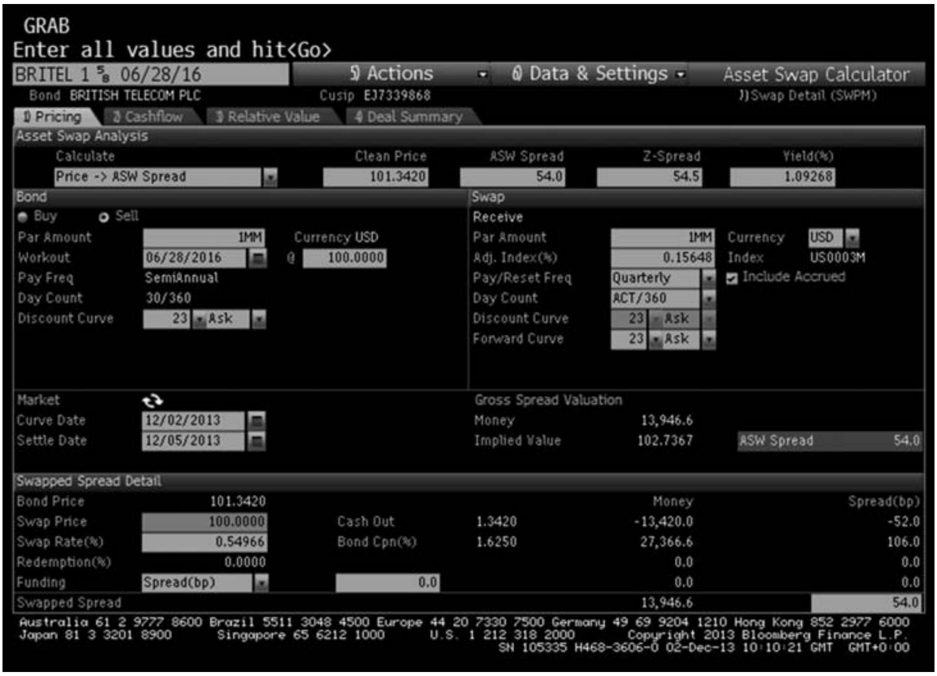

FIGURE 13.4 Example of Bloomberg Asset Swap Calculator Screen

To buy this bond the investor would pay 105.3% of par value. The investor would receive the fixed coupons of 5.5% of par value. Let us assume that the swap rate is 5%. If the investor in this bond enters into an asset swap with a bank in which the investor pays the fixed coupon and receives Libor +/– spread.

The asset swap price (that is, the spread) on this bond has the following components:

1. The value of the excess value of the fixed coupons over the market swap rate is paid to the investor. Let us assume that in this case this is approximately 0.5% when spread into payments over the life of the asset swap.
2. The difference between the bond price and par value is another factor in the pricing of an asset swap. In this case the price premium that is expressed in present value terms should be spread over the term of the swap and treated as a payment by the investor to the bank (if a dirty price is at a discount to the par value, then the payment is made from the bank to the investor). For example, in this case let us assume that this results in a payment from the investor to the bank of approximately 0.23% when spread over the term of the swap.

These two elements result in a net spread of 0.5% – 0.23% = 0.27%. Therefore the asset swap would be quoted as Libor + 0.27% (or Libor plus 27 bps).

NONVANILLA INTEREST-RATE SWAPS

The swap market is very flexible, and instruments can be tailor-made to fit the requirements of individual customers. A wide variety of swap contracts have been traded in the market. Although the most common reference rate for the floating-leg of a swap is six-month Libor, for a semiannual-paying floating leg, other reference rates that have been used include three-month Libor, the prime rate (for dollar swaps), the one-month commercial paper rate, the Treasury bill rate, and the municipal bond rate (again, for dollar swaps). The term of a swap need not be fixed; swaps may be extendable or put-able. In an extendable swap, one of the parties has the right but not the obligation to extend the life of the swap beyond the fixed maturity date, while in a put-able swap one party has the right to terminate the swap ahead of the specified maturity date. It is also possible to transact options on swaps, known as swaptions. A swaption is the right to enter into a swap agreement at some point in the future, during the life of the option. Essentially, a swaption is an option to exchange a fixed-rate bond cash flow for a floating-rate bond cash-flow structure. As a floating-rate bond is valued on its principal value at the start of a swap, a swaption may be viewed as the value on a fixed-rate bond, with a strike price that is equal to the face value of the floating-rate bond.

Constant-Maturity Swap

A constant-maturity swap is a swap in which the parties exchange a Libor rate for a fixed swap rate. For example, the terms of the swap might state that six-month Libor is exchanged for the five-year swap rate on a semiannual basis for the next five years, or for the five-year government bond rate. In the U.S. market, the second type of constant-maturity swap is known as a constant-maturity Treasury swap.

Amortising/Accreting Swap

In a plain-vanilla swap the notional principal remains unchanged during the life of the swap. However, it is possible to trade a swap where the notional principal varies during its life. An accreting (or step-up) swap is one in which the principal starts off at one level and then increases in amount over time. The opposite, an amortising swap, is one in which the notional reduces in size over time. An accreting swap would be useful where, for instance, a funding liability that is being hedged increases over time. The amortising swap might be employed by a borrower hedging a bond issue that featured sinking-fund payments, where a part of the notional amount outstanding is paid off at set points during the life of the bond. If the principal fluctuates in amount—for example, increasing in one year and then reducing in another—the swap is known as a roller-coaster swap. Another application for an amortising swap is as a hedge for a loan that is itself an amortising one. Frequently, this is combined with a forward-starting swap, to tie in with the cash flows payable on the loan. The pricing and valuation of an amortising swap are no different in principle from a vanilla interest-rate swap; a single swap rate is calculated using the relevant discount factors, and at this rate the net present value of the swap cash flows will equal zero at the start of the swap.

Libor-in-Arrears Swap

In a Libor-in-arrears swap (also known as a back-set swap), the setting date is just before the end of the accrual period for the floating-rate setting and not just before the start. Such a swap would be attractive to a counterparty who had a different view on interest rates from the market consensus. For instance, in a rising yield-curve environment, forward rates will be higher than current market rates, and this will be reflected in the pricing of a swap. A Libor-in-arrears swap would be priced higher than a conventional swap. If the floating-rate payer believed that interest rates would, in fact, rise more slowly than forward rates (and the market) were suggesting, she may wish to enter into an arrears swap as opposed to a conventional swap.

Basis Swap

In a conventional swap, one leg comprises fixed-rate payments and the other floating-rate payments. In a basis swap, both legs are floating-rate but linked to different money-market indices. One leg is normally linked to Libor, while the other might be linked to the CD rate, say, or to the commercial-paper rate. This type of swap would be used by a bank in the United States that had made loans that paid at the prime rate, and financed its loans at Libor. A basis swap would eliminate the basis risk between the bank's income and expense cash flows. Other basis swaps have been traded where both legs are linked to Libor, but at different maturities; for instance, one leg might be at three-month Libor and the other at six-month Libor. In such a swap, the basis is different and so is the payment frequency: one leg pays out semiannually while the other would be paying on a quarterly basis. Note that where the payment frequencies differ, there is a higher level of counterparty risk for one of the parties. For instance, if one party is paying out on a monthly basis but receiving semiannual cash flows, it would have made five interest payments before receiving one in return.

Margin Swap

It is common to encounter swaps where there is a margin above or below Libor on the floating leg, as opposed to a floating leg of Libor flat. If a bank's borrowing is financed at Libor+25 bps, it may wish to receive Libor+25 bps in the swap so that its cash flows match exactly. The fixed-rate quote for a swap must be adjusted correspondingly to allow for the margin on the floating side. This is known as a margin swap. In our example, if the fixed-rate quote is, say, 6.00%, it would be adjusted to around 6.25%; differences in the margin quoted on the fixed leg might arise if the day-count convention or payment frequency were to differ between fixed and floating legs. Another reason why there may be a margin is if the credit quality of the counterparty demanded it, so that highly rated counterparties may pay slightly below Libor, for instance.

When a swap is transacted, its fixed rate is quoted at the current market rate for that maturity. Where the fixed rate is different from the market rate, this is an off-market swap, and a compensating payment is made by one party to the other. An off-market rate may be used for particular hedging requirements, for example, or when a bond issuer wishes to use the swap to hedge the bond as well as to cover the bond's issue costs.

Differential Swap

A differential swap is a basis swap but with one of the legs calculated in a different currency. Typically, one leg is floating-rate, while the other is floating-rate but with the reference index rate for another currency, but denominated in the domestic currency. For example, a differential swap may have one party paying six-month sterling Libor, in sterling, on a notional principal of £10 million, and receiving euro-Libor, minus a margin, payable in sterling and on the same notional principal. Differential swaps are not very common and are the most difficult for a bank to hedge. The hedging is usually carried out using what is known as a quanto option.

Forward-Start Swap

A forward-start swap is one where the effective date is not the usual one or two days after the trade date but a considerable time afterwards, for instance; say, six months after trade date. Such a swap might be entered into where one counterparty wanted to fix a hedge or cost of borrowing now, but for a point sometime in the future. Typically, this would be because the party considered that interest rates would rise or the cost of hedging would rise. The swap rate for a forward-starting swap is calculated in the same way as that for a vanilla swap.

SWAPTIONS

Description

Swaptions are options on swaps. The buyer of a swaption has the right but not the obligation to enter into an interest rate swap agreement during the life of the option. The terms of the swaption will specify whether the buyer is the fixed- or floating-rate payer; the seller of the option (the *writer)* becomes the counterparty to the swap if the option is exercised. The convention is that if the buyer has the right to exercise the option as the fixed-rate payer, he has traded a *call swaption,* also known as a *payer swaption,* while if by exercising the buyer of the swaption becomes the floating-rate payer he has bought a *put swaption,* also known as a *receiver swaption.* The writer of the swaption is the party to the other leg. In the sterling market swaptions are referred to in terms similar to FRA terminology, so that a 3/6 or 3-6 payer's swaption is a three-year option to pay fixed on a three-year interest-rate swap.

Swaptions are similar to forward-start swaps up to a point, but the buyer has the *option* of whether or not to commence payments on the effective date. A bank may purchase a call swaption if it expects interest rates to rise, and will exercise the option if indeed rates do rise as the bank has expected. A company will use swaptions as part of an interest-rate hedge for a future exposure. For example, assume that a company will be entering into a five-year bank loan in three months time. Interest on the loan is charged on a floating-rate basis, but the company intends to swap this to a fixed-rate liability after they have entered into the loan. As an added hedge, the company may choose to purchase a swaption that gives it the right to receive Libor and pay a fixed rate, say 10%, for a five-year period beginning in three months time. When the time comes for the company to take out a swap and exchange its interest-rate liability in three months time, (having entered into the loan), if the five-year swap rate is below 10%, the company will transact the swap in the normal way, and the swaption will expire worthless. If the five-year swap rate is above 10%, however, the company will instead exercise the swaption, giving it the right to enter into a five-year swap and paying a fixed rate of 10%. Essentially, the company has taken out protection to ensure that it does not have to pay a fixed rate of more than 10%. Hence swaptions can be used to guarantee a maximum swap rate liability. They are similar to forward-starting swaps, but do not commit a party to enter into a swap on fixed terms. The swaption enables a company to hedge against unfavourable movements in interest rates but also to gain from favourable movements, although there is of course a cost associated with this, which is the premium paid for the swaption.

As with conventional put and call options, swaptions turn in-the-money under opposite circumstances. A call swaption increases in value as interest rates rise, and a put swaption becomes more valuable as interest rates fall. Consider a one-year European call swaption on a five-year semiannual interest-rate swap, purchased by a bank counterparty. The notional value is £10 million, and the "strike price" is 6%, against Libor. Assume that the price (premium) of the swaption is 25 basis points, or £25,000. On expiry of the swaption, the buyer will either exercise it, in which case she will enter into a five-year swap paying 6% and receiving floating-rate interest, or elect to let the swaption expire with no value. If the five-year swap rate for counterparty of similar credit quality to the bank is above 6%, the swaption holder will exercise the swaption, while if the rate is below 6% the buyer will not exercise. The principle is the same for a put swaption, only in reverse.

A company will use swaptions as part of an interest-rate hedge for a future exposure. For example, assume that a company will be entering into a five-year bank loan in three months time. Interest on the loan is charged on a floating-rate basis, but the company intends to swap this to a fixed-rate liability after they have entered into the loan. As an added hedge, the company may choose to purchase a swaption that gives it the right to receive Libor and pay a fixed rate, say 10%, for a five-year period beginning in three months time. When the time comes for the company to take out a swap and exchange its interest-rate liability in three months time (having entered into the loan), if the five-year swap rate is below 10%, the company will transact the swap in the normal way, and the swaption will expire worthless. If, however, the five-year swap rate is above 10%, the company will instead exercise the swaption, giving it the right to enter into a five-year swap and paying a fixed rate of 10%. Essentially the company has taken out protection to ensure that it does not have to pay a fixed rate of more than 10%. Hence swaptions can be used to guarantee a maximum swap rate liability. They are similar to forward-starting swaps, but do not commit a party to enter into a swap on fixed terms. The swaption enables a company to hedge against unfavourable movements in interest rates but also to gain from favourable movements, although there is, of course, a cost associated with this, which is the premium paid for the swaption.

Valuation

Swaptions are typically priced using the Black-Scholes or Black 76 option-pricing models. These are used to value a European option on a swap, assuming that the appropriate swap rate at the expiry date of the option is lognormal. Consider a swaption with the following general terms:

Swap rate on expiry	rs
Swaption strike rate	rX
Maturity	T
Start date	t
Pay basis	F (say quarterly, semiannual, or annual)
Notional principal	M

If the actual swap rate on the maturity of the swaption is *rs*, the pay-off from the swaption is given by:

$$\frac{M}{F}\max(r - r_n, 0)$$

The value of a swaption is essentially the difference between the strike rate and the swap rate at the time it is being valued. If a swaption is exercised, the payoff at each interest date is given by $(rs - rX) \times M \times F$. As a call swaption is only exercised when the swap rate is higher than the strike rate (that is, $rs > rX$), the option payoff on any interest payment in the swap is given by

$$Swaption_{Interest\ Payment} = \text{Max}[0, (rs - rX)] \times M \times F \tag{13.18}$$

It can then easily be shown that the value of a call swaption on expiry is given by

$$PV_{Swaption} = \sum_{n=1}^{N} Df_{(0,n)}(rs - rX) \times M \times F \tag{13.19}$$

where $Df_{(0,\,\mathrm{n})}$ is the spot-rate discount factor for the term beginning now and ending at time t. By the same logic, the value of a put swaption is given by the same expression except that $(rX - rs)$ is substituted at the relevant point above.

Consider then, that a swaption can be viewed as a collection of calls or puts on interest deposits or Libor, enabling us to use the Black model when valuing it. This means that we value each call or put for a single payment in the swap, and then sum these payments to obtain the value of the swaption. The main assumption made when using this model is that the Libor rate follows a lognormal distribution over time, with constant volatility.

Consider a call swaption being valued at time t that matures at time T. We begin by valuing a single payment under the swap (assuming the option is exercised) made at time T_n. The point at time T_n is into the life of the swap, so that we have $T_n > T > t$. At the time of valuation, the option time to expiry is $T - t$ and there is $T_n - t$ until the nth payment. The value of this payment is given by

$$C_t = MFe^{-r(T_n - t)}[rsN(d1) - rXN(d2)] \tag{13.20}$$

where C_t is the price of the call option on a single payment in the swap
r is the risk-free instantaneous interest rate
$N(.)$ is the cumulative normal distribution
σ is the interest-rate volatility

and where

$$d_1 = \frac{1n(rs_t / rX) + \frac{\sigma^2}{2}(T - t)}{\sigma\sqrt{T - t}}$$

$$d_2 = d_1 - \sigma\sqrt{T - t}$$

The remaining life of the swaption ($T - t$) governs the probability that it will expire in-the-money, determined using the lognormal distribution. On the other hand the interest payment itself is discounted (using $e^{-r(Tn-t)}$) over the period $T_n - t$ as it is not paid until time T_n.

Having valued a single interest payment, viewing the swap as a collection of interest payments, we value the call swaption as a collection of calls. Its value is given therefore by

$$PVSwaptuib_t = \sum_{n=1}^{N} MFe^{-r(T_n - t)}[rsN(d_1) - rXN(d_2)] \tag{13.21}$$

where *t*, *T*, and *n* are as before.

If we substitute discrete spot-rate discount factors instead of the continuous form given by (13.21) the expression becomes

$$PVSwaption_t = MF[rsN(d_1) - rXN(d_2)\sum_{n=1}^{N} Df_{t,T_n}] \tag{13.22}$$

Note that vanilla caps and floors are also priced using the Black 76 model, as are European swaptions. This leads to some interesting results.[3] For instance, given our basic assumption that a forward-starting swap is a linear function of forward interest rates, using the Black 76 model for caps or floors assumes that forward *interest rates* follow a lognormal distribution. Using the same model for swaptions means we are assuming that forward *swap* rates are lognormally distributed. This is a contradictory set of assumptions; nevertheless the market uses the same model to price both products.

If we consider a payer's swaption with strike rX and value of PV_F and PV_L for fixed and floating sides respectively, the swaption payoff is given by

$$\max[0, PV_L - PV_F]$$

As the level of interest rates fluctuates during the life of the swaption, the values PV_F and PV_L will also move. This produces a payoff that does not follow the general form for the Black 76 model, which is a stochastic pattern in relation to the fixed level of rX. To counter this the market also uses a "spread option model",[4] which is given in equation (13.23).

$$PV_{Swaption} = PV_L \times N(d_1) - PV_F \times N(d_2) \tag{13.23}$$

where

$$d_1 = [1\text{n}(PV_L / PV_F) + {}^1/_2\ \sigma^2 t] / (\sigma \sqrt{t}), \text{ etc}$$
$$\sigma^2 = (\sigma_L)^2 + (\sigma_F)^2 - 2\sigma_L \sigma_F \rho_{LF}$$

and

σ_L and σ_F are the volatilities of the floating and fixed sides
ρ_{LF} is the correlation between the two sides

See Example 13.1 on swaption pricing.

[3] See, for example, R. Flavell, *Swaps and Other Derivatives* (Chichester, UK: John Wiley & Sons, 2002), Chapter 7.
[4] Ibid., Chapter 7.

EXAMPLE 13.1 SWAPTION PRICING

We present a hypothetical term structure environment in this example to illustrate the basic concepts.[5] This is shown in Table 13.5. We wish to price a forward-starting annual interest swap starting in two years for a term of three years. The swap has a notional of £10 million.

TABLE 13.5 Interest Rate Data for Swaption Valuation

Date	Term (years)	Discount Factor	Par Yield	Zero-Coupon Rate	Forward Rate
18-2-2001	0	1	5	5	5
18-2-2002	1	0.95238095	5.00	5	6.03015
18-2-2003	2	0.89821711	5.50	5.51382	7.10333
18-2-2004	3	0.83864539	6.00	6.04102	6.66173
18-2-2005	4	0.78613195	6.15	6.19602	6.71967
20-2-2006	5	0.73637858	6.25	6.30071	8.05230
19-2-2007	6	0.68165163	6.50	6.58946	8.70869
18-2-2008	7	0.62719194	6.75	6.88862	9.40246
18-2-2009	8	0.57315372	7.00	7.20016	10.1805
18-2-2010	9	0.52019523	7.25	7.52709	5.80396
18-2-2011	10	0.49165950	7.15	7.35361	6.16366

The swap rate is given by

$$rs = \frac{\sum_{t=1}^{N} rf_{(t-1),t} \times Df_{0,t}}{\sum_{t=1}^{N} Df_{0,t}}$$

where rf is the forward rate.

Using the above expression, the numerator in this example is (0.0666 × 0.8386) + (0.0672 × 0.7861) + (0.0805 × 0.7634) or 0.1701.

The denominator is

0.8386 + 0.7861 + 0.7634 or 2.3881

[5] This example follows an approach described in a number of other texts, for example R. Kolb, *Futures, Options and Swaps* (Blackwell, 2000); although here we use discount factors in the calculation whereas in Kolb the illustration uses zero-coupon factors which are (1 + spot rate).

(*continued*)

EXAMPLE 13.1 (*Continued*)

Therefore the forward-starting swap rate is 0.1701/2.3881 or 0.071228 (7.123%).

We now turn to the call swaption on this swap, the buyer of which acquires the right to enter into a three-year swap paying the fixed-rate swap rate of 7.00%. If the volatility of the forward swap rate is 0.20, the d_1 and d_2 terms are

$$d_1 \frac{\ln\left(\frac{rs}{rX}\right)+\frac{\sigma^2}{2}(T-t)}{\sigma\sqrt{T-t}} = \frac{\ln\left(\frac{0.071228}{0.07}\right)+\left(\frac{0.2^2}{2}\times 2\right)}{0.2\sqrt{2}}$$

or 0.2029068.

$$d^2 = d^1 - \sigma\sqrt{T-t} = 0.20290618 - 0.2(1.4142)$$

or −0.079934.

The cumulative normal values are[6]

$N(d_1) = N(0.2029)$ which is 0.580397
$N(d_2) = N(-0.079934)$ which is 0.468145

From above we know that $\sum Df_{t,T_n}$ is 2.3881. So using (13.22) we calculate the value of the call swaption to be

$$\begin{aligned} PVSwaption_t &= MF[rsN(d_1) - rXN(d_2)]\sum_{n-1}^{N} Df_{t,T_n} \\ &= 10{,}000{,}000 \times 1 \times [0.07228 \times 0.580397 - 0.07 \times 0.468145] \\ &\quad \times 2.3881 \\ &= 219{,}250 \end{aligned}$$

or £219,250. Option premiums are frequently quoted as basis points of the notional amount, so in this case the premium is (219,250/10,000,000) or 219.25 basis points.

[6] These values may be found from standard normal distribution tables or using the Microsoft Excel formula =NORMSDIST().

Swaptions for Risk Management

A swaption is another instrument that may be used for interest-rate risk management purposes. Hence they are an alternative to swaps, caps, and floors. In some cases, an institution will have the option of using any of these products. For instance, consider a situation where a corporate entity is aware that it must borrow funds at a future date, for a fixed period of time. The funds are only available at a floating rate. This requirement presents an interest-rate risk, in that rates may rise between now and the start date of the loan, and then rise during the loan. If the Treasury desk of the entity wishes to remove uncertainty and nullify this risk, it may consider the following:

- Trade in a forward-start swap, for the term of the loan, paying fixed and receiving floating. The swap will come into effect at the start of the loan. This removes uncertainty in that the company knows what it will be paying for the term of the loan, thus its interest-rate exposure is known. The downside is that the company cannot benefit if interest rates fall.
- Enter into a forward-start cap. The company can buy a cap, with start date for the start of the loan, and this fixes its upper borrowing rate. It also enables the company to gain if rates fall.
- Buy a swaption. The company can purchase a payers swaption, with expiry date matching the start date of the loan, which if exercised kicks in a pay-fixed swap for the term of the loan. If rates were above the strike rate of the swaption on expiry, the swaption will be exercised, otherwise it would expire worthless.

Alternatively the company could leave the exposure intact and do nothing. It will suffer if rates rise, but it will benefit if rates fall, and it wouldn't have paid any money for the privilege! Shareholders prefer certainty over risk exposure however, so this strategy is rarely followed.

The choice between a cap and a swaption will be influenced by their relative cost. Cap premiums are significantly higher than swaption premiums, both at-the-money or out-of-the-money. This is not unexpected, because a cap provides a greater level of protection for its buyer. It lasts for the entire period of the exposure, and also enables its buyer to benefit from downward movements in rates. The swaption only provides protection for the period leading up to the start of the loan, and then can be exercised only on expiry. If rates have fallen, it will expire unexercised. The cap can be exercised at fixed times during the term of the loan (as each caplet approaches its exercise date). It therefore incorporates more time value than a swaption. So it is a straightforward choice the company is faced with: the higher-priced cap that offers more insurance, or the lower-cost swaption. This is illustrated further in Example 13.2.

OVERNIGHT INTEREST-RATE SWAPS AND EONIA/SONIA SWAPS

This section could also have been placed in Chapter 6, our chapter on money-market securities and derivatives. Overnight-index swaps (OIS) are interest-rate swaps that are traded overwhelmingly in the money markets.

We saw earlier in the chapter that an interest-rate swap contract, which is generally regarded as a capital market instrument, is an agreement between two counterparties to exchange a fixed interest-rate payment in return for a floating

EXAMPLE 13.2 SWAPTION AND CAP PREMIUMS

A corporate will need to enter into a two-year sterling loan in one year's time. The loan rate will be a spread over three-month Libor. To hedge the future risk exposure, it can purchase a forward-start vanilla cap for the term of the loan, a mid-curve cap or a ⅓ payers swaption. We have the following terms:

Trade date	8 January 2001
Start date	9 January 2002
Terms	3-month Libor from 9 Jan 02 to 9 Jan 04
Strike	5.50%
Volatility	12% p.a.

In the sterling market as at January 2001 we observed the following premiums:

Two-year vanilla cap	44.7 bps
Two-year mid-curve cap	36.5 bps
⅓ payer swaption	32.9 bps

If the company exercises the swaption, it will be paying a fixed rate on its loan. This, being the swap rate, is in effect the average of the implied forward rates given by the zero-coupon curve as of January 2001. If Libor falls after the exercise date, the company will not benefit. The vanilla cap is the expensive approach, but it has eight exercise dates—every quarter for the two-year period of the loan. The mid-curve cap has one exercise date, but is made up of eight separate options. The swaption is the cheapest instrument because it is a single option on one exercise date.

interest-rate payment, calculated on a notional swap amount, at regular intervals during the life of the swap. A swap may be viewed as being equivalent to a series of successive FRA contracts, with each FRA starting as the previous one matures. The basis of the floating interest rate is agreed as part of the contract terms at the inception of the trade. Conventional swaps index the floating interest rate to Libor; however, an exciting recent development in the money markets has been the OIS. In the sterling market they are known as sterling overnight interest rate average swaps or SONIA, while euro-currency OIS are known as Eonia. In this section we review OIS swaps, which are used extensively by commercial and investment banks.

SONIA Swaps

SONIA is the average interest rate of interbank (unsecured) overnight sterling deposit trades undertaken before 1530 hours each day between members of the London Wholesale Money Brokers' Association. Recorded interest rates are weighted by volume. A SONIA swap is a swap contract that exchanges a fixed interest rate (the swap rate) against the geometric average of the overnight interest rates that have been recorded during the life of the contracted. Exchange of interest takes place on maturity of the swap. SONIA swaps are used to speculate on or to hedge against interest rates at the very short end of the sterling yield curve; in other words, they can be used to hedge an exposure to overnight interest rates.[7] The swaps themselves are traded in maturities of one week to one year, although two-year SONIA swaps have also been traded.

Conventional swap rates are calculated off the government bond yield curve and represent the credit premium over government yields of interbank default risk. Essentially they represent an average of the forward rates derived from the government spot (zero-coupon) yield curve. The fixed rate quoted on a SONIA swap represents the average level of the overnight interest rates expected by market participants over the life of the swap. In practice, the rate is calculated as a function of the Bank of England's repo rate. This is the two-week rate at which the Bank conducts reverse repo trades with banking counterparties as part of its open market operations. In other words, this is the Bank's base rate. In theory one would expect the SONIA rate to follow the repo rate fairly closely, since the credit risk on an overnight deposit is low. In practice, however, the spread between the SONIA rate and the Bank repo rate is very volatile, and for this reason the swaps are used to hedge overnight exposures.

The daily turnover in SONIA swaps is considerably lower than cash instruments such as gilt repo (£20 billion) or more established derivative instruments such as short sterling (£45 billion); however, it is now a key part of the sterling market. Most trades are between one-week and three-month maturity, and the bid-offer spread has been reported by the BoE as around 2 basis points, which compares favourably with the 1-basis-point spread of short sterling.

Figure 13.5 illustrates the monthly average of the SONIA index minus Bank's repo rate during 1999 and 2000, with the exaggerated spread in December 1999 reflecting millenium bug concerns.

An illustration of the use of OIS to hedge a funding position is given in Example 13.3.

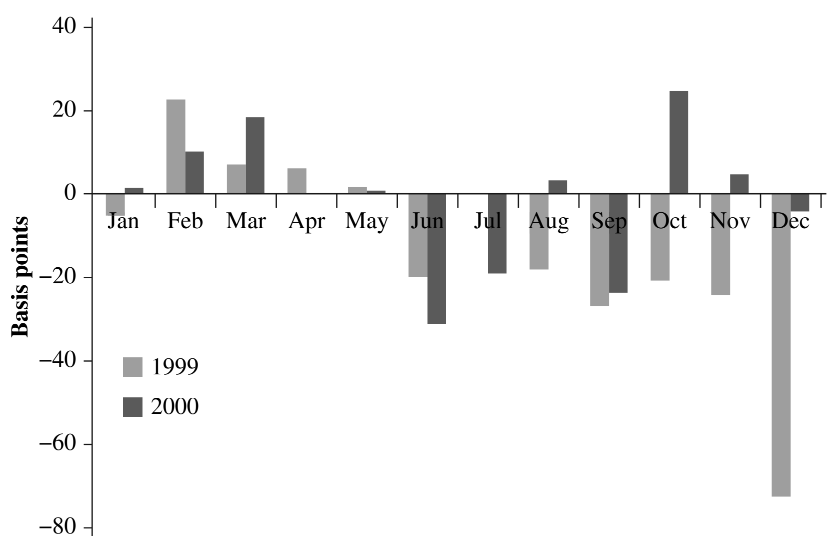

FIGURE 13.5 SONIA Average Rate Minus BoE Repo Rate, 1999-2000
Source: Bank of England.

[7] Traditionally overnight rates fluctuate in a very wide range during the day, depending on the day's funds shortage, and although volatility has reduced since the introduction of gilt repo, it is still unpredictable on occasion.

EXAMPLE 13.3 USING AN OIS SWAP TO HEDGE A FUNDING REQUIREMENT

A structured hedge fund derivatives desk at an investment bank offers a leveraged investment product to a client in the form of a participating interest share in a fund of hedge funds. The client's investment is leveraged up by funds lent to it by the investment bank, for which the interest rate charged is overnight Libor plus a spread. (In other words, for instance for each $25 invested by the client, the investment bank puts up $75 to make a total investment of $100. This gives the investor a leveraged investment in the hedge fund of funds. In most cases, the client would also bear the first $15 of loss of the $100 share of the investment.)

Assume that this investment product has an expected life of at least two years, and possibly longer. As part of its routine asset-liability management operations, the bank's Treasury desk has been funding this requirement by borrowing overnight each day. It now wishes to match the funding requirement raised by this product by matching asset term structure to the liability term structure. Let us assume that this product creates a USD 1 billion funding requirement for the bank.

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11:37 TULLETT & TOKYO PAGE 1 / 1

	USD Cash Deposits	Non-Japanese Bid	Ask	Time		USD Cash Deposits	Japanese Bid	Ask	Time
1)	Spot	1.0000	1.0200	9:33	18)	T/N	1.0000	1.0300	11/07
2)	T/N	1.0100	1.0300	11/07	19)	1 Week	1.0400	1.0600	9:33
3)	1 Week	1.0300	1.0500	9:33	20)	2 Week	1.0500	1.0700	9:33
4)	2 Week	1.0300	1.0500	9:33	21)	3 Week	1.0600	1.0800	9:33
5)	3 Week	1.0300	1.0500	9:33	22)	1 Month	1.0800	1.1000	9:33
6)	1 Month	1.0400	1.0500	9:33	23)	2 Month	1.1800	1.2100	9:33
7)	2 Month	1.1200	1.1400	9:33	24)	3 Month	1.1900	1.2200	9:33
8)	3 Month	1.1300	1.1500	9:33	25)	4 Month	1.2000	1.2300	9:33
9)	4 Month	1.1400	1.1700	9:33	26)	5 Month	1.2100	1.2400	9:33
10)	5 Month	1.1600	1.1900	9:33	27)	6 Month	1.2300	1.2600	9:33
11)	6 Month	1.2000	1.2200	9:33	28)	7 Month	1.2700	1.3000	9:33
12)	7 Month	1.2300	1.2500	9:33	29)	8 Month	1.3100	1.3400	9:33
13)	8 Month	1.2700	1.2900	9:33	30)	9 Month	1.3800	1.4100	9:33
14)	9 Month	1.3300	1.3600	9:33	31)	10 Month	1.4600	1.4900	9:33
15)	10 Month	1.3800	1.4100	9:33	32)	11 Month	1.5300	1.5600	9:33
16)	11 Month	1.4500	1.4800	9:33	33)	12 Month	1.5500	1.5800	9:33
17)	12 Month	1.5000	1.5300	9:33					

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2003 Bloomberg L.P.
G657-802-0 10-Nov-03 11:37:50

FIGURE 13.6 Tullet U.S. Dollar Deposit Rates, 10 November 2003

The current market deposit rates are shown in Figure 13.6. The Treasury desk therefore funds this requirement in the following way:

Assets	$1 billion, > 1-year term
	Receiving overnight Libor + 130 bps
Liability	$350 million, six-month loan
	Pay 1.22%
	$350 million, 12-month loan
	Pay 1.50%
	$300 million, 15-month loan
	Pay 1.70% (not shown in figure 13.6)

EXAMPLE 13.3 (*Continued*)

GRAB Corp ICAU

11:34 USD OIS – ICAU PAGE 1 / 1

	USD OIS	Ask	Bid	Time
1)	1 Month	1.0190	0.9990	9:30
2)	2 Month	1.0240	1.0040	9:30
3)	3 Month	1.0310	1.0110	9:30
4)	4 Month	1.0440	1.0240	10:59
5)	5 Month	1.0710	1.0510	10:59
6)	6 Month	1.1050	1.0850	11:04
7)	7 Month	1.1420	1.1220	10:59
8)	8 Month	1.1920	1.1720	11:00
9)	9 Month	1.2420	1.2220	11:05
10)	10 Month	1.2930	1.2730	11:00
11)	11 Month	1.3580	1.3380	11:00
12)	12 Month	1.4210	1.4000	11:06
13)	15 Month	1.6250	1.6040	11:00
14)	18 Month	1.8090	1.7890	11:00
15)	21 Month	2.0080	1.9880	11:00
16)	24 Month	2.2030	2.1820	11:00

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2003 Bloomberg L.P.
G657-802-0 10-Nov-03 11:34:17

FIGURE 13.7 Garban ICAP U.S. Dollar OIS Rates, 10 November 2003

This matches the asset structure more closely to the term structure of the assets; however, it opens up an interest-rate basis mismatch in that the bank is now receiving an overnight-Libor-based income but paying a term-based liability. To remove this basis mismatch, the Treasury desk transacts an OIS swap to match the amount and term of each of the loan deals, paying overnight floating-rate interest and receiving fixed-rate interest. The rates for OIS swaps of varying terms are shown in Figure 13.7, which shows two-way prices for OIS swaps up to two years in maturity. So for the six-month OIS the hedger is receiving fixed-interest at a rate of 1.085% and for the 12-month OIS he is receiving 1.40%. The difference between what it is receiving in the swap and what it is paying in the term loans is the cost of removing the basis mismatch, but more fundamentally reflects a key feature of OIS swaps versus deposit rates: deposit rates are Libor-related, whereas U.S. dollar OIS rates are driven by the Fed Funds rate. On average, the Fed Funds rate lies approximately 8–10 bps below the dollar deposit rate, and sometimes as much as 15 bps below cash levels. Note that at the time of this trade, the Fed Funds rate was 1% and the market was not expecting a rise in this rate until at least the second half of 2004. This sentiment would have influenced the shape of the USD OIS curve.

The action taken above hedges out the basis mismatch and also enables the Treasury desk to match its asset profile with its liability profile. The net cost to the Treasury desk represents its hedging costs.

Figure 13.8 illustrates the transaction.

(*continued*)

EXAMPLE 13.3 (*Continued*)

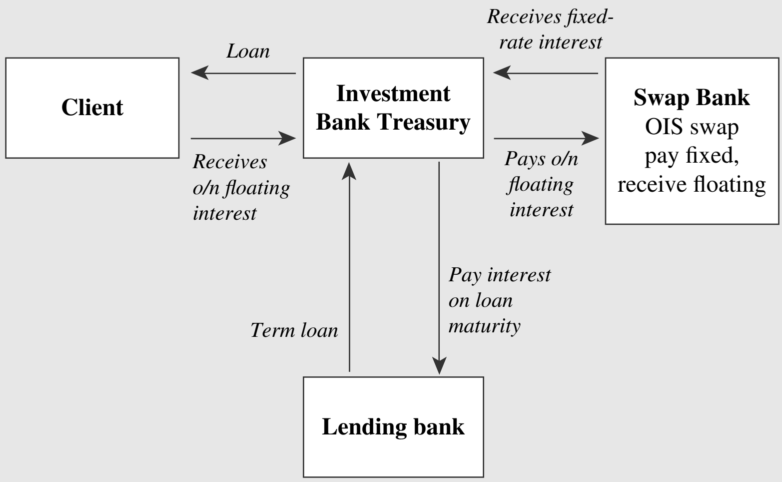

FIGURE 13.8 Illustration of Interest Basis Mismatch Hedging Using OIS Instrument

OIS Swap Terms

To illustrate OISs further, we give here the terms of one of the OIS executed in Example 13.3, the six-month swap. The counterparties to the trade are as labelled in Figure 13.8.

Notional	$350 million
Trade date	10 November 2003
Effective date	12 November 2003
Termination date	12 May 2004
Payment terms	The net interest payment is paid as a bullet amount on maturity

Fixed Amounts		
	Fixed rate payer	OIS swap bank
	Fixed rate period end date	12 May 2004
	Fixed rate	1.085%
	Fixed rate day-count fraction	Act/360

Floating Amounts		
	Floating rate payer	Treasury desk
	Floating rate period end date	12 May 2004
	Floating rate option	USD-Fed Funds

The floating rate is calculated as follows:

$$F_{OIS} = \left[\prod\left(1 + \frac{FedFunds_i \times n_i}{360}\right) - 1\right] \times \frac{360}{d} \tag{13.24}$$

where d_0 is the number of New York banking days in the calculation period
i is a series of whole numbers from 1 to d_0, each representing a New York banking day

$FedFunds_i$ is a reference rate equal to the overnight USD Federal Funds interest rate, as displayed on Telerate page 118 and Bloomberg page BTMM
n_i is the number of calendar days in the calculation period on which the rate is $FedFunds_i$
d is the number of days in the calculation period

Floating rate day-count	Act/360
Reset dates	The last day of each calculation period
Compounding	Inapplicable
Business day convention	Modified following business day
Calculation agent	OIS swap bank

AN OVERVIEW OF INTEREST-RATE SWAP APPLICATIONS

In this section, we review some of the principal uses of swaps as a hedging tool for bond instruments and also how to hedge a swap book.

Corporate Applications

Swaps are part of the over-the-counter (OTC) market, and so they can be tailored to suit the particular requirements of the user. It is common for swaps to be structured so that they match particular payment dates, payment frequencies, and Libor margins, which may characterise the underlying exposure of the customer. As the market in interest-rate swaps is so large, liquid, and competitive, banks are willing to quote rates and structure swaps for virtually all customers, although it may be difficult for smaller customers to obtain competitive piece quotes as notional values below \$10 million or \$5 million.

Swap applications can be viewed as being one of two main types; asset-linked swaps and liability-linked swaps. Asset-linked swaps are created when the swap is linked to an asset such as a bond in order to change the characteristics of the income stream for investors. Liability-linked swaps are traded when borrowers of funds

FIGURE 13.9 Changing Liability from Floating- to Fixed-Rate

wish to change the pattern of their cash flows. Of course, just as with repo transactions, the designation of a swap in such terms depends on from whose point of view one is looking at the swap. An asset-linked swap hedge is a liability-linked hedge for the counterparty, except in the case of swap market-making banks who make two-way quotes in the instruments.

A straightforward application of an interest-rate swap is when a borrower wishes to convert a floating-rate liability into a fixed-rate one, usually in order to remove the exposure to upward moves in interest rates. For instance, a company may wish to fix its financing costs. Let us assume a company currently borrowing money at a floating rate, say six month Libor +100 basis points, fears that interest rates may rise in the remaining three years of its loan. It enters into a three-year semiannual interest-rate swap with a bank, as the fixed-rate payer, paying say 6.75% against receiving six-month Libor. This fixes the company's borrowing costs for three years at 7.75% (7.99 per cent effective annual rate). This is shown in Figure 13.9.

EXAMPLE 13.4 LIABILITY-LINKED SWAP, FIXED- TO FLOATING- TO FIXED-RATE EXPOSURE

A corporate borrows for five years at a rate of $6^1/_4$% and shortly after enters into a swap paying floating-rate, so that its net borrowing cost is Libor +40 bps. After one year, swap rates have fallen such that the company is quoted four-year swap rates as 4.90 –84%. The company decides to switch back into fixed-rate liability in order to take advantage of the lower interest rate environment. It enters into a second swap paying fixed at 4.90% and receiving Libor. The net borrowing cost is now 5.30%. The arrangement is illustrated in Figure 13.10. The company has saved 95 basis points on its original borrowing cost, which is the difference between the two swap rates.

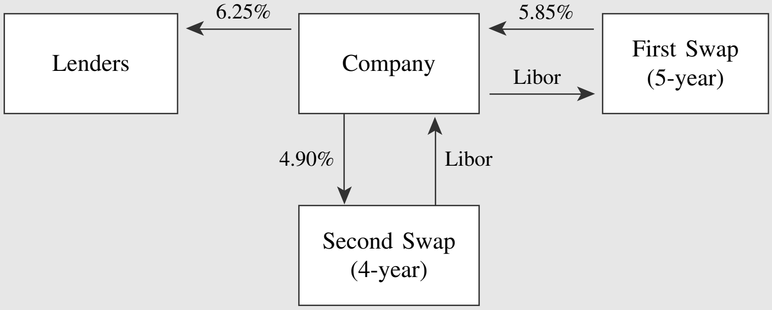

FIGURE 13.10 Changing Interest Basis

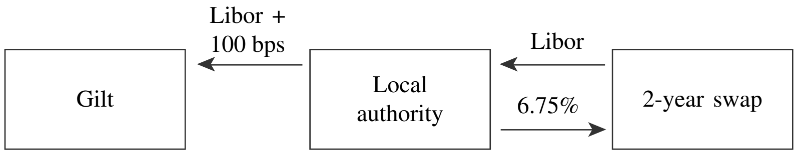

FIGURE 13.11 Transforming Floating-Rate Asset to Fixed-Rate

The use of swaps to hedge liability exposure is illustrated in Example 13.4.

Asset-linked swap structures might be required when, for example, investors require a fixed-interest security when floating-rate assets are available. Borrowers often issue FRNs, the holders of which may prefer to switch the income stream into fixed coupons. As an example, consider a local authority pension fund holding two-year floating-rate UK government gilts. This is an asset of the highest quality, paying the Libor "bid" rate (Libid) minus 12.5 bps. The pension fund wishes to swap the cash flows to create a fixed-interest asset. It obtains a quote for a tailor-made swap where the floating leg pays Libid, the quote being 5.55–5.50%. By entering into this swap, the pension fund has in place a structure that pays a fixed coupon of 5.375%. This is shown in Figure 13.11.

Hedging Bond Instruments Using Interest-Rate Swaps

We illustrate here a generic approach to the hedging of bond positions using interest-rate swaps. The bond trader has the option of using other bonds, bond futures, or bond options, as well as swaps, when hedging the interest-rate risk exposure of a bond position. However, swaps are particularly efficient instruments to use because they display positive convexity characteristics; that is, the increase in value of a swap for a fall in interest rates exceeds the loss in value with a similar magnitude rise in rates. This is exactly the price/yield profile of vanilla bonds.

The primary risk measure we require when hedging using a swap is its present value of a basis point or PVBP.[8] This measures the price sensitivity of the swap for a basis-point change in interest rates. The PVBP measure is used to calculate the hedge ratio when hedging a bond position. The PVBP can be given by

$$PVBP = \frac{\text{Change in swap value}}{\text{Rate change in basis points}} \tag{13.25}$$

which can be written as

$$PVBP = \frac{dS}{dr} \tag{13.26}$$

Using the basic relationship for the value of a swap, which is viewed as the difference between the values of a fixed-coupon bond and equivalent-maturity floating-rate bond (see Table 13.1), we can also write

[8] This is also known as DVBP or dollar value of a basis point in the U.S. market.

$$PVBP = \frac{d\,\text{Fixed bond}}{dr} - \frac{d\,\text{Floating bond}}{dr} \tag{13.27}$$

which essentially states that the basis-point value of the swap is the difference in the basis-point values of the fixed-coupon and floating-rate bonds. The value is usually calculated for a notional £1 million of swap. The calculation is based on the duration and modified-duration calculations used for bonds[9] and assumes that there is a parallel shift in the yield curve.

Figure 13.12 illustrates how (13.26) and (13.27) can be used to obtain the PVBP of a swap. Hypothetical five-year bonds are used in the example. The PVBP for a bond can be calculated using Bloomberg or the MDURATION function on Microsoft Excel. Using either of the two equations above, we see that the PVBP of the swap is £425.00. This is shown below.

Calculating the PVBP using (13.27), we have

$$PVBP_{swap} = \frac{dS}{dr} = \frac{4264 - (-4236)}{20} = 425$$

while we obtain the same result using the bond values

$$\begin{aligned} PVBP_{swap} &= PVBP_{fixed} - PVBP_{floating} \\ &= \frac{1{,}004{,}940 - 995{,}171}{20} - \frac{1{,}000{,}640 - 999{,}371}{20} \\ &= 488.45 - 63.45 \\ &= 425.00 \end{aligned}$$

Interest Rate Swap			
	Term to maturity	5 years	
	Fixed leg	6.50%	
	Basis	Semiannual, act/365	
	Floating leg	6-month Libor	
	Basis	Semiannual, act/365	
	Nominal amount	£1,000,000	
		Present value £	
	Rate change −10 bps	0 bps	Rate change +10 bps
Fixed coupon bond	1,004,940	1,000,000	995,171
Floating rate bond	1,000,640	1,000,000	999,371
Swap	4,264	0	4,236

FIGURE 13.12 PVBP for Interest-Rate Swap

[9] See Chapter 2.

The swap basis-point value is lower than that of the five-year fixed-coupon bond; that is, £425 compared to £488.45. This is because of the impact of the floating-rate bond risk measure, which reduces the risk exposure of the swap as a whole by £63.45. As a rough rule of thumb, the PVBP of a swap is approximately equal to that of a fixed-rate bond that has a maturity similar to the period from the next coupon reset date of the swap through to the maturity date of the swap. This means that a 10-year semiannual paying swap would have a PVBP close to that of a 9.5-year fixed-rate bond, and a 5.50-year swap would have a PVBP similar to that of a five-year bond.

When using swaps as hedge tools, we bear in mind that over time the PVBP of swaps behaves differently from that of bonds. Immediately preceding an interest reset date, the PVBP of a swap will be near-identical to that of the same-maturity fixed-rate bond, because the PVBP of a floating-rate bond at this time has essentially nil value. Immediately after the reset date, the swap PVBP will be near-identical to that of a bond that matures at the next reset date. This means that at the point (and this point only) right after the reset the swap PVBP will decrease by the amount of the floating-rate PVBP. In between reset dates, the swap PVBP is quite stable, as the effect of the fixed- and floating-rate PVBP changes cancel each other out. Contrast this with the fixed-rate PVBP, which decreases in value over time in stable fashion.[10] This feature is illustrated in Figure 13.13. A slight anomaly is that the PVBP of a swap actually increases by a small amount between reset dates; this is because the PVBP of a floating-rate bond decreases at a slightly faster rate than that of the fixed-rate bond during this time.

Hedging bond instruments with interest-rate swaps is conceptually similar to hedging with another bond or with bond futures contracts. If one is holding a long position in a vanilla bond, the hedge requires a long position in the swap: remember that a long position in a swap is to be paying fixed (and receiving floating). This hedges the receipt of fixed from the bond position. The change in the value of the swap will match the change in value of the bond, only in the opposite direction.[11]

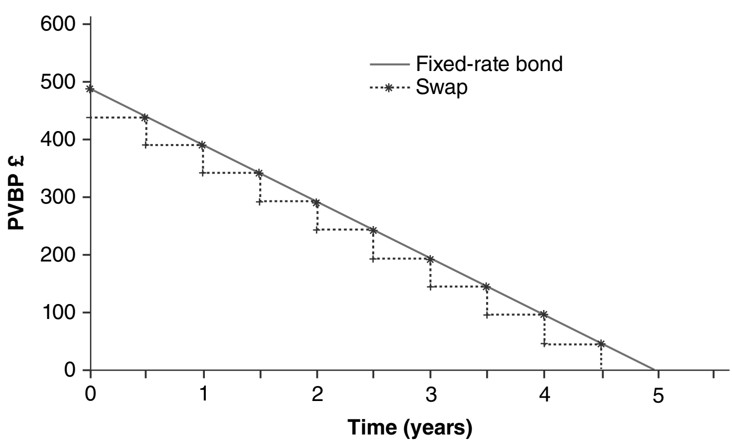

FIGURE 13.13 PVBP of a Five-Year Swap and Fixed-Rate Bond

[10] This assumes no sudden large-scale yield movements.
[11] The change will not be an exact mirror. It is very difficult to establish a precise hedge for a number of reasons, which include differences in day-count bases, maturity mismatches, and basis risk.

The maturity of the swap should match that of the bond as closely as possible. As swaps are OTC contracts, it should be possible to match interest dates as well as maturity dates. If one is short the bond, the hedge is to be short the swap, so the receipt of fixed matches the pay-fixed liability of the short bond position.

The correct nominal amount of the swap to put on is established using the PVBP hedge ratio. This is given as

$$\text{Hedge ratio} = \frac{PVBP_{bond}}{PVBP_{swap}} \tag{13.28}$$

This technique is still used in the market but suffers from the assumption of parallel yield-curve shifts and can therefore lead to significant hedging error at times. More advanced techniques are used by banks when hedging books using swaps, which space does not permit any discussion here.

BLOOMBERG SCREENS

A number of screens on Bloomberg are of value to swaps users and traders. We highlight a selection of some of them here.

Broker Rates Screens

Swap rates can be viewed on a number of brokers' pages. We show those from Tullett as of 9 February 2004.

Figure 13.14 is the USD swaps page SMKR. Rates for semiannual and annual swaps are shown. This page also shows benchmark Treasury yields for selected

GRAB Govt SMKR
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09-Feb-04 12:27 GMT

------------------- Tullett plc USD Medium Term Swaps -------------------

	Price Bid	Price Ask	Mid Yield	Swap Spread Bid	Swap Spread Ask	IRS Semi-Bond	IRS Ann-Actual	When Issued	
2Y	100.070	100.076	1.756	32.50	36.50	2.081-121	2.062-102	2Y	
3Y	101.06+	101.076	2.165	49.75	53.75	2.662-702	2.643-683	3Y	2.304
4Y			2.624	49.25	53.25	3.117-157	3.098-139	5Y	3.107
5Y	100.230	100.240	3.087	39.50	43.50	3.482-522	3.463-503	10Y	4.102
6Y			3.287	47.75	51.75	3.765-805	3.748-788	30Y	
7Y			3.488	50.25	54.25	3.990-030	3.975-015		
8Y			3.690	48.75	52.75	4.177-217	4.162-202		
9Y			3.890	44.75	48.75	4.337-377	4.323-363	Spread to 10Y	
10Y	101.080	101.09+	4.090	38.50	42.50	4.475-515	4.462-502	Bid	Ask
11Y			4.132	46.00	50.00	4.592-632	4.580-620	50.18	54.18
12Y			4.174	52.25	56.25	4.696-736	4.685-726	60.61	64.61
13Y			4.216	57.50	61.50	4.791-831	4.780-820	70.07	74.07
14Y			4.257	62.00	66.00	4.877-917	4.868-908	78.74	82.74
15Y			4.299	63.75	67.75	4.937-977	4.928-968	84.66	88.66
20Y			4.508	63.25	67.25	5.141-181	5.134-174	105.08	109.0
25Y			4.717	49.25	53.25	5.210-250	5.203-244	111.99	115.99
30Y	106.200	106.220	4.927	30.25	34.25	5.229-269	5.222-263		

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
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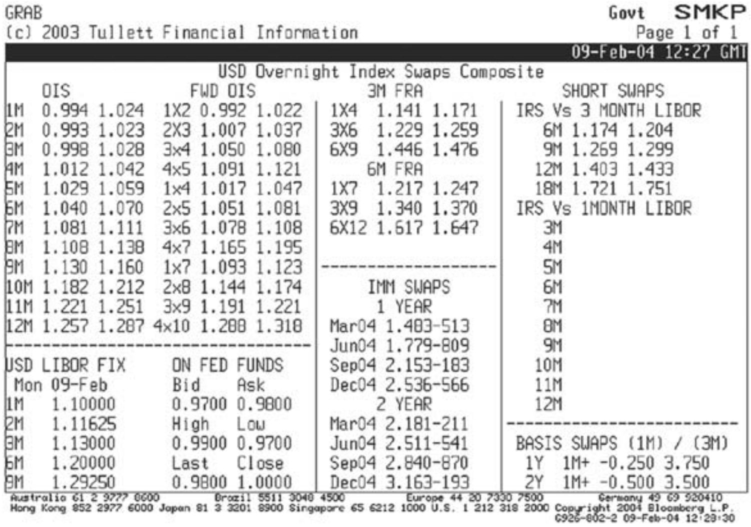
FIGURE 13.14 Screen SMKR on Bloomberg, Tullet USD Swaps Page, 9 February 2004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GRAB                                                          Govt  SMKP
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                                                          09-Feb-04 12:27 GMT
                         USD Overnight Index Swaps Composite
    OIS              FWD OIS            3M FRA              SHORT SWAPS
1M  0.994 1.024  1X2 0.992 1.022 | 1X4  1.141 1.171 | IRS Vs 3 MONTH LIBOR
2M  0.993 1.023  2X3 1.007 1.037 | 3X6  1.229 1.259 |   6M 1.174 1.204
3M  0.998 1.028  3x4 1.050 1.080 | 6X9  1.446 1.476 |   9M 1.269 1.299
4M  1.012 1.042  4x5 1.091 1.121 |     6M FRA       |  12M 1.403 1.433
5M  1.029 1.059  1x4 1.017 1.047 | 1X7  1.217 1.247 |  18M 1.721 1.751
6M  1.040 1.070  2x5 1.051 1.081 | 3X9  1.340 1.370 | IRS Vs 1MONTH LIBOR
7M  1.081 1.111  3x6 1.078 1.108 | 6X12 1.617 1.647 |   3M
8M  1.108 1.138  4x7 1.165 1.195 |                  |   4M
9M  1.130 1.160  1x7 1.093 1.123 |------------------|   5M
10M 1.182 1.212  2x8 1.144 1.174 |    IMM SWAPS     |   6M
11M 1.221 1.251  3x9 1.191 1.221 |     1 YEAR       |   7M
12M 1.257 1.287 4x10 1.288 1.318 | Mar04 1.483-513  |   8M
---------------------------------| Jun04 1.779-809  |   9M
USD LIBOR FIX    ON FED FUNDS    | Sep04 2.153-183  |   10M
 Mon 09-Feb      Bid    Ask      | Dec04 2.536-566  |   11M
1M   1.10000     0.9700 0.9800   |     2 YEAR       |   12M
2M   1.11625     High   Low      | Mar04 2.181-211  |------------------------
3M   1.13000     0.9900 0.9700   | Jun04 2.511-541  | BASIS SWAPS (1M) / (3M)
6M   1.20000     Last   Close    | Sep04 2.840-870  |  1Y  1M+ -0.250 3.750
9M   1.29250     0.9800 1.0000   | Dec04 3.163-193  |  2Y  1M+ -0.500 3.500
Australia 61 2 9777 8600   Brazil 5511 3048 4500   Europe 44 20 7330 7500   Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
                                                                    G926-802-2 09-Feb-04 12:29:30
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FIGURE 13.15 Screen SMKP on Bloomberg, Tullet USD OIS and FRA Rates Page, 9 February 2004

maturities, and the swap spread over the benchmark. Figure 13.15 shows USD OIS and FRA rates.

Figures 13.16–18 shows the same broker's composite pages for AUD, SGD, and TWD currencies. A composite page shows both cash and derivative rates.

Calculation Screens

A number of analytics pages are also used by swap traders and analysts. Figure 13.19 shows the vanilla swap calculator page, BCSW. We illustrate its use to price a five-year Singapore dollar swap where the floating leg has a spread below Libor of 15 bps. As there is a spread, the fixed rate will differ from the standard interbank quote for five-year SGD swaps, which was 2.317% on the calculation date of 22 March 2004. The terms of the swap are:

Trade date	22 March 2004
Effective date	24 March 2004
Nominal	SGD 10,000,000
Floating rate	Libor minus 15 bps
Curve	Singapore government benchmark (Bloomberg number 44)
Floating pay	Semiannual, act/365
Fixed pay	Semiannual, act/365

Figure 13.19A shows the calculation and that the fixed rate for the swap was 2.16750%. This differs from the swap rate because of the spread under Libor on the

```
GRAB                                                            Govt   SMKP
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                                                             09-Feb-04 12:29 GMT
                          AUSTRALIAN DOLLAR COMPOSITE
    IR SWAPS          BASIS SWAPS         FRA           CASH DEPOSITS     SPOT FX
 1Y3 5.58 5.62  | 1Y 7.00   9.00  | 1X4  5.55 5.57 | TN 5.1900 5.2600 |   AUD
 2Y3 5.68 5.72  | 2Y 8.00  10.00  | 2X5  5.55 5.57 | 1W 5.2300 5.2800 | 0.7776/80
 3Y3 5.75 5.79  | 3Y 9.00  11.00  | 3X6  5.58 5.60 | 1M 5.3700 5.4000 |   JPY
 4Y6 5.87 5.91  | 4Y 9.50  11.50  | 4X7  5.58 5.60 | 2M 5.4300 5.4700 | 105.65/69
 5Y6 5.92 5.96  | 5Y 11.00 13.00  | 5X8  5.60 5.62 | 3M 5.4600 5.5000 |   EUR
 7Y6 5.97 6.01  | 6Y 11.00 13.00  | 6X9  5.61 5.63 | 4M 5.4900 5.5400 | 1.2741/43
10Y6 6.04 6.08  | 7Y 11.00 13.00  | 7X10 5.63 5.65 | 5M 5.5000 5.5700 |   GBP
                | 8Y 11.00 13.00  | 8X11 5.64 5.66 | 6M 5.5500 5.6000 | 1.8617/19
                | 9Y 11.00 13.00  | 9X12 5.65 5.67 | 9M 5.5900 5.6500 |   HKD
                |10Y 11.00 13.00  |                | 1Y 5.6600 5.7000 | 7.7680/82
----------------------------INTEREST RATE OPTIONS-------------------------------
                 SWAPTIONS                          CAPS            FLOORS
Ex   1Y    2Y    3Y    4Y    5Y    7Y    10Y  |  1Y 13.50 15.50   1Y 13.50 15.50
1M 15.50 16.60 17.60 17.50 17.40 17.20 17.00  |  2Y 14.90 16.90   2Y 14.90 16.90
2M 15.70 17.00 17.70 17.60 17.50 17.40 17.10  |  3Y 15.20 17.20   3Y 15.20 17.20
3M 15.70 17.40 17.80 17.70 17.60 17.50 17.30  |  4Y 14.90 16.90   4Y 14.90 16.90
6M 16.10 17.30 17.80 17.70 17.50 17.40 17.20  |  5Y 14.70 16.70   5Y 14.70 16.70
1Y 16.40 17.00 17.40 17.20 17.10 17.00 16.90  |  7Y 14.10 16.10   7Y 14.10 16.10
2Y 15.80 16.10 16.90 16.70 16.60 16.60 16.60  | 10Y 13.80 15.80  10Y 13.80 15.80
3Y 15.20 15.60 16.40 16.40 16.40 16.30 16.20  |
Australia 61 2 9777 8600        Brazil 5511 3048 4500        Europe 44 20 7330 7500        Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
                                                                           G926-802-2 09-Feb-04 12:29:59
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FIGURE 13.16 Tullett Australian Dollar Composite Page


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                                                             09-Feb-04 12:29 GMT
                          SINGAPORE DOLLAR COMPOSITE

    IR SWAPS         OIS          SGD/USD SWAPS     CASH DEPOSITS      SPOT FX
1Y   0.98 1.00 | 1M 0.70 0.74 | Sem Bnd/6M (L) | TN 0.6250 0.6250 |   SGD
2Y   1.49 1.52 | 2M 0.70 0.74 | 1Y -3.00 0.00  | SW 0.6250 0.6250 | 1.6842/58
3Y   1.90 1.93 | 3M 0.70 0.74 | 2Y -3.00 0.00  | 1M 0.6250 0.6250 |
4Y   2.21 2.24 | 6M 0.78 0.82 | 3Y -3.00 0.00  | 2M 0.6250 0.6250 |   JPY
5Y   2.51 2.54 | 9M 0.85 0.89 | 4Y -3.00 0.00  | 3M 0.6250 0.6250 | 105.65/69
6Y   2.77 3.81 | 1Y 0.95 0.99 | 5Y -3.00 0.00  | 4M 0.6875 0.6875 |
7Y   3.05 3.08 |              | 7Y -3.00 -1.00 | 5M 0.6875 0.6875 |   EUR
10Y  3.55 3.59 |              | 10Y-3.00 -1.00 | 6M 0.6875 0.6875 | 1.2741/43
12Y  3.61 3.71 |              |                | 9M 0.7500 0.7500 |
15Y  3.72 3.82 |              |                | 1Y 0.8750 0.8750 |
--------------------------------------------------------------------------------
 SIBOR FIX   |  CAPS & FLOORS   |        SWAPTIONS (MID)             |
Mon 09-Feb   |  1Y 56.00 63.50  | Ex  1Y    2Y    3Y    5Y    10Y   |
1M  0.75000  |  2Y 54.50 58.00  | 3M   #     #     #   44.00 34.50  |
2M  0.75000  |  3Y 49.00 52.50  | 6M   #     #     #   41.50 32.50  |
3M  0.75000  |  4Y 44.50 47.50  | 1Y 56.00 47.50 43.00 37.50   #    |
6M  0.81250  |  5Y 40.00 43.50  | 2Y 41.00 38.50 35.50 31.00   #    |
12M 1.00000  |  7Y 33.50 37.50  | 3Y 35.50 32.50 30.50 27.00   #    |
             | 10Y 28.00 32.00  | 5Y 26.00 24.50 24.00 23.25   #    |
Australia 61 2 9777 8600        Brazil 5511 3048 4500        Europe 44 20 7330 7500        Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
                                                                           G926-802-2 09-Feb-04 12:30:33
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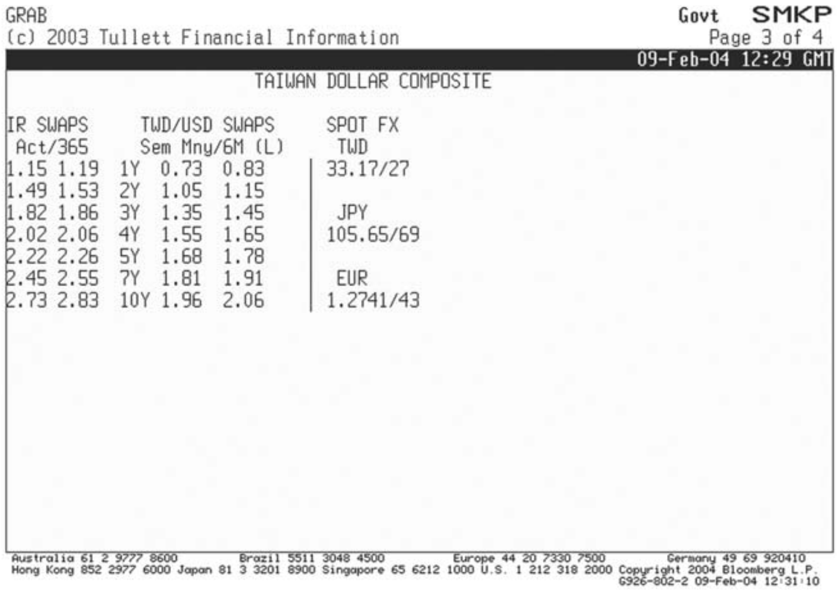
FIGURE 13.17 Tullett Singapore Dollar Composite Page



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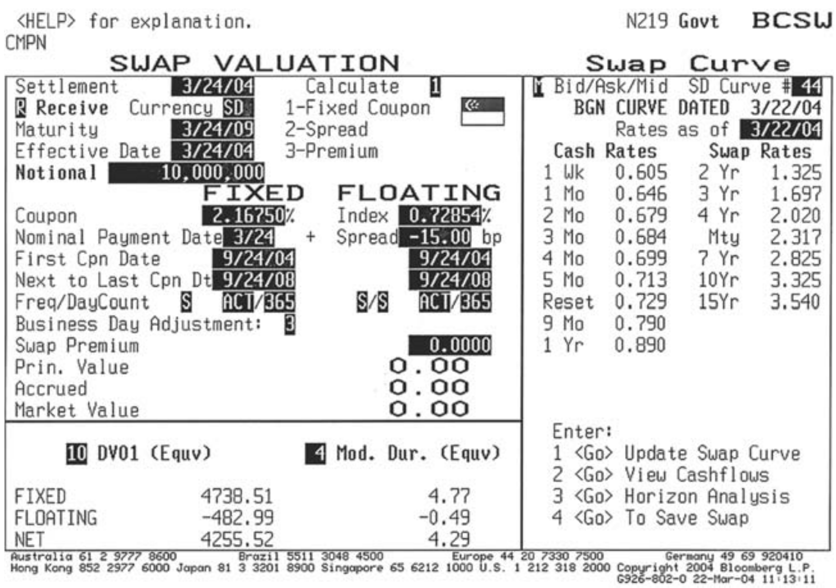
09-Feb-04 12:29 GMT

TAIWAN DOLLAR COMPOSITE

IR SWAPS Act/365			TWD/USD SWAPS Sem Mny/6M (L)		SPOT FX
1.15	1.19	1Y	0.73	0.83	TWD 33.17/27
1.49	1.53	2Y	1.05	1.15	
1.82	1.86	3Y	1.35	1.45	JPY
2.02	2.06	4Y	1.55	1.65	105.65/69
2.22	2.26	5Y	1.68	1.78	
2.45	2.55	7Y	1.81	1.91	EUR
2.73	2.83	10Y	1.96	2.06	1.2741/43

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
G926-802-2 09-Feb-04 12:31:10

FIGURE 13.18 Tullett Taiwan Dollar Composite Page



<HELP> for explanation. — N219 Govt BCSW
CMPN

SWAP VALUATION

Settlement 3/24/04 — Calculate 1
R Receive Currency SD — 1-Fixed Coupon
Maturity 3/24/09 — 2-Spread
Effective Date 3/24/04 — 3-Premium
Notional 10,000,000

	FIXED	FLOATING
Coupon / Index	2.16750%	0.72854%
Nominal Payment Date / Spread	3/24	+ -15.00 bp
First Cpn Date	9/24/04	9/24/04
Next to Last Cpn Dt	9/24/08	9/24/08
Freq/DayCount	S ACT/365	S/S ACT/365

Business Day Adjustment: 3
Swap Premium 0.0000
Prin. Value 0.00
Accrued 0.00
Market Value 0.00

	10 DV01 (Equv)	4 Mod. Dur. (Equv)
FIXED	4738.51	4.77
FLOATING	-482.99	-0.49
NET	4255.52	4.29

Swap Curve

M Bid/Ask/Mid SD Curve # 44
BGN CURVE DATED 3/22/04
Rates as of 3/22/04

Cash Rates		Swap Rates	
1 Wk	0.605	2 Yr	1.325
1 Mo	0.646	3 Yr	1.697
2 Mo	0.679	4 Yr	2.020
3 Mo	0.684	Mty	2.317
4 Mo	0.699	7 Yr	2.825
5 Mo	0.713	10Yr	3.325
Reset	0.729	15Yr	3.540
9 Mo	0.790		
1 Yr	0.890		

Enter:
1 <Go> Update Swap Curve
2 <Go> View Cashflows
3 <Go> Horizon Analysis
4 <Go> To Save Swap

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
G926-802-0 22-Mar-04 11:13:11

FIGURE 13.19A Bloomberg Screen BCSW, Swap Valuation Page, Calculation on 22 March 2004 floating side. The standard swap rate of 2.213% can be seen on the swap rate curve on the right-hand side of the screen. As the swap is being calculated as at its start date, the market value and accrued is zero. Figure 13.19B shows the cash flows for the swap; this is page 2 of the screen.

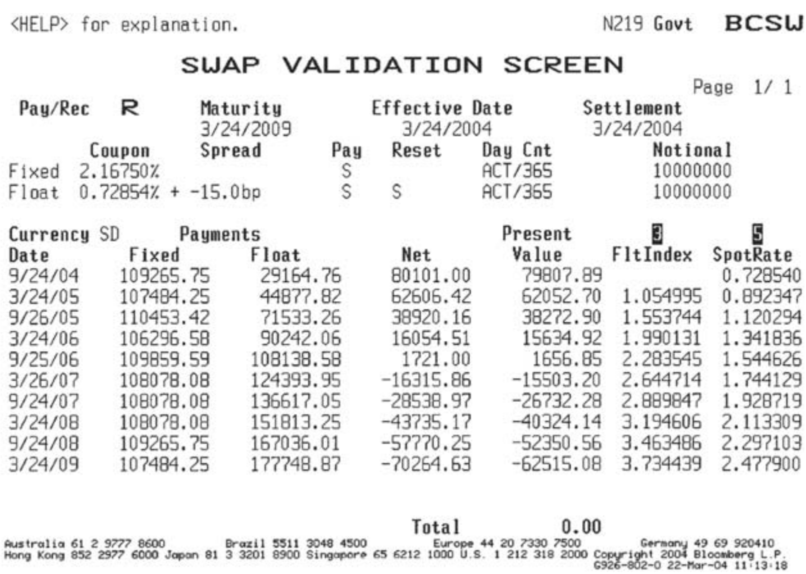
This screen can also be used to obtain the present value of an existing swap, used when swaps are terminated ahead of maturity (and a one-off cash payment is made to close out the swap). The user selects the yield curve against which the swap is valued, usually the generic benchmark curve. For this particular swap, the curve selected was number 44 on the Bloomberg menu, the Singapore government bond yield curve.

Screen SWPM, shown in Figure 13.20A, allows greater flexibility as more of the swap terms can be set by the users themselves. For comparison, we have used the terms for the same Singapore dollar swap calculated using screen BCSW in Figure 13.19. The one difference is that we have elected for the floating leg to pay quarterly, rather than semiannual, payments; the fixed leg still pays semiannually. We can see the difference this causes to the DV01 measure, which was SGD 4,738 in the first swap against SGD 4,507 in this one. The "Index" is also different, which is expected: the first swap seen at Figure 13.19A shows the six-month SGD fix, whereas the rate shown on Figure 13.20A is the three-month rate, since this swap pays floating-rate quarterly. All this can be observed on the screen.

Figure 13.20B is from the same screen, with the user selecting

2 <Go>

to view the swap curve.


<HELP> for explanation. N219 Govt **BCSW**

SWAP VALIDATION SCREEN

Page 1/ 1

Pay/Rec	R	Maturity		Effective Date		Settlement
		3/24/2009		3/24/2004		3/24/2004

	Coupon	Spread	Pay	Reset	Day Cnt	Notional
Fixed	2.16750%		S		ACT/365	10000000
Float	0.72854% +	-15.0bp	S	S	ACT/365	10000000

Currency SD	Payments			Present	3	5
Date	Fixed	Float	Net	Value	FltIndex	SpotRate
9/24/04	109265.75	29164.76	80101.00	79807.89		0.728540
3/24/05	107484.25	44877.82	62606.42	62052.70	1.054995	0.892347
9/26/05	110453.42	71533.26	38920.16	38272.90	1.553744	1.120294
3/24/06	106296.58	90242.06	16054.51	15634.92	1.990131	1.341836
9/25/06	109859.59	108138.58	1721.00	1656.85	2.283545	1.544626
3/26/07	108078.08	124393.95	-16315.86	-15503.20	2.644714	1.744129
9/24/07	108078.08	136617.05	-28538.97	-26732.28	2.889847	1.928719
3/24/08	108078.08	151813.25	-43735.17	-40324.14	3.194606	2.113309
9/24/08	109265.75	167036.01	-57770.25	-52350.56	3.463486	2.297103
3/24/09	107484.25	177748.87	-70264.63	-62515.08	3.734439	2.477900

Total 0.00

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
G926-802-0 22-Mar-04 11:13:18

FIGURE 13.19B Bloomberg Screen BCSW, Page Forward to Show Swap Cash Flows

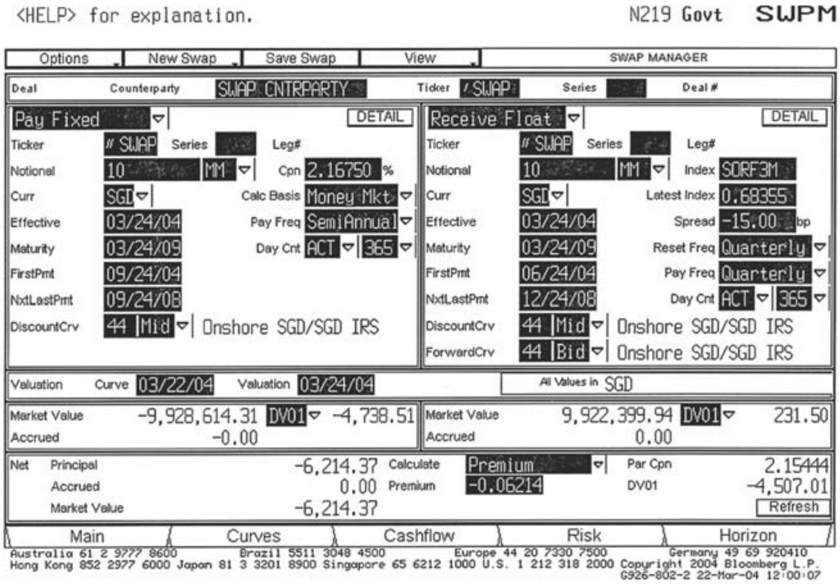


FIGURE 13.20A Bloomberg Screen SWPM, User-Selected Swap Calculator on 22 March 2004 to Analyse Same Swap Shown in Figure 13.19

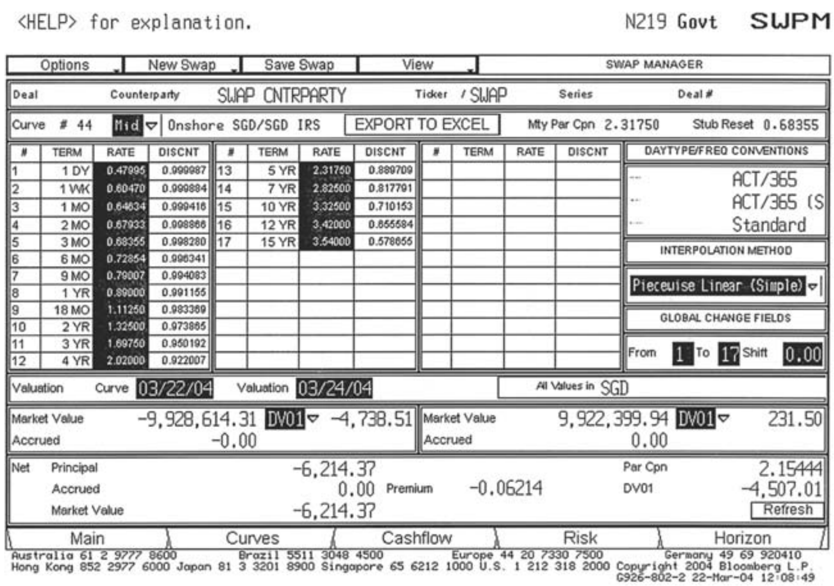


FIGURE 13.20B Bloomberg Screen SWPM, Page Forward to Show Swap Curve

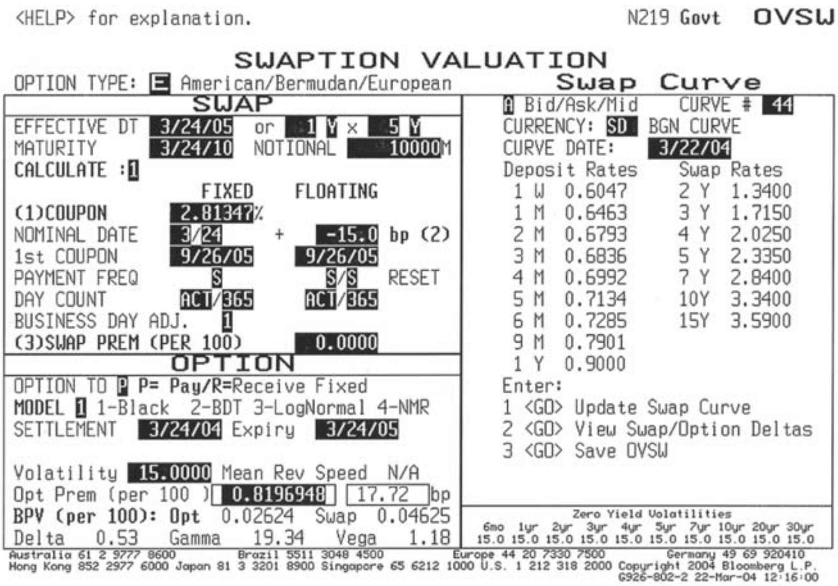


FIGURE 13.21A Bloomberg Screen OVSW, Swaption Valuation Screen for SGD Swap on 22 March 2004, Using Black 76 Model

Figure 13.21 is the swaption valuation page OVSW. We have input details for a one-year option on the same SGD swap analysed in Figure 13.19. The user must select the pricing model as well as other parameters such as the volatility level. We show the same swaption valuation using first the Black 76 model at Figure 13.21A and the Black-Derman-Toy model at Figure 13.21B. As expected, the option premium differs according to the model selected. The zero-yield volatilities are listed on the bottom right-hand of the screen.

Note how the fixed rate calculated for the swap is the five-year forward rate in one year's time.

APPENDIX

APPENDIX 13.1 CALCULATING FUTURES STRIP RATES AND IMPLIED SWAP RATES

A futures strip is a position that contains one each of the futures contracts in a sequence of contract months. For example the one-year short sterling strip in August 2004 would consist of one each of the Sep04, Dec04, Mar05, and Jun05 contracts. The two-year strip would contain these but be followed by one each of the Sep05, Dec05, Mar06, and Jun06 contracts. The three-month forward rates that are implied by the prices of these contracts, together with the initial cash market deposit rate for the *stub* period, may be used to obtain the future value of an investment made today. The expression to calculate this is given in equation (A13.1.1):

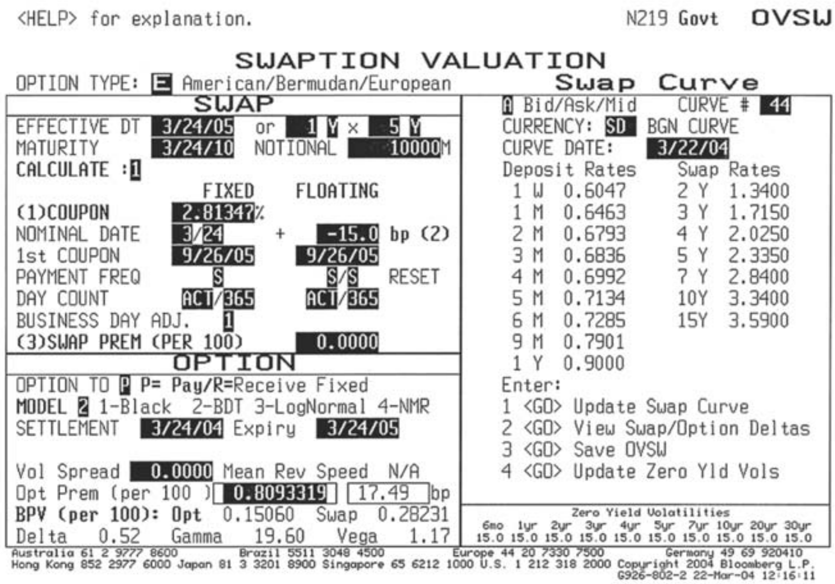


FIGURE 13.21B Bloomberg Screen OVSW, Swaption Valuation Screen Using Black-Derman-Toy Model

$$FV_N = \left(1 + r_0 \times \left(\frac{d_0}{B}\right)\right) \times \left(1 + rf_1 \times \left(\frac{d_0}{B}\right)\right) \times \mathrm{L} \times \left(1 + rf_n \times \left(\frac{d_n}{B}\right)\right) \qquad \text{(A13.1.1)}$$

where FV_N is the future value of £1 invested today for N years

r_0 is the cash market deposit rate (Libor) for the period from today to the expiry date of the first futures contract

rf_1 is the forward rate implied by the price of the front-month futures contract

rf_n is the forward rate implied by the price of the last futures contract in the strip

d_i is the number of days in each futures period, where $i = 0, 1, \ldots n$

B is the year day-base (360 or 365)

The future value of an investment made at futures rates can then be used to calculate implied zero-coupon bond yields. The continuously compounded yield r_{CC} is given by (A13.1.2) while the price $P_{\text{zero-coupon}}$ of a zero-coupon bond of maturity N years is given by (A13.1.3).

$$r_{cc} = ln\left(\frac{FV_N}{N}\right) \qquad \text{(A13.1.2)}$$

$$P_{zero-coupon} = \frac{1}{FV_N} \qquad \text{(A13.1.3)}$$

The forward rates implied by a strip of futures contracts may be used to calculate implied swap rates. As we saw in this chapter, a plain-vanilla interest-rate swap is priced on the basis that it consists of conceptually a short position in a fixed-coupon bond and a long position in a floating-rate bond; that is, a pay fixed- and receive floating-rate swap. The theoretical swap rate is therefore the average of the forward rates given by the futures strip from rf_1 to rf_N where N is the maturity of the swap. The most straightforward way to calculate this is using the discount factors of each spot rate, rather than the forward rates, and this was described in the main body of the text in this chapter.

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CHAPTER 14

Credit Derivatives I: Instruments and Applications

Credit derivatives allow investors to manage the credit-risk exposure of their portfolios or asset holdings, essentially by providing insurance against a deterioration in credit quality of the borrowing entity.[1] If there is a technical default by the borrower[2] or an actual default on the loan itself, and the bond is marked down in price, the losses suffered by the investor can be recouped in part or in full through the payout made by the credit derivative.

CREDIT RISK

Credit risk is the risk that a borrowing entity will default on a loan, either through inability to maintain the interest servicing or because of bankruptcy or insolvency leading to inability to repay the principal itself. When technical or actual default occurs, bondholders suffer a loss as the value of their asset declines, and the potential greatest loss is that of the entire asset. The extent of credit risk fluctuates as the fortunes of borrowers change in line with their own economic circumstances and the macroeconomic business cycle. The magnitude of risk is described by a firm's credit rating. Ratings agencies undertake a formal analysis of the borrower, after which a rating is announced. The issues considered in the analysis include:

- The financial position of the firm itself; for example, its balance-sheet position and anticipated cash flows and revenues.
- Other firm-specific issues such as the quality of the management and succession planning.
- An assessment of the firm's ability to meet scheduled interest and principal payments, both in its domestic and foreign currencies.
- The outlook for the industry as whole, and competition within it.
- General assessments for the domestic economy.

[1] The simplest credit derivative works like an insurance policy, with regular premiums paid by the protection-buyer to the protection-seller, and a payout in the event of a specified credit event. The analogy is not encouraged however, because credit derivatives are "trading book" products for accounting purposes, with daily mark-to-market impact on the balance sheet.

[2] A technical default is a delay in timely payment of the coupon, or nonpayment of the coupon altogether.

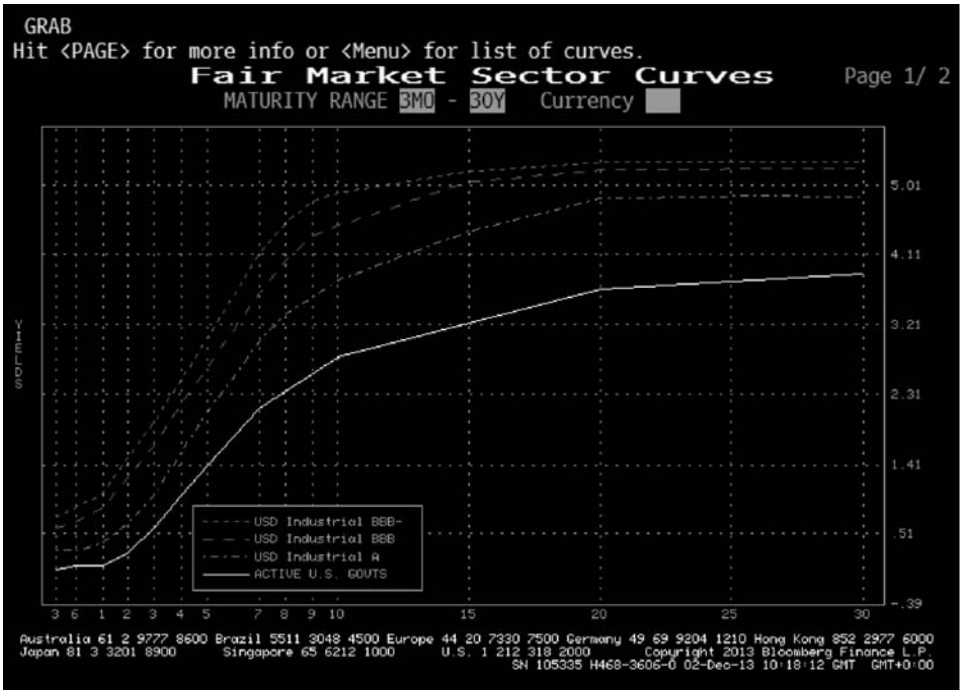

FIGURE 14.1 U.S.-Dollar Bond Yield Curves, December 2013

Another measure of credit risk is the credit-risk premium, which is the difference between yields on the same-currency government benchmark bonds and corporate bonds. This premium is the compensation required by investors for holding bonds that are not default-free. The credit premium required will fluctuate as individual firms and sectors are perceived to offer improved or worsening credit risk, and as the general health of the economy improves or worsens. For example, Figure 14.1 illustrates the yield curves for industrial-sector bonds of various ratings in the U.S.-dollar market, as of December 2013, compared to the benchmark treasury yield curve.

CREDIT RISK AND CREDIT DERIVATIVES

Credit derivatives are financial contracts designed to reduce or eliminate credit-risk exposure by providing insurance against losses suffered due to credit events. A payout under a credit derivative is triggered by a credit event. As banks define default in different ways, the terms under which a credit derivative is executed usually include a specification of what constitutes a credit event.

We focus on credit derivatives as instruments that may be used to manage risk exposure inherent in a corporate or non-AAA sovereign-bond portfolio. They may also be used to manage the credit risk of commercial-loan books. The intense competition among commercial banks, combined with rapid disintermediation, has meant that banks have been forced to evaluate their lending policy, with a view to improving profitability and return on capital. The use of credit derivatives assists banks with restructuring their businesses, because it allows banks to repackage and parcel out credit risk, while retaining assets on balance sheet (when required) and thus maintain client relationships. As the instruments isolate certain aspects of credit risk from the underlying loan or bond and transfer them to another entity, it becomes possible to separate the ownership and management of credit risk from the other features of ownership associated with the assets in question. This means that illiquid assets such as bank loans, and illiquid bonds, can have their credit-risk exposures transferred; the bank owning the assets can protect against credit loss even if it cannot transfer the assets themselves.

The same principles carry over to the credit-risk exposures of portfolio managers. For fixed-income portfolio managers the uses of credit derivatives include the following:

- They can be tailor-made to meet the specific requirements of the entity buying the risk protection, as opposed to the liquidity or term of the underlying reference asset.
- They can be sold short without risk of a liquidity or delivery squeeze, as it is a specific credit risk that is being traded. In the cash market, it is not possible to "sell short" a bank loan, for example, but a credit derivative can be used to establish synthetically the economic effect of such a position.
- As they theoretically isolate credit risk from other factors such as client relationships and interest-rate risk, credit derivatives introduce a formal pricing mechanism to price credit issues only. This means a market can develop in credit only, allowing more efficient pricing, and it becomes possible to model a term structure of credit rates.
- Credit derivatives are off-balance-sheet instruments[3] and as such incorporate tremendous flexibility and leverage, exactly like other financial derivatives. For instance, bank loans are not particularly attractive investments for certain investors because of the administration required in managing and servicing a loan portfolio. However, an exposure to bank loans and their associated return can be achieved by, say, a total-return swap while simultaneously avoiding the administrative costs of actually owning the assets. Hence, credit derivatives allow investors access to specific credits while allowing banks access to further distribution for bank loan credit risk.

Thus, credit derivatives can be an important instrument for bond-portfolio managers as well as commercial banks, who wish to increase the liquidity of their portfolios, gain from the relative value arising from credit pricing anomalies, and enhance portfolio returns. Some key applications are summarised below.

[3] When credit derivatives are embedded in certain fixed-income products, such as structured notes and credit-linked notes, they are then off-balance-sheet but part of a structure that may have on-balance-sheet elements. The practical reality is that irrespective of their form, credit derivatives generate an on-balance sheet cash flow impact.

APPLICATIONS

Diversifying the Credit Portfolio

A bank or portfolio manager may wish to take on credit exposure by providing credit protection on assets that it already owns, in return for a fee. This enhances income on their portfolio. They may sell credit derivatives to enable nonfinancial counterparties to gain credit exposures, if these clients do not wish to purchase the assets directly. In this respect, the bank or asset manager performs a credit-intermediation role.

Reducing Credit Exposure

A bank can reduce credit exposure either for an individual loan or a sectoral concentration, by buying a credit default swap. This may be desirable for assets in their portfolio that cannot be sold for client relationship or tax reasons. For fixed-income managers, a particular asset or collection of assets may be viewed as favourable holdings in the long term, but at risk from short-term downward price movement. In this instance, a sale would not fit in with long-term objectives; however, short-term credit protection can be obtained via credit swap.

Acting as a Credit Derivatives Market Maker

A financial entity may wish to set itself up as a market maker in credit derivatives. In this case, it may or may not hold the reference assets directly, and, depending on its appetite for risk and the liquidity of the market, it can offset derivative contracts as and when required.

CREDIT EVENT

The occurrence of a specified credit event will trigger payment of the default payment by the seller of protection to the buyer of protection. Contracts specify physical or cash settlement. In physical settlement, the protection buyer transfers to the protection seller the deliverable obligation (usually the reference asset or assets), with the total principal outstanding equal to the nominal specified in the default-swap contract. The protection seller simultaneously pays to the buyer 100% of the nominal. In cash settlement, the protection seller hands to the buyer the difference between the nominal amount of the default swap and the final value for the same nominal amount of the reference asset. This final value is usually determined by means of a poll of dealer banks.

The following may be specified as credit events in the legal documentation between counterparties:

- Downgrade in S&P and/or Moody's credit rating below a specified minimum level
- Financial or debt restructuring; for example, occasioned under administration or as required under U.S. bankruptcy protection
- Bankruptcy or insolvency of the reference asset obligor

- Default on payment obligations such as bond coupon and continued nonpayment after a specified time period
- Technical default; for example, the nonpayment of interest or coupon when it falls due
- A change in credit spread payable by the obligor above a specified maximum level

The 1999 International Swaps and Derivativces Association (ISDA) credit default swap documentation specifies bankruptcy, failure to pay, obligation default, debt moratorium, and restructuring to be credit events. Note that it does not specify a rating downgrade to be a credit event.[4]

The precise definition of "restructuring" is open to debate and has resulted in legal disputes between protection buyers and sellers. Prior to issuing its 1999 definitions, ISDA had specified restructuring as an event or events that resulted in making the terms of the reference obligation "materially less favourable" to the creditor (or protection seller) from an economic perspective. This definition is open to more than one interpretation and has caused controversy when determining if a credit event had occurred. The 2001 definitions specified more precise conditions, including any action that resulted in a reduction in the amount of principal. In the European market, restructuring is generally retained as a credit event in contract documentation, but in the U.S. market it is less common to see it included. Instead, U.S. contract documentation tends to include as a credit event a form of modified restructuring, the impact of which is to limit the options available to the protection buyer as to the type of assets it could deliver in a physically settled contract.

RESTRUCTURING, MODIFIED RESTRUCTURING, AND MODIFIED-MODIFIED RESTRUCTURING

The original 1999 ISDA credit definitions defined restructuring among the standard credit events. The five specified definitions included events such as a reduction in the rate of interest payable, a reduction in the amount of principal outstanding, and a postponement or deferral of payment. Following a number of high-profile cases where there was disagreement or dispute between protection buyers and sellers on what constituted precisely a restructuring, the Supplement to the 1999 ISDA limited the term to maturity of deliverable obligations. This was modified restructuring or Mod-R, which was intended to reduce the difference between the loss suffered by a holder of the actual restructured obligation and the writer of a CDS on that reference name. In practice this has placed a maturity limit on deliverable obligations of 30 months.

The 2003 definitions presented further clarification and stated that the restructuring event had to be binding on all holders of the restructured debt.

(*continued*)

[4] The ISDA definitions from 1999, restructuring supplement from 2001, 2003 definitions, and 2008 update are available at www.ISDA.org.

(Continued)

The modified-modified restructuring definition or Mod-Mod-R described in the 2003 ISDA defines the modified restructuring term to maturity date as the later of:

- The scheduled termination date, and
- Sixty months following the restructuring date

in the event that a restructured bond or loan is delivered to the protection seller. If another obligation is delivered, the limitation on maturity is the scheduled maturity date and 30 months following the restructuring date.

CREDIT DERIVATIVE INSTRUMENTS

Credit derivative instruments enable participants in the financial market to trade in credit as an asset, as they isolate and transfer credit risk. They also enable the market to separate funding considerations from credit risk. A number of instruments come under the category of credit derivatives. In this section, we consider the most commonly encountered credit derivative instruments. Irrespective of the particular instrument under consideration, all credit derivatives can be described under the following characteristics:

- The reference entity, which is the asset or name on which credit protection is being bought and sold
- The credit event, or events, which indicate that the reference entity is experiencing or about to experience financial difficulty and which act as trigger events for payments under the credit derivative contract
- The settlement mechanism for the contract, whether cash settled or physically settled
- (Under physical settlement) the deliverable obligation that the protection buyer delivers to the protection seller on the occurrence of a trigger event

Credit derivatives are grouped into funded and unfunded instruments. In a funded credit derivative, typified by a credit-linked note (CLN), the investor in the note is the credit-protection seller and is making an upfront payment to the protection buyer when it buys the note. Thus, the protection buyer is the issuer of the note. If no credit event occurs during the life of the note, the redemption value of the note is paid to the investor on maturity.

If a credit event does occur, then on maturity a value less than par will be paid out to the investor. This value will be reduced by the nominal value of the reference asset that the CLN is linked to. The exact process will differ according to whether cash settlement or physical settlement has been specified for the note. We will consider this later.

In an unfunded credit derivative, typified by a credit default swap, the protection seller does not make an upfront payment to the protection buyer. Credit default swaps have a number of applications and are used extensively for flow trading of single reference-name credit risks or, in portfolio swap form, for trading a basket of reference credits. Credit default swaps and CLNs are used in structured products, in various combinations, and their flexibility has been behind the growth and wide application of the synthetic collateralised debt obligation and other credit hybrid products.

Credit Default Swap

The most common credit derivative is the credit default swap, *credit swap* or default swap.[5] This is a bilateral contract in which a periodic fixed fee or a one-off premium is paid to a protection seller, in return for which the seller will make a payment on the occurrence of a specified credit event. The fee is quoted as a basis-point multiplier of the nominal value. It is usually paid quarterly in arrears.

The swap can refer to a single asset, known as the "reference asset" or "underlying asset", or a basket of assets. The default payment can be paid in whatever way suits the protection buyer or both counterparties. For example, it may be linked to the change in price of the reference asset or another specified asset; it may be fixed at a predetermined recovery rate; or it may be in the form of actual delivery of the reference asset at a specified price. The basic structure is illustrated in Figure 14.2.

The credit default swap enables one party to transfer its credit-risk exposure to another party. Banks may use default swaps to trade sovereign and corporate credit spreads without trading the actual assets themselves. For example, someone who has gone long a default swap (the protection buyer) will gain if the reference asset obligor suffers a rating downgrade or defaults, and can sell the default swap at a profit if he

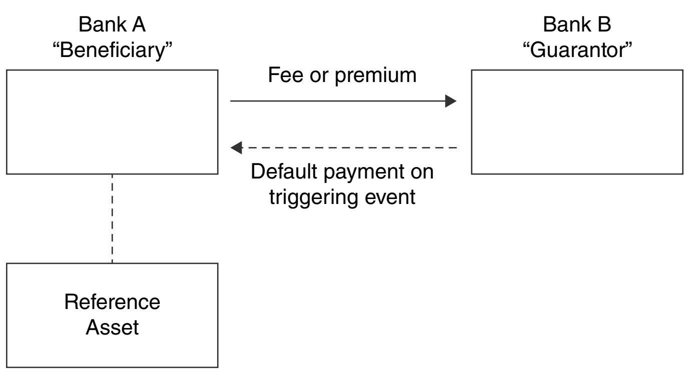

FIGURE 14.2 Credit Default Swap

[5] The author prefers the first term, but the other two terms are common. "Credit swap" does not, we feel, adequately describe the actual purpose of the instrument.

can find a buyer counterparty.[6] This is because the cost of protection on the reference asset will have increased as a result of the credit event. The original buyer of the default swap need never have owned a bond issued by the reference asset obligor.

The maturity of the credit swap does not have to match the maturity of the reference asset and often does not. When a credit event occurs, the swap contract is terminated, and a settlement payment is made by the protection seller or guarantor to the protection buyer. This termination value is calculated at the time of the credit event, and the exact procedure that is followed to calculate the termination value will depend on the settlement terms specified in the contract. This will be either cash settlement or physical settlement.

- **Cash settlement:** The contract may specify a predetermined payout value on occurrence of a credit event. This may be the nominal value of the swap contract. Such a swap is known in some markets as a digital credit derivative. Alternatively, the termination payment is calculated as the difference between the nominal value of the reference asset and its market value at the time of the credit event. This arrangement is more common with cash-settled contracts.[7]
- **Physical settlement:** On occurrence of a credit event, the buyer delivers the reference asset to the seller, in return for which the seller pays the face value of the delivered asset to the buyer. The contract may specify a number of alternative assets that the buyer can deliver; these are known as deliverable obligations. This may apply when a swap has been entered into on a reference name rather than a specific obligation (such as a particular bond) issued by that name. Where more than one deliverable obligation is specified, the protection buyer will invariably deliver the asset that is the cheapest on the list of eligible assets. This gives rise to the concept of the cheapest-to-deliver, as encountered with government-bond futures contracts, and is in effect an embedded option held by the protection buyer.

In theory, the value of protection is identical irrespective of which settlement option is selected. However, under physical settlement the protection seller can gain if there is a recovery value that can be extracted from the defaulted asset; or its value may rise as the fortunes of the issuer improve. Swap market-making banks often prefer cash settlement as there is less administration associated with it. It is also more suitable when the swap is used as part of a synthetic structured product, because such vehicles may not be set up to take delivery of physical assets. Another

[6] Be careful with terminology here. To "go long" of an instrument generally is to purchase it. In the cash market, going long the bond means one is buying the bond and so receiving coupon; the buyer has therefore taken on credit risk exposure to the issuer. In a credit default swap, going long is to buy the swap, but the buyer is purchasing protection and therefore paying premium; the buyer has no credit exposure on the name and has in effect "gone short" on the reference name (the equivalent of shorting a bond in the cash market and paying coupon). So buying a credit default swap is frequently referred to in the market as "shorting" the reference entity.

[7] Determining the market value of the reference asset at the time of the credit event may be a little problematic: the issuer of the asset may well be in default or administration. An independent third-party Calculation Agent is usually employed to make the termination payment calculation.

advantage of cash settlement is that it does not expose the protection buyer to any risks should there not be any deliverable assets in the market; for instance, due to shortage of liquidity in the market. Were this to happen, the buyer may find the value of its settlement payment reduced.

This was particularly true during the build up to the 2008 credit crisis, where due to rampant demand by yield-chasing investors, the amount of CDS written was often more than the total outstanding bonds issued for that particular issuer.

Nevertheless, physical settlement is widely used because counterparties wish to avoid the difficulties associated with determining the market value of the reference asset under cash settlement. Physical settlement also permits the protection seller to take part in the creditor negotiations with the reference entity's administrators, which may result in improved terms for them as holders of the asset.

For illustrative purposes, Figure 14.3 shows investment-grade credit default swap levels during 2013 for U.S. dollar reference entities (average levels taken).

Credit Default Swap: Example XYZ plc credit spreads are currently trading at 120 bps over government for five-year maturities and 195 bps over for 10-year maturities. A portfolio manager hedges a $10 million holding of 10-year paper by purchasing the following credit default swap, written on the five-year bond. This hedge protects for the first five years of the holding and, in the event of XYZ's credit spread widening, will increase in value and may be sold on before expiry at profit. The

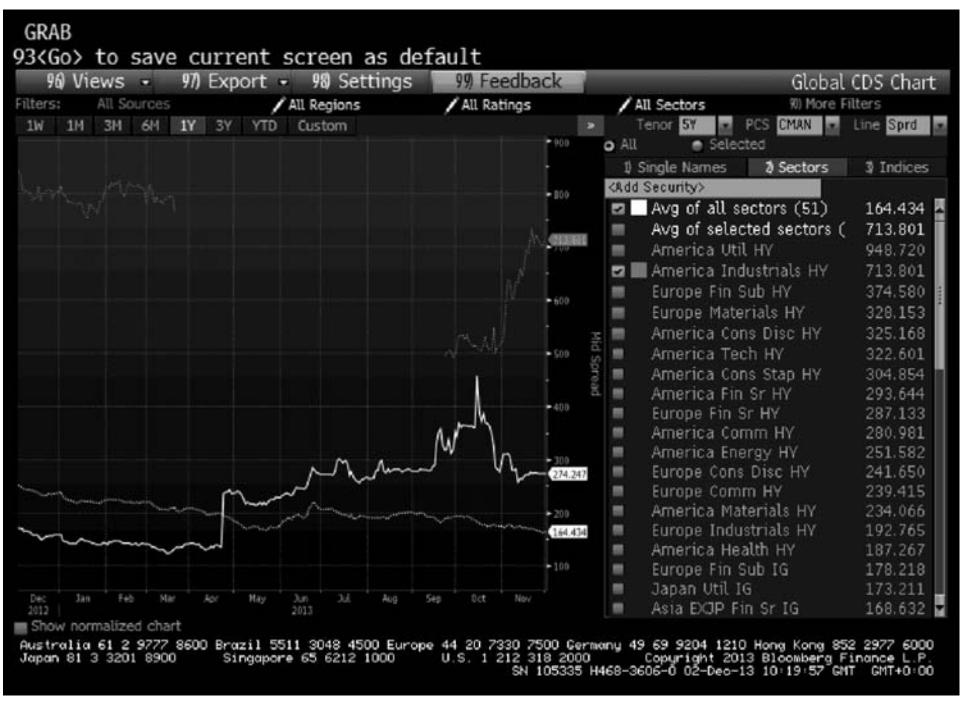

FIGURE 14.3 Investment-Grade Credit Default Swap Levels 2013

10-year bond holding also earns 75 bps over the shorter-term paper for the portfolio manager.

Term	Five Years
Reference credit	XYZ plc five-year bond
Credit event	The business day following occurrence of specified credit event
Default payment	Nominal value of bond × [100 – price of bond after credit event]
Swap premium	3.35%

Assume now that midway into the life of the swap there is a technical default on the XYZ plc five-year bond, such that its price now stands at $28. Under the terms of the swap, the protection buyer delivers the bond to the seller, who pays out $7.2 million to the buyer.

The Total Return Swap

A total return swap (TRS), sometimes known as a total rate of return swap or TR swap, is an agreement between two parties that exchanges the total return from a financial asset between them. This is designed to transfer the credit risk from one party to the other. It is one of the principal instruments used by banks and other financial instruments to manage their credit-risk exposure and, as such, is a credit derivative. One definition of a TRS is given in Francis, Frost, and Whittaker (1999), which states that a TRS is a swap agreement in which the *total return* of a bank loan or credit-sensitive security is exchanged for some other cash flow, usually tied to Libor or some other loan or credit-sensitive security.

In some versions of a TRS, the actual underlying asset is actually sold to the counterparty, with a corresponding swap transaction agreed alongside; in other versions, there is no physical change of ownership of the underlying asset. The TRS trade itself can be to any maturity term; that is, it need not match the maturity of the underlying security. In a TRS, the total return from the underlying asset is paid over to the counterparty in return for a fixed or floating cash flow. This makes it slightly different from other credit derivatives, as the payments between counterparties to a TRS are connected to changes in the market value of the underlying asset, as well as changes resulting from the occurrence of a credit event.

Figure 14.4 illustrates a generic TR swap. The two counterparties are labelled as banks, but the party termed "Bank A" can be another financial institution, including cash-rich fixed-income portfolio managers such as insurance companies and hedge funds. Bank A has contracted to pay the "total return" on a specified reference asset, while simultaneously receiving a Libor-based return from Bank B. The reference or underlying asset can be a bank loan such as a corporate loan or a sovereign or corporate bond. The total return payments from Bank A include the interest payments on the underlying loan as well as any appreciation in the market value of the asset. Bank B will pay the Libor-based return; it will also pay any difference if there is a depreciation in the price of the asset. The economic effect is as if Bank B owned the underlying asset and, as such, TR swaps are synthetic loans

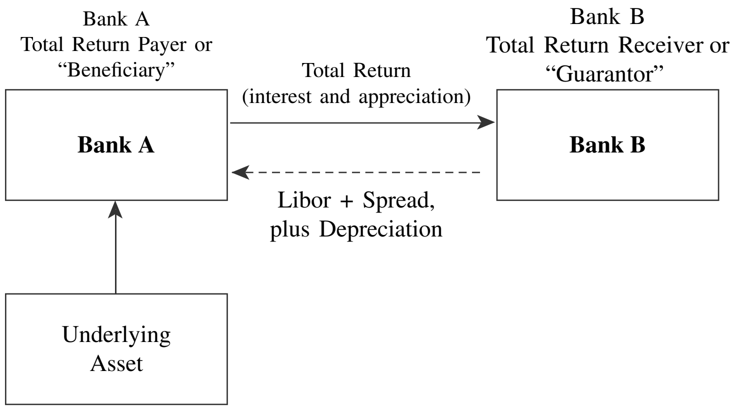

FIGURE 14.4 Total Return Swap

or securities. A significant feature is that Bank A will usually hold the underlying asset on its balance sheet, so that if this asset was originally on Bank B's balance sheet, this is a means by which the latter can have the asset removed from its balance sheet for the term of the TR swap.[8] If we assume Bank A has access to Libor funding, it will receive a spread on this from Bank B. Under the terms of the swap, Bank B will pay the difference between the initial market value and any depreciation, so it is sometimes termed the "guarantor" while Bank A is the "beneficiary".

The total return on the underlying asset is the interest payments and any change in the market value if there is capital appreciation. The value of an appreciation may be cash-settled or, alternatively, there may be physical delivery of the reference asset on maturity of the swap, in return for a payment of the initial asset value by the total-return "receiver". The maturity of the TR swap need not be identical to that of the reference asset, and, in fact, it is rare for it to be so.

The swap element of the trade will usually pay on a quarterly or semiannual basis, with the underlying asset being revalued or marked-to-market on the re-fixing dates. The asset price is usually obtained from an independent third-party source such as Bloomberg or Reuters, or as the average of a range of market quotes. If the *obligor* of the reference asset defaults, the swap may be terminated immediately, with a net present-value payment changing hands according to what this value is, or it may be continued with each party making appreciation or depreciation payments as appropriate. This second option is only available if there is a market for the asset, which is unlikely in the case of a bank loan. If the swap is terminated, each counterparty is liable to the other for accrued interest plus any appreciation or depreciation of the asset. Commonly under the terms of the trade, the guarantor bank has the

[8] Although it is common for the receiver of the Libor-based payments to have the reference asset on its balance sheet, this is not always the case.

option to purchase the underlying asset from the beneficiary bank, and then deal directly with loan defaulter.

There are a number of reasons why portfolio managers may wish to enter into TR swap arrangements. One of these is to reduce or remove credit risk. Using TR swaps as a credit derivative instrument, a party can remove exposure to an asset without having to sell it. In a vanilla TR swap, the total-return payer retains rights to the reference asset, although in some cases servicing and voting rights may be transferred. The total-return receiver gains an exposure to the reference asset without having to pay out the cash proceeds that would be required to purchase it. As the maturity of the swap rarely matches that of the asset, the swap receiver may gain from the positive funding or carry that derives from being able to roll over short-term funding of a longer-term asset.[9] The total-return payer, on the other hand, benefits from protection against market and credit risk for a specified period of time, without having to liquidate the asset itself. On maturity of the swap, the total-return payer may reinvest the asset if it continues to own it, or it may sell the asset in the open market. Thus, the instrument may be considered a synthetic repo. A TR-swap agreement entered into as a credit derivative is a means by which banks can take on unfunded off-balance-sheet credit exposure. Higher-rated banks that have access to Libid funding can benefit by funding on-balance-sheet assets that are credit protected through a credit derivative such as a TR swap, assuming the net spread of asset income over credit-protection premium is positive.

A TR swap conducted as a synthetic repo is usually undertaken to effect the temporary removal of assets from the balance sheet. This may be desired for a number of reasons; for example, if the institution is due to be analysed by credit-rating agencies or if the annual external audit is due shortly. Another reason a bank may wish to temporarily remove lower credit-quality assets from its balance sheet is if it is in danger of breaching capital limits in between the quarterly return periods. In this case, as the return period approaches, lower-quality assets may be removed from the balance sheet by means of a TR swap, which is set to mature after the return period has passed.

Banks have employed a number of methods to price credit derivatives and TR swaps. Essentially, the pricing of credit derivatives is linked to that of other instruments; however, the main difference between credit derivatives and other off-balance-sheet products such as equity, currency, or bond derivatives is that the latter can be priced and hedged with reference to the underlying asset, which can be problematic when applied to credit derivatives. The pricing of credit products uses statistical data on likelihood of default, probability of payout, level of risk tolerance, and a pricing model. With a TR swap, the basic concept is that one party "funds" an underlying asset and transfers the total return of the asset to another party, in return for a (usually) floating return that is a spread to Libor. This spread is a function of:

- The credit rating of the swap counterparty
- The amount and value of the reference asset

[9] This assumes a positively sloping yield curve.

- The credit quality of the reference asset
- The funding costs of the beneficiary bank
- Any required profit margin
- The capital charge associated with the TR swap

The TR swap counterparties must consider a number of risk factors associated with the transaction, which include:

- The probability that the TR beneficiary may default while the reference asset has declined in value
- The reference asset obligor defaults, followed by default of the TR swap receiver before payment of the depreciation has been made to the payer or "provider"

The first risk measure is a function of the probability of default by the TR swap receiver and the market volatility of the reference asset, while the second risk is related to the joint probability of default of both factors as well as the recovery probability of the asset.

THE TOTAL RETURN SWAP AS A FUNDING TOOL

The TRS may be used as a funding tool, as a means of securing off-balance-sheet financing for assets held (for example) on a market-making book. It is most commonly used in this capacity by broker-dealers and securities houses that have little or no access to unsecured or Libor-flat funding. When used for this purpose the TRS is similar to a repo transaction, although there are detail differences. Often a TRS approach is used instead of classic repo when the assets that require funding are less liquid or indeed not really tradable. These can include lower-rated bonds, illiquid bonds such as certain ABS, MBS, and CDO securities, and assets such as hedge fund interests.

Bonds that are taken on by the TRS provider must be acceptable to it in terms of credit quality. If no independent price source is available, the TRS provider may insist on pricing the assets itself.

As a funding tool the TRS is transacted as follows:

- The broker-dealer swaps out a bond or basket of bonds that it owns to the TRS counterparty (usually a bank), who pays the market price for the security or security.
- The maturity of the TRS can be for anything from one week to one year or even longer. For longer-dated contracts, a weekly or monthly reset is usually employed, so that the TRS is repriced and cash flows exchanged each week or month.
- The funds that are passed over by the TRS counterparty to the broker-dealer have the economic effect of being a loan to cover the financing of the underlying bonds. This loan is charged at Libor plus a spread.

(*continued*)

(Continued)

- At the maturity of the TRS, the broker-dealer will owe interest on funds to the swap counterparty, while the swap counterparty will owe the market performance of the bonds to the broker-dealer if they have increased in price. The two cash flows are netted out.
- For a longer-dated TRS that is reset at weekly or monthly intervals, the broker-dealer will owe the loan interest plus any decrease in basket value to the swap counterparty at the reset date. The swap counterparty will owe any increase in value.

By entering into this transaction the broker-dealer obtains Libor-based funding for a pool of assets it already owns, while the swap counterparty earns Libor plus a spread on funds that are in effect secured by a pool of assets. This transaction takes the original assets off the balance sheet of the broker-dealer during the term of the trade, which might also be desirable.

The broker-dealer can add or remove bonds from or to the basket at each reset date. When this happens, the swap counterparty revalues the basket and will hand over more funds or receive back funds as required. Bonds are removed from the basket if they have been sold by the broker-dealer, while new acquisitions can be funded by being placed in the TRS basket.

We illustrate a funding TRS trade using an example. Figure 14.5 shows a portfolio of five hypothetical convertible bonds on the balance sheet of a broker-dealer. The spreadsheet also shows market prices. This portfolio has been swapped out to a TRS provider in a six-month, weekly reset TRS contract. The TRS bank has paid over the combined market value of the portfolio at a lending rate of 1.14125%. This represents one-week Libor plus 7 basis points. We assume the broker-dealer usually funds at above this level, and that this rate is an improvement on its normal funding. It is not unusual for this type of trade to be undertaken even if the funding rate is not an improvement, however, for diversification reasons.

We see from Figure 14.5 that the portfolio has a current market value of approximately USD 151,080,000. This value is lent to the broker-dealer in return for the bonds.

One week later, the TRS is reset. We see from Figure 14.6 that the portfolio has increased in market value since the last reset. Therefore, the swap counterparty pays this difference over to the broker-dealer. This payment is netted out with the interest payment due from the broker-dealer to the swap counterparty. The interest payment is shown as USD 33,526.

Figure 14.7 shows the basket after the addition of new bonds, and the resultant change in portfolio value.

MarketRates	
EUR/USD FX Rate	1.266550
US$ 1W Libor	1.4055

Name	Currency	Nominal Value	Price	Accrued	Amount	FX Rate	ISIN/ CUSIP Code	Market Price	Accrued Interest
ABC Telecom	EUR	16,000,000	111.671%	0.8169%	22,795,534.57	1.2666		111.6713875	0.81693989
XYZ Bank	USD	17,000,000	128.113%	1.7472%	22,076,259.03	1.0000		128.113125	1.74722222
XTC Utility	EUR	45,000,000	102.334%	0.3135%	58,845,000.00	1.2666		102.3337875	0.31352459
SPG Corporation	EUR	30,000,000	100.325		30,000,325.00	1.2666		100.325	0
Watty Exploited	USD	15,000,000	114.997%	0.7594%	17,363,503.13	1.0000		114.9973125	0.759375
					151,080,621.72				

Payments	
Interest($)	
Rate	0.000000%
Principal	151,080,000.00
Interest payable	+0.00
Performance($)	
New portfolio value	151,080,621.72
Old portfolio value	n/a
Performance payment	n/a
Net payment($)	
Broker-dealer receives from swap counterparty	+0.00
New Loan	
Portfolio additions($)	0.00
New loan amount($)	151,080,621.72
New interest rate	1.141250% 1w Libor + 7 bps

FIGURE 14.5 Spreadsheet Showing Basket of Bonds Used in TRS Funding Trade

EUR/USD **1.2431**

Bond	Curr	Nominal Value	Price	Accrued	Amount	FX	ISIN/ CUSIP	Market Price	Accrued
ABC Telecom	EUR	16,000,000	111.500%	0.78%	22,331,239	1.2431		111.5	0.77595628
XYZ Bank	USD	17,000,000	125.000%	1.58%	21,518,931	1		125	1.58194444
XTC Utility	EUR	45,000,000	113.000%	0.28%	63,369,825	1.2431		113	0.28278689
SPG Corporation	EUR	30,000,000	100.750		30,225,000	1.2431		100.75	
Watty Exploited	USD	15,000,000	113.062%	0.63%	17,053,518.2	1		113.0619965	0.628125
					154,498,511.95				
Payments									
Interest									
Rate		1.14125%	1W Libor + 7 bps						
Amount		151,080,000.00	151,113,526.12						
Interest payable		33,526.12							
Performance									
Old portfolio value		151,080,000.00			Old portfolio value:			+151,080,951.67 US$	
New portfolio value		154,498,511.95			Interest rate:			1.14125%	
Performance payment		**(3,418,511.95)**			Interest payable by broker-dealer			+33,526.33 US$	
Swap ctpy pays		**(3,384,985.83)** [if negative, swap counterparty pays, if positive, broker-dealer pays			New portfolio value:			+154,498,511 US$	
					Performance:			3,418,511 US$	
New loan									
Additions		-			Net payment				
New loan amount		**154,498,511.95**							
New interest rate		1.14875%							

FIGURE 14.6 Spreadsheet Showing Basket of Bonds at TRS Re-Set Date plus Performance and Interest Payments Due from Each TRS Counterparty

EUR/USD 1.228

Name	Currency	Nominal	Price	Accrued	Amount	FX	ISIN	Price	Accrued
ABC Telecom	EUR	16,000,000	111.500%	0.78%	22,331,239	1.2431		111.5	0.77595628
XYZBank	USD	17,000,000	125.000%	1.58%	21,518,931	1		125	1.58194444
XTCUtility	EUR	45,000,000	113.000%	0.28%	63,369,825	1.2431		113	0.28278689
SPG Corporation	EUR	30,000,000	100.750		30,225,000	1.2431		100.75	
WattyExploited	USD	15,000,000	113.062%	0.00628125	17,053,518	1		113.061996	0.628125
LloydColeFunding	USD	15,000,000	112.0923%	0.57%	16,899,628.1	1		112.092313	0.571875
					171,398,140.07				

Payments	
Interest	
Rate	1.14875%1W Libor + 7 bps
Amount	154,498,511.95
Interest payable	34,510.03
Performance	
Old portfolio value	154,498,511.95
New portfolio value	171,398,140.07
Performance payment	**(16,899,628.13)**
Swap ctpy pays	(16,865,118.09)
Newloan	
Additions	16,899,628.13
New loan amount	171,398,140.07
New interest rate	1.22750%

FIGURE 14.7 TRS Basket Value after Addition of New Bond

CREDIT-LINKED NOTES

In this section, we investigate the pricing and hedging of credit-linked notes (CLNs). We show the general principles required to accurately represent these structures. We then delve into specific examples of CLNs and how subtle differences between their legal terms can have noticeable effects on the pricing. We show that these structures need a fully dynamic model, containing factors that represent the bank's own credit, the reference credit, interest rates, and anything else relevant to the payoff. If the correlation between these factors is ignored, in particular the correlation between the reference credit and the issuer credit, then the issuer will likely pay too high a coupon on the note.

Background

Credit-linked notes' basic structure is simple:

- If neither the issuer nor the reference credit default, it pays a coupon, and redeems at par at maturity.
- If the issuer defaults, then the note terminates. The noteholder receives a payment equal to the recovery rate of the issuer, multiplied by the face value of the note.
- If the reference entity defaults, the noteholder receives a payment equal to the recovery rate of the reference entity's CDS, less "unwind costs". The unwind costs usually cover the cost to the issuer of breaking the deposit and derivative hedges.

Compared to a vanilla note, the noteholder is taking additional credit risk to the reference entity. Thus, in return, he receives an enhanced coupon.

Imagine an investor who perceives that the real-world probability, adjusted for his personal risk preferences, of companies X and Y defaulting upon its debt is significantly less than the risk-neutral probabilities implied by their CDS and bond prices. He could expose himself to the credit risk to both companies in a CLN issued by X, referencing Y (or vice versa), thus receiving a higher coupon than he could on any vanilla bond issued by either X or Y.

If he thought that the correlation between the default probabilities of X and Y was high, he would reason that the increase in risk by referencing X's bond to Y was low (this is explained in more detail later). For example, an investor could buy an RBS-issued CLN referencing Lloyds.

Many investors perceive their own sovereign debt to be risk free or close to risk free. Many investors will have so much unavoidable exposure to their sovereign that, in the event of its default, an extra loss would be immaterial. This is especially true of investors who are closely linked to their government, such as a sovereign wealth fund or a state-owned entity. Such people may well want to buy CLNs linked to their own sovereign debt: for instance, a French investor could buy an RBS-issued CLN, referencing France.

A company's shareholders will not care about the recovery rate of their bondholders in the event of their company going bankrupt. Hence, the shareholders would be happy to see their company buy a CLN referencing their own company, as this would provide them (in the risky measure representing the preferences of the shareholders) with enhanced coupons at no extra risk. For example, Tesco could buy an RBS-issued CLN, referencing Tesco. (The Tesco bondholders would be less happy with this arrangement, as the recovery they receive upon a default would be reduced by the loss in value of the CLN asset.)

Thus there is a lot of potential demand for CLNs, much of which arises naturally from investor's risk preferences.

We begin our analysis by explaining the effect of correlation on the joint default probability of two names, and how this affects the price of an FTD (first to default) swap, and some related instruments. Then, starting with a simple floating-rate CLN, we show why the effect of this correlation must be considered in the pricing. We then move on to more exotic examples, building on the principles previously explained.

Modelling the Joint Default Probability of Two Credits

The probability of receiving a coupon on a credit-linked note is scaled by E[sp1*sp2], that is, the expectation of the product of the survival probabilities of the two credits.

We know that the survival probability is related to the credit spread and recovery rate by:

$$sp = exp(-CDS * t(1 - RR))$$

Assuming both credit spreads are lognormally distributed, and correlated, we can then write the joint survival probability as:

$$Joint_{sp} = \exp(-(CDS_1 * t)/(1 - RR_1)) - (CDS_2 * t/(1 - RR_2))$$

This expression contains four volatile factors: the two CDS levels and the two default times. (The recovery rates are also volatile, but we shall ignore that complication.)We can make a good analogy to a nuclear decay. In the case of a nuclear decay, the half-life; that is, the time that needs to elapse until there is a 50% probability of the atom decaying, is fixed. This is analogous to having a CDS (and recovery rate) that can never change. Furthermore, there can be no correlation between the default times of different atoms. The probability of an atomic decay (i.e., the default-time) is modelled using a Poisson distribution. This represents the fact that, within a given amount of time, there is a fixed probability that the atom will perform a transition from its current state, to its decayed state. We can apply the same reasoning to the entity represented by the CDS spread; that is, once we know the RR and CDS spread, we know the probability that the company will default within a given time frame.

With a company, we have the added dimension that the half-life itself can change. This is caused by a change in the perceived riskiness of the company; that is a movement in the CDS spread. Although CDS spreads can experience discrete jumps, they normally evolve in a continuous fashion. As a CDS spread cannot be negative, but can become infinitely large, we could approximate its behaviour using a lognormal stochastic process. As exp(0) = 1 and exp(-infinity) = 0, this means that the expected survival probability for a given future time is bounded between zero and one, as expected.

Now that we have a reasonable model for a single credit, we can consider the behaviour of two correlated credits. If we pick two companies/sovereigns (with liquidly traded CDS) at random, it is virtually guaranteed that their historic CDS spreads are positively correlated. This is because general creditworthiness is dependent on many macroeconomic factors that are likely to affect all entities, to some extent. We can consider some specific examples.

The CDS levels of RBS and BNP Paribas are very highly correlated, because they are both European banks. They have similarities in their business models and their portfolios. They operate in similar geographical locations. If Europe as a whole became perceived as being more risky, the CDS spreads of most European companies and sovereigns would widen. If there was a regulatory change, or a decision made on the treatment of European sovereign debt, it would affect both BNP and RBS in a similar way.

RBS and Lloyds are even more highly correlated. They are both UK banks. If a macroeconomic change occurred that affected the UK differently to the rest of Europe, then RBS's and Lloyds's CDS would move together more than BNP's. Furthermore, both are largely owned by the UK Government, and thus there are various decisions the UK government could make that would affect both these banks without directly impacting BNP. In an extreme case, the government could tell them to both wind up certain areas of their business, or withdraw financial support from them entirely. If the UK government's own credit-worthiness were affected, this would have a direct knock-on effect on government-owned banks.

State-owned companies aside, sovereigns are likely to be much more highly correlated with each other, than sovereigns with nonsovereigns. This is particularly true of sovereigns whose financial affairs are closely intertwined, by geographical proximity, trade, or a single currency.

When a CDS spread rises, the half-life of the entity reduces. Thus we do not need to consider any additional correlation between the default times and the CDS spreads. However, we should consider correlations between the default times themselves.

When Lehman collapsed, many other institutions suffered immediate losses because they were owed money by Lehman. Furthermore, most, if not all, of these companies would not have considered the possibility of a Lehman default, and would not have prepared for the event (e.g., by buying CDS protection).This led to the immediate collapse of a number of companies, and the severe weakening of others.

How can we model this behaviour? Without going into detailed mathematical derivations, we can explain qualitatively what properties we might require for a realistic model. The first approach that many banks took was to hold the CDS spreads constant, and correlate the default times directly. The advantage of this model is that it can be written in a simple closed-form solution. Unfortunately, there are a number of disadvantages. The resultant behaviour is clearly unphysical, as real-world CDS levels do change. Furthermore, it seems conceptually odd to say that the default of one entity today somehow affects the expected time until another entity defaults, without the other entity's CDS moving! If traders expected that the second entity would default sooner, following the default of the first entity, then the second entity's CDS would widen in response. Say we wanted to make a price on a trade that depends on the correlation between two entities, and all we have to rely on is historical data (as there are no liquidly traded products that allow us to mark the correlation from their prices). In this case, we can observe historical CDS levels of the two entities, and calculate their correlation from these. However, there is no obvious way to convert this CDS correlation into a default-time correlation. There is no way to observe default-time correlations from historical data either; for one thing, neither of the two entities has defaulted yet!

The alternative to correlating the default-times directly is a more realistic model in which the CDS spreads are volatile and correlated. In this case, the defaults themselves are kept as independent, uncorrelated events, whose probabilities are linked to the CDS levels. This model overcomes the objections listed to the default-time correlation model. However, it has other disadvantages. It does not have a simple analytical solution, and it would require Monte Carlo pricing, which is noisier and slower. If the correlated CDS are lognormal, then it may not be possible to hit market-implied FTD prices (we describe this in more detail later). However, this issue can be overcome by making the spreads "super lognormal". That is, the CDS distributions are skewed further to the right than in a lognormal distribution. One way to do this would be to make the CDS volatilities stochastic and correlated with each other, as well as with their respective CDS levels. This would increase the covariance of the CDSs in a similar way to the stochastic volatility model proposed for the quantos earlier. This would also fit well with what we observe in practice: CDSs become more volatile as credit spreads widen, and there is a high correlation between the volatility of many CDSs.

This brings us on to the question of how to mark CDS volatilities and correlations. Of course, we can use historical data, but do market prices exist that give us information about these quantities?

CDS options rarely trade, and are only really seen on CDS indexes, like iTraxx and CDX. Thus, it is hard to avoid using historical data to mark the volatilities of most credit names. Interestingly, there are actually more market prices relating to the CDS correlations and covariances. The most liquid instrument that gives us information on the correlation is called an FTD, and we investigate this in the following section.

FTDs and Vanishing CDSs

A first to default (FTD) swap has the following payout:

- The protection buyer pays a running fixed coupon for the duration of the swap, as long as none of the reference entities default. If one of the reference entities defaults, then the remaining unpaid coupons are cancelled.
- The protection seller pays (1 – Recovery Rate) of the first entity to default during the duration of the swap. If there is no default, then the protection seller does not make any payments.

An FTD has two or more reference credits. For the purpose of this discussion, we shall only consider the case of two reference credits. If there are more reference credits, we can price the instrument using the same principles, but with additional factors. The two-credit case is of particular relevance to us because it needs to be understood for CLN pricing.

We can simplify the analysis of the two-name FTD by breaking it up into two simpler trades that I shall call "vanishing CDS". The PV of the FTD is equal to the linear sum of the PV of these two vanishing CDSs.

A vanishing CDS has the following payout:

- The protection buyer pays a running fixed coupon for the duration of the swap, as long as neither of the reference entities defaults. If one of the reference entities defaults, then the remaining unpaid coupons are cancelled.
- The protection seller pays (1 – Recovery Rate) if entity one defaults first. If entity two defaults first, then there is no coupon. We shall call the entity that triggers the payout (entity one) the "reference entity".

From the protection buyer's point of view, he gets the same protection by buying an FTD as he gets by buying two vanishing CDSs: one paying out on entity one, and one paying out on entity two. Thus, the coupon on the FTD must equal the sum of the coupons on the two vanishing CDSs.

We shall consider the simpler model of correlating the default times, and then compare this with the model of correlating the CDS spreads. We already covered the basic principles of pricing products that depend on correlated assets when we looked at quantos, and we shall rely on that understanding in what follows.

First of all, let us assume that all the correlations are zero, and think about how the two vanishing-CDS spreads compare to their vanilla equivalents. As there is no correlation, the PVs of both the loss amount and the running coupons are scaled down by the same ratio. This is because the probability of the reference entity defaulting is unaffected by the default of the second entity. This means that the fair price of the spread will equal the vanilla CDS rate of entity one. As the same logic applies to the other vanishing CDS, we can see that the coupon on the FTD will equal the sum of the vanilla CDSs.

Now let us consider the case where the default times are positively correlated. In the limiting case of a 100% correlation, the entity that has the wider CDS level (and the shorter expected default time) must always default first. If the reference entity is the wider credit, then it will always, in the vanishing CDS, default before the second credit. In this case, a default of the second credit can never affect the price of the trade, as there will be no coupons or fees remaining to be paid by the time it defaults! Thus, the fee paid on the vanishing CDS must be equal to the fee paid on the vanilla CDS! Conversely, if the reference entity is the tighter credit, then it must always be the second to default, and thus the loss leg premium will never get paid out. Hence the fee leg must have zero coupons, as the credit insurance is worthless! So, in the case of a 100% default-time correlation, the CDS level paid on the FTD is equal to the wider of the two fees paid on the vanilla CDSs.

Now let us consider what happens between the two limiting cases. To simplify the argument, we consider a one-year FTD with a single coupon fee paid at the end of the year if there have not been any defaults. Let us assign a probability of 30% to credit A defaulting in the year, and a probability of 40% to credit B. The fee leg on the FTD is paid out contingent upon there being no defaults. We then have the following possible outcomes, given no correlation between the default times:

- Only credit A defaults: 18%
- Only credit B defaults: 28%
- Both default: 12%
- Neither defaults: 42%

Let us assume the recovery rates are the same on the two credits, so that the fee is the same, regardless of which credit defaults first. In this case, the same loss amount is paid out 58% of the time.

As the default-time correlation increases, the probabilities of only one credit defaulting both fall, while the probabilities of either both or neither being in default rise. The probability of there being one or more defaults must fall, as the probability of neither defaulting has risen. Hence the total fee payable on the FTD must fall, as the protection leg of the FTD only pays out upon the first default. Once the correlation reaches 100%, the probability of only A defaulting has fallen to zero. The sum of the probabilities of only B defaulting and of both credits defaulting (with B always defaulting first) must equal the probability of B defaulting. Hence, the fair level of the fee on the FTD must equal B's vanilla CSD level, as previously explained. Thus, the fee payable of the FTD falls monotonically from the sum of the vanilla CDS fees to the level of the wider of the two CDS fees, as the correlation increase from zero to 100%.

Initially, as the correlation begins to rise, this reduction in the FTD fee is caused by reductions in the fees of the two constituent vanishing CDSs. This is obvious in the case of A being the reference credit. In the case of B being the reference, we can see that the fee leg is more likely to get paid out as the correlation increases. However, although the probability of B defaulting first *given both credits' default* has increased, the probability of B defaulting *and being the first to default* has actually decreased. This is because it is now more likely that both default, and less likely that only one defaults. Thus, the fee leg is more likely to get paid out, but the trade is less likely to end with a loss leg payment, and hence the vanishing CDS fee must fall. It is only when the correlation becomes very close to 100% that the probability of *B defaulting and being the first to default* starts to increase. Thus, at very high correlations, the fee for the vanishing CDS referencing B hits a minimum, and then rises back up to the level of the vanilla CDS fee on B.

Now, let us consider what the behaviour would be if, instead of correlating default-times, we correlated volatile CDSs themselves. The limiting case of a zero correlation would be the same as before. It must also be true that, as the correlation starts to increase, the levels of the two vanishing CDSs both fall, as above. However, in this model, the behaviour at high correlations will be different. If the correlation is 100%, the CDS that was initially wider will remain wider for all time. However, as the defaults themselves are random and uncorrelated, the entity with the tighter CDS can still default first. Thus we shall no longer observe the unphysical behaviour that we saw in the default-time correlation model where the spread paid on a vanishing CDS referencing the wider credit increases up to the level of the vanilla CDS at 100% correlation. Similarly, the spread on the vanishing CDS referencing the tighter credit will never drop to exactly zero.

Figure 14.8 compares how the spreads on the vanishing CDSs and FTD may look in the two different models.

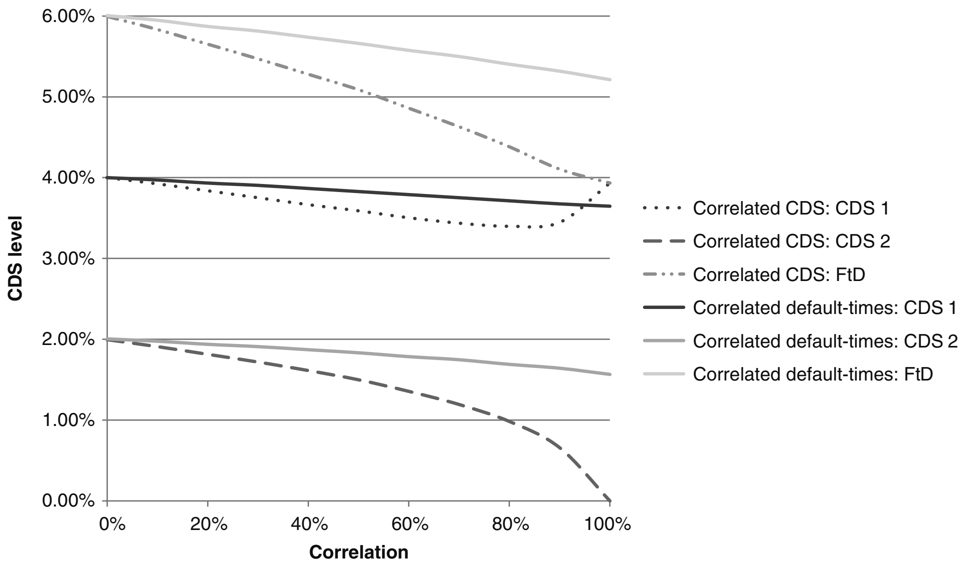

FIGURE 14.8 Comparing FTD Correlation Models

As you can see from Figure 14.8, a potential problem with the correlated-CDS model is that the market spreads of FTDs may actually end up being lower than the spread given by the model at 100% correlation! We could attempt to get round this by increasing the skew on the two credit spreads to be greater than the lognormal skew: for example, by giving the CDS spreads stochastic volatilities and correlating those volatilities. This is similar to the idea mentioned in the section "Quantos, Pricing, and Their Risks" about quantos in Chapter 21 whereby one can increase the effective covariance, and thus the price, on the quanto by correlating the volatilities.

SIDEBAR: UNCOLLATERALISED CDS

We consider an uncollateralised CDS between two counterparties, one of whom is risk free. This is equivalent to the "risk-free" counterparty buying a bond issued by the risky counterparty, with a CDS payoff embedded in the bond. There cannot be any credit risk to the investor, as the other party has his money up front.

We can represent the counterparty credit risk on this type of uncollateralised transaction by scaling the payout by:

$$RR + (1 - RR) \bullet sp$$

This is equivalent to the sum of a collateralised transaction, whose notional is scaled by RR, and a vanishing transaction, whose notional is scaled by $(1 - RR)$.

Thus we can say that:

PV of uncollateralised derivative = *RR* * (*PV of collateralised derivative*) + (1 – *RR*) * (*PV of vanishing derivative*)

As we already know how to price a vanishing and a collateralised CDS, we know how to price an uncollateralised CDS. This will be of use in the CLN pricing that follows.

FLOATING RATE CLN WITH UNWIND COSTS ON REFERENCE DEFAULT

This is the simplest example of a CLN because there is no derivative hedge required. The cash flows are:

- Bank receives the notional on the start date.
- Bank pays the coupon, if there is no default, of Libor + X + Y, where X is the bank's own funding spread, and Y is the CDS spread of the reference entity.
- Bank pays notional on end date, if there is no default.
- If the bank defaults, it pays its own recovery rate on the notional.
- If the reference defaults, the bank pays the recovery rate on the reference entity, less the cost of unwinding its own internal deposit.

Before people started thinking about the differences between collateralised and uncollateralised trades, many people thought this structure could be almost statically hedged, as follow.

Incorrect Cash Flow Diagram for a Floating Rate CLN and Hedges

This issue is illustrated at Figure 14.9.

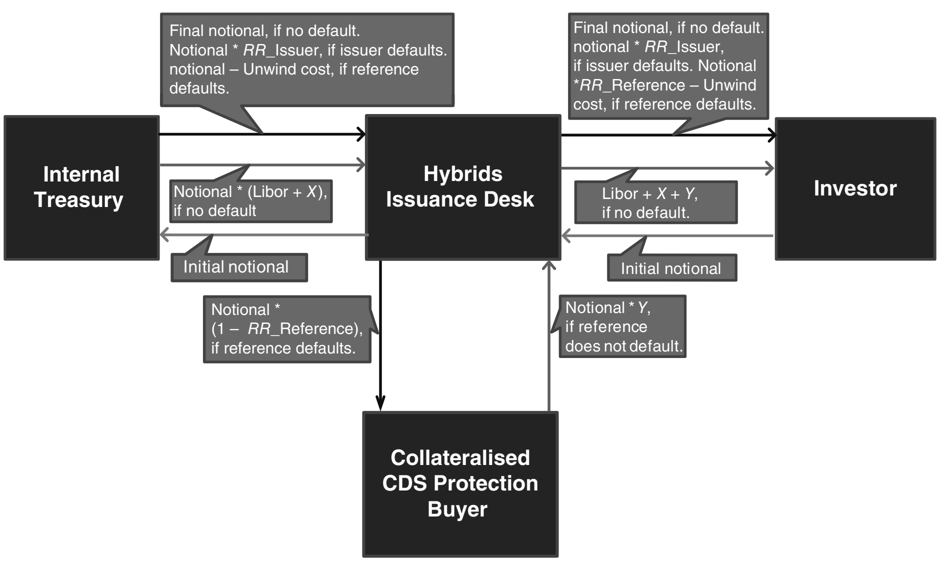

FIGURE 14.9 Incorrect Cash Flow Diagram for FR CLN and Hedge

The only perceived problem was that if the reference defaulted with a low recovery rate when the bank's funding spread was wide, there would not be enough money available to pay for the cost of unwinding the internal deposit. This was seen as a small exposure to a tail event ("tail risk"), and a small reserve would be taken (and amortised over the life of the CLN) to cover it.

As previously explained, we know that we have a perfectly hedged, risk-free portfolio if, regardless of future price changes, no net cash flows are generated. At first glance, it does indeed appear that the trading desk is perfectly hedged. However, we can easily show that this is not true, for example:

- If the reference CDS spread widens, we have to post collateral to our counterparty, upon which we shall receive OIS (or whatever the CSA says).
- We shall have to borrow this money unsecured from our internal treasury, and pay interest equal to our funding spread, which is almost certainly greater than OIS.

There is almost certainly a positive correlation between the reference CDS spread and our funding rate. Thus, the impact of this cash flow mismatch is asymmetrical:

- When the reference's spread widens, the gap between our funding and OIS increases, making the collateral call expensive to finance.
- When the reference's spread contracts, the gap between our funding and OIS decreases, so we receive a reduced rate when we put the collateral we received on an unsecured deposit with the treasury.

Another way to consider the problem is to think about what actually happens when the bank defaults. In this case, the collateralised CDS is probably going to be out-of-the-money for the bank, due to the positive credit correlation. The trading desk will have posted that mark-to-market (Z) to the counterparty, and borrowed it unsecured from the treasury. When the default occurs, the desk has to return Bank Recovery * Z to the treasury, but the collateralised counterparty keeps Z in its entirety. Thus we are not cash flow–matched upon our default.

Notice that we have been assuming that, upon default, the two internal desks in the bank (trading and treasury) pay to each other Recovery on any outstanding, uncollateralised debts. This may seem strange, as they are all part of the same organisation. However, it is still right to consider them as independent entities. For example, say the trading desk issued a vanilla bond to an investor. In this case, he should pay to the investor the same rate that he receives from the treasury on the internal deposit. In his trading ledger, the loan from treasury and the deposit from the customer must have equal and opposite PVs. If we assume the treasury deposit has a different recovery rate than the loan from the customer then, in the risk-neutral measure, the PVs of the two transactions will differ as soon as the at-the-money spread changes.

Bearing this in mind, is it possible to construct a portfolio that statically hedges this CLN?

Correct Cash Flow Diagram for a Floating Rate CLN and Hedges

The corrected approach is illustrated in Figure 14.10.

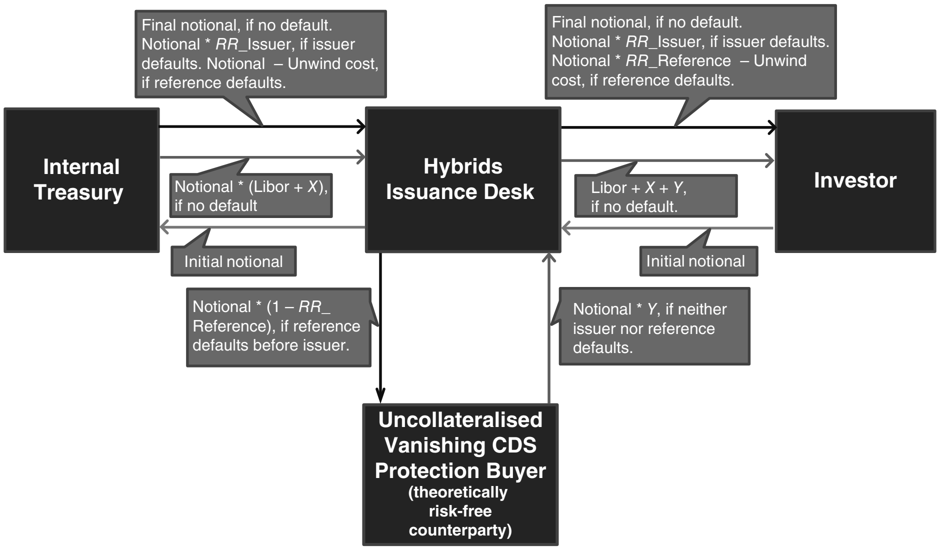

FIGURE 14.10 Correct Approach to FR CLN Hedge

We have replaced the collateralised CDS hedge with an uncollateralised vanishing CDS. We have created a theoretically risk-free (or cash rich) counterparty for the trade, so that exposure to the counterparty's credit risk does not appear. The terms of the vanishing CDS state that:

- The trading desk receives the premium until either the issuer or the reference defaults.
- If the reference defaults before the issuer, the trading desk pays Notional * (1 – Reference's recovery rate).

Now, with the exception of the tail risk mentioned above (which we shall address later), the trading desk is perfectly hedged in all scenarios.

- No collateral needs to be posted at any time.
- If there is no default event, the coupon flows received from treasury and the CDS protection buyer match.
- If the reference defaults, the trading desk pays $(1 - RR_{ref})$ to the protection buyer, trading receives 1 from treasury, less the change in the mark-to-market of the deposit (the unwind costs). Trading passes the net cash flow of RR_{ref} less unwind costs on to the investor.
- If the issuer defaults, the vanishing CDS terminates at a zero mark-to-market. Trading receives RR_{Issuer} from treasury, and passes this amount on to the investor.

The deposit with treasury is a vanilla one (when we break it, trading pays/receives any change in mark-to-market), so the price of that is known. The vanishing CDS can be priced using the model above. Thus we can value the CLN as the sum of these products.

In practice, however, there is a problem. Not only does the theoretically risk-free counterparty not exist, but we are unlikely to be able to trade the uncollateralised vanishing CDS. Thus we have to try to dynamically replicate its value using vanilla market instruments. If we have valued the unhedged parameters correctly (i.e., the credit correlation), then the dynamic portfolio rebalancing will be expected to have no net PV impact, on average, over time.

Dynamic Replication of Uncollateralised Vanishing CDS

The liquid instruments we have access to are:

- Collateralised OIS swaps
- Treasury loans and deposits
- CDS on the reference, collateralised at OIS

In order to dynamically replicate the uncollateralised vanishing CDS hedge, let us consider that trade in isolation, and try to create a dynamically risk-free portfolio containing that and the above hedges.

On day one, our portfolio must look like that illustrated in Figure 14.11.

The collateralised CDS has a larger coupon than the uncollateralised vanishing CDS. The collateralised CDS will have a shorter duration than the uncollateralised vanishing CDS, so that the CR01 to the reference are matched. The trading desk loses money on the uncollateralised vanishing CDS if the treasury funding rate increases. To hedge this, money is borrowed today from the internal treasury, and repaid over time. This makes the trading desk long cash today. This is put on overnight deposit, and the interest earned will go towards repaying the debt to treasury. The overnight deposit has almost no CR01 to the bank.

Now, let us imagine that both the credit spreads of the reference and the bank widen. If this results in a realised correlation that is less than the one used to price the

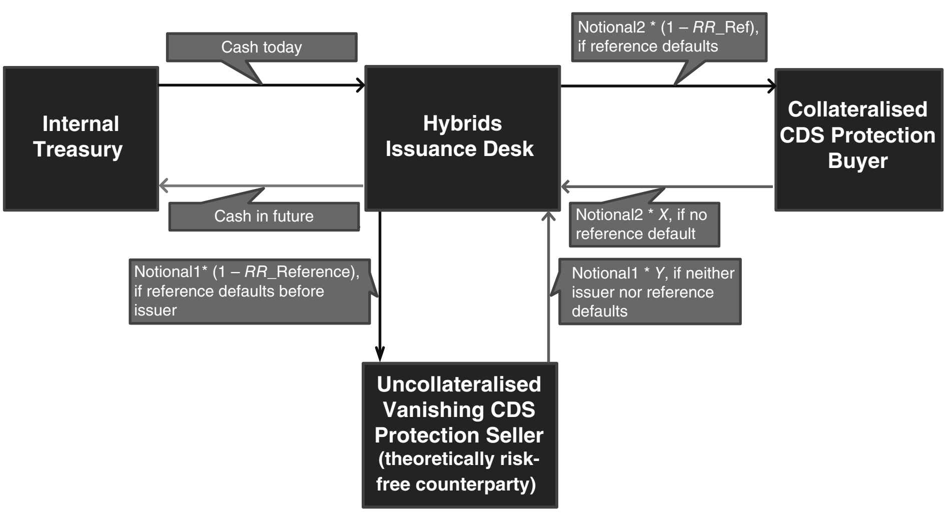

FIGURE 14.11 Day 1 Portfolio: Dynamic Replication of Uncollateralised Vanishing CDS

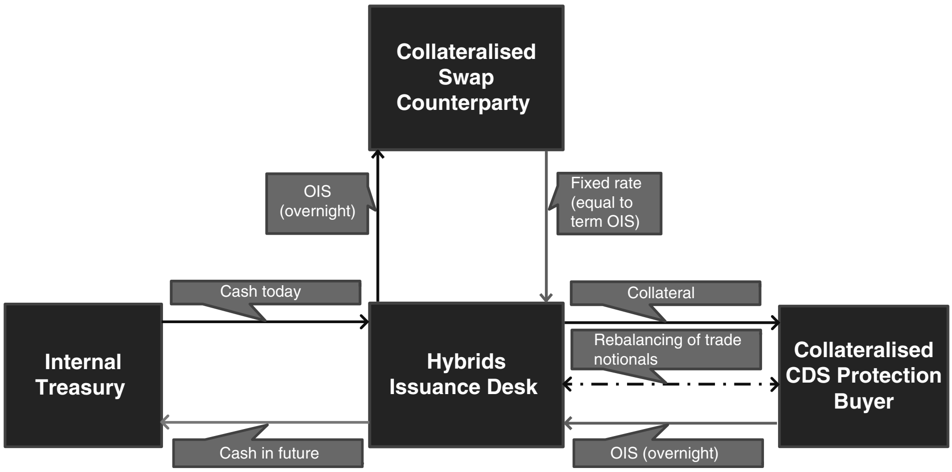

FIGURE 14.12 Portfolio Rehedging

uncollateralised vanishing CDS, then the trading desk gains money on the resultant rehedging. If it results in a higher correlation, then the trading desk loses money. (The reasons for this have been explained previously, in a different example.)

When the credit spreads widen, the bank has to undertake the rehedges shown in Figure 14.12 (the cash flows for the initial hedges have been omitted).

The uncollateralised vanishing CDS has increased in value due to the widening of the reference spread. The positive PV creates additional exposure to the bank spread. This is hedged by borrowing more money today, to return in the future (the present value of the PV gain is borrowed). There is also a gain made on the original money that was borrowed from the treasury, which again results in a positive PV to be hedged by borrowing more cash. Thus, the present values of both gains are monetised today. As the net portfolio value has not changed, the additional cash borrowed has to be posted as collateral to cover the loss on the collateralised CDS. The counterpart must pay OIS overnight on this cash. We are still not exactly hedged. The loss on the collateralised trade has occurred because the expectation of future cash flows in that trade have moved against us (i.e., the probability of paying the future coupons has increased, while the probability of receiving the loss leg has decreased). This results in net risk to the level of OIS discounting until the payment dates of these cash flows. To hedge this final piece of risk, we enter into a collateralised OIS swap, where the trading desk pays out OIS overnight, and in return receives a fixed rate equal to the expectation of the terminal OIS level. Finally, the exposure to the reference CDS may have changed (in size or term structure). Thus we can rebalance the notional of the CDS hedge. As this is an at-market collateralised swap transaction, it does not create immediate cash flows.

If the realised correlation had been higher than the implied, then the trading desk would have found that, after rehedging to cancel out the risks to the bank and reference credit spreads, they were short an amount of cash equal to the realised loss. This loss would have to be funded by overnight borrowing from the internal treasury ad infinitum. Thus the funding cost on the loss would cause its size to accrete up at the bank's funding rate, over time. This makes sense because, as explained previously, future cash is discounted at the rate at which it costs to fund it over time.

If the bank's funding spread had increased by just a bit more than "expected" by the correlation, then the uncollateralised trade would still have increased in value, but not by enough to balance the losses on the other trades in the dynamic portfolio. In the extreme case of a large increase in the bank's spread, the uncollateralised trade could actually reduce in value, as the probability of the reference defaulting before the bank approaches zero.

If we combine this dynamic hedging strategy with the static hedges in Figure 14.10, we have thus developed a dynamic risk-free portfolio containing the CLN.

Floating Rate CLN with Unwind Costs on Both Issuer and Reference Defaults

The difference between this example and the previous case is that now, upon default by the issuer, the issuer returns RR_{Issuer} * *notional* * (1 – *mark-to-market of reference CDS*), instead of simply RR_{Issuer} * *notional*. We can construct a static hedge portfolio (again ignoring the "tail" risk), as shown in Figure 14.13.

As above, we have the problem that the uncollateralised CDS required as a hedge does not exist in the market. We can represent an uncollateralised trade as:

$$(RR_{Issuer} + (1 - RR_{Issuer}) * sp_{Issuer}) * Collateralised_{MtM}$$

In this case, this is equivalent to:

$$(RR_{Issuer} * Collateralised_{CDS} + (1 - RR_{Issuer}) * Vanishing_{CDS})$$

From the above, we already know how to dynamically replicate a vanishing CDS, and thus this example should not cause further difficulty.

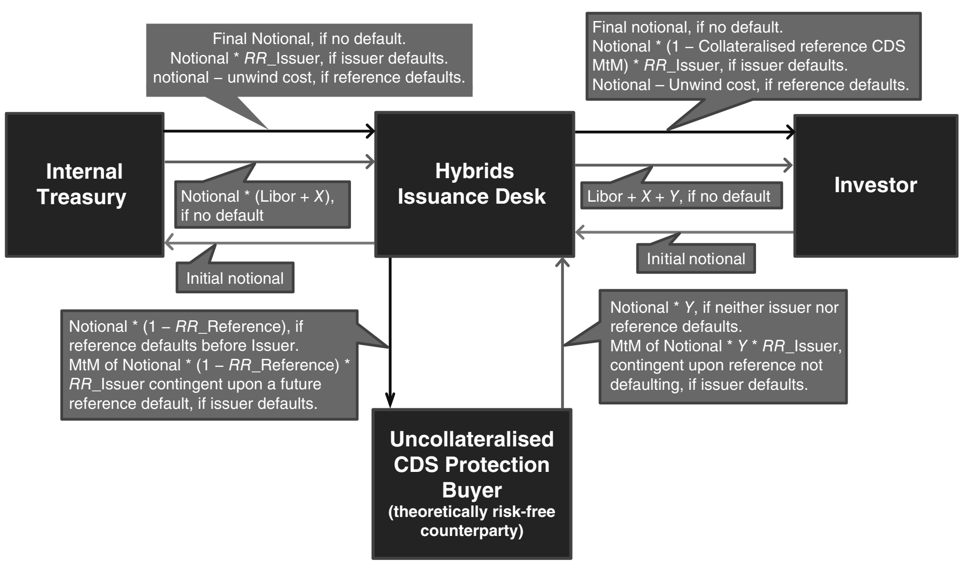

FIGURE 14.13 FR CLN and Unwind Costs on Issuer and Reference Default

Monetisation of Difference between Vanishing and Uncollateralised Cases

If the CDS unwind costs are not included when the issuer defaults, the CLN noteholder has been preferenced over other *pari passu* creditors, as the CLN investor will, in effect, receive more than recovery × mark-to-market. Thus, if the investor demands a reduced coupon for the vanishing case (i.e., pays for his preferential treatment), this results in an overall loss for other bondholders, but a net gain for the shareholders, who would not receive anything on default anyway, but now pay out lower coupons while the company survives.

Thus, the vanishing CLN appears to be a better trade to do because, as discussed earlier, we should consider the impact of our trades on the shareholders as a separate group, as well as the shareholders and bondholders combined.

However, in reality, the investor would not enter into a CLN if he thought the probability of the issuer (or reference) defaulting was high. Hence, in the investor's risky measure, the exact size of the recovery he receives upon the issuer's default is less significant than in the risk-neutral measure. Thus, he is unlikely to be prepared to accept a lower coupon in the vanishing case.

Therefore, by trading the vanishing CLN, we would create a potential loss for our other bondholders, with no offsetting profit for the shareholders. Thus we should aim to include the unwind costs in the CLN's documentation. (In reality, the issuer would expect to make a risk-neutral profit on the CLN issuance, so the vanishing case would simply result in a reduced profit for the trading desk.)

A final point worth noting is that, in general, a vanilla bondholder will not receive recovery on the risk-free mark-to-market of the remaining cash flows, upon a default. Instead, he will receive recovery × notional. However, as the loss of the recovery on the future coupons is roughly compensated for by receiving the recovery on the notional immediately, instead of on the original maturity date, we can ignore this distinction.

Tail Risk upon Reference Default

In the example CLNs above, the issuer deducts the cost (or adds the benefit) of unwinding a theoretical internal treasury deposit, matching the size and maximum duration of the note, upon a default by the reference entity. It must be noted that, as we showed above, the actual deposit that trading places with the treasury is likely to have a different size and duration, due to the dynamic rehedging of the risks. Due to the positive correlation between the credit spreads, it is likely that this theoretical deposit, which was initially worth 100, will be worth significantly less when the reference entity defaults. Thus the addition of this clause is favourable to the issuer, allowing a higher coupon to be paid on the CLN.

In the examples above, we ignored the risk that the recovery rate on the reference could be insufficient to pay for the cost of unwinding the theoretical deposit. We can represent this risk mathematically by saying that, *contingent upon the default of the reference*, the issuer is short the option:

$$Max(100 - Deposit_{MtM} - RR_{Reference}, 0)$$

In essence, this is a knock-in spread option (the knock in being the reference defaulting before the issuer) on the deposit mark-to-market and the reference entity's recovery rate. To fully value this option, we would need to dynamically model the two credit spreads, the risk-free discount rate, and the recovery rate itself. In practice, there is probably no models that allow for a volatile recovery rate. Even if such a model were used, there would be no market data available to model the recovery rate's distribution.

In reality, recovery rates are commonly marked at a constant value between 25% and 40%. When an entity's credit spread starts to widen, recovery locks often begin to trade liquidly. These products pay out the difference between the auction recovery rate and the strike, contingent upon the entity defaulting within a specified time. Thus the recovery rate has a very strange distribution. Its implied volatility is virtually zero when credit spreads are tight. Then, if the recovery locks start to trade, it becomes volatile, as the at-the-money strike moves around. Upon the default event itself, the recovery rate is fixed dependent upon the auction price of the outstanding senior obligations. This auction price is often far from where the locks were previously trading. One reason may be that, upon default, suddenly a lot more information becomes available on the like net value of the company. Thus the recovery rate will likely experience a sudden jump in value upon the default. (The actual recovery rate on the bonds may only be determined years later, when the company is finally wound up by the liquidators. This may well be at a completely different level to the auction rate, but this is irrelevant to our CLN.)

Bearing this in mind, an approximate method must be found to value the tail risk. One suggestion is to dynamically model the two credits and interest rate, and then to apply a static haircut to the level of $RR_{Reference}$, to compensate for the "missing" volatility that the Bank is short. Alternatively, the tail risk option could be priced multiple times, each with a different recovery rate, and then a weighted average of these prices used as the final valuation, to represent a possible spread in recovery rates.

Including the price of the tail risk affects the day one hedges that the trading desk must execute. The tail risk must make the trading desk short the credit spreads of both the issuer and the reference, and short the recovery rate of the reference. Thus, the inclusion of the tail risk means that the duration and/or size of the treasury deposit is reduced, and less reference CDS protection is sold. The trading desk should also buy some recovery locks on the reference, if a market is available.

As the tail risk reduces the present value of the unwind costs clause, the resultant hedge profile must lie somewhere between the hedge profile ignoring the tail risk, and the hedge profile of a CLN with no unwind costs clause. We shall examine the latter case below.

Floating Rate CLN with No Unwind Costs

The payoff is defined to be:

- Bank receives the notional on the start date.
- Bank pays the coupon, if there is no default, of Libor + X.
- Bank pays notional on end date, if there is no default
- If the bank defaults, it pays its own recovery rate on the notional.
- If the reference defaults, the bank pays the recovery rate on the reference entity.

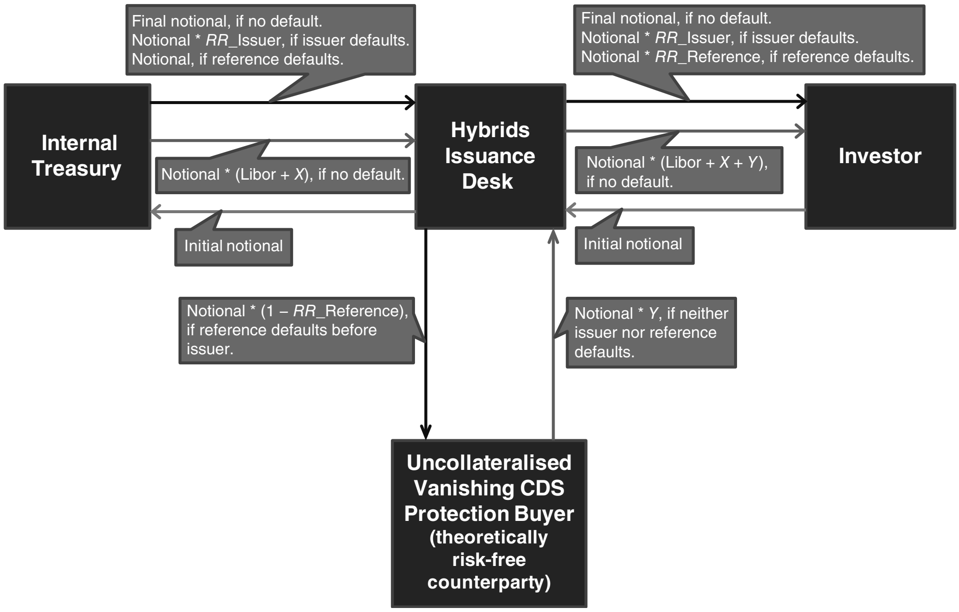

FIGURE 14.14 Static Hedge Portfolio, Cash Flow Deposit with Treasury Desk

This is identical to the first payoff we considered, except for the cash flow upon the reference default. Our static hedge portfolio must now involve a credit-linked deposit with our bank's internal treasury desk, as shown in Figure 14.14.

We already know how to dynamically replicate the uncollateralised vanishing CDS, so let us consider the vanishing treasury deposit. A deposit and a CDS are very similar. One could think of a CDS as an "unfunded" deposit. If the risk-free funding rate is Libor + X, then the coupon on the deposit is theoretically (ignoring any CDS/bond credit basis) Libor + X + Y, where Y is the CDS spread.

In Figure 14.15 we show a dynamically hedged portfolio where we unwind the credit-linked treasury deposit, and replace it with some vanilla hedges. To distinguish between the two deposits, we have defined "trading treasury" to be our counterparty to the structured (credit-linked) deposit. Of course, it is the same desk (although it may be different people within the department)—the bank's Treasury desk, also known as the asset-liability management or ALM desk.

Not only are the three notionals likely to be different at various points in time, but they will also have different roller-coaster term structures. Any mismatch in notionals 1 and 2 on day one will result in some overnight lending/borrowing to/from the internal treasury (we could just consider this as meaning that their sum at spot always has to be zero).

Now consider, as we did before, a widening of both the reference and issuer spreads, and the rehedges this generates, as shown in Figure 14.16.

Profit is made on the collateralised CDS, resulting in the posting of collateral to us, on which we have to pay OIS overnight. The OIS risk is hedged by use of a term OIS swap. The collateral received is placed on deposit with the internal treasury, and

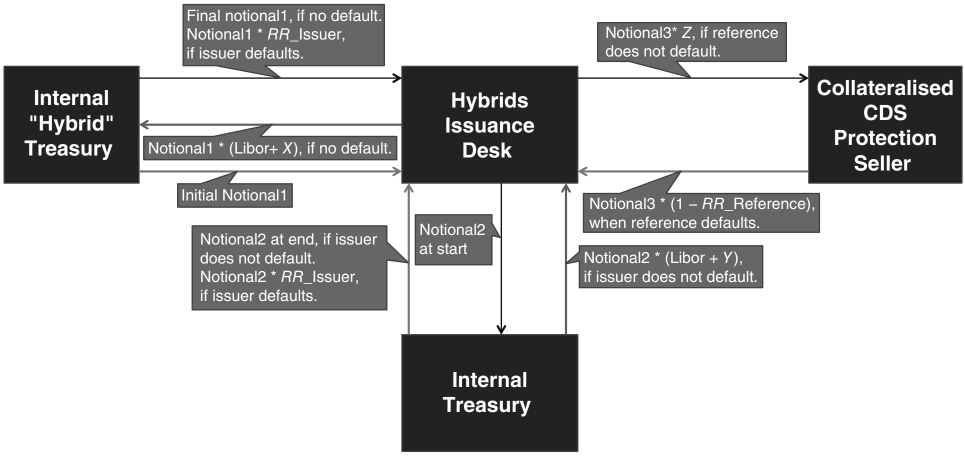

FIGURE 14.15 Dynamically Hedged Portfolio

the term structure of the deposit is rebalanced to rehedge the changed risk profile of the "structured" loan in the previous diagram.

The absence of the clause that covers the deposit unwind costs is disadvantageous to the trading desk, resulting in a lower CLN coupon being paid to the investor. As explained previously, the investor is likely to prefer to take an increased loss upon the reference default, in return for an enhanced note coupon. Thus this variant is unlikely to trade.

Callable CLN

A common feature requested on CLNs, and notes in general, is an issuer call. The call means that the issuer can cancel the note when the coupons appear to be large compared to the equivalent par coupons on a newly issued vanilla note. The issuer is likely to call the note, returning the full notional to the investor, when either credit

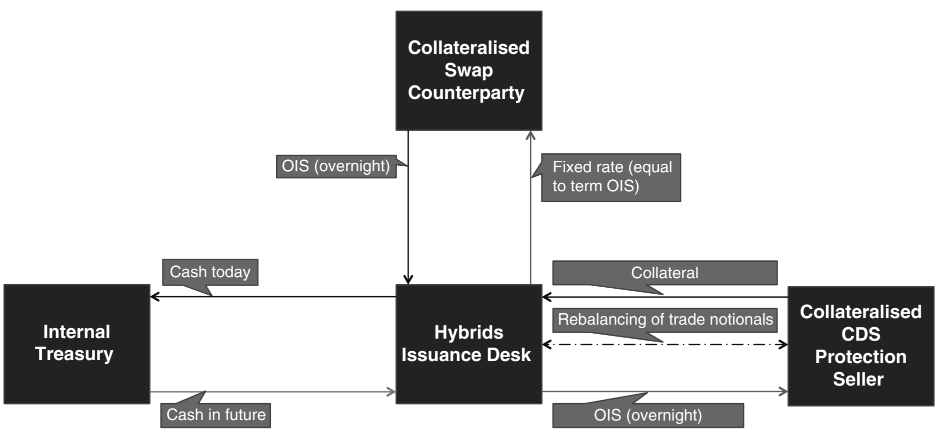

FIGURE 14.16 Spread Widening Impact

spread tightens, or interest rates fall compared to the exotic coupon levels. (A floating rate CLN will have virtually no exposure to interest rates.)

This trade is harder to dynamically hedge than the previous examples, due to the presence of the call. The noncallable CLNs all have some exposure to credit volatility, through the correlation effects, and the tail risk option. Both effects make the trading desk short the volatility, as they are short the tail-risk option, and the positive correlation is disadvantageous to them. In the callable case, the Bermudan option results in a positive credit volatility exposure for the trading desk, on both the bank's funding spread and the reference spread. If the note is issued close to par, then the positive vega exposures are likely to far outweigh the negative effects. Over time, if credit spreads widen, the call will become worthless, and the negative vega from the tail risk will dominate. With no liquid hedge available for the credit vega, the trading desk will have to manage the exposure by dynamically rehedging the treasury deposit and the vanilla reference CDS. The possibility of cancellation reduces the duration (and possibly size) of those hedges, as the delta exposure to both credits is reduced.

The trading desk also has to manage the volatility on the exotic coupon swap. To do this, they could either sell a Bermudan option to that swap counterparty, or try to manage the volatility independently, by trading European options on the underlying asset. If they use the former approach, it could be hard to determine how to structure the callability, because the presence of the credit effects in the note is likely to make the note and swap call profiles extremely different. There is likely to be a lot of future hedge rebalancing required in both cases. Bearing this in mind, it is probably more efficient to hedge the callability using separate European options, as the bid/offer costs will be reduced.

Upon a default by the reference, the unwind costs could contain no reference to the exotic coupon swap, or refer to the unwind of a noncallable coupon swap, or refer to the unwind of a Bermudan callable coupon swap. Although the latter disadvantageously increases the volatility of the tail-risk option that trading is short, it is probably beneficial to include, as the value of a Bermudan callable swap will always be positive to the call holder, once the initial noncallable period has expired. The other possible disadvantage of referring to the unwind of an exotic swap is that it may be harder to establish a market price for it.

THE iTRAXX INDEX NOTE

The details below are of a contract for a CDS written on the iTraxx index. The hypothetical bank, XYZ Securities, is the protection seller, and the counterparty is an hypothetical hedge fund called ABC Fund.

iTraxx is the name of a range of credit indices developed by a group of credit derivatives market makers. The index is available in funded form, as the iTraxx note, or in unfunded form as a CDS written on the index. The index can be traded in the following forms:

- A portfolio of 100 equally weighted European corporate and financial entities. These were a selection of corporates and financial entities, with a Moody's diversity score of 61 and average rating of Baal/Baa2. At launch in September 2003 the CLN paid quarterly Euribor plus 39 basis points.

- The top 100 euro-denominated issuers weighted by market value and duration; at launch these notes paid Euribor plus 90 basis points, at an issue price of €99.87. The average rating was Baa1, and they had a Moody's diversity score of 33.
- The iTraxx 100 excluding financial entities. This note had a diversity score of 23 and paid Euribor plus 125 basis points on launch, issue price par. The average rating was Baa2 across 62 reference entities.

The first index is aimed at providing an index for investors who are seeking to access a diversified market exposure, whereas the second index is for investors who wish to track the market overall rather than diversify. The third index exhibits higher beta and volatility compared to the other two. The main advantage of these index products for investors is the standardisation they represent. Investors can now trade in a standard note across the market, and one that has a ready set of market makers available at all times supporting it.

The funded form of the indices are CLNs tied to the index level and are known as iTraxx notes. They are all 5.25-year notes, with quarterly coupon for the first index and semiannual coupon for the other two indices. The bonds are issued by an entity incorporated in Ireland, iBond Securities plc.

Investors trading in the funded credit derivative for iTraxx, via the iTraxx note, can opt for a fixed coupon bond set at inception, whose price fluctuates as the value of the reference names fluctuate.

Figure 14.17 shows Bloomberg page DES for the iTraxx Index Note.

FIGURE 14.17 Bloomberg Screen DES for iTraxx Note, 2 December 2013

Investors can also trade in the unfunded instrument tied to the index. Details of a CDS traded on the iTraxx are shown on the iTraxx Index Term Sheet shown below.

The "fixed rate" refers to the coupon on the index at its inception. There is a fixed coupon set at the start of trading in the index; this is set at "fair value" based on the aggregate value of all the single-name CDS prices for each reference entity in the index (100 names in all). Over the time the traded value of the index fluctuates to reflect changes in the value of underlying names. The current price is calculated by taking the cash flows representing the underlying CDSs and discounting them at the risky rate.

To arbitrage between the index and the underlying names, a trader will put on a position in the index contract and the opposite position in single-name CDSs for each of the constituent names. This would be undertaken when the "traded value" of the index CDS was felt to not reflect the fair value given the current price of each of the underlying names. Given that this would involve administration for 101 CDS contracts, as well as paying the bid-offer spread, the amount of divergence from fair value would need to be sufficiently high to make the trade worthwhile, say 10 bps or more.

iTraxx Index Term Sheet

Counterparty	**ABC Fund LLC**
Trade date	24 February 2004
Effective date	25 February 2004
Schedule termination date	20 March 2009
Floating rate payer (seller)	Counterparty
Fixed rate payer (buyer)	XYZ Securities Ltd
Calculation Agent:	XYZ Securities Ltd
Index	iTraxx.HVOL ®
Traded rate	107.5 bps
Fixed rate	125 bps
XYZ receives (pays)	$256,517
Amount traded	$25,000,000

APPLICATIONS FOR PORTFOLIO MANAGERS

Applications Overview

Credit derivatives have allowed market participants to separate and disaggregate credit risk, and thence to trade this risk in a secondary market. Initially, portfolio managers used them to reduce credit exposure. Subsequently, they have been used in the management of portfolios, to enhance portfolio yields and in the structuring of synthetic collateralised-debt obligations. We summarise portfolio managers' main uses of credit derivatives below.

Enhancing Portfolio Returns

Asset managers can derive premium income by trading credit exposures in the form of derivatives issued with synthetic structured notes. The multitranching aspect of structured products enables specific credit exposures (credit spreads and outright default), and their expectations, to be sold to specific areas of demand. By using structured notes such as credit-linked notes, tied to the assets in the reference pool of the portfolio manager, the trading of credit exposures is crystallised as added yield on the asset manager's fixed-income portfolio. In this way, the portfolio manager has enabled other market participants to gain an exposure to the credit risk of a pool of assets but not to any other aspects of the portfolio, and without the need to hold the assets themselves.

Reducing Credit Exposure

Consider a portfolio manager that holds a large portfolio of bonds issued by a particular sector (say, utilities) and believes that spreads in this sector will widen in the short term. Previously, in order to reduce its credit exposure it would have to sell bonds; however, this may crystallise a mark-to-market loss and may conflict with its long-term investment strategy. An alternative approach would be to enter into a credit default swap, purchasing protection for the short term; if spreads do widen, these swaps will increase in value and may be sold at a profit in the secondary market. Alternatively, the portfolio manager may enter into total-return swaps on the desired credits. It pays the counterparty the total return on the reference assets, in return for Libor. This transfers the credit exposure of the bonds to the counterparty for the term of the swap, in return for the credit exposure of the counterparty.

Consider now the case of a portfolio manager wishing to mitigate credit risk from a growing portfolio (say, one that has just been launched). Figure 14.18 shows an example of an unhedged credit exposure to a hypothetical credit-risky portfolio. It illustrates the manager's expectation of credit risk building up to $250 million as the portfolio is ramped up, and then reducing to a more stable level as the credits become more established. A three-year credit default swap entered into shortly after provides protection on half of the notional exposure, shown as the broken line. The net exposure to credit events has been reduced by a significant margin.

Credit Switches and Zero-Cost Credit Exposure

Protection buyers utilising credit default swaps must pay premium in return for laying off their credit-risk exposure. An alternative approach for an asset manager involves the use of credit switches for specific sectors of the portfolio. In a credit switch, the portfolio manager purchases credit protection on one reference asset or pool of assets, and simultaneously sells protection on another asset or pool of assets.[10] So, for example, the portfolio manager would purchase protection for a particular fund

[10] A pool of assets would be concentrated on one sector, such as utility company bonds.

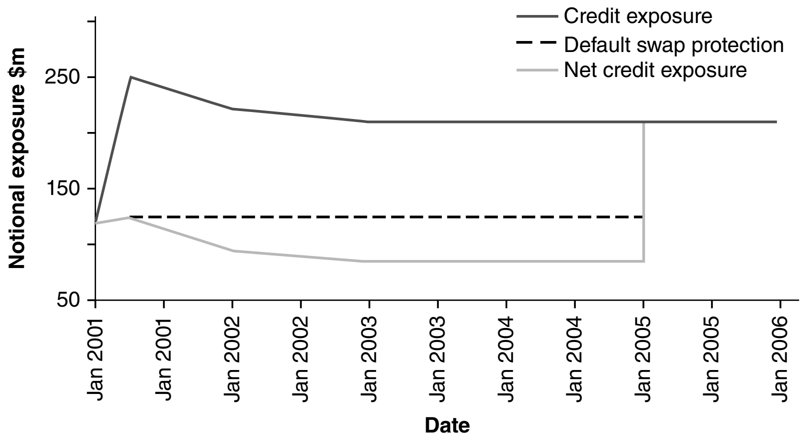

FIGURE 14.18 Reducing Credit Exposure

and sell protection on another. Typically the entire transaction would be undertaken with one investment bank, which would price the structure so that the net cash flows would be zero. This has the effect of synthetically diversifying the credit exposure of the portfolio manager, enabling it to gain and/or reduce exposure to sectors it desires.

Exposure to Market Sectors

Investors can use credit derivatives to gain exposure to sectors for which they do not wish a cash-market exposure. This can be achieved with an index swap, which is similar to a TR swap, with one counterparty paying a total return that is linked to an external reference index. The other party pays a Libor-linked coupon or the total return of another index. Indices that are used might include the government-bond index, a high-yield index, or a technology-stocks index. Assume that an investor believes that the bank loan market will outperform the mortgage-backed bond sector; to reflect this view the investor enters into an index swap in which he pays the total return of the mortgage index and receives the total return of the bank loan index.

Another possibility is synthetic exposure to foreign currency and money markets. Again we assume that an investor has a particular view on an emerging-market currency. If he wishes, he can purchase a short-term (say, one-year) domestic coupon-bearing note, whose principal redemption is linked to a currency factor. This factor is based on the ratio of the spot value of the foreign currency on issue of the note to the value on maturity. Such currency-linked notes can also be structured so that they provide an exposure to sovereign credit risk. The downside of currency-linked notes is that if the exchange rate goes the other way, the note will have a zero return; in effect, a negative return once the investor's funding costs have been taken into account.

Credit Spreads

Credit derivatives can be used to trade credit spreads. Assume that an investor has negative views on a certain emerging-market government-bond credit spread relative to UK gilts. The simplest way to reflect this view would be to go long a credit default swap on the sovereign, paying X basis points. Assuming that the investor's view is correct and the sovereign bonds decrease in price as their credit spread widens, the premium payable on the credit swap will increase. The investor's swap can then be sold into the market at this higher premium.

Capital-Structure Arbitrage

A capital-structure arbitrage describes an arrangement whereby investors exploit mispricing between the yields received on two different loans by the same issuer. Assume that the reference entity has both a commercial bank loan and a subordinated bond issue outstanding, but that the former pays Libor plus 330 basis points while the latter pays Libor plus 315 basis points. An investor enters into a total-return swap in which it effectively is purchasing the bank loan and selling short the bond. The nominal amounts will be at a ratio, for argument's sake let us say 1.25:1, as the bonds will be more price-sensitive to changes in credit status than the loans.

The trade is illustrated in Figure 14.19. The investor receives the "total return" on the bank loan, while simultaneously paying the return on the bond in addition to Libor plus 30 basis points, which is the price of the TR swap. The swap generates a net spread of 67.5 basis points, given by [(330 bps × 1.25) – (315 + 30)]

Synthetic Repo

A portfolio manager believes that a particular bond that it does not hold is about to decline in price. To reflect this view the portfolio manager may do one of the following.

- *Sell the bond in the market and cover the resulting short position in repo:* The cash flow out is the coupon on the bond, with capital gain if the bond falls in price. Assume that the repo rate is floating, say Libor plus a spread. The manager must be aware of the funding costs of the trade, so that unless the bond can be covered in repo at general collateral rates,[11] the funding will be at a loss. The yield on the bond must also be lower than the Libor plus spread received in the repo.

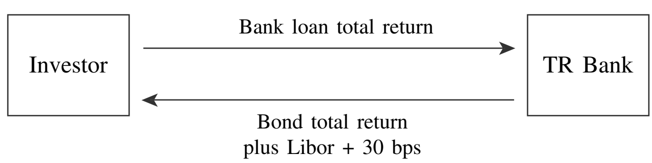

FIGURE 14.19 Total Return Swap in Capital-Structure Arbitrage

[11] That is, the bond cannot be *special*. A bond is special when the repo rate payable on it is significantly (say, 20–30 basis points or more) below the *general collateral* repo rate, so that covering a short position in the bond entails paying a substantial funding premium.

- *As an alternative, enter into a TR swap:* The portfolio manager pays the total return on the bond and receives Libor plus a spread. If the bond yield exceeds the Libor spread, the funding will be negative. However, the trade will gain if the trader's view is proved correct and the bond falls in price by a sufficient amount. If the break-even funding cost (which the bond must exceed as it falls in value) is lower in the TR swap, this method will be used rather than the repo approach. This is more likely if the bond is special.

Total-return swaps are increasingly used as synthetic repo instruments, most commonly by investors that wish to purchase the credit exposure of an asset without purchasing the asset itself. This is conceptually similar to what happened when interest-rate swaps were introduced, which enabled banks and other financial institutions to trade interest-rate risk without borrowing or lending cash funds.

Under a TR swap, an asset such as a bond position may be removed from the balance sheet. In order to avoid adverse impact on regular internal and external capital and credit-exposure reporting a bank may use TR swaps to reduce the amount of lower-quality assets on the balance sheet. This can be done by entering into a short-term TR swap with, say, a two-week term that straddles the reporting date. Bonds are removed from the balance sheet if they are part of a sale-plus-TR-swap transaction. This is because legally the bank selling the asset is not required to repurchase bonds from the swap counterparty, nor is the total-return payer obliged to sell the bonds back to the counterparty (or indeed sell the bonds at all on maturity of the TR swap).

RISKS IN CREDIT DEFAULT SWAPS

Unintended Risks

As credit derivatives can be tailored to specific requirements in terms of reference exposure—term to maturity, currency, and cash flows—they have enabled market participants to establish exposure to specific entities without the need for them to hold the bond or loan of that entity. This has raised issues of the different risk exposure that this entails compared to the cash equivalent. A recent Moody's special report highlights the unintended risks of holding credit exposures in the form of default swaps and credit-linked notes.[12] Under certain circumstances it is possible for credit default swaps to create unintended risk exposure for holders, by exposing them to greater frequency and magnitude of losses compared to that suffered by a holder of the underlying reference credit.

In a credit default swap, the payout to a buyer of protection is determined by the occurrence of credit events. The definition of a credit event sets the level of credit-risk exposure of the protection seller. A wide definition of "credit event" results in a higher level of risk. To reduce the likelihood of disputes, counterparties can adopt the ISDA Credit Derivatives definitions to govern their dealings. The Moody's paper states that the current ISDA definitions do not unequivocally separate and isolate

[12] Jeffrey Tolk, "Understanding the Risks in Credit Default Swaps," *Moody's Investors Service Special Report,* March 16, 2001.

credit risk, and, in certain circumstances, credit derivatives can expose holders to additional risks. A reading of the paper would appear to suggest that differences in definitions can lead to unintended risks being taken on by protection sellers. Two examples from the paper are cited below as illustrations.

Extending Loan Maturity

The bank debt of Conseco, a corporate entity, was restructured in August 2000. The restructuring provisions included deferment of the loan maturity by three months, higher coupon, corporate guarantee, and additional covenants. Under Moody's definition, as lenders received compensation in return for an extension of the debt, the restructuring was not considered to be a "diminished financial obligation", although Conseco's credit rating was downgraded one notch. However, under the ISDA definition, the extension of the loan maturity meant that the restructuring was considered to be a credit event, and thus triggered payments on default swaps written on Conseco's bank debt. Hence, this was an example of a loss event under ISDA definitions that was not considered by Moody's to be a default.

Risks of Synthetic Positions and Cash Positions Compared

Consider two investors in XYZ, one of whom owns bonds issued by XYZ while the other holds a credit-linked note (CLN) referenced to XYZ. Following a deterioration in its debt situation, XYZ violates a number of covenants on its bank loans, but its bonds are unaffected. XYZ's bank accelerates the bank loan, but the bonds continue to trade at 85 cents on the dollar, coupons are paid, and the bond is redeemed in full at maturity. However, the default swap underlying the CLN cites "obligation acceleration" (of either bond or loan) as a credit event, so the holder of the CLN receives 85% of par in cash settlement, and the CLN is terminated. However, the cash investor receives all the coupons and the par value of the bonds on maturity.

These two examples illustrate how, as credit default swaps are defined to pay out in the event of a very broad range of definitions of a "credit event", portfolio managers may suffer losses as a result of occurrences that are not captured by one or more of the ratings agencies' rating of the reference asset. This results in a potentially greater risk for the portfolio manager compared to the position were it to actually hold the underlying reference asset. Essentially, therefore, it is important for the range of definitions of a "credit event" to be fully understood by counterparties, so that holders of default swaps are not taking on greater risk than is intended.

BIG BANG IN CDS

After the credit crisis, the CDS was criticised as the single most dangerous financial innovation that contributed to the crisis. This is a slightly exaggerated view to take. Nonetheless reformers, regulators, and popular journalists liken it to a financial weapon of mass destruction and call for tighter controls of its use. A hidden danger of a CDS deal is in its counterparty risks—not only is a buyer of a CDS exposed to the underlying obligor, it is also exposed to the risk of the seller defaulting on its obligation when a credit event occurs. Before the crisis, this aspect was not well understood because the interbank environment is assumed to be non–credit risky (hence the traditional Libor discounting practice). Chief among the policy recommendations was for the creation of centralised credit counterparties (CCP), or regulated clearing houses. This will move counterparty risks away from individual players to the CCP where the usual safeguards (margins, reserves, etc.) exist.

As a forerunner to the formation of CCPs, a treaty called the Big Bang for CDS took place in April 2009. Under this agreement, all banks are encouraged to migrate to a global standard approach for CDS trading. The proposed new conventions that would support central clearing are:

- Formation of event determination committee—which would decide on the formal trigger of credit and succession events.
- Hardwiring of auction mechanism—an agreed process to decide on a binding and standard cash settlement in the event of a credit event.
- Simplification/streamlining of restructuring clauses.
- Rolling event effective date—every open position has the same effective date regardless of the original trade date.
- Fixed coupons—all CDS contracts will be benchmarked to standardised coupons (100 bps or 500 bps) thereby making it easier to offset contracts in a CCP.
- A standardised accrual is introduced into the first premium period, which, along with standardised coupons, allows efficient offset of contracts in a CCP.

The last two points can best be understood by looking at the example in Figure 14.20.

The trade is a five-year CDS for Portugal Telecom—long CDS protection for amount \$10 million. The quoted five-year spread is 471.42 bps. Since the CDS is standardised

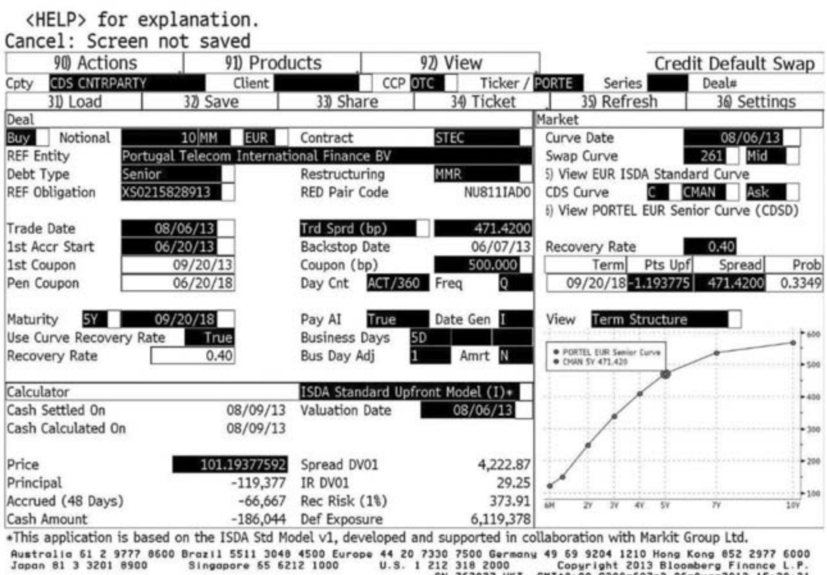

FIGURE 14.20 Bloomberg CDS Pricing Screenshot

with a nearest benchmark coupon of 500 bps, the buyer will pay a quarterly premium of 500 bps; that is, overpay by 28.58 bps over the quoted (fair) spread. The overpayment translates into an above-par price (at 101.19377), hence, the buyer will *receive* an up-front payment of $119,377 as compensation. The trade date (6 August 2013) falls in between IMM quarterly dates (20 June 2013 and 20 September 2013). Hence, there is an accrual for 48 days from 20 June to 6 August. Since a long CDS is equivalent to a short bond, the CDS buyer is expected to *receive* an accrual of $66,667 also paid up front.

CONCLUSION

Credit derivatives are well established instruments in the fixed-income markets, and their flexibility mirrors that of an earlier generation of derivatives such as swaps and options. This chapter has highlighted how they may be used both to hedge credit-risk exposure as well as to enhance portfolio returns. It is clear, though, that users need to be aware of the risks involved in writing credit swaps, such that the legal terms underpinning each contract are clearly defined and communicated.

APPENDIX

APPENDIX 14.1 BOND CREDIT RATINGS

Fitch	Moody's	S&P	Summary Description
Investment Grade			
AAA	Aaa	AAA	Gilt edged, prime, maximum safety, lowest risk, and when sovereign borrower considered "default-free"
AA+	Aa1	AA+	
AA	Aa2	AA	High-grade, high credit quality
AA–	Aa3	AA–	
–A+	Al	A+	
A	A2	A	Upper-medium grade
A–	A3	A-	
BBB+	Baa1	BBB+	
BBB	Baa2	BBB	Lower-medium grade
BBB–	Baa3	BBB–	
–Speculative	*Grade*		
BB+	Ba1	BB+	
BB	Ba2	BB	Low grade; speculative
BB–	Ba3	BB–	
–B+	Bl		

Fitch	Moody's	S&P	Summary Description
B	B	B	Highly speculative
B–	B3		
Predominantly Speculative, Substantial Risk or in Default			
CCC+		CCC+	
CCC	Caa	CCC	Substantial risk, in poor standing
CC	Ca	CC	May be in default, very speculative
C	C	C	Extremely speculative
		CI	Income bonds—no interest being paid
DDD			
DD			Default

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CHAPTER 15

Credit Derivatives II: Pricing, Valuation, and the Basis

INTRODUCTION

In this second chapter on credit derivatives we look at the approaches used in their pricing and valuation. We consider generic techniques and also compare prices obtained using different models. In addition, we highlight the difference in the market between the cash and synthetic spread levels for the same reference name. This is known as the basis and should be observed closely in the market.

The pricing of credit derivatives should aim to provide a "fair value" for the credit-derivative instrument. In the sections below, we discuss the pricing models currently proposed by the industry. The effective use of pricing models requires an understanding of the models' assumptions, the key pricing parameters, and a clear understanding of the respective limitations of the models. Issues to consider when carrying out credit-derivative pricing include:

- The implementation and selection of appropriate modelling techniques
- The estimation of parameters
- The quality and quantity of data to support parameters and calibration
- The calibration to market instruments for risky debt

For credit derivative contracts in which the payout is on credit events other than default, the modelling of the credit evolutionary path is critical. If, however, a credit derivative contract does not pay out on intermediate stages between the current state and default, then the important factor is the probability of default from the current state.

We begin by considering the asset-swap pricing method, which was commonly used at the inception of the credit derivatives market.

ASSET SWAP PRICING

Credit derivatives were originally valued using the asset swap pricing technique. In addition to its use by dealers, risk-management departments who wished to independently price such swaps also adopted this technique. The asset swap market remains a reasonably reliable indicator of the returns required for individual credit exposures, and provides a mark-to-market framework for reference assets as well as a hedging mechanism.

A par-asset swap typically combines the sale of an asset such as a fixed-rate corporate bond to a counterparty, at par and with no interest accrued, with an interest-rate swap. The coupon on the bond is paid in return for Libor, plus a spread if necessary. This spread is the asset-swap spread and is the price of the asset swap. In effect, the asset swap allows market participants that pay Libor-based funding to receive the asset-swap spread. This spread is a function of the credit risk of the underlying bond asset, which is why it, in effect, becomes the cornerstone of the price payable on a credit-default swap written on that reference asset.

The generic pricing is given by (15.1).

$$Y_a = Y_b - ir \tag{15.1}$$

where Y_a is the asset-swap spread
Y_b is the asset spread over the benchmark
ir is the interest-rate swap spread

The asset spread over the benchmark is simply the bond (asset) redemption yield over that of the government benchmark. The interest-rate swap spread reflects the cost involved in converting fixed-coupon benchmark bonds into a floating-rate coupon during the life of the asset (or default swap), and is based on the swap rate for that maturity.

Note that formula (15.1) is only an approximate pricing (which is often not good enough). The reason is the asset's coupon dates may not be identical to the asset swap cash flow dates. And the asset is typically not priced at par, while the asset swap is. Hence, a proper pricing requires cash flow discounting using the risky curve.

Asset Swap Pricing: An Example

XYZ plc is a Baa2-rated corporate. The seven-year asset swap for this entity is currently trading at 93 basis points; the underlying seven-year bond is hedged by an interest-rate swap with an Aa2-rated bank. The risk-free rate for floating-rate bonds is Libid minus 12.5 basis points (assume the bid-offer spread is six basis points). This suggests that the credit spread for XYZ plc is 111.5 basis points. The credit spread is the return required by an investor for holding the credit of XYZ plc. The protection seller is conceptually long the asset, and so would short the asset as a hedge of its position. This is illustrated in Figure 15.1. The price charged for the default swap is the price of shorting the asset, which works out as 111.5 basis points each year.

Therefore, we can price a credit default written on XYZ plc as the present value of 111.5 basis points for seven years, discounted at the interest-rate swap rate of 5.875%. This computes to a credit-swap price of 6.25%.

Reference	XYZ plc
Term	Seven years
Interest-rate swap rate	5.875%
Asset swap	Libor plus 93 bps

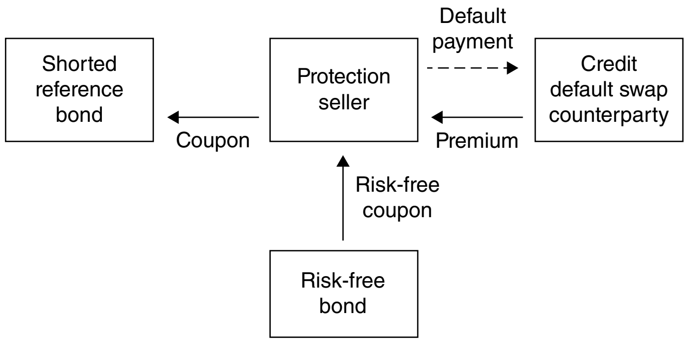

FIGURE 15.1 Credit Default Swap and Asset-Swap Hedge

Credit default swap pricing:

Benchmark rate	Libid minus 12.5 bps
Margin	6 bps
Credit-default swap	111.5 bps
Default-swap price	6.252%

There are a number of reasons why this approach is no longer applied except perhaps by risk managers or middle-office staff as an independent check.[1] These reflect the respective natures of asset swaps and credit-default swaps as market instruments.

We now consider a number of pricing models as used in the credit derivative markets.

PRICING MODELS

Pricing models for credit derivatives fall into two classes:

1. Structural models
2. Reduced form models

We discuss these models next.

Structural Models

Structural models are characterised by modelling the firm's value in order to provide the probability of the firm's default. The Black-Scholes-Merton option-pricing framework is the foundation of the structural-model approach. The default event is assumed to occur when the firm's assets fall below the book value of the debt.

[1] See M. Choudhry, "Some Issues in the Asset Swap Pricing of Credit Default Swaps," *Derivatives Week*, Euromoney Publications, 2 December 2001.

Merton applied option-pricing techniques to the valuation of corporate debt. By extension, the pricing of credit derivatives based on corporate debt may in some circumstances be treated as an option on debt (which is therefore analogous to an option on an option model).

Merton models have the following features:

- Default events occur predictably when a firm has insufficient assets to pay its debt.
- A firm's assets evolve randomly. The probability of a default is determined using the Black-Scholes-Merton option-pricing theory.

Some practitioners argue that Merton models are more appropriate than reduced-form models when pricing credit default swaps on high-yield bonds, due to the higher correlation of high-yield bonds with the underlying equity of the issuer firm.

The constraint of structural models is that the behaviour of the value of assets and the parameters used to describe the process for the value of the firm's assets are not directly observable and the method does not consider the underlying market information for credit instruments.

Reduced-Form Models

Reduced-form models are a form of no-arbitrage model. These models can be fitted to the current term structure of risky bonds to generate no-arbitrage prices. In this way, the pricing of credit derivatives using these models will be consistent with the market data on the credit-risky bonds traded in the market. These models allow the default process to be separated from the asset value and are more commonly used to price credit derivatives.

Some key features of reduced-form models include:

- Complete and arbitrage-free credit-market conditions are assumed.
- Recovery rate is an input into the pricing model.
- Credit-spread data is used to estimate the risk-neutral probabilities.
- The use of transition probabilities from credit agencies can be accommodated in some of these models. The formation of the risk-neutral transition matrix from the historical transition matrix is a key step.
- Default can take place randomly over time, and the default probability can be determined using the risk-neutral transition matrix.

When implementing reduced-form models, it is necessary to consider issues such as the illiquidity of underlying credit-risky assets. Liquidity is often assumed to be present when we develop pricing models. However, in practice, there may be problems when calibrating a model to illiquid positions, and in such cases the resulting pricing framework may be unstable and provide the user with spurious results. Another issue is the relevance of using historical credit-transition data, which are used to project future credit-migration probabilities. In practice, it is worthwhile reviewing the sensitivity of price to the historical credit-transition data when using the model.

Recent models that provide a detailed modelling of default risk include those presented by Jarrow, Lando, and Turnbull (1997); Das and Tufano (1996); and Duffie and Singleton (1995).We consider these models in this section.

The Jarrow, Lando, and Turnbull (JLT) Model This model focuses on modelling default and credit migration. Its data and assumptions include the use of

- A statistical rating transition matrix, which is based on historic data.
- Risky bond prices from the market used in the calibration process.
- An assumption of a constant recovery rate. The recovery amount is assumed to be received at the maturity of the bond.
- A credit-spread assumption for each rating level.

It also assumes no correlation between interest rates and credit-rating migration.

The statistical transition matrix is adjusted by calibrating the expected risky-bond values to the market values for risky bonds. The adjusted matrix is referred to as the "risk-neutral transition matrix". This matrix is key to the pricing of several credit derivatives.

The JLT model allows the pricing of default swaps, as the risk-neutral transition matrix can be used to determine the probability of default. The JLT model is sensitive to the level of the recovery-rate assumption and the statistical rating matrix. It has a number of advantages: as the model is based on credit migration, it allows the pricing of derivatives for which the payout depends on such credit migration. In addition, the default probability can be explicitly determined and may be used in the pricing of credit-default swaps.

The disadvantages of the model include the fact that it depends on the selected historical-transition matrix. The applicability of this matrix to future periods needs to be considered carefully: whether, for example, it adequately describes future credit-migration patterns. In addition, it assumes all securities with the same credit rating have the same spread, which is restrictive. For this reason, the spread levels chosen in the model are a key assumption in the pricing model. Finally, the constant-recovery rate is another practical constraint as, in practice, the level of recovery will vary.

The Das-Tufano Model The Das-Tufano (DT) model is an extension of the JLT model. The model aims to produce the risk-neutral transition matrix in a similar way to the JLT model; however, this model uses stochastic recovery rates. The final risk-neutral transition matrix should be computed from the observable term structures. The stochastic recovery rates introduce more variability in the spread volatility. Spreads are a function of factors that may not only be dependent on the rating level of the credit, as in practice credit spreads may change even though credit ratings have not changed. Therefore, to some extent, the DT model introduces this additional variability into the risk-neutral transition matrix.

Various credit derivatives may be priced using this model; for example, credit-default swaps, total-return swaps, and credit-spread options. The pricing of these products requires the generation of the appropriate credit-dependent cash flows at each node on a lattice of possible outcomes. The fair value may be determined by discounting the probability-weighted cash flows. The probability of the outcomes would be determined by reference to the risk-neutral transition matrix.

The Duffie-Singleton Model The Duffie-Singleton modelling approach considers the three components of risk for a credit-risky product; namely, the risk-free rate, the hazard rate, and the recovery rate. It is the basis for most credit default swap pricing models used by banks in the market.

The hazard rate characterises the instantaneous probability of default of the credit-risky underlying exposure. As each of the components may not be static over time, a pricing model may assume a process for each of these components of risk. The process may be implemented using a lattice approach for each component. The constraint on the lattice formation is that this lattice framework should agree to the market pricing of credit-risky debt.

Here we demonstrate that the credit spread is related to risk of default (as represented by the hazard rate) and the level of recovery of the bond. We assume a zero-coupon risky bond maturing in a small time element Δt where:

λ is the annualised hazard rate

φ is the recovery value

r is the risk-free rate

s is the credit spread

and where its price P is given by

$$P = e^{-\Delta t}((1 - \lambda\Delta t) + (\lambda\Delta t)\varphi) \tag{15.2}$$

Alternatively, P may be expressed as:

$$P \cong e^{-\Delta t(r+\lambda(1-\varphi))} \tag{15.3}$$

However, as the usual form for a risky zero-coupon bond is

$$P = e^{-\Delta t(r+s)} \tag{15.4}$$

Therefore, we have shown that:

$$s \cong \lambda(1 - \varphi) \tag{15.5}$$

This would imply that the credit spread is closely related to the hazard rate (that is, the likelihood of default) and the recovery rate.

This relationship between the credit spread, the hazard rate, and the recovery rate is intuitively appealing. The credit spread is perceived to be the extra yield (or return) the investor requires for credit risk assumed. For example:

- As the hazard rate (or instantaneous probability of default) rises, then the credit spread increases; and
- As the recovery rate decreases, the credit spread increases.

A "hazard-rate" function may be determined from the term structure of credit. The hazard-rate function has its foundation in statistics and may be linked to the instantaneous default probability.

The hazard-rate function ($\lambda(s)$) can then be used to derive a probability function for the survival function $S(t)$:

$$S(t) = \exp^{-\int_0^t \lambda(s)ds} \tag{15.6}$$

The hazard-rate function may be determined by using the prices of risky bonds. The lattice for the evolution of the hazard rate should be consistent with the hazard-rate function implied from market data. An issue when performing this calibration is the volume of relevant data available for the credit.

Recovery Rates The recovery rate usually takes the form of the percentage of the par value of the security recovered by the investor.

The key elements of the recovery rate include:

- The level of the recovery rate
- The uncertainty of the recovery rate based on current conditions specific to the reference credit
- The time interval between default and the recovery value being realised

Generally, recovery rates are related to the seniority of the debt. Therefore, if the seniority of debt changes, then the recovery value of the debt may change. Also, recovery rates exhibit significant volatility.

The recovery rate is an input rather than an output of the pricing model. Although the recovery rate can be estimated econometrically using past observations of recovery, most banks just assume a recovery rate depending on the seniority and rating of an issuer. A typical recovery rate input for a corporate bond is in the range of 25–40%. But the actual recovery rate can vary from single digits to something in the 90% range. As an example, the actual recovery of Lehman Brothers, which defaulted during the credit crisis, was approximately 8%. So in essence one of the primary model inputs is an assumption that may, in practice, prove to be materially inaccurate.

Credit-Spread Modelling

Although spreads may be viewed as a function of default risk and recovery risk, spread models do not attempt to break down the spread into its default-risk and recovery-risk components.

The pricing of credit derivatives that pay out according to the level of the credit spread would require that the credit-spread process be adequately modelled. In order to achieve this, a stochastic process for the distribution of outcomes for the credit spread is an important consideration.

An example of the stochastic process for modelling credit spreads, which may be assumed, includes a mean-reverting process such as:

$$ds = k(\mu - s)dt + \sigma s dw \tag{15.7}$$

where ds is the change in the value of the spread over an element of time (dt)
dt is the element of time over which the change in spread is modelled
s is the credit spread
k is the rate of mean reversion
m is the mean level of the spread
dw is the Wiener increment (sometimes written dW, dZ, or dz)
σ is the volatility of the credit spread

In this model, when s rises above a mean level of the spread, the drift term ($\mu - s$) will become negative, and the spread process will drift towards (revert to) the mean level. The rate of this drift towards the mean is dependent on k the rate of mean reversion.

The pricing of a European spread-option requires the distribution of the credit spread at the maturity (T) of the option. The choice of model affects the probability assigned to each outcome. The mean-reversion factor reflects the historic economic features over time of credit spreads to revert to the average spreads after larger-than-expected movements away from the average spread.

Therefore, the European option price may be reflected as:

$$\text{Option price} = E[e^{-rT}(\text{Payoff}(s, X)] = e^{-rT}\int_0^{\infty} f(s, X)p(s)ds \tag{15.8}$$

where X is the strike price of the spread option
$p(s)$ is the probability function of the credit spread
$E[]$ denotes the expected value
$f(s,X)$ is the payoff function at maturity of the credit spread

More complex models for the credit-spread process may take into account factors such as the term structure of credit and possible correlation between the spread process and the interest process.

The pricing of a spread option is dependent on the underlying process. As an example, we compare the pricing results for a spread-option model including mean reversion to the pricing results from a standard Black Scholes model in Table 15.1.

Tables 15.1 and 15.2 show the sensitivity on the pricing of a spread option to changes to the underlying process. Comparing the two tables shows the impact of time to expiry increasing by six months. In a mean-reversion model, the mean level and the rate of mean reversion are important parameters that may significantly affect the probability distribution of outcomes for the credit spread and, hence, the price.

Credit Default Swaps

The pricing of a credit default swap that has a payout on an underlying risky bond (issued by company ABC plc) involves the following key factors when pricing:

- Risk-free interest-rate term structure.
- Risky term structure. Ideally we would determine this by considering the term of the bonds issued by company ABC. However, if a wide term is not available,

TABLE 15.1 Model Results and Difference, 6-Month Expiry

Expiry in 6 Months **Risk free rate = 10%** **Strike = 70 bps** **Credit spread = 60 bps** **Volatility = 20%**	**Mean-Reversion Model Price**	**Standard Black-Scholes Price**	**% Difference between Standard Black-Scholes and Mean-Reversion Model Price**
Mean level = 50 bps			
K = .2			
Put	0.4696	0.5524	17.63%
Call	10.9355	9.7663	11.97%
Mean level = 50 bps			
K = .3			
Put	0.3510	0.5524	57.79%
Call	11.2031	9.7663	14.12%
Mean level = 80 bps			
K = .2			
Put	0.8729	0.5524	58.02%
Call	8.4907	9.7663	15.02%
Mean level = 80 bps			
K = .3			
Put	0.8887	0.5524	60.87%
Call	7.5411	9.7663	29.51%

TABLE 15.2 Model Results and Differences, 12-Month Expiry

Expiry in 12 Months **Risk free rate = 10%** **Strike = 70 bps** **Credit spread = 60 bps** **Volatility = 20%**	**Mean-Reversion Model Price**	**Standard Black-Scholes Price**	**% Difference between Standard Black-Scholes and Mean-Reversion Model Price**
Mean level = 50 bps			
K = .2			
Put	0.8501	1.4331	68.58%
Call	11.2952	10.4040	8.56%
Mean level = 50 bps			
K = .3			
Put	0.7624	1.4331	87.97%
Call	12.0504	10.4040	15.82%
Mean level = 80 bps			
K = .2			
Put	1.9876	1.4331	38.69%
Call	7.6776	10.4040	35.51%
Mean level = 80 bps			
K = .3			
Put	2.4198	1.4331	68.85%
Call	6.7290	10.4040	54.61%

then the bonds of similar credit-risky companies may be used to create a more complete term structure of credit.
- The recovery rate.

For the risk-free interest-rate term structure, the observable money-market and swap curve may be best choice. However, a risk-free interest-rate term structure may also be built from government-bond prices.

The risky term structure and the recovery rate can be used to estimate the risk-neutral probability of default.

$$rs_{Risky} = rs_{Riskfree} * [((1-p)*1)+(p*R)] \tag{15.9}$$

where rs_{Risky} is the risky zero-coupon rate (from the risky term structure)
$rs_{Riskfree}$ is the risk-free zero-coupon rate (from the risk-free term structure)
p is the risk-neutral default probability
R is the recovery rate

The unknown p can be implied out of equation (15.9):

$$p = [1/(1-R)]*[1-(rs_{risky}/rs_{riskfree})] \tag{15.10}$$

This equation shows that the probability of default is related to the risky term structure, risk-free term structure, and the recovery rate. In order to determine p, we will make assumptions for a suitable recovery value to be used in the above equation.

The expected value of a single-period credit swap may take the form of the expected payout:

$$CDS_1 = rs_{Riskfree} * [(0*(1-p))+(p*(1-R)] \tag{15.11}$$

This reflects the fact that there is no payout on survival of the credit; that is, in the event of no default.

Key issues in the pricing of credit-default swaps include:

- Determining an appropriate assumption for the recovery rate.
- Selecting an appropriate risky debt required for calibration and the necessary adjustments to allow for liquidity and embedded options of the risky debt.
- Selecting an appropriate risk-free curve.
- Allowing for the credit risk of the counterparty and correlation of the underlying credit with the counterparty. We would expect that the cost of credit protection is cheaper if it is purchased from a "high" risk counterparty.

The level of sensitivity of the pricing of credit default swaps (for example, a one-period credit default swap pricing analysis) is examined in Table 15.3. The table compares the pricing of a default swap based on the inputs in the table. The discussion that follows compares the pricing outputs and attempts to identify sensitivities of the pricing to changes in the inputs.

TABLE 15.3 Credit-Default Swap Pricing Variations

	A	B	C	D
rs_{Risky}	0.95	0.949	0.95	0.95
$rs_{Riskfree}$	0.97	0.97	0.969	0.97
R	0.5	0.5	0.5	0.3
p	0.041237	0.043299	0.039216	0.029455
Credit default swap (CDS) price	0.02	0.021	0.019	0.02

The pricing analysis of the data in Table 15.3 is as follows:

(i) Credit default swap price column A/column B = 0.02/0.021 = 0.952381. As a result of changing the risky zero-coupon price by approximately 0.1% (compare column A inputs to column B inputs), we have had a CDS price change of approximately 5%. Clearly, this shows the sensitivity of the CDS price to a change in the risky zero-coupon price. This emphasises the risk in the estimates of the zero-coupon risky price and the impact of possible adjustments relating to illiquidity and embedded options.

(ii) Credit default swap price column A/column C = 0.02/0.019 = 1.052632. As a result of changing the risk-free zero-coupon price by approximately 0.1% (compare column A to column C), we have had a CDS price change of approximately 5%. This shows the sensitivity of the CDS price to a change in the risk-free zero-coupon price. However, given the technology and existing research into yield curves, this is perhaps a smaller risk than (i).

(iii) Credit default swap price column A/column D = 0.02/0.02 = 1. The CDS pricing is relatively insensitive to the change in recovery price (column D's recovery is 0.3). However, we note that the risk-neutral default probability (p) is sensitive to the change in the recovery-price assumption. This shows that there is a compensating effect in the pricing of a CDS between the probability of default and the level of recovery. However, care should be taken if the default payoff is fixed (for example, a digital swap) as this compensating effect is removed.

Comparing Cash and Synthetic Markets

The rapid growth of the credit default swap has resulted in a liquid market in credit default swaps across the credit curve. Figure 15.2 shows the growth in credit-default swap volumes from before to after the global bank crash.

The liquidity of the credit derivative market frequently exceeds that available for the same reference market in the cash market. It is this feature that has enabled fund managers to exploit their expertise in credit trading by originating synthetic CDO vehicles that enable them to arbitrage between cash and synthetic markets. These structures are considered in the next chapter. As well as greater liquidity, the synthetic market also offers other potential advantages to investors who would generally consider only the cash markets. These advantages are summarised in Table 15.4.

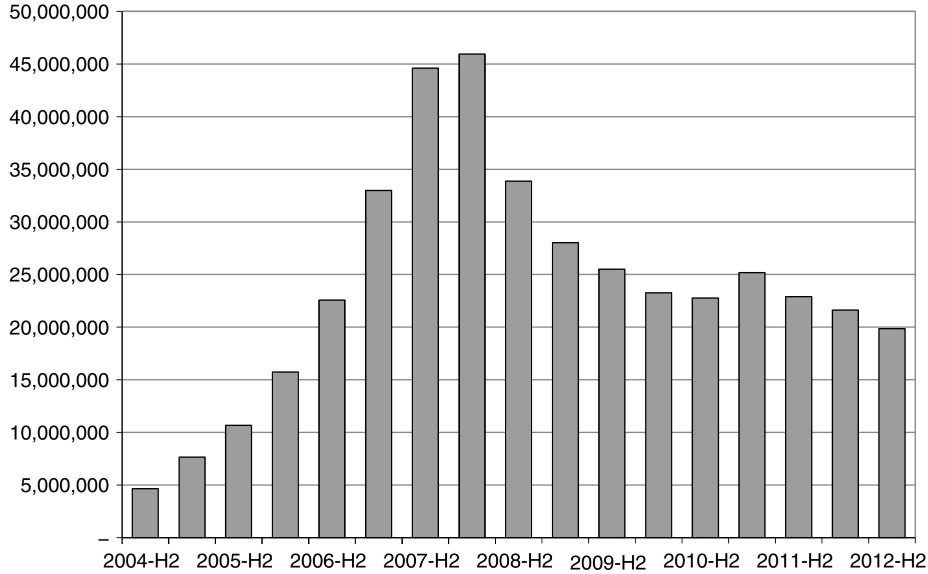

FIGURE 15.2 Credit-Default Swap Gross Notional Outstanding
Source: BIS.

The difference in pricing between the cash and synthetic markets has already been noted. Because credit derivatives isolate and trade credit as their sole asset, they will be priced at a different level from the asset swap on the same reference asset. This difference is known as the *basis*. The basis is:

Credit-default spread – The asset-swap spread

A positive basis occurs when the credit derivative trades higher than the asset-swap price, and is common. A negative basis describes when the credit derivative trades tighter than the cash-bond asset swap spread.

TABLE 15.4 Comparing Cash and Derivatives Markets for Investors

	Cash Bonds	Credit Derivatives
Corporate (issuer) names available	Existing issuers only	Any name required
Liquidity	Variable liquidity	No limit to size of trade
Bid-offer spread	Greater below AAA names	Smaller
Maturity	Fixed dates	Any date required
Principal guaranteed	Rare	Available if required
Coupon	Typically fixed	Fixed or floating
Yield	Lower	Higher

The factors leading to a positive basis include:

- The delivery option afforded to the protection buyer, when a basket of deliverable obligations is specified in physical settlement, which enables the delivery of the cheapest-to-deliver asset names.
- Liquidity resulting in credit default swaps being the most common means by which a reference name is shorted in the market.
- A quoted spread that is always above Libor, despite the fact that AAA and some AA names are funded in the asset-swap market at sub-Libor.
- The inability for credit default swap spreads to be negative.

Figure 15.3 shows the performance of the basis for the bank Credit Agricole across the term structure, in 2007 and again in 2010. Note the altered relationship post-crisis of 2008. For more detail on the CDS basis please refer to the author's book *Structured Credit Products* (2010).

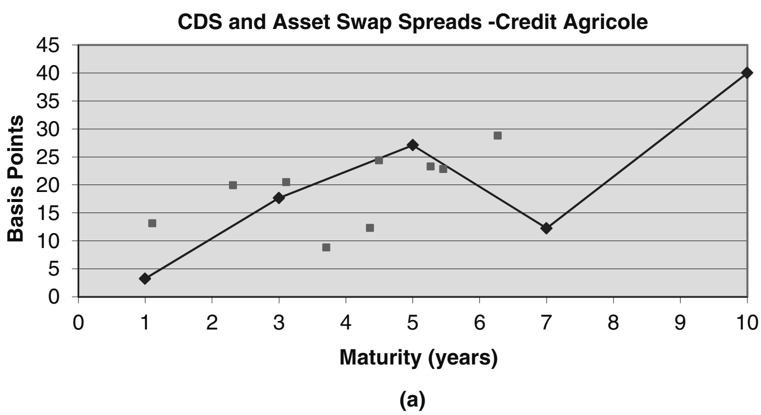

FIGURE 15.3(i) Credit Agricole CDS Basis, August 2007

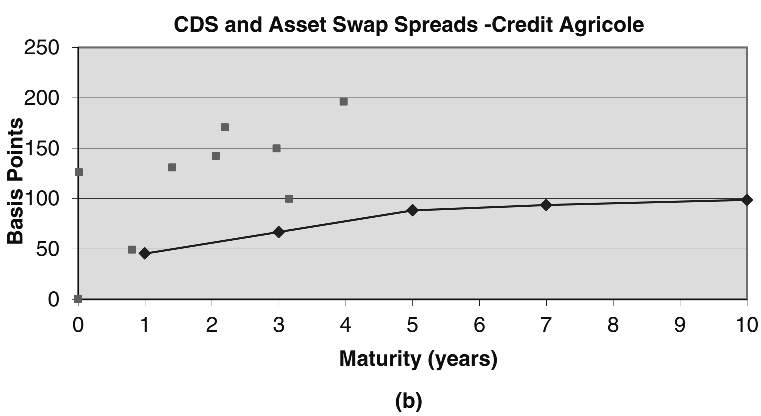

FIGURE 15.3(ii) Credit Agricole CDS Basis, November 2010

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CHAPTER 16

Options I

As a risk-management tool, options allow banks and corporates to hedge market exposure but also to gain from upside moves in the market. This makes them unique among hedging instruments. Options have special characteristics that make them stand apart from other classes of derivatives. As they confer a right to conduct a certain transaction, but without imposing an obligation to do so, their payoff profile is different from other financial assets, both cash and off-balance-sheet. This makes an option more of an insurance policy than a pure hedging instrument, as the person who has purchased the option for hedging purposes need only exercise it if required. The price of the option is, in effect, the insurance premium that has been paid for peace of mind. Options are also used for purposes other than hedging. As part of speculative and arbitrage trading, and option market makers generate returns from managing the risk on their option books profitably. The range of combinations of options that can be dealt today, and the complex-structured products of which they form a part, is constrained only by imagination and customer requirements. Virtually all participants in capital markets will have some requirement that may be met by the use of options.

This chapter is a large one as we first introduce the basics of options and then consider the complex topic of option pricing. A subsequent chapter looks at the main sensitivity measures used in running an option book, and the uses to which options may be put.

BACKGROUND

An option is a contract in which the buyer has the right, but not the obligation, to buy or sell an underlying asset at a predetermined price during a specified period of time. The seller of the option, known as the writer, grants this right to the buyer in return for receiving the price of the option, known as the premium. An option that grants the right to buy an asset is a call option, while the corresponding right to sell an asset is a put option. The option buyer has a long position in the option and the option seller has a short position in the option.

Before looking at the other terms that define an option contract, we'll discuss the main feature that differentiates an option from all other derivative instruments, and from cash assets. Because options confer on a buyer the right to effect a transaction, but not the obligation (and correspondingly on a seller the obligation, if requested by the buyer, to effect a transaction), their risk/reward characteristics are different from other financial products. The payoff profile from holding an option is nonlinear,

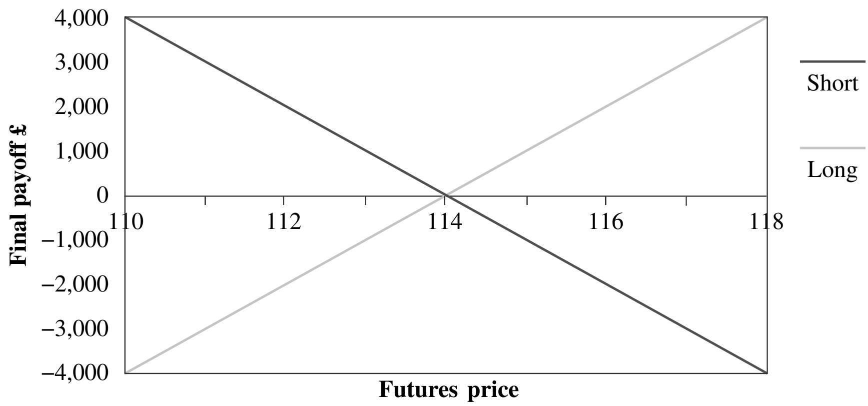

FIGURE 16.1 Payoff Profile for a Bond Futures Contract

unlike the profile of any other financial instrument. Let us consider the payoff profiles for a vanilla call option and a gilt futures contract. Suppose that a trader buys one lot of the gilt futures contract at 114.00 and holds it for one month before selling it. On closing the position, the profit made will depend on the contract sale price. If it is above 114.00, the trader will have made a profit, and if below 114.00 she will have made a loss. On one lot, this represents a £1,000 gain for each point above 114.00. The same applies to someone who had a short position in the contract and closed it out—if the contract is bought back at any price below 114.00 the trader will realise a profit. The profile is shown in Figure 16.1.

This profile is the same for other derivative instruments such as forward-rate agreements (FRAs) and swaps, and, of course, for cash instruments such as bonds or equity. The payoff profile therefore has a linear characteristic, and it is linear whether one has bought or sold the contract.

The profile for an option contract differs from the conventional one. Because options confer a right but not an obligation to one party (the buyer), and an obligation but not a right to the seller, the profile will differ according to whether one is the buyer or seller. Supposing now that our trader buys a call option that grants the right to buy a gilt futures contract at a price of 114.00 at some point during the life of the option, her resulting payoff profile will be like that shown in Figure 16.2. If during the life of the option the price of the futures contract rises above 114.00, the trader will exercise her right to buy the future, under the terms of the option contract. This is known as exercising the option. If, on the other hand, the price of the future falls below 114.00, the trader will not exercise the option and, unless there is a reversal in price of the future, it will eventually expire worthless, on its maturity date. In this respect, it is exactly like an equity or bond warrant. The seller of this particular option has a very different payout profile. If the price of the future rises above 114.00 and the option is exercised, the seller will bear the loss equal to the profit that the buyer is now benefiting from. The seller's payoff profile is also shown in Figure 16.2, as the dashed line. If the option is not exercised and expires, the trade will have generated premium income for the seller, which is revenue income that contributes to the P/L account.

The holders of long and short positions in options do not have the same symmetrical payoff profile. The buyer of the call option will benefit if the price of the

FIGURE 16.2 Payoff Profile for Call-Option Contract

underlying asset rises, but will not lose if the price falls (except the funds paid for purchasing the rights under the option). The seller of the call option will suffer loss if the price of the underlying asset rises, but will not benefit if it falls (except realising the funds received for writing the option). The buyer has a right but not an obligation, while the seller has an obligation if the option is exercised. The premium charged for the option is the seller's compensation for granting such a right to the buyer.

Originally, options were written on commodities such as wheat and sugar. Nowadays, these are referred to as options on physicals, while options on financial assets are known as financial options. Today, one can buy or sell an option on a wide range of underlying instruments, including financial products such as foreign exchange, bonds, equities, and commodities, and derivatives such as futures, swaps, equity indices, and other options.

Option Terminology

Let us now consider the basic terminology used in the options markets.

A call option grants the buyer the right to buy the underlying asset, while a put option grants the buyer the right to sell the underlying asset. There are, therefore, four possible positions that an option trader may put on: long a call or put, and short a call or put. The payoff profiles for each type are shown in Figure 16.3.

The strike price describes the price at which an option is exercised. For example, a call option to buy ordinary shares of a listed company might have a strike price of £10.00. This means that if the option is exercised, the buyer will pay £10 per share. Options are generally either American- or European-style, which defines the times during the option's life when it can be exercised. There is no geographic relevance to these terms, as both styles trade can be traded in any market. There is also another type, Bermudan-style options, which can be exercised at pre-set dates.[1]

[1] I'm told because Bermuda is midway between Europe and America. A colleague once informed me that it's "Asian" for average-rate options because these originated in Japanese commodity markets.

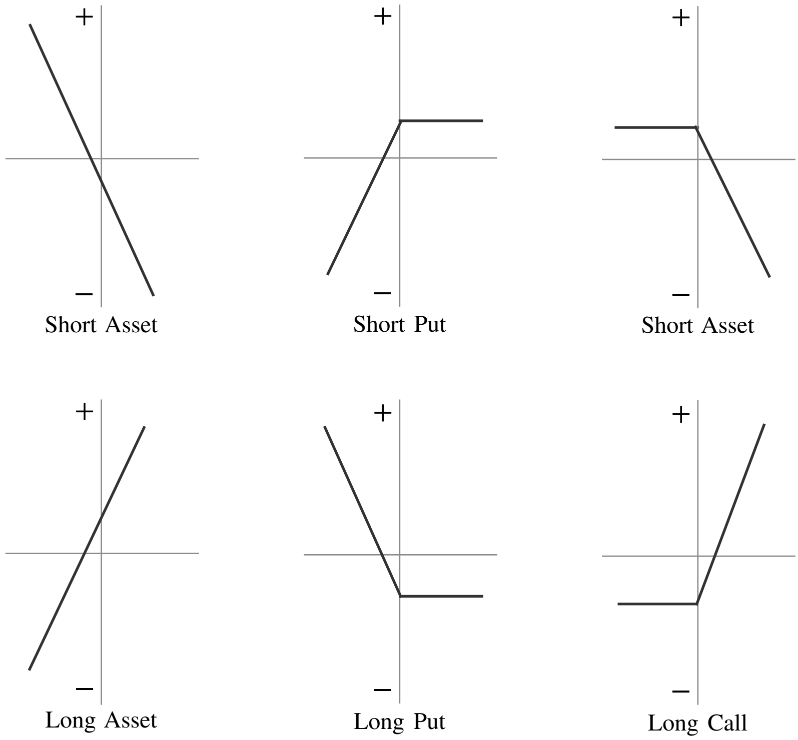

FIGURE 16.3 Basic Option Payoff Profiles

For reasons that we shall discuss later, it rare for an American option to be exercised ahead of its expiry date, so this distinction has little impact in practice, although of course the pricing model being used to value European options must be modified to handle American options. The holder of a European option cannot exercise it prior to expiry. If she wishes to realise its value, however, she will sell it in the market.

The premium of an option is the price at which the option is sold. Option premium is made up of two constituents, intrinsic value and time value.

The intrinsic value of an option is the value of the option if it is exercised immediately, and it represents the difference between the strike price and the current underlying asset price. If a call option on a bond futures contract has a strike price of 100.00 and the future is currently trading at 105.00, the intrinsic value of the option is 5.00, as this would be the immediate profit gain to the option holder if it were exercised. Since an option will only be exercised if there is benefit to the holder from so doing, its intrinsic value will never be less than zero. So, in our example, if the bond future was trading at 95.00, the intrinsic value of the call option would be zero, not –5.00. For a put option, the intrinsic value is the amount by which the current underlying price is below the strike price. When an option has intrinsic value, it is described as being *in-the-money.* When the strike price for a call option is higher than the underlying price (or for a put option is lower than the underlying price) and has no intrinsic value it is said to be *out-of-the-money*. An option for which the strike price is equal to the current underlying price is said to be *at-the-money*. This term is normally used at the time the option is first traded, in cases where the strike price is set to the current price of the underlying asset.

The time value of an option is the amount by which the option value exceeds the intrinsic value. An option writer will almost always demand a premium that is higher than the option's intrinsic value, because of the risk that the writer is taking on. This reflects the fact that over time the price of the underlying asset may change sufficiently to produce a much higher intrinsic value. During the life of an option, the option writer has nothing more to gain over the initial premium at which the option was sold. However, until expiry there is a chance that the writer will lose if the markets move against her, hence the inclusion of a time-value element. The value of an option that is out-of-the-money is composed entirely of time value.

Table 16.1 summarises the main option terminology that we have just been discussing.

TABLE 16.1 Basic Option Terminology

Call	The right to buy the underlying asset
Put	The right to sell the underlying asset
Buyer	The person who has purchased the option and has the right to exercise it if she wishes
Writer	The person who has sold the option and has the obligation to perform if the option is exercised
Strike price	The price at which the option may be exercised, also known as the *exercise price*
Expiry date	The last date on which the option can be exercised, also known as the maturity date
American	The style of option; an American option can be exercised at any time up to the expiry date
European	An option which may be exercised on the maturity date only, and not before
Premium	The price of the option, paid by the buyer to the seller
Intrinsic value	The value of the option if it was exercised today, which is the difference between the strike price and the underlying asset price
Time value	The difference between the current price of the option and its intrinsic value
In-the-money	The term for an option that has intrinsic value
At-the-money	An option for which the strike price is identical to the underlying asset price
Out-of-the-money	An option that has no intrinsic value

OPTION INSTRUMENTS

Options are traded both on recognised exchanges and in the over-the-counter (OTC) market. The primary difference between the two types is that exchange-traded options are standardised contracts and essentially plain-vanilla instruments, while OTC options can take on virtually any shape or form. Options traded on an exchange are often options on a futures contract; so, for example, gilt option on London International Financial Futures Exchange (LIFFE) in London is written on the exchange's gilt futures contract. The exercise of a futures options will result in a long position in a futures contract being assigned to the party that is long the option, and a short position in the future to the party that is short the option. Note that exchange-traded options on U.S. Treasuries are quoted in option ticks that are half the bond tick; that is, 1/64th rather than 1/32nd.

Like OTC options, those traded on an exchange can be either American- or European-style. For example, on the Philadelphia Currency Options Exchange both versions are available, although on LIFFE most options are American-style. Exchange-traded options are available on the following:

- *Ordinary shares:* These are traded on major exchanges including the New York Stock Exchange, LIFFE, Eurex, the Chicago Board Options Exchange (CBOE), and SIMEX (in Singapore trade options on corporation ordinary shares).
- *Options on futures:* Most exchanges trade an option contract written on the futures that are traded on the exchange, which expires one or two days before the futures contract itself expires. In certain cases such as those traded on the Philadelphia exchange, cash settlement is available so that if, for example, the holder of a call exercises, she will be assigned a long position in the future as well as the cash value of the difference between the strike price and the futures price. One of the most heavily traded exchange-traded options contracts is the Treasury bond option, written on the futures contract, traded on the Chicago Board of Trade options exchange.
- *Stock index options:* These are equity-market instruments that are popular for speculating and hedging; for example, the FTSE-100 option on LIFFE and the S&P 500 on CBOE. Settlement is in cash and not the shares that constitute the underlying index, much like the settlement of an index futures contract.
- *Bond options:* Options on bonds are invariably written on the bond futures contract; for example, the aforementioned Treasury bond option or LIFFE's gilt option. Options written on the cash bond must be traded in the OTC market.
- *Interest-rate options:* These are also options on futures, as they are written on the exchange's 90-day interest-rate futures contract.
- *Foreign-currency options:* These are rarer among exchange-traded options, and the major exchange is in Philadelphia. Its sterling option contract, for example, is for an underlying amount of £31,250.

Option trading on an exchange is similar to that for futures and involves transfer of margin on a daily basis. Individual exchanges have their own procedures; for example, on LIFFE the option premium is effectively paid via the variation margin. The amount of variation margin paid or received on a daily basis for each position reflects the change in the price of the option. So if, for example, an option were to expire on maturity with no intrinsic value, the variation-margin payments made during its life would be equal to the change in value from the day it was traded to zero. The option trader does not pay a separate premium on the day the position is put on. On certain other exchanges, though, it is the other way around, and the option buyer will pay a premium on the day of purchase but then pay no variation margin. Some exchanges allow traders to select either method. Margin is compulsory for a party that writes options on the exchange.

The other option market is the OTC market, where there is a great variety of instruments traded. As with products such as swaps, the significant advantage of OTC options is that they can be tailored to meet the specific requirements of the buyer. Hence, they are ideally suited as risk-management instruments for corporate and financial institutions, because they can be used to structure hedges that match perfectly the risk exposure of the buying party. Some of the more ingenious structures are described in a later chapter on exotic options.

OPTION PRICING: SETTING THE SCENE

The price of an option is a function of six different factors, which are:

1. The strike price of the option
2. The current price of the underlying
3. The time to expiry
4. The risk-free rate of interest that applies to the life of the option
5. The volatility of the underlying asset's price returns
6. The value of any dividends or cash flows paid by the underlying asset during the life of the option.

We review the basic parameters next.

Pricing Inputs

Possibly the two most important parameters are the current price of the underlying and the strike price of the option. The intrinsic value of a call option is the amount by which the strike price is below the price of the underlying, as this is the payoff if the option is exercised. Therefore, the value of the call option will increase as the price of the underlying increases, and will fall as the underlying price falls. The value of a call will also decrease as the strike price increases. All this is reversed for a put option.

Generally, for bond options a higher time to maturity results in higher option value. All other parameters being equal, a longer-dated option will always be worth at least as much as one that had a shorter life. Intuitively, we would expect this because the holder of a longer-dated option has the same benefits as someone holding a shorter-dated option, in addition to a longer time period in which the intrinsic value may increase. This rule is always true for American options and usually true for European options. Certain factors, however, such as the payment of a coupon during the option life, may cause a longer-dated option to have only a slightly higher value than a shorter-dated option.

The risk-free interest rate is the rate applicable to the period of the option's life. So for our table of gilt options in the previous section, the option value reflected the three-month rate. The most common rate used is Libor, as a proxy for the "risk-free" rate, rather than the T-bill rate, because the market making bank cannot actually fund at the T-bill rate. For government bond options it is common to see the appropriate repo rate being used. A rise in interest rates will increase the value of a call option, although not always for bond options. A rise in rates lowers the price of a bond, because it decreases the present value of future cash flows. In the equity markets, however, it is viewed as a sign that share-price growth rates will increase. Generally, however, the relationship is the same for bond options as equity options. The effect of a rise in interest rates for put options is the reverse: they cause the value to drop.

A coupon payment made by the underlying during the life of the option will reduce the price of the underlying asset on the ex-dividend date. This will result in a fall in the price of a call option and a rise in the price of a put option.

OPTION PRICING

Interest-rate products described in this book so far, both cash and derivatives, can be priced using rigid mathematical principles, because on maturity of the instrument there is a defined procedure that takes place such that one is able to calculate a fair value. This does not apply to options because there is uncertainty as to what the outcome will be on expiry; an option seller does not know whether the option will be exercised or not. This factor makes options more difficult to price than other financial-market instruments. In this section, we review the parameters used in the pricing of an option, and introduce the Black-Scholes pricing model.

Pricing an option is a function of the probability that it will be exercised. Essentially, the premium paid for an option represents the buyer's expected profit on the option. Therefore, as with an insurance premium, the writer of an option will base his price on the assessment that the payout on the option will be equal to the premium, and this is a function on the probability that the option will be exercised. Option pricing, therefore, bases its calculation on the assessment of the probability of exercise and derives from this an expected outcome and, hence, a fair value for the option premium. The expected payout, as with an insurance company premium, should equal the premium received.

The following factors influence the price of an option:

- **The behaviour of financial prices:** One of the key assumptions made by the Black-Scholes model (B-S) is that asset prices follow a lognormal distribution. Although this is not strictly accurate, it is considered close enough of an approximation to allow its use in option pricing. In fact, observation shows that while prices themselves are not normally distributed, asset returns are, and we define returns as In $\left(\frac{P_{t+1}}{P_t}\right)$ where P_t is the market price at time t and P_{t+1} is the price one period later. The distribution of prices is called a lognormal distribution because the logarithm of the prices is normally distributed. The asset returns are defined as the logarithm of the price relatives and are assumed to follow the normal distribution. The expected return as a result of assuming this distribution is given by $E\left[\ln\left(\frac{P_t}{P_0}\right)\right] = rt$ where $E[\ ]$ is the expectation operator and r is the annual rate of return. The derivation of this expression is given in Appendix 16.2.

- **The strike price:** The difference between the strike price and the underlying price of the asset at the time the option is struck will influence the size of the premium, as this will have an impact on the probability that the option will be exercised. An option that is deeply in-the-money has a greater probability of being exercised.
- **Volatility:** The volatility of the underlying asset will influence the probability that an option will be exercised, as a higher volatility indicates a higher probability of exercise. This is considered in detail below.
- **The term to maturity:** A longer-dated option has greater time value and a greater probability of eventually being exercised.
- **The level of interest rates:** The premium paid for an option in theory represents the expected gain to the buyer at the time the option is exercised. It is paid up front so it is discounted to obtain a present value. The discount rate used, therefore, has an effect on the premium, although it is less influential than the other factors presented here.

The volatility of an asset measures the variability of its price returns. It is defined as the annualised standard deviation of returns, where variability refers to the variability of the returns that generate the asset's prices, rather than the prices directly. The standard deviation of returns is given by (16.1).

$$\sigma = \sqrt{\sum_{i=1}^{N} \frac{(x-\mu)^2}{N-1}} \tag{16.1}$$

where x_i is the ith price relative, μ the arithmetic mean of the observations, and N is the total number of observations. The value is converted to an annualised figure by multiplying it by the square root of the number of days in a year, usually taken to be 250 working days. Using this formula from market observations it is possible to calculate the *historic volatility* of an asset. The volatility of an asset is one of the inputs to the B-S model. Of the inputs to the B-S model, the variability of the underlying asset, or its volatility is the most problematic. The distribution of asset prices is assumed to follow a lognormal distribution, because the logarithm of the prices is normally distributed (we assume lognormal rather than normal distribution to allow for the fact that prices cannot—as could be the case in a normal distribution—have negative values). The range of possible prices starts at zero and cannot assume a negative value.

Note that it is the asset price *returns* on which the standard deviation is calculated, and not the actual prices themselves. This is because using prices would produce inconsistent results, as the actual standard deviation itself would change as price levels increased.

However, calculating volatility using the standard statistical method gives us a figure for historic volatility. What is required is a figure for *future* volatility, since this is relevant for pricing an option expiring in the future. By definition, future volatility cannot be measured directly. Market-makers get around this by using an option-pricing model "backwards". An option-pricing model calculates the option price from volatility and other parameters. Used in reverse, the model can calculate the volatility implied by the option price. Volatility measured in this way is called *implied volatility*. Evaluating implied volatility is straightforward using this method and generally more appropriate than using historic volatility, as it provides a clearer measure of an option's fair value. Implied volatilities of deeply in-the-money or out-of-the-money options tend to be relatively high.

THE BLACK-SCHOLES OPTION MODEL

Most option-pricing models are based on one of two methodologies, although both types employ essentially identical assumptions. The first method is based on the resolution of the partial differentiation equation of the asset-price model, corresponding to the expected payoff of the option security. This is the foundation of the Black-Scholes model. The second type of model uses the martingale method, and was first introduced by Harrison and Kreps (1979) and Harrison and Pliska (1981), where the price of an asset at time 0 is given by its discounted expected future payoffs, under the appropriate probability measure, known as the risk-neutral probability. There is a third type that assumes lognormal distribution of asset returns but follows the two-step binomial process.

In order to employ the pricing models, we accept a state of the market that is known as a *complete market,*[2] one where there is a viable financial market. This is where the rule of no-arbitrage pricing exists, so that there is no opportunity to generate risk-free arbitrage due to the presence of, say, incorrect forward interest rates. The fact that there is no opportunity to generate risk-free arbitrage gains means that a zero-cost investment strategy that is initiated at time t will have a zero maturity value. The martingale property of the behaviour of asset prices states that an accurate estimate of the future price of an asset may be obtained from current price information. Therefore, the relevant information used to calculate forward asset prices is the latest price information. This was also a property of the semistrong and strong-form market-efficiency scenarios described by Fama (1970).

In this section, we describe the Black-Scholes option model in accessible fashion. More technical treatments are given in the relevant references listed in the bibliography.

Assumptions

The Black-Scholes model describes a process to calculate the fair value of a European call option under certain assumptions, and apart from the price of the underlying asset S and the time t, all the variables in the model are assumed to be constant, including—most crucially—the volatility. The following assumptions are made:

- There are no transaction costs, and the market allows short selling
- Trading is continuous

[2] First proposed by Arrow and Debreu (1954).

- Underlying asset prices follow geometric Brownian motion, with the variance rate proportional to the square root of the asset price
- The asset is a non-dividend-paying security
- The interest rate during the life of the option is known and constant
- The option can only be exercised on expiry

The B-S model is neat and intuitively straightforward to explain, and one of its many attractions is that it can be readily modified to handle other types of options such as foreign-exchange or interest-rate options. The assumption of the behaviour of the underlying asset price over time is described by (16.2), which is a generalised Weiner process, and where a is the expected return on the underlying asset and b is the standard deviation of its price returns.

$$\frac{\mathrm{d}S}{S} = a\mathrm{d}t + b\mathrm{d}W \tag{16.2}$$

The Black-Scholes Model and Pricing Derivative Instruments

We assume a financial asset is specified by its terminal payoff value. Therefore, when pricing an option, we require the fair value of the option at the initial time when the option is struck, and this value is a function of the expected terminal payoff of the option, discounted to the day when the option is struck. In this section, we present an intuitive explanation of the B-S model, in terms of the normal distribution of asset-price returns.

From the definition of a call option, we can set the expected value of the option at maturity T as:

$$E(C_T) = E[\max(S_T - X, 0)] \tag{16.3}$$

where S_T is the price of the underlying asset at maturity T
X is the strike price of the option

From (16.3) we know that there are only two possible outcomes that can arise on maturity: either the option will expire in-the-money and the outcome is $S_T - X$, or the option will be out-of-the-money and the outcome will be 0. If we set the term p as the probability that on expiry $S_T > X$, equation (16.3) can be rewritten as (16.4).

$$E(C_T) = p \times (E[S_T | S_T > X] - X) \tag{16.4}$$

where $E[S_T \mid S_T > X]$ is the expected value of S_T given that $S_T > X$. Equation (16.4) gives us an expression for the expected value of a call option on maturity. Therefore, to obtain the fair price of the option at the time it is struck, the value given by (16.4) must be discounted back to its present value, and this is shown as (16.5).

$$C = p \times e^{-rt} \times (E[S_T | S_T > X] - X) \tag{16.5}$$

where r is the continuously compounded risk-free rate of interest, and t is the time from today until maturity. Therefore, to price an option we require the probability

p that the option expires in-the-money, and we require the expected value of the option given that it does expire in-the-money, which is the last term of (16.5). To calculate p we assume that asset prices follow a stochastic process, which enables us to model the probability function.

The B-S model is based on the resolution of the following partial differential equation:

$$\tfrac{1}{2}\sigma^2 S^2\left(\frac{\partial^2 C}{\partial S^2}\right)+rS\left(\frac{\partial C}{\partial S}\right)+\left(\frac{\partial C}{\partial t}\right)-rC=0 \tag{16.6}$$

under the appropriate parameters. The parameters refer to the payoff conditions corresponding to a European call option, which we considered above. We do not present a solution to the differential equation in (16.6), which is beyond the scope of the book, but we can consider now how the probability and expected-value functions can be solved. For a fuller treatment, readers may wish to refer to the original account by Black and Scholes (1973). Other good accounts are given in Ingersoll (1987), Neftci (1996), and Nielsen (1999), among others.

We wish to find the probability p that the underlying asset price at maturity exceeds X is equal to the probability that the return over the time period for which the option is held exceeds a certain critical value. Remember that we assume normal distribution of asset-price returns. As asset returns are defined as the logarithm of price relatives, we require p such that:

$$p = prob[S_T > X] = prob\left[return > \ln\left(\frac{X}{S_0}\right)\right] \tag{16.7}$$

where S_0 is the price of the underlying asset at the time the option is struck. Generally, the probability that a normally distributed variable x will exceed a critical value x_c is given by (16.8).

$$p[x > x_c] = 1 - N\left(\frac{x_c - \mu}{\sigma}\right) \tag{16.8}$$

where μ and σ are the mean and standard deviation of x, respectively, and $N()$ is the cumulative normal distribution. We know from our earlier discussion of the behaviour of asset prices that an expression for μ is the natural logarithm of the asset-price returns. We already know that the standard deviation of returns is $\sigma\sqrt{t}$. Therefore, with these assumptions, we may combine (16.7) and (16.8) to give us (16.9), that is,

$$p = prob[S_T > X] = prob\left[return > \ln\left(\frac{X}{S_0}\right)\right] = 1 - N\left[\frac{\ln\left(\frac{X}{S_0}\right)-\left(\frac{r-\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right] \tag{16.9}$$

Under the conditions of the normal distribution, the symmetrical shape means that we can obtain the probability of an occurrence based on l – $N(d)$ being equal to $N(-d)$. Therefore, we are able to set the following relationship, in (16.10).

$$p = prob[S_T > X] = N\left[\frac{\ln\left(\frac{S_0}{X}\right) + \left(r - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}\right] \tag{16.10}$$

Now we require a formula to calculate the expected value of the option on expiry, the second part of the expression in (16.5). This involves the integration of the normal distribution curve over the range from X to infinity. This is not shown here, but the result is given in (16.11).

$$E[S_T \mid S_T > X] = S_0 e^{rt} \frac{N(d_1)}{N(d_2)} \tag{16.11}$$

where

$$d_1 = \frac{\ln\left(\frac{S_0}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$$

and

$$d_2 = \frac{\ln\left(\frac{S_0}{X}\right) + \left(r - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} = d_1 - \sigma\sqrt{t}$$

We now have expressions for the probability that an option expires in-the-money as well as the expected value of the option on expiry, and we incorporate these into the expression in (16.5), which gives us (16.12).

$$C = N(d_2) \times e^{-rt} \times \left[S_0 e^{-rt} \frac{\mid N(d_1)}{N(d_2)} - X\right] \tag{16.12}$$

Equation (16.12) can be rearranged to give (16.13), which is the famous and well-known Black-Scholes option-pricing model for a European call option.

$$C = S_0 N(d_1) - Xe^{-rt} N(d_2) \tag{16.13}$$

where S_0 is the price of the underlying asset at the time the option is struck
X is the strike price
R is the continuously compounded risk-free interest rate
T is the maturity of the option

and d_1 and d_2 are as before.

What the expression in (16.13) states is that the fair value of a call option is the expected present value of the option on its expiry date, assuming that prices follow a lognormal distribution.

$N(d_1)$ and $N(d_2)$ are the cumulative probabilities from the normal distribution of obtaining the values d_1 and d_2, given above. $N(d_1)$ is the delta of the option. The term $N(d_2)$ represents the probability that the option will be exercised. The term e^{-rt} is the present value of one unit of cash received t periods from the time the option is struck. Where $N(d_1)$ and $N(d_2)$ are equal to 1, which is the equivalent of assuming complete certainty, the model is reduced to:

$$C = S - Xe^{-rt}$$

which is the expression for Merton's lower bound for continuously compounded interest rates, and which we introduced in intuitive fashion in the previous chapter. Therefore, under complete certainty, the B-S model reduces to Merton's bound.

Example 16.1 illustrates a first-principles application of the B-S model.

The Put-Call Parity Relationship

Up to now, we have concentrated on calculating the price of a call option. However, the previous section introduced the boundary condition for a put option, so it should be apparent that this can be solved as well. In fact, the price of a call option and a put option are related via what is known as the put-call parity theorem. This is an important relationship and obviates the need to develop a separate model for put options.

Consider a portfolio Y that consists of a call option with a maturity date T and a zero-coupon bond that pays X on the expiry date of the option. Consider also a second portfolio Z that consists of a put option also with maturity date T and one share. The value of portfolio A on the expiry date is given by (16.14):

$$MV_{Y,T} = \max[S_T - X, 0] + X = \max[X, S_T] \tag{16.14}$$

The value of the second portfolio Z on the expiry date is:

$$MV_{Z,T} = \max[X - S_T, 0] + S_T = \max[X, S_T] \tag{16.15}$$

Both portfolios have the same value at maturity. Therefore, they must also have the same initial value at start time t, otherwise there would be an arbitrage opportunity. Prices must be arbitrage-free; therefore, the following put-call relationship must hold:

$$C_t - P_t = S_t - Xe^{-r(T-t)} \tag{16.16}$$

If the relationship in (16.15) did not hold, then arbitrage would be possible. So, using this relationship, the value of a European put option is given by the B-S model as shown below, in (16.17).

$$P(S,T) = -SN(-d_1) + Xe^{-rT}N(-d2) \tag{16.17}$$

EXAMPLE 16.1 THE BLACK-SCHOLES MODEL

Here, we illustrate a simple application of the B-S model. Consider an underlying asset, assumed to be a non-dividend-paying equity, with a current price of 25, and volatility of 23%. The short-term risk-free interest rate is 5%. An option is written with strike price 21 and a maturity of three months. Therefore, we have:

$$S = 25$$
$$X = 21$$
$$r = 5\%$$
$$T = 0.25$$
$$\sigma = 23\%$$

To calculate the price of the option, we first calculate the discounted value of the strike price, as follows:

$$Xe^{-rT} = 21e^{-0.05(0.25)} = 20.73913$$

We then calculate the values of d_1 and d_2.

$$\begin{aligned} d_1 &= \frac{1\mathrm{n}(25/21) + [0.05 + (0.5)(0.23)^2]0.25}{0.23\sqrt{0.25}} = \frac{0.193466}{0.115} \\ &= 1.682313 \\ d_2 &= d_1 - 0.23\sqrt{0.25} = 1.567313 \end{aligned}$$

We now insert these values into the main price equation:

$$C = 25N(1.682313) - 21e^{-0.05(0.25)}N(1.567313)$$

Using the approximation of the cumulative normal distribution at the points 1.68 and 1.56, the price of the call option is:

$$C = 25(0.9535) - 20.73913(0.9406) = 4.3303$$

or 4.3303.

What would be the price of a put option on the same stock? The values of $N(d_1)$ and $N(d_2)$ are 0.9535 and 0.9406; therefore, the put price is calculated as:

$$P = 20.7391(1 - 0.9406) - 25(1 - 0.9535) = 0.06943$$

(*continued*)

EXAMPLE 16.1 (*Continued*)

If we use the call price and apply the put-call parity theorem, the price of the put option is given by:

$$\begin{aligned} P &= C - S + Xe^{-rT} \\ &= 4.3303 - 25 + 21e^{-0.05(0.25)} \\ &= 0.069434 \end{aligned}$$

This is exactly the same price that was obtained by the application of the put-option formula in the B-S model above.

As we noted early in this chapter, the premium payable for an option will increase if the time to expiry, the volatility, or the interest rate is increased (or any combination is increased). Thus, if we keep all the parameters constant but price a call option that has a maturity of six months or $T = 0.5$, we obtain the following values:

$$d_1 = 1.3071, \text{ giving } N(d_1) = 0.9049$$
$$d_2 = 1.1445, \text{ giving } N(d_2) = 0.8740$$

The call price for the longer-dated option is 4.7217.

THE BLACK-SCHOLES MODEL AS AN EXCEL SPREADSHEET

In Appendix 16.3, we show the spreadsheet formulae required to build the Black-Scholes model into Microsoft Excel. By setting up the cells in the way shown, the fair value of a vanilla call or put option may be calculated. The put-call parity is used to enable calculation of the put price.

Black-Scholes and the Valuation of Bond Options

In this section, we illustrate the application of the B-S model to the pricing of an option on a zero-coupon bond and a plain-vanilla fixed-coupon bond.

For a zero-coupon bond, the theoretical price of a call option written on the bond is given by (16.18).

$$C = PN(d_1) - Xe^{-rT}N(d_2) \tag{16.18}$$

where P is the price of the underlying bond and all other parameters remain the same. If the option is written on a coupon-paying bond, it is necessary to subtract the present value of all coupons paid during the life of the option from the bond's price. Coupons sometimes lower the price of a call option because a coupon makes it more attractive to hold a bond rather than an option on the bond. Call options on bonds are often priced at a lower level than similar options on zero-coupon bonds.

The approach is illustrated in Example 16.2.

EXAMPLE 16.2 B-S MODEL AND BOND-OPTION PRICING

Consider a European call option written on a bond that has the following characteristics:

Price	£98
Coupon	8.00% (semiannual)
Time to maturity	Five years
Bond price volatility	6.02%
Coupon payments	£4 in three months and nine months
Three-month interest rate	5.60%
Nine-month interest rate	5.75%
One-year interest rate	6.25%

The option is written with a strike price of £100 and has a maturity of one year. The present value of the coupon payments made during the life of the option is £7.78, as shown below.

$$4e^{-0.056\times 0.25} + 4e^{-0.057\times 0.25} = 3.9444 + 3.83117 = 7.77557$$

This gives us $P = 98 - 7.78 = £90.22$

Applying the B-S model we obtain:

$$d_1 = [1n(90.22/100) + 0.0625 + 0.001812]/0.0602 = -0.6413$$
$$d_2 = d_1 - (0.0602 \times 1) = -0.7015$$
$$C = 90.22N(-0.6413) - 100e^{-0.0625}N(-0.7015) = 1.1514$$

Therefore, the call option has a value of £1.15, which will be composed entirely of time value. Note also that a key assumption of the model is constant interest rates, yet it is being applied to a bond price—which is essentially an interest rate—that is considered to follow a stochastic price process.

INTEREST-RATE OPTIONS AND THE BLACK MODEL

In 1976, Fischer Black presented a slightly modified version of the B-S model, using similar assumptions, to be used in pricing forward contracts and interest-rate options. The Black model is used in banks today to price instruments such as swaptions, in addition to bond and interest-rate options like caps and floors.

In this model, the spot price $S(t)$ of an asset or a commodity is the price payable for immediate delivery today (in practice, up to two days forward) at time t. This price is assumed to follow a geometric Brownian motion. The theoretical price for a futures contract on the asset, $F(t,T)$, is defined as the price agreed today for delivery of the asset at time T, with the price agreed today but payable on delivery. When $t = T$, the futures price is equal to the spot price. A futures contract is cash-settled every day via the clearing mechanism, whereas a forward contract is a contract to buy or sell the asset where there is no daily mark-to-market and no daily cash settlement.

Let us set f as the value of a forward contract, u as the value of a futures contract, and C as the value of an option contract. Each of these contracts is a function of the futures price $F(t,T)$, as well as additional variables. So we may write at time t the values of all three contracts as $f(F,t)$, $u(F,t)$ and $C(F,t)$. The value of the forward contract is also a function of the price of the underlying asset S at time T and can be written $f(F,t,S,T)$. Note that the value of the forward contract f is not the same as the price of the forward contract. The forward price at any given time is the delivery price that would result in the contract having a zero value. At the time the contract is transacted, the forward value is zero. Over time, both the price and the value will fluctuate. The futures price, on the other hand, is the price at which a forward contract has a zero current value. Therefore, at the time of the trade the forward price is equal to the futures price F, which may be written as:

$$f(F,tF,T) = 0 \tag{16.19}$$

Equation (16.19) simply states that the value of the forward contract is zero when the contract is taken out and the contract price S is always equal to the current futures price, $F(t,T)$.[3]

The principal difference between a futures contract and a forward contract is that a futures contract may be used to imply the price of forward contracts. This arises from the fact that futures contracts are repriced each day, with a new contract price that is equal to the new futures price. Hence, when F rises, such that $F > S$, the forward contract has a positive value, and when F falls, the forward contract has a negative value. When the transaction expires and delivery takes place, the futures price is equal to the spot price, and the value of the forward contract is equal to the spot price minus the contract price or the spot price.

$$f(F,T,S,T) = F - S \tag{16.20}$$

On maturity, the value of a bond or commodity option is given by the maximum of zero, and the difference between the spot price and the contract price. Since at that date the futures price is equal to the spot price, we conclude that:

$$C(F,T) = \begin{cases} F - S & \text{if } F \geq S_T \\ 0 & \text{else} \end{cases} \tag{16.21}$$

[3] This assumption is held in the market but does not hold good over long periods, due chiefly to the difference in the way futures and forwards are marked-to-market, and because futures are cash-settled on a daily basis while forwards are not.

The assumptions made in the Black model are that the prices of futures contracts follow a lognormal distribution with a constant variance, and that the Capital Asset Pricing Model applies in the market. There is also an assumption of no transaction costs or taxes. Under these assumptions, we can create a risk-free hedged position that is composed of a long position in the option and a short position in the futures contract. Following the B-S model, the number of options put on against one futures contract is given by $[\partial C(F,t)/\partial F]$, which is the derivative of $C(F, t)$ with respect to F. The change in the hedged position resulting from a change in price of the underlying is given by (16.22) below.

$$\partial C(F,t) - [\partial C(F,t)/\partial F]\partial F \tag{16.22}$$

Due to the principle of arbitrage-free pricing, the return generated by the hedged portfolio must be equal to the risk-free interest rate, and this together with an expansion of $\partial C(F,t)$ produces the following partial differential equation:

$$\left[\frac{\partial C(F,t)}{\partial t}\right] = rC(F,t) - \tfrac{1}{2}\sigma^2 F^2\left[\frac{\partial^2 C(F,t)}{\partial F^2}\right] \tag{16.23}$$

which is solved by setting the following:

$$\tfrac{1}{2}\sigma^2 F^2\left[\frac{\partial^2 C(F,t)}{\partial F^2}\right] - rC(F,t) + \left[\frac{\partial C(F,t)}{\partial t}\right] = 0 \tag{16.24}$$

The solution to the partial differential equation (16.23) is not presented here.

The result, by denoting $T = t - T$ and using (16.10) and (16.23), gives the fair value of a commodity option or option on a forward contract as shown in (16.25).

$$C(F,t) = e^{-rT}[FN(d_1) - S_T N(d_2)] \tag{16.25}$$

where

$$d_1 = \frac{1}{\sigma\sqrt{t}}\left[\ln\left(\frac{F}{S_T}\right) + \tfrac{1}{2}\sigma^2\right)T\right]$$

$$d_2 = d_1 - \sigma\sqrt{t}$$

There are a number of other models that have been developed for specific contracts; for example, the Garman and Kohlhagen (1983) and Grabbe (1983) models, used for currency options, and the Barone-Adesi and Whaley or BAW model (1987) used for commodity options. For the valuation of American options, on dividend-paying assets, another model has been developed by Whaley (1981). Additionally, the Black-Derman-Toy model (1990) has been used to price exotic options. A survey of these, though very interesting, is outside the scope of this book.

COMMENT ON THE BLACK-SCHOLES MODEL

The introduction of the B-S model was one of the great milestones in the development of the global capital markets, and it remains an important pricing model today. Many of the models introduced later for application to specific products are still based essentially on the B-S model. Subsequently, academics have presented some weaknesses in the model that stem from the nature of the main assumptions behind the model itself, which we will summarise here. The main critique of the B-S model appears to centre on:

- Its assumption of frictionless markets. This is at best only approximately true for large market counterparties.
- The assumption of a constant interest rate. This is possibly the most unrealistic assumption. Interest rates over even the shortest time frame (the overnight rate) fluctuate considerably. In addition to a dynamic short rate, the short-end of the yield curve often moves in the opposite direction to moves in underlying asset prices, particularly so with bonds and bond options.
- The assumption of lognormal distribution. This is accepted by the market as a reasonable approximation but not completely accurate, and also misses out most extreme moves or market shocks.
- It is a European option only. Although it is rare for American options to be exercised early, there are situations when it is optimal to do so, and the B-S model does not price these situations.
- For stock options, the assumption of a continuous constant dividend yield is clearly not realistic, although the trend in the U.S. markets is for ordinary shares to cease paying dividends altogether.

These points notwithstanding, the Black-Scholes model paved the way for the rapid development of options as liquid tradable products and is widely used today.

Stochastic Volatility

The B-S model assumes a constant volatility and for this reason, and because it is based on mathematics, often fails to pick up on market "sentiment" when there is a large downward move or shock. This is not a failing limited to the B-S model, however. For this reason, though, it undervalues out-of-the-money options, and to compensate for this market makers push up the price of deep in- or out-of-the-money options, giving rise to the volatility smile. This is considered in the next chapter.

The effect of stochastic volatility not being catered for then is to introduce mispricing, specifically the undervaluation of out-of-the-money options and the overvaluation of deeply in-the-money options. This is because when the price of the underlying asset rises, its volatility level also increases. The effect of this is that assets priced at relatively high levels do not tend to follow the process described by geometric Brownian motion. The same is true for relatively low asset prices and price volatility, but in the opposite direction. To compensate for this, stochastic volatility models have been developed, such as the Hull-White model (1987).

Implied Volatility

The volatility parameter in the B-S model, by definition, cannot be observed directly in the market as it refers to volatility going forward. It is different from historic volatility, which can be measured directly, and this value is sometimes used to estimate implied volatility of an asset price. Banks therefore use the value for implied volatility, which is the volatility obtained using the prices of exchange-traded options. Given the price of an option and all the other parameters, it is possible to use the price of the option to determine the volatility of the underlying asset implied by the option price. The B-S model, however, cannot be rearranged into a form that expresses the volatility measure σ as a function of the other parameters. Generally, therefore, a numerical iteration process is used to arrive at the value for σ given the price of the option; this is usually the Newton-Raphson method.

The market uses implied volatilities to gauge the volatility of individual assets relative to the market. Volatility levels are not constant, and fluctuate with the overall level of the market, as well as for stock-specific factors. When assessing volatilities with reference to exchange-traded options, market makers will use more than one value, because an asset will have different implied volatilities depending on how in-the-money the option itself is. The price of an at-the-money option will exhibit greater sensitivity to volatility than the price of a deeply in- or out-of-the-money option. Therefore, market makers will take a combination of volatility values when assessing the volatility of a particular asset.

A FINAL WORD ON OPTION MODELS

We have only discussed the B-S model and the Black model in this chapter. Other pricing models have been developed that follow on from the pioneering work done by Messrs. Black and Scholes. The B-S model is essentially the most straightforward and the easiest to apply, and subsequent research has focused on easing some of the restrictions of the model in order to expand its applicability. Areas that have been focused on include a relaxation of the assumption of constant volatility levels for asset prices, as well as work on allowing for the valuation of American options and options on dividend-paying stocks. However, in practice some of the newer models often require input of parameters that are difficult to observe or measure directly, which limits their application as well. Often there is a difficulty in calibrating a model due to the lack of observable data in the marketplace. The issue of calibration is an important one in the implementation of a pricing model, and involves inputting actual market data and using this as the parameters for calculation of prices. So, for instance, a model used to calculate the prices of sterling-market options would use data from the UK market, including money-market, futures, and swaps rates to build the zero-coupon yield curve, and volatility levels for the underlying asset or interest rate (if it is valuing options on interest-rate products, such as caps and floors). What sort of volatility is used? Some banks use actual historical volatilities, but, more usually, it is volatilities implied by exchange-traded option prices that are used. Another crucial piece of data for multifactor models (following Heath-Jarrow-Morton and other models based on this) is the correlation coefficients between forward rates on the term structure. This is used to calculate volatilities using the model itself.

The issue of calibrating the model is important, because incorrectly calibrated models will produce errors in option valuation. This can have disastrous results, which may be discovered only after significant losses have been suffered. If data is not available to calibrate the model, it may be that a simpler one needs to be used. The lack of data is not an issue for products priced in, say, dollars, sterling, or euros, but may be in other currency products if data is not so readily available. This might explain why the B-S model is still widely used today, although markets observe an increasing use of models such as the Black-Derman-Toy (1990) and Brace-Gatarek-Musiela (1994) for more exotic option products.

Many models, because of the way that they describe the price process, are described as Gaussian interest-rate models. The basic process is described by an Itô process:

$$\mathrm{d}P_T / P_T = \mu_T \mathrm{d} + \sigma_T \mathrm{d}W$$

where P_T is the price of a zero-coupon bond with maturity date T and W is a standard Weiner process. The basic statement made by Gaussian interest-rate models is that:

$$P_T(t) = E_T \exp\left[-\int_t^T r(s)ds\right]$$

Models that capture the process in this way include Cox-Ingersoll-Ross and Harrison and Pliska. We are only summarising here but, essentially, such models state that the price of an option is equal to the discounted return from a risk-free instrument. This is why the basic B-S model describes a portfolio of a call option on the underlying stock and a cash deposit invested at the risk-free interest rate. This was reviewed earlier. We then discussed how the representation of asset prices as an expectation of a discounted payoff from a risk-free deposit does not capture the real-world scenario presented by many option products. Hence the continuing research into developments of the basic model.

Following on from B-S, under the assumption that the short-term spot rate drives bond and option prices, the basic model can be used to model an interest-rate term structure, as given by Vasicek and Cox-Ingersoll-Ross. The short-term spot rate is assumed to follow a diffusion process

$$\mathrm{d}r = \mu \mathrm{d}t + \sigma \mathrm{d}w$$

which is a standard Weiner process. From this, it is possible to model the complete term structure based on the short-term spot rate and the volatility of the short-term rate. This approach is modified by Heath, Jarrow, and Morton (1992), which was reviewed and referenced in Chapter 4 as an interest-rate model.

APPENDICES

APPENDIX 16.1 SUMMARY OF BASIC STATISTICAL CONCEPTS

The arithmetic mean μ is the average of a series of numbers. The variance is the sum of the squares of the difference of each observation from the mean, and from the variance we obtain the standard deviation σ, which is the square root of the variance. The probability density of a series of numbers is the term for how likely any of them is to occur. In a normal probability density function, described by the normal distribution, the probability density is given by:

$$\frac{1}{\sqrt{2\pi e}^{\frac{x2}{2}}}$$

Most option-pricing formulae assume a normal probability density function, specifically that movements in the natural logarithm of asset prices follow this function. That is,

$$1\text{n}\left[\frac{\text{today's price}}{\text{yesterday's price}}\right]$$

is assumed to follow a normal probability density function. This relative price change is equal to:

$$\left[1 + r \times \frac{\text{Days}}{\text{Year}}\right]$$

where r is the rate of return being earned on an investment in the asset. The value

$$1\text{n}(1 + r \times \text{Days/Years})$$

is equal to $r \times$ Days/Years where r is the continuously compounded rate of return. Therefore, the value 1*n*(Today's Price/Yesterday's price) is equal to the continuously compounded rate of return on the asset over a specified holding period.

APPENDIX 16.2 LOGNORMAL DISTRIBUTION OF RETURNS

In the distribution of asset-price returns, returns are defined as the logarithm of price relatives and are assumed to follow the normal distribution, given by:

$$\text{In}\left(\frac{P_1}{P_0}\right) \sim N\left(rt, \sigma\surd t\right) \quad \text{(A16.2.1)}$$

where
- P_t is the price at time t
- P_0 is the price at time 0
- $N(m,s)$ is a random normal distribution with mean m and standard deviation s
- r is the annual rate of return
- σ is the annualised standard deviation of returns

From (A16.2.1) we conclude that the logarithm of the prices is normally distributed, due to (A16.2.2) where P_0 is a constant

$$\text{In}(P_t) \sim \text{In}(P_0) + N(rt, \sigma\sqrt{t}) \quad \text{(A16.2.2)}$$

We conclude that prices are normally distributed and are described by the relationship,

$$\frac{P_1}{P_0} \sim e^{N(rt,\sigma\sqrt{t})}$$

and from this relationship we may set the expected return as *rt*.

APPENDIX 16.3 BLACK-SCHOLES MODEL IN MICROSOFT EXCEL

To value a vanilla option under the following parameters, we can use Microsoft Excel to carry out the calculation as shown in Figure A16.1.

	Price of underlying	100	
	Volatility	0.0691	
	Maturity of option	3 months	
	Strike price	99.5	
	Risk-free rate	5%	
Cell	C	D	
8	**Underlying price, *S***	100	
9		Volatility %	0.0691
10	Option maturity years	0.25	
11	**Strike price, *X***	99.50	
12	Risk-free interest rate %	0.05	
13			
14			
15			Cell formulae:
16	ln (S/X)	0.005012542	=LN (D8/D11)
17	Adjusted return	0.0000456012500	=((D12-D9)^2/2)*D10
18	Time-adjusted volatility	0.131434394	=(D9*D10)^0.5
19	d_2	0.038484166	=(D16+D17)/D18
20	$N(d_2)$	0.515349233	=NORMSDIST(D19)
21			
22	d_1	0.16991856	=D19+D18
23	$N(d_1)$	0.56746291	=NORMSDIST(D22)
24	e^{-rt}	0.9875778	=EXP(-D10*D12)
25			
26	**CALL**	6.106018498	=D8*D23-D11*D20*D24
27	**PUT**	4.370009648	* =D26-D8+D11*D24

*By put-call parity, $P = C - S + Xe^{-rt}$

FIGURE A16.3.1 Microsoft Excel Calculation of Vanilla Option Price

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CHAPTER 17

Options II

We continue with options in this chapter with a look at how options behave in response to changes in market conditions. To start, we consider the main issues that a market maker in options must consider when writing options. We then review "the Greeks", the measures by which the sensitivity of an option book is calculated. We conclude with a discussion on an important set of interest-rate options in the market, *caps* and *floors*. Options may seem complex at first sight, but there are many analogies in the behaviour of option price sensitivity to bond sensitivity measures.

BEHAVIOUR OF OPTION PRICES

As we noted in the previous chapter, the value of an option is a function of five factors:

1. The price of the underlying asset; Notation: S
2. The strike price of the option; Notation: K
3. The time to expiry of the option; Notation: T
4. The volatility level of the underlying asset price returns; Notation: σ
5. The risk-free interest rate applicable to the life of the option. Notation: r

The Black-Scholes model assumes that the level of volatility and interest rates stays constant, so that changes in these will impact on the value of the option. On the expiry date, the price of the option will be a function of the strike price and the price of the underlying asset. However, for pricing purposes an option trader must take into account all the factors above. From Chapter 16 we know that the value of an option is composed of intrinsic value (IV) and time value (TV); intrinsic value is apparent straightaway when an option is struck, and a valuation model is essentially pricing time value of the option.

ASSESSING TIME VALUE

The time value of an option reflects the fact that it is highest for at-the-money options and also higher for an in-the-money option than for an out-of-the-money option. This can be demonstrated by considering the hedge process followed by a market maker in options. A deep out-of-the-money call option for instance, presents the lowest probability of exercise for the market maker; therefore she may not even

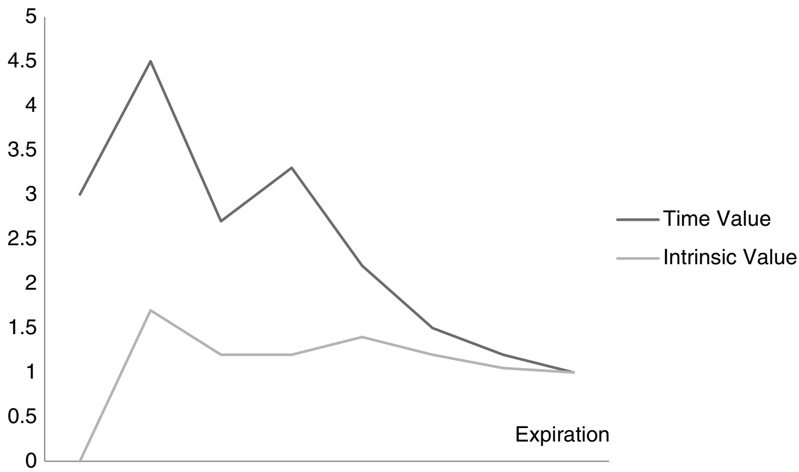

FIGURE 17.1 Time Value versus Intrinsic Value

hedge such a position. Should the price of the underlying rise sufficiently to make the option in-the-money, the market maker would have to purchase the asset in the market, thereby suffering a loss. This is a risk that must be considered by the market maker, but deeply out-of-the-money options are often not hedged. So the risk to the market maker is lowest for this type of option, which means that the time value is also lowest for such an option. Usually time value decreases as the option approaches maturity.

Time value is illustrated conceptually in Figure 17.1.

An in-the-money call option carries a greater probability that it will be exercised. A market maker writing such an option will therefore hedge the position, either with the underlying asset, with futures contracts or via a *risk reversal*. This is a long or short position in a call option that is reversed to the same position in a put by selling or buying the position forward (and vice versa). The risk with hedging using the underlying is that if the spot price falls, it will cause the option not to be exercised and force the market maker to dispose of the underlying at a loss. However, this risk is lowest for deeply in-the-money options, and this is reflected in the time value for such options, which diminishes the more in-the-money the option is.

The highest risk lies in writing an at-the-money option. In fact the majority of over-the-counter (OTC) options are struck at-the-money. The risk level reflects the fact that there is greatest uncertainty with this option, because there is an even chance of it being exercised or not. The decision on whether to hedge is therefore not as straightforward. As an at-the-money option carries the greatest risk for the market maker in terms of hedging it, the time value for it is the highest.

AMERICAN OPTIONS

In the previous chapter we discussed the Black-Scholes (B-S) and other models in terms of European options, and also briefly referred to a model developed for American options on dividend-paying securities. In theory, an American option will have greater value than an equivalent European option, because of the early exercise option. This added feature implies a higher value for the American option. In theory this is correct, but in practice it carries a lower weight because American options are rarely exercised ahead of expiry. A holder of an American option must assess if it is ever optimal to exercise it ahead of the expiry date, and usually the answer to this is "no". This is because, by exercising an option, the holder realises only the intrinsic value of the option. However, if the option is traded in the market, that is, sold, then the full value will be realised, including the time value. Therefore it is rare for an American option to be exercised ahead of the expiry date; rather, it will be sold in the market to realise the full value.

As the chief characteristic that differentiates American options from European options is rarely employed, in practical terms, they do not have significantly greater value than European ones, unless that particular option market is characterised by very low liquidity. However, an option-pricing model, calculating the probability that an option will be exercised, will determine under certain circumstances that the American option has a higher probability of being exercised and assign it a higher price.

Under certain circumstances it is optimal to exercise American options early. The most notable is when an option has negative time value. An option can have negative time value when, for instance, a European option is deeply in-the-money and very near to maturity. The time value will be small positive; however, the potential value in deferring cash flows from the underlying asset may outweigh this, leading to a negative time value. The best example of this is for a deeply in-the-money option on a futures contract. By deferring exercise, the opportunity to invest the cash proceeds from the profit on the futures contract (remember, futures are cash settled daily via the margin process) is lost, and this is potential interest income foregone. In such circumstances, it would be optimal to exercise an option ahead of its maturity date, assuming it is an American one. Therefore, when valuing an American option, the probability of its being exercised early is considered, and if it is deeply in-the-money, this probability will be at its highest.

MEASURING OPTION RISK: THE GREEKS

It is apparent from a reading of the previous chapter that the price sensitivity of options is different to other financial market instruments. This is clear from the variables that are required when pricing an option, which we presented by way of recap at the start of this chapter. The value of an option is sensitive to changes in one or any combination of the five variables that are used in the valuation.[1] This makes risk management of an option book more complex compared to other instruments. For example, the value of a swap is sensitive to one variable only, the swap rate. The relationship between the change in value of the swap and the change in the swap rate also is a linear one. A bond futures contract is priced as a function of the current spot price of the cheapest-to-deliver bond and the current money-market repo rate. Options, on the other hand, react to moves in any of the variables used in pricing them; more importantly, the relationship between the value of the option and the

[1] Of course, the strike price for a plain-vanilla option is constant.

change in a key variable is not a linear one. The market uses a measure for each of the variables, and in some cases for a derivative of these variables, which are termed the "Greeks" as they are called after letters in the ancient Greek alphabet.[2] In this section, we review these sensitivity measures and how they are used.

DELTA

The *delta* of an option is a measure of how much the value or premium of the option changes with changes in the price of the underlying asset. That is,

$$\delta = \frac{\Delta C}{\Delta S}$$

Mathematically the delta of an option is the partial derivative of the option premium with respect to the underlying, given by (17.1.)

$$\delta = \frac{\partial C}{\partial S}$$

or

$$\partial = \frac{\partial P}{\partial S} \tag{17.1}$$

Delta is illustrated in Figure 17.2.

In fact the delta of an option is given by the $N(d_1)$ term in the B-S equation. It is closely related to but not equal to the probability that an option will be exercised. If an option has a delta of 0.6 or 60%, this means that a £100 increase in the value of the underlying will result in a £60 increase in the value of the option. Delta is probably the most important sensitivity measure for an option, as it measures the

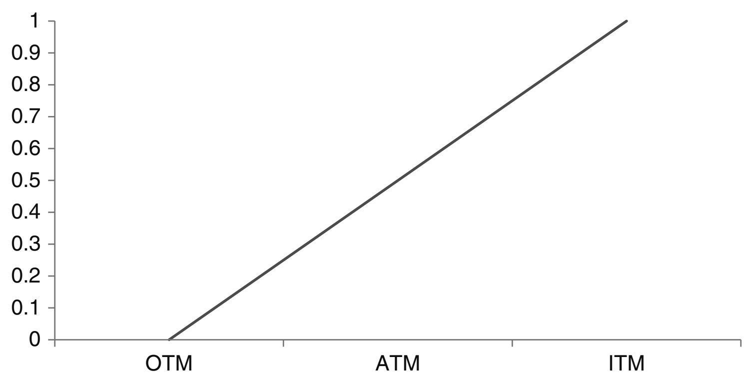

FIGURE 17.2 Delta for a Call Option

[2] All but one; the term for the volatility sensitivity measure, *vega*, is not a Greek letter. In certain cases one will come across the use of the term *kappa* to refer to volatility, and this is a Greek letter. However, it is more common for volatility to be referred to by the term vega.

TABLE 17.1 Delta-Neutral Hedging for Changes in Underlying Price

Option	Rise in Underlying Asset Price	Fall in Underlying Asset Price
Long call	Rise in delta: sell underlying	Fall in delta: buy underlying
Long put	Fall in delta: sell underlying	Rise in delta: buy underlying
Short call	Rise in delta: buy underlying	Fall in delta: sell underlying
Short put	Fall in delta: buy underlying	Rise in delta: sell underlying

sensitivity of the option price to changes in the price of the underlying, and this is very important for option market makers. It is also the main hedge measure. When an option market maker wishes to hedge a sold option, she may do this by buying a matching option, by buying or selling another instrument with the same but opposite value as the sold option, or by buying or selling the underlying. If the hedge is put on with the underlying, the amount is governed by the delta. So, for instance, if the delta of an option written on one ordinary share is 0.6 and a trader writes 1,000 call options, the hedge would be a long position in 600 of the underlying shares. This means that of the value of the shares rises by £1, the £600 rise in the value of the shares will offset the £600 loss in the option position. This is known as *delta hedging*. As we shall see later on, this is not a static situation, and the fact that delta changes, and is also an approximation, means that hedges must be monitored and adjusted, so-called *dynamic hedging*.

The delta of an option measures the extent to which the option moves with the underlying asset price; at a delta of zero the option does not move with moves in the underlying, while at a delta of 1 it will behave identically to the underlying.

A positive delta position is equivalent to being long the underlying asset, and can be interpreted as a bullish position. A rise in the asset price results in profit, as in theory a market maker could sell the underlying at a higher price, or in fact sell the option. The opposite is true if the price of the underlying falls. With a positive delta, a market maker would overhedge if running a delta-neutral position. Table 17.1 shows the effect of changes in the underlying price on the delta position in the option book; to maintain a delta-neutral hedge, the market maker must buy or sell delta units of the underlying asset, although in practice futures contracts may be used.

GAMMA

In a similar way that the modified duration measure becomes inaccurate for larger yield changes, due to the nature of its calculation, there is an element of inaccuracy with the delta measurement and with delta hedging an option book. This is because the delta itself is not static, and changes with changes in the price of the underlying. A book that is *delta neutral* at one level may not be as the underlying price changes. To monitor this, option market makers calculate *gamma*. The gamma of an option is a measure of how much the delta value changes with changes in the underlying price. It is given by:

$$\Gamma = \frac{\Delta\delta}{\Delta S}$$

Mathematically gamma is the second partial derivative of the option price with respect to the underlying price, that is:

$$\frac{\partial^2 C}{\partial S^2} \quad \text{or} \quad \frac{\partial^2 P}{\partial S^2}$$

and is given by (17.2).

$$\Gamma = \frac{N(d_1)}{S\sigma\sqrt{T}} \tag{17.2}$$

The delta of an option does not change rapidly when an option deeply in or out-of-the-money, so that in these cases the gamma is not significant. However, when an option is close to or at-the-money, the delta can change very suddenly, and at that point the gamma is very large. The value of gamma is positive for long call and put options, and negative for short call and put options. Gamma is similar to the convexity of a bond.

An option with a high absolute value gamma causes the most problems for market makers, as the delta hedge must be adjusted constantly, which will lead to high transaction costs. The higher the gamma, the greater is the risk that the option book is exposed to loss from sudden moves in the market. A negative gamma exposure is the highest risk, and this can be hedged only by putting on long positions in other options. A perfectly hedged book is gamma neutral, which means that the delta of the book does not change.

When gamma is positive, a rise in the price of the underlying asset will result in a higher delta. Adjusting the hedge will require selling the underlying asset or futures contracts, which may generate hedging profits. That is why market makers prefer having a positive gamma book as opposed to a negative. The reverse applies if there is a fall in the price of the underlying.

However, with a negative gamma, an increase in the price of the underlying will reduce the value of the delta, so therefore to adjust the delta hedge, the market maker must buy more of the underlying asset or futures equivalents. However, when the underlying asset price falls, the delta will rise, necessitating selling of the underlying asset to rebalance the hedge. In this scenario, irrespective of whether cash or off-balance-sheet instruments are being used, the hedge involves selling assets in a falling market, which will generate losses even as the hedge is being put on. Negative gamma is, therefore, a high-risk exposure in a rising market. Managing an option book that has negative gamma is more risky if the underlying asset price volatility is high. In a rising market, the market maker becomes short and must purchase more of the underlying, which may produce losses. The same applies in a falling market. If the desk is pursuing a delta-neutral strategy, running a positive gamma position should enable generation of profit in volatile market conditions. Under the same scenario, a negative gamma position would be risky and would be excessively costly in terms of dynamically hedging the book.

Gamma is the only one of the major Greeks that does not measure the sensitivity of the option premium, instead it measures the change in delta. The delta of an option is its hedge ratio, and gamma is a measure of how much this hedge ratio changes for movements in the price of the underlying. This is why a high absolute

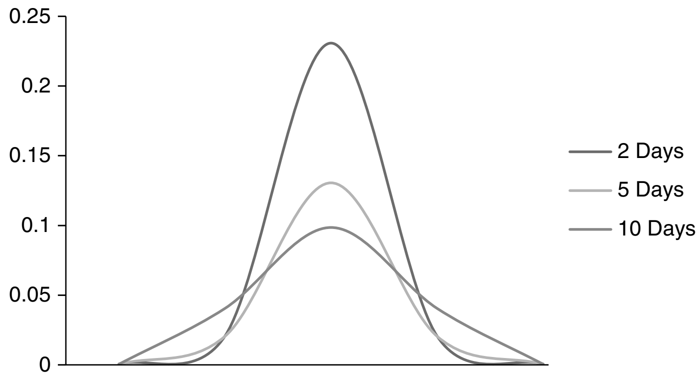

FIGURE 17.3 Gamma for ATM Options

gamma value makes hedging an option book difficult, as the hedge ratio is always changing. This ties in with our earlier comment that at-the-money options have the highest value, because they present the greatest uncertainty and hence the highest risk. The relationship is illustrated by the behaviour of gamma, which follows that of the delta. As you can see, the closer the option is to expiry, the higher the gamma is for the buyer.

Gamma is illustrated in Figure 17.3.

To adjust an option book so that it is gamma neutral, a market maker must put on positions in an option on the underlying or on the future. This is because the gamma of the underlying and the future is zero. It is common for market makers to use exchange-traded options. Therefore, a book that needs to be made gamma-neutral must be rebalanced with options; however, by adding to its option position, the book's delta will alter. Therefore to maintain the book as delta neutral, the market maker will have to rebalance it using more of the underlying asset or futures contracts. The calculation made to adjust gamma is a snapshot in time, and as the gamma value changes dynamically with the market, the gamma hedge must be continually rebalanced, like the delta hedge, if the market maker wishes to maintain the book as gamma neutral. Because of the complex nature of options, it's difficult to maintain a perfectly neutral position, so traders will keep their books within accepted ranges vetted by risk management department.

THETA

The *theta* of an option measures the extent of the change in value of an option with change in the time to maturity, also known as the time decay of an option. It is illustrated in Figure 17.4. Different values that result across different option tenors are illustrated in Figure 17.5.

Theta is given by

$$\Theta = \frac{\Delta C}{\Delta T}$$

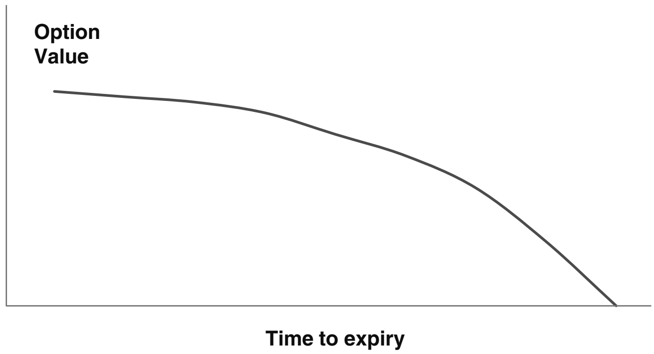

FIGURE 17.4 Option Time Decay

or

$$-\frac{\partial C}{\partial T} \quad \text{or} \quad -\frac{\partial P}{\partial T}$$

and, from the formula for the B-S model, mathematically it is given for a call option as (17.3).

$$\Theta = -\frac{S\sigma}{2\sqrt{2\pi T}} e^{-\frac{d_1^2}{2}} - Xre^{-rT} N(d_2) \tag{17.3}$$

Theta is a measure of time decay for an option. A holder of a long option position suffers from time decay because as the option approaches maturity, its value is made up increasingly of intrinsic value only, which may be zero as the option approaches expiry. For the writer of an option, the risk exposure is reduced as a result of time decay, so it is favourable for the writer if the theta is high. There is also a relationship between theta and gamma, however; when an option gamma is high, its theta is also high, and this results in the option losing value more rapidly as it approaches maturity. Therefore a high theta option, while welcome to the writer, has a downside because it is also high gamma. There is therefore in practice no gain to be high theta for the writer of an option.

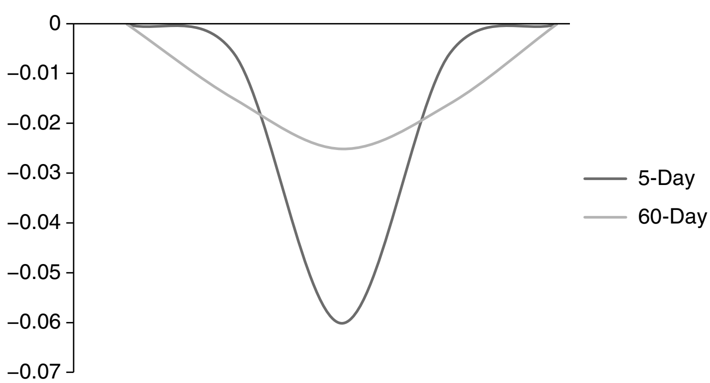

FIGURE 17.5 Theta for Different Expiration Dates

The theta value impacts certain option strategies. For example, it is possible to write a short-dated option and simultaneously purchase a longer-dated option with the same strike price. This is a play on the option theta; if the trader believes that the time value of the longer-dated option will decay at a slower rate than the short-dated option, the trade will generate a profit.

VEGA

The *vega* of an option measures how much its value changes with changes in the volatility of the underlying asset. It is also known as *epsilon* (ε), *eta* (η), or *kappa* (κ).

We define vega as:

$$v = \frac{\Delta C}{\Delta \sigma}$$

or

$$v = \frac{\partial C}{\partial \sigma} \quad \text{or} \quad \frac{\partial P}{\partial \sigma}$$

and mathematically from the B-S formula it is defined as equation (17.4) for a call or put.

$$v = \frac{S\sqrt{\dfrac{T}{2\pi}}}{e^{\frac{d_1^2}{2}}} \tag{17.4}$$

It may also be given by (17.5).

$$v = S\sqrt{\Delta T}N(d_1) \tag{17.5}$$

An option exhibits its highest vega when it is at-the-money, and decreases as the underlying and strike prices diverge. Options with only a short time to expiry have a lower vega compared to longer-dated options. An option with positive vega generally has positive gamma. Vega is also positive for a position composed of long call and put options, and an increase in volatility will then increase the value of the options. A vega of 12.75 means that for a 1% increase in volatility, the price of the option will increase by 0.1275. Buying options is the equivalent of buying volatility, while selling options is equivalent to selling volatility. Market makers generally like volatility and set up their books so that they are positive vega. The basic approach for volatility trades is that the market maker will calculate the implied volatility inherent in an option price, and then assess whether this is accurate compared to her own estimation of volatility. Just as positive vega is long call and puts, if the trader feels the implied volatility in the options is too high, she will put on a short vega position of short calls and puts, and then reverse the position out when the volatility declines.

Table 17.2 shows the response to a delta hedge following a change in volatility.

Managing an option book involves trade-offs between the gamma and the vega, much as there are between gamma and theta. A long option means long vega and

TABLE 17.2 Dynamic Hedging as a Result of Changes in Volatility

Option Position	Rise in Volatility	Fall in Volatility
Long Call		
ATM	No adjustment to delta	No adjustment to delta
ITM	Rise in delta, buy underlying	Rise in delta, sell underlying
OTM	Fall in delta, sell underlying	Fall in delta, buy underlying
Long Put		
ATM	No adjustment to delta	No adjustment to delta
ITM	Fall in delta, sell underlying	Rise in delta, buy underlying
OTM	Rise in delta, buy underlying	Fall in delta, sell underlying
Short Call		
ATM	No adjustment to delta	No adjustment to delta
ITM	Fall in delta, sell underlying	Rise in delta, buy underlying
OTM	Rise in delta, buy underlying	Fall in delta, sell underlying
Short Put		
ATM	No adjustment to delta	No adjustment to delta
ITM	Rise in delta, buy underlying	Rise in delta, sell underlying
OTM	Fall in delta, sell underlying	Fall in delta, buy underlying

long gamma, which is not conceptually difficult to manage; however, if there is a fall in volatility levels, the market maker can either maintain positive gamma, depending on her view of whether the fall in volatility can be offset by adjusting the gamma in the direction of the market, or she can sell volatility (that is, write options) and set up a position with negative gamma. In either case the costs associated with rebalancing the delta must compensate for the reduction in volatility.

RHO

The *rho* of an option is a measure of how much its value changes with changes in interest rates. Mathematically this is:

$$\frac{\partial C}{\partial r} \quad \text{or} \quad \frac{\partial P}{\partial r}$$

and the formal definition, based on the B-S model formula, is given in (17.6) for a call option.

$$\rho = Xte^{-rT}N(d_2) \tag{17.6}$$

The level of rho tends to be higher for longer-dated options. It is probably the least used of the sensitivity measures because market interest rates are probably the least variable of all the parameters used in option pricing.

LAMBDA

The lambda of an option is similar to its delta in that it measures the change in option value for a change in underlying price. However, lambda measures this sensitivity as a percentage change in the price for a percentage change in the price of the underlying. Hence lambda measures the gearing or leverage of an option. This in turn gives an indication of the expected profit or loss for changes in the price of the underlying. From Figure 17.6 we note that in-the-money options have a gearing of a minimum of five, and sometimes the level is considerably higher. This means that if the underlying was to rise in price, the holder of the long call could benefit by a minimum of five times more than if he had invested the same cash amount in the underlying instead of in the option.

This has been a brief review of the sensitivity measures used in managing option books. They are very useful to market makers and portfolio managers because they enable them to see what the impact of changes in market rates is on an entire book. A market maker need take only the weighted sum of the delta, gamma, vega, and theta of all the options on the book to see the impact of changes on the portfolio. Therefore the combined effect of changes can be calculated, without having to re-price all the options on the book. The Greeks are also important to risk managers and those implementing value-at-risk systems.

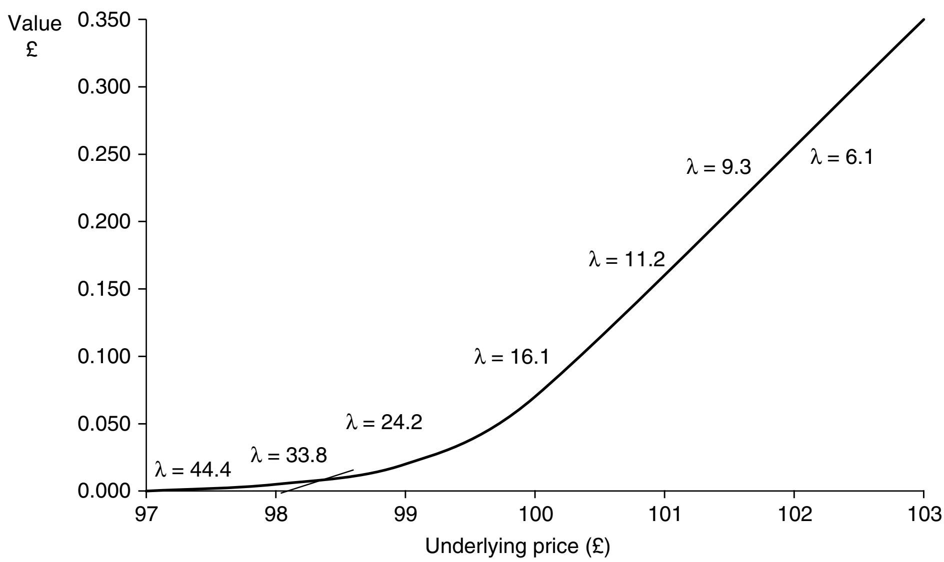

FIGURE 17.6 Option Lambda, Nine-Month Bond Option

THE OPTION SMILE

Our discussion on the behaviour and sensitivity of options prices will conclude with an introduction to the option *smile*. Market makers calculate a measure known as the volatility smile, which is a graph that plots the implied volatility of an option as a function of its strike price. The general shape of the smile curve is given in Figure 17.7. What the smile tells us is that out-of-the-money and in-the-money options both tend to have higher implied volatilities than at-the-money options. We define an at-the-money option as one whose strike price is equal to the forward price of the underlying asset.

Under the B-S model assumptions, the implied volatility should be the same across all strike prices of options on the same underlying asset and with the same expiry date. However, implied volatility is usually observed in the market as a convex function of exercise price, shown in generalised form as Figure 17.7 (in practice, it is not a smooth line or even often a real smile). The observations confirm that market makers price options with strikes that are less than *S*, and those with strikes higher than *S*, with higher volatilities than options with strikes equal to *S*. The existence of the volatility smile curve indicates that market makers make more complex

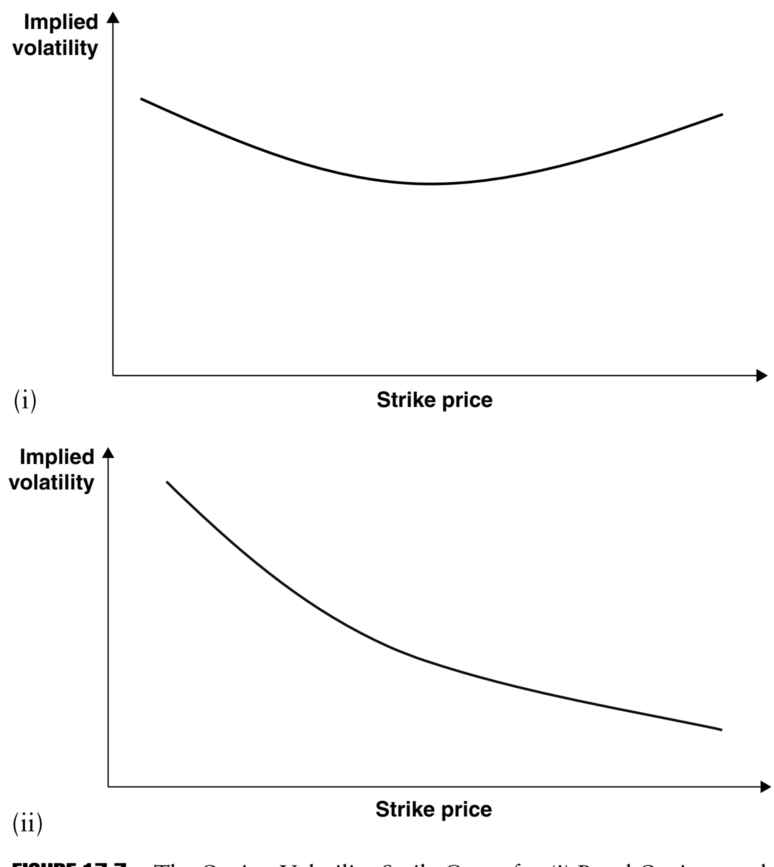

FIGURE 17.7 The Option Volatility Smile Curve for (i) Bond Options and (ii) Equity Options

assumptions about the behaviour of asset prices than can be fully explained by the geometric Brownian motion (GBM) model. As a result, market makers attach different probabilities to terminal values of the underlying asset price than those that are consistent with a lognormal distribution. The extent of the convexity of the smile curve indicates the degree to which the market-price process differs from the lognormal function contained in the B-S model. In particular, the more convex the smile curve, the greater the probability the market attaches to extreme outcomes for the price of the asset on expiry, S_T. This is consistent with the observation that in reality asset price returns follow a distribution with fatter tails than that described by the lognormal distribution. In addition the direction in which the smile curve slopes reflects the skew of the price process function; a positively sloped implied volatility smile curve results in a price-returns function that is more positively skewed than the lognormal distribution. The opposite applies for a negatively sloping curve. The existence of the smile suggests asset price behaviour that is more accurately described by nonstandard price processes, such as the jump diffusion model, or a stochastic volatility, as opposed to constant volatility model.

Considerable research has gone into investigating the smile. The book references in this and the previous chapter are good starting points on this subject.

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Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

PART Four

Bond Trading and Hedging

In the last part of the book we present some insights on trading compiled by the authors based on their experience working in the bond markets. The topics covered include implied spot yields and market zero-coupon yields, yield curve spread trading and butterfly yields. There is a new chapter on interest-rate derivatives trading and hedging, and the associated collateral and correlation issues.

We begin, however, with an introduction to risk management for a bond or credit trader, and a look at the value-at-risk risk measurement tool.

Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

CHAPTER 18

Value-at-Risk and Credit VaR

In this chapter, we review the main risk-measurement tool used in banking, known as value-at-risk (VaR). The review looks at the three main methodologies used to calculate VaR, as well as some of the key assumptions used in the calculations, including those on the normal distribution of returns, volatility levels, and correlations. The methodology is worth understanding, irrespective of its flaws, because national regulators have adopted it as the tool with which bank regulatory capital is calculated. We also discuss the use of the VaR methodology with respect to credit risk.

INTRODUCING VALUE-AT-RISK

The introduction of value-at-risk as an accepted methodology for quantifying market risk and its adoption by bank regulators was a part of the evolution of risk management. The application of VaR has been extended from its initial use in securities houses to commercial banks and corporates, following its introduction in October 1994 when JPMorgan launched RiskMetrics free over the Internet.

VaR is a measure of the quantile loss that a firm may suffer over a period of time that has been specified by the user, under normal market conditions and a specified level of confidence. This measure may be obtained in a number of ways, using a statistical model or by computer simulation. We define VaR as follows:

VaR is a measure of market risk. It is the minimum loss that can occur with X percent confidence over a holding period of n days. Put another way, there is a (1–X) percent chance of a loss larger than VaR.

VaR is the expected loss of a portfolio over a specified time period for a set level of probability. For example, if a daily VaR is stated as £100,000 to a 95% level of confidence, this means that during the day there is a only a 5% chance that the loss the next day will be *greater* than £100,000. VaR measures the potential loss in market value of a portfolio using estimated volatility and correlation. The "correlation" referred to is the correlation that exists between the market prices of different instruments in a bank's portfolio. VaR is calculated within a given confidence interval, typically 95% or 99%; it seeks to measure the possible losses from a position or portfolio under "normal" circumstances. The definition of normality is critical and is essentially a statistical concept that varies by firm and by risk-management system. Put simply, however, the most commonly used VaR models assume that the prices of assets in the financial markets follow a normal distribution. To implement VaR, all of a firm's positions data must be gathered into one centralised database. Once this is complete, the overall risk has to be calculated by aggregating the risks from individual instruments across the entire portfolio. The potential move in each instrument (that is, each risk factor) has to be inferred from past daily price movements over a given observation period. For regulatory purposes, this period is at least one year. Hence, the data on which VaR estimates are based should capture all relevant daily market moves over the previous year.

The main assumption underpinning VaR—and which in turn may be seen as its major weakness—is that the distribution of future price and rate changes will follow past variations. Therefore, the potential portfolio loss calculations for VaR are worked out using distributions from historic price data in the observation period.

There is no one VaR number for a single portfolio, because different methodologies used for calculating VaR produce different results. The VaR number captures only those risks that can be measured in quantitative terms. It does not capture risk exposures such as operational risk, liquidity risk, regulatory risk, or sovereign risk. It is important to be aware of what precisely VaR attempts to capture and what it clearly makes no attempt to capture. Also, VaR is not "risk management".

A risk-management department may choose to use a VaR-measurement system in an effort to quantify a bank's risk exposure; however, the application itself is merely a tool. Implementing such a tool in no way compensates for inadequate risk management procedures and rules in the origination of risky assets or the management of a trading book.

Assumption of Normality

A distribution is described as normal if there is a high probability that any observation from the population sample will have a value that is close to the mean, and a low probability of having a value that is far from the mean. The normal distribution curve is used by many VaR models, which assume that asset returns follow a normal pattern. A VaR model uses the normal curve to estimate the losses that an institution may suffer over a given time period. Normal distribution tables show the probability of a particular observation moving a certain distance from the mean.

If we look along a normal distribution table we see that at –1.645 standard deviations, the probability is 5%. This means that there is a 5% probability that an observation will be at least 1.645 standard deviations below the mean. This level is used in many VaR models.

Calculation Methods

There are three different methods for calculating VaR. They are

1. The variance/covariance (or *correlation* or *parametric* method)
2. Historical simulation
3. Monte Carlo simulation

Variance-Covariance Method This method assumes that the returns on risk factors are normally distributed, the correlations between risk factors are constant, and the delta (or price sensitivity to changes in a risk factor) of each portfolio constituent is constant. Using the correlation method, the volatility of each risk factor is extracted from the historical observation period. Historical data on investment returns is therefore required. The potential effect of each component of the portfolio on the overall portfolio value is then worked out from the components' delta (with respect to a particular risk factor) and that risk factor's volatility.

There are different methods of calculating the relevant risk factor volatilities and correlations. Two alternatives are

1. Simple *historic volatility:* This is the most straightforward method, but the effects of a large one-off market move can significantly distort volatilities over the required forecasting period. For example, if using 30-day historic volatility, a market shock will stay in the volatility figure for 30 days until it drops out of the sample range and correspondingly causes a sharp drop in (historic) volatility 30 days *after* the event. This is because each past observation is equally weighted in the volatility calculation.
2. *To weight past observations unequally:* This is done to give more weight to recent observations so that large jumps in volatility are not caused by events that occurred some time ago. One method is to use exponentially weighted moving averages.

Historical Simulation Method The historical simulation method for calculating VaR is the simplest and avoids some of the pitfalls of the correlation method. Specifically, the three main assumptions behind correlation (normally distributed returns, constant correlations, constant deltas) are not needed in this case. For historical simulation, the model calculates potential losses using actual historical returns in the risk factors and so captures the non-normal distribution of risk-factor returns. This means rare events and crashes can be included in the results. As the risk-factor returns used for revaluing the portfolio are actual past movements, the correlations in the calculation are also actual past correlations. They capture the dynamic nature of correlation as well as scenarios when the usual correlation relationships break down.

Monte Carlo Simulation Method The third method, Monte Carlo simulation, is more flexible than the previous two. As with historical simulation, Monte Carlo simulation allows the risk manager to use actual historical distributions for risk-factor returns rather than having to assume normal returns. A large number of randomly generated simulations are run forward in time using volatility and correlation estimates chosen by the risk manager. Each simulation will be different, but, in total, the simulations will aggregate to the chosen statistical parameters (that is, historical distributions and volatility and correlation estimates). This method is more realistic than the previous two models and therefore is more likely to estimate VaR more accurately. However, its implementation requires powerful computers, and there is also a trade-off in that the time required to perform calculations is longer.

The level of confidence in the VaR estimation process is selected by the number of standard deviations of variance applied to the probability distribution. A standard deviation selection of 1.645 provides a 95% confidence level (in a one-tailed test) that the potential estimated price movement will not be more than a given amount based on the correlation of market factors to the position's price sensitivity.

EXPLAINING VALUE-AT-RISK

Correlation

Measures of correlation between variables are important to practitioners interested in reducing their risk exposure through diversifying their portfolio. Correlation is a measure of the degree to which a value of one variable is related to the value of another. The correlation coefficient is a single number that compares the strengths and directions of the movements in two instruments' values. The sign of the coefficient determines the relative directions that the instruments move in, while its value determines the strength of the relative movements. The value of the coefficient ranges from –1 to +1, depending on the nature of the relationship. So if, for example, the value of the correlation is 0.5, this means that one instrument moves in the same direction by half of the amount that the other instrument moves. A value of zero means that the instruments are uncorrelated, and their movements are independent of each other.

Correlation is a key element of many VaR models, including parametric models. It is particularly important in the measurement of the variance (hence, volatility) of a portfolio. If we take the simplest example, a portfolio containing just two assets, (18.1) following gives the volatility of the portfolio based on the volatility of each instrument in the portfolio (x and y) and their correlation with one another.

$$\sigma_{port} = \sqrt{\sigma_x^2 + \sigma_y^2 + 2\sigma_x\sigma_y\rho} \tag{18.1}$$

where σ_x is the volatility of asset x
σ_y is the volatility of asset y

The correlation coefficient between two assets uses the covariance between the assets in its calculation. The standard formula for covariance is shown in (18.2):

$$Cov = \frac{\sum_{i=1}^{n} (xi - \overline{x})(yi - \overline{y})}{(n-1)} \tag{18.2}$$

where the sum of the distance of each return value x and y from their mean is divided by the number of observations minus one. The covariance calculation enables us to calculate the correlation coefficient, shown as (18.3):

$$\rho = \frac{Cov(x,y)}{\sigma_x\sigma_y} \tag{18.3}$$

where σ is the standard deviation of each asset.

Equation (18.1) may be modified to cover more than two instruments. In practice, correlations are usually estimated on the basis of past historical observations. This is an important consideration in the construction and analysis of a portfolio, as the associated risks will depend to an extent on the correlation between its constituents.

It should be apparent that from a portfolio perspective a positive correlation increases risk. If the returns on two or more instruments in a portfolio are positively correlated, strong movements in either direction are likely to occur at the same time. The overall distribution of returns will be wider and flatter, as there will be higher joint probabilities associated with extreme values (both gains and losses). A negative correlation indicates that the assets are likely to move in opposite directions, thus reducing risk.

It has been argued that in extreme situations, such as market crashes or large-scale market corrections, correlations cease to have any relevance, because all assets will be moving in the same direction. However, under most market scenarios, using correlations to reduce the risk of a portfolio is considered satisfactory practice, and the VaR number for diversified portfolio will be lower than that for an undiversified portfolio.

Simple VaR Calculation

To calculate the VaR for a single asset, we would calculate the standard deviation of its returns, using either its historical volatility or implied volatility. If a 95% confidence level is required, meaning we wish to have 5% of the observations in the left-hand tail of the normal distribution, this means that the observations in that area are 1.645 standard deviations away from the mean. This can be checked from standard normal tables. Consider the following statistical data for a government bond, calculated using one year's historical observations.

Nominal:	£10 million
Price:	£100
Average return:	7.35%
Standard deviation:	1.99%

The VaR at the 95% confidence level is 1.645 × 0.0199 or 0.032736. The portfolio has a market value of £10 million, so the VaR of the portfolio is 0.032736 × 10,000,000 or £327,360. So this figure is the minimum loss the portfolio may sustain over one year for 5% of the time.

We may extend this analysis to a two-stock portfolio. In a two-asset portfolio, we stated in (18.1) that there is a relationship that enables us to calculate the volatility of a two-asset portfolio; this expression is used to calculate the VaR, and is shown in (18.4):

$$VaR_{port} = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2\rho_{1,2}} \tag{18.4}$$

where w_1 is the weighting of the first asset
w_2 is the weighting of the second asset
σ_1 is the standard deviation or volatility of the first asset
σ_2 is the standard deviation or volatility of the second asset
$\rho_{1,2}$ is the correlation coefficient between the two assets

In a two-asset portfolio, the undiversified VaR is the weighted average of the individual standard deviations; the diversified VaR, which takes into account the correlation between the assets, is the square root of the variance of the portfolio. In practice, banks will calculate both diversified and undiversified VaR. The diversified VaR measure is used to set trading limits, while the larger undiversified VaR measure is used to gauge an idea of the bank's risk exposure in the event of a significant correction or market crash. This is because in a crash situation, liquidity dries up as market participants all attempt to sell off their assets. This means that the correlation relationship between assets ceases to have any impact on a book, as all assets move in the same direction. Under this scenario, then, it is more logical to use an undiversified VaR measure.

Although the description given here is very simple, it nevertheless explains what the essence of the VaR measure is. VaR is essentially the calculation of the standard deviation of a portfolio, which is used as an indicator of the volatility of that portfolio. A portfolio exhibiting high volatility will have a high VaR number. An observer may then conclude that the portfolio has a high probability of making losses. Risk managers and traders may use the VaR measure to help them to allocate capital to more efficient sectors of the bank, as return on capital can now be measured in terms of return on risk capital. Regulators may use the VaR number as a guide to the capital-adequacy levels that they feel the bank requires.

Variance-Covariance Value-at-Risk

Calculation In the previous section we showed how VaR could be calculated for a two-stock portfolio. Here, we illustrate how this is done using matrices.

Consider the following hypothetical portfolio, invested in two assets, as shown in Table 18.1. The standard deviation of each asset has been calculated on historical

TABLE 18.1 Two-Asset Portfolio VaR

Assets	Bond 1	Bond 2
Standard deviation	11.83%	17.65%
Portfolio weighting	60%	40%
Correlation coefficient		0.647
Portfolio value		£10,000,000
Variance		0.016506998
Standard deviation		12.848%
95% C.I. standard deviations		1.644853
Value-at-Risk		0.211349136
Value-at-Risk £		£2,113,491

observation of asset returns. Note that returns are returns of asset prices, rather than the prices themselves; they are calculated from the actual prices by taking the ratio of closing prices. The returns are then calculated as the logarithm of the price relatives. The mean and standard deviation of the returns are then calculated using standard statistical formulae. This would then give the standard deviation of daily price relatives, which is converted to an annual figure by multiplying it by the square root of the number of days in a year, usually taken to be 250.

The standard equation (shown as (18.4)) is used to calculate the variance of the portfolio, using the standard deviations of the individual assets and the asset weightings. The VaR of the book is the square root of the variance. Multiplying this figure by the current value of the portfolio gives us the portfolio VaR, which is £2,113,491.

The RiskMetric VaR methodology uses matrices to obtain the same results that we have shown here. This is because once a portfolio starts to contain many assets, the method we described above becomes unwieldy. Matrices allow us to calculate VaR for a portfolio containing many hundreds of assets, which would require assessment of the volatility of each asset and correlations of each asset to all the others in the portfolio. We can demonstrate how the parametric methodology uses variance and correlation matrices to calculate the variance, and hence standard deviation, of a portfolio. The matrices are shown in Table 18.2. Note that the multiplication of matrices carries with it some unique rules; readers who are unfamiliar with matrices should refer to a standard mathematics textbook.

As shown in Table 18.2, using the same two-asset portfolio described, we can set a 2×2 matrix with the individual standard deviations inside; this is labelled the "variance" matrix. The standard deviations are placed on the horizontal axis of the matrix, and a zero entered in the other cells. The second matrix is the correlation matrix, and the correlation of the two assets is placed in cells corresponding to the other asset. That is why a "1" is placed in the other cells, as an asset is said to have a

TABLE 18.2 RiskMetrics Grid Points

Grid Points
1 month
3 months
6 month
1 year
2 years
3 years
4 years
5 years
7 years
9 years
10 years
15 years
20 years
30 years

correlation of 1 with itself. The two matrices are then multiplied to produce another matrix, labelled "VC matrix" in Figure 18.1.[1]

The VC matrix is then multiplied with the Variance matrix to obtain the variance-covariance matrix or VCV matrix. This shows the variance of each asset; for Bond 1 this is 0.01399, which is expected since that is the square of its standard deviation, which we were given at the start. The matrix also tells us that Bond 1 has a covariance of 0.0135 with Bond 2. We then set up a matrix of the portfolio weighting of the two assets, and this is multiplied by the VCV matrix. This produces a 1×2 matrix, which we need to change to a single number, so this is multiplied by the W matrix, reset as a 2×1 matrix, which produces the portfolio variance. This is 0.016507. The standard deviation is the square root of the variance, and is 0.1284795 or 12.848%, which is what we obtained before. In our illustration it is important to note the order in which the matrices were multiplied, as this will obviously affect the result. The volatility matrix contains the standard deviations along the diagonal, and zeros are entered in all the other cells. So if the portfolio we were calculating has 50 assets in it, we would require a 50×50 matrix and enter the standard deviations for each asset along the diagonal line. All the other cells would have a zero in them. Similarly for the weighting matrix; this is always one row, and all the weights are entered along the row. To take the example just given, the result would be a 1×50 weighting matrix.

The matrix method for calculating the standard deviation is more effective than the first method we described, because it can be used for a portfolio containing a large number of assets. In fact, this is exactly the methodology used by RiskMetrics

	Variance matrix		**Correlation matrix**		**VC matrix**	
			Bond 1	Bond 2		
Bond 1	11.83%	0	1	0.647	0.1183	0.07654
Bond 2	0	17.65%	0.647	1	0.114196	0.1765
	VC matrix		**Variance matrix**		**VCV matrix**	
	0.1183	0.07654	11.83%	0	0.013995	0.013509
	0.1141955	0.1765	0	17.65%	0.013509	0.031152
	Weighting matrix		**VCV matrix**		**WVCV**	
	60%	40%	0.013995	0.013509	0.013801	0.020566
			0.013509	0.031152		
	WVCV		**W**		**WVCVW**	
	0.013801	0.020566	60%		0.016507	
			40%			
				Standard deviation 0.1284796		

FIGURE 18.1 Matrix Variance-Covariance Calculation for a Two-Asset Portfolio

[1] Microsoft Excel has a function for multiplying matrices that may be used for any type of matrix. The function is "=MMULT()" typed in all the cells of the product matrix.

and the computer model used for the calculation will be set up with matrices containing the data for hundreds, if not thousands, of different assets.

The variance-covariance method captures the diversification benefits of a multi-product portfolio because of the correlation coefficient matrix used in the calculation. For instance, if the two bonds in our hypothetical portfolio had a negative correlation, the VaR number produced would be lower. It was also the first methodology introduced, by JPMorgan in 1994. To apply it, a bank would require data on volatility and correlation for the assets in its portfolio. This data is available from the RiskMetrics website (and other sources), so a bank does not necessarily need its own data. It should be noted that RiskMetrics uses the exponentially weighted moving averages (EWMA) weighting scheme for the competition of its volatilities. A bank may wish to use its own data sets however, should it have them, to tailor the application to its own use. The advantages of the variance-covariance methodology are that:

- It is simple to apply, and fairly straightforward to explain;
- Data sets for its use are immediately available.

The drawbacks of the variance-covariance are that it assumes stable correlations and measures only linear risk; it also places excessive reliance on the normal distribution, and returns in the market are widely believed to have "fatter tails" than a true-to-normal distribution. This phenomenon is known as leptokurtosis, that is, the non-normal distribution of outcomes. Another disadvantage is that the process requires mapping. To construct a weighting portfolio for the RiskMetric tool, cash flows from financial instruments are mapped into precise maturity points, known as grid points. We will review this issue later in the chapter. However, in most cases, assets do not fit into neat grid points, and complex instruments cannot be broken down accurately into cash flows. The mapping process makes assumptions that frequently do not hold in practice.

Mapping

The cornerstone of variance-covariance methodologies such as RiskMetrics is the requirement for data on volatilities and correlations for assets in the portfolio. The RiskMetrics data set does not contain volatilities for every maturity possible, as that would require a value for very period from 1 day to over 10,950 days (30 years) and longer, and correlations between each of these days. This would result in an excessive amount of calculation. Rather, volatilities are available for set maturity periods, and these are shown in Table 18.2.

If a bond is maturing in six years' time, its redemption cash flow will not match the data in the RiskMetrics data set, so it must be mapped to two periods; in this case, being split to the five-year and seven-year grid point. This is done in proportions so that the original value of the bond is maintained once it has been mapped. More importantly, when a cash flow is mapped, it must split in a manner that preserves the volatility characteristic of the original cash flow. Therefore, when mapping cash flows, if one cash flow is apportioned to two grid points, the share of the two new cash flows must equal the present value of the original cash flows, and the combined volatility of the two new assets must be equal to that of the original asset. A simple demonstration is given in Example 18.1.

EXAMPLE 18.1 CASH FLOW MAPPING

A bond trading book holds £1 million nominal of a gilt strip that is due to mature in precisely six years' time. To correctly capture the volatility of this position in the bank's RiskMetrics VaR estimate, the cash flow represented by this bond must be mapped to the grid points for five years and seven years, the closest maturity buckets for which the RiskMetrics data set holds volatility and correlation data. The present value of the strip is calculated using the six-year zero-coupon rate, which RiskMetrics obtains by interpolating between the five-year rate and the seven-year rate. The details are shown in Table 18.3.

TABLE 18.3 Bond Position to be Mapped to Grid Points

Gilt strip nominal (£)	1,000,000
Maturity (years)	6
5-year zero-coupon rate	5.35%
7-year zero-coupon rate	5.50%
5-year volatility	24.50%
7-year volatility	28.95%
Correlation coefficient	0.979

Note that the correlation between the two interest rates is very close to 1. This is expected because five-year interest rates generally move very closely in line with seven-year rates.

We wish to assign the single cash flow to the five-year and seven-year grid points (also referred to as vertices). The present value of the bond, using the six-year interpolated yield, is £728,347. This is shown in Table 18.4, which also uses an interpolated volatility to calculate the volatility of the six-year cash flow. However, we wish to calculate a portfolio volatility based on the apportionment of the cash flow to the five-year and seven-year grid points. To do this, we need to use a weighting to allocate the cash flow between the two vertices. In the hypothetical situation used here, this presents no problem because six years falls precisely between five years and seven years. Therefore, the weightings are 0.5 for year five and 0.5 for year seven. If the cash flow had fallen in a less obvious maturity point, we would have to calculate the weightings using the formula for portfolio variance. Using these weightings, we calculate the variance for the new "portfolio", containing the two new cash flows, and then the standard deviation for the portfolio. This gives us a VaR for the strip of £265,853.

TABLE 18.4 Cash Flow Mapping and Portfolio Variance

Interpolated yield	0.05425
Interpolated volatility	0.26725
Present value	728,347.0103
Weighting 5-year grid point	0.5
Weighting 7-year grid point	0.5
Variance of portfolio	0.070677824
Standard deviation	0.265853012
VaR £	265,853

Confidence Intervals

Many models estimate VaR at a given confidence interval, under normal market conditions. This assumes that market returns generally follow a random pattern but one that approximates over time to a normal distribution. The level of confidence at which the VaR is calculated will depend on the nature of the trading book's activity and what the VaR number is being used for. The Market Risk amendment to the Basel Capital Accord stipulates a 99% confidence interval and a 10-day holding period if the VaR measure is to be used to calculate the regulatory capital requirement. However, certain banks prefer to use other confidence levels and holding periods; the decision on which level to use is a function of the asset-types in the portfolio, the quality of market data available, and the accuracy of the model itself, which will have been tested over time by the bank.

For example, a bank may view a 99% confidence interval as providing no useful information, as it implies that there should only be two or three breaches of the VaR measure over the course of one year. That would leave no opportunity to test the accuracy of the model until a longer period of time had elapsed, and, in the meantime, the bank would be unaware if the model was generating inaccurate numbers. A 95% confidence level implies that the VaR level is being exceeded around one day each month, if a year is assumed to contain 250 days.[2] If a VaR calculation is made using 95% confidence, and a 99% confidence level is required for, say, regulatory purposes, we need to adjust the measure to take account of the change in standard deviations required. For example, a 99% confidence interval corresponds to 2.32 standard deviations, while a 95% level is equivalent to 1.645 standard deviations. Thus, to convert from 95% confidence to 99% confidence, the VaR figure is divided by 1.645 and multiplied by 2.32.

In the same way, there may be occasions when a firm will wish to calculate VaR over a different holding period to that recommended by the Basel Committee. The holding period of a portfolio's VaR calculation should represent the period of time required to unwind the portfolio; that is, sell off the assets on the book. A 10-day holding period is recommended but would be unnecessary for a highly liquid portfolio—for example, one holding government bonds.

To adjust the VaR number to fit it to a new holding period, we simply scale it upwards or downwards by the square root of the time period required. For example, a VaR calculation measured for a 10-day holding period will be $\sqrt{10}$ times larger than the corresponding 1-day measure.

The time scaling relies strictly on the normal distribution assumption. So in a crisis situation, this assumption fails, and the VaR number will be underestimated. A competent risk manager should be aware of this limitation of the tool.

Historical VaR Methodology

The historical approach to value-at-risk is a relatively simple calculation, and it is also easy to implement and explain. To implement it, a bank requires a database record of its past profit/loss figures for the total portfolio. The required confidence interval is then applied to this record, to obtain a cutoff of the worst-case scenario.

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[2] For the 99% confidence level, $250 \times 1\% = 2.5$ days in one year, while 95% confidence is $250 \times 5\%$ or 12.5 days.

For example, to calculate the VaR at a 95% confidence level, the fifth percentile value for the historical data is taken, and this is the VaR number. For a 99% confidence level measure, the 1% percentile is taken. The advantage of the historical method is that it uses the actual market data that a bank has recorded (unlike RiskMetrics, for example, for which the volatility and correlations are not actual values, but estimated values calculated from average figures over a period of time, usually the last five years), and so produces a reasonably accurate figure. Its main weakness is that as it is reliant on actual historical data built up over a period of time, generally at least one year's data is required to make the calculation meaningful. Therefore, it is not suitable for portfolios whose asset weightings frequently change, as another set of data would be necessary before a VaR number could be calculated.

To overcome this drawback banks use a method known as historical simulation. This calculates VaR for the current portfolio weighting, using the historical data for the securities in the current portfolio. To calculate historical simulation VaR for our hypothetical portfolio considered earlier, comprising 60% of bond 1 and 40% of bond 2, we require the closing prices for both assets over the specified previous period (usually three or five years). We then calculate the value of the portfolio for each day in the period assuming constant weightings.

Again, the time scaling relies on the normal distribution assumption, and again in a crisis situation this assumption is unrealistic.

Simulation Methodology

The most complex calculations use computer simulations to estimate value-at-risk. The most common of these is the Monte Carlo method. To calculate VaR using a Monte Carlo approach, a computer simulation is run to generate a number of random scenarios, which are then used to estimate the portfolio VaR. The method is probably the most realistic, if we accept that market returns follow a similar random walk pattern. However, Monte Carlo simulation is best suited to trading books containing large option portfolios, whose price behaviour is not captured very well with the RiskMetrics methodology. The main disadvantage of the simulation methodology is that it is time-consuming and uses a substantial amount of computer resources.

A Monte Carlo simulation generates simulated future prices, and it may be used to value an option as well as for VaR applications. When used for valuation, a range of possible asset prices are generated, and these are used to assess what intrinsic value the option will have at those asset prices. The present value of the option is then calculated from these possible intrinsic values. Generating simulated prices, although designed to mimic a "random walk", cannot be completely random because asset prices, although not a pure normal distribution, are not completely random either. The simulation model is usually set to generate very few extreme prices. Strictly speaking, it is asset price *returns* that follow a normal distribution, or rather a lognormal distribution. Monte Carlo simulation may also be used to simulate other scenarios; for example, the effect on option Greeks for a given change in volatility, or any other parameters. The scenario concept may be applied to calculating VaR as well. For example, if 50,000 simulations of an option price are generated, the 95th-lowest value in the simulation will be the VaR at the 95% confidence level. The correlation between assets is accounted for by altering the random selection programme to reflect relationships.

Example 18.2 illustrates both variance-covariance and simulation methods.

EXAMPLE 18.2 PORTFOLIO VOLATILITY USING VARIANCE-COVARIANCE AND SIMULATION METHODS

A simple two-asset portfolio is composed of the following instruments:

	Gilt Strip	FTSE100 Stock
Number of units	£100 million	£5 million
Market value	£54.39 million	£54 million
Daily volatility	£0.18 million	£0.24 million

The correlation between the two assets is 20%. Using equation (18.4) we calculate the portfolio VaR as follows:

$$\text{Vol} = \sqrt{\sigma^2_{bond} + \sigma^2_{stock} + 2\sigma_{bond}\sigma_{stock}\rho_{bond,stock}}$$

$$\begin{aligned}\text{Vol} &= \sqrt{0.18^2 + 0.24^2 + (2 \times 0.18 \times 0.24 \times 0.2)} \\ &= 0.327\end{aligned}$$

We have ignored the weighting element for each asset because the market values are roughly equal. The calculation gives a portfolio volatility of £0.327 million. For a 95% confidence level VaR measure, which corresponds to 1.645 standard deviations (in a one-tailed test), we multiply the portfolio volatility by 1.645, which gives us a portfolio value-at-risk of £0.538 million.

In a Monte Carlo simulation, we also calculate the correlation and volatilities of the portfolio. These values are used as parameters in a random-number simulation to throw out changes in the underlying portfolio value. These values are used to reprice the portfolio, and this value will be either a gain or loss on the actual mark-to-market value. This process is repeated for each random number that is generated. In Table 18.5, we show the results for 15 simulations of our two-asset portfolio. From the results, we read off the loss level that corresponds to the required confidence interval.

This profit/loss is the difference between the simulated portfolio value and the base value of [108.39 (= 54.39 + 54)].

TABLE 18.5 Monte Carlo Simulation Results

Simulation	Market Value: Bond	Market Value: Stock	Portfolio Value	Profit/Loss
1	54.35	54.9	109.25	0.86
2	54.64	54.02	108.66	0.27
3	54.4	53.86	108.26	–0.13
4	54.25	54.15	108.4	0.01
5	54.4	54.17	108.57	0.18
6	54.4	54.03	108.43	0.04

(*continued*)

EXAMPLE 18.2 (*Continued*)

TABLE 18.5 (*Continued*)

Simulation	Market Value: Bond	Market Value: Stock	Portfolio Value	Profit/Loss
7	54.31	53.84	108.15	–0.24
8	54.3	53.96	108.26	–0.13
9	54.46	54.11	108.57	0.18
10	54.32	53.92	108.24	–0.15
11	54.31	53.97	108.28	–0.11
12	54.47	54.08	108.55	0.16
13	54.38	54.03	108.41	0.02
14	54.71	53.89	108.6	0.21
15	54.29	54.05	108.34	–0.05

As the number of trials is increased, the results from a Monte Carlo simulation approach those of the variance-covariance measure. This is shown in Figure 18.2.

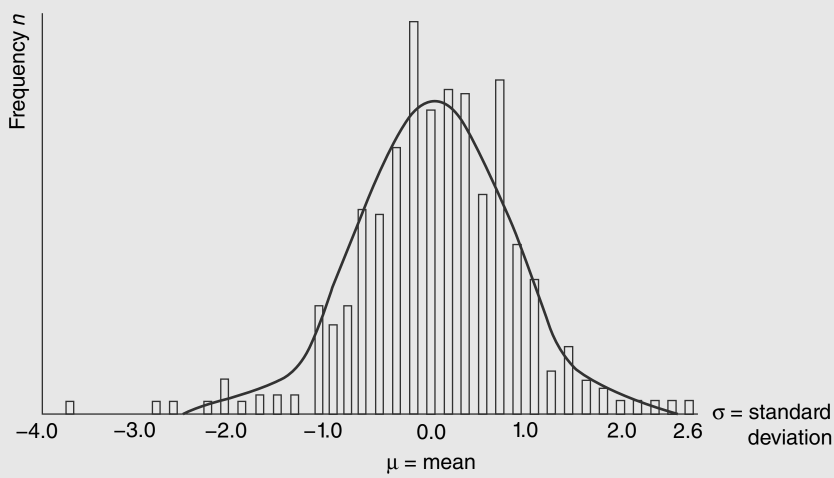

FIGURE 18.2 The Normal Approximation of Returns

VALUE-AT-RISK FOR FIXED-INCOME DERIVATIVE INSTRUMENTS

Perhaps the most straightforward instruments to which VaR can be applied are foreign-exchange and interest-rate instruments such as money-market products, bonds, forward-rate agreements, and swaps. In this section, we review the calculation of VaR for a simple portfolio of bonds.

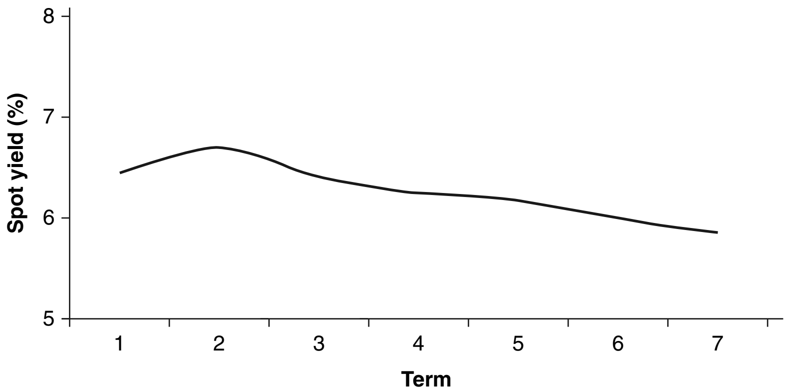

FIGURE 18.3 Term Structure Used for Valuation

Sample Bond Portfolio

Table 18.6 details the bonds that are in our portfolio. For simplicity, we assume that all the bonds pay an annual coupon and have full years left to maturity. In order to calculate the value-at-risk, we first need to value the bond portfolio itself. The bonds are valued by breaking them down into their constituent cash flows; the present value of each cash flow is then calculated, using the appropriate zero-coupon interest rate. Note from Figure 18.3 that the term structure is inverted.

Table 18.7 shows the present values for each of the cash flows. The total portfolio value is also shown.

We then use the volatility for each period rate to calculate the VaR. Note that *price volatility* is used here (unless mentioned otherwise). If yield volatility is quoted instead, we will need to convert the present value to PV01 before multiplying with volatility. Data on interest-rate volatility for all major currencies is available, for example, from the RiskMetrics website. The volatility levels for our hypothetical currency are relatively low in this example. The VaR for each maturity period is then obtained by multiplying the total present value of the cash flows for that period by its volatility level. This is shown in Table 18.8. By adding together all the individual values, we obtain an undiversified VaR for the portfolio. The total VaR is £1.77 million, for a portfolio with a market value of £23.1 million.

The figure just calculated is the undiversified VaR for the bond portfolio. To obtain the diversified VaR for the book, we require the correlation coefficient of each interest rate with the other interest rates (the correlation will be very close to unity, although the shorter-dated rates will be closer in line with each other than they will be with long-dated rates). We may then use the standard variance-covariance approach, using a matrix of the undiversified VaR values and a matrix with the

TABLE 18.6 Sample Three-Bond Portfolio

	Bond 1	Bond 2	Bond 3
Nominal value	10,000,000	3,800,000	9,700,000
Coupon	5%	7.25%	6%
Maturity	5	7	2

TABLE 18.7 Bond Portfolio Valuation

Period	Cash Flows: Bond 1	Bond 2	Bond 3	Zero-Coupon Rates	Discount Factor	Present Values		
1	500,000	275,500	582,000	6.45	0.939408173	469,704	258,807	546,736
2	500,000	275,500	10,282,000	6.7	0.878357191	439,179	241,987	9,031,269
3	500,000	275,500		6.4	0.830185447	415,093	228,716	
4	500,000	275,500		6.25	0.784664935	392,332	216,175	
5	10,500,000	275,500		6.18	0.740945722	7,779,930	204,131	
6		275,500		5.98	0.705759136	0	194,437	
7		4,075,500		5.87	0.670794678	0	2,733,824	
					Totals	9,496,238	4,078,077	9,578,004
				Portfolio Value		23,152,319		

TABLE 18.8 Bond Portfolio Undiversified VaR

Period	Cash Flows	Present Value	Volatility	Value-at-Risk at 84% C.I.
1	1,357,500.00	1,275,246.59	0.0687	87,609.44
2	11,057,500.00	9,712,434.64	0.0695	675,014.21
3	775,500.00	643,808.81	0.07128	45,890.69
4	775,500.00	608,507.66	0.0705	42,899.79
5	10,775,500.00	7,984,060.63	0.08501	678,724.99
6	275,500.00	194,436.64	0.08345	16,225.74
7	4,075,500.00	2,733,823.71	0.08129	222,232.53
		Undiversified VaR 1,768,597.39		

correlation values. However, the diversification benefit of a portfolio of bonds will be small, mainly because their volatilities will be closely correlated.

Forward-Rate Agreements

The VaR calculation for a forward-rate agreement (FRA) follows the principles reviewed in the previous section. An FRA is a notional loan or deposit for a period starting at some point in the future; in effect, it is used to fix a borrowing or lending rate. The derivation of an FRA rate is based on the principle of what it would cost for a bank that traded one to hedge it; this is known as the break-even rate. So a bank that has bought 3v6 FRA (referred to as a "threes-sixes FRA") has effectively borrowed funds for three months and placed the funds on deposit for six months. Therefore, an FRA is best viewed as a combination of an asset and a liability, and that is how one is valued. So a long position in a 3v6 FRA is valued as the present value of a three-month cash-flow asset and the present value of a six-month cash-flow liability, using the three-month and six-month deposit rates. The net present value is taken, of course, because one cash flow is an asset and the other a liability.

Consider a 3v6 FRA that has been dealt at 5.797%, the three-month forward-forward rate. The value of its constituent (notional) cash flows is shown in Table 18.9. The three-month and six-month rates are cash rates in the market, while the interest-rate volatilities have been obtained from RiskMetrics. The details are summarised in Table 18.9.

The undiversified VaR is the sum of the individual VaR values, and is £34,537. It has little value in the case of an FRA, however, and would overstate the true VaR, because an FRA is made up of a notional asset and liability, so a fall in the value of one would see a rise in the value of the other. Unless a practitioner was expecting

TABLE 18.9 Undiversified VaR for a 3v6 FRA Contract

Cash Flow	Term (days)	Cash Rate	Interest Rate Volatilities	Present Value	Undiversified VaR at 84% C.I.
10,000,000	91	5.38%	0.14%	9,867,765	13,815
−10,144,536	182	5.63%	0.21%	−9,867,765	20,722

three-month rates to go in an opposite direction to six-month rates, there is an element of diversification benefit. There is a high correlation between the two rates, so the more logical approach is to calculate a diversified VaR measure.

For an instrument such as an FRA, the fact that the two rates used in calculating the FRA rate are closely positively correlated will mean that the diversification effect will be to reduce the VaR estimate, because the FRA is composed notionally of an asset and a liability. From the values in Table 18.19, therefore, the six-month VaR is actually a negative value (if the bank had sold the FRA, the three-month VaR would have the negative value). To calculate the diversified VaR, then, requires the correlation between the two interest rates, which may be obtained from the RiskMetrics data set. This is observed to be 0.87. This value is entered into a 2×2 correlation matrix and used to calculate the diversified VaR in the normal way. The procedure is:

- Transpose the weighting VaR matrix, to turn it into a 2×1 matrix
- Multiply this by the correlation matrix
- Multiply the result by the original 1×2 weighting matrix
- This gives us the variance; the VaR is the square root of this value

The result is a diversified VaR of £11,051.

Interest-Rate Swaps

To calculate a variance-covariance VaR for an interest-rate swap, we use the process described earlier for an FRA. There are more cash flows that go to make up the undiversified VaR, because a swap is essentially a strip of FRAs. In a plain-vanilla interest-rate swap, one party pays fixed-rate basis on an annual or semiannual basis, and receives floating-rate interest, while the other party pays floating-rate interest payments and receives fixed-rate interest. Interest payments are calculated on a notional sum, which does not change hands, and only interest payments are exchanged. In practice, it is the net difference between the two payments that is transferred.

The fixed rate on an interest-rate swap is the break-even rate that equates the present value of the fixed-rate payments to the present value of the floating-rate payments. As the floating-rate payments are linked to a reference rate such as LIBOR, we do not know what they will be, but we use the forward rate applicable to each future floating payment date to calculate what it would be if we were to fix it today. The forward rate is calculated from zero-coupon rates today. A "long" position in a swap is to pay fixed and receive floating, and is conceptually the same as being short in a fixed-coupon bond and being long in a floating-rate bond. In effect, the long is "borrowing" money, so a rise in the fixed rate will result in a rise in the value of the swap. A "short" position is receiving fixed and paying floating, so a rise in interest rates results in a fall in the value of the swap. This is conceptually similar to a long position in a fixed-rate bond and a short position in a floating-rate bond.

Describing an interest-rate swap in conceptual terms of fixed- and floating-rate bonds gives some idea as to how it is treated for value-at-risk purposes. The coupon on a floating-rate bond is reset periodically in line with the stated reference rate, usually LIBOR. Therefore, the duration of a floating-rate bond is very low, and conceptually the bond may be viewed as being the equivalent of a bank deposit, which receives interest payable at a variable rate. For market-risk purposes,[3] the risk exposure of a bank deposit is nil, because its present value is not affected by changes in market interest rates. Similarly, the risk exposure of a floating-rate bond is very low and to all intents and purposes its VaR may be regarded as zero. This leaves only the fixed-rate leg of a swap to measure for VaR purposes.

Table 18.10 shows the fixed-rate leg of a five-year interest-rate swap. To calculate the undiversified VaR, we use the volatility rate for each term interest rate; this may be obtained from RiskMetrics. Note that the RiskMetrics data set supports only liquid currencies; for example, data on volatility and correlation is not available for certain emerging-market economies. Below, we show the VaR for each payment; the sum of all the payments constitutes the undiversified VaR. We then require the correlation matrix for the interest rates, and this is used to calculate the diversified VaR. The weighting matrix contains the individual term VaR values, which must be transposed before being multiplied by the correlation matrix.

Using the volatilities and correlations supplied by RiskMetrics, the diversified VaR is shown to be £10,325. This is very close to the undiversified VaR of £10,528. This is not unexpected because the different interest rates are very closely correlated.

Using VaR to measure market-risk exposure for interest-rate products enables a risk manager to capture nonparallel shifts in the yield curve, which is an advantage over the traditional duration measure and interest-rate gap measure. Therefore, estimating a book's VaR measure is useful not only for the trader and risk manager, but also for senior management, who by using VaR will have a more accurate idea of the risk-market exposure of the bank. Value-at-risk methodology captures pivotal shifts in the yield curve by using the correlations between different maturity interest rates. This reflects the fact that short-term interest rates and long-term interest rates are not perfectly positively correlated.

TABLE 18.10 Fixed-Rate Leg of Five-Year Interest-Rate Swap and Undiversified VaR

Pay Date	Swap Rate	Principal (£)	Coupon (£)	Coupon Present Value (£)	Volatility	Undiversified VaR at 84% C.I.
7-Jun-00	6.73%	10,000,000	337,421	327,564	0.05%	164
7-Dec-00	6.73%	10,000,000	337,421	315,452	0.05%	158
7-Jun-01	6.73%	10,000,000	335,578	303,251	0.10%	303
7-Dec-01	6.73%	10,000,000	337,421	294,898	0.11%	324
7-Jun-02	6.73%	10,000,000	335,578	283,143	0.20%	566
9-Dec-02	6.73%	10,000,000	341,109	277,783	0.35%	972
9-Jun-03	6.73%	10,000,000	335,578	264,360	0.33%	872
8-Dec-03	6.73%	10,000,000	335,578	256,043	0.45%	1,152
7-Jun-04	6.73%	10,000,000	335,578	248,155	0.57%	1,414
7-Dec-04	6.73%	10,000,000	337,421	242,161	1.90%	4,601
					Total	**10,528**

[3] We emphasise for *market-risk* purposes; the credit-risk exposure for a floating-rate bond position is a function of the credit quality of the issuer.

Derivative Products and Value-at-Risk

The variance-covariance methodology for calculating value-at-risk is considered adequate for trading books that mostly contain products that have a linear payoff profile. This covers money-market interest-rate instruments; however, the price/yield relationship for bonds exhibits a curved relationship, which gives rise to the convexity property. A trading book with convex instruments will have an added convexity risk exposure, and while most VaR methodologies are able to capture convexity risks adequately, an adjustment to the basic calculation has to be made. Such an adjusted measure is known as the *delta-gamma VaR*. Option products, however, have a non-linear payoff profile, and it is more difficult to capture risks associated with option trading books using the variance-covariance approach. In this section, we review the delta-gamma approach and its application to bonds and options. The specific risk measures used by option traders are considered in the next section.

Bond Convexity

The duration and convexity of fixed income instruments were reviewed in Chapter 2. To recap, the *modified duration* of a bond is an indicator of its sensitivity to interest rates; it measures the change in price of the bond for a 1% change in yield. The higher the level of modified duration in a bond, the more sensitive it is to changes in market interest rates. However, the relationship between price and yield in a bond is not a straight-line one; for changes in yield much above 50 basis points, the result given by the modified duration measure becomes increasingly inaccurate. Thus, the measure given by a bond's modified duration is only an approximation. The extent of the approximation of modified duration is given by *convexity,* which might be said to be a measure of the error made in using modified duration. Bonds with greater convexity perform differently under the same conditions compared to low convex bonds. For option products, the *delta* of an option is a measure of how much its price moves with respect to changes in the price of the underlying asset. Therefore, delta for an option is a similar risk measure to modified duration for bonds. Due to the way it is calculated, the delta figure is also an approximation, and the delta of an option changes as the price of the underlying changes. To measure the change in an option delta with respect to changes in the price of the underlying, traders calculate the *gamma* of the option.

The convex relationship between bond price and yield illustrates that the change in prices for a given change in interest rates is not constant, nor is it identical, for all but very small amounts, for both upward and downward changes in yield. This feature makes it a difficult property to capture in VaR calculations. Generally the slope of the convex curve flattens out as the yield increases, while it steepens as yields decrease. As this property cannot be captured, and differs among individual bonds, risk managers often overcome the problem with modelling it by simply assuming that the relationship is constant, and using modified duration as the usual risk measure. Under this assumption, we would calculate the VaR for a bond portfolio as described in Example 18.3.

The drawback in assuming constant changes in the price/yield relationship is that it is very inaccurate for large changes in yield. One of the main objectives of risk management is to provide accurate management information on the risk exposure of

EXAMPLE 18.3 SIMPLIFIED VAR FOR A BOND PORTFOLIO

A portfolio consists of several conventional bonds. The portfolio modified duration is 6.794, and it has a market value of £368 million. Market yields are expected to rise during the year, and the worst-case scenario, to 95% confidence, is a rise in yields of 75 basis points. As the interest-rate scenario is already given to the required degree of confidence, the VaR of the portfolio is given by:

$$368 \text{ million} \times -6.794 \times 0.75\%$$

or £18,751,440.

the bank, particularly under volatile conditions such as a market correction. As these are exactly the types of situation where there are large-scale moves in interest rates, the information contained in a bank's risk reports would be of questionable value.

To overcome this problem, it is necessary to make a convexity adjustment to the duration-based VaR measure. Convexity is the second derivative of price change with respect to yield, and is derived using a Taylor expansion (this is discussed in greater detail in Chapter 1). The convexity measure is used to provide a more accurate measure of modified duration, and is given by:

$$\tfrac{1}{2} \times CV \times (\Delta r)^2 \tag{18.5}$$

The convexity adjustment is added to the modified duration value. This is demonstrated in Example 18.4.

EXAMPLE 18.4 APPLYING THE CONVEXITY ADJUSTMENT FOR A BOND

The analysis is conducted on a hypothetical bond, the 6% 2021, which pays an annual coupon on an actual/actual basis. Its redemption date is 24 January 2021, so assuming today is 24 January 2001, the bond has precisely 20 years to maturity. The details are listed in Table 18.11.

TABLE 18.11 Modified Duration and Convexity of 6% 2021 Bond

Bond	6% 2021
Yield	6.15%
Price	98.30027
Duration	12.08839
Modified duration	11.38803
Convexity	184.21158

(continued)

EXAMPLE 18.4 (*Continued*)

The bond has a modified duration of 11.38803, a high value and not unexpected as the bond is very long dated. The modified duration value is used to estimate the price of the bond resulting from a 1% upward move in yields; we see from Table 18.12 that already, using only modified duration, the new price is overestimated by approximately one point. The convexity adjustment, using equation (18.5) is 0.92106%. However the overestimation is pronounced for a large change, as shown by the 3% upward move in yields. Using the convexity adjustment, which is 8.28952%, the estimated price is considerably closer to the actual price. Note that we are estimating prices for a rise in yields; this means that the bond price will fall, so the modified duration measure is given a negative value. When making the convexity adjustment, we add the value for convexity to the (negative) modified duration value, thus reducing it. For a downward move in yield, modified duration is a positive value, as prices will rise. Adding the convexity adjustment will then increase the value for modified duration.

TABLE 18.12 Price Estimate Using Convexity Adjustment

Yield	7.15%	9.15%	5.00%
Price	87.95764	71.54983	112.46221
MD estimate	86.91224	64.13618	111.39650
MD + *CV* estimate	87.8333	72.42570	113.46889

Convexity is a valuable property in a bond, and fund managers frequently look to invest in high-convexity bonds; conversely there is usually a convexity premium in a bond, so that it will trade at a lower yield to a bond of similar duration but lower convexity. The attraction of convexity is apparent if the price/yield profile for two bonds is plotted; the bond with higher convexity will outperform the other bond no matter what happens to interest rates. That is, it will rise in price by more for a given fall in yields, and it will drop in price by less for a given rise in yields. Equally therefore, to be short convexity, as a result of running a short position in a bond, can significantly add to a trader's risk. This is because, being short, even a quite small fall in interest rates can substantially increase the risk exposure of the book. This property is even more pronounced with option products.

Option Gamma

The gamma measurement for an option is conceptually similar to convexity for a bond. Convexity is a measure of the error made in using modified duration; that is, the curvature of the price/yield relationship. Gamma is the second derivative of an option's delta, so in effect it measures the same thing as convexity. As with convexity, it is important for a trader to be aware of the gamma exposure of his book, as at a high gamma level, even very small changes in the price of the underlying asset may lead to substantial mark-to-market losses. A trader who writes options, whether put or call options, is effectively short gamma.

The gamma effect on an option book cannot be captured accurately by most VaR models. This is because the relationship between gamma and the price of the underlying is nonlinear. To approximate the VaR measure for an option book, a *delta-gamma* calculation is made, and although it is still not completely accurate, it is a better estimate than the conventional delta-normal approach. However, although intuitively delta-gamma is similar to the convexity adjustment for a bond portfolio, it is not as good an approximation as the convexity measure. This is because behaviour of an option is more unpredictable than that of a bond. A bond instrument may be broken down into a series of zero-coupon bonds, so that volatility and other data maybe adjusted for convexity with relative ease. This is not as easy for options, and becomes particularly acute as an option approaches maturity. For example, an at-the-money option will experience extreme movement in its gamma as it approaches maturity, in a way that unpredictable. It is difficult to capture this effect in a VaR model. Nevertheless, the delta-gamma measure is recognised as a close approximation of option book risk, short of using simulation-type VaR models.

The gamma effect has an impact on the distribution of returns from an option book. This transforms the distribution from normal to one with slightly skewed tails, as illustrated by Figure 18.4.

To illustrate the gamma adjustment, consider a position in a bond instrument and a put option on foreign exchange. The details are set out in Table 18.13. The interest-rate and FX volatility and correlation data may be obtained from RiskMetrics. Using these, we calculate the undiversified VaR in the normal manner, multiplying the market value of the instrument by the volatility value to obtain VaR. For the option, we also multiply the value and the volatility by the delta (that is, $1{,}507{,}000 \times 0.54 \times 6.10\%$). The delta adjustment is required because the price of the option does not "tick-for-tick" with the underlying, but by 0.54 for each unit change in the underlying. The undiversified VaR is 73,753.

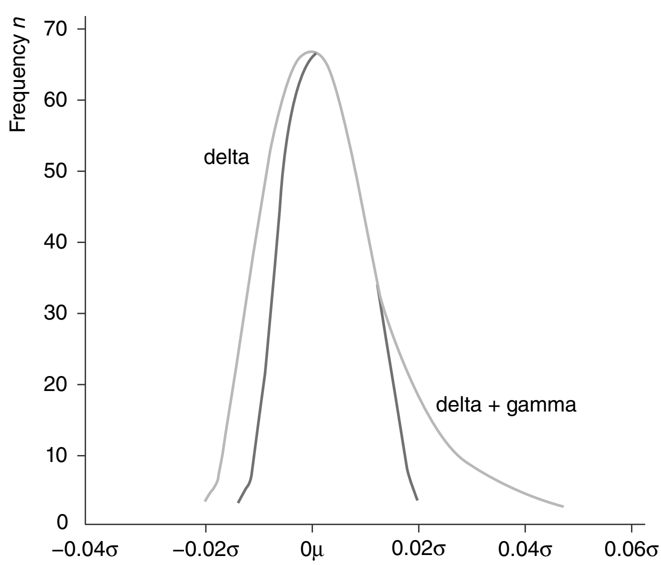

FIGURE 18.4 Delta + Gamma Effect

TABLE 18.13 Hypothetical Portfolio and Undiversified VaR

Bond nominal	2,000,000
Maturity (years)	2
Market value	1,507,000
Volatility	1.60%
Undiversified VaR at 84% C.I.	24,112
Nominal value FX option	1,507,000
Delta	0.54
Gamma	3.9
FX volatility of spot	6.10%
Undiversified VaR at 84% C.I.	49,641
Correlation coefficient	–0.31

To calculate the undiversified VaR we require the portfolio variance, which would normally be done in the conventional way using matrices; here there are only two assets so we may use the standard variance equation (18.1). The square root of this is the VaR, which is calculated as:

$$\begin{aligned} VaR_{port} &= \sqrt{24{,}112^2 + 49{,}641^2 + [(2 \times 24{,}112 \times 49{,}641 \times (-0.31)]} \\ &= 41{,}498 \end{aligned}$$

Although the diversified VaR is more realistic a measure, it will not take into account the gamma effect of the option. Previously, we allowed for the delta of the option, which was used to modify the volatility level, which changed from 6.10% to 3.294%. The gamma adjustment is made by using equation (18.5), which in this case gives a gamma adjustment of 0.7256%. The delta-gamma approximation for the volatility is therefore 2.568%. Multiplying this by the weighting (the option value) we have a new diversified VaR for the option of 38,700. If we use the same portfolio variance equation we obtain a delta-gamma adjusted diversified VaR of 27,488.

The delta-gamma adjustment is only an approximation of an option book's gamma risk exposure, and it is not as close as a convexity adjustment. This is due mainly to the unpredictable behaviour of gamma as an option approaches maturity, more so if it is at-the-money.

STRESS TESTING

Risk-measurement models and their associated assumptions are not without limitation. It is important to understand what will happen should some of the model's underlying assumptions break down. Stress testing is a process whereby a series of scenario analyses or simulations are carried out to investigate the effect of extreme market conditions on the VaR estimates calculated by a model. It is also an analysis of the effect of violating any of the basic assumptions behind a risk model. If carried out efficiently, stress testing will provide clearer information on the potential exposures at risk due to significant market corrections, which is why the Basel Committee recommends that it be carried out.

Simulating Stress

There is no standard way to undertake stress testing. It is a means of experimenting with the limits of a model. It is also a means to measure the residual risk that is not effectively captured by the formal risk model, thus complementing the VaR framework. If a bank uses a confidence interval of 99% when calculating its VaR, the losses on its trading portfolio due to market movements should not exceed the VaR number on more than one day in 100. For a 95% confidence level the corresponding frequency is one day in 20, or roughly one trading day each month. The question to ask is "What are the expected losses on those days?" Also, what can an institution do to protect itself against these losses? Assuming that returns are normally distributed provides a workable daily approximation for estimating risk, but when market moves are more extreme these assumptions no longer add value. The 1% of market moves that are not used for VaR calculations include events such as the October 1987 crash, the bond market collapse of February 1994, and the Mexican peso crisis at the end of 1994. In these cases, market moves were much larger than any VaR model could account for; in fact, the October 1987 crash was a 20 standard deviation move. Under these circumstances, correlations between markets also increase well above levels normally assumed in models.

An approach used by risk managers is to simulate extreme market moves over a range of different scenarios. One method is to use Monte Carlo simulation. This allows dealers to push the risk factors to greater limits. For example, a 99% confidence interval captures events up to 2.33 standard deviations from the mean asset-return level. A risk manager can calculate the effect on the trading portfolio of a 10 standard deviation move. Similarly, risk managers may want to change the correlation assumptions under which they normally work. For instance, if markets all move down together, something that happened in Asian markets from the end of 1997 and emerging markets generally from July 1998 after the Russian bond technical default, losses will be greater than if some markets are offset by other negatively correlated markets.

Only by pushing the bounds of the range of market moves that are covered in the stress-testing process can financial institutions have an improved chance of identifying where losses might occur and, therefore, a better chance of managing their risk effectively.

Stress Testing in Practice

For effective stress testing, a bank has to consider nonstandard situations. The Basel policy group has recommended certain minimum standards in respect of specified market movements. The parameters chosen are considered large moves to overnight marks, and include:

- Parallel yield-curve shifts of 100 basis points up and down
- Steepening and flattening of the yield curve (2-year to 10-year) by 25 basis points

- Increase and decrease in three-month yield volatilities by 20%
- Increase and decrease in equity index value by 10%
- Increase and decrease in swap spread by 20 basis points

These scenarios represent a starting point for a framework for routine stress testing.

Banks agree that stress testing must be used to supplement VaR models. The main problem appears to be difficulty in designing appropriate tests. The main issues are:

- Difficulty in "anticipating the unanticipated"
- Adopting a systematic approach, with stress testing carried out by looking at past extremes and analysing the effect on the VaR number under these circumstances
- Selecting 10 scenarios based on past extreme events and generating portfolio VaRs based on reruns of these scenarios

The latest practice is to adapt stress tests to suit the particular operations of a bank itself. On the basis that one of the main purposes of stress testing is to provide senior management with accurate information of the extent of a bank's potential risk exposure, more valuable data will be gained if the stress test is particularly relevant to the bank. For example, an institution such as Standard Chartered Bank, which has a relatively high level of exposure to exotic currencies, may design stress tests that take into account extreme movements in, say, regional Asian currencies. A mortgage book holding option positions only to hedge its cash book—say, one of the former UK building societies that subsequently converted to banks—may have no need for excessive stress testing on, perhaps, the effect of extreme moves in derivatives liquidity levels.

Issues in Stress Testing

It is to be expected that extreme market moves will not be captured in VaR measurements. The calculations will always assume that the probability of events such as the Mexican peso devaluation is extremely low when analysing historical or expected movements of the currency. Stress tests need to be designed to model for such occurrences. Back-testing a firm's qualitative and quantitative risk-management approach for actual extreme events often reveals the need to adjust reserves, increase the VaR factor, adopt additional limits and controls, and expand risk calculations. With back-testing, a firm will take, say, its daily VaR number, which we will assume is computed to 95% degree of confidence. The estimate will be compared to the actual trading losses suffered by the book over a 20-day period, and, if there is a significant discrepancy, the firm will need to go back to its model and make adjustments to parameters. Frequent and regular back-testing of the VaR model's output with actual trading losses is an important part of stress testing. To conduct back-testing efficiently, a firm would need to be able to strip out its intraday profit-and-loss figures, so it could compare the actual change in P/L to what was forecast by the VaR model.

The procedure for stress testing in banks usually involves:

- Creating hypothetical extreme scenarios
- Computing corresponding hypothetical P/Ls

One method is to imagine *global* scenarios. If one hypothesis is that the euro appreciates sharply against the dollar, the scenario needs to consider any related areas, such as the effect, if any, on the Swiss franc and Norwegian krone rate, or the effect on the yen and interest rates. Another method is to generate many *local* scenarios and so consider a few risk factors at a time. For example, given an FX option portfolio, a bank might compute the hypothetical P/L for each currency pair under a variety of exchange rate and implied volatility scenarios. There is then the issue of amalgamating the results: one way would be to add the worst-case results for each of the subportfolios, but this ignores any portfolio effect and cross-hedging. This may result in an overestimate that is of little use in practice.

Nevertheless, stress testing is one method to account for the effect of extreme events that occur more frequently than would be expected were asset returns to follow a true normal distribution. For example, five-standard-deviation moves in a market in one day have been observed to occur twice every 10 years or so, which is considerably more frequent than given by a normal distribution. Testing for the effects of such a move gives a bank an idea of its exposure under these conditions.

VALUE-AT-RISK METHODOLOGY FOR CREDIT RISK

The credit crisis in 2007 was a painful stress test of the way the world models credit risk. Risk models used by banks for the purpose of setting capital buffers to protect against credit risks were found to be wholly inadequate, and underestimated the extent of actual losses experienced by banks. Before the credit crisis broke, the nature of the various facets of credit risk were not fully understood nor appreciated by modellers. Most bankers are aware that a credit-risky instrument is exposed to credit spread risk and default risk (today we know there are other less obvious but equally devastating risks), but the work to combine the two has been a challenge. As a start, they have very different risk horizons: spreads are marked daily in the trading book, but default risk is seen more as a banking book phenomenon affecting portfolio of loans over the course of a business year. This could be the reason why precrisis the Basel capital charge for traded credit dealt only with spread VaR, while most of the portfolio credit risk models were originally developed for the banking book; there was no requirement under pillar I of Basel rules for a default risk capital then.

The bull market leading up to the crisis (2000–2007) saw the growth in innovation of credit products. Banks were monetising and securitising credit risks originated from bonds and mortgage pools, and selling them to investors as structured products (CDS, CLNs, etc.). The exponential growth in this space caused a compression in credit spreads generally, that led to even more leveraged form of securitisation (CDOs) to chase after diminishing yields. This boom created proved unsustainable and the bubble burst when the U.S. subprime mortgage fiasco caused the default of a few financial institutions and triggered a general loss of confidence in the markets.

The subsequent crisis revealed a lot about the true nature of credit risk. Nobody anticipated the extent of the carnage; the interconnectedness of the global banking system was brought home: fears of counterparty defaults and the hoarding of liquidity conspired to cause the worst crisis since the Great Depression (1929). After the crisis, regulatory reform and Basel III required that the modelling of liquidity risk and counterparty risk also be key elements of the credit risk complex. The interplay of these risks is a topic of much research and development in Basel risk modelling. For example, Basel III now requires the modelling of an Incremental Risk Charge (IRC) and counterparty risk VaR (or CVA VaR). But first, let's talk about the most basic form of credit risk.

Credit Risk

There are two main types of credit risk:

1. Credit spread risk
2. Credit default risk

Credit-Spread Risk Credit spread is the excess premium, over and above government or risk-free risk, required by the market for taking on a certain assumed credit exposure. Credit-spread risk is the risk of financial loss resulting from changes in the level of credit spreads used in the marking-to-market of a product. It is exhibited by a portfolio for which the credit spread is traded and marked. Changes in observed credit spreads affect the value of the portfolio and can lead to losses for investors.

Credit Default Risk This is the risk that an issuer of debt (obligor) is unable to meet its financial obligations. Where an obligor defaults, a firm generally incurs a loss equal to the amount owed by the obligor less any recovery amount that the firm recovers as a result of foreclosure, liquidation, or restructuring of the defaulted obligor. All portfolios of exposures exhibit credit default risk.

The value-at-risk measurement methodology was first applied for credit risk by JPMorgan, which introduced the CreditMetrics tool in 1995. The measurement of credit risk requires a slightly different approach from that used for market risk, because the distribution of credit losses follows a different pattern. In the following sections we describe the approach used to measuring such risk.

Modelling Credit Risk

Credit-risk VaR methodologies take a portfolio approach to credit-risk analysis. This means that:

- Credit risks to each obligor across the portfolio are restated on an equivalent basis and aggregated in order to be treated consistently, regardless of the underlying asset class.
- Correlations of credit-quality moves across obligors are taken into account.

This allows portfolio effects—the benefits of diversification and risks of concentration—to be quantified.

The portfolio risk of an exposure is determined by four factors:

1. Size of the exposure
2. Maturity of the exposure
3. Probability of default of the obligor
4. Systematic or concentration risk of the obligor

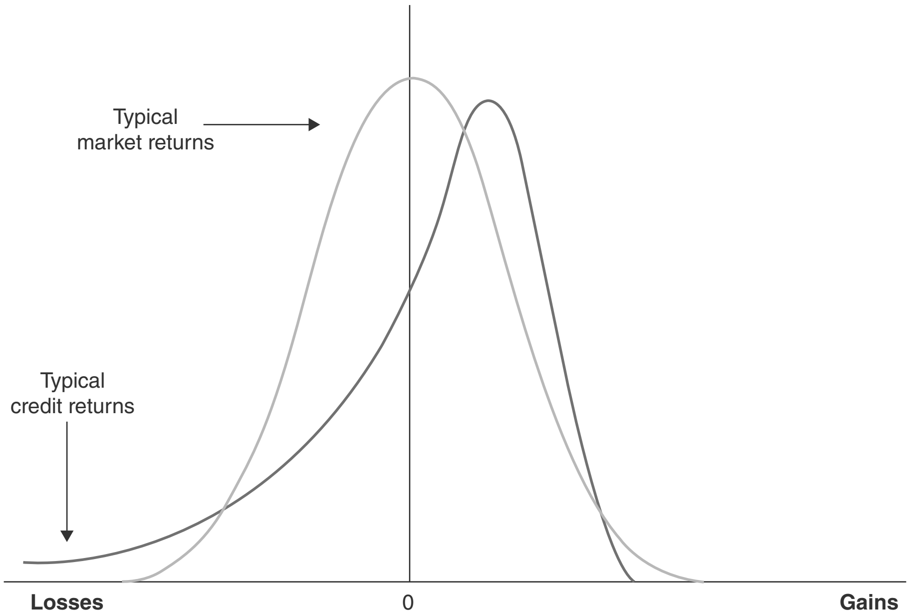

FIGURE 18.5 Comparison of Distribution of Market Returns and Credit Returns

Credit VaR, like market-risk VaR, considers (credit) risk in a mark-to-market framework. It arises from changes in value due to credit events—that is, changes in obligor credit quality including defaults, upgrades, and downgrades.

Nevertheless, credit risk is different in nature from market risk. Typically, market-return distributions are assumed to be relatively symmetrical and approximated by normal distributions. In credit portfolios, value changes will be relatively small upon minor up/downgrades, but can be substantial upon default. This remote probability of large losses produces skewed distributions, with heavy downside tails that differ from the more normally distributed returns assumed for market VaR models. This is shown in Figure 18.5.

This difference in risk profiles does not prevent us from assessing risk on a comparable basis. Analytical market-VaR models consider a time horizon and estimate value-at-risk across a distribution of estimated market outcomes. Credit VaR models similarly look to a horizon and construct a distribution of value given different estimated credit outcomes.

When modelling credit risk the two main measures of risk are:

1. *Distribution of loss:* Obtaining distributions of loss that may arise from the current portfolio. This considers the question of what the expected loss is for a given confidence level.
2. *Identifying extreme or catastrophic outcomes:* This is addressed through the use of scenario analysis and concentration limits.

To simplify modelling, no assumptions are made about the causes of default. Mathematical techniques used in the insurance industry are used to model the event of an obligor default.

Time Horizon The choice of time horizon will not be shorter than the time frame over which risk-mitigating actions can be taken. Practitioners originally suggested two alternatives:

1. A constant time horizon such as one year
2. A hold-to-maturity time horizon

The constant time horizon is similar to the CreditMetrics approach and also to that used for market-risk measures. It is more suitable for trading desks. The hold-to-maturity approach is used by institutions such as portfolio managers.

Data Inputs Modelling credit risk requires certain data inputs. For example, CreditRisk + uses the following:

- Credit exposures
- Obligor default rates
- Obligor default-rate volatilities
- Recovery rates

These data requirements present some difficulties. There is a lack of comprehensive default and correlation data, and assumptions need to be made at certain times. The most accessible data is compiled by the credit ratings agencies such as Moody's.

We now consider the CreditMetrics model for measuring credit value at risk.

CreditMetrics

CreditMetrics was JPMorgan's portfolio model for analysing credit risk, and provides an estimate of value-at-risk due to credit events caused by upgrades, downgrades, and default. A software package known as CreditManager is available that allows users to implement the CreditMetrics methodology.

Methodology There are two main frameworks in use for quantifying credit risk. One approach considers only two states: default and no default. This model constructs a binomial tree of default versus no default outcomes until maturity. This approach is shown in Figure 18.6.

The other approach, sometimes called the RAROC (Risk-Adjusted Return on Capital) approach holds that risk is the observed volatility of corporate-bond values within each credit-rating category, maturity band, and industry grouping. The idea is to track a benchmark corporate bond (or index) that has observable pricing. The resulting estimate of volatility of value is then used to proxy the volatility of the exposure (or portfolio) under analysis.

The CreditMetrics methodology sits between these two approaches. The model estimates portfolio VaR at the risk horizon due to credit events that include upgrades and downgrades, rather than just defaults. Thus, it adopts a mark-to-market framework. As shown in Figure 18.7, bonds within each credit rating category have volatility of value due to day-to-day credit-spread fluctuations. The figure shows the loss distributions for bonds of varying credit quality. CreditMetrics assumes that all credit migrations have been realised, weighting each by a migration likelihood.

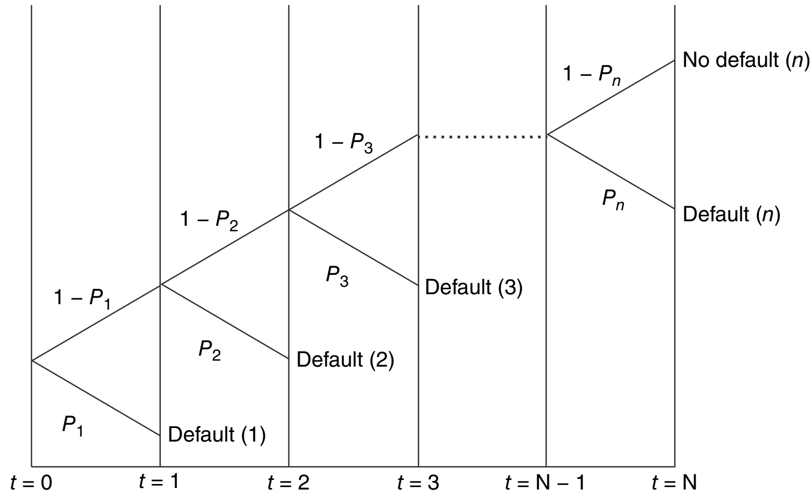

FIGURE 18.6 A Binomial Model of Credit Risk

Time Horizon CreditMetrics adopts a one-year risk horizon. The justification given in its technical document[4] is that this is because much academic and credit agency data is stated on an annual basis. This is a convenient convention similar to the use of annualised interest rates in the money markets. The risk horizon is adequate as long as it is not shorter than the time required to perform risk-mitigating actions. Users must therefore adopt their risk-management and risk-adjustment procedures with this in mind.

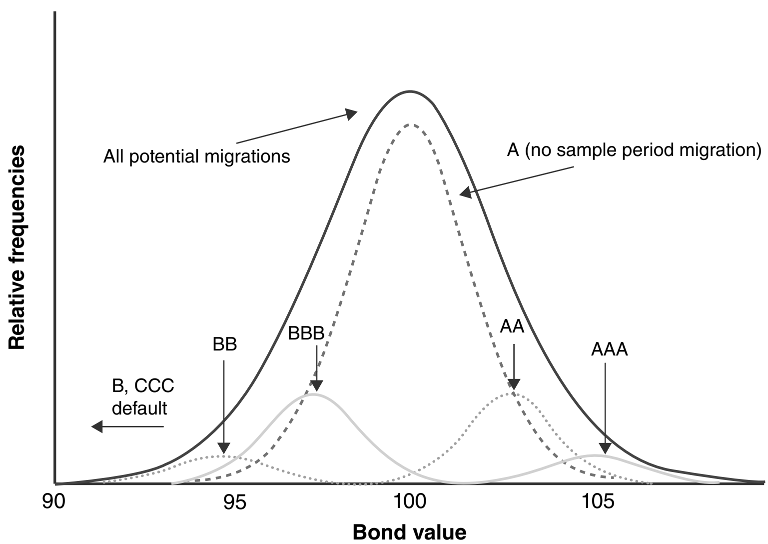

FIGURE 18.7 Distribution of Credit Returns by Rating

[4] JPMorgan, *Introduction to CreditMetrics* (JPMorgan & Co., 1997).

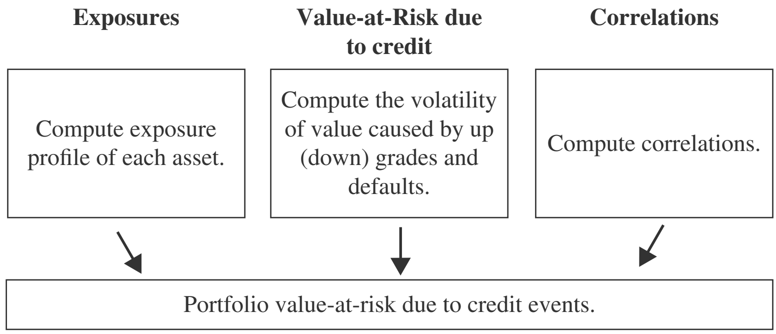

FIGURE 18.8 Analytics Road Map for CreditMetrics
Source: JPMorgan 1997.

The steps involved in CreditMetrics measurement methodology are shown in Figure 18.8, described by JPMorgan as its analytical "road map".

The elements in each step are:

Exposures

- User portfolio
- Market volatilities
- Exposure distributions

VaR due to credit events

- Credit rating
- Credit spreads
- Rating change likelihood
- Recovery rate in default
- Present-value bond revaluation
- Standard deviation of value due to credit-quality changes

Correlations

- Ratings series
- Models (for example, correlations)
- Joint credit-rating changes

Calculating the Credit VaR CreditMetrics methodology assesses individual and portfolio VaR due to credit in three steps:

- *Step 1:* It establishes the exposure profile of each obligor in a portfolio.
- *Step 2:* It computes the volatility in value of each instrument caused by possible upgrade, downgrade, and default.
- *Step 3:* Taking into account correlations between each of these events, it combines the volatility of the individual instruments to give an aggregate portfolio risk.

Step 1—Exposure Profiles CreditMetrics incorporates the exposure of instruments such as bonds (fixed- or floating-rate) as well as other loan commitments and market-driven instruments such as swaps. The exposure is stated on an equivalent basis for all products. Products covered include:

- Receivables (or trade credit)
- Bonds and loans
- Loan commitments
- Letters of credit
- Market-driven instruments

Step 2—Volatility of Each Exposure from Up(Down)grades and Defaults The levels of likelihood are attributed to each possible credit event of upgrade, downgrade, and default. The probability that an obligor will change over a given time horizon to another rating is calculated. Each change (migration) results in an estimated change in value (derived from credit-spread data and—in default—recovery rates). Each value outcome is weighted by its likelihood to create a distribution of value across each credit state, from which each asset's expected value and volatility (standard deviation) of value are calculated.

There are three steps to calculating the volatility of value in a credit exposure:

1. The senior unsecured credit rating of the issuer determines the chance of either defaulting or migrating to any other possible credit-quality state in the risk horizon.
2. Revaluation at the risk horizon can be by either (i) the seniority of the exposure, which determines its recovery rate in case of default; or (ii) the forward zero-coupon curve (spot curve) for each credit-rating category, which determines the revaluation upon up(down)grade.
3. The probabilities from the two steps above are combined to calculate volatility of value due to credit-quality changes.

Step 3—Correlations Individual value distributions for each exposure are combined to give a portfolio result. To calculate the portfolio value from the volatility of individual asset values requires estimates of correlation in credit-quality changes. CreditMetrics itself allows for different approaches to estimating correlations, including a simple constant correlation. This is because of frequent difficulty in obtaining directly observed credit-quality correlations from historical data.

The entire process is illustrated in Example 18.5.

Applications of Credit VaR

One purpose of a risk-management system is to direct and prioritise actions. When considering risk-mitigating actions, there are various features of risk worth targeting, including obligors having:

- The largest absolute exposure
- The largest percentage level of risk (volatility)
- The largest absolute amount of risk

EXAMPLE 18.5

An example of calculating the probability step is illustrated in Figure 18.9. The probabilities of all possible credit events on an instrument's value must be established first. Given this data, the volatility of value due to credit-quality changes for this one position can be calculated. The process is shown in Figure 18.9.

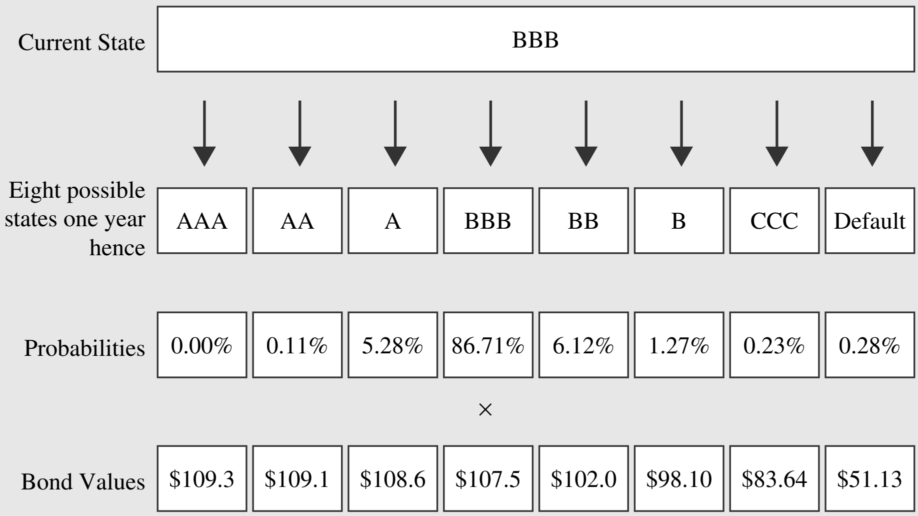

FIGURE 18.9 Constructing the Distribution Value for a BBB-Rated Bond
Source: JPMorgan, 1997.

A CreditMetrics-type methodology helps to identify these areas and allow the risk manager to prioritise risk-mitigating action.

Exposure Limits Within bank trading desks, credit-risk limits are often based on intuitive, but arbitrary, exposure amounts. This is not a logical approach because resulting decisions are not risk-driven. Limits should ideally be set with the help of a quantitative analytical framework.

Risk statistics used as the basis of VaR methodology can be applied to limit setting. Ideally, such a quantitative approach should be used as an aid to business judgment and not as a standalone limit-setting tool.

A credit committee considering limit setting can use several statistics such as marginal risk and standard deviation or percentile levels. Figure 18.10 illustrates how marginal risk statistics can be used to make credit limits sensitive to the trade-off between risk and return. The lines on Figure 18.10 represent risk/return trade-offs for different credit ratings, all the way from AAA to BBB. The diagram shows how marginal contribution to portfolio risk increases geometrically with exposure size of an individual obligor, noticeably so for weaker credits. To maintain a constant

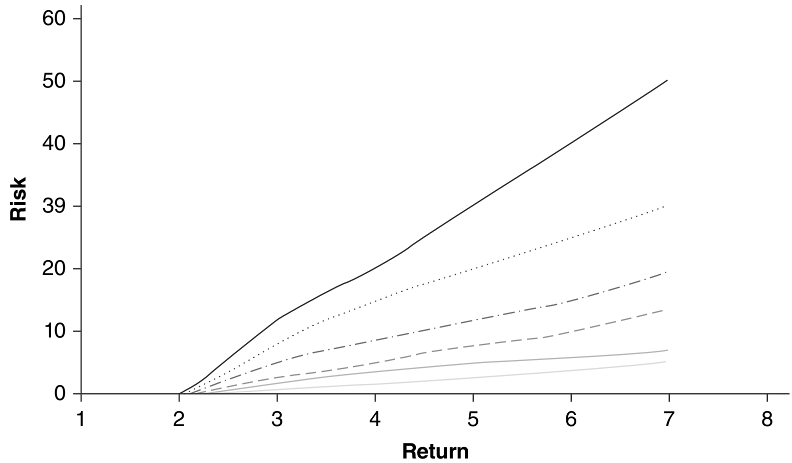

FIGURE 18.10 Size of Total Exposure to Obligor—Risk/Return Profile

balance between risk and return, proportionately more return is required with each increment of exposure to an individual obligor.

Standard Credit-Limit Setting In order to equalise a firm's risk appetite between obligors as a means of diversifying its portfolio, a credit-limit system could aim to have a large number of exposures with equal expected losses. The expected loss for each obligor can be calculated as Default rate × (Exposure amount – Expected recovery).

This means that individual credit limits should be set at levels that are inversely proportional to the default rate corresponding to the obligor rating.

Concentration Limits Concentration limits identified by CreditRisk+-type methodologies have the effect of trying to limit the loss from identified scenarios and are used for managing "tail" risk.

Integrating Credit Risk and Market Risk Functions

It is logical for banks to integrate credit-risk and market-risk management for the following reasons:

- The need for comparability between returns on market and credit risk
- The convergence of risk-measurement methodologies
- The transactional interaction between credit and market risk
- The emergence of hybrid-credit and market-risk product structures

The objective is for returns on capital to be comparable for businesses involved in credit and market risk, to aid strategic allocation of capital.

We have shown that as VaR was being adopted as a market-risk measurement tool, the methodologies behind it were steadily applied to the next step along the risk continuum, that of credit risk. Recent market events, such as bank trading losses in emerging markets and the meltdown of the Long Term Capital Management hedge fund in summer 1998, have illustrated the interplay between credit risk and market risk. The ability to measure market and credit risk in an integrated model would allow for a more complete picture of the underlying risk exposure. (We would add that adequate senior management understanding and awareness of a third type of risk—liquidity risk—would almost complete the risk-measurement picture.)

Market-risk VaR measures can adopt one of the different methodologies available; in all of them there is a requirement for the estimation of the distribution of portfolio returns at the end of a holding period. This distribution can be assumed to be normal, which allows for analytical solutions to be developed. The distribution may also be estimated using historical returns. Finally, a Monte Carlo simulation can be used to create a distribution based on the assumption of certain stochastic processes for the underlying variables. The choice of methodology is often dependent on the characteristics of the underlying portfolio, plus other factors. For example, risk managers may wish to consider the degree of leptokurtosis in the underlying asset-returns distribution, the availability of historical data or the need to specify a more sophisticated stochastic process for the underlying assets. The general consensus is that Monte Carlo simulation, while the most IT-intensive methodology, is the most flexible in terms of specifying an integrated market and credit model.

The preceding paragraphs in this section have shown that credit-risk measurement models generally fall into two categories. The first category includes models that specify an underlying process for the default process. In these models, firms are assumed to move from one credit rating to another with specified probabilities. Default is one of the potential states that a firm could move to. The CreditMetrics model is of this type. The second type of model requires the specification of a stochastic process for firm value. Here, default occurs when the value of the firm reaches an externally specified barrier. In both models, when the firm reaches default, the credit exposure is impacted by the recovery rate. Again, market consensus would seem to indicate that the second type of methodology, the firm value model, most easily allows for development of an integrated model that is linked not only through correlation but also the impact of common stochastic variables.

Basel III Requirement: The Incremental Risk Charge (IRC)

Under the old Basel II, a bank's specific risk capital is computed purely by looking a credit spreads fluctuation and disregarding default risk. The incremental risk charge (IRC) was introduced by Basel 2.5 (a precursor to Basel III) in recognition that the default risk of illiquid credit products is not well-reflected under Basel II VaR. Basel stipulated that the IRC should capture default risks, rating migration risks and default correlation risks at a horizon of one year at the 99.9% confidence level (in contrast to credit spread VaR, which captures continuous spread movement over a 10-day horizon at the 99% confidence level). Basel recommended high-level principles for the IRC without prescribing specific models. For example, the IRC should have a soundness standard comparable to the IRB (internal ratings based) approach already used in the banking book to avoid possible "regulatory arbitrage" between the trading and banking books. Since the IRB model is based on a one-factor Gaussian copula model, banks naturally develop the IRC models in a similar vein. Another requirement is for the modelling of cross-correlation between default, migration, and other risk factors.

A novel feature is the introduction of multiple time horizons as a way to address liquidity risk. Firstly, the IRC must assume a constant level of risk over a one-year risk horizon. The reason: banks can seldom respond efficiently in the short-term when in distress, so a one-year period seems appropriate for crisis response. Also, consistent with a "going concern" view of a bank, the bank must continue to trade to support its income-producing activities despite losses—this suggests the concept of rebalancing of positions to maintain a constant level of risk.

Secondly, the IRC must differentiate between credit products of different liquidity by assigning them a differing liquidity horizon. In theory, this horizon represents the shortest amount of time taken to sell off/hedge a position in a stressed market without adversely affecting prices. In practice, the liquidity horizon for a product is determined subjectively. The liquidity horizon must be at least three months by Basel rule. A natural way to implement the IRC is via a multistep simulation approach (in quarterly time steps) where products are grouped into 3-, 6-, 9-, and 12-month horizons. This classification is typically done in consultation with traders.

To model default, it is convenient to use a Merton-style approach whereby default occurs when the asset (for a specific issuer) falls below some threshold. For the purpose of this simulation, the asset return is modelled using a one-factor Gaussian copula model. The correlation parameter is typically difficult to calibrate from the market because this is a multistep approach (where such observations are seldom observed), so typically, the regulator stipulates a conservative number to use.

The IRC must also reflect optionality if the credit trading book contains credit options. This is often implemented using a gamma adjustment. A bank must also ensure that concentration risks are captured; in practice, this means that the risk-factor mapping must be granular enough (at the very least spanning the dimensions of ratings, sector, region) to capture any lumpy positions that a bank may hold. At the time of this writing, most major banks would have implemented an approved IRC model according to the implementation timeline set out by global regulators.

APPENDIX 18.1 ASSUMPTION OF NORMALITY

The RiskMetrics assumption of conditional multivariate normality is open to criticism that financial series tend to produce fat tails (leptokurtosis). That is, in reality there is a greater occurrence of non-normal returns than would be expected for a purely normal distribution. This is shown in Figure A18.1. There is evidence that fat tails are a problem for calculations. The RiskMetrics technical document defends its assumptions by pointing out that if volatility changes over time there is a greater likelihood of incorrectly concluding that the data is not normal when in fact it is. In fact, conditional distribution models can generate data that possesses fat tails.

Higher Moments of the Normal Distribution

The skewness of a price data series is measured in terms of the third moment about the mean of the distribution. If the distribution is symmetric, the skewness is zero. The measure of skewness is given by

$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\overline{x})^3}{S^3} \qquad \text{(A18.1.1)}$$

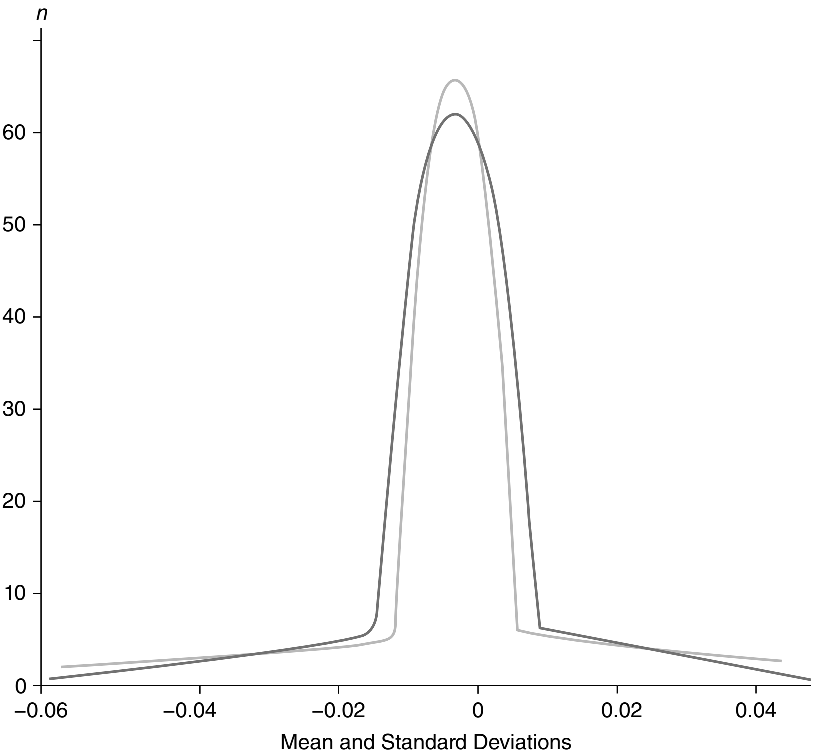

FIGURE A18.1 Leptokurtosis

where x is the return variable, n is the number of observations, and S is the standard deviation of x.

The kurtosis describes the extent of the peak of a distribution; that is, how peaked it is. It is measured by the fourth moment about the mean. A normal distribution has a kurtosis of three. The kurtosis is given by

$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{S^4} \tag{A18.1.2}$$

Distributions with a kurtosis higher than three are commonly observed in asset-market prices and are called leptokurtic. A leptokurtic distribution has higher peaks and fatter tails than the normal distribution. A distribution with a kurtosis lower than three is known as platykurtic.

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CHAPTER 19

Government Bond Analysis, the Yield Curve, and Relative-Value Trading

This chapter examines a number of issues relevant to participants in the sovereign fixed-income markets. As it is based on AAA-rated government trading experience, the analysis is confined to generic bonds that are for practical purposes default-risk free. There is no consideration of factors that would apply to corporate bonds, asset- and mortgage-backed bonds, convertibles, and other nonvanilla securities, and issues such as credit risk or prepayment risk.

The chapter is broken down as follows: first we consider the redemption yield and duration. This is followed by observations on the implied spot rate and market zero-coupon yields, relative-value trading, and butterfly trades.

THE DETERMINANTS OF YIELD

The yield at which a fixed-interest security is traded is determined by the market. This determination is a function of three factors: the term-to-maturity of the bond, the liquidity of the bond, and its credit quality. Government securities such as Treasuries are default free, and so this factor drops out of the equation. Under "normal" circumstances, the yield on a bond is higher the greater its maturity, reflecting both the expectations-hypothesis and liquidity-preference theories. Intuitively, we associate higher risk with longer-dated instruments, for which investors must be compensated in the form of higher yield. This higher risk reflects greater uncertainty with longer-dated bonds, both in terms of default and future inflation and interest-rate levels. However, for a number of reasons the yield curve assumes an inverted shape and long-dated yields become lower than short-dated ones.[1] Long-dated yields generally are expected to be less volatile over time than short-dated yields. This is mainly because incremental changes to economic circumstances or other technical considerations generally have an impact for only short periods of time, which affects the shorter end of the yield curve to a greater extent.

The liquidity of a bond also influences its yield level. The liquidity may be measured by the size of the bid-offer spread, the ease with which the stock may be transacted in size, and the impact of large-size bargains on the market. It is also measured by the extent of any specialness in its repo rate. Supply and demand for an individual stock, and the amount of stock available to trade are the main drivers of

[1] For a summary of term-structure theories, see Choudhry (2001, Chapter 6).

liquidity.[2] The general rule is that there is a yield premium for transacting business in lower-liquidity bonds.

In the analysis that follows, we assume satisfactory levels of liquidity and that it is straightforward to deal in large sizes without moving the market.

PRACTICAL USES OF REDEMPTION YIELD AND DURATION

The drawbacks of the traditional gross-redemption yield (henceforth referred to simply as "yield") and duration measures for bond analysis are well documented. That different bonds, even vanilla government securities, can have their yield measured in a number of ways hints, however, at the lack of a satisfactory acceptable measure of return. When assessing the opportunities available in a market, investors will often use the market convention for yield in their analysis. However, the different methods available to calculate yield mean that the comparison of rates of return for different bonds becomes problematic (for example, see Figure 19.1 for the alternative yield measures available for a U.S. Treasury security on the Bloomberg "YA" page).

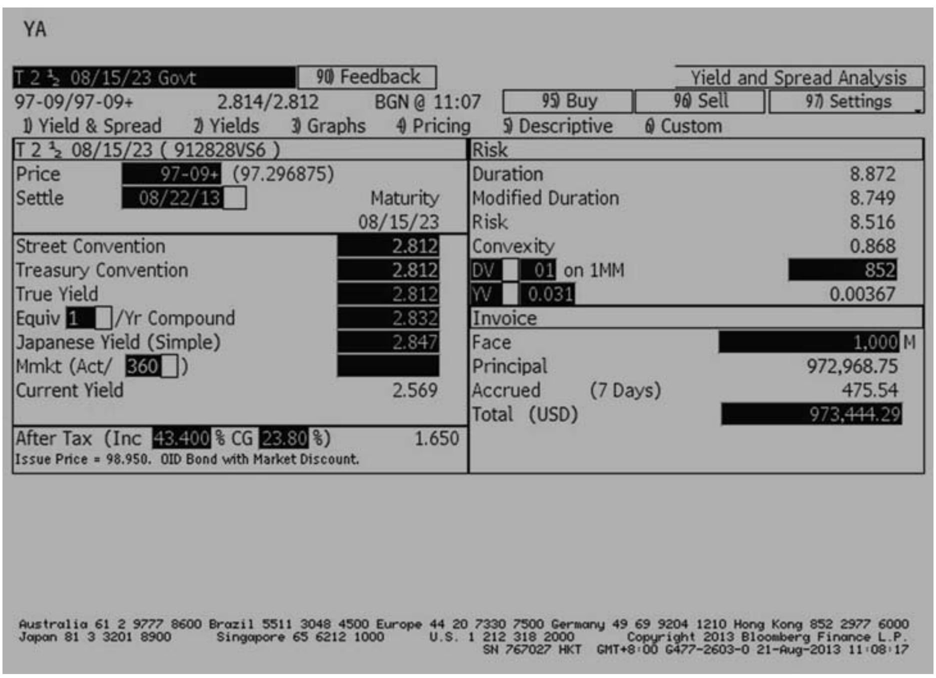

FIGURE 19.1 Yield Calculations for Treasury 2.5 Percent 2023 on Bloomberg "YA" Page

[2] The amount of stock issued and the amount of stock available to trade are not the same thing. If a large amount of a particular issue has been locked away by institutional investors, this may impede liquidity. However, the existence of a large amount at least means that some of the paper may be made available for lending in the stock loan and repo markets. A small issue size is a good indicator of low liquidity.

Duration is another measure that can be defined in more than one way, again making comparison between different bonds an issue.

In this section, we look at how the problems inherent in using yield and duration can be mitigated, and how the analysis should proceed when doing this.

The Concept of Yield

Ideally the yield of an instrument should measure the return achieved from holding it; this would make it a function of the value of the initial investment, the period of the investment and the value of the matured investment. If investment income received during the period is capitalised, this should also be captured by the yield measure, so that we are measuring compounded interest. Under these properties, a yield on a simple instrument such as a T-bill can be defined as shown now.

In the sterling market, the fixed-income interest basis is semiannual. With this in mind, consider a T-bill with a maturity of m days and a price of P. The true yield rm of the bill is given by

$$P = \frac{100}{(1 + \frac{1}{2} rm)n} \tag{19.1}$$

where n is the number of interest periods from value date until maturity. On an actual/365 basis, one interest period in the gilt market is a half-year or 182.5 days;[3] so, for a 90-day T-bill priced at 98.379 the true yield can be computed by solving

$$98.379 = \frac{100}{(1 + \frac{1}{2} rm)^{90/182.5}}$$

which gives rm equal to 0.067389 or 6.739%. That said, the yield quoted on T-bills, when computed using the market price, will often differ from the "true" yield. This is because often yield quotes for bills assume simple interest, rather than compound interest in their calculation. Nevertheless, we are interested in this definition of the true yield because we wish to apply it to longer-dated coupon bonds. Readers are familiar with the definition of bond yield, which involves discounting all the bond's cash flows (its coupons) at a uniform interest rate; yield here is then that interest rate that equates the sum of all the discounted cash flows equal to the observed market price. Let us consider this further.

With a vanilla bond such as a gilt, there are m future cash flows of value C, which are the bond interest payments calculated as one-half of the bond coupon rate. The ith payment is termed C_i, with m_i being, the days from value date to maturity. The number of interest periods from today to a cash-flow date or to maturity is simply the days divided by 182.5 and so is denoted $n_i = m_i/182.5$. At a discount interest rate of r, the value of the bond's discounted cash flows would be given by (19.2).

$$PV = \frac{C_1}{(1 + \frac{1}{2} r)^{n_1}} + \frac{C_2}{(1 + \frac{1}{2} r)^{n_2}} + \ldots\ldots + \frac{C_m}{(1 + \frac{1}{2} r)^{n_m}} \tag{19.2}$$

[3] The accrued basis in the gilt market is now actual/actual and not actual/365.

This leads easily to the true yield definition for the bond, which is the discount interest rate that equates the current market price of the bond to the discounted value of the cash flows. The market price is the dirty price, which includes accrued interest. We replace *PV* in (19.2) with *P* plus accrued interest, shown in (19.3).

$$P + AI = \frac{C_1}{(1+\frac{1}{2}rm)^{n_1}} + \frac{C_2}{(1+\frac{1}{2}rm)^{n_2}} + \ldots\ldots + \frac{C_m}{(1+\frac{1}{2}rm)^{n_m}} \tag{19.3}$$

We now look at how the true yield for a gilt may differ from the quoted yield.

Yield Comparisons

Let's look at a market example. Price quotes are in ticks, or fractions of 32nds. One-half of a tick is denoted by "+". On 10th May 1994, the $10^1/_4$% 1995 gilt, with a maturity date of 21st July 1995, was quoted at a price of 104–28+. To calculate the true yield, and indeed any other yield such as the conventional yield, we use the dirty price of the bond, together with the discounted value of all remaining cash flows.

The bond pays coupon on 21st January and 21st July. For value 11th May 1994, the accrued was 109 days, which means that the accrued interest was

$$(10.25 \times {}^1/_2) \times 109/365$$

or 1.53048. The dirty price of the bond was (104-28+ plus 1.53048) or 106.421105.

The remaining bond cash flows were £5.125 on 21st July 1994 and 21st January 1995, and 105.125 on 21st July 1995. However, 21st January 1995 is a Saturday, so the cash flow was not actually made until Monday, 23rd January. The period from value date to receipt of cash flows is

21st July 1994	71 days
23rd January 1995	230 days
21st July 1995	436 days

The number of interest periods in this term for each cash-flow date was

(71/182.5) or 0.38904

(230/182.5) or 1.26027

(436/182.5) or 2.38904.

Therefore, the true yield is computed as follows:

$$106.421105 = \frac{5.125}{(1+\frac{1}{2}rm)^{0.38904}} + \frac{5.125}{(1+\frac{1}{2}rm)^{1.26027}} + \frac{5.125}{(1+\frac{1}{2}rm)^{2.38904}}$$

which solves to give rm = 0.073241 or 7.324%.

The conventional yield calculation will almost invariably be different. This is because it ignores delays in the receipt of cash flows when payment dates fall on nonbusiness days, and so the number of interest periods between cash flows are deemed to be in exact one-half year amounts. So the conventional or *consortium* yield for the same bond would be calculated as

$$106.421105 = \frac{5.125}{(1+\frac{1}{2}rm)^{0.38904}} + \frac{5.125}{(1+\frac{1}{2}rm)^{1.24932}} + \frac{5.125}{(1+\frac{1}{2}rm)^{2.38904}}$$

which solves to rm = 0.0732577 or 7.325%. This shows the true yield to be 0.001% below the consortium yield. This is because under the consortium-yield method the cash flows are received earlier than in the true-yield treatment. Table 19.1 shows this difference in yields for gilts as of 10th May 1994 for value the next day. Note that the difference in the two measures declines to zero from the 6% 1999 gilt onwards. This is because for longer-dated bonds, the discounting of future cash flows is not materially affected by small delays in the treatment of cash-flow receipts, which are only ever a few days if at all, and so there is no difference when computing the true yield.

Measuring True Return on a Bond

Although we may be satisfied with our true-yield measure, it is not as straightforward as the true-yield measure given for the T-bill earlier. With a single-cash-flow instrument, the actual return generated is straightforward: the maturity value is known, so the return on investment is calculated easily as the increase in value from start to maturity. An investment in a 90-day T-bill at a yield of 5% means that the initial value will increase in value by 5% over 90 days, using semiannual compounding. We are not able to say this with certainty when confronted with a yield quote for a coupon-bearing bond. A 5% yield on a 90-day T-bill is the return associated with a 90-day maturity, and gives no indications (nor does it attempt to) on the value of the bill after, say, 60 days or the yield available for reinvestment on maturity of the bill. Is there a way to view a bond yield in the same fashion as that for a T-bill? Certainly, it would assist investors if there was a way to analyse a bond as if the instrument was a single cash-flow security. This is because investors often buy bonds as

TABLE 19.1 Gilt Yields as of 10th May 1994

Gilt	True Yield%	Consortium Yield	% Difference
10% 15 November 1996	7.171	7.173	0.002
10.5% 21 February 1997	7.379	7.391	0.012
7.25% 30 March 1998	7.780	7.781	0.001
6% 10 August 1999	7.900	7.900	0.000
9% 3 March 2000	8.191	8.191	0.000
7% 6 November 2001	8.220	8.220	0.000
9.75% 27 August 2002	8.484	8.482	-0.002
8% 10 June 2003	8.359	8.359	0.000
6.75% 26 November 2004	8.243	8.243	0.000

assets against liabilities that they are required to discharge on known future dates. If the true return on a bond were known, there would be comfort to the investor that the return achieved would meet requirements. Put very simply, this is the concept of immunisation.

Therefore, if possible we would wish to view bond return in the same way as that for a single cash-flow security: a known initial investment is placed in the market for a set period of time, and the future value of the investment on maturity is calculated. The return on the bond would measure the average rate of capital gain in the investment period. The problem with a bond instrument is that its future value is not known with certainty, and is dependent at the rate at which the investor is able to reinvest interim cash flows; this rate cannot be predicted. There are a number of approaches to get around this problem, which we consider now in simple interest-rate environments.

The first scenario assumes, somewhat unrealistically, a flat yield-curve environment. Further, we expect one movement only in the yield curve, a parallel shift upwards or downwards. A bond instrument is viewed as a package of zero-coupon securities, which would make its theoretical price the sum of the discounted values of each zero-coupon security. This is significant because each cash flow would then be discounted at the interest rate for that specific term, rather than one single "internal rate of return". Under our simple assumptions, we can then measure the return on capital generated from a bond instrument.

The price of the bond, given the redemption yield, is given by (19.3). This is straightforward and conventional. Example 19.1 illustrates the use of this relationship.

EXAMPLE 19.1 CONVENTIONAL BOND PRICING

We have a hypothetical 5% 2002 semiannual coupon bond that matures on 8th December 2002. For value on 8th December 2000, the bond has precisely four interest periods to maturity and no accrued interest; its cash flows are 2.50, 2.50, 2.50, and 102.50. If the yield curve on 7th December is flat at 5% (using semiannual compounding), then the price of the bond is given by

$$
\begin{aligned}
PV + AI &= \frac{c_1}{(1+\frac{1}{2}rm)^{n_1}} + \frac{c_2}{(1+\frac{1}{2}rm)^{n_2}} + \frac{c_3}{(1+\frac{1}{2}rm)^{n_3}} + \frac{c_4}{(1+\frac{1}{2}rm)^{n_4}} \\
&= \frac{2.50}{(1+0.025)^1} + \frac{2.50}{(1+0.025)^2} + \frac{2.50}{(1+0.025)^3} + \frac{2.50}{(1+0.025)^4} \\
&= 2.4390 + 2.3879 + 2.3215 + 92.8600 \\
&= 100.00
\end{aligned}
$$

This is the simplest approach, viewing the bond as a bundle of four zero-coupon securities; however, as the yield curve is flat we can use a uniform discount rate. This is shown and, at a rate of 5% the price of the bond is par.

We now assume a one-off parallel shift in the yield curve to a new level of rm_2, which results in a change in the expected future value of the bond. If there are s interest periods from the value date to a specified "horizon date" this is given by (19.4),

$$P(rm_2, s) = \frac{C_1}{(1+\tfrac{1}{2}rm_2)^{n_1-s}} + \frac{C_2}{(1+\tfrac{1}{2}rm_2)^{n_2-s}} + \ldots\ldots + \frac{C_m}{(1+\tfrac{1}{2}rm_2)^{n_m-s}} \tag{19.4}$$

which expresses the fact that the ith cash flow contributes

$$\frac{C_i}{(1+rm_2)^{n_i-s}}$$

to the future value of the bond. If this cash flow is received ahead of the horizon date, then $n_i - s$ will have a negative value, which means we compound C_i to the horizon date at the discount rate rm_2. Otherwise $n_i - s$ will be positive, and C_i is discounted backwards to the horizon date at the same rate. So, the same rate rm_2 is used to compound or discount payments received ahead of or after the horizon date.

Remember, we wish to measure the rate of return, or at best approximate it, for a bond. This return is influenced by changes in the yield curve after initial purchase. Let us consider this now. If we have a short-term horizon period and thus small s, a majority of the bond's cash flows will take place after the horizon date. A higher rate rm_2 will produce a lower future value $P(rm_2,s)$. The opposite occurs if s is large; most of the cash flows will then take place before the horizon date, and a higher value rm_2 will actually increase the value of the cash flows, as they can be reinvested at a higher rate of interest. If the value of s is sufficiently high, this reinvestment gain may match and exceed the loss suffered due to revaluation of the bond at the higher rate, which would result in a higher future value $P(rm_2,s)$. The reverse effect can occur if $rm_2 < rm$. This is the reinvestment risk borne by the bondholder.

There is a horizon date in between the short-term and long-term dates where the net effect of the change in reinvestment rate on the bond's future value is close to zero. At this horizon date, the bond behaves like a single-cash-flow or zero-coupon security, and so its future value can be predicted with more certainty, irrespective of the change in the yield curve after we have purchased it. Let us call this horizon date s_H. At this date the bond will resemble a zero-coupon security that has s_H interest periods to maturity. We denote P_H as the future value of the bond s_H periods after the purchase date. It can be shown that the rate of return on this bond up to this specific horizon date is the value for rm_H that solves (19.5).

$$P + AI = \frac{P_H}{(1+\tfrac{1}{2}rm_H)^{s_H}} \tag{19.5}$$

In (19.5) the market price is given as clean price plus accrued interest, as before. In fact, it can be shown that the return rm_H is identical to the initial yield value rm.

The value that will work for a stable future value is, in fact, the bond's Macaulay duration value. At this point, under the restrictive assumptions we have listed, the effect of a change in yields will not affect the future value of the bond. At this point, the bond's cash flows are said to be immunised, and the instrument could be used to match a liability that existed at that date.

The restrictions that we imposed limit the usefulness of the analysis. We assume one change only in rates after purchase, and this change is uniform across the term structure. Therefore, in practice an investor must continually adjust his portfolio to maintain its immunisation properties.

IMPLIED SPOT RATES AND MARKET ZERO-COUPON YIELDS[4]

That the duration measure has a number of limitations when used in portfolio management is well known. There are further considerations that limit its use. For instance, the limiting value on duration means that most gilts have duration measures of under 12 years.[5] This makes portfolio immunisation difficult when liabilities are very long dated. The need for constant rebalancing of the portfolio also adds to the problem. However, investors in zero-coupon bonds do not face these problems, which makes such bonds potentially very attractive securities for investors. Zero-coupon bonds do not suffer from the duration-based drawbacks of coupon bonds, as their duration value is identical to their term to maturity. This makes portfolio matching easier. A five-year zero-coupon bond has a duration of five years when purchased, and after two years its duration will be three years, irrespective of the change in interest rates in that time. A long-dated zero-coupon bond can then be used to match a long-dated investment liability. However, how should the yield on a zero-coupon bond be compared to that on a coupon bond? It is easy to see that we compare a two-year zero-coupon bond with a coupon bond of two years' duration; however, this becomes impractical when dealing with very long-dated zero-coupon bonds, for which no equivalent coupon gilt is usually available. The way around this can be observed in the U.S. Treasury market, where the technique of stripping coupon Treasuries enables us to calculate implied zero-coupon rates with which we can compare actual strip-market yields.

In this section, we attempt to describe the relationship between spot interest rates, the actual market yields that would be observed on zero-coupon bonds, and the yields on coupon bonds. We illustrate how an implied spot-rate curve can be derived from the redemption yields and prices observed on coupon bonds, and discuss how this curve may be used to assess relative value in bond yields.

[4] This section is based on an unpublished paper originally written in July 1997 when the author (Moorad Choudhry) was working on the sterling proprietary trading desk in the Treasury division at Hambros Bank Limited. As first written, it predates the introduction of zero-coupon strips in the UK gilt market in December 1997. It was updated as part of a guest lecture delivered at the City University Business School in 2000.

[5] For instance, see Blake (1990), paragraph 5.8.1.

Spot (Zero-Coupon) Yields[6] and Coupon Bond Prices

A gilt is a collection of cash flows, each of which represents an obligation on the part of HM Treasury to service the debt represented by the gilt issue on a semiannual basis. We can view a gilt as a bundle of individual zero-coupon securities, each maturing on their respective payment date. Viewed in this way, the present value of the gilt becomes the sum of the present values of all the constituent cash flows. Let us consider this first.

Assume that we can observe the spot rates for various maturities and that these rates are $r_1, r_2, r_3, \ldots, r_N$. If a bond pays coupon C annually from period 1 to period N its present value using the series of spot rates is given by

$$P = \frac{C_1/2}{(1+\frac{1}{2}r_1)} + \frac{C_2/2}{(1+\frac{1}{2}r_2)^2} + \frac{C_3/2}{(1+\frac{1}{2}r_3)^3} + \ldots + \frac{C_{N-1}/2}{(1+\frac{1}{2}r_{N-1})^{N-1}} + \frac{C_N/2+100}{(1+\frac{1}{2}r_N)^{2N}} \tag{19.6}$$

which is, of course, different from the conventional redemption-yield formula with which we are very familiar. Each cash flow is discounted by the specific spot rate that corresponds to the maturity period of the cash flow. It follows that, in order to value a bond in this way, we must know the spot-rate term structure. However, this may not always be readily observable. Gilt prices, on the other hand, are readily observable, and so we may use them to derive spot interest rates.

Spot Yields and Bond Yields

In Table 19.2 we show the redemption yields for gilts as of 25th June 1997, together with implied spot rates, and the corresponding graph of these yields in Figure 19.2.

The redemption-yield curve in Figure 19.2 is a conventional positively sloping curve with a dip at the long end of the term structure at the point of the longest-dated bond. Why is there a spread between the two curves? Remember that the spot yield curve is the theoretical return on a zero-coupon instrument; that is, one that has no coupons paid during its life. However, the redemption-yield curve measures the expected return on instruments that pay a coupon at regular intervals during their life. The rising shape of the curve implies that medium-dated zero-coupon bonds, if they represented actual instruments, would offer a better yield than short-dated zero-coupon bonds. This positively sloping part of the curve influences the relationship between spot yields and bond yields at that part of the curve. For instance, the five-year spot rate is 7.131%, so a five-year zero-coupon bond being priced on 24th June 1997 for settlement the following day in theory should pay 7.131%. The actual five-year coupon bond would pay a series of coupons before the final coupon and

[6] The term "spot rate" is usually considered to be synonymous with "zero-coupon rate", and it is common to see this written in textbooks (including this one!). In this chapter, the two terms are not used thus, as we make the distinction between a spot rate derived from coupon bond prices and a zero-coupon rate that is observed on a zero-coupon bond trading in the market.

TABLE 19.2 Conventional and Spot Yields, 25 June 1997

Term (years)	Gilt	Conventional Yield %	Spot Yield %	Spread%
0.25	8.75% 1/9/1997	6.508	6.508	0.000
1	7.25% 30/3/1998	6.763	6.770	–0.007
1.5	12% 20/11/1998	6.968	6.978	–0.01
2	6% 10/8/1999	6.956	6.972	–0.016
3	8% 7/12/2000	7.133	7.143	–0.01
4	7% 6/11/2001	7.158	7.179	–0.021
5	7% 7/6/2002	7.118	7.131	–0.013
6	8% 10/6/2003	7.173	7.184	–0.011
7	6.75% 27/11/2004	7.151	7.167	–0.016
8	8.50% 7/12/2005	7.188	7.196	–0.008
9	7.50% 7/12/2006	7.189	7.207	–0.018
10	7.25% 7/12/2007	7.170	7.180	–0.01
11	9% 13/10/2008	7.220	7.239	–0.019
12	8% 25/9/2009	7.235	7.271	–0.036
13	6.25% 25/11/2010	7.255	7.312	–0.057
14	9% 12/7/2011	7.254	7.289	–0.035
15	9% 6/8/2012	7.265	7.307	–0.042
16	8% 27/9/2013	7.254	7.289	–0.035
18	8% 7/12/2015	7.222	7.215	0.007
21	8.75% 25/8/2017	7.267	7.318	–0.051
24	8% 7/6/2021	7.209	7.172	0.037

Source: Bloomberg.

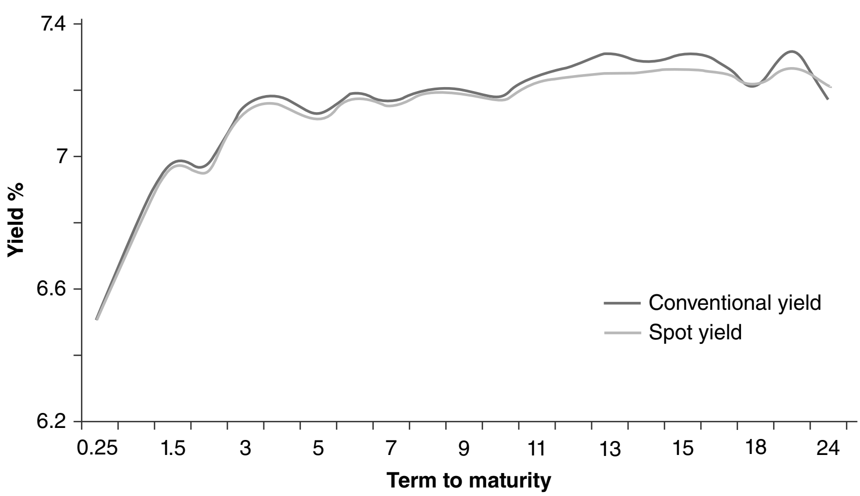

FIGURE 19.2 Gilt Market Yields, 25 June 1997
Source: Williams de Broe and Hambros Bank.

principal amount on maturity. This last coupon and principal would indeed be valued at 7.131% on 24th June, as they represent what are in effect zero-coupon payments in five years' time.[7] However, the earlier coupon payments would be valued at lower spot yields, as given by the positively sloping spot-yield curve. We know from earlier in this chapter that the yield on a coupon bond is, in effect, the average of the yields payable on the bundle of spot obligations that constitute the bond, so that the final yield on the bond must be below 7.131% when priced for value on 25th June. The yield is, in fact, 7.118%. This is why the spot yields lie above the redemption yields at the short- to medium-end of the curve.

We also note that at the very long-end of the curve the demand for bonds is such that the yields are depressed for the 8% 2015 and 8% 2021 gilts. Again these bonds pay regular coupon but, at the longer-end, their coupons are valued higher by the market, so that these coupons are, in effect, priced at lower yields. This is confirmed by the implied spot-rate curve. The 18- and 24-year spot rates, which would be the theoretical yields on identical-maturity zero-coupon bonds, are 7.215% and 7.172%, respectively. However, the 18- and 24-year bonds themselves are priced at 7.222% and 7.209%.

The implication of this property for potential investors in zero-coupon gilts or strips is clear: in a positively sloping yield-curve environment there is a yield pickup on similar maturity strips, although these will be instruments of higher modified duration.

Consider now gilt yields for 10th November 1996, shown in Table 19.3 and Figure 19.3.

TABLE 19.3 Conventional and Spot Yields, 10 November 1996

Term (years)	Gilt	Conventional Yield %	Spot Yield %	Spread %
0.25	T-Bill	6.06	6.06	0.000
1.5	7.25% 30/3/1998	6.71	6.714	–0.004
3	6% 10/8/1999	6.95	6.979	–0.029
4	8% 7/12/2000	7.22	7.255	–0.035
5	7% 6/11/2001	7.31	7.364	–0.054
6	8% 10/6/2003	7.47	7.542	–0.072
8	6.75% 27/11/2004	7.59	7.690	–0.100
9	8.50% 7/12/2005	7.66	7.762	–0.102
10	7.50% 7/12/2006	7.67	7.780	–0.11
15	9% 12/7/2011	7.90	8.129	–0.229
19	8% 7/12/2015	7.91	8.086	–0.176
25	8% 7/6/2021	7.91	8.046	–0.136

[7] We call this is a "five-year" bond as it is the benchmark bond for that period, while accepting that the actual maturity is slightly less than five years. An equivalent zero-coupon bond would also be priced on 24th June 1997 for maturity on 7th June 2002 rather than 25th June 2002, although we would call it a five-year security.

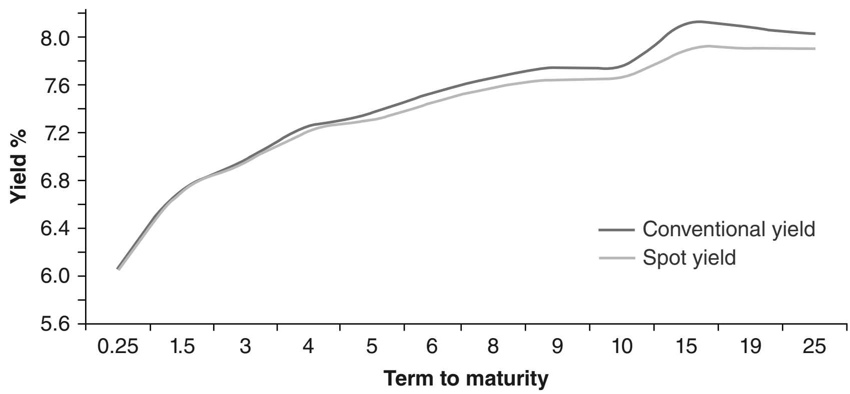

FIGURE 19.3 Gilt Yield Curves, 10 November 1996
Source: ABN Amro, Bloomberg.

At this time, the bond yield curve was sloping upwards all the way out the long-bond. The difference in the yields of the 8% 2015 and 8% 2021 is much less pronounced with the curves for June 1997. The effect of this is that although the implied spot rates decrease at the very long end, they do not fall below the bond yield curve. This means that on this earlier date the market was not placing as high a premium on coupon payments made after 19 years as it would be seen to do several months later. This illustration leads nicely on to the next section, when we look at the yields of strips compared to their theoretical yields.

Implied Spot Yields and Zero-Coupon Bond Yields

In an environment in which there is a liquid market in actual zero-coupon securities or strips, we can observe market spot yields from the prices at which the strips are trading. In the previous section, we showed how the prices and redemption yields on coupon bonds can be used to derive an implied spot-yield curve. But how do these implied rates compare to actual strip yields, and how should we expect them to behave vis-à-vis each other? We discuss some of the key issues in this section.

A spot rate could be said to be the rate of return at purchase on a single-cash-flow security held to maturity. It can also, as the previous sections have shown, be viewed as the rate payable on a coupon bond that is viewed as a bundle of individual zero-coupon cash flows, with the bond yield viewed as a complex average of the spot yields on the individual payments. Spot yields as such cannot be actually observed in the market. Zero-coupon securities, on the other hand, are actual market instruments that have been created from stripping individual cash flows from coupon bonds, and trading the resultant cash flows separately.[8] Therefore, the yields observed on zero-coupon bonds will reflect actual market supply and demand for specific securities. This makes zero-coupon yields different from spot yields in practice, notwithstanding the fact that they should be viewed as identical in theory.

[8] For more information on this procedure with regard to the gilt strips market, see Choudhry (1999).

This reflects how spot yields are derived from other prices and yields, and not observed on market instruments.

How do the two rates differ? If a particular zero-coupon bond is in demand, its price will rise, and its yield will fall. The opposite occurs if the bond is not sought after and is being sold by investors and market makers. The effect of this is that at any time the zero-coupon on yield can and does differ from the equivalent-maturity spot yield, be it higher or lower. If investors value an individual zero-coupon bond lower when it is a stripped security compared to when it is part of a package of zero-coupon cash flows (in a coupon bond), the yield on the zero will be higher than the equivalent-maturity spot rate. The opposite will happen if investors prefer to hold the zero-coupon security. Despite this reflection of supply and demand, implied spot rates are still important because they allow investors to assess relative value for both zero-coupon bonds and coupon bonds.

Consider Table 19.4 and Figure 19.4; the latter shows the implied spot yields derived from gilt prices on 2nd March 1999, together with the yield for coupon

TABLE 19.4 Gilt Market Gross Redemption True Yields and Implied Spot Yields on 2 March 1999

Term	Gilt	True Yield	Spot Yields	Coupon Strip Yield	Principal Strip Yield
0.5	6% 10/8/1999	5.033	5.033	5.068	
1.5*	8% 7/12/2000	5.027	4.977	4.973	5.036
2.5	7% 6/11/2001	4.963	4.927	4.949	
3*	7% 7/6/2002	4.878	4.836	4.91	4.867
4	8% 10/6/2003	4.845	4.789	4.824	
4.5*	6.50% 7/12/2003	4.735	4.687	4.786	4.691
5	6.75% 26/11/2004	4.709	4.658	4.798	
6*	8.50% 7/12/2005	4.78	4.728	4.774	4.74
7*	7.50% 7/12/2006	4.8	4.77	4.781	4.775
8*	7.25% 7/12/2007	4.759	4.721	4.753	4.723
9	9% 13/10/2008	4.72	4.644	4.735	
10*	5.75% 7/12/2009	4.604	4.542	4.723	4.518
11	6.25% 25/11/2010	4.695	4.665	4.711	
12	9% 12/7/2011	4.751	4.721	4.713	
13	9% 6/8/2012	4.787	4.727	4.713	
14	8% 27/9/2013	4.763	4.75	4.71	
16*	8% 7/12/2015	4.711	4.659	4.705	4.670
18	8.75% 25/8/2017	4.729	4.695	4.682	
22*	8% 7/6/2021	4.679	4.609	4.635	4.612
29*	6% 7/12/2028	4.537	4.376	4.402	4.365

* Indicates strippable gilts.
Source: Bloomberg.

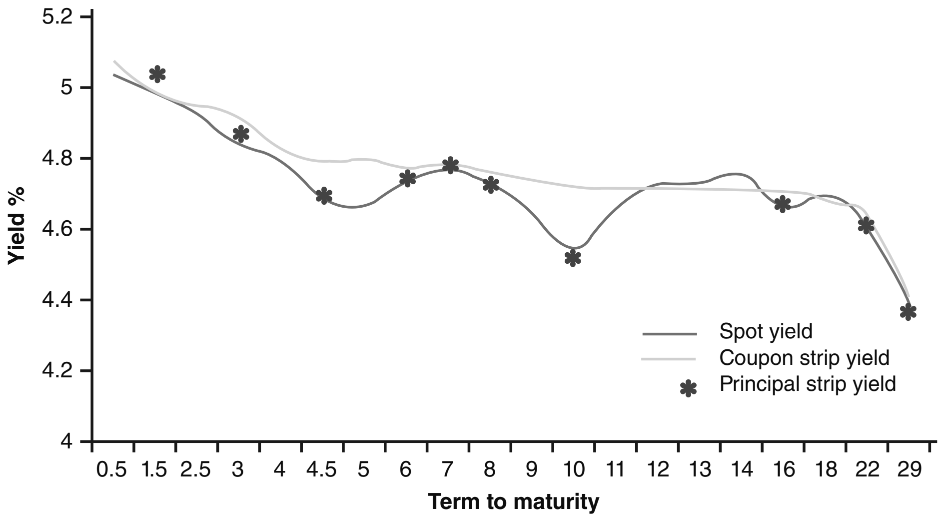

FIGURE 19.4 Spot and Strip Yields on 2 March 1999
Source: Bloomberg.

strips and principal strips as at the same date. The yields on principal strips are not joined as a curve, as the comparatively long period between strips would make any conclusions from the resulting curve problematic. These are the yields on the principal or residual cash flow when a bond is stripped. The other cash flows are known as coupon strips.

Two observations are worth making. First, that zero-coupon bonds traded cheap to the spot curve throughout the term structure, indicating that investors were not prepared to hold strips without a premium to the theoretical value. This probably reflected the inverted yield curve, which meant that strips would be trading expensive to coupon bonds of the same maturity. However, strips traded expensive to the spot curve at the 11- to 15-year point of the curve. Second, principal strips trade at lower yields than coupon strips of the same maturity. This reflects the liquidity and demand considerations for principal strips, which investors prefer to hold instead of coupon strips.

For reference, we graph the curve for the bond yields alongside the spot and zero-coupon yields at Figure 19.5.

YIELD-SPREAD TRADES[9]

Relative-value trades are common among investors who do not wish to put on a naked directional position but rather believe that the yield curve will change shape and flatten or widen between two selected points. Such a trade would involve simultaneous positions in bonds of different maturity. Another relative-value trade

[9] Originally written by the author (Moorad Choudhry) in June 1997 as an internal paper when at Hambros Bank Limited. The principles are still very valid in 2014, however! Prices quoted are tick prices, that is fractions of a 32nd.

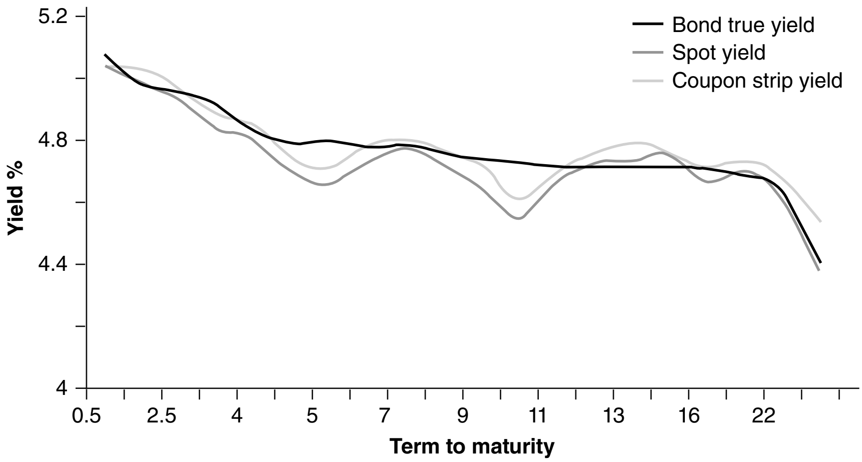

FIGURE 19.5 Gilt Bond, Spot, and Coupon Strip Yields, 2 March 1999
Source: Bloomberg.

may involve high-coupon bonds against low coupon bonds of the same maturity, as a tax-related trade. These trades are concerned with the change in yield spread between two or more bonds rather than a change in absolute interest-rate level. The key factor is that changes in spread are not conditional upon directional change in interest-rate levels; that is, yield spreads may narrow or widen whether interest rates themselves are rising or falling.

Typically, spread trades will be constructed as a long position in one bond against a short position in another bond. If it is set up correctly, the trade will only achieve a profit or incur a loss if there is change in the shape of the yield curve. This is regarded as being first-order risk-neutral, which means that there is no interest-rate risk in the event of change in the general level of market interest rates. In this section, we examine some common yield-spread trades.

Spread-Trade Risk Weighting

A relative-value trade usually involves a long position set up against a short position in a bond of different maturity. The trade must be weighted so that the two positions are first-order neutral, which means the risk exposure of each position nets out when considered as a single trade, but only with respect to a general change in interest-rate levels. If there is a change in yield spread, a profit or loss will be generated.

A common approach to weighting spread trades is to use the basis-point value (BPV) of each bond.[10] Table 19.5 shows price and yield data for a set of benchmark gilts for value date 17th June 1997. The BPV for each bond is also shown, per £100 of stock. For the purposes of this discussion, we quote mid-prices only and assume that the investor is able to trade at these prices.

[10] This is also known as “dollar value of a basis point” (DVBP), “present value of a value point” (PVBP) or simply DV01.

TABLE 19.5 Gilt Prices and Yields for Value 17th June 1997

Term	Bond	Price	Accrued	Dirty Price	Yield %	Modified Duration	BPV	Per £1m Nominal
2-yr	6% 10/8/1999	98–17	127	100.62	6.753	1.689	0.016642	166.42
5-yr	7% 7/6/2002	100–10	10	100.50	6.922	3.999	0.040115	401.15
10-yr	7.25% 7/12/2007	101–14	10	101.64	7.052	6.911	0.070103	701.03
25-yr	8% 7/6/2021	110–01	10	110.25	7.120	11.179	0.123004	1,230.04

Source: Williams de Broe and Hambros Bank Limited.

An investor believes that the yield curve will flatten between the 2-year and 25-year sectors of the curve and that the spread between the 6% 1999 and the 8% 2021 will narrow from its present value of 0.367%. To reflect this view, he buys the 25-year bond and sells short the 2-year bond, in amounts that leave the trade first-order risk-neutral. If we assume the investor buys £1 million nominal of the 8% 2021 gilt, this represents an exposure of £1230.04 loss (profit) if there is a one-basis-point increase (decrease) in yields. Therefore, the nominal amount of the short position in the 6% 1999 gilt must equate this risk exposure. The BPV per £1 million nominal of the two-year bond is £166.42, which means that the investor must sell

$$(1{,}230.04/166.42)$$

or £7.3912 million of this bond, given by a simple ratio of the two-basis-point values. We expect to sell a greater nominal amount of the shorter-dated gilt because its risk exposure is lower. This trade generates cash because the short sale proceeds exceed the long-buy purchase funds, which are respectively

Buy £1 m 8% 2021	–£1,102,500
Sell £7.39 m 6% 1999	+£7,437,025

What are the possible outcomes of this trade? If there is a parallel shift in the yield curve, the trade neither gains nor loses. If the yield spread narrows by, say, 15 basis points, the trade will gain either from a drop in yield on the long side or a gain in yield in the short side, or a combination of both. Conversely, a widening of the spread will result in a loss. Any narrowing spread is positive for the trade, while any widening is harmful.

The trade would be put on the same ratio if the amounts were higher, which is known as scaling the trade. So, for example, if the investor had bought £100 million of the 8% 2021, he would need to sell short £739 million of the two-year bonds. However, the risk exposure is greater by the same amount, so that in this case the trade would generate 100 times the risk. As can be imagined, there is a greater potential reward but at the same time a greater amount of stress in managing the position.

Using BPVs to risk-weight a relative-value trade is common but suffers from any traditional duration-based measure because of the assumptions used in the analysis. Note that when using this method, the ratio of the nominal amount of the bonds must equate the reciprocal of the bonds' BPV ratio. So, in this case the BPV ratio is

$$(166.42/1230.04)$$

or 0.1353, which has a reciprocal of 7.3912. This means that the nominal values of the two bonds must always be in the ratio of 7.39:1. This weighting is not static, however; we know from Chapter 2 that duration measures are a snapshot estimation of dynamic properties such as yield and term to maturity. Therefore, for anything but very short-term trades the relative values may need to be adjusted as the BPVs alter over time.

Another method to weigh trades is by duration-weighting, which involves weighting in terms of market values. This compares to the BPV approach, which provides a weighting ratio in terms of nominal values. In practice, the duration approach does not produce any more accurate risk weighting.

A key element of any relative-value trade is the financing cost of each position. This is where the repo market in each bond becomes important. In the example just described, the financing requirement is:

- Repo out the 8% 2021, for which £1.1 million of cash must be borrowed to finance the purchase; the trader pays the repo rate on this stock;
- Reverse repo the 6% 1999 bond, which must be borrowed in repo to cover the short sale; the trader earns the repo rate on this stock.

If the repo rate on both stocks is close to the general repo rate in the market, there will be a bid-offer spread to pay, but the greater amount of funds lent out against the 6% 1999 bond will result in a net financing gain on the trade whatever happens to the yield spread. If the 8% 2021 gilt is special, because the stock is in excessive demand in the market (for whatever reason), the financing gain will be greater still. If the 6% 1999 is special, the trade will suffer a financing loss.

Figures 19.6A and 19.6B show the funding considerations for this trade for value on 17 June 1997, calculated on the Bloomberg screen RRRA. Both stocks were traded as "general collateral" in the market that day, that is, not special. The financing rate for the 6% 1999 was 6.00%, that for the 8% 2021 stock 6.125%. This reflects the bid-offer spread for GC at that time.

The cash value of the short sale far exceeds that of the long buy, hence over any time horizon the net funding for this trade is positive. From Figure 19.6 we see the net funding gain for a one-month trade is £29,698. This is a bonus for the position, as it means there is no "break-even" level for the spread should the yield curve move in the direction anticipated. In fact the funding gain will cushion against a small move in the wrong direction.

Identifying Yield-Spread Trades

Yield-spread trades are a type of relative-value position that a trader can construct when the objective is to gain from a change in the spread between two points on

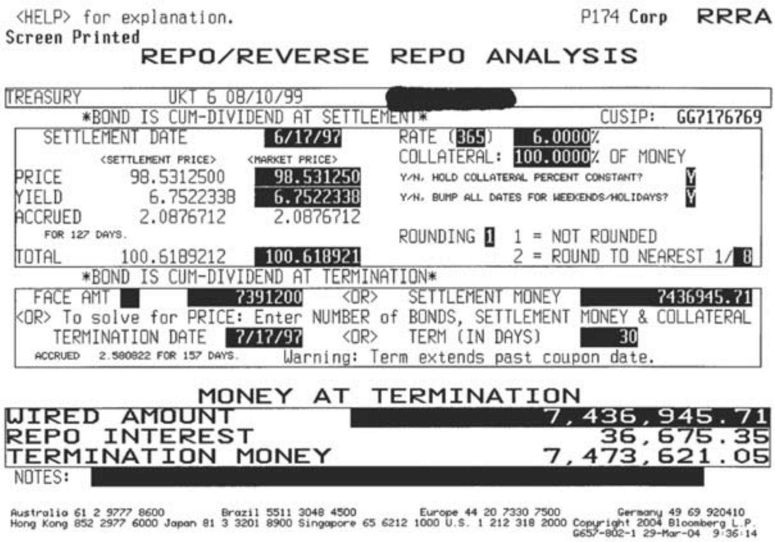

FIGURE 19.6A Bloomberg Screen RRRA Showing Funding Calculation for One-Month Holding in 6 Percent 1999, Value 17 June 1997, Short Sale Funding

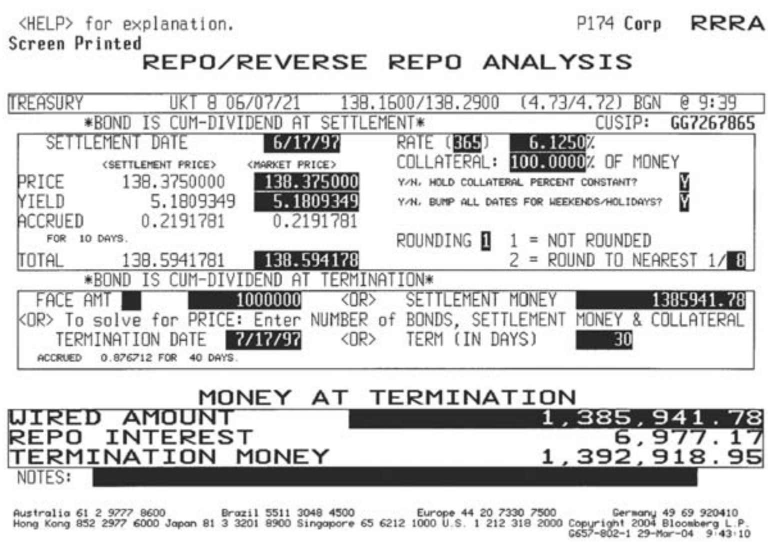

FIGURE 19.6B Bloomberg Screen RRRA Showing Funding Calculation for One-Month Holding in 8 Percent 2021, Value 17 June 1997, Long Position Funding

TABLE 19.6 Yield Levels and Yield Spreads during November 1996 to November 1997

Changes in Yield Levels						
	3-Month	**1-Year**	**2-Year**	**5-Year**	**10-Year**	**25-Year**
11/10/96	6.06	6.71	6.83	7.31	7.67	7.91
7/10/97	6.42	6.96	7.057	7.156	7.025	6.921
Change	0.36	0.25	0.227	–0.154	–0.645	–0.989
11/10/97	7.15	7.3	7.09	6.8	6.69	6.47
Change	0.73	0.34	0.033	–0.356	–0.335	–0.451

Changes in Yield Spread						
	3m/1y	**1y/2y**	**2y/5y**	**5y/10y**	**5y/25y**	**10y/25y**
11/10/96	–0.65	–0.12	–0.48	–0.36	–0.6	–0.24
7/10/97	–0.54	–0.457	–0.099	0.131	0.235	0.104
Change	0.11	–0.337	0.381	0.491	0.835	0.344
11/10/97	–0.15	0.21	0.29	0.11	0.33	0.22
Change	0.39	0.667	0.389	–0.021	0.095	0.116

Source: ABN Amro, Hambros Bank Limited.

the yield curve. The decision on which sectors of the curve to target is an important one and is based on a number of factors. An investor may naturally target, say, the 5- and 10-year areas of the yield curve to meet investment objectives and have a view on these maturities. Or a trader may draw conclusions from studying the historical spread between two sectors.

Yield spreads do not move in parallel, however, and there is not a perfect correlation between the changes of short-, medium-, and long-term sectors of the curve. The money-market yield curve can sometimes act independently of the bond curve. Table 19.6 shows the change in benchmark yields during 1996–1997. There is no set pattern in the change in both yield levels and spreads. It is apparent that one segment of the curve can flatten while another is steepening, or remain unchanged.

Another type of trade is where an investor has a view on one part of the curve relative to two other parts of the curve. This can be reflected in a number of ways, one of which is the butterfly trade, considered in the next section.

COUPON SPREAD

Coupon spreads are becoming less common in the Treasury and Gilt markets because of the disappearance of high-coupon or other exotic bonds and the concentration on liquid benchmark issues. However, they are genuine spread trades. The basic principle behind the trade is a spread of two bonds that have similar maturity or similar duration but different coupons.

Table 19.7 shows the yields for a set of high-coupon and low(er)-coupon gilts for a specified date in May 1993 and the yields for the same gilts six months later. From the yield curves, we see that general yield levels decline by approximately 80–130 basis points. The last column in the table shows that, apart from the earliest pair of gilts (which do not have strictly comparable maturity dates), the performance of the lower-coupon gilt exceeded that of the higher-coupon gilt in every instance. Therefore, buying the spread of the low coupon versus the high coupon should, in theory, generate a trading gain in an environment of falling yields. One explanation for this is that the lower-coupon bonds are often the benchmark, which means the demand for them is higher. In addition, during a bull market, more bonds are considered to be "high" coupon as overall yield levels decrease.

The exception noted in Table 19.7 is the outperformance by the 14% Treasury 1996 over the lower-coupon 10% 1995 stock. This is not necessarily conclusive, because the bonds are six months apart in maturity, which is a significant amount for short-dated stock. However, in an environment of low or falling interest rates, shorter-dated investors such as banks and insurance companies often prefer to hold

TABLE 19.7 Yield Changes on High- and Low-Coupon Gilts from May 1993 to November 1993

Stock	**Term**	**10-May-1993**	**12-Nov-1993**	
Gilt	1	5.45	5.19	
10Q 95	2	6.39	5.39	
10 96	3	6.94	5.82	
10H 97	4	7.13	5.97	
9T 98 and 7Q 98	5	7.31	6.14	
10Q 99	6	7.73	6.55	
9 00	7	7.67	6.54	
10 01	8	8.01	6.82	
9T 02	9	8.13	6.95	
8 03	10	8.07	6.85	
9 08	15	8.45	7.18	
9 12	20	8.55	7.23	
8T 17	30	8.6	7.22	
Gilt	**Maturity**	**10/05/1993 Yield %**	**12/11/1993 Yield %**	**Yield Change %**
10Q 95	21-Jul-1995	6.393	5.390	–1.003
14 96	10-Jan-1996	6.608	5.576	–1.032
15Q 96	3-May-1996	6.851	5.796	–1.055
13Q 96	15-May-1996	6.847	5.769	–1.078
13Q 97	22-Jan-1997	7.142	5.999	–1.143
10H 97	21-Feb-1997	7.131	5.974	1.157
7 97	6-Aug-1997	7.219	6.037	–1.182

TABLE 19.7 (*Continued*)

Gilt	MatuCrity	10/05/1993 Yield %	12/11/1993 Yield %	Yield Change %
8T 97	1-Sep-1997	7.223	6.055	–1.168
15 97	27-Oct-1997	7.294	6.113	–1.161
9T 98	19-Jan-1998	7.315	6.102	–1.213
7Q 98	30-Mar-1998	7.362	6.144	–1.218
6 99	10-Aug-1999	7.724	6.536	–1.188
10Q 99	22-Nov-1999	7.731	6.552	–1.179
8 03	10-Jun-2003	8.075	6.854	–1.221
10 03	8-Sep-2003	8.137	6.922	–1.215

Source: ABN Amro Hoare Govett Sterling Bonds Ltd; Bloomberg.

very high-coupon bonds because of the high income levels they generate. This may explain the demand for the 14% 1996 stock,[11] although the evidence at the time was only anecdotal.

BUTTERFLY TRADES

Butterfly trades are another method by which traders can reflect a view on changing yield levels without resorting to a naked bet on interest rates. They are another form of relative-value trade. Among portfolio managers, they are viewed as a means of enhancing returns. In essence, a butterfly trade is a short position in one bond against a long position of two bonds, one of shorter maturity and the other of longer maturity than the short-sold bond. Duration-weighting is used so that the net position is first-order risk-neutral, and nominal values are calculated such that the short-sale and long-purchase cash flows net to zero, or very close to zero.

This section reviews key aspects of butterfly trades.

Basic Concepts

A butterfly trade is a yield-curve trade par excellence. If the average return on the combined long position is greater than the return on the short position (which is a cost) during the time the trade is maintained, the strategy will generate a profit. It reflects a view that the short-end of the curve will steepen relative to the "middle" of the curve while the long-end will flatten. For this reason, higher-convexity stocks are usually preferred for the long positions, even if this entails a loss in yield. However, the trade is not "risk-free" for the same reasons that a conventional two-bond yield spread is not. Although in theory a butterfly is risk-neutral with respect to parallel changes in the yield curve, changes in the shape of the curve can result in losses. For

[11] This stock also has a special place in the author's heart, although he was No. 2 on the desk when the Treasury head put on a very large position in it . . . ! Well, large for that bank's balance sheet and trading limit—a mere £100 million. In the era of the "London Whale", that is almost pocket change for some banks.

this reason, the position must be managed dynamically and monitored for changes in risk relative to changes in the shape of the yield curve.

In a butterfly trade, the trader is long a short-dated and long-dated bond, and short a bond of a maturity that falls in between these two maturities. A portfolio manager with a constraint on running short positions may consider this trade as a switch out of a long position in the medium-dated bond and into duration-weighted amounts of the short-dated and long-dated bond. However, it is not strictly correct to view the combined long position as an exact substitute for the short position; because of liquidity and other reasons the two positions will behave differently for given changes in the yield curve. In addition, one must be careful to compare like with like, as the yield change in the short position must be analysed against yield changes in *two* bonds. This raises the issue of portfolio yield.

Putting on the Trade

We begin by considering the calculation of the nominal amounts of the long positions, assuming a user-specified starting amount in the short position. In Table 19.8 we show three gilts as of 25th April 1997. The trade we wish to put on is a short position in the five-year bond, the 7% Treasury 2002, against long positions in the two-year bond, the 6% Treasury 1999, and the 10-year bond, the 7% Treasury 2007. Assuming £10 million nominal of the five-year bond, the nominal values of the long positions can be calculated using duration, modified duration or basis-point values (the last two, unsurprisingly, will generate identical results). The more common approach is to use basis-point values.

In a butterfly trade, the net cash flow should be as close to zero as possible, and the trade must be basis-point value-neutral. Using the following notation,

P_1 the dirty price of the short-position

P_2 the dirty price of the long position in the 2-year bond

TABLE 19.8 Bond Values for Butterfly Strategy, 25 April 1997

	2-Year Bond	5-Year Bond	10-Yyear Bond
Gilt	6% 1999	7% 2002	7.25% 2007
Maturity date	10 Aug 1999	07 Jun 2002	07 Dec 2007
Price	98–08	99–27	101–06
Accrued interest	2.30137	0.44110	0.45685
Dirty price	100.551	100.285	101.644
GRY %	6.913	7.034	7.085
Duration	1.969	4.243	7.489
Modified duration	1.904	4.099	7.233
Basis point value	0.01914	0.0411	0.07352
Convexity	0.047	0.204	0.676

P_3 the dirty price of the long position in the 10-year bond

M_1 the nominal value of short-position bond, with M_2 and M_3 the long position bonds

BPV_1 the basis-point value of the short-position bond if applying basis-point values, the amounts required for each stock are given by

$$M_1 P_1 = M_2 P_2 + M_3 P_3 \tag{19.7}$$

while the risk-neutral calculation is given by

$$M_1 BPV_1 = M_2 BPV_2 + M_3 BPV_3 \tag{19.8}$$

The value of M_1 is not unknown, as we have set it at £10 million. The equations can be rearranged, therefore, to solve for the remaining two bonds, which are

$$\begin{aligned} M_2 &= \frac{P_1 BPV_3 - P_3 BPV_1}{P_2 BPV_3 - P_3 BPV_2} M_1 \\ M_3 &= \frac{P_2 BPV_1 - P_1 BPV_2}{P_2 BPV_3 - P_3 BPV_2} M_1 \end{aligned} \tag{19.9}$$

Using the dirty prices and BPVs from Table 19.8, we obtain the following values for the long positions. The position required is short £10 million 7% 2002 and long £5.347 million of the 6% 1999 and £4.576 million of the 7% 2007. With these values, the trade results in a zero net cash flow and a first-order risk-neutral interest-rate exposure. Identical results would be obtained using the modified-duration values, and using the duration measures. If using Macaulay duration the nominal values are calculated using

$$D_1 = \frac{MV_2 D_2 + MV_3 D_3}{MV_2 + MV_3} \tag{19.10}$$

where D and MV represent duration and market value for each respective stock.

Yield Gain

We know that the gross-redemption yield for a vanilla bond is that rate r where

$$P_d = \sum_{i=1}^{N} C_i e^{-rn} \tag{19.11}$$

The right-hand side of (19.11) is simply the present value of the cash-flow payments C to be made by the bond in its remaining lifetime. Expression (19.12) gives the continuously compounded yields to maturity. In practice, users define a yield with compounding interval m, that is

$$r = (e^{rmn} - 1)/m \tag{19.12}$$

Treasuries and gilts compound on a semiannual basis.

In principle, we may compute the yield on a portfolio of bonds exactly as for a single bond, using (19.11) to give the yield for a set of cash flows, which are purchased today at their present value. In practice, the market calculates portfolio yield as a weighted average of the individual yields on each of the bonds in the portfolio. This is described, for example, in Blake (2000) and his description points out the weakness of this method. An alternative approach is to weight individual yields using the bonds' basis-point values, which we illustrate here in the context of the earlier butterfly trade.

In this trade we have

- Short £10 million 7% 2002
- Long £5.347 million 6% 1999 and £4.576 million $7^1/_4$% 2007

Using the semiannual adjusted form of (19.11), the true yield of the long position is 7.033%. To calculate the portfolio yield of the long position using market-value weighting, we may use

$$r_{port} = (MV_2 / MV_{port})r_2 + (MV_3 / MV_{port})_{r_3} \tag{19.13}$$

which results in a portfolio yield for the long position of 6.993%. If we weight the yield with basis-point values we use

$$r_{port} = \frac{BPV_2M_2r_2 + BPV_3M_3r_3}{BPV_2M_2 + BPV_3M_3} \tag{19.14}$$

Substituting the values from Table 19.8, we obtain

$$r_{port} = \frac{(1,914)(5.347)(6.913) + (7,352)(4.576)(7.085)}{(1,914)(5.347) + (7,352)(4.576)}$$

or 7.045%.

We see that using basis-point values produces a seemingly more accurate weighted yield, closer to the true yield computed using the expression above. In addition, using this measure, a portfolio manager switching into the long butterfly position from a position in the 7% 2002 would pick up a yield gain of 1.2 basis points, compared to the 4 basis points that an analyst would conclude had been lost using the first yield measure.[12]

The butterfly trade, therefore, produces a yield gain in addition to the capital gain expected if the yield curve changes in the anticipated way.

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[12] The actual income gained on the spread will depend on the funding costs for all three bonds, a function of the specific repo rates available for each bond. Shortly after the time of writing, the 6% Treasury 1999 went special, so the funding gain on a long position in this stock would have been excessive. However, buying the stock outright would have necessitated paying a yield premium, as demand for it increased as a result of it going special. In the event, the premium was deemed high, an alternative stock was nominated, the $10^1/_4$% Conversion 1999, a bond with near-identical modified-duration value.

Convexity Gain

In addition to yield pick-up, the butterfly trade provides, in theory, a convexity gain that will outperform the short position irrespective of which direction interest rates moved in, provided we have a parallel shift. This is illustrated in Table 19.9. This shows that the changes in value of the 7% 2002 as interest rates rise and fall, together with the change in value of the combined portfolio.

We observe from Table 19.9 that whatever the change in interest rates, up to a point, the portfolio value will be higher than the value of the short position, although the effect is progressively reduced as yields rise. The butterfly will always gain if yields fall, and protects against downside risk if yields rise to a certain extent. This is the effect of convexity; when interest rates rise, the portfolio value declines by less than the short position value, and when rates fall, the portfolio value increases by more. Essentially, the combined long position exhibits greater convexity than the short position. The effect is greater if yields fall, while there is an element of downside protection as yields rise, up to the +150-basis-point parallel shift.

Portfolio managers may seek greater convexity, whether or not there is a yield pick-up available from a switch. However, the convexity effect is only material for large changes in yield; so, if there was not a corresponding yield gain from the switch, the trade may not perform positively. As we noted, this depends partly on the funding position for each stock. The price/yield profile for each stock is shown in Figure 19.7.

Essentially, by putting on a butterfly as opposed to a two-bond spread or a straight directional play, the trader limits the downside risk if interest rates fall, while preserving the upside gain if yields fall.

TABLE 19.9 Changes in Bond Values with Changes in Yield Levels

Yield Change Value (bps)	7% 2002 Value (£)	Portfolio Value* (£)	Difference (£)	BPV 7% 2002 (5-Year)	BPV 6% 1999 (2-Year)	BPV 7.25% 2007 (10-Year)
+250	9,062,370	9,057,175	–5,195	0.0363	0.0180	0.0584
+200	9,246,170	9,243,200	–2,970	0.0372	0.0182	0.0611
+150	9,434,560	9,435,200	640	0.0381	0.0184	0.0640
+100	9,627,650	9,629,530	1,880	0.0391	0.0187	0.0670
+ 50	9,825,600	9,828,540	2,940	0.0401	0.0189	0.0702
0	10,028,500	10,028,500	0	0.0411	0.0191	0.0735
– 50	10,236,560	10,251,300	14,740	0.0421	0.0194	0.0770
–100	10,450,000	10,483,800	33,800	0.0432	0.0196	0.0808
–150	10,668,600	10,725,700	57,100	0.0443	0.0199	0.0847
–200	10,893,000	10,977,300	84,300	0.0454	0.0201	0.0888
–250	11,123,000	11,240,435	117,435	0.0466	0.0204	0.0931

* Combined value of long positions in 6% 1999 and 7.25% 2007. Values rounded. Yield change is parallel shift.

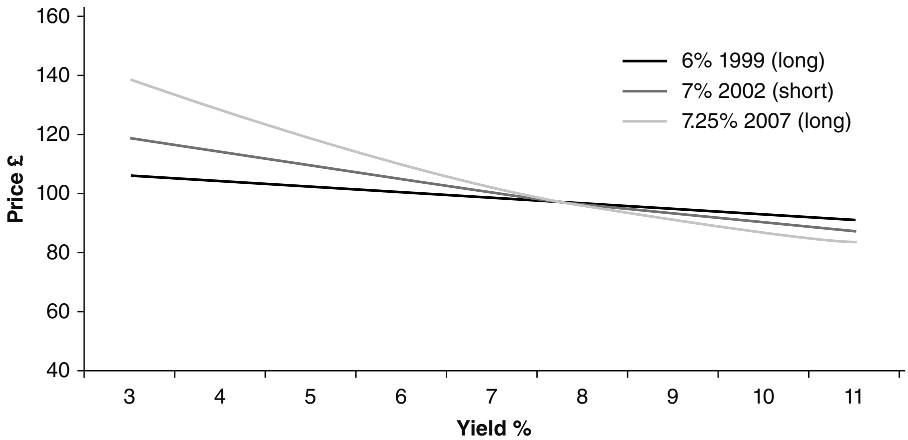

FIGURE 19.7 Illustration of Convexity for Each Stock in Butterfly Trade, 27 June 1997

To conclude the discussion of butterfly-trade strategy, we describe the analysis if using the "BBA" screen on Bloomberg. The trade is illustrated in Figure 19.8.

Using this approach, the nominal values of the two long positions are calculated using BPV ratios only. This is shown under the column "Risk Weight", and we note that the difference is zero. However, the nominal value required for the two-year bond is much greater, at £10.76 million, and for the 10-year bond much lower at £2.8 million. This results in a cash outflow of £3.632 million. The profit profile is, in theory, much improved, however; at the bottom of the screen, we observe the results of a 100-basis-point parallel shift in either direction, which is a profit. Positive results

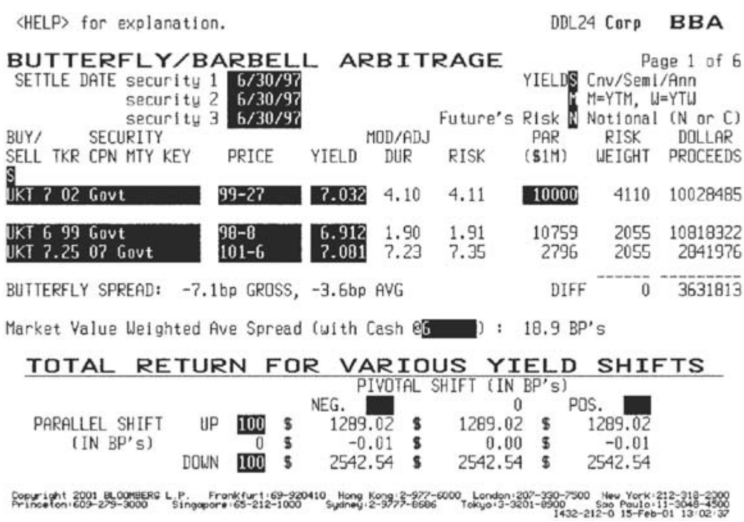

FIGURE 19.8 Butterfly-Trade Analysis on 27 June 1997, on Screen BBA

FIGURE 19.9 Butterfly-Trade Spread History
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were also seen for 200- and 300-basis-point parallel shifts in either direction. This screen incorporates the effect of a (uniform) funding rate, input on this occasion as 6.00%.[13] Note that the screen allows the user to see the results of a pivotal shift; however, in this example a 0-basis-point pivotal shift is selected.

This trade, therefore, created a profit whatever direction interest rates moved in.

The spread history for the position up to the day before the trade is shown in Figure 19.9, a reproduction of the graph on Bloomberg screen BBA.

CASE STUDY: SPREAD TRADE

To conclude this chapter we present a further "real-world" analysis conducted when a spread was observed. It concerns two UK gilt benchmark securities, but the principles may be applied to all markets and sectors. The analysis is conducted using standard Bloomberg screens.

On 23 March 2004 we observe that the spread between the two-year and five-year securities is at its lowest since June 2003. The bonds concerned are the 7.75% 2006 and the 5.75% 2009. This is shown in Figure 19.10, the screen SS used to plot spread histories. The widest spread during that time is shown to be 42.3 bps, the narrowest 14.3 bps.

[13] In reality, the repo rate will be slightly different for each stock, and there will be a bid-offer spread to pay. But as long as none of the stocks are special, the calculations should be reasonably close.

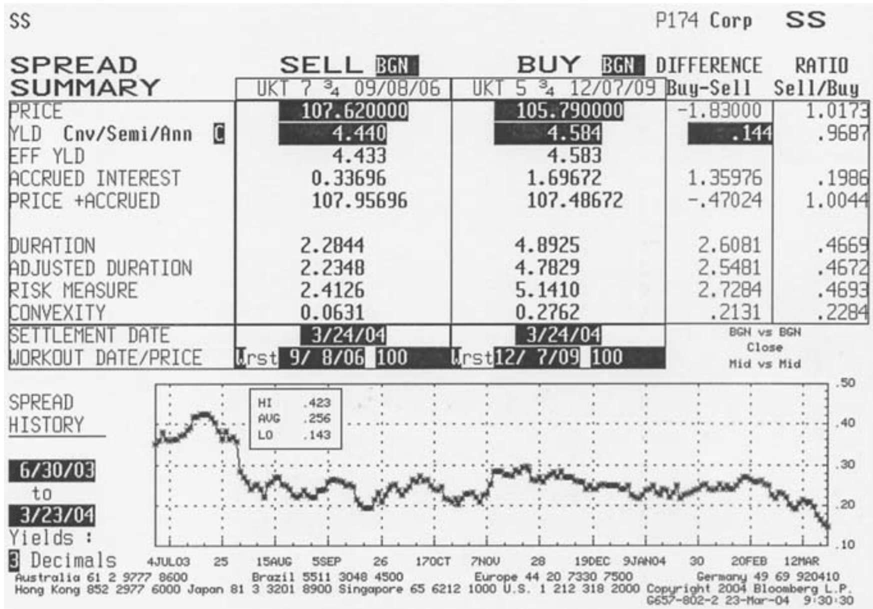

FIGURE 19.10 Spread Summary, 7.75 Percent 2006 Versus 5.75 Percent 2009 Gilts, 23 March 2004

This suggests that a spread-widening trade may be worthwhile, with an expectation of widening. Therefore we

- Sell the UKT 7.75% 2006 stock
- Buy the 5.75% 2009 stock

in duration-weighted ratios.

We conduct a linear regression analysis to determine the relationship between the two securities; that is, the sensitivity of a selected security to a change in the value of another security. As both bonds are UK gilts, we expect a high degree of correlation, and this is shown in Figure 19.11, the screen HRA from Bloomberg. A close relationship between two security price sensitivities is a useful feature in a first-order risk-neutral spread trade, which this is. Figure 19.11 shows that the independent variable has a coefficient of 0.76, a high explanatory value, while the R^2 is 0.974, a very high degree of near-perfect correlation.

We then conduct the cost of carry analysis. Figure 19.12 shows the repo rates for UK gilts, the HBOS plc prices screen. We enter the respective bid and offer rates into screen CCS, the two-security carry screen. We use a one-month term for this calculation. This is shown in Figure 19.13.

Finally, we calculate the required number of securities. This is shown in Figure 19.14, page PDH2 from Bloomberg. From the respective basis point value, a long position of £10 million of the 2009 stock is balanced with a sale of £21.3 million of the 2006 stock. From Figure 19.14 we see that this produces a zero-basis-point value to the spread at the time of the trade. For a 30-basis-point

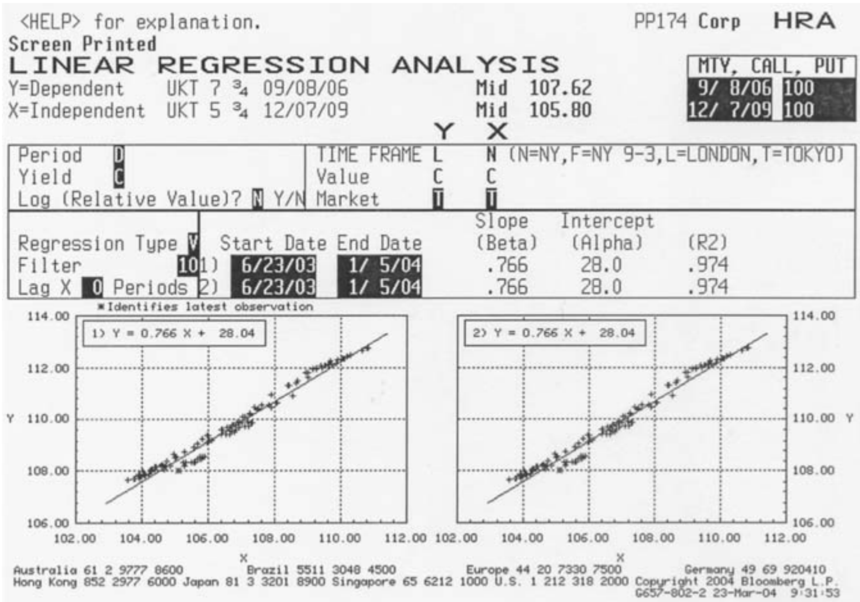

FIGURE 19.11 Stock Linear Regression Analysis

4 P300 a Corp HBOS
HBOS Treasury Services

Page 1 of 1
09:48 GMT
23-Mar-04

	EUR Bid	EUR Offer		GBP Bid	GBP Offer
TN	2.03	1.99	1W	4.15	4.00
SN	2.03	1.99	2W	4.00	3.90
SW	2.04	2.00	3W	4.00	3.90
SMTH	2.01	1.97	1M	4.02	3.92
2MTH	1.98	1.94	2M	4.04	3.94
3MTH	1.97	1.93	3M	4.06	3.96
4MTH	1.941	1.907	4M	4.10	4.00
5MTH	1.931	1.897	5M	4.14	4.04
6MTH	1.921	1.887	6M	4.18	4.08
7MTH	1.921	1.887	9M	4.30	4.20
8MTH	1.921	1.887	1Y	4.40	4.30
9MTH	1.910	1.876			
10MTH	1.910	1.876			
11MTH	1.920	1.886			
YR	1.940	1.917			

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 920410
Hong Kong 852 2977 6000 Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2004 Bloomberg L.P.
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FIGURE 19.12 HBOS gilt repo rates, 23 March 2004

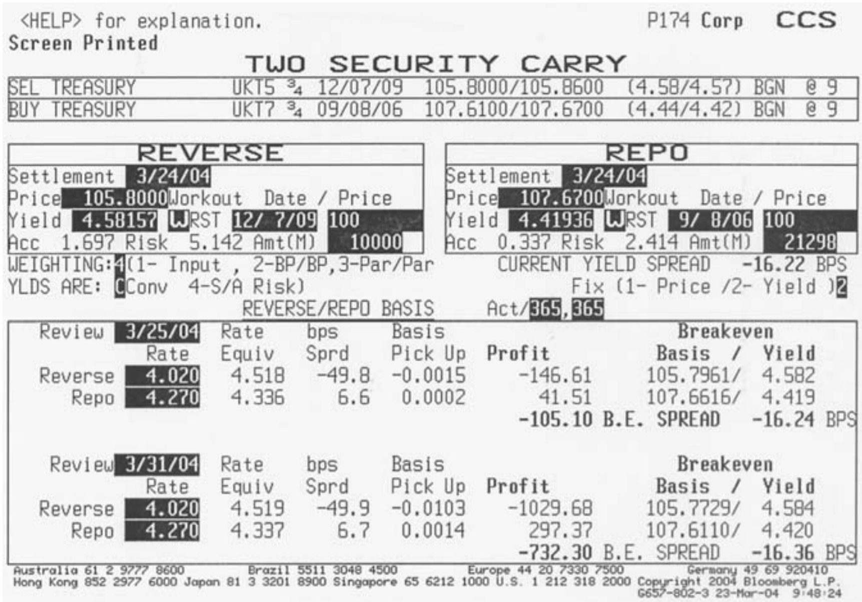

FIGURE 19.13 Two-Security Carry Analysis

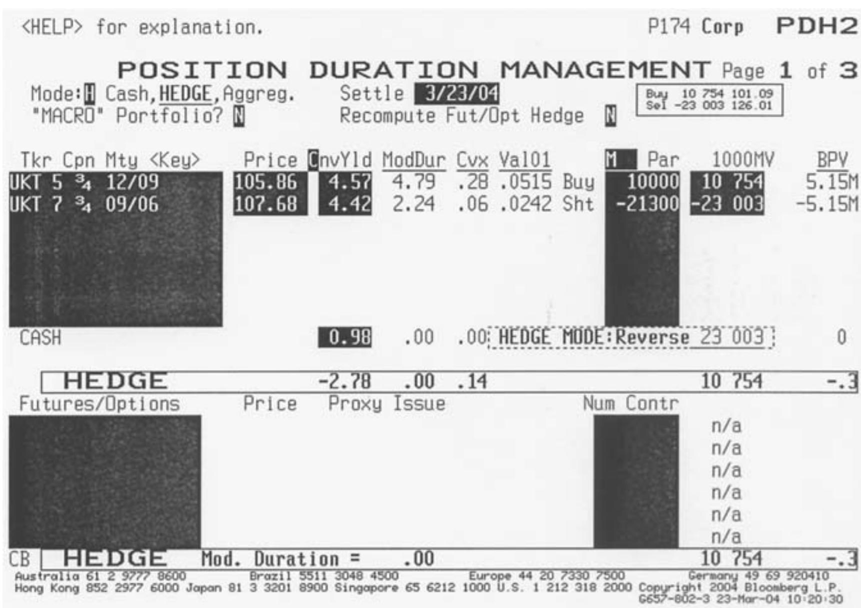

FIGURE 19.14 Screen PDH2, the Calculation of the Securities Amounts

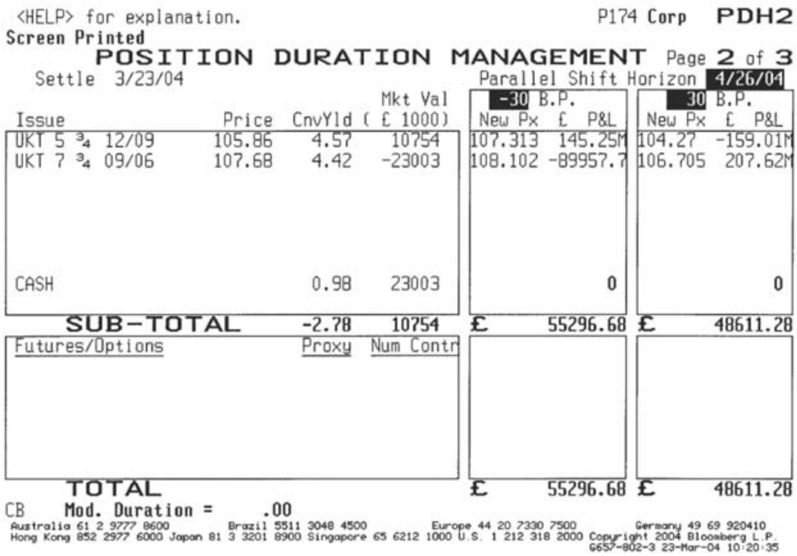

FIGURE 19.15 Impact of Parallel Shift in Yield Curve

parallel shift in the yield curve, in either direction, this spread produces a profit (for a down-shift the profit is £55,296 and for an upshift the gain is £48,611). This is shown in Figure 19.15, page 2 from screen PDH2, while page 3 from the same screen showing the favourable price-yield profile for this spread is shown in Figure 19.16. From this last figure, we note that the price-yield profile remains favourable in either direction, a very desirable position. Of course, this is liable to change with pivotal shifts in the yield curve, but at the time of the trade this spread looks very positive and first-order risk-neutral.

The Sovereign Debt Crisis: Impact on Government Bond Investing

Up to the time of the 2008 crash, the sovereign bonds of developed countries were generally regarded as safe assets and free from the risk of default, irrespective of their credit rating. For this reason, government bonds, starting with U.S. Treasuries were always favoured safe havens during periods of flight to quality. We witnessed this during the credit crisis in 2007–2008 where sharp sell-offs in equities were accompanied by steep rallies in Treasuries, as investible funds moved from risky markets to Treasuries. However, the subsequent sovereign debt crisis has changed much of this thinking.

Following the crash, economies of developed countries embarked on coordinated global fiscal stimulus in order to spend their way out of the recession; at the cost of a rise in public debt during 2009–2011. The policy of easy money and quantitative easing by the U.S. Federal Reserve and other G7 central banks caused a gradual deterioration of the credit quality of sovereigns and potentially undermined the strength of their currencies as a store of value. In August 2011 S&P downgraded the credit rating of U.S. government debt from AAA to AA+. The concerted fiscal

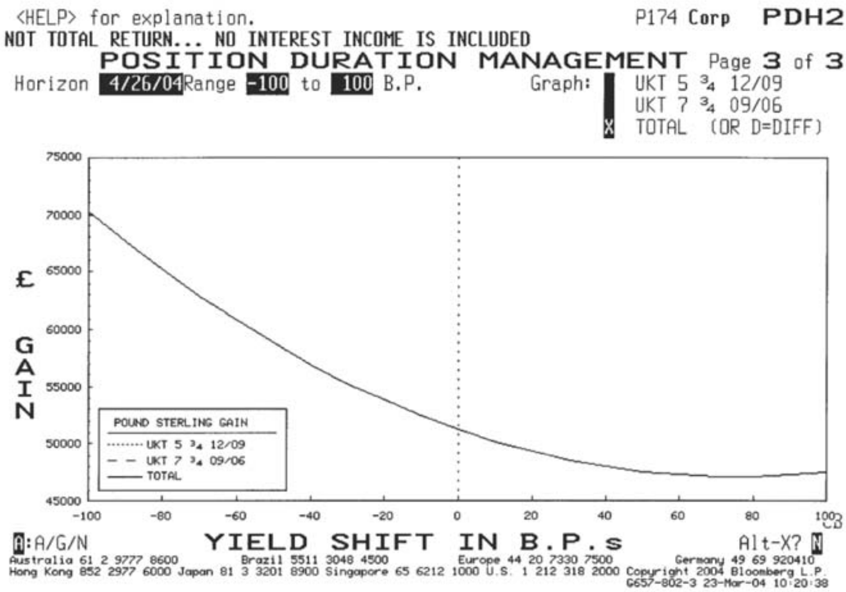

FIGURE 19.16 Price/Yield Profile for Spread Position, Showing Positive Impact of Yield Change in Either Direction, as of 23 March 2004

stimulus in the Eurozone led to a European sovereign debt crisis; weaker countries and economies that were highly leveraged were the first to experience a loss of confidence and contagion, notably Portugal, Italy, Spain, and Greece. In the past, bond investors such as pension and insurance funds would generally favour Italian bonds for example as these gave higher yields for the same perceived (sovereign) credit quality. Hence, the fall in these government bond prices caught many institutional investors by surprise.

The Eurozone debt crisis is complex as it is intricately linked to confidence in the euro. Under monetary union, constituent countries lose the independence of their foreign currency rates, which previously acted as a pressure outlet to compensate any imbalance in a country's capital account position. Without this corrective mechanism, unemployment rose dramatically in the weaker states within the Eurozone.

The sovereign debt crisis has influenced the behaviour and strategy of bond portfolio managers, who no longer view sovereign bonds as risk-free assets. First, the asset allocation strategy has been reconsidered. Traditionally, managers benchmark their funds' performance to bond indices, which are mostly market-capital weighted. This procedure is now flawed: during the crisis, Greece issued more and more bonds to refinance its unsustainable debt burden, which would increase its weighting in the bond index. This adverse selection would encourage passive investors to invest in more indebted countries. At some point, when Greece was downgraded and excluded from key indices, passive funds would then be forced to liquidate holdings of Greek bonds in a falling market.

New proposals in the industry included the creation of bond indices, which are weighted by a credit risk measure, or weighted by country GDP. The rationale is to reward countries with high income and low bond volatility. Bruder, Hereil, and Roncalli (2011) suggested one possible formulation.

The volatility of the bond is first derived from the volatility of its sovereign CDS spreads at time t or $S_i(t)$, thus:

$$\sigma_i^B = D_i \sigma_i^S S_i(t) \tag{19.15}$$

This formula is just a yield-to-price conversion for volatilities, where D_i is the (portfolio average) duration of bonds of country i, the vectors (of length i) σ^B and σ^S are the bond and spread volatilities respectively. The total volatility of a portfolio of sovereign bonds is computed using the usual variance-covariance method we encountered in Chapter 18:

$$R(x) = \sqrt{x^T (\rho_{i,j} \sigma_i^B \sigma_i^B) x} \tag{19.16}$$

where ρ_{ij} is the correlation matrix of spreads of different country CDS, x is the vector of country-weights in the portfolio, and x^T is the transpose of x. Once R is computed, we can derive the (marginal) risk contribution:

$$RC_i = x_i \frac{\partial R(x)}{\partial x_i} \tag{19.17}$$

In other words, the contribution can be computed by shifting x_i marginally by dx_i and observing its incremental effect on R. In this way, a portfolio manager can modify his country allocation (or vector x) based on his risk appetite for a particular country (the vector RC).

A more sophisticated method involves the use of GDP information. A manager can allocate the weights (x) such that the risk contribution by country i is equal to the GDP contribution of country i to the total GDP of all the countries in the portfolio. That means,

$$RC_i = \frac{GDP_i}{\sum_i GDP_i} \tag{19.18}$$

Another outcome of the sovereign debt crisis is in terms of risk management. The market has made increasing use of sovereign CDS (or SCDS). As per Table 19.10, since 2009, the deterioration in the quality of sovereign debt of advanced countries has led to rising demands in SCDS for hedging purposes. The major players are banks and hedge funds. According to data from the BIS, dealer banks are typically net long SCDS protection to hedge their large sovereign bond inventory. Also these primary dealers do not, as a conventional practice, demand collateral from their sovereign counterparties for derivative trades—hence, SCDS are convenient hedges for sovereign counterparty risk also. Nondealer banks and security firms have been net sellers of SCDS, mainly to pick up yield on top of the cash government bond they already hold. Hedge funds have been net sellers of SCDS prior to 2010, and hence would have been hurt by the crisis, but have turned long SCDS after 2010.

TABLE 19.10 Gross Notional Outstanding of Sovereign CDS (Top 10, End 2012)

Country	Economy	Amount (USD/billion)
Italy	Advance	388
Spain	Advance	212
France	Advance	177
Brazil	EM	156
Germany	Advance	154
Turkey	EM	137
Mexico	EM	117
Russia	EM	109
Korea	Advance	85
Japan	Advance	79

Sources: Depository Trust and Clearing Corporation (DTCC); and IMF staff calculations.

The continuing debt crisis in Europe will continue to influence the way sovereign bonds are traded and hedged as an asset class.

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Moorad Choudhry, David Moskovic and Max Wong.

CHAPTER 20

Approaches to Trading and Hedging

The term *trading* covers a wide range of activities. Market makers who are quoting two-way prices to market participants may be tasked with providing a customer service, building up retail and institutional volume, or with purely running the book at a profit and trying to maximise return on capital. The nature of the market that is traded will also have an impact on their approach. In a highly transparent and liquid market such as the U.S. Treasury or the UK gilt market the price spreads are fairly narrow, although increased demand has reduced this somewhat in both markets. However, this means that opportunities for profitable trading as a result of mispricing of individual securities, while not completely extinct, are rare. It is much more common for traders in such markets to take a view on relative-value trades, such as the yield spreads between individual securities or the expected future shape of the yield curve. This is also called spread trading. A large volume of trading on derivatives exchanges is done for hedging purposes, but speculative trading is also prominent. Very often, bond and interest-rate traders will punt using futures or options contracts, based on their view of market direction. Ironically, market makers who have a low level of customer business—perhaps because they are newcomers to the market, or for historical reasons or because they do not have the appetite for risk that is required to service high-quality customers—tend to speculate on the futures exchanges to relieve tedium, often with unfortunate results.

Speculative trading is undertaken on the basis of the views of the trader, desk, or head of the department. This view may be an in-house view; for example, the collective belief of the economics or research department, or the individual trader's view, which will be formulated as a result of fundamental analysis and technical analysis. The former is an assessment of macroeconomic and microeconomic factors affecting not just the specific bond market itself but the economy as a whole. Those running corporate-debt desks will also concentrate heavily on individual sectors and corporations and their wider environment, because the credit spread—and what drives the credit spread—of corporate bonds is of course key to the performance of the bonds. Technical analysis or charting is a discipline in its own right and has its adherents. It is based on the belief that over time the patterns displayed by a continuous time series of asset prices will repeat themselves. Therefore, detecting patterns should give a reasonable expectation of how asset prices should behave in the future. Many traders use a combination of fundamental and technical analysis, although chartists often say that for technical analysis to work effectively, it must be the only method adopted by the trader.

A review of technical analysis is presented in Chapter 63 of the author's book, *The Bond and Money Markets: Strategy, Trading, Analysis*.

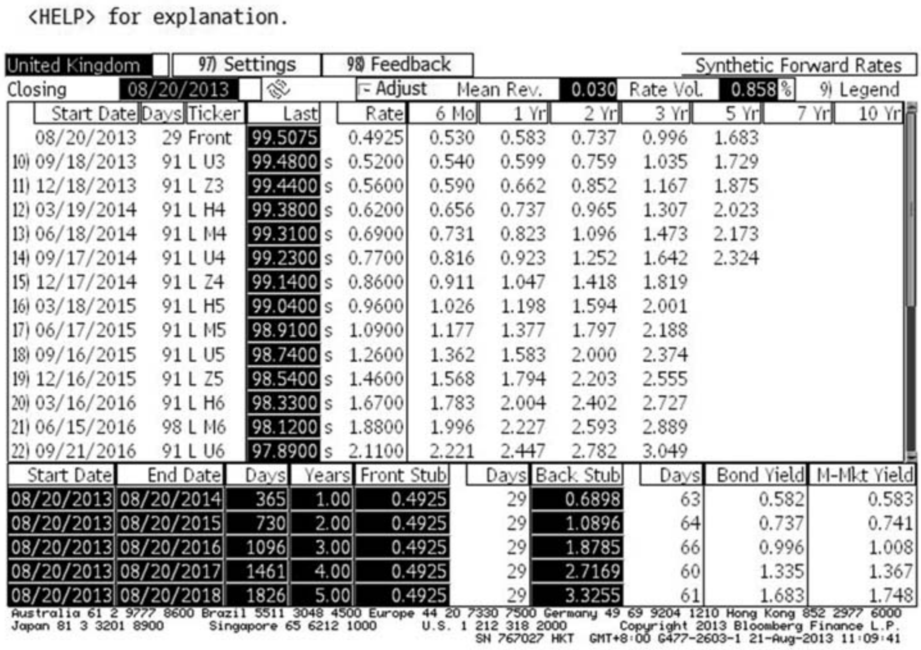
In this chapter, we introduce some common methods and approaches, and some not so common, that might be employed on a fixed-interest desk.

FUTURES TRADING

Trading with derivatives is often preferred, for both speculative or hedging purposes, to trading in the cash markets mainly because of the liquidity of the market and the ease and low cost of undertaking transactions. The essential features of futures trading are volatility and leverage. To establish a futures position on an exchange, the level of margin required is very low proportional to the notional value of the contracts traded. For speculative purposes, traders often carry out open—that is, uncovered—trading which is a directional bet on the market. So, therefore, if a trader believed that short-term sterling interest rates were going to fall, he could buy a short sterling contract on LIFFE. This may be held for under a day (in which case, if the price rises the trader will gain), or for a longer period, depending on his view. The tick value of a short sterling contract is £12.50; so, if he bought one lot at 92.75 (that is, 100 – 92.75 or 7.25%) and sold it at the end of the day for 98.85, he made a profit of £125 on his one lot, from which brokerage will be subtracted. The trade can be carried out with any futures contract. The same idea could be carried out with a cash-market product or a forward-rate agreement (FRA), but the liquidity, narrow price spread and the low cost of dealing make such a trade easier on a futures exchange. It is much more interesting, however, to carry out a spread trade on the difference between the rates of two different contracts. Consider Figures 20.1 and 20.2, which relate to the prices for the LIFFE short-sterling futures contract on 20 August 2013.

Futures exchanges use the letters H, M, U, and Z to refer to the contract months for March, June, September, and December. So the June 2014 contract is denoted by "M4". From Chapter 3 we know that forward rates can be calculated for any term, starting on any date. In Figure 20.1, we see the future prices on that day, and the interest rate that the prices imply. The "stub" is the term for the interest rate from today to the expiry of the first futures contract, which is called the front month contract (in this case the front month contract is the September 2013 contract) and is 29 days. Figure 20.2 lists the forward rates from the spot date to six months, one year, and so on. It is possible to trade a strip of contracts to replicate any term, out to the maximum maturity of the contract. This can be done for hedging or speculative purposes. Note from Figure 20.2 that there is a spread between the cash curve and the futures curve. A trader can take positions on cash against futures, but it is easier to transact only on the futures exchange.

Short-term money-market interest rates often behave independently of the yield curve as a whole. A money-markets trader may be aware of cash-market trends—for example, an increased frequency of borrowing at a certain point of the curve—as well as other market intelligence that suggests that one point of the curve will rise or fall relative to others. One way to exploit this view is to run a position in a cash instrument such as a CD against a futures contract, which is a *basis spread* trade.



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United Kingdom | 97) Settings | 98) Feedback | Synthetic Forward Rates

Closing 08/20/2013 | Adjust | Mean Rev. 0.030 | Rate Vol. 0.858 % | 9) Legend

	Start Date	Days	Ticker	Last		Rate	6 Mo	1 Yr	2 Yr	3 Yr	5 Yr	7 Yr	10 Yr
	08/20/2013	29	Front	99.5075		0.4925	0.530	0.583	0.737	0.996	1.683		
10)	09/18/2013	91	L U3	99.4800	s	0.5200	0.540	0.599	0.759	1.035	1.729		
11)	12/18/2013	91	L Z3	99.4400	s	0.5600	0.590	0.662	0.852	1.167	1.875		
12)	03/19/2014	91	L H4	99.3800	s	0.6200	0.656	0.737	0.965	1.307	2.023		
13)	06/18/2014	91	L M4	99.3100	s	0.6900	0.731	0.823	1.096	1.473	2.173		
14)	09/17/2014	91	L U4	99.2300	s	0.7700	0.816	0.923	1.252	1.642	2.324		
15)	12/17/2014	91	L Z4	99.1400	s	0.8600	0.911	1.047	1.418	1.819			
16)	03/18/2015	91	L H5	99.0400	s	0.9600	1.026	1.198	1.594	2.001			
17)	06/17/2015	91	L M5	98.9100	s	1.0900	1.177	1.377	1.797	2.188			
18)	09/16/2015	91	L U5	98.7400	s	1.2600	1.362	1.583	2.000	2.374			
19)	12/16/2015	91	L Z5	98.5400	s	1.4600	1.568	1.794	2.203	2.555			
20)	03/16/2016	91	L H6	98.3300	s	1.6700	1.783	2.004	2.402	2.727			
21)	06/15/2016	98	L M6	98.1200	s	1.8800	1.996	2.227	2.593	2.889			
22)	09/21/2016	91	L U6	97.8900	s	2.1100	2.221	2.447	2.782	3.049			

Start Date	End Date	Days	Years	Front Stub	Days	Back Stub	Days	Bond Yield	M-Mkt Yield
08/20/2013	08/20/2014	365	1.00	0.4925	29	0.6898	63	0.582	0.583
08/20/2013	08/20/2015	730	2.00	0.4925	29	1.0896	64	0.737	0.741
08/20/2013	08/20/2016	1096	3.00	0.4925	29	1.8785	66	0.996	1.008
08/20/2013	08/20/2017	1461	4.00	0.4925	29	2.7169	60	1.335	1.367
08/20/2013	08/20/2018	1826	5.00	0.4925	29	3.3255	61	1.683	1.748

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000
Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2013 Bloomberg Finance L.P.
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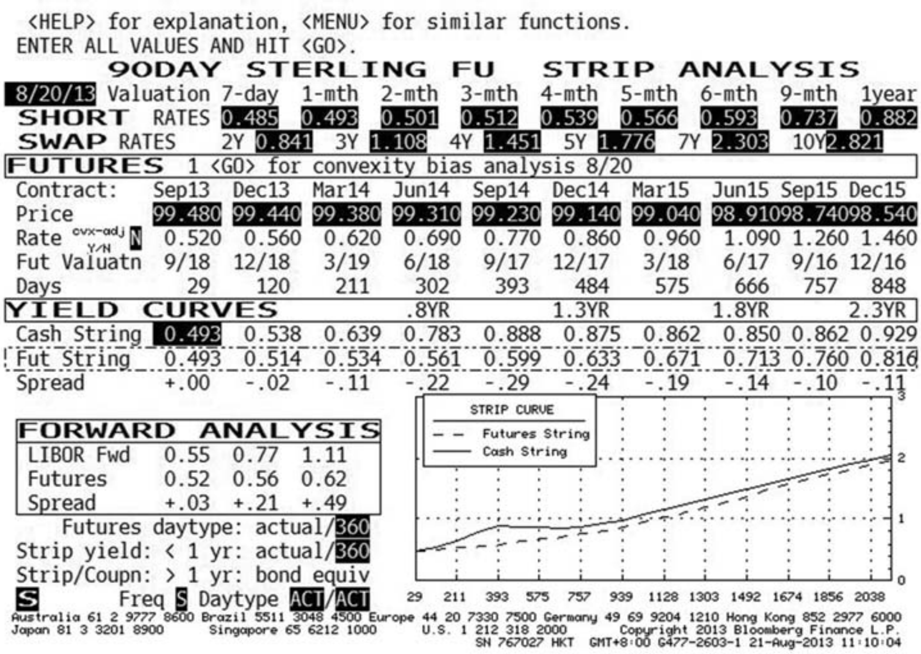
FIGURE 20.1 LIFFE Short-Sterling Contract Analysis, 20 August 2013



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90DAY STERLING FU STRIP ANALYSIS

8/20/13 Valuation	7-day	1-mth	2-mth	3-mth	4-mth	5-mth	6-mth	9-mth	1year
SHORT RATES	0.485	0.493	0.501	0.512	0.539	0.566	0.593	0.737	0.882

SWAP RATES 2Y 0.841 3Y 1.108 4Y 1.451 5Y 1.776 7Y 2.303 10Y 2.821

FUTURES 1 <GO> for convexity bias analysis 8/20

Contract:	Sep13	Dec13	Mar14	Jun14	Sep14	Dec14	Mar15	Jun15	Sep15	Dec15
Price	99.480	99.440	99.380	99.310	99.230	99.140	99.040	98.910	98.740	98.540
Rate cvx-adj Y/N N	0.520	0.560	0.620	0.690	0.770	0.860	0.960	1.090	1.260	1.460
Fut Valuatn	9/18	12/18	3/19	6/18	9/17	12/17	3/18	6/17	9/16	12/16
Days	29	120	211	302	393	484	575	666	757	848

YIELD CURVES .8YR 1.3YR 1.8YR 2.3YR

Cash String	0.493	0.538	0.639	0.783	0.888	0.875	0.862	0.850	0.862	0.929
Fut String	0.493	0.514	0.534	0.561	0.599	0.633	0.671	0.713	0.760	0.816
Spread	+.00	-.02	-.11	-.22	-.29	-.24	-.19	-.14	-.10	-.11

FORWARD ANALYSIS			
LIBOR Fwd	0.55	0.77	1.11
Futures	0.52	0.56	0.62
Spread	+.03	+.21	+.49

Futures daytype: actual/360
Strip yield: < 1 yr: actual/360
Strip/Coupn: > 1 yr: bond equiv
S Freq S Daytype ACT/ACT

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000
Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2013 Bloomberg Finance L.P.
SN 767027 HKT GMT+8:00 G477-2603-1 21-Aug-2013 11:10:04

FIGURE 20.2 LIFFE Short-Sterling Forward Rates Analysis, 20 August 2013

However, the best way to trade on this view is to carry out a spread trade, shorting one contract against a long position in another trade. Consider Figure 20.1; if we feel that three-month interest rates in September 2014 will be higher than where they are implied by the futures price today, but that September 2015 rates will be lower, we will short the U4 contract and long the U5 contract. This is not a market-directional trade; rather, it's a view on the relative spread between two contracts. The trade must be carried out in equal weights; for example, 100 lots of the U4 against 100 lots of the U5. If the rates do move in the direction that the trader expects, the trade will generate a profit. There are similar possibilities available from an analysis of Figure 20.2, depending on our view of forward interest rates.

Spread trading carries a lower margin requirement than open-position trading, because there is no directional risk in the trade. It is also possible to arbitrage between contracts on different exchanges. If the trade is short the near contract and long the far contract (that is, the opposite of our example), this is known as buying the spread, and the trader believes the spread will widen. The opposite is shorting the spread and is undertaken when the trader believes the spread will narrow. Note that the difference between the two price levels is not limitless, because the theoretical price of a futures contract provides an upper limit to the size of the spread or the basis. The spread or the basis cannot exceed the cost of carry; that is, the net cost of buying the cash security today and then delivering it into the futures market on the contract expiry. The same principle applies to short-dated interest-rate contracts; the net cost is the difference between the interest cost of borrowing funds to buy the "security" and the income accruing on the security while it is held before delivery. The two associated costs for a short-sterling spread trade are the notional borrowing and lending rates from having bought one and sold another contract. If the trader believes that the cost of carry will decrease, she could sell the spread to exercise this view.

The trader may have a longer time horizon and trade the spread between the short-term interest-rate contract and the long bond future. This is usually carried out only by the proprietary trading desk, because it is unlikely that one person would be trading both three-month and 10-year (or 20-year, depending on the contract specification) interest rates. A common example of such a spread trade is a yield-curve trade. If a trader believes that the sterling yield curve will steepen or flatten between the three-month and the 10-year terms, she can buy or sell the spread by using the LIFFE short-sterling contract and the long gilt contract. To be first-order risk-neutral, however, the trade must be duration-weighted, as one short-sterling contract is not equivalent to one gilt contract. The tick value of the gilt contract is £10, however, although the gilt contract represents £100,000 of a notional gilt and the short-sterling contract represents a £500,000 time deposit. We use equation (20.1) to calculate the hedge ratio, with £1,000 being the value of a 1% change in the value of the gilt contract against £1,250 for the short sterling contract.

$$h = \frac{(100 \times tick) \times P_b \times D}{(100 \times tick) \times P_f} \tag{20.1}$$

where *tick* is the tick value of the contract
D is the duration of the bond represented by the long bond contract

P_b is the price of the bond futures contract
P_f is the price of the short-term deposit contract

The notional maturity of a long bond contract is always given in terms of a spread; for example, for the long gilt it is $8^3/_4$–13 years. Therefore, in practice, one would use the duration of the cheapest-to-deliver bond.

A butterfly spread is a spread trade that involves three contracts, with the two spreads between all three contracts being traded. This is carried out when the middle contract appears to be mispriced relative to the two contracts on either side of it. The trader may believe that one or both of the outer contracts will move in relation to the middle contract. If the belief is that only one of these two will shift relative to the middle contract, then a butterfly will be put on if the trader is not sure which of these will adjust.

For example, consider Figure 20.2 again. Assume the trader believes that the cash curve is reflecting true market expectation, and the futures strip is underpricing the forward yield especially around the 393 days tenor (at the September 2014 contract). In this case he should short the September 2014 contract, but this simple strategy would expose him to the risks of an overall fall in rate level and also changes in the slope of the yield curve. Is there a way to take advantage of the (perceived) mispricing without taking on these two types of risks?

To express this view the trader puts on a *butterfly spread* in which he shorts two contracts of September 2014 (the body of the butterfly) and longs one contract each on the wings; that is, one March 2014 and one March 2015. For contracts that are not too far apart in expiry, this strategy is approximately duration neutral, meaning that the trader will not be affected by parallel movement in the yield curve or changes in curve slope. If the wings are further apart, a proper duration weighting will be required; that is, buying more contracts with shorter expiry to offset the duration risk of the longer-dated contracts.

YIELD CURVES AND RELATIVE VALUE

Bond-market participants take a keen interest in both cash and zero-coupon (spot) yield curves. In markets where an active zero-coupon bond market exists, much analysis is undertaken into the relative spreads between derived and actual zero-coupon yields. In this section, we review some of the yield-curve analysis used in the market.

The Determinants of Government Bond Yields

Market makers in government-bond markets will analyse various factors in the market in deciding how to run their book. Customer business apart, decisions to purchase or sell securities will be a function of their views on:

- Market direction itself; that is, the direction in which short-term and long-term interest rates are headed
- Which maturity point along the entire term structure offers the best value
- Which specific issue within a particular maturity point offers the best value

All three areas are related but will react differently to certain pieces of information. A report on the projected size of the government's budget deficit, for example, will not have much effect on two-year bond yields, whereas if the expectations come as a surprise to the market, it could have an adverse effect on long-bond yields. The starting point for analysis is, of course, the yield curve—both the traditional coupon curve plotted against duration and the zero-coupon curve. Figure 20.3 illustrates the traditional yield curve for gilts on 20 August 2013.

For a first-level analysis, many market practitioners will go no further than Figure 20.3. An investor who has no particular view on the future shape of the yield curve or the level of interest rates may well adopt a neutral outlook and hold bonds that have a duration that matches their investment horizon. If investors believed interest rates were likely to remain stable for a time, they might hold bonds with a longer duration in a positive-sloping yield-curve environment, and pick up additional yield but with higher interest-rate risk. Once the decision has been made on which part of the yield curve to invest in or switch into, the investor must decide on the specific securities to hold, which then brings us to relative-value analysis. For this, the investor will analyse specific sectors of the curve, looking at individual stocks. This is sometimes called looking at the "local" part of the curve.

An assessment of a local part of the yield curve will include looking at other features of individual stocks in addition to their duration. This recognises that the yield of a specific bond is not simply a function of its duration, and that two bonds with near-identical duration can have different yields. The other determinants of yield are liquidity of the bond and its coupon. To illustrate the effect of coupon on

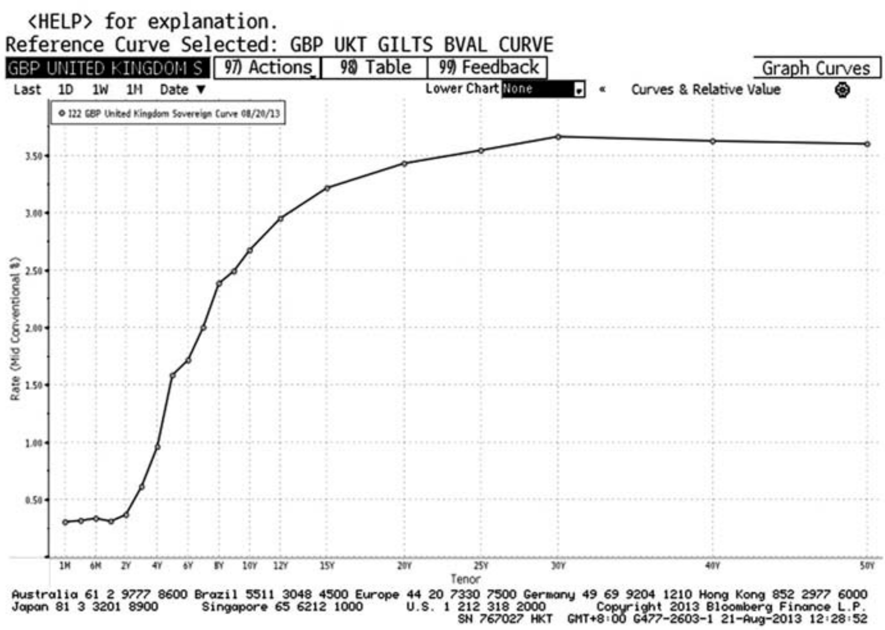

FIGURE 20.3 Yield and Duration of Gilts, 20 August 2013
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TABLE 20.1 Duration and Yield Comparisons for Bonds in a Hypothetical Inverted Curve Environment

Coupon	Maturity	Duration	Yield
8%	20-Feb-02	1.927	5.75%
12%	5-Feb-02	1.911	5.80%
10%	20-Jun-10	7.134	4.95%
6%	1-Jul-10	7.867	4.77%

yield, consider Table 20.1. This shows that, where the duration of a bond is held roughly constant, a change in coupon of a bond can have a significant effect on the bond's yield.

In the case of the long bond, under this scenario an investor could both shorten duration and pick up yield, which is not the first thing that an investor might expect. However, an anomaly of the markets is that, liquidity issues aside, the market does not generally like high-coupon bonds, so they usually trade cheap to the curve.

The other factors affecting yield are supply and demand, and liquidity. A shortage of supply of stock at a particular point in the curve will have the effect of depressing yields at that point. A reducing public-sector deficit is the main reason why such a supply shortage might exist. In addition, as interest rates decline—ahead of or during a recession, say—the stock of high-coupon bonds increases, as the newer bonds are issued at lower levels, and these "outdated" issues can end up trading at a higher yield. Demand factors are driven primarily by the investor's views on the country's economic prospects, but also by government legislation; for example, the Minimum Funding Requirement in the United Kingdom, which compelled pension funds to invest a set minimum amount of their funds in long-dated gilts. This kept yields artificially low until the requirement was removed during 2002.

Liquidity often results in one bond having a higher yield than another, despite both having similar durations. Institutional investors prefer to hold the benchmark bond, which is the current 2-year, 5-year, 10-year, or 30-year bond, and this depresses the yield on the benchmark bond. A bond that is liquid also has a higher demand, and thus a lower yield, because it is easier to convert into cash if required. This can be demonstrated by valuing the cash flows on a six-month bond with the rates obtainable in the Treasury-bill market. We could value the six-month cash flows at the six-month bill rate. The lowest obtainable yield in virtually every market[1] is the T-bill yield. Therefore, valuing a six-month bond at the T-bill rate will produce a discrepancy between the observed price of the bond and its theoretical price implied by the T-bill rate as the observed price will be lower. The reason for this is simple: because the T-bill is more readily realisable into cash at any time, it trades at a lower yield than the bond, even though the cash flows fall on the same day.

We have therefore determined that a bond's coupon and liquidity level, as well as its duration, will affect the yield at which it trades. These factors can be used in

[1] The author is not aware of any market where there is a yield lower than its shortest-maturity T-bill yield, but that does not mean such a market doesn't exist!

conjunction with other areas of analysis, which we look at next, when deciding which bonds carry relative value over others.

Characterising the Complete Term Structure

As many readers would have gathered, the yield-versus-duration curve illustrated in Figure 20.3 is an ineffective technique with which to analyse the market.

This is because it does not highlight any characteristics of the yield curve other than its general shape; this does not assist in the making of trading decisions. To facilitate a more complete picture, we might wish to employ the technique described here. Figure 20.4 shows the bond par-yield curve and T-bill yield curve for gilts in March 2004. Figure 20.5 shows the difference between the yield on a bond with a coupon that is 100 basis points below the par-yield level, and the yield on a par bond. The other curve in Figure 20.5 shows the level for a bond with a coupon that is 100 basis points above the par yield. These two curves show the "low-coupon" and "high-coupon" yield spreads. Using the two figures together, an investor can see the impact of coupons, the shape of the curve and the effect of yield on different maturity points of the curve.

Identifying Relative Value in Government Bonds

Constructing a zero-coupon yield curve provides the framework within which a market participant can analyse individual securities. In a government-bond market, there is no credit-risk consideration (unless it is an emerging-market government market) and, therefore, no credit spreads to consider. There are a number of factors that can be assessed in an attempt to identify relative value.

The objective of much of the analysis that occurs in bond markets is to identify value, and identifying which individual securities should be purchased and which sold. At the overview level, this identification is a function of whether one thinks interest rates are going to rise or fall. At the local level, though, the analysis is more concerned with a specific sector of the yield curve: whether this will flatten

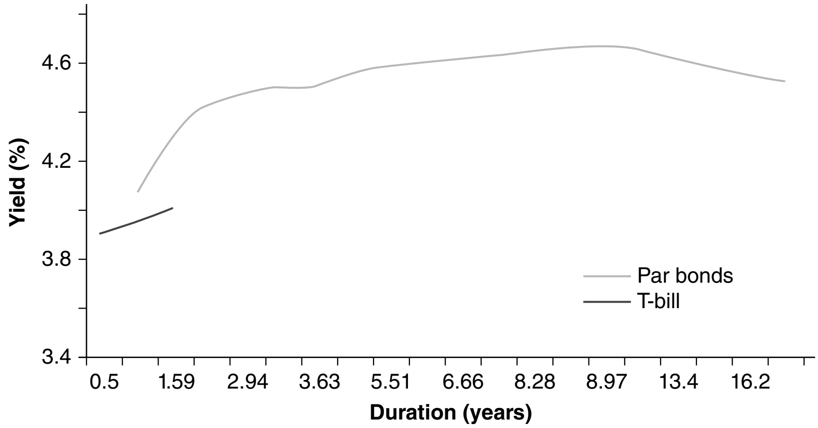

FIGURE 20.4 T-Bill and Par-Yield Curve, March 2004

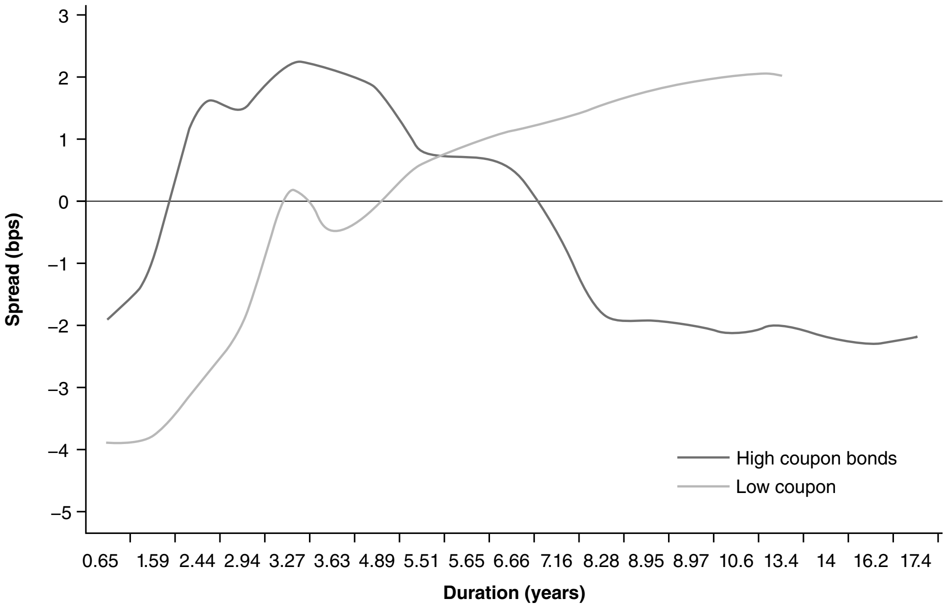

FIGURE 20.5 Structure of Bond Yields, March 2004

or steepen, whether bonds of similar duration are trading at enough of a spread to warrant switching from one into another. The difference in these approaches is one of identifying which stocks have absolute value, and which have relative value. A trade decision based on the expected direction of interest rates is based on assessing absolute value, whether interest rates themselves are too low or too high. Yield-curve analysis is more a matter of assessing relative value. On (very!) rare occasions, this process is fairly straightforward. For example, if the three-year bond is trading at 5.75% when two-year yields are 5.70% and four-year yields are at 6.15%, the three-year would appear to be overpriced. However, this is not a real-life situation. Instead, a trader might find himself assessing the relative value of the three-year bond compared to much shorter- or longer-dated instruments. That said, there is considerable difference between comparing a short-dated bond to other short-term securities and comparing, say, the two-year bond to the 30-year bond. Although it looks like it on paper, the space along the x-axis should not be taken to imply that the smooth link between one-year and five-year bonds is repeated from the five-year out to the 30-year bonds. It is also common for the very short-dated sector of the yield curve to behave independently of the long end of the curve.

One method used to identify relative value is to quantify the coupon effect on the yields of bonds. The relationship between yield and coupon is given by (20.2):

$$rm = rm_p + c \cdot max(C_{PD} - rm_p, 0) + d \cdot min(C_{PD} - rm_p, 0) \tag{20.2}$$

where rm is the yield on the bond being analysed
rm_P is the yield on a par bond of specified duration

C_{PD} is the coupon on an arbitrary bond of similar duration to the par bond

and c and d are coefficients.

The coefficient c reflects the effect of a high coupon on the yield of a bond. If we consider a case where the coupon rate exceeds the yield on the similar-duration par bond ($C_{PD} > rm_p$), equation (20.2) reduces to (20.3):

$$rm = rm_p + c \cdot (C_{PD} - rm_p) \tag{20.3}$$

Equation (20.3) specifies the spread between the yield on a high-coupon bond and the yield on a par bond as a linear function of the spread between the first bond's coupon and the yield and coupon of the par bond. In reality, this relationship may not be purely linear: for instance, the yield spread may widen at a decreasing rate for higher coupon differences. Therefore (20.3) is an approximation of the effect of a high coupon on yield where the approximation is more appropriate for bonds trading close to par. The same analysis can be applied to bonds with coupons lower than the same-duration par bond.

The value of a bond may be measured against comparable securities or against the par or zero-coupon yield curve. In certain instances, the first measure may be more appropriate when, for instance, a low-coupon bond is priced expensive to the curve itself but fair compared to other low-coupon bonds. In that case, the overpricing indicated by the par-yield curve may not represent unusual value but, rather, a valuation phenomenon that was shared by all low-coupon bonds. Having examined the local structure of a yield curve, the analysis can be extended to the comparative valuation of a group of similar bonds. This is an important part of the analysis, because it is particularly informative to know the cheapness or dearness of a single stock compared to the whole yield curve, which might be somewhat abstract. Instead, we would seek to identify two or more bonds, one of which was cheap and the other dear, so that we might carry out an outright switch between the two, or put on a spread trade between them. Using the technique, we can identify excess positive or negative yield spread for all the bonds in the term structure. This has been carried out for five gilts, together with other less-liquid issues as of March 2004 and the results are summarised in Table 20.2.

From the table, as we might expect, the benchmark securities are all expensive to the par curve, and the less-liquid bonds are cheap. Note that the 6.25% 2010 appears cheap to the curve, but the 5.75% 2009 offers a yield pick-up for what is a shorter-duration stock; this is a curious anomaly and one that had disappeared a few days later.[2]

What this section has introduced is the concept of relative value for individual securities, and how the simple duration/yield analysis can be extended to assess other determinants of a bond's yield. Other types of analysis, such as assessing the value of coupon bonds relative to the zero-coupon yield, are assessed in Chapter 19.

We now look at the issues involved in putting on a spread trade.

[2] In other words, we've missed the opportunity! This analysis used mid-prices, which would not be available in practice.

TABLE 20.2 Yields and Excess Yield Spreads for Selected Gilts, 25th March 2004

Bond	Duration (years)	Yield %	Excess Yield Spread (bps)
6.75% 26-Nov-2004	0.651	4.07	–1.05
8.5% 7-Dec-2005	1.585	4.307	2.08
7.5% 7-Dec-2006	2.453	4.434	–0.08
4.5% 7-Mar-2007	2.782	4.453	0.02
7.25% 7-Dec-2007	3.267	4.502	1.3
5% 7-Mar-2008	3.626	4.51	1.6
5.75% 7-Dec-2009	4.888	4.591	1.7
6.25% 25-Nov-2010	5.506	4.589	3.19
5% 7-Sep-2014	8.28	4.645	–5.6
4.25% 7-Jun-2032	16.172	4.555	–7.7

YIELD-SPREAD TRADES

In the earlier section on futures trading, we introduced the concept of spread trades, which are not market-directional trades but rather the expression of a viewpoint on the shape of a yield curve or, more specifically, the spread between two particular points on the yield curve. Generally, there is no analytical relationship between changes in a specific yield spread and changes in the general level of interest rates. That is to say, the yield curve can flatten when rates are both falling or rising, and equally may steepen under either scenario as well. The key element of any spread trade is that it is structured so that a profit (or any loss) is made only as a result of a change in the spread, and not due to any change in overall yield levels. That is, spread trading eliminates market-directional or first-order market risk.

Bond-Spread Weighting

Table 20.3 shows data for our selection of gilts but with additional information on the basis-point value (BPV) for each point. This is also known as the "dollar value of a basis point" or DV01.

If a trader believed that the yield curve was going to flatten, but had no particular strong feeling about whether this flattening would occur in an environment of falling or rising interest rates, and thought that the flattening would be most pronounced in the 2-year versus 10-year spread, he could put on a spread consisting of a short position in the 2-year and a long position in the 10-year. This spread must be duration-weighted to eliminate first-order risk. At this stage, we must point out, and it is important to be aware of, the fact that basis-point values, which are used to weight the trade, are based on modified-duration measures. From Chapter 2, we know that this measure is an approximation and will be inaccurate for large changes in yield. Therefore, the trader must monitor the spread to ensure that the weights are not going out of line, especially in a volatile market environment.

TABLE 20.3 Bond Basis-Point Value, 25th March 2004

Bond	Duration (yrs)	Yield %	Price	Basis Point Value (BPV)
6.75% 26-Nov-2004	0.651	4.07	101.74	0.00664
8.5% 7-Dec-2005	1.585	4.307	106.79	0.01697
7.5% 7-Dec-2006	2.453	4.434	107.71	0.02639
4.5% 7-Mar-2007	2.782	4.453	100.12	0.02738
7.25% 7-Dec-2007	3.267	4.502	109.26	0.03561
5% 7-Mar-2008	3.626	4.51	101.75	0.03618
5.75% 7-Dec-2009	4.888	4.591	105.88	0.05142
6.25% 25-Nov-2010	5.506	4.589	109.44	0.06004
5% 7-Sep-2014	8.28	4.645	102.91	0.08349
4.25% 7-Jun-2032	16.172	4.555	95.18	0.15252

To weight the spread, we use the ratios of the BPVs of each bond to decide on how much to trade. In our example, assume the trader wants to purchase £10 million of the 10-year. In that case, he must sell [(0.08349/0.02639) × 10,000,000] or £31,636,983 of the two-year bond. It is also possible to weight a trade using the bonds' duration values, but this is rare. It is common practice to use the BPV.

The payoff from the trade will depend on what happens to the 2-year vis-à-vis the 10-year spread. If the yields on both bonds move by the same amount, there will be no profit generated, although there will be a funding consideration. If the spread does indeed narrow, the trade will generate profit. Note that disciplined trading calls for both an expected target spread as well as a fixed time horizon. Say, for example, that the current spread is 59.7 basis points; the trader may decide to take the profit if the spread narrows to 50 basis points, with a three-week horizon. If at the end of three weeks the spread has not reached the target, the trader should unwind the position anyway, because that was the original target. On the other hand, what if the spread has narrowed to 48 basis points after one week and looks like narrowing further—what should the trader do? Again, disciplined trading suggests the profit should be taken. If, contrary to expectations, the spread starts to widen, if it reaches 64.5 basis points the trade should be unwound, this "stop-loss" being at the half-way point of the original profit target.

The financing of the trade in the repo markets is an important aspect of the trade and will set the trade's break-even level. If the bond being shorted (in our example, the two-year bond) is special, this will have an adverse impact on the financing of the trade. The repo considerations are considered in Chapter 6.

Types of Bond Spreads

A bond spread has two fundamental characteristics. In theory, there should be no P/L effect due to a general change in interest rates, and any P/L should only occur as a result of a change in the specific spread being traded. Note that this assumes that all moves in the yield curve are a parallel shift. Most bond-spread trades are yield-curve trades where a view is taken on whether a particular spread will widen or narrow. Therefore, it is important to be able to identify which sectors of the curve to sell. Assuming that a trader is able to transact business along any part of the yield curve, there are a number of factors to consider. In the first instance, there is the historic spread between the two sectors of the curve. To illustrate in simplistic fashion, if the two-10-year spread has been between 40 and 50 basis points over the last six months but very recently has narrowed to less than 35 basis points, this may indicate imminent widening. Other factors to consider are demand and liquidity for individual stocks relative to others, and any market intelligence that the trader gleans. If there has been considerable customer interest in certain stocks relative to others, because investors themselves are switching out of certain stocks and into others, this may indicate a possible yield-curve play. It is a matter of individual judgment.

An historical analysis requires that the trader identify some part of the yield curve within which he expects to observe a flattening or steepening. It is, of course, entirely possible that one segment of the curve will flatten while another segment is steepening. In fact, this phenomenon is quite common. This reflects the fact that different segments respond to news and other occurrences in different ways. From Figure 20.6 we see that the 2-year versus 10-year spread was at its narrowest in March 2004 during the last 12 months, having reached a high point in July 2003. This might indicate that spread widening was imminent.

A more exotic type of yield curve spread is a *curvature* trade. Consider, for example, a trader who believes that three-year bonds will outperform, on a relative basis, both two-year and five-year bonds. That is, he believes that the two-year/three-year spread will narrow relative to the three-year/five-year spread or, in other words, that the curvature of the yield curve will decrease. This is also known as a butterfly/barbell trade. In our example, the trader will buy the three-year bond, against short sales

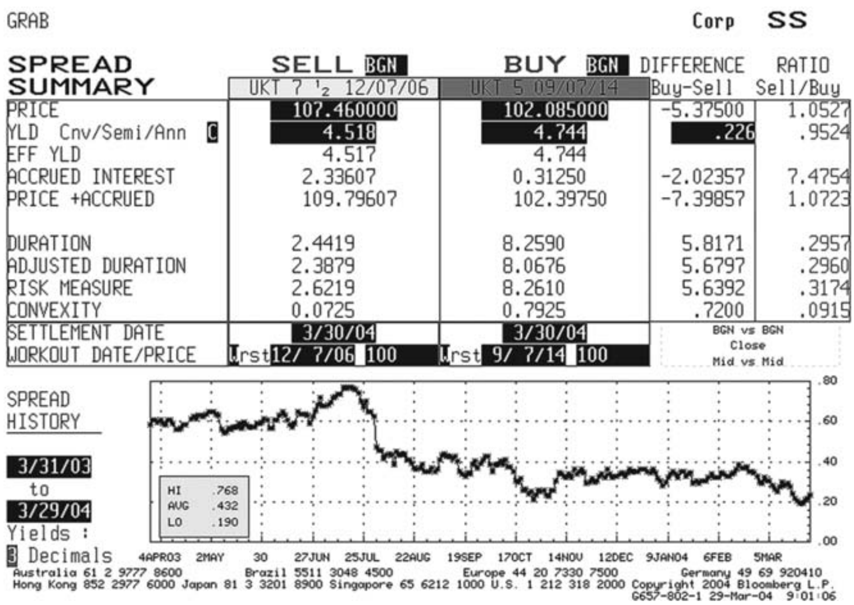

FIGURE 20.6 Two-Year and 10-Year Spread History, UK Gilt Market, 29 March 2004

of both the two-year and the five-year bonds. All positions are duration-weighted. The principle is exactly the same as the butterfly trade we described in the previous section on futures trading.

HEDGING BOND POSITIONS

Hedging is a straightforward concept to understand or describe. However, it is very important that be undertaken as accurately as possible. Therefore, the calculation of a hedge is critical. A hedge is a position in a cash or off-balance-sheet instrument that removes the market-risk exposure of another position. For example, a long position in 10-year bonds can be hedged with a short position in 20-year bonds, or with futures contracts. That is the straightforward part; the calculation of the exact amount of the hedge is where complexities can arise. In this section, we review the basic concepts of hedging, and a case study at the end illustrates some of the factors that must be considered.

Simple Hedging Approach

The hedge calculation that first presents itself is the duration-weighted approach. From the sample of gilts in Table 20.3, it is possible to calculate the amount of one bond required to hedge an amount of any other bond, using the ratio of the BPVs. This approach is very common in the market; however, it suffers from two basic flaws that hinder its effectiveness. First, the approach implicitly assumes comparable volatility of yields on the two bonds; and, secondly, it also assumes that yield changes on the two bonds are highly correlated. Where one or both of these factors do not apply, the effectiveness of the hedge will be compromised.

The assumption of comparable volatility becomes increasingly unrealistic the more the bonds differ in terms of market risk and market behaviour. Consider a long position in two-year bonds hedged with a short position in five-year bonds. Using the bonds from Table 20.3, if we had a position of £1 million of the two-year, we would short £513,225 of the five-year. Even if we imagine that yields between the two bonds are perfectly correlated, it may well be that the amount of yield change is different because the bonds have different volatilities. For example, if the yield on the five-year bond changes only by half the amount that the two-year does, if there was a five-basis-point rise in the two-year, the five-year would have risen only by 2.5 basis points. This would indicate that the yield volatility of the two-year bond was twice that of the five-year bond. This suggests that a hedge calculation that matched nominal amounts, due to BPV, on the basis of an equal change in yield for both bonds would be incorrect. In our illustration, the short position in the five-year bond would be effectively hedging only half of the risk exposure of the two-year position.

The implicit assumption of perfectly correlated yield changes can also lead to inaccuracy. Across the whole term structure, it is not always the case that bond yields are even positively correlated all the time (although most of the time there will be a close positive correlation). Therefore, using our illustration again, imagine that the two-year and the five-year bonds possess identical yield volatilities, but that changes in their yields are uncorrelated. This means that knowing that the yield on the two-year bond rose or fell by one basis point does not tell us anything about the change in the yield on the five-year bond. If yield changes between the two bonds are indeed uncorrelated, this means that the five-year bonds cannot be used to hedge two-year bonds, at least not with accuracy.

Hedge Analysis

From the foregoing we note that there are at least three factors that will affect the effectiveness of a bond hedge; these are the basis-point value, the yield volatility of each bond, and the correlation between changes in the two yields of a pair of bonds. Considering volatilities and correlations, Table 20.4 shows the standard deviations and correlations of weekly yield changes for a set of gilts during the nine months to March 2004. The standard deviation of weekly yield changes was, in fact, highest for the short-date paper, and actually declined for longer-dated paper. From the table we also note that changes in yield were imperfectly correlated. We expect correlations to be highest for bonds in the same segments of the yield curve, and to decline between bonds that are in different segments. This is not surprising, and indeed 2-year bond yields are more positively correlated with 5-year bonds and less so with 30-year bonds.

We can use the standard relationship for correlations and the effect of correlation to adjust a hedge. Consider two bonds with nominal values M_1 and M_2; if the yields on these two bonds change by Δr_1 and Δr_2, the net value of the change in position is given by:

$$\Delta PV = M_1 BPV_1 \Delta r_1 + M_2 BPV_2 \Delta r_2 \tag{20.4}$$

The uncertainty of the change in the net value of a two-bond position is dependent on the nominal values, the volatility of each bond and the correlation between these yield changes. Therefore, for a two-bond position we set the standard deviation of the change in the position as (20.5):

$$\sigma_{pos} = \sqrt{M_1^2 BPV_1^2 \sigma_1^2 + M_2^2 BPV_2^2 \sigma_2^2 + 2M_1 M_2 BPV_1 BPV_2 \sigma_1 \sigma_2 \rho} \tag{20.5}$$

TABLE 20.4 Yield Volatility and Correlations, Selected Gilts, March 2004

	Segment					
	2-Year	**3-Year**	**5-Year**	**10-Year**	**20-Year**	**30-Year**
Volatility (bps)						
Correlation	20.5	19.5	20.2	20.0	20.1	20.3
2-year	1.000	0.973	0.949	0.919	0.887	0.879
3-year	0.973	1.000	0.961	0.935	0.901	0.889
5-year	0.949	0.961	1.000	0.968	0.951	0.945
10-year	0.919	0.935	0.968	1.000	0.981	0.983
20-year	0.887	0.901	0.951	0.981	1.000	0.987
30-year	0.879	0.889	0.945	0.983	0.987	1.000

where ρ is the correlation between the yield volatilities of bonds 1 and 2. We can rearrange equation (20.5) to set the optimum hedge value for any bond using equation (20.6):

$$M_2 = -\frac{\rho BPV_1 \sigma_1}{BPV_2 \sigma_2} M_1 \tag{20.6}$$

so that M_2 is the nominal value of any bond used as a hedge given any nominal value M_1 of the first bond, and using each bond's volatility and the correlation. The derivation of (20.6) is given in Appendix 20.1. A lower correlation leads to a smaller hedge position, because where yield changes are not closely related, this implies greater independence between yield changes of the two bonds. In a scenario where the standard deviation of two bonds is identical, and the correlation between yield changes is 1, (20.6) reduces to:

$$M_2 = \frac{BPV_1}{BPV_2} M_1 \tag{20.7}$$

which is the traditional hedge calculation based solely on basis point values.

BOND ANALYSIS USING SPOT RATES AND FORWARD RATES IN CONTINUOUS TIME

This section analyses further the relationship between spot and forward rates and briefly discusses how this can be applied in bond analysis.

The Relationship between Spot and Forward Rates

In the discussion to date, we have assumed discrete time intervals and interest rates in discrete time. Here, we consider the relationship between spot and forward rates in continuous time. For this, we assume the mathematical convenience of a continuously compounded interest rate.

The rate r is compounded using e^r and an initial investment M earning $r(t,T)$ over the period $T - t$, initial investment at time t and for maturity at T, where $T > t$, would have a value of $Me^{r(t,T)((T-t)}$ on maturity.[3] If we denote the initial value M_t and the maturity value M_T, then we can state $M_t e^{r(t,T)(T-t)} = M_T$ and therefore the continuously compounded yield, defined as the continuously compounded interest rate $r(t,T)$ can be shown to be

$$r(t,T) = \frac{log(M_T/M_t)}{T-t} \tag{20.8}$$

[3] e is the mathematical constant 2.7182818 . . . and it can be shown that an investment of £1 at time t will have grown to e on maturity at time T (during the period $T - t$) if it is earning an interest rate of $1/(T - t)$ continuously compounded.

We can then formulate a relationship between the continuously compounded interest rate and yield. It can be shown that

$$M_T = M_t e \int_t^T r(s)ds \tag{20.9}$$

where *r(s)* is the instantaneous spot interest rate and is a function of time. It can further be shown that the continuously compounded yield is actually the equivalent of the average value of the continuously compounded interest rate. In addition, it can be shown that

$$r(t,T) = \frac{\int_t^T r(s)ds}{T-t} \tag{20.10}$$

In a continuous time environment, we do not assume discrete time intervals over which interest rates are applicable but, rather, a period of time in which a borrowing of funds would be repaid instantaneously. So we define the forward rate $f(t,s)$ as the interest rate applicable for borrowing funds where the deal is struck at time t; the actual loan is made at s (with $s > t$) and repayable almost instantly. In mathematics, the period $s - t$ is described as infinitesimally small. The spot interest rate is defined as the continuously compounded yield or interest rate $r(t,T)$. In an environment of no arbitrage, the return generated by investing at the forward rate $f(t,s)$ over the period $s - t$ must be equal to that generated by investing initially at the spot rate $r(t, T)$. So we may set

$$e^{\int_t^T f(t,s)ds} = e^{r(t)dt} \tag{20.11}$$

which enables us to derive an expression for the spot rate itself, which is

$$r(t,T) = \frac{\int_t^T f(t,s)ds}{T-t} \tag{20.12}$$

The relationship described by (20.12) states that the spot rate is given by the *arithmetic* average of the forward rates $f(t,s)$, where $t < s < T$. How does this differ from the relationship in a discrete time environment? From Chapter 3 we know that the spot rate in such a framework is the *geometric* average of the forward rates,[4] and this is the key difference in introducing the continuous time structure. Equation (20.12) can be rearranged to

$$r(t,T)(T-t) = \int_t^T f(t,s)ds \tag{20.13}$$

[4] To be precise, if we assume annual compounding, the relationship is one plus the spot rate is equal to the geometric average of one plus the forward rates.

and this is used to show (by differentiation) the relationship between spot and forward rates:

$$f(t,s) = r(t,T) + (T-t)\frac{dr(t,T)}{dT} \tag{20.14}$$

If we assume we are dealing today (at time 0) for maturity at time T, then the expression for the spot rate becomes

$$r(0,T) = \frac{\int_0^T f(0,s)ds}{T} \tag{20.15}$$

so we can write

$$r(0,T) \bullet \mathrm{T} = \int_0^T f(0,s)ds \tag{20.16}$$

This is illustrated in Figure 20.7, which is a diagrammatic representation showing that the spot rate $r(0,T)$ is the average of the forward rates from 0 to T, using the hypothetical value of 5% for $r(0,T)$. Figure 20.7 also shows the area represented by (20.16), to the period $0.7T$.

What (20.14) implies is that if the spot rate increases, then, by definition, the forward rate (or "marginal" rate as has been suggested that it be called[5]) will be greater. From (20.14) we deduce that the forward rate will be equal to the spot rate plus a value that is the product of the *rate* of increase of the spot rate and the time period ($T - t$). In fact, the conclusions simply confirm what we already discovered

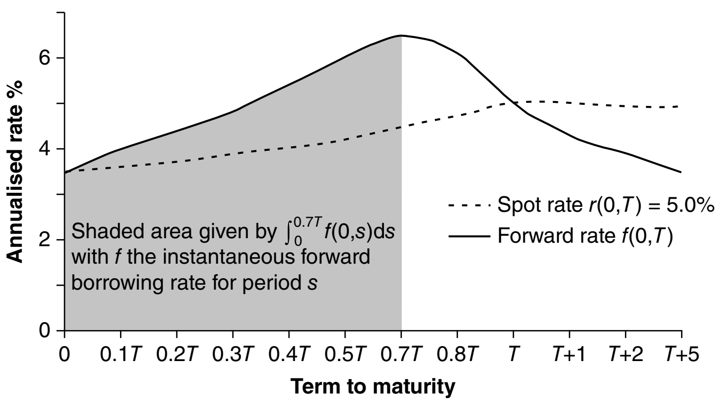

FIGURE 20.7 Diagrammatic Representation of the Relationship between Spot and Forward Rate
The spot rate $r(t, T)$ is the average of the forward rates between t and T.

[5] For example, see Section 10.1 of Campbell, Lo, and MacKinlay (1997), Chapter 10 of which is an excellent and accessible study of the term structure, and provides proofs of some of the results discussed here. This book is written in a very readable style and is worth purchasing for Chapter 10 alone.

in the discrete time analysis in Chapter 3, that the forward rate for any period will lie above the spot rate if the spot-rate term structure is increasing, and will lie below the spot rate if it is decreasing. In a constant spot-rate environment, the forward rate will be equal to the spot rate.

However, it is not as simple as that. An increasing spot-rate term structure only implies that the forward rate lies above the spot rate, but not that the forward-rate structure is itself also *increasing*. In fact, one can observe the forward-rate term structure to be increasing or decreasing while spot rates are increasing. As the spot rate is the average of the forward rates, it can be shown that in order to accommodate this, forward rates must in fact be *decreasing* before the point at which the spot rate reaches its highest point. This confirms market observation. An illustration of this property is given in Appendix 20.2. As Campbell, Lo, and MacKinlay (1997) state, this is a property of average and marginal cost curves in economics. An introduction to the integration function is given in Appendix B.

BOND PRICES AS A FUNCTION OF SPOT AND FORWARD RATES

In this section, we describe the relationship between the price of a zero-coupon bond and spot and forward rates. We assume a risk-free zero-coupon bond of nominal value £1, priced at time t and maturing at time T. We also assume a money-market bank account of initial value $P(t,T)$ invested at time t. The money-market account is denoted M. The price of the bond at time t is denoted $P(t,T)$ and if today is time 0 (so that $t > 0$), then the bond price today is unknown and a random factor (similar to a future interest rate). The bond price can be related to the spot rate or forward rate that is in force at time t.

Consider the scenario below, used to derive the risk-free zero-coupon bond price.[6]

The continuously compounded *constant* spot rate is r as before. An investor has a choice of purchasing the zero-coupon bond at price $P(t,T)$, which will return the sum of £1 at time T or of investing this same amount of cash in the money-market account, and this sum would have grown to £1 at time T. We know that the value of the money-market account is given by $Me^{r(t,T)(T-t)}$. If M must have a value of £1 at time T, then the function $e^{-r(t,T)(T-t)}$ must give the present value of £1 at time t and, therefore, the value of the zero-coupon bond is given by

$$P(t,T) = e^{-r(t,T)(T-t)} \tag{20.17}$$

If the same amount of cash that could be used to buy the bond at t, invested in the money-market account, does *not* return £1 then arbitrage opportunities will result. If the price of the bond exceeded the discount function $e^{-r(t,T)(T-t)}$ then the investor could short the bond and invest the proceeds in the money-market account. At time T, the bond position would result in a cash outflow of £1, while the money-market account would be worth £1. However, the investor would gain because in the first place $P(t,T) - e^{-r(t,T)(T-t)} > 0$. Equally, if the price of the bond was below $e^{-r(t,T)(T-t)}$, then the investor would borrow $e^{-r(t,T)(T-t)}$ in cash and buy the bond at price $P(t,T)$.

[6] This approach is also used in Campbell et al. (q.v.).

On maturity, the bond would return £1, which proceeds would be used to repay the loan. However, the investor would gain because $e^{-r(t,T)(T-t)} - P(t,T) > 0$. To avoid arbitrage opportunities we must therefore have

$$P(t,T) = e^{-r(t,T)(T-t)} \tag{20.18}$$

Following the relationship between spot and forward rates, it is also possible to describe the bond price in terms of forward rates.[7] We show the result only here. First we know that

$$P(t,T)e^{-\int_t^T f(t,s)ds} = 1 \tag{20.19}$$

because the maturity value of the bond is £1, and we can rearrange (20.19) to give

$$P(t,T) = e^{-\int_t^T f(t,s)ds} \tag{20.20}$$

Expression (20.20) states that the bond price is a function of the range of forward rates that apply for all $f(t,s)$; that is, the forward rates for all time periods s from t to T (where $t < s < T$, and where s is infinitesimally small). The forward rate $f(t,s)$ that results for each s arises as a result of a random or stochastic process that is assumed to start today at time 0. Therefore, the bond price $P(t,T)$ also results from a random process; in this case, all the random processes for all the forward rates $f(t,s)$.

The zero-coupon bond price may also be given in terms of the spot rate $r(t,T)$, as shown in (20.18). From our earlier analysis we know that

$$P(t,T)e^{r(t,T)(T-t)} = 1 \tag{20.21}$$

which is rearranged to give the zero-coupon bond price equation

$$P(t,T) = e^{-r(t,T)(T-t)} \tag{20.22}$$

as before.

Equation (20.22) describes the bond price as a function of the spot rate only, as opposed to the multiple processes that apply for all the forward rates from t to T. As the bond has a nominal value of £1, the value given by (20.22) is the discount factor for that term; the range of zero-coupon bond prices would give us the discount function.

What is the importance of this result for our understanding of the term structure of interest rates? First, we see (again, but this time in continuous time) that spot rates, forward rates, and the discount function are all closely related, and given one we can calculate the remaining two. More significantly, we may model the term structure either as a function of the spot rate only, described as a stochastic process, or as a function of all of the forward rates $f(t,s)$ for each period s in the period $(T-t)$, described by multiple random processes. The first yield-curve models adopted the first approach, while a later development described the second approach.

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[7] For instance, see *ibid.*, Section 4.2.

APPENDICES

APPENDIX 20.1 SUMMARY OF DERIVATION OF OPTIMUM HEDGE EQUATION

From equation (20.5) we know that the variance of a net change in the value of a two-bond portfolio is given by

$$\sigma_{pos}^2 = M_1^2 BPV_1^2 \sigma_1^2 + M_2^2 BPV_2^2 \sigma_2^2 + 2M_1 M_2 BPV_1 BP_1V_2 \sigma_1 \sigma_2 \rho \qquad \text{(A20.1.1)}$$

Using the partial derivative of the variance σ^2 with respect to the nominal value of the second bond, we obtain

$$\frac{\partial \sigma^2}{\partial^2 M^2} = 2M_2 BPV_2^2 \sigma_2^2 + 2M_1 BPV_1 BPV_2 \sigma_1 \sigma_2 \rho \qquad \text{(A20.1.2)}$$

If (A20.1.1) is set to zero and solved for M_2 we obtain (A20.1.3), which is the hedge quantity for the second bond.

$$M_2 = -\frac{\rho BPV_1 \sigma_1}{BPV_2 \sigma_2} M_1 \qquad \text{(A20.1.3)}$$

APPENDIX 20.2 FORWARD-RATE STRUCTURE IN CONVENTIONAL YIELD-CURVE ENVIRONMENT

This is a formal statement of the forward-rate structure when the spot-rate structure is increasing. We assume the spot rate $r(0,T)$ is a function of time and is increasing to a high point at $\bar{T}$. It is given by

$$r(0,T) = \frac{\int_0^T f(0,s)ds}{T} \qquad \text{(A20.2.1)}$$

At its high point, the function is neither increasing nor decreasing, so we may write

$$\frac{dr(0,\bar{T})}{dT} = 0 \qquad \text{(A20.2.2)}$$

and therefore the second derivative with respect to T will be

$$\frac{d^2 r(0,\bar{T})}{dT^2} < 0 \qquad \text{(A20.2.3)}$$

From (20.14) and (A20.2.2) we may state

$$f(0,\bar{T}) = r(0,\bar{T}) \tag{A20.2.4}$$

and from (A20.2.3) and (A20.2.4) the second derivative of the spot rate is

$$\frac{d^2 r(0,\bar{T})}{dT^2} = \left[\frac{df(0,\bar{T})}{dT} - \frac{dr(0,\bar{T})}{dT}\right]\frac{1}{T} < 0 \tag{A20.2.5}$$

From (A20.2.2) we know the spot-rate function is zero at T, so the derivative of the forward rate with respect to T would therefore be

$$\frac{df(0,\bar{T})}{dT} < 0 \tag{A20.2.6}$$

So, in this case, the forward rate is decreasing at the point T when the spot rate is at its maximum value. This is illustrated hypothetically at Figure 20.7 and it is common to observe the forward-rate curve decreasing as the spot rate is increasing.

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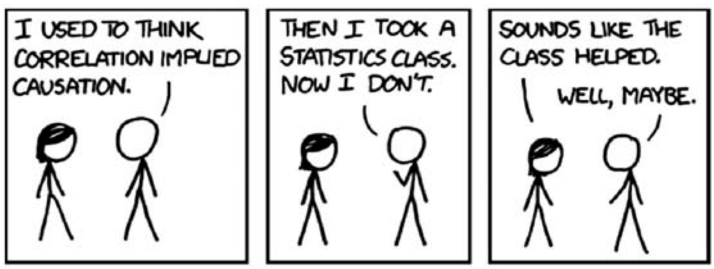
CHAPTER 21

Derivatives Risk Management: Convexity, Collateral, and Correlation



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ABSTRACT

In this chapter, we introduce the concept of correlation and explain its relevance to finance. We start with a discussion of how it can be measured. We then move on to some simple products that allow us to trade the correlation and covariance. We explain how these factors lead to "convexity", and how we can trade that. We then propose a method to bring all these ideas and prices together into a single consistent modelling framework. Finally, we discuss the relevance of these ideas to bond pricing.

WHY CORRELATION OCCURS IN THE MARKETS

Assets and other observable market instruments can be correlated for various reasons. We discuss the major causes of correlations below.

Macroeconomic Reasons

If economic conditions change, this is likely to impact many assets in the same way, due to their exposure to systematic risk (referred to as *Beta* in the Capital Asset Pricing Model). The more closely related two assets are, the higher their correlation is likely to be. For example, almost all stock prices will have some kind of positive correlation. Changes in the values of shares of companies that are in the same country or business area will be particularly highly correlated. This is because there will exist specific risks that affect these closely related companies all to a large extent, while having less or no effect on the rest of the economy. For example, price moves of the shares of Mitsubishi and Nissan, which are both Japanese auto manufacturers, will be higher correlated than price moves of the shares of Nissan and Tesco.

Trading Axes

Actual trading by market participants that are trying to hedge their exposures can create correlations that cannot fully, or sometimes cannot at all, be explained by economic reasons. First of all, we shall show how volatility can be created or destroyed by trading axes. Then we shall show the same effect on correlation.

Imagine a derivative is being traded in the market, perhaps embedded into a bond as its coupon formula, which has the following PV as a function of an underlying (X), in Figure 21.1.

The investors who have purchased the bonds containing this embedded derivative have decided to take a view that X will increase, and that they will thus make a profit. They have no intention of trying to hedge or mitigate their exposure to X through hedging or any other means.

The derivative performs well over time, and the market for this derivative becomes extremely large. The banks that provided the derivative used a risk-neutral model, with risk-neutral probabilities, to price it. They sold the investments at a price that is expected to be profitable to them. In this simple example, the model of the PV requires two basic inputs: the expected value of X on each coupon payment date and its volatility until that date. The former can be deduced from the traded prices of swaps, and the latter from the traded price of vanilla options (calls/puts). A bank that is paying out the coupons loses money if X appreciates in value, and also if the volatility of X increases. The bank is short the volatility because the payout it has to make has a positive curvature from the point of view of the investor. For example, if X = 4 and there was no volatility, then the payout would always be 16. However, if the expectation of X were 4, but it was volatile, and reached the value 3 or 5, with 50% probabilities, then the expectation of the payout would be 17.

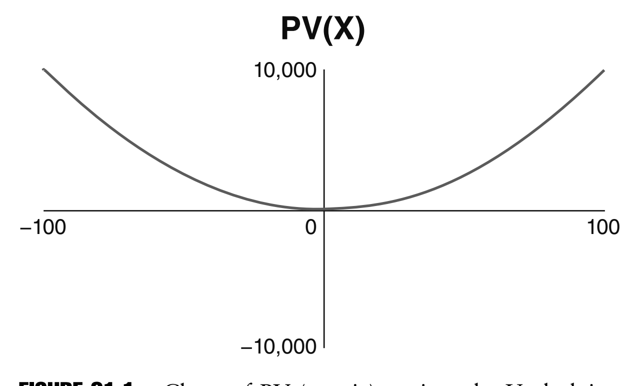

FIGURE 21.1 Chart of PV (y-axis) against the Underlying X (X-axis)

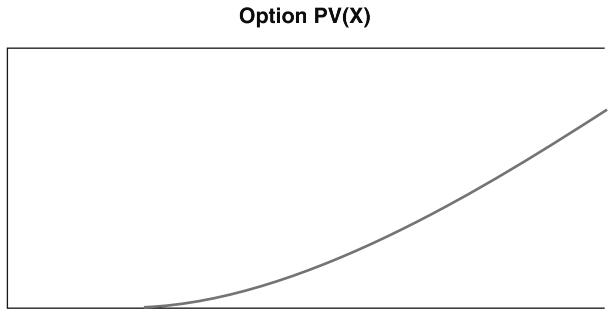

FIGURE 21.2 Chart of Option PV (*Y*-axis) against the Underlying X (*X*-axis)

Let us imagine that the banks hedge their exposure by buying call options on X, and that they can statically replicate the PV by selling options at various strikes to perfectly replicate the derivative payout. This means that, every time one of these bonds is sold, there is demand to buy volatility on X. This will push up the price of the options, resulting in the risk-neutral volatility, implied by this price, to rise.

These option hedges will be sold by the vanilla option trading desks at various banks. The vanilla option desks have ended up net short volatility. They probably cannot find investors who will do trades with them that will cover this short position. Thus, they will be forced to warehouse the volatility risk. Consider the PV profile of one of these options (from the point of view of the person who is long the option), in Figure 21.2.

As we can see from this diagram, the person who is short this option has a delta to the outright value of X, and that changes as X moves. If X falls in price, the delta becomes less negative, forcing the trader to sell back hedges he has already bought on X. This action will force the price to fall even further. If X rises in price, the delta becomes more negative, forcing the trader to buy more of X. This action forces the price even higher. We can see that a positive feedback mechanism has developed: the delta hedging done by the trader who is short volatility results in volatility rising even higher, even though he is not trying to hedge the volatility directly. Thus, it is not only the risk-neutral price of the volatility that is increased by the purchasing of options, but also the realised volatility that is increased by the delta hedging of them! Of course this is great news for the investor, who will see the value of his product increase by virtue of the banks trying to mitigate their risks!

The same process can happen in reverse, if the payout of the bond is reversed. Imagine the coupons are now of the form shown in Figure 21.3.

In this case, the bond seller is long volatility, and will statically hedge his position by selling options. This forces the risk-neutral value of the volatility to fall. The options desks are now net long the volatility.

As before, let us consider a single option payout. In this case, the options trader has to sell X every time it rises, pushing the price down, and buy X every time it falls, pushing the price back up. This action results in a negative feedback that reduces the realised volatility of X. This is again good news for the investor!

To conclude this example, we can briefly consider a case where the initial option hedge is not static. In this case, the bond trader will have to rebalance his hedge

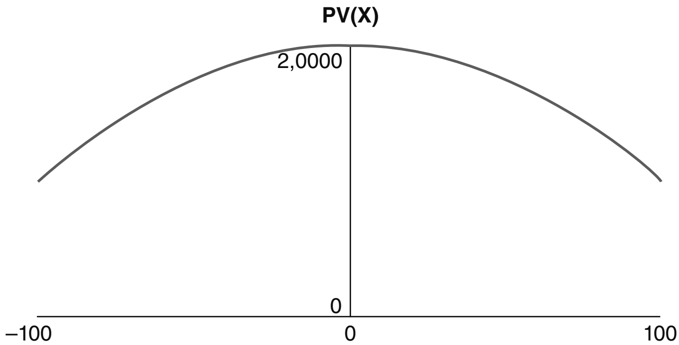

FIGURE 21.3 Chart of PV (*Y*-axis) against the Underlying X (*X*-axis)

portfolio when the level and/or volatility of X change. If he has to rebalance the portfolio when X moves, for instance, by buying volatility as X rises, that will result in a positive correlation between X and its volatility, because the volatility of X will be forced up by demand, as the level of X rises. If he has to sell more options as the volatility itself rises, this will push the volatility back down, and reduce the volatility of the volatility of X.

Now, let us consider an example where trading axes can influence the correlation between two different assets. A lot of investors in the United States want to buy the Nikkei. The simple way to do this is to sell their dollar cash, and buy yen. They can then invest in the Nikkei using this yen. The problem with this strategy is that the investor is trying to express his view that the Nikkei will appreciate, without taking any other risks. In this case, if the Nikkei appreciates by 10%, but the value of the yen falls by 20% against the dollar, then he will actually have lost money in dollar terms.

To avoid this risk, he can find a derivatives desk who will allow him to trade a quanto. At a predetermined point in the future, the quanto swap will pay out the change in the Nikkei, times the investment size in USD. For example, say the investor buys an exposure of 100USD per 1-point move in the Nikkei, on a two-year time horizon. Let us assume the Nikkei starts at 8,500 at the time of investment, and finishes at 8,600 in two years' time. The investor will have made a profit of 10,000USD, regardless of the value of the yen.

The investor has expressed his view on the Nikkei perfectly, and will do no further trades. However, the bank that sold him this investment will have to hedge its position. On day one, the bank knows its exposure to the Nikkei, in yen, is 100USD * USDJPY. For example, if USDJPY is 80, then the bank will sell 8,000 units of the Nikkei as a hedge. We imagine four possible scenarios:

1. USDJPY rises, that is, that the dollar appreciates against the yen. This increases the bank's yen exposure to the Nikkei, and the bank has to buy more Nikkei to compensate. This additional demand will cause the Nikkei to rise as well.
2. USDJPY falls, that is, the dollar depreciates against the yen. This decreases the bank's yen exposure to the Nikkei, and the bank has to sell more Nikkei to compensate. This additional supply will cause the Nikkei to depreciate as well.

3. The Nikkei rises. This increases the amount, in dollars, that the bank owes to the customer. The bank will sell yen and buy dollars. This demand for dollars can cause USDJPY to increase.
4. The Nikkei falls. This decreases the amount, in dollars, that the bank owes the customer. The bank sells dollars, and buys yen, creating downward pressure on USDJPY.

In all four cases, the trading activity that was triggered by a move in one asset led to a change in value of the other asset. This created an increased covariance between them (i.e., an increase in both the correlation between the assets, and in the volatilities of the assets). As before, this trading-generated effect acts in the favour of the customer. If the bank is aware that the market is of a sufficient size to cause this feedback mechanism to occur, it should charge the customer an increased fee to execute the trade, to pay for the consequences.

Correlations and Volatilities in a Crisis

It is a well-known fact that both volatilities and correlations increase in times of financial stress. In normal market conditions, the value of an asset is determined by a combination of systematic (macro) factors that affect the market as a whole, and idiosyncratic factors that affect that asset in particular. Investors and other market participants seek to diversify their risk by taking out a wide range of investments.

In a crisis, the systemic factors start to dominate, compared to the idiosyncratic ones. For example, a default by a central bank will lead to correlated defaults by commercial banks in that country, who hold a large proportion of their assets in the central bank's bonds. This, in turn, will lead to the default of domestic companies who have deposited their assets in the commercial banks. Policy makers (e.g., governments, central banks) intervene to a greater extent in the markets during a crisis. Under normal market conditions, there is a general desire to allow the economy to grow under its own steam, without taking any interventionist action. In a crisis, governments will intervene to try to avert problems. These interventions can take many forms, including the printing of money, support for failing companies and banks, and changes in monetary and fiscal policy. These actions are more likely to have a systemic than an idiosyncratic effect.

It is also true that free market participants (e.g., investors) behave in a more correlated fashion during a crisis. People become naturally more risk averse, and will move their assets to perceived safe havens. They will prefer holding bonds to equities, government bonds to commercial bonds, and bonds of large robust economies rather than small volatile ones. As the crisis worsens, low-risk assets will appreciate compared to high-risk assets. As it alleviates, high-risk assets will appreciate compared to low-risk ones. Thus large groups of assets start behaving in an extremely correlated and volatile fashion.

CORRELATION AND CAUSATION

It is very easy to find things that are correlated but have no causal relationship. Previously, we state that the movements in the share prices of Mitsubishi and Nissan are highly correlated. However, this does not mean that a movement in the price of Mitsubishi shares will give rise to a movement in Nissan's. In fact, if all other economic (and other relevant) factors remained constant, but the Nissan share price fell (this could happen due to any idiosyncratic reason, like a design flaw on a new model), you may actually expect the Mitsubishi share price to rise, as they capture market share from their competitor! The reason their share price movements are highly correlated is because the same set of external factors affect both companies in similar ways.

Correlation only implies causation if the correlation remains after the impact of all other factors has been accounted for. For example, if all economic, environmental, and regulatory conditions (etc.) remained constant over a six-month period, but the share prices still moved with a high correlation, then we would be pushed towards the conclusion that the movement in one price was causing the movement in the other (although we still would not know which was the cause and which the effect)!

It is very hard to prove causation in economics, finance, and social science, because it is extremely hard, if not impossible, to perform a controlled experiment where other variables are held constant. It is much easier to prove causation in subjects like physics and medicine. Fortunately, from the point of view of a trader or risk manager, understanding the causation is less important than understanding the correlation (although if you understand the causation, this may help you search from highly correlated assets that can be used to hedge each other).

HOW CAN WE DEFINE CORRELATION?

Absolute Levels or Changes in Levels?

A common mistake that people make is to look at the correlation between the absolute levels of two assets, instead of the correlation between the changes in these assets. In finance and economics, we normally think about price changes, rather than absolute price levels. So it makes sense to talk about the volatility of a price change, or the correlation between two price changes.

We can give a trivial example to show why considering the volatility of the asset levels is wrong. Consider a share whose values measured every month follow the pattern: $PV_{t+1} = PV_t * 1.01$.

This theoretical stock has no volatility at all! Its price is simply drifting upwards at a constant rate of 1% per month. If you measure the volatility of the price changes, you will get result zero. However, if you look at the standard deviation of the levels, you will get a nonzero level.

We can easily extend our example to correlation. Consider the two lognormal assets below:

Asset 1	Asset 2	Log Change in:	
100	100	Asset 1	Asset 2
272	272	100%	100%
272	272	0%	0%

Asset 1	Asset 2	Log Change in:	
272	272	0%	0%
272	272	0%	0%
272	272	0%	0%
100	739	–100%	100%
100	739	0%	0%
100	739	0%	0%
100	739	0%	0%
100	739	0%	0%

In the first period, there was a large correlated change in both assets. In subsequent periods, there were no changes, apart from one perfectly anticorrelated log-change. Both log-changes are of the same magnitude. The correlation between the levels (measured using Pearson's formula) is –72%, but the correlation between the log-changes is zero! The zero level makes sense, if the only two changes that occurred were equal and opposite. Why is the correlation between the levels –72% then? This is because the second, anticorrelated change occurred when the absolute levels of the two assets were much larger, and thus had a much higher weight on the correlation of the levels than the initial change had.

While facile, this example serves well to demonstrate the dangers of looking at the wrong correlations.

Which Correlation Measure Do We Apply to the Changes?

We have established that we should observe the changes in the assets, and not their absolute price levels. However, there is still the question of what we do to the changes in these levels. Do we look at the absolute changes, percentage changes, the logs of the changes, or something else? Then, what correlation measure do we apply to these, for instance, Pearson or Spearman. (Note that this is not a statistics book, so a comprehensive discussion of these, and other correlation measures, is beyond its scope.)

Some factors, like interest rates, and inflation rates, can take negative values (although in practice, this happens relatively rarely). Others, like share prices, bond prices, and FX rates, must take positive values. When deciding how to manipulate the changes in the asset, it is best to go for something relatively simple, as adding extra complexity is unlikely to yield additional accuracy. Thus, we should choose between observing the logs of the returns, the percentage returns, or the absolute returns.

Log and percentage returns do not have a sensible meaning for something that can flip sign or take a zero value. Thus, for things like interest rates, we should use the absolute returns (I am using "returns" synonymously with "change in levels/values").

Now let us consider something like an FX rate, which cannot take a negative value. Should we use percentage returns or log-returns? If we believe that the FX rate is equally likely to double as it is to half, then it probably makes more sense to consider the log-returns. Imagine an FX that doubles in one period, and halves the next. Its percentage returns will be +100%, –50%. Its log returns will be +0.69

and –0.69. Thus, in log-space, the two moves appear to be of equal magnitude, whereas the percentage returns are not. If something is equally likely to half or double, it can be described as being lognormally distributed and thus taking log-returns seems more appropriate.

In reality, the FX rate probably is not equally likely to double or halve. Its distribution may well be skewed away from the lognormal, even though it cannot go below zero. For example, the risk-neutral distribution of the AUDJPY FX rate has negative skew, compared to the lognormal distribution. That is, it is more likely to decrease by (X/(1 + X))% than it is to increase by X%. Most stock prices exhibit this negative skew. In the case of stocks, this is mainly for the macroeconomic reason that, as prices start to fall, economic, and thus price uncertainty usually increases. We could choose either to ignore this effect and continue to use log-returns, or to account for it by using Spearman's Rank correlation.

Pearson's correlation formula assumes that the joint distribution of the two variables (the variable being the logs of the returns) is bivariate normal (this is not the same as saying each individual variable is normal). For example, if we have two variables related by $Y_i = X_i^2$, for $i = 1$ to 100, then they will exhibit a Pearson correlation that is slightly less than 100 percent. To calculate the Spearman's Rank correlation, we first replace the values of their variables with their ranks, where the lowest value is replaced by rank 1, and the highest by rank n ($n = 100$, in the above example). We then calculate the Pearson correlation between the ranks. Applied to the above, we can clearly see that a correlation of 100% is now returned.

If you calculated the Spearman's Rank correlation between two joint-normally distributed assets, then you would find that the correlation coefficient was not exactly the same as the Pearson correlation. To account for this, we have to convert Spearman correlations into Pearson ones using the formula:

$$\rho_{Pearson} = sin\left(\frac{\pi * \rho_{Spearman}}{6}\right)$$

In practice, if the fluctuations in the asset are small, then there is little practical difference between using log, percentage, or even absolute returns to calculate the volatilities, and then the correlation. Furthermore, the skew on the two assets is rarely so extreme that it matters whether we use Pearson's or Spearman's method. Thus, we could just use Pearson's correlation of the log-returns "by convention", although we should be aware of the circumstances where this approximation breaks down. Note that we still need to use the returns, and not the absolute levels, as using the absolute levels will typically result in a material error in the correlation.

MEASURING CORRELATION

As with all market parameters, we can estimate correlation in three ways:

1. From the traded prices of instruments sensitive to the correlation
2. From historical data
3. Using economics and market knowledge to form an opinion on its value

From the point of view of a trader, the first method, using tradable prices, is the best. This is the price at which the risk to the correlation can be hedged in the market. In the risk-neutral world, the trader should always mark his parameters to tradable prices. Unfortunately, there are not many liquid instruments, that are traded inter-bank, that are sensitive to correlation. Most correlation-sensitive instruments are not only traded with investors, but are quite often traded in the same direction. That is, all the customer trades will result in the trader having the same directional exposure to the correlation parameter. Thus there is no price at which the bank can easily hedge its correlation exposure. Furthermore, these correlation-sensitive trades usually have exposure to other unhedgable parameters, and thus we have too many degrees of freedom to accurately determine the risk-neutral correlation from their prices.

Examples of potentially liquid instruments that have significant correlation exposure are quantos, spread options, and correlation swaps. We look at some of these later.

In the absence of liquidly traded instruments, we have to rely on historical data and market knowledge. There are three major problems associated with using historical correlations to mark our models:

1. The behaviour of assets can change over time, and thus future behaviour may not be the same as previous behaviour.
2. There are sampling errors associated with the choice of a specific time series. If we try to reduce the impact of sampling error by extending our sample further into the past, then we are assigning more weight to events further in the past that have less and less association with current conditions.
3. Even if the past behaviour is expected to continue for the foreseeable future, this has not told us anything about the risk-neutral price where market participants may be prepared to trade the correlation. It has only told us about what we expect the real-world outcome to be.

When picking historical data, we have to ensure our fixing sources are as accurate as possible. A composite closing price of an illiquid instrument like CDS is likely to have a lot of measurement noise, as it is the average price reported by several market participants who all have wide bid-offers, and who may calculate their price at slightly different times of the day. However, the daily fixing of the USD 10-year swap rate, determined by ISDA, at a specific time, is likely to be very accurate and representative of a firm tradable price at that moment. Any uncertainty adds noise to the calculation and will, on average, push the observed correlation closer to zero.

Another point to be aware of is to ensure the time-lag between the fixings of the two assets is as small as possible.

Imagine we are observing two assets, correlated at 60%. This correlation refers to the instantaneous changes in the two assets. Say that we observe each asset daily, but one at 1200 hrs, and the other at 1600 hrs. Consider a single observation period. Between 1200 hrs and 1600 hrs, we do not record any change in the second asset. Between 1600 hrs and 1200 hrs the next day, we record the changes on both assets. Between 1200 hrs and 1600 hrs, we only record the changes in the second asset. Hence, there is a four-hour window for each asset where there is no corresponding change in the other asset. Assuming there are no correlations between the moves in the two assets at two different times (which is a fair assumption, as otherwise markets would exhibit a predictive quality), this means that, out of the 24 hours, there was a 4-hour

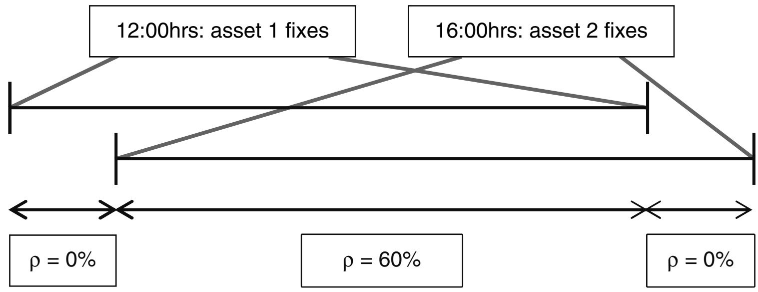

FIGURE 21.4 Effect of Asynchronous Fixings on a Correlation Swap

period where no correlation is observed. The total correlation observed over many periods will thus be (60% * 20 + 0% * 4)/24 = 50%. (We have made the assumption that there were also observations at the weekends, just to make the example simpler.) The situation is illustrated in Figure 21.4.

We can get round this problem in two ways. If we know what the time lag is, we can use the above analysis to deduce the impact of the zero-correlation period, and thus adjust the final answer to the level implied for synchronous observations. This has the disadvantage of introducing more random noise into the data, especially if the time lag is large. Alternatively, we could increase the time between observations, to reduce the relative size of the lag compared to the observation gaps. This has the disadvantage of decreasing the number of observation points. For example, if we observe correlation weekly, compared to daily, we have five times fewer observations. This problem can be circumvented by using a rolling observation window of, say, five business days.

Figure 21.5 shows the historical correlation between the changes in the S&P 500 index, and the EURUSD exchange rate. The correlations are measured using a 90-business-day rolling window of synchronous daily observations. The correlation has been high and positive since the 2008 credit crisis began, until the end of 2012. Before the credit crisis, there was no material correlation. In 2013, the correlation fell. This can be explained economically. Between 2008 and 2012, the euro had been seen as a much riskier and more vulnerable than the U.S. dollar. Thus, when the crisis deepened, EURUSD was likely to fall, whereas it was likely to rise as the crisis alleviated. Clearly the S&P index would fall at times of crisis and rise during times of recovery. At the end of 2012, the United States had budgetary problems (including "the fiscal cliff"). Europe began to recover from its economic woes. Thus, the euro stopped being seen as a lot more risky than the U.S. dollar, and thus the correlation abated.

If we were looking at the historical behaviour of this correlation, with a view to using it to price a trade, we would have to think carefully about whether we believed that the economic conditions that led to the historical correlation levels would persist going forward.

This is a good example of the third numbered point made at the start of this section. If we do not understand the reasons behind an observed historical correlation, then assuming it is a good indicator for future behaviour is extremely dangerous!

Is it possible to observe correlations between the movements in assets at different times?

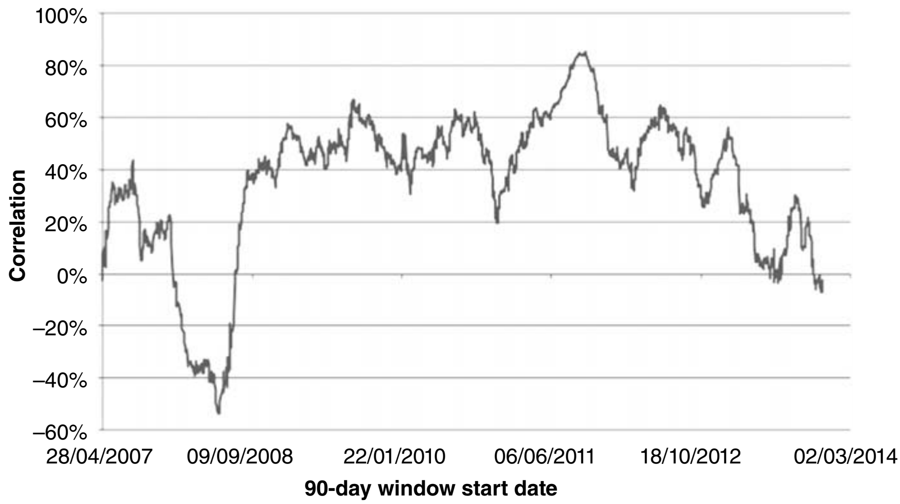

FIGURE 21.5 Correlation between S&P and EURUSD

Above, we have used the historic data to deduce the instantaneous correlation between two assets. However, could there be a correlation between the movement of asset 1 now, and the movement in asset 1, say, five minutes later? If there was behaviour like this, there would be several interesting results:

1. Markets would be predictive:
   We could observe the change in the first asset, and from that we would get some knowledge of the likely change of the second asset in the future. This would create an arbitrage, as we could buy the second asset in the knowledge that it was more likely to go up rather than down.
2. Correlations would be larger over longer time periods:
   Assume that the move in the first asset over the previous five minutes had a 50% correlation with the move in the second asset over the next five minutes. Assume also that the instantaneous (i.e., simultaneous) moves in the two assets also had a zero correlation. If we observed the two assets over simultaneous five-minute intervals, we would expect to measure a zero correlation. However, if we observed the two assets over 1-hour intervals, then the first 55 minutes of asset 1 would be 50% correlated with the last 55 minutes of asset 2. The remaining five minutes of each asset would exhibit no correlation with the moves of the other asset within that hour. Thus, we would now measure a correlation of (50% * 55 + 0% * 5)/60 = 46%.

So, over long time-scales, the result of the laggy and nonlaggy correlations would be identical, but over short time-scales the results would be very different.

What could cause behaviour like this?

To answer this, we should return to the situation where the market in a specific correlation is driven by a large trading interest or “axe”, rather than by economic forces. Trading desks managing complex derivatives often only calculate their risk once a day, due to the numerical complexity involved, and limiting processing power. Usually the trader will see his risk when he arrives at work in the morning. If he is long the correlation of the two assets, and one of them has appreciated the previous day, he will now have to buy the second asset, to hedge his outright exposure to it. By doing so, he will force up the price of the second asset. If a large number of traders are behaving in this way, then there will appear a correlation between the movements in the two assets the previous day, and their movements during the current day!

CORRELATION SWAPS, PRICING, AND THEIR RISKS

The simplest way to establish the tradable price of a market variable/asset is to trade a product that either solely depends on the price of that variable, or depends on the price of that variable as well as some other variables whose risk-neutral prices are known. For example, the price of an option on a stock tells you the value of the volatility of that stock at the traded expiry and strike, given that the expectation of the forward asset price is already known from the prices of other instruments (e.g., futures or forwards). Thus, one simple way to determine the price of the correlation between two assets is to trade a correlation swap.

Correlation swaps are relatively new products and are still traded infrequently. Trading in the interbank market helps participants to exchange/hedge risk with each other and to establish where to mark their correlation parameters. A client like a pension fund may wish to trade a correlation to hedge an exposure generated by other investments (e.g., quantos, which were touched on above, and are covered in more detail later). Hedge funds may trade correlation swaps to take views on realised correlations, especially where the bank has a large exposure to a specific correlation, and is willing to dispose of it at an attractive price.

As an example, let's consider a one-year daily correlation swap between the EURUSD FX rate (i.e., the number of USD per one EUR) and the S&P 500 index. Bank A buys the correlation from Bank B at a price of 40%. Bank A receives from Bank B:

$$100{,}000 * (\rho - 40\%) * 100$$

Note that the payment may be negative or positive.

We shall observe the prices of the two assets on every day where both trade. We record the closing levels of the S&P, and the EURUSD rate fixing at 4 p.m. New York time. The value of the correlation is determined to be:

$$\rho = \frac{\sum_{i=1}^{i=n}(X_i - \bar{X})(Y_i - \bar{Y})}{\left(\sum_{i=1}^{i=n}(X_i - \bar{X})^2 \sum_{i=1}^{i=n}(Y_i - \bar{Y})^2\right)^{1/2}}$$

Where The number of observations is $n + 1$

$X_i = ln(x_i) - ln(x_{i+1})$, and x_i are the observations of EURUSD

$Y_i = ln(y_i) - ln(y_{i+1})$, and y_i are the observations of S&P

Now let us consider the risks we are exposed to on a correlation swap whose first fixing is next week. It would appear, from the above, that the only thing we are exposed to is the correlation itself. However, things are not quite so simple. Imagine the unlikely scenario where, over the next year, EURUSD and S&P are volatile on alternate months. That is, in the first month, EURUSD is highly volatile, but S&P hardly changes. In the second month, EURUSD has a low volatility, but S&P is very volatile, and so on. Even if the moves between the two assets are 100% correlated, the observed correlation in the correlation-swap will be close to zero. This is because the large changes in one asset occur when the other asset is hardly moving. Thus there is actually no correlation between the material movements in the two assets. To account for this effect, we should really integrate over the volatility term-structures of the two assets, for the next year, to get a more accurate prediction of what correlation we should expect to observe. The exact method to do this is outside the scope of this book, so in the discussion below, we make the approximation that the volatility term-structures are flat. In that limiting case, the forward-starting correlation swap truly does only have exposure to the value of the correlation.

Now, let us consider what the risks are on the correlation swap after a single fixing has occurred. To simplify the math, we shall assume that the expectations of X and Y (i.e., the daily drift) are both zero. This assumption does not change the conclusions, but it does slightly simplify the maths. It is also a reasonable assumption to make for almost all assets.

We rewrite the correlation formula, separating the first fixing out from the sum, and then replace the values of the summations over the future fixings with the known market volatilities and correlation. Thus:

$$\rho_{total} = \frac{\rho\sigma_X\sigma_Y t + X_1Y_1}{\left(\left(\sigma_X^2 t + X_1^2\right)\left(\sigma_Y^2 t + Y_1^2\right)\right)^{1/2}}$$

Where the volatilities are annualised, and the time is in years.

By partial differentiation, we can work out how the price of the trade changes with respect to X_1 and Y_1, to second order, and then convert that into exposures to x_2 and y_2.

We get the following results:

$$\frac{\partial\rho_{total}}{\partial X_1} = \sigma_X t \frac{\sigma_X Y_1 - \rho\sigma_Y X_1}{((\sigma_X^2 t + X_1^2)^3(\sigma_Y^2 t + Y_1^2))^{1/2}}$$

$$\frac{\partial\rho_{total}}{\partial Y_1} = \sigma_Y t \frac{\sigma_Y X_1 - \rho\sigma_X Y_1}{((\sigma_X^2 t + X_1^2)(\sigma_Y^2 t + Y_1^2)^3)^{1/2}}$$

$$\frac{\partial^2\rho_{total}}{\partial X_1\partial Y_1} = \sigma_X\sigma_Y t \frac{\sigma_X\sigma_Y t + X_1Y_1}{((\sigma_X^2 t + X_1^2)(\sigma_Y^2 t + Y_1^2))^{3/2}}$$

$$\frac{\partial^2\rho_{total}}{\partial X_1^2} = -\sigma_X t \frac{3\sigma_X XY - 2\rho\sigma_Y X_1^2 + \rho\sigma_X^2\sigma_Y t}{((\sigma_X^2 t + X_1^2)^5(\sigma_Y^2 t + Y_1^2))^{1/2}}$$

$$\frac{\partial^2\rho_{total}}{\partial Y_1^2} = -\sigma_Y t \frac{3\sigma_Y XY - 2\rho\sigma_X Y_1^2 + \rho\sigma_Y^2\sigma_X t}{((\sigma_Y^2 t + Y_1^2)^5(\sigma_X^2 t + X_1^2))^{1/2}}$$

If we assume that the daily moves in the assets are small compared to the expected moves over the life of the trade, then we can simplify the preceding to:

$$\frac{\partial \rho_{total}}{\partial x_2} = \frac{\sigma_X ln\left(\frac{y_2}{y_1}\right) - \rho\sigma_Y ln\left(\frac{x_2}{x_1}\right)}{x_2 \sigma_X^2 \sigma_Y t}$$

$$\frac{\partial \rho_{total}}{\partial y_2} = \frac{\sigma_Y ln\left(\frac{x_2}{x_1}\right) - \rho\sigma_X ln\left(\frac{y_2}{y_1}\right)}{y_2 \sigma_Y^2 \sigma_X t}$$

$$\frac{\partial^2 \rho_{total}}{\partial x_2 \partial y_2} = \frac{1}{x_2 y_2 \sigma_X \sigma_Y t}$$

$$\frac{\partial^2 \rho_{total}}{\partial x_2^2} = -\frac{\rho}{x_2 \sigma_X^2 t}$$

$$\frac{\partial^2 \rho_{total}}{\partial y_2^2} = -\frac{\rho}{y_2 \sigma_Y^2 t}$$

We could rewrite in terms of the PV of the trade. However, as the PV is simply: $100{,}000 * (\rho - 40\%) * 100 * df$ (where *df* is the discount factor on the payment date), there is no point in this exercise.

This makes a lot of sense empirically. If x increases then, to maximise the observed correlation, we either want y to increase, or x to decrease. The deltas to x and y both increase approximately linearly; that is, the gammas and cross-gamma are approximately constants. The scaling by the volatilities and time are logical, as you would expect the materiality of the movements to depend on the expected changes in the assets over the life of the trade. Notice that the gamma terms are scaled by the expected correlation. A move by a single asset in one period pushes the observed correlation towards zero. Thus, in the case of a zero expected correlation, there is no gamma.

As long as we have risk to the movements of the two assets, our trade price is no longer solely a function of the future correlation. However, if we frequently delta-hedge, for instance, if x has increased and y has not, we sell x and buy y, we can cancel out the exposure to the asset levels themselves. By doing this, we should find that the final profit/loss on our portfolio of the correlation swap and these hedges become a much more accurate measure of the realised correlation. Notice that if our correlation swap had weekly, instead of daily, observations, then we would expect to build up much larger exposures to the absolute movements of the two assets between fixings, if we did not rehedge. This is another way of saying that we expect a trade with less fixings is a less accurate measure of the realised correlation. Also note that, as we can rehedge as often as we want, we can actually reduce the noise level on the weekly fixing trade to that of the daily fixing one!

We could continue the above analysis to predict what our risks will look like as the trade seasons. We give a quick summary, without going into the mathematical details. To calculate the deltas and gammas, we should use the same technique as before, but we would split the remaining sum (i.e., the bit excluding the current fixing) into a part representing the future fixings and a part representing the past fixings, whose values are known. We also find that, if the realised volatilities and covariances are not the same as their predicted forward values, then the correlation swap will start to have vega, as well as correlation, exposure. For example, imagine we are halfway through the trade. The realised correlation has come in at a level much lower than our mark. We are still marked at the higher level of correlation, because we can see market prices for new correlation swaps that are trading at that level. In this case, we shall find ourselves long vegas. This is because, if the volatilities of the assets for the second part of the trade are much higher than the realised volatilities for the first half, then the weighting of the future fixings in the final correlation calculation will be much greater than the weighting of the realised fixings. As the future correlation is expected to be higher than the realised correlation, we want the future fixings to have as much weight as possible, and thus increase our average correlation at the end of the trade.

A final subtlety is to consider what happens when a fixing occurs. In this case, the current period becomes a past period, and our deltas to the two assets depend on how far the spots have moved from the new fixing. At the moment of the fixing itself, the spot rate will be exactly at the fixing level, and thus our deltas will immediately disappear. If we had been accurately delta-hedging during the last period, we shall now have to unwind all those hedges to become delta flat again.

In some cases, the fixings on the two assets do not happen at identical times. Imagine a case where x fixes a bit before y, and we have now two fixings on x and one on y. Now, the PV impact from the first period can only be affected by movements in y_2. However, although x_2 is fixed, x_3 is variable. Under the reasonable assumption that $y_2 = y_3$, this means that the PV impact from the second period can only be impacted by changes in x_3. This means that, while our gammas remain, they refer to different periods; our cross-gamma has completely disappeared. Once the second fixing occurs on y, then the cross-gamma reappears, and the gammas both refer to the second period. Notice also that, when x fixes, the delta to x jumps to zero, but the delta to y does not jump, and vice versa.

QUANTOS, PRICING, AND THEIR RISKS

A quanto is a trade that contains exposure to a "quantity-adjusted" forward. According to www.margrabe.com, the term was coined by Lee Thomas of Goldman Sachs in 1986 (although we cannot find any corroborating evidence to support or challenge this claim).

Earlier, we gave an example of NKY quantoed into USD. Below, we shall stick with our example of the correlation between S&P and EURUSD, and thus we shall consider S&P quantoed into EUR.

Consider the following payoff, one year in the future:

$$PV = (S\&P_{1y} - Strike) * 100\ EUR$$

This is equivalent to receiving in USD:

$$(S\&P_{1y} - Strike) * 100 * EURUSD_{1y}/EURUSD_{t=0} = 0$$

For the PV on day one to be zero, should the strike be higher or lower than the strike of a vanilla S&P forward? The answer depends on the correlation. If the correlation between EURUSD and S&P is positive, then the strike should be higher. This is because the equivalent USD notional increases when the payout is positive; that is, when S&P increases, and decreases when the payout is negative; that is, when the S&P decreases.

We can also consider the costs of rehedging. First, we calculate the day one delta hedges for the person receiving the PV of the above trade.

You make 100EUR (discounted) every time S&P rises by one point. So your initial vanilla hedge required is to sell 100*$\text{EURUSD}_{\text{spot}}$ on a dollar notional, of the one-year S&P future. How about your exposure to EURUSD? Well, as the deal has a zero PV, a change in EURUSD will not change the PV from, as all it does is to scale the final cash flow. Hence, no hedge to EURUSD is required.

Now, let us imagine that the positive correlation between EURUSD and S&P causes them to both increase. The increase in EURUSD has increased the effective USD notional, and thus we need to sell more dollar-denominated S&P contracts. As the S&P has risen, we sell these additional contracts at an increased price. The PV of the future cash flow is positive now, and so we are now exposed to the movement in EURUSD. If the EUR weakens, the PV of that cash flow is reduced. To hedge this, we sell some EURUSD (sell EUR, buy USD).

Subsequently, both S&P and EURUSD fall back to their initial levels, simultaneously. To rebalance our hedges, we should buy back the extra S&P we sold, and buy back all the EURUSD we sold. We can see that, as a result of hedging the correlated move, we have locked in a profit, as we sold high, then bought low on both assets. The same argument would work in reverse, if both S&P and EURUSD had fallen, and then risen.

Thus we can see that the fair strike of the quanto contract must be higher than the fair strike of the vanilla contract, due to the positive correlation. If the correlation were negative, the fair strike would be lower for the quanto, and if the correlation were zero, the fair strike would be identical.

So far, we have only thought about the correlation. However, it is actually the covariance between the assets that is a better measure of the amount by which the fair strike should be shifted. As we can see from the example where we locked in hedging profits, the size of those profits must be proportional to the size of co-movements in the assets. The expected co-movements over the life of the trade are scaled by the correlation, the two volatilities, and the time till expiry; that is, the covariance.

If we make a number of simplifying assumptions about the two assets, including the fact that the movements in both assets are lognormally distributed, and that the logs of their movements are joint-normally distributed, then we can say that the fair strike of the quanto is:

$$Strike_{Quanto} = Strike_{Vanilla} * \exp(\rho\sigma_{S\&P}\sigma_{EURUSD}t)$$

Where the volatilities refer to the movements in the terminal forwards of the two assets, and the correlation is between those movements.

From the above, we can see that, as well as having to rehedge our deltas to S&P and EURUSD, we also have to sell some volatility on the two forwards, for example using one-year options. Note that the vega hedges, like the delta hedges, will need to be rebalanced over the life of the trade. The quanto has what is referred to as "strike-less" vega. A change in the level of EURUSD has no impact on the amount of EURUSD vega (on a fixed EUR notional). Similarly, a change in S&P does not change the amount of S&P vega (on a fixed S&P notional). However, if we hedged by selling at-the-money S&P and EURUSD options, then, as the forwards moved away from the strikes, the vega on the options would reduce towards zero. Thus we have hedged strikeless-vega with vanilla strike-dependent-vega, and we are not statically hedged.

Another reason why we cannot be statically hedged is that, if the expected correlation changed sign, the sign of our vega exposures would also flip! Similarly, our S&P vega is scaled by the EURUSD volatility, and vice versa.

These observations can lead us to predict some other subtle risk exposures we would get from this quanto that are not captured in the Black model.

First of all, let us consider the effect of changing the skew of one of the assets. When equities are low, we expect volatility to be high. What effect does this have on the quanto price?

$$PV_{USD} = \left(S\,\&\,P_{1y} - \text{Strike}\right) * \text{EURUSD}_{1y} * \exp\left(\rho\sigma_{S\&P}\sigma_{EURUSD}t\right)$$

$$\Rightarrow Vega_{S\&P} = S\&P_{1y} * \text{EURUSD}_{1y} * \rho\sigma_{EURUSD}t * \exp\left(\rho\sigma_{S\&P}\sigma_{EURUSD}t\right)$$

$$And \ \Rightarrow Delta_{S\&P} = EURUSD_{1y} * \exp\left(\rho\sigma_{S\&P}\sigma_{EURUSD}t\right)$$

Let us consider our dynamic rehedges. On day one, we bought the quanto, and we sold some volatility on both assets, and we sold some S&P forward. Now, the S&P rises, and its volatility falls. The rising S&P has increased our S&P vega. It is true that the falling S&P volatility has reduced our S&P vega, but this is a second-order effect and can be ignored. To remain hedged, we need to sell additional S&P volatility, when the volatility is low. The falling S&P volatility has reduced our S&P delta, so we have to buy back some S&P when it is expensive. Thus our rehedges are both locking in losses.

We can think about this in another way. The quanto gives us strikeless vega, as discussed above. Our vanilla option hedges have strike-dependent vega. Thus, to be better hedged, we should sell options at a variety of strikes, from out-of-the-money to in-the-money. If S&P volatility is high when the S&P is high, then we receive higher premia for selling the high-strike options, than we receive for selling the low-strike options. This gain in value, in USD terms, on the high-strike premia outweighs the loss in value from the low-strike premia. Thus, we can lock in more money from our initial vega hedges if the skew is positive. More generally, because the PV of the quanto is a function of the product of the S&P vol and the S&P forward (just Taylor-expand the exponential to see this), then a positive correlation must cause the expectation of the PV to increase.

By similar analysis, we can draw the same conclusion about our exposure to the skew on EURUSD. We can also see that we would prefer it if the volatilities of the two assets were themselves highly correlated, and also correlated to the asset-correlation!

While we are not aware of any stochastic-correlation models, there do exist stochastic-volatility models that calibrate to the asset skews, and allow for the skews (caused by the volatility of the volatility of each asset) to be correlated to each other.

In reality, asset volatilities are generally positively correlated, as most volatilities, and correlations, increase in a crisis situation.

A detailed discussion of these more complex models is beyond the scope of this work, but at least we are in a position to understand the limitations of the simpler models, and we have developed an understanding of the dynamics of the quanto swap.

DIFFERENCES BETWEEN INSTANTANEOUS AND TERMINAL VOLATILITIES AND CORRELATIONS

We have touched on this topic before, when we mentioned the Black quanto model. Without going into technical details, we give below a brief summary of the differences between instantaneous and terminal volatilities and correlations.

When looking at both the correlation swap and the quanto, we ignored the differences between instantaneous and terminal correlations. This is not correct. For example, our deltas and gammas in the correlation swap arise due to the movements of the asset price on the next fixing date. Thus the correlation and volatilities we should be considering are the instantaneous ones, which we then need to integrate over to get quantities like the terminal correlation or covariance. In the quanto, our deltas and cross-gamma are to the terminal forward asset prices. However, we still care about the term structure of the instantaneous volatilities. For example, imagine that the movements in the two assets were highly correlated, but they had volatility at different times (e.g., one had a volatility of 20% for six months, then 1% for six months, whereas the other had a volatility of 1% then 20%). In this case, the resultant terminal correlation would be very low.

Vanilla option prices only give us information on the terminal volatilities of the assets. To work out the instantaneous volatility term structure, we need to get a strip of option prices, with expiries up to the trade end date. We then work out the instantaneous volatility term-structure we need to bootstrap all these terminal volatilities.

The instantaneous correlation is much easier to measure historically than the terminal correlation. Analysis of a historical time series is telling you what the correlation between the instantaneous (daily) movements of the assets was. To measure the correlation in Black's model, we would require a one-year time series of the S&P and EURUSD forwards to a fixed date one year from the start of the series. That would allow us to measure the correlation between the decreasing-time forward asset prices (i.e., on the first measurement they are 365-day forwards, then they become 364-day forwards, etc.).

We shall give a more realistic example of the difference between instantaneous and terminal correlations. Let us consider the relationship between an FX spot rate, its forward value, and the foreign and domestic interest rates: $EURUSD_{fwd} = EURUSD_{spot} * \frac{df_{EUR}}{df_{USD}}$.

Now, let us imagine that the only nonzero instantaneous correlation is between the spot rate and the EUR interest rate—and that this correlation is positive. Thus, when EURUSD rises, the EUR discount factors tend towards zero. What does this mean for the correlation between the EURUSD forward rate and the EUR interest rates? Well, as $df_t = (1+r)^{-t}$, short-dated discount factors will be close to 100% and can hardly change, even if rates move a lot. Hence the short-dated FX forward will be very highly correlated with the spot, and thus must be positively correlated to EUR rates. However, how about the long-dated forwards? These are heavily dependent on rates. When EUR rates rise, the long-dated FX forward will have to fall, unless the spot rate rose by an astronomical amount! Thus, although a positive correlation between the spot and EUR rates will tend to make the correlation between the forward and rates more positive than it otherwise would have been, a sufficiently long-dated FX forward is always going to be negatively correlated to the foreign interest rate. If we naively observed a historic instantaneous correlation, and plugged that into Black's formula, we would get the wrong sign of effect from the correlation for a long-dated trade!

HEDGING A PORTFOLIO OF CORRELATION SWAPS AND QUANTOS

Now, we shall bring together what we learned about the correlation swap and the quanto, to consider how we may hedge one with the other. We shall work out the hedges below using the Black model. As discussed, a more complex model will give rise to different hedges, but the principles will be the same. We should also remember that these hedges will need to be dynamically rebalanced over time.

Imagine we have bought a one-year correlation swap between EURUSD and S&P, and one fixing has just occurred (so the spots are at the fixing levels). The payoff is 100USD per 1% move in correlation from the strike. For simplicity, we assume a discount rate of 100%. The correlation swap has the following risks:

- Correlation delta of 100USD per 1%
- Cross-gamma of $10^4 / (EURUSD * S\&P * \sigma_{EURUSD}\sigma_{S\&P}t)$
- EURUSD gamma of $-10^4 * \rho / \left(EURUSD^2\sigma^2_{EURUSD}t\right)$
- S&P gamma of $-10^4 * \rho / (S\&P\sigma^2_{S\&P}t)$

The only other liquid instrument we have access to that can hedge the correlation exposure is a quanto. The notional, in USD, of the quanto, that we need to sell is:

$$-10^4 / \left(\sigma_{EURUSD}\sigma_{S\&P}t * S\&P_{fwd}EURUSD_{fwd} * exp\left(\rho\sigma_{EURUSD}\sigma_{S\&P}t\right)\right)$$

This leaves us with some additional risks: cross-gamma and vegas:

- Cross-gamma of $-10^4 / \left(EURUSD * S\&P * \sigma_{EURUSD}\sigma_{S\&P}t\right)$
- EURUSD vega of $-10^4\rho / \sigma_{EURUSD}$
- S&P vega of $-10^4\rho / \left(EURUSD * S\&P * \sigma_{S\&P}\right)$

By hedging the correlation exposure, we have also managed to hedge the cross-gamma, too! However, we are still left with the vegas and gammas. We can hedge these with vanilla options. We shall only look at hedging the EURUSD risks, as the same argument will apply to the S&P.

If we buy an ATM call on EURUSD, deriving from Black-Scholes logic the ratio of its vega to its gamma will be:

$$EURUSD^2 \sigma_{EURUSD} t$$

From the above, we can see that this is the same as the ratio of the vega from the quanto to the gamma on the correlation swap! Thus, by hedging or vega using ATM options, we also end up hedging our gamma. Thus, in the Black model, we can construct a dynamically risk-neutral (i.e., instantaneously hedged) portfolio consisting of ATM options, a quanto, and a correlation swap.

UNCOLLATERALISED DERIVATIVES AND NOTE DISCOUNTING

In this section, we look at the pricing and risk management of uncollateralised derivatives and unsecured deposits/bonds. We start by explaining the implications of the equivalence between senior debt and uncollateralised derivatives. We bring to light the deficiencies in the common methods of pricing these trades, and how this can lead to disaster for the banks. We move on to propose a method that attempts to accurately model the credit risk embedded in these products. Finally, we apply our method to a number of examples, and highlight instances where the incorrect approach would lead to either an over- or underestimation of the true value of the trade.

Before the collapse of Lehman Brothers, most banks were able to fund themselves close to the Libor rate, which was often taken as the proxy for the Black-Scholes "risk-free" rate. Following that event, many financial institutions now have CDS levels in the region of hundreds of basis points. Figure 21.6 shows a typical example of a spread blow-out, following the Lehman collapse.

Not only were the "pre-Lehman" credit spreads tight (often in the single figures), but they were also fairly nonvolatile. Thus, ignoring or simplifying the effect of the spread, its volatility, and the possibility of the bank's default had little, if any, material impact upon pricing.

Some institutions have yet to catch up with recent events, and are now using pricing methods that are inappropriate. In ignoring the bank's credit risk (often referred to as "Own Credit Risk"), not only does the bank end up with mismatches in its funding, but it also ends up with incorrect hedges for the derivatives embedded in these products. In fact, the credit risk is often managed by completely different trading desks to those who manage the derivative risk, and the interaction between the derivative price and the creditworthiness of the bank is completely ignored. Not only are existing trade portfolios being mismanaged, but new deals are being executed at prices that are resulting in day one losses on new deals.

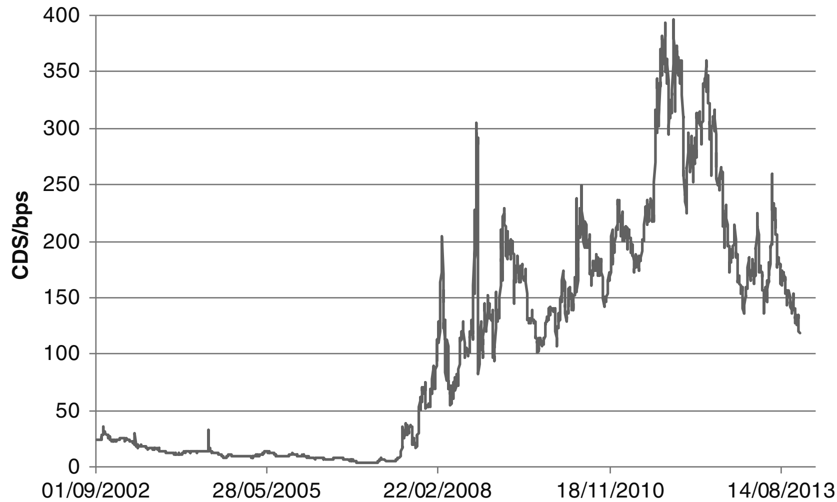

FIGURE 21.6 RBS CDS Price History, 2002–2013.
Source: Bloomberg L.P.

EXPLANATION OF THE RISK-NEUTRAL MEASURE

There are various different ways in which financial transactions can be valued. It may be appropriate to use different valuation methods in different circumstances. However, the only method that should be used to value a trading book is the risk-neutral measure (although we may take other considerations into account when deciding whether to enter into a deal on day one). Before getting to an explanation of the risk-neutral measure, let us briefly examine another three pricing methods, and explain why they are inappropriate.

Accrual Accounting

In this case, we simply ignore all future cash flows and potential cash flows, and only value cash as it is paid to us. Using this pricing method would result in us entering into any deal that allowed us to receive a large up-front premium, or large initial coupons, even if there were later cash flows whose negative value far outweighed the early benefits. The fact that many people (outside trading desks) do use this method means that a lot of swaps and bonds exist where the counterparty gets large initial coupons in return to smaller coupons or, in the case of a swap, negative coupons, towards the end of the trade.

Going Concern Basis

This is the method by which most company accounts are drawn up. Its assumption is that the entity itself will not cease to exist in the foreseeable future. Thus, the state of the world where the company defaults is, in effect, assigned a zero probability. Imagine you were issuing two bonds, identical in every way, except that one paid out the regular recovery rate upon your default, while the other explicitly paid out nothing. Using the going concern basis, you would value these bonds identically, even though the purchaser certainly would not. Although your shareholders may not care about payments upon your own default, your bondholders do; this method completely ignores the preferences of your bondholders. More fundamentally, if you do not consider any scenario where your company defaults, you cannot explain why you would ever have to include the effect of your own credit risks on your uncollateralised derivatives and internal deposits. This leads to problems, as explained later.

Risky Investor

This person wants to invest a certain sum of money. He has some views of probabilities that various future states of the world will occur. If he is risk averse, he will show preference for investments that have a homogenous payout over all these possible future states, over investments that may realise very high or very low returns. Thus, he will demand a higher rate of return from the risky investment compared to the nonrisky investment, and value them accordingly. This makes sense for him, as he is not going to try to hedge against the possible negative outcomes. As he will not be hedging in the future, he does not need to compare his valuation to those of other market participants. Thus, the same investment will have different values to people with different levels of risk aversion. If a trading portfolio were to be valued in this way, then it would be impossible for that value to be verified by a third party.

Risk-Neutral Measure

The underlying assumptions to this pricing method are that there exists a *complete market*, with *no arbitrage opportunities*. A complete market is one in which the complete set of gambles on future states of the world can be constructed with existing assets without friction (transaction costs). An arbitrage opportunity would result if you could construct the same payoff at two different prices.

In reality, neither of these assumptions is entirely true. There are always transaction costs, like bid-offer and brokerage fees. It usually impossible to statically replicate the payoff of an exotic derivative using a portfolio of liquid instruments. The problem of transaction costs can be mitigated by valuing the portfolio of trades using mid-prices. In the absence of a static replication portfolio, we have to resort to dynamic replication, where we rebalance our hedges over time. The cost of this dynamic replication is taken into account by marking illiquid parameters in our model, such as correlation. Such a market is *dynamically* complete. The more we have to rely on dynamic rehedging and the larger the transaction costs get, the more uncertain our valuations become, due to the breakdown of our assumptions.

Under these assumptions, the market price of risk must be zero, because you can always mitigate your risk exposure at a zero cost. As an example, consider two games that you play for an up-front fee. In game one, you receive £2 if a fair coin lands heads up. In game two, you receive £2 if a fair coin lands tails up. We can consider each game to be a "derivative" on the state of the coin. Let us imagine that the market price of risk is greater than zero, and you are considering playing each game, on the result of a single coin-toss. By playing a single game, you are taking a risk. As a risk-averse participant, you will demand a premium over the expected return of each game in order to play. Thus, the sum of the premia you will pay to play both games will be less than £2. Furthermore, the person taking the other side of the bets will demand a return in excess of the expectation, and thus he will demand premia whose sum is in excess of £2.

In a complete and arbitrage-free market, the sum of the premia must equal exactly £2. This is because, if they summed to more than £2, you could lock in an automatic profit by receiving the two premia from two different market recipients, and if they summed to less than £2, you could lock in a profit by paying the premia on both games, on the same coin toss.

However, it does not follow that each premium has to be exactly £1! Imagine that there has been a lot of trading on a single coin toss, and a person who paid the premium many times on game one (so his total exposure may be £1,000, rather than £2), all on the same future coin toss, wants to mitigate his risk by receiving the premium several times. In this case, if there are not many willing market participants who want to pay him the premium, he may push the market price of the premium lower, in order to sell it. However, if the premium for game one is reduced to 70p, then the premium of game 2, to avoid arbitrage, must increase to 130p. This maintains the zero market price of risk, as someone can simultaneously choose to receive the premia on both games, and end up with a risk-free portfolio.

Thus we see another important point: the risk-neutral price may differ from the real-world expectation of an event. Clearly, we have given a trivial example where this is not very likely to happen, but it is this principle that can drive a lot of the trades between banks' derivatives desks and their customers. When the risk-neutral price is much lower, in the investor's eyes, than where he believes the real world price should be, then the trade looks very attractive to him. For example, a bank may have to pay 3% above Libor to raise money on a 10-year bond issue, but the real-world probability of it defaulting, as perceived by the investor, may be a lot lower than the risk-neutral probability implied by the risk-neutral bond price. In this case he will buy the bond.

In the risk-neutral measure, all securities must yield the same risk-free return. This is because you can always turn your portfolio into a risk-free one at no cost, at will—if there was a security yielding more than the risk-free rate, then you could hedge out the risk and be left with a guaranteed profit. Of course, the bank bond will pay more than the risk-free rate, but this premium will be (theoretically) equal to the premium you have to pay to insure against the loss of your principal upon the bank's default. Hence all securities should be discounted by the risk-free rate. If a £100 risk-free (with a guaranteed payout) deposit returned £102 in a year's time, then risk-free rate must be 2%, because we know that the present value of £102 is £100.

The present value of our portfolio is the sum of the products of the payout in every future state of the world multiplied by its risk-neutral probability of occurrence, discounted to today at the risk-free rate of return.

To calculate the correct value of an illiquid exotic derivative, we construct a risk-free portfolio. This contains the exotic derivative plus any number of liquidly traded assets. This portfolio may have to be dynamic or static. In either case, if this portfolio results in no net cash flows in any possible future state of the world, then we know that its total value must be zero. In the static case, we know that the value of the exotic derivative must simply be the sum of the market values of the hedging derivatives. In the dynamic case, we need to add to this the cost/benefit of the dynamic rehedging to come to the value of the exotic derivative. Usually this is done by marking illiquid model parameters (e.g., mean reversion, correlation) at sensible levels.

WHAT DOES DISCOUNTING REALLY REPRESENT?

We consider now the issue of discounting, which we briefly touched upon earlier. In order to value a portfolio containing many different assets, we need to express their value in terms of a common denominator: cash today, in a single currency. If the portfolio contains net future cash flows, these need to be turned into an equivalent amount of present cash. In order to make a trade truly risk-free with respect to the credit risk of your counterparty, we need to engineer a way to ensure that a default by either counterparty does not result in a loss to the other. We can do this through the dynamic posting of collateral.

Every collateralised derivative transaction (or, more usually, all transactions between the same two counterparties) is backed up by a credit support annex (CSA). In its simplest form, the CSA states that the derivatives be valued at the end of each day. Cash, in a predetermined currency, representing the net value of the derivatives, is passed between the two counterparties. The recipient of the cash has to pay interest to its counterparty, at the rate determined by the CSA. The rate is usually a form of overnight rate, such as SONIA, OIS, or EONIA, for GBP, USD, and EUR, respectively. For example, SONIA (Sterling OverNight Index Average) is calculated on each London business day as the weighted average rate of all unsecured overnight sterling transactions brokered in London by the Wholesale Markets Brokers' Association (WMBA) members between 00:00 and 15.15 GMT in a minimum deal size of 25 million GBP with all counterparties listed under Section 43 of the Financial Services Act 1986.

Let us examine a simple example. Say that we enter into a derivative transaction, under a SONIA CSA, under which we pay 10mGBP on 01/03/2016 if the GBPUSD FX rate on 01/02/2016 is above 1.6. It is now 29/02/2016, and the FX rate had been observed to be 1.7. Thus, we owe 10mGBP on 01/03/2016, in all states of the world. The present value of the derivative must be 10mGBP, discounted at SONIA by one day. Suppose SONIA is currently 36.5 bps, A/365. This means that if we post 9,999,900GBP tonight, this will accrue 100GBP of interest (ignoring rounding). Thus, when 01/03 is reached, we shall have a total of 10mGBP on deposit. The transaction will then terminate, and our counterparty will keep the 10mGBP of collateral, in lieu of payment of the coupon. We have just shown that the correct amount to post, one day before maturity, was 10mGBP discounted at SONIA. We could then consider how much we would need to post on 28/02/2016. By the same logic, it must be 9,999,900GBP, discounted at SONIA. By continual application of this argument, it can be clearly seen that, at any point in the past, the value of collateral posted, which must be equal to the present value of the liability, is the final payment amount (or, more generally, for instance, before the FX rate fixed, the expectation of the final payment amount), discounted at SONIA.

As a result, we can state that:

The risk-free discount rate represents the cost of funding a risk-free derivative until its maturity, and this is why we have to discount cash flows.

Interestingly, rather than having one risk-free rate, we actually have many, as there is no reason why SONIA, OIS, or any of the other overnight rates have to be equivalent. In fact, SONIA, EONIA, OIS, and the others are all defined with reference to *unsecured* overnight rates. Thus, when credit riskiness increases, these overnight rates will also increase. We can only consider these rates to be risk free in the context of a CSA where the transaction is fully collateralised, with interest being paid at the referenced rate. Some CSAs actually allow the counterparty posting the collateral to choose (usually from GBP, EUR, and USD) which currency to post. Interest is then paid in SONIA, EONIA, or OIS, respectively. The choice may be on the full amount posted in total, or just on the variation margin (the change in the amount posted from last night). This results in a cheapest-to-deliver option that is virtually impossible to value! A CSA could, in theory, simply say that the rate of funding on the collateral is 5% (or any other fixed rate). In this case, the risk-free rate would just be 5%, although I am not sure if there are any practical examples of such CSAs.

Unfortunately, the existence of a CSA does not completely mitigate counterparty credit risk. It is possible the counterparties disagree on the value of their derivatives, resulting in an incorrect amount being posted. More dangerously, say you bought CDS protection on RBS from Lehman, before Lehman collapsed. Even if you had agreed upon the value of this derivative the day before, upon the Lehman default, most CDS rates instantly spiked. Thus, the cost of replacing this trade would have exceeded the value of collateral you had on deposit, and you would have incurred a loss. This type of risk, caused by a sudden change in market prices, is known as *gap risk*, and is notoriously difficult to value (not to be confused with a bank's treasury desk use of the term "gap risk", which refers to asset-liability funding mismatch risk).

One way to mitigate it would be to require an overcollateralisation of the derivative, but your counterparty is unlikely to agree to that (especially in a case where it is not clear which direction the derivative value is likely to gap in the event of a default)!

The arguments above may sound very unsatisfactory to a student of theoretical economics. Such a person is likely of the opinion that there should be a single well-defined risk-free rate, against which we can compare the rates of return of various risky deposits. The concept of the CSA is somewhat strange, in that it allows us to artificially define the risk-free rate to be anything that we want. The "true" risk-free rate should be the rate offered on deposits by a counterparty who actually cannot default. In reality, no such counterparty exists. The general consensus is that the closest we can get to a risk-free deposit is to give our money to the U.S. Federal Reserve overnight. The probability that they will default is so close to zero that we can discard it, and thus we can define the risk-free overnight rate to be Fed Funds, which is equivalent to OIS. OIS swaps exist, which pay OIS on one leg against fixed on the other leg. We can use these traded instruments to calculate the level of OIS on future dates, and thus build a "true" risk-free discount curve in USD. By the further use of cross-currency swaps, we can get the equivalent rate in any other currency, and thus define a single, true risk-free rate.

Although overnight OIS/Fed Funds does incorporate the theoretical risk of the Federal Reserve defaulting on its obligations, an OIS rate to term (e.g., the level on a five-year OIS swap) does not directly represent the risk that the Fed will default over five years. A default by the U.S. government would likely result in some kind of debt restructuring. Following the restructuring event, institutions would still deposit overnight funds with the Federal Reserve. An OIS swap, unlike a CDS, would continue to exist after such a default and would not create a windfall payment upon default. Thus, although term OIS must incorporate some element of credit risk to the Fed, it should trade below the coupons on U.S. treasuries. Even if a default occurred and the Fed/U.S. government ceased to exist afterwards, this would still theoretically trigger a CDS windfall payment, while it would be unclear what would happen to the OIS swap, in the absence of any future Fed Funds rates being published!

At this point it is worth clearing up some subtle points surrounding OIS, Fed Funds, and the U.S. government. As stated above, OIS is identical to the Fed Funds rate. An OIS swap represents the projected levels of OIS over the term of the swap, just as a Libor swap will trade at the projected level of Libor over the swap's term. The Federal Reserve does not set the Fed Funds rate directly. This is unlike the Bank of England, which directly sets the Base Rate. The Federal Reserve sets the Fed Funds Target Rate. They then use open market operations to make the actual realised Fed Funds rate as close as possible to the target. For example, if Fed Funds is too high, they can increase the money supply by buying back Treasury Bonds, or lending out money against Treasuries using repos. The Federal Reserve is very unlikely to default on its domestic obligations because it can print money at will (either physically or, more likely, by increasing the balance in its own bank account). Of course, it may still choose to default, rather than create more money, but it will never be in a situation where it has to default.

The U.S. government (of which the U.S. Treasury is part) is separate from the Federal Reserve. It is not able to print/create money at will. If it needs money, it has to issue debt; that is, Treasury bills or bonds. These bonds can be purchased by other central banks, private individuals, or by the Fed itself. Theoretically, the Fed could always choose to print enough money to buy a new bond issue, thus making it unlikely the U.S. government will default on its domestic debt. However, the Fed is not obliged to do this, so it is more likely that the U.S. government would default than the Fed. A default could happen if that was seen to be preferable to creating the hyperinflation associated with printing huge amounts of money. In mid-2011 the possibility of a U.S. government default appeared, if Congress had been unable to approve an increase in the government debt ceiling. Had that happened, the U.S. government would have been unable to pay its obligations as they fell due, and would technically have been in default. In this case, the Fed would still have remained solvent.

Although precrash it was often viewed as a proxy for the "risk-free" rate, Libor is not a risk-free discount rate. Libor (London Inter Bank Offered Rate) is officially defined as:

The rate at which an individual Contributor Panel bank could borrow funds, were it to do so by asking for and then accepting inter-bank offers in reasonable market size, just prior to 11.00 London time.

This is much less risk-free than OIS, as it is a term, rather than overnight, rate. Furthermore, it is paid on deposits to banks that are more likely to default than the U.S. government is. Finally, note that it does not actually represent an actual market rate that was paid on a real deposit. Based on the above, the reader might now assume that, if a collateralised transaction is discounted at the risk-free rate, then an uncollateralised transaction must be discounted at some kind of uncollateralised, or risky, rate (often known as the *cost of funds* or COF). We shall go on to show that a more subtle method is actually required.

PRICING AN UNCOLLATERALISED CASH FLOW

We stated previously that, in risk-neutral pricing, we evaluate the cash flows that can arise in all future states of the world, and discount them by their respective risk-neutral probabilities of occurrence. Imagine that company X owes us (company Y) 10mUSD in a year's time. Using the price of X's recovery lock, we can deduce the expected recovery rate (RR). Combining this with the price of X's CDS, we can deduce the probability X survives for another year: $sp_x(t = 1)$. Conversely, the probability it defaults within the next year must be $(1 - sp_x\ (t = 1))$. The risk-free (OIS) discount fact for one year is $df\_riskfree(t = 1)$. If X defaults, we have a claim on the full risk-free amount of the debt, discounted at the risk-free rate. Thus, by summing over all of probability space, we find:

$$PV = 10mUSD * df\_riskfree_{t=1} * \left(sp_{t=1} + \left(1 - sp_{t=1}\right) * \mathrm{RR}\right)$$

However, this is not the end of the story: we have not yet considered the impact of our own probability of default. Before doing so, we require a small diversion.

Set-Off and Netting

In certain jurisdictions, set-off (or netting) is enforceable by law. For example, imagine that there are two transactions between parties A and B. Under one of them, A owes B 5mGBP; under the other, B owes A 7mGBP. If B defaults on its debts, and netting is enforceable, then A has a claim on B for 2mGBP, on which A will receive B's recovery rate. If netting is not enforceable, then A will have to pay the liquidators of B 5mGBP and subsequently make a claim for 7mGBP. Say the roles were reversed, and it was B who was insolvent. Under netting, A would be owed 2mGBP by B, and B would have to pay the liquidators, in full, on the due date of the contract. However, A could be a lot smarter than that! Before B defaults, A could go to the market and buy either B's bonds with a face value of 2mGBP, or CDS protection on a 2mGBP notional. Note that the CDS protection has to have physical delivery of B's bonds as the settlement mechanism (rather than cash settlement) to create a subsequent debt from B to A. Now, when B defaults, A will have a total claim of 5mGBP on the derivative, and 2mGBP on the bonds. A can then net this with the 7mGBP he owes B, resulting in a final payment of zero. As A would have paid something close to B's recovery rate to buy the bonds, this trick has saved A 2mGBP * (1 – RR_B).

Note that not all transactions will be subject to netting, even in jurisdictions where it is inscribed in law. For example, if party X defaults, then all senior bonds it has issued will be nettable with any uncollateralised derivative transactions into which it has entered. However, if X has bought a bond issued by Y, and thus is owed money by Y, Y cannot net that bond's face value off against a bond issued by X and owned by Y. Furthermore, some transactions can be specifically ring-fenced, so as to be excluded from general netting.

There are various other variations in the law of set-off, and these have been continually tested (and revised) in the courts. However, as this is not intended to be a legal text, we shall leave the discussion at this point.

Pricing Uncollateralised Derivatives under Set-Off

Let us return to our initial example and now add in the effect of our own credit risk. We have already written down the formula for the pricing assuming we do not default. If we simply multiply that by our own survival probability, then we have got the first part of our final formula. Now we have to consider the case where we do default, with probability (1-sp_Y). In this case, our counterparty will use set-off to ensure that he only has to pay us our own recovery rate times the mark-to-market of the derivative. Note that the mark-to-market is still reduced by X's own credit risk, and thus he requires less than 10mUSD of bonds to cover the mark-to-market. Thus, we can finally conclude that the true mark-to-market of this derivative is:

$$PV = 10mUSD * df_{riskfree_{t=1}} * \left(\mathrm{sp}_{\mathrm{x}t=1} + \left(1 - \mathrm{sp}_{\mathrm{x}t=1}\right) * \mathrm{RR}_{\mathrm{x}}\right) * \left(\mathrm{sp}_{y_{t=1}} + \left(1 - \mathrm{sp}_{y_{t=1}}\right) * \mathrm{RR}_{y}\right)$$

We look at the effect of the correlation between sp_{x} and sp_{y} later.

Pricing Uncollateralised Derivatives without Set-Off

How should we price the same transaction if netting and set-off do not apply in law? In this case, we should simply ignore our own credit risk, as the event of our default has no economic impact on the value of what X has to pay. This sounds like a very surprising statement. Surely the cost of funding this position for a year depends on our own credit spread? This common misconception is not true because, if we borrow funds unsecured for a year, we shall not have a risk-free portfolio!

Imagine we borrow unsecured the right amount of money today to convert the PV of the debt from X into cash today. If we subsequently default tomorrow, we are still owed the money from X, but we only have to pay our recovery on the unsecured loan. Hence, we have made a windfall profit upon our own default. Thus we do not have a risk-free portfolio, and the debt from X cannot have an identical value to the money we have borrowed. What we could do instead would be to pledge the value of the debt from Y as collateral, and borrow money at the risk-free rate against it. Now we really do have a risk-free portfolio, and it is clear that our own credit risk should not affect the pricing.

In practice, this is not the case. The debt from X is not a liquid tradable instrument, and we could not find someone to accept it as collateral to lend us money at the risk-free rate. Thus our requirement of a complete market has broken down. If we attempt to hedge our position with unsecured borrowing, we shall lose a small amount of money if we do not default, from the interest payments on the loan. If we do default, we make a windfall gain. This is of course not "real-world", to operate on the basis that if the firm defaults we will record a "gain"!

We should still value our positions in the risk-neutral measure, though. However, when we enter into the transaction with X, we should also include the potential cost of managing the unhedgable position, assuming we do not default. This is discussed in the section "Making Money for Whom?".

Things are further complicated if we consider an uncollateralised derivative whose underlying value may switch sign due to other market conditions. Let us say our derivative is a physical FX swap where we (Y) receive 1.6mUSD in return for paying 1mGBP.

In the case of set-off, then we can simply say:

$$PV(for\ Y) = \left(1.6m\mathrm{USD} * df_{USD} - 1m\mathrm{GBP} * df_{GBP}\right) * \left(\mathrm{sp_x} + (1-\mathrm{sp_x}) * \mathrm{RR_x}\right) * \left(\mathrm{sp}_y + (1-\mathrm{sp}_y) * \mathrm{RR}_y\right)$$

However, if there is no set-off, then the value is not symmetrical if the FX moves from below 1.6 to above. If the FX rate is below 1.6, then only a default by X has an impact on the payout, as Y will be paid in full following Y's own default. The situation is reversed if the FX is above 1.6. Pricing has now become extremely complex! There is no simple way in which we can write out the PV any more. One way to price it would be to use a Monte Carlo simulation with one factor for each of: the FX rate, the USD and the GBP risk-free rates, Y's credit risk, and X's credit risk. We can simulate the year using daily time steps (more practically, we might use monthly steps, at the cost of some granularity). If, on a given path, there is no default by X or Y, then the payout on that path is the FX swap at the end of a year. If, on a given path, a default occurs, we check the level of the FX rate on the day the default occurred. If X owes Y money, and X defaults, then we scale by X's recovery. If Y defaults instead, then X owes the value of the swap, scaled by the effect of X's credit risk. Note that, in this case, the swap must be settled based on the value at the time of Y's default. Should we allow the swap to run to maturity, then the FX rate could move, and Y could end up owing X money that it cannot pay!

We can briefly consider some limiting cases of the payoff, in which the optionality caused by the asymmetry is immaterial.

- *FX rate very high*. Now X's credit risk becomes irrelevant. The PV will be close to that of an uncollateralised swap between Y and a theoretically risk-free counterparty. If X defaults, Y will have to settle the value, taking Y's credit risk into account. If Y defaults, X gets Y's recovery.
- *FX rate very low.* Now, it is Y's credit risk that is irrelevant, and we simply have the reverse of the situation above.
- *FX rate close to 1.6*. Now the PV is no longer linear with respect to the FX rate, assuming the CDS spreads (or recovery rates) of X and Y are different.

GRAPH OF PV FROM DIFFERENT CSAs

In Figure 21.7, we represent the PV of a derivative transaction, between A and B, whose collateralised PV, to A, is equal to X. We show the value of the derivative under different legal conditions. The fully collateralised PV (X) is plotted on the x-axis, and the PV, under the actual trade conditions, is plotted on the y-axis.

We have chosen:

$$RR_A + sp_A * (1 - RR_A) = 0.8$$
$$RR_B + sp_B * (1 - RR_B) = 0.5$$

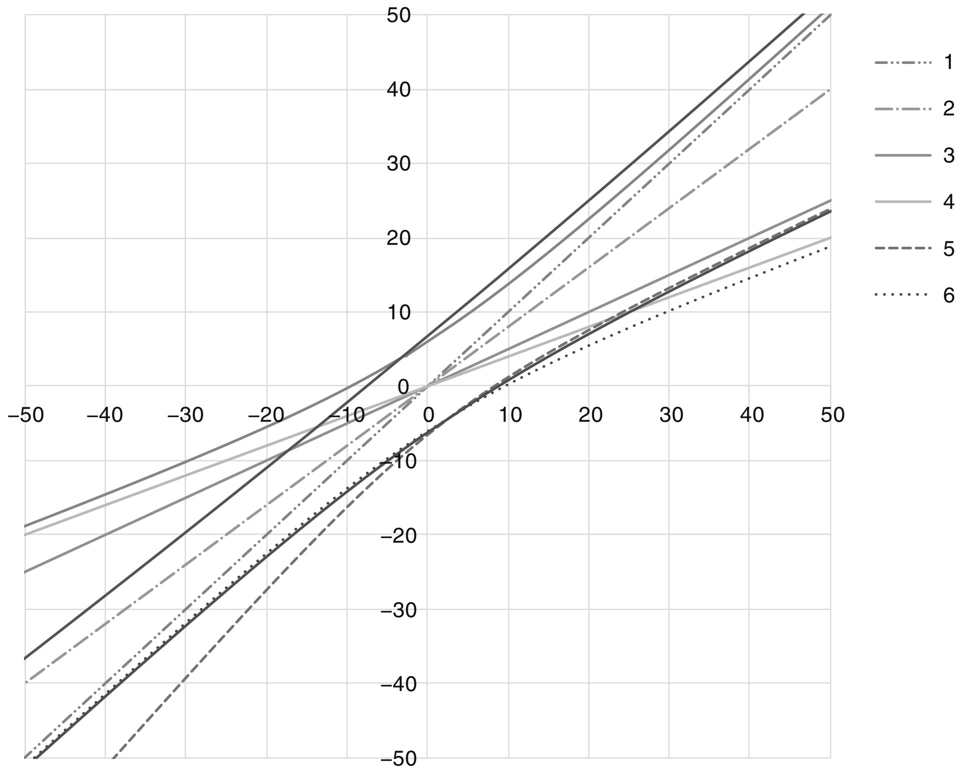

FIGURE 21.7 The Effect of the CSA's Terms on a Derivative's PV

We have assumed there is no correlation between sp_A and sp_B, and thus:

$$\left(RR_A + sp_A * \left(1 - RR_A\right)\right) * \left(RR_B + sp_B * \left(1 - RR_B\right)\right) = 0.4$$

If there were a positive correlation between the survival probabilities, then the product above would have been greater than 0.4, but still less than 0.5. This would merely have altered some of the gradients on the graph. The effect of this correlation is considered in a later section.

The different cases we have plotted are described below:

1. *Collateralised*. The PV is a straight line of gradient 1.
2. *Uncollateralised, between A and a theoretically risk-free counterparty.* The PV is a straight line of gradient 0.8.
3. *Uncollateralised, with offset, between B and a theoretically risk-free counterparty.* The PV is a straight line of gradient 0.5.
4. *Uncollateralised, with offset, between A and B*. The PV is a straight line of gradient 0.4.
5. *Uncollateralised, without offset, between A and B*. The PV is a curved line, asymptotic to 0.5 when positive, and asymptotic to 0.8 when negative.
6. *One-way CSA: only A posts collateral.* Offset applies. The PV is a curved line, asymptotic to 0.4 when positive, and asymptotic to 1 when negative.

7. *One-way CSA: only B posts collateral.* Offset applies. The PV is a curved line, asymptotic to 1 when positive, and asymptotic to 0.4 when negative.
8. *One-way CSA: only A posts collateral.* No offset. The PV is a curved line, asymptotic to 0.5 when positive, and asymptotic to 1 when negative.
9. *One-way CSA: only B posts collateral.* No offset. The PV is a curved line, asymptotic to 1 when positive, and asymptotic to 0.8 when negative.

We can see that, in the cases where set-off does not apply, and for the one-way CSAs, the payout is a curved line, which no longer crosses the origin. This means that there must be some embedded optionality to X. We can see that this makes sense, because the delta to X is not the same when X is large and positive compared to when X is large and negative. If the derivative is uncollateralised but there is no set-off, then A loses out more from B's credit risk than vice-versa, because B is more likely to default. Hence B is long optionality to X, and A is short. The more volatile X, the more negative (to A) the PV is when X = 0. In the case of the one-way CSA, whoever receives the collateral is long the optionality.

SENIOR UNSECURED BONDS: EQUIVALENCE TO UNCOLLATERALISED DERIVATIVES

In general, senior unsecured debt ranks *pari passu* to uncollateralised derivatives (although there will be specific exceptions). This means that if a company defaults on a senior bond, it must simultaneously default on its unsecured derivatives. Furthermore, the same recovery rate must apply to the derivatives and the bonds. Thus, we can hedge both the derivatives and the bonds with CDS referencing a specific senior bond. It must therefore be true that we can apply the same survival probabilities and recovery rates when pricing these instruments. (We have made a small simplification here. In general, there exists a basis between the CDS spread and the risk premium on senior debt, caused by supply and demand and liquidity imbalances. Often the risk premium on the bonds will be greater due to people's reluctance to part with cash. However, the basis cannot get too large, or cash-rich investors could simply trade the bonds against the CDS to lock in a profit.)

Upon a default event, a bondholder will either have a claim on the notional of the bond, or a claim on its mark-to-market. This depends on the legal terms in the bond issuance programme, and the specific term sheet for the bond itself. In general, for vanilla bonds, the claim will be on the notional, payable immediately (although in practice, it can take a long time to get the money). For more exotic bonds, it is likely that the claim would be on the mark-to-market (for example, the final notional repayment may contain an embedded FX option, making it unclear what the notional actually is!) of the future cash flows, discounted at the risk-free rate.

Bonds whose recovery is on the mark-to-market should be priced identically to the uncollateralised derivatives, with the proviso that the credit risk of the bondholder is ignored. Bonds whose recovery is on the notional should be priced similarly to a CDS; all the future coupons and the final notional repayment vanish upon a default, and recovery multiplied by the notional is received immediately. The pricing of CDS and bonds is discussed in more detail in Part Three and Part One, respectively.

Common Erroneous Methods to Price One's Own Credit Risk

The preceding arguments make it clear that the only truly correct way to price uncollateralised derivatives is to use the same funding/credit curve as for the bank's senior bonds. Not all banks do this! Here we explore some of the common errors made in pricing uncollateralised derivatives and bonds.

Completely Ignore Own Credit Risk on Uncollateralised Derivatives Probably the most common error is a lack of understanding of the importance of one's own credit risk. A common refrain is: "Why do I care what happens after we default?" This argument was simply that the bank should not care what happens to it after it defaults. This is an error for a number of reasons.

1. It is saying we should ignore an entire state of the world (after our own default). This is at odds with the principles of risk-neutral pricing. All the models used by trading desks use the risk-neutral measure. Should we use risk-neutral pricing for everything except our own credit risk? Or price everything in the bank-risky measure (i.e., the probability space where the bank does not default)? If the former, we are being inconsistent. If the latter, then we have to rewrite all our derivative models.
2. Everyone outside the bank clearly does care about its credit risk as it is a potential cost (or, in the case of set-off, benefit) to them. Thus, if we ignore our own credit risk, we shall be unable to agree on a price with any of our counterparties.
3. We shall suffer real losses every time we trade an uncollateralised derivative that is in-the-money to us. For example, say we buy an FX option (from a theoretically risk-free counterparty), and pay an up-front fee. If we ignored our own credit risk then we would believe that the perfect hedge would be to sell a collateralised FX option where we receive an identical up-front fee. Assuming the collateralised option was traded at the mid-price, we would have to immediately post back the fee as collateral! Thus, we find ourselves short a cash amount, equal to the PV of the option, for the entire duration of the option (although the amount will fluctuate as the option price changes). We have to then borrow back this cash amount, paying our unsecured rate, for the duration of the option, which is a real cost to us.

Assume We Can Fund Our Uncollateralised Derivative Position Overnight Anecdotally we have observed proposed discounting of USD-denominated derivatives at USD Libor + 40 bps, and an equivalent rate (i.e., with cross-currency basis) for other currencies. The error with this approach is evident when one realises that the net PV of that bank's uncollateralised trades was positive, and its term funding rates were closer to 300 bps. By discounting at a reduced rate, the bank would be overvaluing these assets. It also reflects that, for sound ALM purposes, one cannot simply fund uncollateralised derivatives in the overnight, and keep rolling the overnight funding.

This is because:

1. The logic for unsecured funding should not differ from the logic for any other funding. Most swap curves are upward sloping. We can choose to enter long-dated payer swaps, and hedge the parallel interest rate delta with shorter-dated

receivers. The yield curve is not flat, and so we cannot always fund at the short-dated swap rate, and enter into this calendar spread to make an infinite amount of money.

2. We cannot assume, as 2008 showed us, that we can always fund ourselves overnight at Libor +40. If all banks were guaranteed to be able to do this, then Northern Rock would not have become bankrupt.
3. If the bank was able to fund all its long-dated assets over night at Libor +40 forever, it would not have to issue senior bonds at 10-year maturity, at a spread of 300 over Libor.

The real mistake of the bank in question was a simple one. It confused risk-neutral pricing with a trading view. All banks, to some extent, fund their long-dated assets with shorter-dated borrowing. That is the nature of banking, termed *maturity transformation*. This is an example of a carry trade. Say your one-year funding spread is 30 bps, but your two-year funding spread is 60 bps. This implies that your one-year-forward one-year spread, in the risk-neutral world, is about 90 bps. If you believe that, in the real world, it is likely that your one-year spread will stay below 90 bps, then it makes sense to run a carry trade where you borrow money on a one-year rolling basis, and use that to fund assets that have a two-year lifespan. If your view turns out to be correct, for example, if the one-year rate stays at 30 bps, you will save 60 bps in the second year. However, if your view is wrong, then you will lose money. Most of the time, this carry trade is very profitable for the banks, especially as a lot of money deposited in instant access accounts is, in reality, left there for the long term. However, if there is a run on a bank (like Northern Rock), or if liquidity dries up, then the overnight borrowing rate can quickly spike.

There is nothing wrong, within reason, with putting on a trading position that represents a view of the market. However, you should still mark your position to market. If you are correct in your view, then you will realise profit over time. In this case, the risk-neutral expectation is that the one-year spread will rise by about 1/6th of a bp a day (to go from 30 bps to 60 bps in a year). Every day it does not rise, you realise a small profit. By marking your position to market, you also correctly represent the risk incurred by your position, allowing this to be correctly monitored by both front office and control functions.

Ignore Own Credit Risk on Bond Issuance This is a very commonly made mistake, which can have various undesirable consequences, depending on the nature of the bond. When a bank's derivatives desk issues a structured/exotic bond, the cash flows usually look something like Figure 21.8.

Alternatively, the issuance may be done by the treasury function, and the internal (or external) derivatives desk used as a swap hedge, as shown in Figure 21.9.

In this simple example, let us assume that there was no day-one P/L on the transaction.

We start by considering the bank's internal derivatives trading desk in the first diagram. If we discounted all our cash flows at the risk-free rate, and ignored our own credit risk, then it would appear that the derivatives desk has a completely flat book (i.e., a risk-free portfolio). However, the reality is that this is not true at all! Imagine that, subsequent to trading, the value of the structured coupon decreased, but the value of the Libor plus funding was unchanged. The derivatives desk would

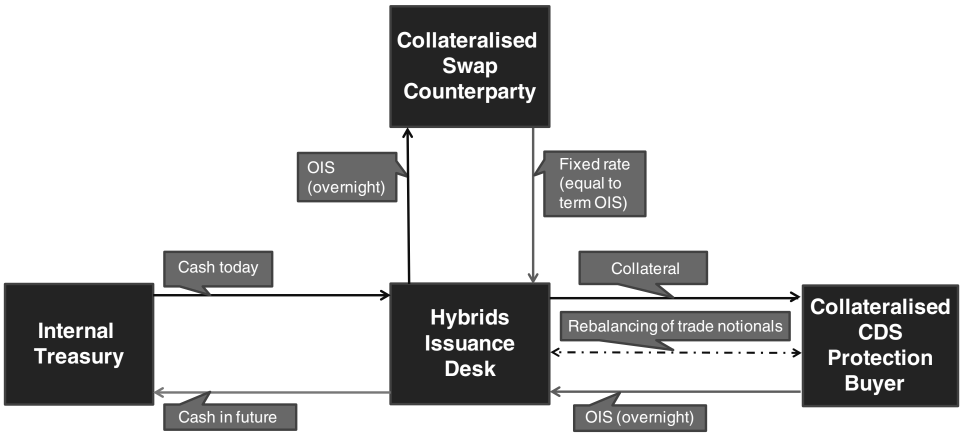

FIGURE 21.8 Cash Flow Diagram for a Structured Bond Issued by a Derivatives Desk

need to post the PV move as collateral to the swap hedge counterparty, and in return receive OIS (or equivalent). If the portfolio had been truly risk-free, this could not have happened. At first, we might think that this is not really a problem, as we are getting the fair market interest rate for the collateral posted. We have to fund our cash position by borrowing uncollateralised, which is more expensive than OIS. However, the derivative price could have moved in our favour, and so the effect should net out. Furthermore, we ourselves have argued that, in the risk-neutral measure, the funding is not actually a cost—it just creates an unhedged risk position where we lose a little money every day we do not default (from the funding gap), but make a large profit in the unlikely event that we default.

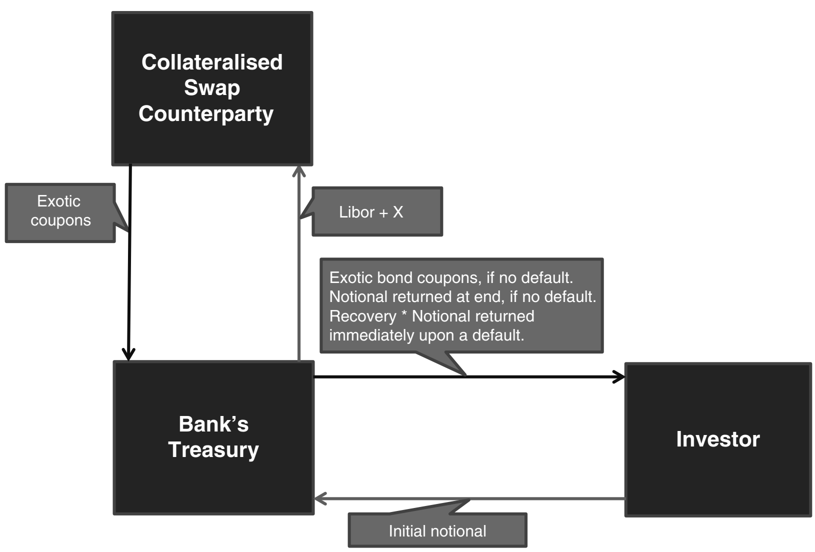

FIGURE 21.9 Cash Flow Diagram for a Structured Bond Issued by a Bank's Treasury

However, let's say that there is a negative correlation between the PV of the structured coupon and our funding spread. That is, when the structured coupon increases in value, we expect our funding spread to tighten.

1. *Structured coupon increases in value, and funding tightens.* We receive collateral from the swap counterparty and pay them OIS. We put these additional funds on deposit with our treasury at a reduced rate.
2. *Structured coupon decreases in value, and the funding widens.* We post collateral to the swap counterparty and receive OIS. We borrow the money unsecured from our treasury at a higher rate.

Due to this correlation, we can no longer dynamically rehedge our portfolio at zero cost. For example, if the structured coupon increases in value and then later decreases, we end up putting money on deposit at a low rate, followed by unwinding the deposit, at a loss, when the funding spread is wider. By ignoring the effect of credit in the pricing, we have no way to account for this, resulting in an unexplained bleed from our trading book as we rehedge (note that, if the correlation was of the opposite sign, we would have an unexplained profit).

Unfortunately, things get even worse. The way most banks are set up actually means that the bleed is not seen in the derivatives trading book! This is because the collateral posting is invariably handled by a separate funding desk, with the loss/benefit never being accounted back to the derivatives desk.

Whenever a profit is made on the collateralised trade, the collateral is posted to the funding desk, which pays OIS. The trading desk, which is discounting all cash flows at the risk-free rate (often incorrectly assumed to be Libor, rather than OIS) will see offsetting profits and losses on the structured note and derivative hedge. The funding desk will find itself long cash, and lend this out at the unsecured overnight rate (or use it to fund other negative cash positions).

Whenever a loss is made on the collateralised derivative, the trading desk will show a net flat P/L, but the funding desk will have to post out cash, receiving OIS. They will have to borrow this cash unsecured at an expensive rate.

Thus the P/L bleed occurs at the funding desk and is often not realised at the source.

In the second example, where treasury issues the note directly, the situation is slightly different, but the end result is pretty much the same. The treasury function will announce to The Street that they want to issue a note with a certain structured coupon, and also announce what their funding levels are. The derivative desks at various banks (including their own) will show quotes on the structured coupon levels that can be swapped for the quoted funding level in a collateralised trade. The treasury will enter into a swap that they believe leaves them with a flat book. Thus we are left with a situation that is essentially the same as before. The treasury does not have a flat book, and, because the funding costs do not accrue to the issuance desk, they never become aware of the problem.

In these two examples, the consequences for the note issuance desk were not necessarily too severe. However, there are two types of structures where the problem can be serious.

In the example in Figure 21.10, the derivatives trading desk has issued a deep-discount bond. Note that the structured coupon is very low, in order to compensate for

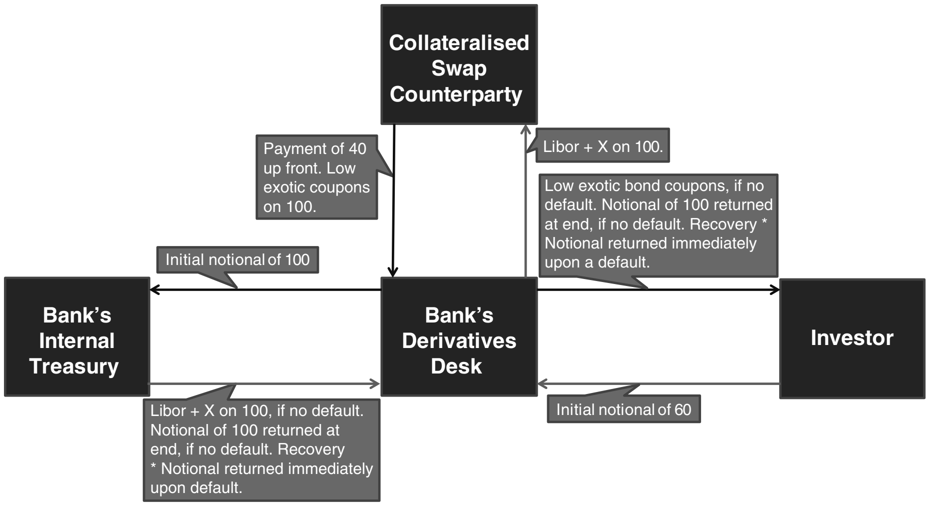

FIGURE 21.10 Deep-Discount Bond Issued by a Derivatives Desk

the discount on the bond. As before, the desk appears to be cash-flow flat. The collateralised swap had to be transacted at-the-money, once the up-front fee of 40 is included in the price. Hence, after the fee is paid, the swap will be 60 in-the-money to the derivatives desk, and thus the 60 has to be immediately posted back as collateral! Without any market moves, the bank already has a cash outflow of 60. If there are no further market moves, we expect the collateral on deposit to whittle down to zero as the high funding coupons are exchanged for the low structured coupons. Hence, the bank expects its collateral to be on deposit for roughly half the lifetime of the note. The bank will have to fund this at their unsecured rate, resulting in a huge loss over time accruing to the funding desk. In effect, the derivatives desk deposited money it did not have at treasury, and another internal desk had to pay the cost of financing the missing cash.

In this final example, illustrated in Figure 21.11, the bond is callable, at the discretion of the issuer. The derivatives desk has transacted a collateralised swap with an external counterparty, where the counterparty owns the call. The derivatives desk has deposited the up-front cash from the bond sale at their treasury. The treasury has paid them a reduced spread over Libor, but in return the derivatives desk is free to call the deposit with them and have the 100 returned to them. Again, the derivatives desk believes it is completely flat. If the calls on the swap and bond are never exercised, then we have the same situation as in the first example, which we already know is slightly wrong. If the swap counterparty exercises their call, then the derivatives desk calls both the bond and the treasury deposit, unwinding the structure. Again, they believe they are completely flat risk. However, there are five main problems with this.

1. There is the problem mentioned previously that stems from a possible correlation between the structured coupon and the funding level.
2. The derivatives desk has bought two calls: one from the investor and one from treasury. They have sold only one call, on the swap. Common sense would suggest that buying a call on a bond and on a deposit cannot hedge a call sold on one swap!

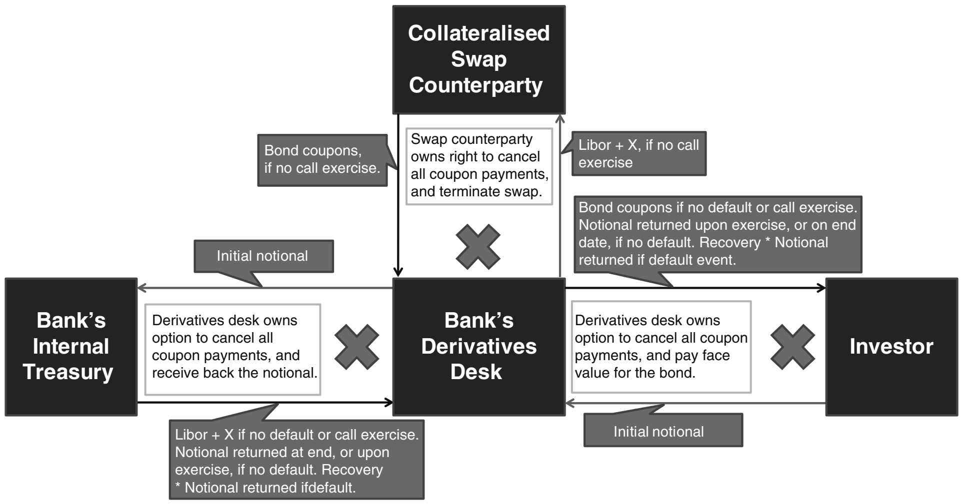

FIGURE 21.11 Callable Bond Issued by a Derivatives Desk

3. Imagine the structured coupon increases in value at the same time as the bank's funding spread widens. The collateralised counterparty will call the swap. This is fine for the derivatives desk, which just calls both the bond and the deposit. However, the treasury will now have to replace those funds at a much higher rate, thus accruing a large loss in the treasury. The derivatives desk will have lost money for the bank by voluntarily returning funds to an investor that were raised when funding was cheap, when funding is expensive! Alternatively, if the derivatives desk does not call the bond, they will have to enter into a new callable swap. As the structured coupon is now worth a lot more money, they will have to pay a much larger spread.
4. If the bank's funding tightens significantly and the structured coupon happens to be small, this will not entice the collateralised counterparty to call the swap. So we have the reverse of the situation above. Either the treasury loses out by paying out large coupons on a deal that should have been cancelled, or the derivatives desk calls the bond and the deposit, and has to pay a fee to the swap counterparty to cancel the collateralised trade.
5. Instead of using the call to optimise the cost of funding the bank, the derivatives desk may find that, because they have received much lower spreads from their treasury, that the callable trade actually offers a worse coupon than the noncallable trade. (As explained, these reduced coupons do not necessarily insulate the treasury from a loss!)

The correct way to price and dynamically hedge these and other example structures will be investigated later.

Ignore the Recovery Rate A commonly taught concept is the idea of a "cost of funds" or "internal rate of return". The cost of funds is the rate at which a company can borrow money (usually unsecured). The internal rate of return is the discount rate that, when applied to set of cash flows, returns a net value of zero. For example, say a company is considering whether it is worth investing 10mGBP up front in a project that will return 1mGBP for 10 years, and have a residual value of 2m at the end of that period. The internal rate of return is the rate (r) for which

$$\left[\sum_{i=1}^{10}(1+r)^{-i}\right]+2(1+r)^{-10}+10=0$$

The company should only enter into this transaction if its cost of funds is less than r. There is absolutely nothing wrong with using this approach for a risky investor who is operating in the risky measure in which his company does not default; the above approach evaluates all the cash flows that occur while the company survives. This method even takes into account the company's credit-riskiness through the adjusted discount rate, r. However, in the risk-neutral world, this approach is not quite right.

If a company defaults, any counterparty that has entered into an uncollateralised derivative transaction with it will have a claim upon the mark-to-market of that derivative. An investor who has purchased a bond from the company will either have a claim upon the notional of the bond (common for vanilla bonds), or a claim upon the risk-free mark-to-market of the remaining cash flows of the bond (common for structured bonds; that is, those with nonvanilla coupons). The type of claim will usually be determined by either the bond's own term sheet or the issuance programme.

First of all, we show that discounting by the cost of funds rate is equivalent to using the risk-free rate in conjunction with the survival probabilities, in the special case of a zero recovery rate.

If recovery on the event of default is on the mark-to-market, then, for a fixed cash flow:

$$PV = X * df_{risky}$$

using the cost of funds approach.

We can rewrite this as $PV = X * df_{riskfree} * sp$ because the ratio of $df_{risky}/df_{riskfree}$ represents the probability that the cash flow is paid (sp).

$$PV = X * \left(sp + RR * (1-sp) * df_{riskfree}\right)$$

in the risk-neutral measure.

Thus, when $RR = 0$, we have: $X * sp * df_{riskfree}$.

This is the same as we had from the cost of funds approach. This is unsurprising as, if there is no cash flow upon the company's default, then ignoring all cash flows post-default has no impact on the pricing.

Now, let's consider the case of a nonzero recovery rate:

$$sp = exp\left(-spread * t \big/ (1-RR)\right)$$

where the spread is the difference between the cost of funds and the risk-free rate.

Expanding the exponential to first order gives:

$$sp = 1 - spread * t \big/ (1-RR)$$

$$\Rightarrow PV = X * (1 - spread * t) * df_{riskfree}$$

Thus, to first order, the PV does not depend on the recovery rate. However, if we take the second order term of the expansion, we get:

$$PV = X * \left(1 - spread * t + \frac{(spread * t)^2}{(1 - RR)} \right) * df_{riskfree}$$

So, for larger spreads, long trades, or high recovery rates, the error in the cost of funds method starts to become material.

Now, let us consider the case of a bond where the recovery is on the notional. Without going into detailed maths, we can explain that the impact of the recovery rate is even greater than in the fore-mentioned case.

First of all, let us consider a bond whose fair value is currently 100. The reason the fair value is 100 is because the interest on the bond coupons exactly offsets the combination of the credit-risk and the risk-free discounting that arises from receiving the notional back at a future date. If the coupons exactly offset the risky discounting on the principal, then receiving recovery on a principal payment that is accelerated to the default time is economically equivalent to receiving the recovery rate on the present value of all the future coupon payments and the future principal.

Now let us consider a bond whose fair value has risen to 120. The reason the price is above 100 is because the present value of the future interest payments overcompensate the investor for receiving the notional of 100 at maturity, compared to 100 today. As the present value of these future cash flows is greater than 100, the investor would rather receive recovery on that present value, than recovery on 100. Conversely, if the bond's value has fallen below 100, the investor would rather receive recovery on the 100 notional, than recovery on the present value of the future cash flows.

Typically, bonds will be issued either with at-the-money coupons or at a discount. Furthermore, callable bonds will tend to have coupons that are below the at-the-money rate for a vanilla fixed bond because they would have been called if the coupons were higher than those of an at-the-money bond. The value of these bonds is underestimated by the cost-of-funds discounting approach. Hence using this method will result in bond being issued at too low a price, thus costing the bank real money.

Almost all the credit models in use only allow for a fixed recovery rate that is neither volatile nor a function of time. In reality, both these assumptions are false. An example of a case where the expectation of the recovery rate is likely to have a strong term structure is that of a (high tech) start-up. At inception, this company is likely to be provided with a large cash surplus. It will slowly burn through this cash, investing in R&D, with no revenue stream. If the research proves fruitless, and no revenue is forthcoming, the company will become insolvent and be wound up. As the cash reserves are reducing over time, the expected recovery rate will tend from one to zero.

Ignoring the Correlation between the Derivative (Bond) Payoff and Our Own Credit Risk This effect was mentioned above, in the context of ignoring the bank's own credit risk entirely. However, below we shall discuss it again, in the context of someone who is already accounting for their own credit spread. Imagine the bank enters into an uncollateralised FX forward, with a theoretically risk-free counterparty, at the at-the-money forward collateralised FX rate. The payoff of the trade, in five years' time is 10mUSD * (EURUSD – 1.4).

The PV, from the bank's point of view, of this cash flow (assuming no offset is applicable) is:

$$10mUSD * (1.4 - EURUSD) * \left(sp_{t=5} + RR * (1 - sp_{t=5})\right) * df_{riskfree_{t=5}}$$

Ignoring any correlations between the bank's survival and the level of EURUSD, we can simply evaluate the (liquidly traded) collateralised FX forward and scale the result by:

$$sp_{t=5} + RR * (1 - sp_{t=5})$$

As the forward was struck at-the-money, this value will also be zero.

At the time of writing, the euro is expected to lose value as financial conditions worsen, so we expect that when the bank's survival probability is low (i.e., credit spread is high), EURUSD will also be low. Applied to the above transaction, this means that:

- As the probability that the bank will survive to pay/receive the cash flow increases, the expected value of the cash will move in favour of the counterparty.
- As the probability that the bank will survive to pay/receive the cash flow decreases, the expected value of the cash will move in favour of the bank.

Thus, although the trade was struck at the collateralised forward rate, it actually has a negative value to the bank. This loss will be realised through the cost of rehedging and funding the position through the life of the trade.

On day one, let us assume that:

- The collateralised forward: $EURUSD_{t=5} = 1.4$ (we stated that the trade was at-the-money)
- $sp_{t=5} + RR * (1 - sp_{t=5}) = sp' = 0.7$
- $df_{riskfree_{t=5}} = 0.9$

If we are using a zero correlation assumption then:

- Our expectation of the PV is zero, regardless of the level of sp', hence no hedge of the bank's credit risk is required, as there is no exposure to it.
- We have an exposure to the *EURUSD* forward (and df_{riskfree}) that can be hedged by entering into an at-the-money collateralised trade, with a PV to the bank of:

$$7mUSD * (EURUSD - 1.4) * df_{riskfree_{t=5}}$$

If there were truly no correlation, then either of the following could occur:

- sp' changes while *EURUSD* does not. If sp' decreased from 0.7 to 0.5, we would unwind 2mUSD of our hedge, at the same rate at which we put it on. If sp' then returned to 0.7, we would increase our hedge notional by 2mUSD, again at the same rate.

- *EURUSD* changes while *sp′* does not. If *EURUSD* decreased from 1.4 to 1.2, then the PV of the uncollateralised derivative would be $2m * sp' * df_{riskfree}$. This would be hedged by borrowing $2m * 0.7 * 0.9USD = 1.26mUSD$ from the bank's treasury today, in return for repaying 2mUSD in five years. If *EURUSD* returned to 1.4, then we would unwind the loan at the same rate at which it was entered into.

If there were a positive correlation between *EURUSD* and *sp′*, then the following could occur:

- *sp′* decreases from 0.7 to 0.5, at the same time as *EURUSD* decreases from 1.4 to 1.2.
  - To rehedge the change in the FX exposure, caused by the move in sp′, we need to unwind 2mUSD of our 7mUSD hedge, struck at 1.4. However, as the at-the-money strike has moved, we shall have to pay a forward premium of $2mUSD * (1.2 - 1.4) = -0.4mUSD$. Alternatively, we could pay a spot premium of $-0.4mUSD * 0.9 = -0.36mUSD$.
  - To rehedge the change in the exposure to *sp′*, caused by the move in the FX, we need to borrow $10mUSD * (1.4 - 1.2) * 0.5 * 0.9 = 0.9mUSD$ today, returning 2mUSD in five years.
- *sp′* returns to 0.7, while *EURUSD* returns to 1.4.
  - To rehedge the change in the FX exposure, caused by the move in *sp′*, we need to reinstate the 2mUSD of our hedge that we unwound. We do so at the at-the-money strike of 1.4, thus returning the size of our FX hedge to its original level.
  - To rehedge the change in the exposure to our own credit risk, caused by the move in the FX, we need to unwind the loan we took out from treasury. In order for them to give us the 2mUSD in five years' time, we shall have to pay them $2mUSD * 0.7 * 0.9 = -1.26mUSD$ today.

The net result of these transactions is that we have paid out –0.36mUSD rehedging the FX exposure, and $0.9mUSD - 1.26mUSD = -0.36mUSD$ rehedging the exposure to our own credit risk. Thus we have realised a total loss of –0.72mUSD, which was not accounted for by our correlation-free model.

Now consider the converse example, where the bank is paying USD and receiving EUR uncollateralised. In this case, the uncollateralised value of the trade is actually in excess of the collateralised value. After the theoretical moves described above, we would have made a profit of 0.72mUSD.

CVA Is a Cost, DVA a Benefit CVA and DVA stand for Credit/Debit Valuation Adjustment. CVA refers to the impact on the price of an uncollateralised transaction due to the counterparty's credit risk. DVA refers to the impact caused by one's own credit risk. As the names Credit and Debit suggest, CVA is often viewed as a cost, and DVA as a benefit. The logic behind DVA is that, if CVA is a cost accruing to us, due to our counterparty's credit risk, then there must be an equal and opposite benefit that accrues to our counterparty from what he sees as his own credit risk. People then started trying to find ways to monetise this apparent benefit. In actual fact, the assumption that counterparty risk is a cost and own credit risk is a benefit, is a flawed one. If netting is allowed, then both DVA and CVA have the same direction of impact; that is, depending on whether the net portfolio is in-the-money or out-of-the-money, both CVA and DVA are either a cost or a benefit. It is only in the absence of netting that the potential benefit of CVA, and cost of DVA, disappear (although, in practice, the cost of DVA will remain for the bank's shareholders, as explained previously).

The idea that DVA is a benefit takes on even more dangerous proportions when we consider its treatment in statutory accounts. At the time of writing, many banks have shown huge accounting gains due to the write-down of their own debt. The logic is simple: the debt is a liability. Using mark-to-market accounting, we should show our liabilities at their market values in our accounts. In theory, if a bank issued debt when its credit spread was 100 over the risk-free rate, and then the bond later traded at a level implying a spread of 300, the bank could immediately buy back this debt in the secondary market, and make an immediate profit. So we represent this profit in our accounts as DVA. In reality, the bank raised debt because it needed the cash to fund its operations. If it has invested the cash in long-term assets, then it cannot buy back the debt. We should still write down the debt to its market level, but we should also write down the long-term assets, by discounting them at the new risky-funding level (ideally using a credit model rather than a cost-of-funds curve). The DVA gain on our own debt will then be offset by the loss on the write-down of the assets. Arguably, unless the assets are completely illiquid, we could sell them at a market value that is unrelated to our cost of funding them. However, if we intend to hold the assets, the cost of funding them will become a real-world cost to the company. Given that our accounts are written on a *going concern basis*, this cost should be included.

It is hardly surprising that no bank has considered doing this, given that this would almost certainly result in a large accounting loss rather than a gain. Typically, banks have long-term assets (e.g., loans) that they fund by borrowing short term, cheaply. When the bank's credit spread widens, the back end typically increases faster than the front end. Due to the smaller relative move at the front end, as well as the reduced credit sensitivity of the shorter-dated instruments, the bank will realise a much smaller gain on its debt than the loss on its assets. Furthermore, many banks may still be holding sovereign debt at face value, rather than taking what is effectively CVA risk to the counterparty to the loan—the sovereign—defaulting.

FUNDING A TRADING BOOK

Now that we understand how to value and discount collateralised and uncollateralised derivatives, and unsecured notes that we have issued, we should bring all this understanding together and conclude on the best way to fund a bank's trading book. The problem at many banks is that the posting of collateral, management of counterparty risk, note issuance, the funding of uncollateralised derivatives, and the pricing and management of the market risk on the trades are all handled by separate desks, all using their own methodologies. It is logical that different desks, with different expertise, handle these different functions, for risk-control purposes. However, it is not logical to do it in an inconsistent manner. First of all, we shall identify the problems caused by inconsistencies, and then propose a remedy for these.

Every night, the bank must contact all the counterparties to its collateralised derivatives and agree on amounts of collateral to post to or receive from each of them. Each counterparty will have its own collateral agreement with the bank, paying a different rate (e.g., OIS, SONIA, EONIA). If the trade PVs that have generated these collateral positions are not attributed back to the trading desk to whom each trade belongs, then a problem is created, because no one in the bank will know the funding rate to apply to the resulting cash balances in each trading book. In fact, the situation may be even worse, as it may not be possible to identify the PVs resulting from uncollateralised trades separately from those resulting from collateralised ones. The maturity of the cash flows in the trades that are leading to the collateral postings will also not be known. The only information available will be the average funding rate paid/received in each currency, by the bank as a whole, at the tenor the bank chose to fund its position. The only way to allocate this cost/benefit to each trading desk is to decide on bank-wide funding rates in each currency, and to fund, probably overnight, the cash positions in each trading book at these rates.

This method will give rise to a number of problems. For example:

- Uncollateralised and collateralised trades are funded at the same rates, from the perspective of the trading desk. This means that it no longer makes sense for a trading desk to discount a collateralised trade at OIS and an uncollateralised trade at the bank's cost-of-funds, because the two trades, from the point of view of the trading desk, will be treated identically.
- If the bank is borrowing funds overnight to fund what may be very long-dated cash inflows, then they are poorly hedged. A sudden spike in overnight funding rates will cause the bank to lose a lot of money due to a risk caused by a funding mismatch that was never even known about.
- Correlations between uncollateralised derivative or note PVs with the bank's funding rate will not be priced or risk-managed.
- Trading desks will hedge uncollateralised derivatives with equal notionals of collateralised derivatives. In fact, the notional of the collateralised hedge should be less, because the discounting of the final cash flow is less.

The way to solve these problems is actually very simple (in theory) and will also reveal to us the true cost of funding our trading book.

First of all, we should realise that we can consider the funding of each trade individually, instead of thinking about the overall funding rate on the whole trading book. (In reality, one-way-CSAs, lack of set-off/netting and other subtleties mean that we have to consider some trades in groups, but we shall ignore that for now.) Any cash flows on uncollateralised trades or notes we have issued have to be funded to the maturity of the cash flow. If we are discounting all these trades at our cost-of-funds rate, then we shall see the maturity of our risk on the trading desk. The trading desk can then cover this risk internally by lending/borrowing cash from the bank's treasury function to the designated maturity, in the currency of the cash flow. If there is a future cash inflow on a collateralised trade, then the present value of that cash has already been posted to the bank. This cash should be internally deposited to the trading book that owns the derivative, and that trading desk must pay overnight interest, at the collateralised rate (let's say OIS), to the collateral management desk, which then passes it on (in aggregate, over all trades) to the counterparty. The trading desk will see it has risk to paying OIS overnight, and receiving OIS to the term of the cash flow (from the discounting risk of that cash flow). It can then enter into an OIS swap where it receives OIS overnight against a fixed rate, to the term of the cash flow being hedged. The trading desk is now dynamically hedged to a movement in OIS and the cost-of-funds rate. (If the values of the trades change, then rehedging is required.)

A final point, which bothers many people, is how to fund cash positions generated by trading profits or losses. Let us imagine that a trading desk took a view on the markets, on a combination of some collateralised and uncollateralised trades, and has lost money. This will lead to a negative cash balance in the trading book. To what tenor should this be funded and at what rate? Some people argue that, because the loss is on a combination of collateralised and uncollateralised trades, it should be funded to the average maturity of those trades at a "blended" rate, somewhere between the collateralised and cost-of-funds levels. However, the correct answer is simpler. Once a loss has been made, then the money has been lost forever! Assuming the market does not move again (as that would just lead to a new profit or loss from our current position), the loss made on these trades, and the resulting negative cash balances does not suddenly disappear at the maturity of the trades. It is there permanently. Hence there is no "term" to which this loss must be funded. The cash balance will sit in the trading ledger and be funded overnight, unsecured. Every quarter (for example), the bank's treasury should take the loss away from the trading desk and move it into a central ledger by making a payment to the trading book equal to the value of the loss. The central treasury can then decide what to do with that loss. It could cover it by issuing more capital in the form of shares or bonds, or keep funding it overnight. Conversely, if a profit was made, it could pay that out in dividends to shareholders, use it to redeem existing capital, or use it to fund further operations.

In conclusion:

- Collateralised trades are funded at the rate specified in the CSA because the present value of any future cash flows is posted in return for interest payments at that rate.
- Uncollateralised trades and notes are funded at the appropriate tenor cost-of-funds rate of the bank (with additional costs/benefits associated with counterparty credit risk).
- Profits and losses do not require term funding, assuming they are "paid away" immediately.

Separately to this, the bank may wish to impose some internal rules stating that the trading operation should aim to make a return exceeding a certain amount given the capital (from shareholders and bondholders) required to keep it running. This leads to a target return on equity or return on investment, which can be translated into profit targets for individual trading desks. The higher these targets are, the higher the target annual profits of the trading desks. However, these targets cannot be directly converted into funding rates for the trading books. Confusion of these targets with risk-neutral pricing of derivatives probably has contributed to some of the confusions described above.

A DIVERSION: ST. PETERSBURG PARADOX AND CREDIT RISK

This is a very famous mathematical puzzle, and we show that the answer is actually all down to credit risk!

The question is: How much money would you pay to play the following game?

A coin is tossed repeatedly, until it lands heads up. The payout I receive from you is 2^T, where T is the number of tails. For example, if the sequence was *TTH*, the payoff would be 4. If it were H, the payoff would be 1.

The first thing to observe is the apparent paradox that the payout appears to be infinite. Summing over the value and probability of each outcome, we get:

$$PV = 0.5*1 + 0.25*2 + 0.125*4 + \ldots + 2^{-n}*2^{(n-1)} = 0.5 + 0.5 + 0.5 + \ldots = \infty$$

However, no one in the world has infinite money with which to pay me out! Let us say you have 1,048,576 in wealth. This means that, for all sequences with 21 tails or more, I receive exactly 1,048,576. That means that the payout now becomes:

$$\begin{aligned} PV &= 21*0.5(\textit{for } 0 \text{ to } 20 \textit{ tails}) + 2^{20}*\left(2^{-21} + 2^{-22} + 2^{-n}\right) \\ &= 10.5 + 2^{20}*2^{-21}*2 = 11.5 \end{aligned}$$

Thus, the amount you would be prepared to pay up front must increase as the wealth of your counterparty increases. To draw the analogy to credit risk, imagine you bought a 10-year FX option, from an uncollateralised counterparty, paying an up front premium. The tighter your counterparty's credit spread (i.e., the more wealth he has, to some extent), the greater the premium you will be prepared to pay.

I have seen this in interviews a few times. The first time, I was being interviewed for a summer internship at JPMorgan. The interviewer, it soon became clear, considered the problem to be a true paradox, and wanted me to come up with a bid offer spread to play the game. He didn't realise that my bid, as a poor student with large potential future earnings (thus making me fairly risk-neutral), should potentially be higher than my offer! I asked the interviewer how much money I should assume JPM had to pay out on the deal. He looked at me as if I was an idiot and scoffed, "You can assume we have an infinite amount with which to pay you". In retrospect, attempting to explain to him why he was wrong was probably not the best strategy.

Subsequently, I asked the question myself to applicants for RBS credit trading roles, with little success. One person did not understand how to sum the infinite series!

MAKING MONEY FOR WHOM?

As a trader, my primary role is to make money for my employer, while minimising risk. However, how exactly should we define "money" and "employer" in these circumstances, and why is this important?

Money

If I work for a UK bank who publishes its accounts in GBP, then my performance should be measured in the GBP value of my trading portfolio. At a U.S. bank, I should be measured in USD and at a European bank in euros. More precisely, using the UK example, if I have perfectly hedged my risk and locked in a "risk-free" profit, I should have a GBP cash amount sitting in my trading book. Overnight, this money is deposited unsecured with my group treasury function, earning me some interest. (It should be swept completely to treasury at month-end.) Note that this definition of a risk-free profit is measure-dependent. GBP could change in value against other currencies, or my employer could go bankrupt overnight. Many Japanese people, particularly of the older generation, saw their risk-free deposit as a box of yen bank notes buried in their garden. Following the tsunami in 2011, hundreds of billions of yen washed up on Japan's shores.

Employer

The answer to this question is not as easy as one might assume. My employer has many stakeholders, for example equity holders, senior debt holders, preference shareholders, depositors, HMRC, employees, and society at large, to name but a few. These people have widely differing risk appetites and shares in the bank's success. To simplify matters, we shall consider the only the two groups with the main financial interest and investment in the bank: the shareholders and the bondholders. We shall further simplify things to consider all bonds to be *pari passu* senior unsecured debt.

My opinion is that I should trade in a way to ensure that the following groups, each in their own right, does not make a financial loss:

1. The bank as a whole; that is, the shareholders and bondholders combined.
2. The shareholders.

Imagine a trade in which I pay an up front premium and, in return receive a windfall payment upon the default of the bank. This trade may make a profit for the bank as a whole. However, it will make a loss for the shareholders of the exact size of the up front premium. Clearly I should not execute a trade of this nature.

Imagine the opposite trade; that is, one in which I receive an up front premium, but have to pay out a cash settlement upon my default. As long as the risk-neutral value of these cash flows is positive, it will satisfy the first criterion. It has to satisfy the second criterion, regardless of the size of the final payout. Hence, I should engage in such a trading strategy, even though it has a negative expectation for the bondholders.

You may wonder why I have chosen to omit the bondholders as a group in their own right for whom each decision I make should be profitable.

Firstly, a large part of a trader's remuneration is the performance-related bonus he receives. The size of this bonus depends on the profit made by the trader and the overall profit made by the bank. A trading strategy that is loss-making while the bank is solvent, and profit-making upon its default will not generate a large bonus for the trader.

Secondly, it is the shareholders who are taking the largest risks with their investments, and it is they who have the primary say (through their voting rights) in the bank's policies.

Thirdly, as long as the bank is solvent and being run as a going concern, it is not necessary to consider the bondholders as an individual group. Note that it is no coincidence that the traders are remunerated in a way to act primarily in the shareholders' best interest: it is the shareholders who have the primary say in the bank's policies!

Why It Is Important

As shown above, a trade (or portfolio of trades) may be profitable in the risk-neutral measure (i.e., to the bank as a whole), but not to the shareholders. Such a trade should not be executed. Thus we reach the surprising realisation that we need to look beyond the risk-neutral measure when pricing up a new trade. We shall see later, in the discussion of CLNs and SPVs, some examples where this poses us problems. Below we give a more everyday example.

Imagine I go to my bank to ask for a 100kGBP, three-year mortgage on my house (with no early repayments allowed). My house is worth 1mGBP. The bank comes to the opinion that this is effectively a risk-free loan as it is so overcollateralised that it is practically impossible for my house (which is insured and has no other outstanding loans upon it) to end up with a value of less than 100kGBP. The bank decides they wish to make a profit of 5kGBP on this transaction, which they do by charging an up front fee of that amount. If the three-year risk-free interest rate is 1% per year, but the bank's unsecured borrowing rate is 4%, what rate of interest do they need to charge me?

In the risk-neutral measure, they should charge 1% a year. However, in the real world, this will generate a loss for the shareholders. My house has no rehypothecation value, and thus cannot be used to secure borrowing by the bank. Thus, in order to fund the loan made to me, the bank has to borrow unsecured at 4%. Thus, every day the bank does not default, they have lost roughly (4%–3%)/365*100kGBP. If they do default, then they (the bondholders) make a windfall gain, as they only pay recovery on the unsecured debt, but keep the full value of the loan to me of 100kGBP.

The only way to hedge this exposure with a liquid asset is for the bank to sell CDS protection on itself. Ignoring any bond-CDS spread, the collateralised CDS rate will be the difference between the risky and risk-free borrowing rates; that is, 3%. Assume the recovery rate (RR) is known to be 40%. Say the bank sold uncollateralised protection on itself to a theoretically risk-free counterparty (so we can ignore the counterparty credit risk for simplicity). Upon default, it will owe a payment of notional * (1 – RR), of which it will only pay out its recovery (i.e., 100kGBP*60%*40%). Hence, the counterparty will pay a premium of 3%*40% for the protection. Thus, the bank will sell protection on a notional of 100kGBP/40%, which will generate coupons of 3%*40%/40%*100kGBP = 3kGBP, and a loss on default of 100kGBP*(1 – 40%)*40%/40% = 60kGBP.

With the addition of this CDS trade, we have created a risk-neutral portfolio that makes no net profit or loss upon default or survival of the bank. Under these circumstances, we could offer the loan at 1%. However, in practice, due to legal and reputational issues, it is virtually impossible to sell one's own CDS. In that case, the bank remains with net risk to its own default, and the only way to ensure that the loan does not generate a loss on a day-to-day basis is to charge 4% interest.

CHAPTER SUMMARY

- Correlation occurs in financial markets due to both macro-economic effects, and due to the actions of traders themselves trying to hedge their axes.
- A correlation often, but not always, implies causation,
- In finance we should always look at the correlations between the changes in asset, and not the correlation between the absolute levels of the assets.
- There are different ways to measure correlation of changes (e.g. Spearman vs Pearson, log vs absolute) but, in practice, the differences between the measurement methods are usually not material.
- When measuring correlation, we need to ensure that we sample the assets at the same times.
- Historical correlations are only a useful guide to future correlations if economic conditions in the past are similar to those in the future.
- If our view on the correlation price is different to the market price, we can take advantage of this by putting on a correlation-sensitive trade. Two possible ways to do this are either using a correlation swap, or using a package of a quanto and vanilla options (building up a correlation position from covariance- and variance-sensitive assets).
- Financial transactions can be valued in different measures. Derivative books are always valued in the risk-neutral measure, where we assume we have access to a complete market, with no arbitrage opportunities.
- Discounting represents the cost of funding future cash flows. Collateralised derivatives are discounted at the rate set out in the CSA.
- The discounting of uncollateralized derivatives depends on the risk-adjusted funding rates of the two parties in the transaction. It also depends on the nature of the legal recourse in the event of a default, e.g. set-off.
- The effect of uncollateralized discounting on the derivative depends on the correlations between the credit risk of the two entities, and how these credit risks are correlated with the price of the equivalent collateralized derivative. These correlations are very often far from zero, and greatly affect the pricing and hedging of the portfolio.
- Uncollateralised derivatives and senior notes are usually *pari passu*, and thus should be priced similarly.

The different stakeholders in a bank will use different risky measures to value their stake. Thus, the trading desk has to be aware that a portfolio of trades that retains residual risk (particularly on the event of the bank defaulting on its debt), valued in the risk-neutral measure, might be losing money for a group of stakeholders.

Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

APPENDIX A

Statistical Concepts

MEAN AND STANDARD DEVIATION

In finance, two fundamental concepts are the mean and standard deviation of a series of numbers. Imagine that there are five dot-com companies that are suffering cash-flow problems. The year-end operating losses for these companies are £1,000, £2,000, £3,000, £5,000, and £9,000. The mean or average loss suffered by the companies is

$$(1{,}000 + 2{,}000 + 3{,}000 + 5{,}000 + 9{,}000)/5 \text{ or } £4{,}000$$

An analyst of this particular dot-com sector may wish to know how much variation (or perhaps "dispersion") away from the mean value has occurred. Therefore, we require a measure of the variance of the raw data. This is a measure of the dispersion for each item away from the mean (here denoted as X). Individual measures may be either positive or negative, and so to remove the effects of the sign we take the square of each deviation before adding them together. This is shown below.

X	$(X - \bar{X})$	$(X - \bar{X})^2$
1,000	–3,000	9,000,000
2,000	–2,000	4,000,000
3,000	–1,000	1,000,000
5,000	1,000	1,000,000
9,000	5,000	25,000,000
		40,000,000

If we are calculating variance for a sample of the population, rather than the entire population itself, then the total of the squared sums is divided by $(n - 1)$ where n is the number of observations (so here $n = 5$). The reason for this can be found in any textbook on statistics. So, in our example, the variance is [40,000,000/4] or 10,000,000. In fact, this value measures £-squared, which in our particular example does not in itself tell us very much. We, therefore, use a measure known as *standard deviation.* The standard deviation is the square root of the variance, but as it measures (here) actual £ units, it is more illustrative. The standard deviation is given by the square root of the variance. Here, the standard deviation is $\sqrt{10{,}000{,}000}$ or

£3,162.28. This means that the average (or mean) loss is £4,000, and the standard deviation is £3,162. This indicates the extent of the dispersion around the mean of the underlying data, which in this case is quite high. Statistically, the mean and variance of a set of data are given in (AA.1) and (AA.2).

$$x = \frac{\sum_{i=1}^{n} x_i}{n} \tag{AA.1}$$

$$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})}{n-1} \tag{AA.2}$$

If we are calculating variance for the population rather than a sample of the population, the denominator of (AA.2) is simply *n*.

The statistics used in value-at-risk calculations are based on well-established concepts. There are standard formulae for calculating the mean and standard deviation of a set of values. If we assume that *X* is a random variable with particular values *x*, we can apply the basic formula to calculate mean and standard deviation. Remember that the mean is the average of the set of values or observations, while the standard deviation is a measure of the dispersion away from the mean of the range of values. In fact, the standard deviation is the square root of the variance, but the variance, being the sum of squared deviations of each value from the mean, divided by the number of observations, has little use for us.

We say that the random variable is *X*, so the mean is *E*(*X*). In a time series of observations of historical data, the probability values are the frequencies of the observed values. The mean is

$$E(X) = \left[\sum_i x_i\right] / n$$

where the assigned probability to a single value among *n* is 1/*n* and where *n* is the number of observations.

The standard deviation of the set of values is

$$\sigma(X) = (1/n)\sqrt{\sum_i [x_i - E(X)]^2}$$

The probability assigned to a set of values is given by the type of distribution and, in fact, from a distribution we can determine mean and standard deviation depending on the probabilities p_i assigned to each value x_i of the random variable *X*. The sum of all probabilities must be 100 percent. From probability values, then, the mean is given by

$$E(X)\left[\sum_i p_i x_i\right] / n$$

The variance is the weighted average by the probabilities of the squared deviations from the mean; so, of course, the standard deviation—which we now call volatility—is the square root of this value. The volatility is given by:

$$\sigma(X) = \sqrt{\sum_i p_i [x_i - E(X)]^2}$$

In the example below at Table AA.1, we show the calculation of mean, variance and standard deviation as calculated from an Excel spreadsheet. The expectation is the mean of all the observations, while the variance is, as we noted earlier, the sum of squared deviations from the mean. The standard deviation is the square root of the variance.

What happens when we have observations that can assume any value within a range, rather than the discrete values we have seen in our example? When there is a probability that a variable can have a value of any measure between a range of specified values, we have a continuous distribution.

Probability Distributions

A probability distribution is a model for an actual or empirical distribution. If we are engaged in an experiment in which a coin is tossed a number of times, the number of heads recorded will be a discrete value of 0, 1, 2, 3, 4, or so on, depending on the number of times we toss the coin. The result is called a discrete random variable. Of course, we know that the probability of throwing a head is 50 percent, because there

TABLE AA.1 Mean, Variance, and Standard Deviation

Dates	Observations	Deviations from Mean	Squared Deviation
1	22	4.83	23.36
2	15	–2.17	4.69
3	13	–4.17	17.36
4	14	–3.17	10.03
5	16	–1.17	1.36
6	17	–0.17	0.03
7	16	–1.17	1.36
8	19	1.83	3.36
9	21	3.83	14.69
10	20	2.83	8.03
11	17	–0.17	0.03
12	16	–1.17	1.36
Sum	206	Sum	85.67
Mean	17.17	Variance	7.788
		Standard deviation	2.791

are only two outcomes in a coin-toss experiment, heads or tails. We may throw the coin three times and get three heads (it is unlikely but by no means exceptional); however, performing the experiment a great number of times should produce something approaching our 50 percent result. So an experiment with a large number of trials would produce an empirical distribution that would be close to the theoretical distribution as the number of tosses increases.

This example illustrates a discrete set of outcomes (0, 1, 2, 3); in other words, a discrete probability distribution. It is equally possible to have a continuous probability distribution: for example, the probability that the return on a portfolio lies between 3 and 7 percent is associated with a continuous probability distribution because the final return value can assume any value between those two parameters.

The Normal Distribution

A very commonly used theoretical distribution is the normal distribution, which is plotted as a bell-shaped curve and is familiar to most practitioners in business. The theoretical distribution actually looks like many observed distributions such as the height of people, shoe sizes, and so on. The distribution is completely described by the mean and standard deviation. The normal distribution $N(\mu,\sigma)$ has mean μ and standard deviation σ. The probability function is given by

$$P(X = x) = (1/\sigma\sqrt{2\pi}) \exp [-(x - \mu)^2/2\sigma^2]$$

The distribution is standardised as $N(0, 1)$ with mean of 0 and standard deviation of 1. It is possible to obtain probability values for any part of the distribution by using the standardised normal distribution curve and converting variables to this standardised distribution; thus the variable $Z = (X - \mu)/\sigma$ follows the standardised normal distribution $N(0, 1)$ with probability:

$$P(Z = Z) = (1/\sqrt{2\pi})\ exp\ [-Z^2/2]$$

The Central Limit Theorem (known also as the law of large numbers) is the basis for the importance of the normal distribution in statistical theory, and, in real life, a large number of distributions tend towards the normal, provided that there is a sufficient number of observations. This explains the importance of the normal distribution in statistics. If we have large numbers of observations—for example, the change in stock prices, or closing prices in government bonds—it makes calculations straightforward if we assume that they are normally distributed.

Often, as we have seen in the discussions on value-at-risk, it is convenient to assume that the returns from holding an asset are normally distributed. It is often convenient to define the return in logarithmic form as:

$$\ln\left(\frac{P_t}{P_{t-1}}\right)$$

where P_t is the price today
P_{t-1} is the previous price

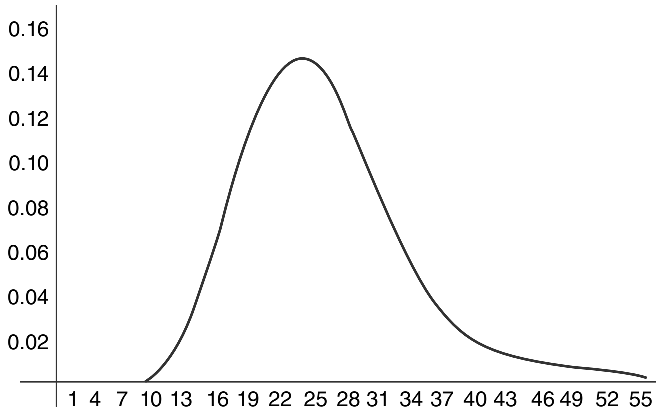

FIGURE AA.1 The Lognormal Distribution

If this is assumed to be normally distributed, then the underlying price will have a lognormal distribution. The lognormal distribution never goes to a negative value, unlike the normal distribution, and hence is intuitively more suitable for asset prices. The lognormal distribution is illustrated as Figure AA.1.

The normal distribution is assumed to apply to the returns associated with stock prices and, indeed, all financial time series observations. However, it is not strictly accurate, as it implies extreme negative values that are not observed in practice. For this reason the lognormal distribution is used instead, in which case the logarithm of the returns is used instead of the return values themselves; this also removes the probability of negative stock prices. In the lognormal distribution, the logarithm of the random variable follows a normal distribution. The lognormal distribution is asymmetric, unlike the normal curve, because it does not have negatives at the extreme values.

Confidence Intervals

Assume an estimate x of the average of a given statistical population where the true mean of the population is μ. Suppose that we believe that on average $\bar{x}$ is an unbiased estimator of μ. Although this means that on average $\bar{x}$ is accurate, the specific sample that we observe will almost certainly be above or below the true level. Accordingly, if we want to be reasonably confident that our inference is correct, we cannot claim that μ is precisely equal to the observed $\bar{x}$.

Instead, we must construct an interval estimate or confidence interval of the following form:

$$\mu = \bar{x} \text{ +/– sampling error}$$

The crucial question is: how wide must this confidence interval level be? The answer of course will depend on how much $\bar{x}$ fluctuates. We first set our requirements for level of confidence; that is, how certain we wish to be statistically. If we wish to be incorrect only 1 day in 20—that is we wish to be right 19 days each month (a month is assumed to have 20 working days)—that would equate to a 95 percent

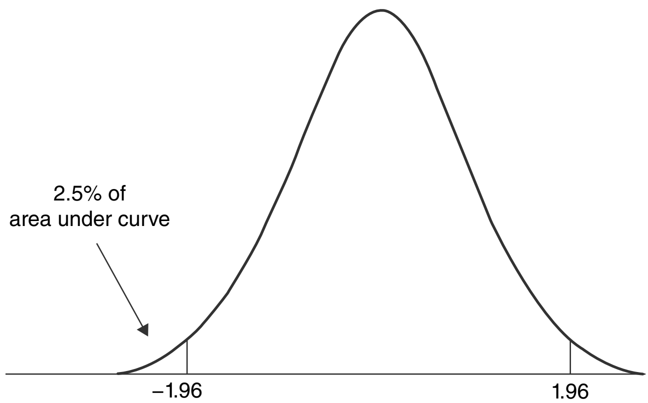

FIGURE AA.2 Normal Distribution

confidence interval that our estimate is accurate. We also assume that our observations are normally distributed. In that case we would expect that the population would be distributed along the lines portrayed in Figure AA.2.

In the normal distribution, 2.5 percent of the outcomes are expected to fall more than 1.96 standard deviations from the mean. So that means 95 percent of the outcomes would be expected to fall within +/–1.96 standard deviations. That is, there is a 95 percent chance that the random variable $\bar{x}$ will fall between μ–1.96 standard deviations and μ +1.96 standard deviations. This would be referred to as a "two-sided" (or "two-tailed") confidence interval. It gives the probability of a move upwards or downwards by the random variable outside the limits we are expecting.

In the financial markets, we do not, however, expect negative prices, so that values below zero are not really our concern. In this scenario, it makes sense to consider a one-sided test if we are concerned with the risk of loss: a move upward into profit is of less concern (certainly to a risk manager anyway!). From the statistical tables associated with the normal distribution we know that 5 percent of the outcomes are expected to fall more than 1.645 (rounded to 1.65) standard deviations from the mean. This would be referred to as a one-sided confidence interval.

APPENDIX B

Basic Tools

In this appendix, we provide a very brief summary overview of some the main statistical tools that are mentioned in the text. There are no derivations or proofs but interested readers can access any number of text books on statistics and econometrics.

SUMMATION AND PRODUCT OPERATORS

Summation is indicated by the use of the Greek capital letter Σ, pronounced sigma. Thus, we have

$$\sum_{i=1}^{N} x_i = x_1 + x_2 + x_3 + \ldots\ldots + x_N$$

The double summation operator is also used; thus

$$\sum_{i=1}^{n}\sum_{j=1}^{m} x_{ij} = \sum_{i=1}^{n}(x_{i1} + x_{i2} + \ldots\ldots + x_{im})$$

The product operator is given by

$$\prod_{i=1}^{n} x_i = x_1 \cdot x_2 \cdot x_3 \cdot \ldots\ldots \cdot x_N$$

STANDARD BROWNIAN MOTION AND THE DYNAMICS OF THE ASSET-PRICE PROCESS

A normal random variable, also termed a Gaussian random variable, with mean 0 and variance l is defined by a probability density function of the form

$$f(x) = \frac{1}{\sqrt{2\pi}}\exp\left[-\frac{x^2}{2}\right] \qquad \text{(AB.1)}$$

where

$$\int_{-\infty}^{+\infty} f(x)\mathrm{d}x = 1$$

A normally distributed variable x with mean 0 and variance 1 is denoted by $x \to N(0,1)$. Given two real numbers a and b, then $a + bx$ would be normally distributed with a mean of a and standard deviation of b. This is expressed as

$$a + bx \to N(a, b^2) \tag{AB.2}$$

Standard Brownian motion (Z, $t \geq 0$), sometimes written as W_t, is a continuous time stochastic process. For a given state of the world ω, the sample path $t \mapsto Z(t,\omega)$ is a function of time. Standard Brownian motion exhibits the following properties:

- $Z_0 = 0$
- For all $t \geq 0$ and $h \geq 0$, the value of $Z_{t+h} - Z_t$ is independent of all previous values of the process
- For all $t \geq 0$ and $h \geq 0$, $Z_{t+h} - Z_t \to N(0,h)$
- The distribution of Z_h is identical to the distribution of $Z_{t+h} - Z_t$; that is, it is a Gaussian variable with mean 0 and variance h

A *random walk* is defined thus: if Z_t, $t \geq 0$ is a standard Brownian motion and $T > 0$, then the value of Z_T is regarded as an infinite sum of infinitely small independent Gaussian increments. Assume that $A > 0$ and the infinitesimally small time increment $\Delta t = /A$ and that the time interval $[0, T]$ is divided into A sub-intervals each of duration Δt. We have the following

$$Z_T \sim \sum_{i=1}^{A} \chi_i \sqrt{\Delta t} \tag{AB.3}$$

where

$$\chi_i \to N(0,1)$$

and where χ_i is independent. We then have

$$E(Z_T) = \sum_{i=1}^{A} E(\chi_i)\sqrt{\Delta t} = 0 \tag{AB.4}$$

and

$$\begin{aligned} \mathrm{var}(Z_T) &= \sum_{i=1}^{A} \mathrm{var}(\chi_i \sqrt{\Delta t}) + 2\sum_{i<j} \mathrm{cov}(\chi_i \sqrt{\Delta t}, \chi_j \sqrt{\Delta t}) \\ &= \sum_{i=1}^{A} \Delta t + 0 = A\Delta t = T \end{aligned} \tag{AB.5}$$

Therefore, we have

$$Z_t \to N(0,1)$$

As each subinterval decreases in duration to 0, at the limit of $A \to \infty$ we obtain Brownian motion Z, which is also known as a Weiner process.

To define a generalisation of standard Brownian motion Z_t, assume a process X given by

$$\forall t \geq 0, X_t = X_0 + bt + \sigma Z_t \tag{AB.6}$$

We then have

$$X_t \to N(x_0 + bt, \sigma^2 t) \tag{AB.7}$$

The value of X now, or at time $t = 0$ is given by X_0. In (AB.7) the drift term of the process is b, and its volatility is σ2. Informally, we may then write

$$\mathrm{d}X_t = b\mathrm{d}t + \sigma \mathrm{d}Z_t \tag{AB.8}$$

where X is a generalised Brownian motion.

In continuous-time pricing models, the dynamics of an asset price P at time t is described by

$$\frac{\mathrm{d}P_t}{P_t} = \mu \mathrm{d}t + \sigma \mathrm{d}Z_t \tag{AB.9}$$

where Z is the standard Brownian motion or Weiner process. Equation (AB.9) is also written as

$$\mathrm{d}P_t = \mu P_t d_t + \sigma P_t \mathrm{d}Z_t \tag{AB.10}$$

and it can be shown that

$$\frac{\Delta P_t}{P_t} \to N(\mu \Delta t, \sigma^2 \Delta t) \tag{AB.11}$$

where $\Delta P_t / P_t$ is the ratio of the change in price of the asset between time t and $t + \Delta t$; in other words, the return on the asset during this time. A diagram of the generalised price process is at Figure AB.1.

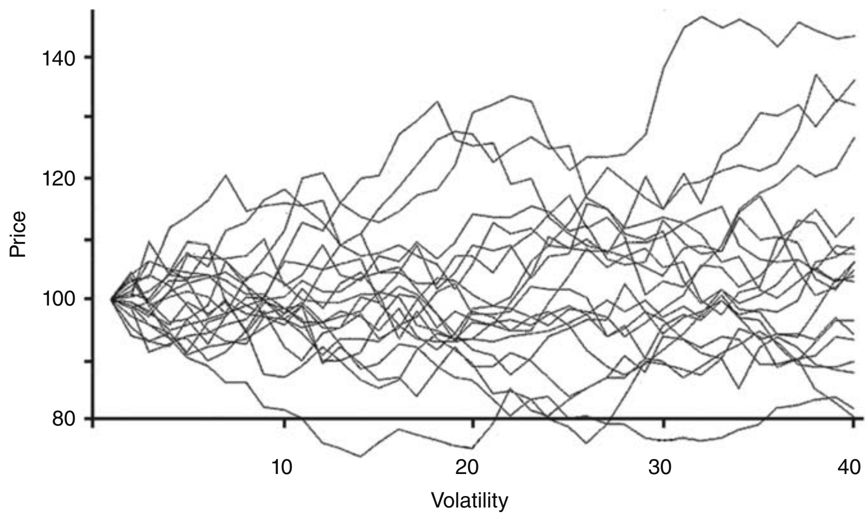

FIGURE AB.1 The Generalised Price Process

THE INTEGRAL CALCULUS

The process of differentiation is demonstrated by

$$y = 6x^3 + 2x^2 + 3x$$

where

$$\frac{dy}{dx} = 18x^2 + 4x + 3$$

The reverse of this process is integration, indicated by

$$\int 18x^2 + 4x + 3dx$$

where dx is used to indicate the original differentiation with respect to x.

If we know that

$$\int g(x)dx = f(x)$$

then

$\int_a^b g(x)dx$ is defined as $f(a) - f(b)$ and is the definite integral between a and b of the function $g(x)$.

Figure AB.2 shows part of the function $y = f(x)$ between the points a and b on the x-axis. Imagine that we wish to find the area A between the curve, the x-axis, and the lines $x = a$ and $x = b$. We divide the area A into strips of width δx and height y. These strips are approximately rectangles.

Again approximating, we can say that the area of each strip $\delta A = y\delta x$, so that

$$\frac{\delta A}{dx} \approx y \qquad \text{(AB.12)}$$

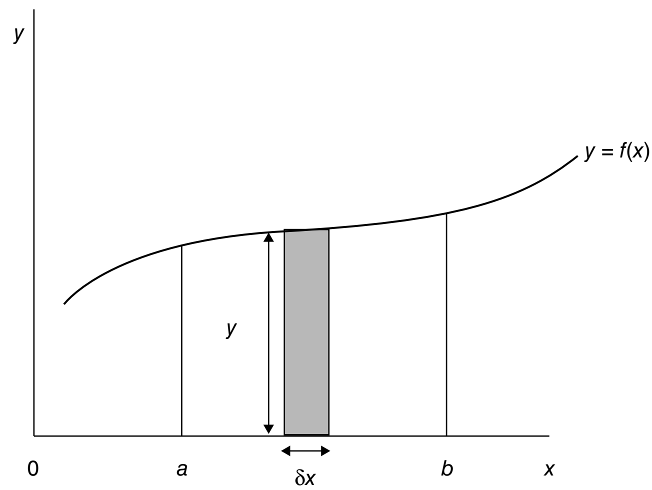

FIGURE AB.2 Function $Y = f(X)$

As the value of δx becomes progressively smaller and approaches (but does not reach) zero, any strips become closer to a rectangle, and the approximation given by (AB.12) approaches an equality, so that we then get

$$\frac{dA}{dx} \approx y \tag{AB.13}$$

However, $y = f(x)$, which means we can state

$$\frac{dA}{dx} \approx f(x) \tag{AB.14}$$

Therefore, we can state

$$A = \int f(x)dx$$

We let the integral be

$$A = g(x)$$

At the point a on the x-axis $A = 0$, which also means that $g(a) = 0$, and at the point b on the x-axis we would have A equal to the whole area, which would allow us to set

$$A = g(b)$$

But as $g(a) = 0$ at this point, we can set

$$A = g(b) - g(a) \tag{AB.15}$$

The expression in (AB.15) is actually

$$\int_a^b f(x)\mathrm{d}x$$

which is usually written

$$\int_a^b y\mathrm{d}x$$

Hence, using the definite integral allows us to calculate the area under the graph.

STOCHASTIC INTEGRALS

If X is a generalised Brownian motion described by

$$X_t = X_0 + bt + \sigma Z_t \tag{AB.16}$$

at time t, for the time interval $t_0 = 0 < t_1 < t_2 <\ldots\ldots< t_{A-1} < t_A = t$ we have

$$X_t = X_0 + \sum_{i=1}^{A}\left[b(t_i - t_{i-1}) + \sigma(Z_{ti} - Z_{ti} - 1)\right] \tag{AB.17}$$

At the limit where A goes to infinity, we obtain

$$X_t = X_0 + \int_0^T \left[bdt + \sigma dZ_t\right] \tag{AB.18}$$

If the variables b and Σ are functions of t and X_t and are not constant, then the process X can be shown to be defined as

$$X_t = X_0 + \int_0^T \left[b(t, X_t)dt + \sigma(t, X_t)dZ_t\right] \tag{AB.19}$$

which describes the limit position as A approaches infinity. At this point, we have

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dZ_t \tag{AB.20}$$

and X is described as a *diffusion* process. This process has the following properties

- The value of the process at time $t = 0$ is X_0
- The instantaneous *drift* of the process at time t is given by $b(t, X_t)$
- The instantaneous standard deviation or *volatility* of the process at time t is given by $\sigma(t,X_t)$.

The integral $\int_0^T \sigma(t, X_t)dZ_t$ is called a *stochastic integral*.

Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

APPENDIX C

Introduction to the Mathematics of Fixed-Income Pricing[1]

Appendix B introduced the techniques for handling the dynamics and calculus of stochastic variables such as interest rates. In this section we introduce the fundamentals of mathematical finance with respect to fixed-income pricing. An extended discussion of the content of this section can be found in Choudhry (2004).

To begin we need to state the following sets of assumptions, generally adopted from Merton's[2] pricing method:

- There are no transaction cost or taxes.
- There exists an exchange market for borrowing and lending at the same rate of interest (no bid-offer spread).
- The term structure is "flat" and known with certainty.
- There is a rational and competitive market.
- Market participants prefer to increase wealth.
- There are no arbitrage opportunities.

The main prerequisite of mathematical finance that is imperative in understanding fixed income are risk-neutral valuation and arbitrage pricing theory. In this introduction we will establish the probabilistic setting in which these concepts are formulated.

As stated in Musiela and Rutkowski (1997), an economy is a family of filtered space $\{(\Omega, I, \mu) : \mu \in P\}$[3], where the filtration satisfies the usual conditions,[4] and P is a collection of mutually equivalent probability measures on the measurable space.[5] We model the subjective market uncertainty of each investor by associating to each investor a probability measure from P. Investors with more risky tolerance will be represented by probability measures that weight unfavourable events relatively lower, whereas conservative investors are characterised by probability measures that

[1] This section was co-written with Mohamoud Dualeh.

[2] Robert C. Merton, "Continuous-Time Finance," 1998.

[3] To forecast a random variable, one utilises some information denoted by the symbol I_t. See more of this in Neftci (2000, p. 97).

[4] See Karatzas and Shreve (1991, section 2.7) and Korn and Korn (2000, p. 18).

[5] For a definition of measurable space and how to construct this see Jeffrey S. Rosenthal (2000, section 2).

weight unfavourable relatively higher. Moreover, it is assumed that investment information is revealed to each investor simultaneously as events in the filtration.

Since the measures in P are mutually equivalent, the investors agree on the events that have and have not occurred. We refer the reader to Neftci (2000) for an excellent example of this. It is convenient to further assume that investors initially have no other information; that is, the filtration is trivial with respect to each probability measure in P. This assumption asserts that the initial information available to investors is objective.

The foundation of a working knowledge of fixed-income finance rests on an understanding of the inherent relationship between the various interest rates and bonds. Consider the economy $\{(\Omega,I,\mu):\mu\in P\}$ on the interval $[0,T]$ and a Markov process[6] X_t with $I \equiv \sigma(X_{:0\le s\le t})$. Implicit in this statement is the assumption that the state variable[7] probability $P \equiv P^x$ associated with X_t belongs to P for some fixed elements X of the state space X_t.

Setting the scene further, a zero coupon or discount bond of maturity T is a security that pays the holder one unit of currency at time T. The prices of government and corporate discount bonds at time $t \le T$ are denoted $B(t,T)$ and $\tilde{B}(t,T)$ respectively. The local expectation hypothesis (L-EH) relates the discount bond to the instantaneous interest rate, or the spot rate for borrowing of the loan over the time interval $[t,t+\mathrm{d}t]$.

Denote the riskless spot rate by $r_t = r(X_t)$ and assume that it is a non-negative, adapted process with almost all sample paths integrable on the $[0,T]$ with respect to the Lebesgue measure.

The L-EH asserts that

$$B(t,T) = E_p(\exp(-\int_t^T r(Xs)ds|I_t)$$

As defined in Musiela and Rutkowski (1998, p. 283), the economic interpretation of this hypothesis is that ". . . the current bond price equals the expected value . . . of the bond price in the next (infinitesimal) period, discounted at the current short-term rate".

This statement is better understood in a discrete time setting. In fact, using a left sum approximation to the integral with partition $\{t_i\}_{t=0}^{n}$ of $[0,T]$ yields

$$\begin{aligned} B(0,T) &= E_p(\exp(-\sum_{i=L}^{n} r(X_{t_{i-l}})\Delta t_i)) \\ &= E_p(\exp(-r(X_{t_o})\Delta t_L)\exp(-\sum_{i=2}^{n} r(X_{t_{i-l}})\Delta t)) \\ &= (\exp(-r(X)t_L)E_P(E_P(\exp(-\sum_{i=2}^{n} r(X_{t_{i-l}})\Delta t_t|I_{t_l})) \\ &= (\exp(-r(X_{t_o})t_L)E_P^X(B(t,T)) \end{aligned}$$

[6] See Choudhry (2004, p. 185) for a lucid definition of Markov process.

[7] See Choudhry (2004, p. 144).

Under the assumption of no arbitrage, it can be shown that above equation holds under the risk-neutral measure.[8] Naturally, a similar relationship holds between the risky bond and the risky spot rate.

BOND PRICING

The process B_t is referred to as an accumulation factor or savings account.[9] B_t represents the price of a riskless security that continuously compounds at the spot rate. More precisely it is the amount of cash at time t that accumulates by investing \$1 initially, and continually rolling over a bond with an infinitesimal time to maturity. See Musiela and Rutkowski (1998, p. 268) for more detail on this.

Therefore an adapted process B_t of finite variation with continuous sample path is given by

$$B_t = \exp(\int_0^t r(X_s)\mathrm{d}s$$

When security S_t is divided by the savings account, the resultant process is the price process of the security discounted at the riskless rate.[10]

We consider next a coupon-bearing bond, with fixed coupon payments $c_1, \ldots, c_n$ at predetermined times $T_1, \ldots, T_n$ with $T_n = T$.

The price of the coupon bond is simply the present value of the sum of these cash flows. Denoting the price of a riskless coupon bond at time by $cB(t,T)$, we have

$$B_c(t,T) = \sum_{i=1}^{n} c_i B(t,T)$$

A similar relationship holds for the risky coupon bond.

However the coupons are typically structured by setting $cj = c$ for $j = 1, \ldots, n-1$ and $c_n = N + c$, where N is the principal or face value, and c is a fixed amount that is generally quoted as a percentage of N called the coupon rate. A problem that arises in comparing coupon bonds is that uncertainty about the rate at which the coupons will be reinvested causes uncertainty in the total return of the coupon bond. Hence, coupon bonds of different coupon rates and payment dates are not directly comparable. The standard way to overcome this problem is to extend the notion of a yield to maturity to coupon bearing bonds.

YIELD TO MATURITY (YTM)

In Musiela and Rutkowski (1998) the continuously compounded riskless yield to maturity $Y_c(t) = Y_c(t; c_1, \ldots, c_n.T1, \ldots T_n)$ is derived as the unique exposition to the equation

[8] We refer to the interested reader to Ingersoll (1987).

[9] See Pliska (1997, Chapter 1).

[10] In other words, the bank account is the Numeraire.

$$B_c(t) = \sum_{T_i > t} c_i e^{-Y_c(t)(T_i - t)}$$

and stands for the total return on the coupon bond under the assumption that each of the coupon payments occurring after t is reinvested at the rate $Y_c(t)$. The risky yield to maturity is defined in a similar fashion.

EXPECTATION HYPOTHESIS

There are number of excellent textbooks that the reader is encouraged to read which provides the necessary background, in particular Ingersoll (1987) and Choudhry (2004).

The yield to maturity expectation hypothesis (YTM-EH) relates the riskless YTM and the riskless spot rate. Musiela and Rutkowski (1998) state that this hypothesis as the assertion that

> *the [continuously compounded] yield from holding any [discount] bond is equal to the [continuously compounded] yield expected from rolling over a series of single period [discount] bonds.*

To gain a better understanding of this statement, we first observe that the YTM of a discrete time setting with the partition $\{t_i\}_{i=0}^{n}$ of $[t, T]$, we have that the yield of a discount bond $B(t_{i-1}, t_i)$ is given by

$$Y(t_{i-1}, t_i) = r(X_{t_{i-1}})$$

from which we deduce that the bond price is given by

$$B(t_{i-1}, t_i) = \exp(-r(X_{t_{i-1}})\Delta t_i$$

Since the YTM-EH asserts that the yield of $B(t,T)$ is equal to the yield expected from rolling over a series of discount bonds $B(t_{i-1}, t_i)$, it follows that

$$\begin{aligned} Y(t,T) = \frac{1}{T-t} \ln(B(t,T)) &= -\frac{1}{T-t} E_p\left[\ln\left(\sum_{i-L}^{n} B(t_{i-L}, t_i) \right) \mid I_t \right] \\ &= -\frac{1}{T-t} E_p\left[\sum_{i-L}^{n} r(X_{t_{i-1}})\Delta t_i | I_t \right] \end{aligned}$$

Taking the limit, as the mesh of the partition tends to zero, we obtain the continuous time discount bond price and YTM under the YTM-EH:

$$B(t,T) = \exp\left[-E_p \left(\int_t^T r(X_s) \mathrm{d}s \right) \mid I_t \right]$$

$$Y(t,T) = \frac{1}{T-t}\left[E_p \left(\int_t^T r(X_s) \mathrm{d}s \right) \mid I_t \right]$$

The last interest rate that we will consider is the instantaneous forward interest rate, or forward rate for borrowing or lending over the time interval $[s,s+ds]$ as seen from time $[t \le s]$. This will be denoted by $f(t,s)$ in the riskless case and $\tilde{f}(t,s)$ in the risky case.

If the dynamics of the process $\{f(t,s)\}_{t \le s \le T}$ are specified, then the price of the discount bond is defined by

$$B(t,T) = \exp(-\int_t^T f(s,t) ds$$

Alternatively, if the dynamics of the discount bond are known, then we have

$$f(t,T) = \frac{\partial}{\partial T} \ln B(t,T)$$

provided that this derivation exists!

Therefore the YTM-EH asserts that the forward rate is an unbiased estimate of the spot rate under the state variable probability measure P. See Choudhry (2004) Chapter 2, equation (2.18). For the relationship between the spot and forward rate we refer the reader to Chapter 3 of Choudhry (2004).

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Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

APPENDIX D

About the Companion Website

We detail here the material that can be found on the companion website for *Fixed Income Markets*, second edition. The purpose of the site is to complement the technical detail in the book by making available spreadsheet models that can be used by readers to undertake some of the calculations and analysis that we describe.

The Wiley URL is www.wiley.com/go/fim2e

The password is moorad567.

DESCRIPTION OF THE CONTENT

The website holds the following files:

Excel Spreadsheets

The Monte Carlo instructional spreadsheet was written by Abukar Ali.

The relative value funding trade repo calculator was written by Didier Joannas. This spreadsheet calculates the net funding gain or loss for a two-bond relative value position. In effect, it shows the net break-even, in terms of basis points, that the trade must make to be profitable. It is currently set up for US Treasury securities; however, the user may easily adjust the relevant cells to bring the worksheet up to date. The list of nonbusiness days is maintained in the Visual Basic module. The user enters the bond price, coupon, and maturity date in columns B, F, and G. Long positions are entered separately to short positions. The repo rate applicable to the bond position is entered at column Q. The net funding gain or loss is shown for each bond against all the other bonds in column S. Note that this means a long position against all short positions, and vice versa.

The credit default swap pricing calculator was written by Zhuoshi Liu. This model enables the user to value any CDS contract using standard parameter inputs.

Yield Curve Models

There are two yield curve models available, written by Zhuoshi Liu. The model named YCF 3.0 is a cubic spline model; the other file is a parametric curve interpolation model based on Svensson 94.

The website also holds a PDF file titled "Yield Curve Tool Instruction" that gives full instructions on how to use these models.

Bond Pricing Models

There are two bond market pricing models, both written by Michele Lizzio. One may be used to value bonds with embedded options. The other model is used more generally to value convertible bonds. An instruction guide written in PowerPoint is also available on the site as a PDF file.

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Glossary

A

30/360 An accrued interest calculation method that assumes 30 days in each month and 360 days in the year.

Accreting An accreting principal is one that increases during the life of the deal. See **amortising, bullet.**

Accreting swap Swap whose notional amount increases during the life of the swap (opposite of **amortising** swap).

Accrued benefit obligation Present value of pension benefits that have been earned by an employee to date, whether vested in the employee or not. It can be an important measure when managing the asset/liability ration of pension funds.

Accrued interest The proportion of interest or coupon earned on an investment from the previous coupon-payment date until the value date.

Accumulated value The same as **future value.**

ACT/360 A day/year count convention taking the number of calendar days in a period and a "year" of 360 days.

ACT/365 A day/year convention taking the number of calendar days in a period and a "year" of 365 days. Under the ISDA definitions used for interest-rate swap documentation, ACT/365 means the same as ACT/ACT.

ACT/ACT A day/year count convention taking the number of calendar days in a period and a "year" equal to the number of days in the current coupon period multiplied by the coupon frequency. For an interest-rate swap, that part of the interest period falling in a leap year is divided by 366 and the remainder is divided by 365.

Add-on factor Simplified estimate of the potential future increase in the replacement cost, or market value, of a derivative transaction.

All-in price See **dirty price.**

All or nothing Digital option. This option's put (call) pays out a predetermined amount ("the all") if the index is below (/above) the strike price at the option's expiration. The amount by which the index is below (/above) the **strike** is irrelevant; the payout will be all or nothing.

American option An option that may be exercised at any time during its life.

Amortising An amortising principal is one that decreases during the life of a deal, or is repaid in stages during a loan. Amortising an amount over a period of time also means accruing for it pro rata over the period. See **accreting, bullet.**

Annuity An investment providing a series of (generally equal) future cash flows.

Appreciation An increase in the market value of a currency in terms of other currencies. See **depreciation, revaluation.**

Arbitrage The process of buying securities in one country, currency, or market, and selling identical securities in another to take advantage of price differences. When this is carried out simultaneously, it is, in theory, a risk-free transaction. There are many forms of arbitrage transactions. For instance, in the cash market a bank might issue a money-market instrument in one money centre and invest the same amount in another centre at a higher rate, such as an issue of three-month U.S.-dollar CDs in the United States at 5.5 percent and a purchase of three-month Eurodollar CDs at 5.6 percent. In the futures market, arbitrage might involve buying three-month contracts and selling forward six-month contracts.

Arbitrageur Someone who undertakes arbitrage trading.

ARCH (Stands for autoregressive conditional heteroscedasticity.) A discrete-time model for a random variable. It assumes that variance is stochastic and is a function of the variance of previous time steps and the level of the underlying.

Arithmetic mean The average.

Asian option An Asian option depends on the average value of the underlying over the option's life.

Ask See **offer.**

Asset Probable future economic benefit obtained or controlled as a result of past events or transactions. Generally classified as either current or long-term.

Asset allocation Distribution of investment funds within an asset class or across a range of asset classes for the purpose of diversifying risk or adding value to a portfolio.

Asset and Liability Management (ALM) The practice of matching the term structure and cash flows of an organisation's asset and liability portfolios to maximise returns and minimise risk.

Asset-backed security A security that is collaterised by specific assets—such as mortgages—rather than by the intangible creditworthiness of the issuer.

Asset-risk benchmark Benchmark against which the riskiness of a corporation's assets may be measured. In sophisticated corporate risk-management strategies, the dollar risk of the liability portfolio may be managed against an asset-risk benchmark.

Asset securitisation The process whereby loans, receivables, and other illiquid assets in the balance sheet are packaged into interest-bearing securities that offer attractive investment opportunities.

Asset-sensitivity estimates Estimates of the effect of risk factors on the value of assets.

Asset swap An interest-rate swap or currency swap used in conjunction with an underlying asset such as a bond investment. See **Liability swap.**

At-the-money (ATM) An option is at-the-money if the current value of the underlying is the same as the strike price. See **In-the-money, Out-of-the-money.**

Auction A method of issue where institutions submit bids to the issuer on a price or yield basis. Auction rules vary considerably across markets.

Average cap Also known as an average rate cap, a cap on an average interest rate over a given period rather than on the rate prevailing at the end of the period. See also **Average price (rate) option**.

Average price (rate) option Option on a currency's average exchange rate or a commodity's average spot price in which four variables have to be agreed between buyer and seller: the premium, the **strike** price, the source of the exchange rate or commodity price data and the sampling interval (each day, for example). At the end of the life of the option, the average spot exchange rate is calculated and compared with the strike price. A cash payment is then made to the buyer of the option that is equal to the face amount of the option times the difference between the two rates (assuming the option is **In-the-money**; otherwise it expires worthless).

Average spot exchange rate The average of the spot FX rate for any given currency pair over a specified period of time.

Average worst-case exposure The expression of an exposure in terms of the average of the worst-case exposure over a given period.

B

Back-testing The validation of a model by feeding it historical data and comparing the results with historical reality.

Backwardation The situation when a forward of futures price for something is lower than the spot price (the same as forward discount in foreign exchange). See **contango.**

Balance sheet Statement of the financial position of an enterprise at a specific point in time, **giving** assets, liabilities, and stockholders' equity.

Band The Exchange Rate Mechanism (ERM II) of the European Union links the currencies of Denmark and Greece in a system that limits the degree of fluctuation of each currency against the Euro within a band (of 15 percent for the drachma and 2¼ percent for the krone) either side of an agreed par value.

Banker's acceptance See **bill of exchange**.

Banking book As described under Bank rules and the EU capital adequacy directives (CAD), a bank's outstanding transactions that relate to customer lending or long-term investments.

Barrier option A barrier option is one that ceases to exist, or starts to exist, if the underlying reaches a certain barrier level. See **knock in/out**.

Base currency Exchange rates are quoted in terms of the number of units of one currency (the variable or counter currency) that corresponds to one unit of the other currency (the base currency).

Basis The underlying cash-market price minus the futures price. In the case of a bond futures contract, the futures price must be multiplied by the conversion factor for the cash bond in question.

Basis points In interest-rate quotations, 0.01 percent.

Basis rate The rate for the difference between two different interest rates bases, e.g., Libor and Fed Funds.

Basis risk A form of market risk that arises whenever one kind of risk exposure is hedged with an instrument that behaves in a similar, but not necessarily identical, way. For instance, a bank trading desk may use three-month interest-rate futures to hedge its commercial-paper or euronote programme. Although eurocurrency rates, to which futures prices respond, are well correlated with commercial-paper rates, they do not always move in lock-step. If, therefore, commercial-paper rates move by 10 basis points but futures prices dropped by only seven basis points, the three bp gap would be the basis risk.

Basis swap An interest-rate swap where both legs are based on floating-rate payments.

Basis trade Buying the basis means selling a futures contract and buying the commodity or instrument underlying the futures contract. Selling the basis is the opposite.

Basket option Option based on an underlying basket of bonds, currencies, equities, or commodities.

Bearer bond A bond for which physical possession of the certificate is proof of ownership. The issuer does not know the identity of the bondholder. Traditionally, the bond carries detachable coupons, one for each interest-payment date, which are posted to the issuer when payment is due. At maturity, the bond is redeemed by sending in the certificate for repayment. These days, bearer bonds are usually settled electronically, and while no register of ownership is kept by the issuer, coupon payments may be made electronically.

Bear spread A spread position taken with the expectation of a fall in value in the underlying.

Benchmark A bond whose terms set a standard for the market. The benchmark usually has the greatest liquidity, the highest turnover and is usually the most frequently quoted. It also usually trades expensive to the yield curve, due to higher demand for it among institutional investors.

Beta The sensitivity of a stock relative to swings in the overall market. The market has a beta of one, so a stock or portfolio with a beta greater than one will rise or fall more than the overall market, whereas a beta of less than one means that the stock is less volatile.

Bid The price at which a market maker will buy bonds. A tight bid-offer spread is indicative of a liquid and competitive market. The bid rate in a **repo** is the interest rate at which the dealer will borrow the **collateral** and lend the cash. See **offer**.

Bid/ask The two-way price at which a market maker or "dealer" will transact in a financial instrument or interest rate.

Bid figure In a foreign-exchange quotation, the exchange rate omitting the last two decimal places. For example, when EURUSD is 1.1910/20, the big figure is 1.19. See **points**.

Bid-offer The two-way price at which a market will buy and sell stock.

Bilateral netting The ability to offset amounts owed to a counterparty under one contract against amounts owed to the same counterparty under another contract—for example, where both transactions are governed by one master agreement. Also known as cherry-picking.

Bill A **bill of exchange** is a payment order written by one person (the drawer) to another, directing the latter (drawee) to pay a certain amount of money at a future date to a third party. A bill of exchange is a bank draft when drawn on a bank. By accepting the draft, a bank agrees to pay the face value of the obligation if the drawer fails to pay, hence the term "bankers acceptance". A **Treasury bill** is short-term government paper of up to one year's maturity, sold at a discount to principal value and redeemed at par.

Bill of exchange A short-term, **zero-coupon** debt issued by a company to finance commercial trading. If it is guaranteed by a bank, it becomes a banker's acceptance.

Binomial pricing model A tool for valuing an option based on building a **binomial tree** of all the possible paths, both up and down, that the underlying asset price might take, from start until expiry. It assumes each up or down move is by a given amount.

Binomial tree A mathematical model to value options, based on the assumption that the value of the underlying can move either up or down a given extent over a given short time. This process is repeated many times to give a large number of possible paths (the "tree") that the value could follow during the option's life.

BIS Bank for International Settlements.

Black-Scholes A widely used option-pricing formula devised by Fischer Black and Myron Scholes and published in 1973.

Blended interest-rate swap Result of adding forward swap to an existing swap and blending the rates over the total life of the transaction.

Bloomberg The trading, analytics, and news service produced by Bloomberg LP; also used to refer to the terminal itself.

Bond basis An interest rate is quoted on a bond basis if it is on an **ACT/365**, **ACT/ACT** or 30/360 basis. In the short term (for accrued interest, for example), these three are different. Over a whole (non-leap) year, however, they all equate to 1. In general, the expression "bond basis" does not distinguish between them and is calculated as ACT/365. See **money-market basis**.

Bond-equivalent yield The yield that would be quoted on a U.S. Treasury bond that is trading at par and that has the same economic return and maturity as a given Treasury bill.

Bond futures Contracts traded on a recognised futures exchange that are standardised agreements to buy or sell a fixed nominal amount of a government bond. The contract is based on a "notional" bond, and a specified basket of actual bonds may be delivered against the contract.

Bootstrapping Building up successive zero-coupon yields from a combination of coupon-bearing yields.

BPV Basis-point value. The price movement due to a one-basis-point change in yield.

Bräss/Fangmeyer A method for calculating the yield of a bond similar to the **Moosmüller** method, but in the case of bonds that pay coupons more frequently than annually, using a mixture of annual and less-than-annual compounding.

Break forward A product equivalent to a straightforward option, but structured as a forward deal at an off-market rate, which can be reversed at a penalty rate.

Broken date A maturity date other than the standard ones (such as 1 week, 1, 2, 3, 6, and 12 months) normally quoted. Also known as a "cock-date" by FRA traders.

Broker-dealers Members of the London Stock Exchange who may intermediate between customers and market makers; may also act as principals, transacting business with customers from their own holdings of stock.

Bulldog Sterling domestic bonds issued by non–UK-domiciled borrowers. These bonds trade under a similar arrangement to gilts and are settled via the Central Gilts Office (now CREST).

Bullet A loan/deposit has a bullet maturity if the principal is all repaid at maturity. See **amortising**.

Bull spread A spread position taken with the expectation of a rise in value in the underlying.

Business intelligence tools Term used to describe the latest generation of access tools, which are expected to support both data extraction and subsequent analysis.

Butterfly Either an option spread that comprises the purchase of a call (or put) combined with the purchase of another call (or put) at a different strike, plus the sale of two calls at a mid-way strike, **OR**, a bond spread of one short position and a long position in two other bonds.

Buy/sell-back Opposite of **sell/buy-back**.

C

Cable The exchange rate for sterling against the U.S. dollar.

CAD The European Union's Capital Adequacy Directive.

Calendar spread The simultaneous purchase/sale of a futures contract for one date and the sale/purchase of a similar futures contract for a different date. See **spread**.

Callable bond A bond that provides the borrower with an option to redeem the issue before the original maturity date. In most cases, certain terms are set before the issue, such as the date after which the bond is callable and the price at which the issuer may redeem the bond.

Call option An option to purchase the commodity or instrument underlying the option. See **put**.

Call price The price at which the issuer can call in a bond or preferred bond.

Cancellable swap Swap in which the payer of the fixed rate has the option, usually exercisable on a specified date, to cancel the deal (see also **swaption**).

Cap A series of short-dated interest rate options, designed to protect a borrower against rising interest rates on each of a series of dates.

Capital adequacy ratio A ratio calculated to meet banking regulators' requirements, and made up of the size of a bank's own funds (available capital and reserves) as a proportion of its risky assets (the funds it has lent to credit-risky borrowers).

Capital market Long-term market (generally longer than one year) for financial instruments. See **money market**.

Capped option Option where the holder's ability to profit from a change in value of the underlying is subject to a specified limit.

Caption Option on a **cap**.

Cash See **cash market**.

Cash-and-carry An arbitrage trade in which a trader sells a bond futures contract and simultaneously buys the CTD bond, to lock in perceived mispricing in the implied future price of the bond. A reverse cash-and-carry is a purchase of futures against a sale of cash bonds. The key measure to analyse is the repo rate for the CTD, and whether the CTD is expected to be unchanged on futures expiry.

Cash market The market for trading an actual financial instrument for its true value. The cash market is the underlying market for derivatives contracts.

CBOT The Chicago Board of Trade, one of the two futures exchanges in Chicago, and one of the largest in the world.

CD See **certificate of deposit**.

Cedel Centrale de Livraison de Valeurs Mobilieres; a clearing system for euro-currency and international bonds. Cedel is located in Luxembourg and is jointly owned by a number of European banks.

Ceiling The same as **cap**.

Central bank repo A central bank repo is when the central bank lends funds (provide liquidity) to the market; as such it is a reverse repo in market terms.

Central Gilts Office The office of the Bank of England that runs the computer-based settlement system for gilt-edged securities and certain other securities (mostly **bulldogs**) for which the Bank acts as Registrar.

Central line theorem The assertion that as sample size, n, increases, the distribution of the mean of a random sample taken from almost any population approaches a normal distribution.

Certificate of deposit (CD) A money-market instrument of up to one year's maturity (although CDs of up to five years have been issued) that pays a bullet interest payment on maturity. After issue, CDs can trade freely in the secondary market, the ease of which is a function of the credit quality of the issuer.

CGBCR Central Government Net Cash Requirement.

CGBR Central Government Borrowing Requirement.

CGO reference prices Daily prices of gilt-edged and other securities held in CGO, which are used by CGO in various processes, including revaluing stock-loan transactions, calculating total consideration in a repo transaction, and DBV assembly.

Cheapest to deliver (Or CTD.) In a bond futures contract, the one underlying bond among all those that are deliverable, which is the most price-efficient for the seller to deliver.

Cherry-picking See **bilateral netting**.

Classic repo Repo is short for "sale and repurchase agreement"—a simultaneous spot sale and forward purchase of a security, equivalent to borrowing money against a loan of collateral. A reverse repo is the opposite. The terminology is usually applied from the perspective of the repo dealer. For example, when a central bank does repos, it is lending cash (the repo dealer is borrowing cash from the central bank).

Clean deposit The same as **time deposit**.

Clean price The price of a bond excluding accrued coupon. The price quoted in the market for a bond is generally a clean price rather than a **dirty price**.

Close-out netting The ability to net a portfolio of contracts with a given counterparty in the event of default. See also **bilateral netting**.

Closing leg The second leg of a repo transaction, which occurs on maturity.

CMO Central Money-Markets Office, which settles transactions in Treasury bills and other money-market instruments, and provides a depository.

CMTM Current **mark-to-market** value. See **current exposure** and **replacement cost**.

Collar The simultaneous sale of a **put** (**or call**) **option** and purchase of a call (or put) at different strikes—typically both **out-of-the-money**.

Collateral An asset of value, often of good credit quality, such as a government bond, given temporarily to a counterparty to enhance a party's creditworthiness. In a **repo**, the collateral is sold temporarily by one party to the other and bought back on maturity. Also given up under a CSA agreement to cover for mark-to-market differences in a derivative transaction, by the party that is negative mark-to-market.

Commercial paper A short-term security issued by a company or bank, generally with a zero coupon.

Competitive bid A bid for the stock at a price stated by a bidder in an auction. Noncompetitive bid is a bid where no price is specified; such bids are allotted at the weighted-average price of successful competitive bid prices.

Compound interest When some interest on an investment is paid before maturity and the investor can reinvest it to earn interest on interest, the interest is said to be compounded. Compounding generally assumes that the reinvestment rate is the same as the original rate. Also known as compounding. See **simple interest**.

Compound option Option on an option, the first giving the buyer the right, but not the obligation, to buy the second on a specific date at a predetermined price. There are two kinds.

One, on currencies, is useful for companies tendering for overseas contracts in a foreign currency. The interest-rate version comprises **captions** and **floortions**.

Consideration The total price paid in a transaction, including taxes, commissions and (for bonds) accrued interest.

Contango The situation when a forward or futures price for something is higher than the spot price (the same as forward premium in foreign exchange). See **backwardation**.

Contingent option Option where the premium is higher than usual but is only payable if the value of the underlying reaches a specified level. Also known as a contingent premium option.

Continuous compounding A mathematical, rather than practical, concept of compound interest where the period of compounding is infinitesimally small.

Contract date The date on which a transaction is negotiated. See **value date**.

Contract for differences A deal such as an **FRA** and some futures contracts, where the instrument or commodity effectively bought or sold cannot be delivered; instead, a cash gain or loss is taken by comparing the price dealt with the market price, or an index, at maturity.

Conventional gilts (included double-dated) Gilts on which interest payments and principal repayments are fixed.

Conversion factor In a bond futures contract, a factor to make each deliverable bond comparable with the contract's notional bond specification. Defined as the price of one unit of the deliverable bond required to make its yield equal the notional coupon. The price paid for a bond on delivery is the futures settlement price times the conversion factor.

Convertible currency A currency that may be freely exchanged for other currencies.

Convexity A measure of the curvature of a bond's price/yield curve (mathematically, $\frac{d^2P}{dr^2}$ / dirty price).

Correlation matrices Statistical constructs used in the **value-at-risk** methodology to measure the degree of relatedness of various market forces.

Corridor The same as **collar**.

Cost of carry The net running cost of holding a position (which may be negative)—for example, the cost of borrowing cash to buy a bond, less the coupon earned on the bond while holding it.

Cost volatility Volatility relating to operational errors or the fines and losses a business unit may incur. Reflected in excess costs and penalty charges posted to the profit and losses. See also **revenue volatility**.

Counterparty risk The risk that the other side to a transaction will default on payments owed by it during the transaction and/or on maturity.

Counterparty risk weighting See **risk weighting**.

Country risk The risks, when business is conducted in a particular country, of adverse economic or political conditions arising in that country. More specifically, the credit risk of a financial transaction or instrument arising from such conditions.

Coupon The interest payment(s) made by the issuer of security to the holders, based on the coupon rate and the face value.

Coupon swap An interest-rate swap in which one leg is fixed rate and the other floating rate. See **basis rate**.

Cover To cover an exposure is to deal in such a way as to remove the risk—either reversing the position or hedging it by dealing in an instrument with a similar but opposite risk profile. Also the amount by how much a bond auction is subscribed.

Covered call/put The sale of a covered call option is when the option writer also owns the underlying. If the underlying rises in value so that the option is exercised, the writer is protected by his position in the underlying. Covered puts are defined analogously. See **naked**.

Covered-interest arbitrage Creating a loan/deposit in one currency by combining a loan/deposit in another with a forward foreign-exchange swap.

CP See **commercial paper**.

Credit (or default) risk The risk that a loss will be incurred if a counterparty to a derivatives transaction does not fulfill its financial obligations in a timely manner.

Credit derivatives Financial contracts that involve a potential exchange of payments in which at least one of the cash flows is linked to the performance of a specified underlying credit-sensitive asset or liability.

Credit-equivalent amount As part of the calculation of the risk-weighted amount of capital the Bank for International Settlements (BIS) advises each bank to set aside against derivative credit risk, banks must compute a credit-equivalent amount for each derivative transaction. The amount is calculated by summing the current replacement cost, or market value, of the instrument and an **add-on factor**.

Credit risk Risk that a borrower of funds will default, thereby causing loss of capital and/or interest due to the lender. Usually quantified as a default probability or decribed qualitatively by a credit rating.

Credit-risk (or default-risk) exposure The value of the contract exposed to default. If all transactions are marked-to-market each day, such positive market value is the amount of previously recorded profit that might have to be reversed and recorded as a loss in the event of counterparty default.

Credit spread The interest-rate spread between two debt issues of similar duration and maturity, reflecting the relative creditworthiness of the issuers.

Credit swaps Agreement between two counterparties to exchange disparate cash flows, at least one of which must be tied to the performance of a credit-sensitive asset or to a portfolio or index of such assets. The other cash flow is usually tied to a **floating-rate index** (such as **Libor**) or a fixed rate or is linked to another credit-sensitive asset.

Credit value-at-risk (CVaR) See **value-at-risk (VaR)**.

CREST The paperless share-settlement system through which trades conducted on the London Stock Exchange can be settled. The system is operated by CRESTCo and was introduced in 1996.

CRND Commissioners for the Reduction of the National Debt, formally responsible for investment of funds held within the public sector—for example, the National Insurance Fund.

Cross See **cross-rate**.

Cross-currency repo A repo in which cash and stock are denominated in different currencies.

Cross-rate Generally an exchange rate between two currencies, neither of which is the U.S. dollar. In the American market, spot cross is the exchange rate for U.S. dollars against Canadian dollars in its direct form.

CSA Credit support annexe. The annexe to the standard International Swaps and Derivatives Association (ISDA) legal agreement governing the legal treatment of derivative transactions, and which describes the mechanism for passing collateral between derivative counterparties.

CTD See **cheapest to deliver**.

Cum-dividend Literally "with dividend", stock that is traded with interest or dividend accrued included in the price.

Cumulative default rate See **probability of default**.

Currency option The option to buy or sell a specified amount of a given currency at a specified rate at or during a specified time in the future.

Currency swap An agreement to exchange a series of cash flows determined in one currency, possibly with reference to a particular fixed or floating interest-payment schedule, for a series of cash flows based in a different currency. See **interest-rate swap**.

Current assets Assets that are expected to be used or converted to cash within one year or one operating cycle.

Current exposure The current risk exposure on the balance sheet.

Current liabilities Obligations that the firm is expected to settle within one year or one operating cycle.
Current yield Bond coupon as a proportion of clean price per 100; does not take principal gain/loss or time-value of money into account. See **yield to maturity**, **simple yield to maturity**.
Curve fitting Plotting or estimating the yield curve from market-observed yield data.
Cylinder The same as **collar**.

D

DAC-RAP Delivery against collateral-receipt against payment. Same as **DVP**.
Daily range The difference between the high and low points of a single trading day.
Day-count The convention used to calculate accrued interest on bonds and interest on cash. For UK gilts, the convention changed to actual/actual from actual/365 on 1st November 1998. For cash, the convention in sterling markets is actual/365.
DBV (delivery by value) A mechanism whereby a CGO member may borrow from or lend money to another CGO member against overnight gilt collateral. The CGO system automatically selects and delivers securities to a specified aggregate value on the basis of the previous night's CGO reference prices; equivalent securities are returned the following day. The DBV functionality allows the giver and taker of collateral to specify the classes of security to be included within the DBV. The options are the following: all classes of security held within CGO, including strips and bulldogs; coupon-bearing gilts and bulldogs; coupon-bearing gilts and strips; only coupon-bearing gilts.
DEaR Daily earnings at risk.
Debenture In the U.S. market, an unsecured domestic bond, backed by the general credit quality of the issuer. Debentures are issued under a trust deed or indenture. In the UK market, a bond that is secured against the general assets of the issuer.
Debt Management Office (DMO) An executive arm of the UK Treasury, responsible for cash management of the government's borrowing requirement. This includes responsibility for issuing government bonds (gilts), a function previously carried out by the Bank of England. The DMO began operations in April 1998.
Default correlation The degree of covariance between the probabilities of default of a given set of counterparties. For example, in a set of counterparties with positive default correlation, a default by one counterparty suggests an increased probability of a default by another counterparty.
Default probability See **probability of default**.
Default risk See **credit risk**.
Default-risk exposure See **credit-risk exposure**.
Default start options Options purchased before their "lives" actually commence. A corporation might, for example, decide to pay for a deferred-start option to lock into what it perceives as current advantageous pricing for an option that it knows it will need in the future.
Deferred-strike option Option where the strike price is established at future date on the basis of the spot foreign-exchange price prevailing at that future date.
Delegation costs Incentive costs incurred by banks in delegating monitoring activities.
Deliverable bond One of the bonds that is eligible to be delivered by the seller of a bond futures contract at the contract's maturity, according to the specifications of that particular contract.
Delivery Transfer of gilts (in settlements) from seller to buyer.
Delivery versus payment (DVP) The simultaneous exchange of securities and cash. The assured payment mechanism of the CGO achieves the same protection.
Delta (δ) The change in an option's value relative to a change in the underlying's value.
Delta neutral An option portfolio contracted to have zero delta.

Depreciation A decrease in the market value of a currency in terms of other currencies. See **appreciation, devaluation**.

Derivative Strictly, any financial instrument whose value is derived from another, such as a forward foreign-exchange rate, a futures contract, an option, an interest-rate swap, etc. Forward deals to be settled in full are not always called derivatives, however.

Devaluation An official one-off decrease in the value of a currency in terms of other currencies. See **revaluation, depreciation**.

Digital option Unlike simple European and American options, a digital option has fixed payouts and, rather like binary digital circuits, which are either on or off, pays out either this amount or nothing. Digital options can be added together to create assets that exactly mirror index price movements anticipated by investors. See **one-touch all-or-nothing**.

Direct An exchange-rate quotation against the U.S. dollar in which the dollar is the **variable currency** and the other currency is the **base currency**.

Dirty price The price of a bond including accrued interest. Also known as the "all-in" price.

Discount The amount by which a currency is cheaper, in terms of another currency, for future delivery than for spot, is the forward discount (in general, a reflection of interest-rate differentials between two currencies). If an exchange rate is "at a discount" (without specifying to which of the two currencies this refers), this generally means that the **variable currency** is at a discount. See **premium**.

Discount factor A factor by which one multiplies a future known cash flow, to obtain its present value.

Discount house In the UK money market, originally securities houses that dealt directly with the Bank of England in T-bills and bank bills, or discount instruments; hence the name. Most discount houses were taken over by banking groups, and the term is not generally used, as the BoE now also deals directly with clearing banks and securities houses.

Discount rate The method of market quotation for certain securities (U.S. and UK Treasury bills, for example), expressing the return on the security as a proportion of the face value of the security received at maturity—as opposed to a **yield**, which expresses the yield as a proportion of the original investment.

Discount swap Swap in which the fixed-rate payments are less than the internal rate of return on the swap, the difference being made up at maturity by a balloon payment.

Diversified A portfolio that has been invested across a range of assets such that its credit risk is minimised. This is achieved by having a mixture of assets whose individual credit risks are uncorrelated with each other.

Dividend discount model Theoretical estimate of market value that computes the economic or the net present value of future cash flows due to an equity investor.

DMO The UK Debt Management Office.

Down-and-in option Barrier option where the holder's ability to exercise is activated if the value of the underlying drops below a specified level. See also **up-and-in option**.

Down-and-out-option Barrier option where the holder's ability to exercise expires if the value of the underlying drops below a specified level.

Dual-currency option Option allowing the holder to buy either of two currencies.

Dual-currency swap Currency swap where both the interest rates are fixed rates.

Dual-strike option Interest-rate option, usually a **cap** or a **floor**, with one floor or ceiling rate for part of the option's life and another for the rest.

Duration A measure of the weighted-average life of a bond or other series of cash flows, using the present values of the cash flows as the weights. See **modified duration**.

Duration gap Measurement of the interest-rate exposure of an institution.

Duration weighting The process of using the modified-duration value for bonds to calculate the exact nominal holdings in a spread position. This is necessary because £1 million nominal of a two-year bond is not equivalent to £1 million of, say, a five-year bond. The modified-duration value of the five-year bond will be higher, indicating that its "basis-point value" (bpv) will be greater and that, therefore, £1 million worth of this bond represents greater sensitivity to a move in interest rates (risk). As another example, consider a fund manager holding £10 million of five-year bonds. The fund manager wishes to switch into a holding of two-year bonds with the same overall risk position. The basis-point values of the bonds are 0.041583 and 0.022898, respectively. The ratio of the bpvs are 0.041583/0.022898 = 1.816. The fund manager, therefore, needs to switch into £10m × 1.816 = £18.160 million of the two-year bond.

DV01 An acronym for "dollar value of an 01", meaning price value of a basis point. The change in value of a bond or derivative for a one-basis-point change in its yield. Also known as "Dollar value of a basis point" or DVBP.

DVP Delivery versus payment, in which the settlement mechanics of a sale or loan of securities against cash is such that the securities and cash are exchanged against each other simultaneously through the same clearing mechanism and neither can be transferred unless the other is.

E

Early exercise The exercise or assignment of an option prior to expiration.

ECU The European Currency Unit, a basket composed of European Union currencies, now defunct following introduction of the euro currency.

Effective rate An effective interest rate is the rate which, earned as simple interest over one year, gives the same return as interest paid more frequently than once per year and then compounded. See **nominal rate**.

Efficient-frontier method Technique used by fund managers to allocate assets.

Embedded option Interest-rate-sensitive option in debt instrument that affects its redemption. Such instruments include **mortgage-backed securities** and **callable bonds**.

End-end A money-market deal commencing on the last working day of a month and lasting for a whole number of months, maturing on the last working day of the correlation.

Epsilon (ε) The same as **vega**.

Equity The residual interest in the net assets of an entity that remains after deducting the liabilities.

Equity-linked swap Swap where one of the cash flows is based on an equity instrument or index, when it is known as an equity-index swap.

Equity options Options on shares of an individual common stock.

Equity warrant Warranty, usually attached to a bond, entitling the holder purchase share(s).

Equivalent life The weighted-average life of the principal of a bond where there are partial **redemptions**, using the **present values** of the partial redemptions as the weights.

Equivalent rate The interest rate that returns the same amount as another quoted interest rate but at a different compounding basis.

ERA See **exchange-rate agreement**.

Eta (η) The same as **vega**.

Euribor The reference rate for the euro currency, set in Frankfurt.

Euro The name for the domestic currency of the European Monetary Union. Not to be confused with **eurocurrency**.

Euroclear An international clearing system for international securities. Euroclear is based in Brussels and managed by Morgan Guaranty Trust Company.

Eurocurrency A eurocurrency is a currency owned by a non-resident of the country in which the currency is legal tender. Not to be confused with **euro**.

Euro-issuance The issue of gilts (or other securities) denominated in euro.

Euromarket The international market in which **eurocurrencies** are traded.

European option A European **option** is one that may be exercised only at **expiry**. See **American option**.

Exchange controls Regulations restricting the free convertibility of a currency into other currencies.

Exchange-rate agreement A **contract for differences** based on the movement in a **forward-forward** foreign-exchange swap price. Does not take account of the effect of **spot** rate changes as an **FXA** does. See **SAFE**.

Exchange-traded **Futures** contracts are traded on a futures exchange, as opposed to **forward** deals, which are **OTC. Option** contracts are similarly exchange-traded rather than **OTC**.

Ex-dividend (xd) date A bond's record date for the payment of coupons. The coupon payment will be made to the person who is the registered holder of the stock on the xd date. For UK gilts, this is seven working days before the coupon date.

Exercise To exercise an **option** (by the **holder**) is to require the other party (the **writer**) to fulfill the underlying transaction. Exercise price is the same as **strike** price.

Exotic option An option that is not plain vanilla, any complex option.

Expected (credit) loss Estimate of the amount a derivatives counterparty is likely to lose as a result of default from a derivatives contract, with a given level of probability. The expected loss of any derivative position can be derived by combining the distributions of credit exposures, rate of recovery, and probabilities of default.

Expected default rate Estimate of the most likely rate of default of a counterparty expressed as a level of probability.

Expected rate of recovery See **rate of recovery**.

Expiry An option's expiry is the time after which it can no longer be **exercised**.

Exposure Risk to market movements.

Exposure profile The path of worst-case or expected exposures over time. Different instruments reveal quite differently shaped exposure profiles arising from the interaction of the diffusion and amortisation effects.

Extinguishable option Option in which the holder's right to exercise disappears if the value of the underlying passes a specified level. See also **barrier option**.

Extrapolation The process of estimating a price or rate for a particular value date, from other known prices, when the value date required lies outside the period covered by the known prices. See **interpolation**.

F

Face value The principal amount of a security generally repaid ("redeemed") all at maturity, but sometimes repaid in stages, on which the **coupon** amounts are calculated.

Fence The same as **collar**.

Fixing See **Libor fixing**.

Floating rate An interest rate set with reference to an external index. Also, an instrument paying a floating rate is one where the rate of interest is refixed in line with market conditions at regular intervals, such as every three or six months. In the current market, an exchange rate determined by market forces with no government intervention.

Floating-rate CD CD on which the rate of interest payable is refixed in line with market conditions at regular intervals (usually six months).

Floating-rate gilt Gilt issued with an interest rate adjusted periodically in line with market interbank rates.

Floating-rate index An index like Libor against which a floating-rate asset or liability is linked, for the purposes of calculating interest due.

Floating-rate note **Capital market** instrument on which the rate of interest payable is refixed in line with market conditions at regular intervals (usually six months).

Floor A series of lender's **IRG**s, designed to protect an investor against falling interest rates on each of a series of dates.

Floortion Option on a **floor**.

Forward In general, a deal for value later than the normal value date for that particular commodity or instrument. In the foreign-exchange market, a forward price is the price quoted for the purchase or sale of one currency against another where the value date is at least one month after the **spot** date. See **short date**.

Forward band Zero-cost collar; that is, one in which the premium payable as a result of buying the **cap** is offset exactly by that obtained from selling the **floor**.

Forward break See **break forward**.

Forward-exchange agreement A **contract for differences** designed to create exactly the same economic result as a foreign-exchange cash **forward-forward** deal. See **ERA**, **SAFE**.

Forward-forward Short-term exchange of currency deposits. (See also forward-forward deposit).

Forward-forward yield curve A yield curve of zero-coupon rates for periods starting at a future point, say one month or one year from today.

Forward-rate agreement (FRA) Short-term interest-rate **hedge**. Specifically, a contract between buyer and seller for an agreed interest rate on a notional deposit of a specified maturity on a predetermined future date. No principal is exchanged. At maturity, the seller pays the buyer the difference if rates have risen above the agreed level, and vice versa.

Forward-rate swap An interest-rate swap where the pay-fixed leg is of the relevant tenor forward rate.

Forward swap **Swap** arranged at the current rate but entered into at some time in the future.

FRA See **forward-rate agreement**.

FRCD See **floating-rate CD**.

FRN See **floating-rate note**.

Framework document Sets out the DMO's responsibilities, objectives and targets, its relationship with the rest of the Treasury, and its accountability as an executive agency.

Fraption Option on a forward-rate agreement. Also known as an interest-rate guarantee.

FSA The Financial Services Authority, the body responsible for the regulation of investment business, and the supervision of banks and money-market institutions in the UK. The FSA took over these duties from nine "self-regulatory organisations" that had previously carried out this function, including the Securities and Futures Authority (SFA), which had been responsible for regulation of professional investment business in the City of London. The FSA commenced its duties in 1998.

FTSE-100 Index comprising 100 major UK shares listed on The International Stock Exchange in London. Futures and options on the index are traded at the London International Financial Futures and Options Exchange (**LIFFE**).

Funding reserve A specified (say, 10 bps) multiple of the aggregate value of the funding gap, across the maturity structure.

Fungible A financial instrument that is equivalent in value to another, and easily exchanged or substituted. The best example is cash money, as a £10 note has the same value and is directly exchangeable with another £10 note. A bearer bond also has this quality.

Future A futures contract is a contract to buy or sell securities or other goods at a future date at a predetermined price. Futures contracts are usually standardised and traded on an exchange.

Future exposure See **potential exposure**.

Future value The amount of money achieved in the future, including interest, by investing a given amount of money now. See **time value of money**, **present value**.

Futures contract A deal to buy or sell some financial instrument or commodity for value on a future date. Unlike a **forward** deal, futures contracts are traded only on an exchange (rather than **OTC**), have standardised contract sizes and value dates, and are often only **contract for differences** rather than deliverable.

FXA See **forward-exchange agreement**.

G

G7 The "Group of Seven" countries the United States, Canada, the UK, Germany, France, Italy, and Japan.

Gamma (Γ) The change in an option's delta relative to a change in the underlying's value.

Gap The difference in the maturity profile of assets versus liabilities by time bucket.

Gapping Feature of commodity markets whereby there are large and very rapid price movements to new levels followed by relatively stable prices.

Gap ratio Ratio of interest-rate-sensitive assets to interest-rate-sensitive liabilities; used to determine changes in the risk profile of an institution with changes in interest-rate levels.

GDP Gross domestic product, the value of total output produced within a country's borders.

GEMM A gilt-edged market maker, a bank or securities house registered with the Bank of England as a market maker in gilts. A GEMM is required to meet certain obligations as part of its function as a registered market maker, including making two-way price quotes at all times in all gilts and taking part in gilt auctions. The Debt Management Office now makes a distinction between conventional gilt GEMMs and index-linked GEMMs, known as IG GEMMs.

General collateral (GC) Securities, which are not "special", used as collateral against cash borrowing. A repo buyer will accept GC at any time that a specific stock is not quoted as required in the transaction. In the gilts market, GC includes DBVs.

GIC Guaranteed investment contract.

Gilt A UK Government sterling-denominated, listed security issued by HM Treasury with initial maturity of over 365 days when issued. The term "gilt" (or "gilt-edged") is a reference to the primary characteristic of gilts as an investment: their security.

Gilt-edged market-maker See **GEMM**.

GMRA Global Master Repurchase Agreement, the industry-standard legal agreement describing repo transactions. Issued under the auspices of the Bond Market Association in the United States and the International Securities Market Association (**ISMA**).

GNP Gross national product, the total monetary value of a country's output, as produced by citizens of that country.

Gold warrant Naked or attached warrant exercisable into gold at a predetermined price.

Gross-redemption yield The same as **yield to maturity;** "gross" because it does not take tax effects into account.

GRY See **gross-redemption yield**.

H

Haircut The value of collateral required over and above the loan value in a repo, required by the lender of cash. Also called **margin**.

Hedge To mitigate or neutralise risk exposure.

Hedge ratio The ratio of the size of the position it is necessary to take in a particular instrument as a hedge against another, to the size of the position being hedged.

Hedging Protecting against the risks arising from potential market movements in exchange rates, interest rates, or other variables. See **cover**, **arbitrage**, **speculation**.

Herstatt risk See **settlement risk**.

High-coupon swap Off-market coupon swap where the coupon is higher than the market rate. The floating-rate payer pays a front-end fee as compensation. Opposite of low-coupon swap.

Historical simulation methodology Method of calculating **value-at-risk** (**VAR**) using historical data to assess the likely effect of market moves on a portfolio.

Historic rate rollover A **forward-rate swap** in FX where the settlement exchange rate for the near date is based on a historic **off-market** rate rather than the current market rate. This is prohibited by many central banks.

Historic volatility The actual **volatility** of an instrument or asset, recorded in market prices over a particular period in the past.

Holder The holder of an **option** is the party that has purchased it.

I

IDB Interdealer broker; in this context a broker that provides facilities for dealing in bonds between market makers.

IG Index-linked gilt whose coupons and final redemption payment are related to the movements in the Retail Price Index (RPI).

Immunisation This is the process by which a bond portfolio is created that has an assured return for a specific time horizon irrespective of changes in interest rates. The mechanism underlying immunisation is a portfolio structure that balances the change in the value of a portfolio at the end of the investment horizon (time period) with the return gained from the reinvestment of cash flows from the portfolio. As such, immunisation requires the portfolio manager to offset interest-rate risk and reinvestment risk.

Implied repo rate The break-even interest rate at which it is possible to sell a bond **futures contract**, buy a **deliverable bond**, and **repo** the bond out. See **cash-and-carry**.

Implied volatility The **volatility** used by a dealer to calculate an **option** price; conversely, the volatility implied by the price actually quoted.

Index A reference group of stocks, bonds, or interest rates against which an asset value may be linked.

Indexed notes Contract whereby the issuer usually assumes the risk of unfavourable price movements in the instrument, commodity, or index to which the contract is linked, in exchange for which the issuer can reduce the cost of borrowing (compared with traditional instruments without the risk exposure).

Index option An **option** whose **underlying** security is an index. Index options enable a trader to bet on the direction of the index.

Index swap Sometimes the same as a **basis swap**. Otherwise, a swap like an **interest-rate swap** where payments on one or both of the legs are based on the value of an index—such as an equity index, for example.

Indirect An exchange-rate quotation against the U.S. dollar in which the dollar is the **base currency** and the other currency is the **variable currency**.

Initial margin The excess either of cash over the value of securities, or of the value of securities over cash in a repo transaction at the time it is executed and, subsequently, after margin calls.

Interbank The market in unsecured lending and trading between banks of roughly similar credit quality.

Interest-rate cap See **cap**.

Interest-rate floor See **floor**.

Interest-rate guarantee An **option** on a **forward-rate agreement (FRA)**.

Interest-rate option Option to pay or receive a specified rate of interest on or from a predetermined future date.

Interest-rate swap An agreement to exchange a series of cash flows determined in one currency, based on fixed- or **floating**-interest payments on an agreed **notional** principal, for a series of cash flows based in the same currency but on a different interest rate. May be combined with a bond to form an asset swap.

Intermarket spread A spread involving futures contracts in one market spread against futures contracts in another market.

Internal rate of return The yield necessary to discount a series of cash flows to an **NPV** of zero.

Interpolation The process of estimating a price or rate for value on a particular date by comparing the prices actually quoted for value dates either side. See **extrapolation**.

Intervention Purchases or sales of currencies in the market by central banks in an attempt to reduce exchange-rate fluctuations or to maintain the value of a currency within a particular band, or at a particular level. Similarly, central bank operations in the money markets to maintain interest rates at a certain level.

In-the-money A **call** (**put**) **option** is in-the-money if the underlying is currently more (less) valuable than the **strike** price. See **at-the-money**, **out-of-the-money**.

Intrinsic value The amount by which an option is in-the-money.

IRG See **interest-rate guarantee**.

IRR See **internal rate of return**.

IRS See **interest-rate swap**.

ISMA The International Securities Market Association. This association compiled, with the PSA (now renamed the Bond Market Association), the PSA/ISMA Global Master Repurchase Agreement.

Issuer risk Risk to an institution when it holds debt securities issued by another institution. (See also **credit risk**.)

Iteration The mathematical process of estimating the answer to a problem by seeing how well an estimate fits the data, adjusting the estimate appropriately and trying again, until the answer is close to the actual. Used, for example, in calculating a bond's **yield** from its price.

J

Junk bonds The common term for high-yield bonds; higher-risk, low-rated debt.

K

Kappa (κ) An alternative term to refer to volatility; see **vega**.

Kick-in note An index-linked hybrid bond whose enhanced return is triggered if the index reaches a certain level above or below where it is when the note is issued.

Knock in/out A knock out/in **option** ceases to exist (starts to exist) if the underlying reaches a certain trigger level. See **barrier option**.

L

Lambda (λ) The same as **vega**.

Large exposure A risk exposure to a bank caused by having a large part of lending made to one counterparty. Under EU CAD, an extra risk number must be allocated for this risk.

Lender option Floor on a single-period forward-rate agreement.

Leptokurtosis The non-normal distribution of asset-price returns. Refers to a probability distribution that has a fatter tail and a sharper hump than a normal distribution.

Level payment swap Evens out those fixed-rate payments that would otherwise vary; for example, because of the amortisation of the principal.

Leverage The ability to control large amounts of an underlying variable for a small initial investment.

Liability Probable future sacrifice of economic benefit due to present obligations to transfer assets or provide services to other entities as a result of past events or transactions. Generally classed as either current or long-term.

Liability swap An interest-rate swap or currency swap used in conjunction with an underlying liability such as a borrowing. See **asset swap**.

LIBID The London Interbank Bid Rate, the rate at which banks will pay for funds in the interbank market. See **Libor**.

Libor The London Interbank Offered Rate, the lending rate for all major currencies up to one-year maturity set at 1100 hours each day by the British Bankers' Association.

Libor fixing The Libor rate "fixed" by the British Bankers' Association (BBA) at 1100 hours each day, for maturities up to one year.

LIFFE The London International Financial Futures and Options Exchange, the largest futures exchange in Europe.

LIMEAN The arithmetic average of Libor and Libid rates.

Limit up/down **Futures** prices are generally not allowed to change by more than a specified total amount in a specified time, in order to control risk in very volatile conditions. The maximum movements permitted are referred to as limit up and limit down.

Liquidation Any transaction that closes out or offsets a futures or options position.

Liquidity A word describing the ease with which one can undertake transactions in a particular market or instrument. A market where there are always ready buyers and sellers willing to transact at competitive prices is regarded as liquid. In banking, the term is also used to describe the requirement that a portion of a bank's assets be held in short-term risk-free instruments, such as government bonds, T-bills, and high-quality certificates of deposit.

Loan-equivalent amount Description of derivative exposure that is used to compare the credit risk of derivatives with that of traditional bonds or bank loans.

Lognormal A variable's **probability distribution** is lognormal if the logarithm of the variable has a normal distribution.

Lognormal distribution The assumption that the log of today's interest rate, for example, minus the log of yesterday's rate is normally distributed.

Long A long position is a surplus of purchases over sales of a given currency or asset, or a situation that naturally gives rise to an organisation benefitting from a strengthening of that currency or asset. To a money-market dealer, however, a long position is a surplus of borrowings taken in over money lent out, (which gives rise to a benefit if that currency weakens rather than strengthens). See **short**.

Long-dated forward Forward foreign-exchange contract with a maturity of greater than one year. Some long-dated forwards have maturities as great as 10 years.

Long-term assets Assets that are expected to provide benefits and services over a period longer than one year.

Long-term liabilities Obligations to be repaid by the firm more than one year later.

Lookback option Option that allows the purchaser, at the end of a given period of time, to choose as the rate for exercise any rate that has existed during the option's life.

Low-coupon swap Tax-driven swap, in which the fixed-rate payments are significantly lower than current market interest rates. The floating-rate payer is compensated by a front-end fee.

LSE London Stock Exchange.

LTV Loan-to-value, the ratio of the loan amount over the value of the asset. A lending risk ratio calculated by dividing the total amount of the mortgage or loan by the appraised value of the asset.

M

Macaulay duration See **duration**.

Manufactured dividend A payment from the repo buyer to the repo seller during the term of the trade, representing the coupon or dividend received by the temporary owner (repo buyer) of the security being repo'd. Also applies in a stock loan transaction.

Mapping The process whereby a treasury's derivative positions are related to a set of risk "buckets".

Margin Initial margin is **collateral**, placed by one party with a counterparty at the time of the deal, against the possibility that the market price will move against the first party, thereby leaving the counterparty with a credit risk. Variation margin is a payment or extra collateral transferred subsequently from one party to the other because the market price has moved. Variation margin payment is either, in effect, a settlement of profit/loss (for example, in the case of a **futures** contract) or the reduction of credit exposure (for example, in the case of a **repo**). In gilt **repos**, variation margin refers to the fluctuation band or threshold within which the existing collateral's value may vary before further cash or **collateral** needs to be transferred. In a loan, margin is the extra interest above a **benchmark** (for example, a margin of 0.5 percent over **Libor**) required by a lender to compensate for the credit risk of that particular borrower.

Margin call A request following marking-to-market of a repo transaction for the initial margin to be reinstated or, where no initial margin has been taken, to restore the cash/securities ratio to parity.

Margin default rate See **probability of default**.

Margin transfer The payment of a **margin call**.

Market comparables Technique for estimating the fair value of an instrument for which no price is quoted by comparing it with the quoted prices of similar instruments.
Market maker Market participant who is committed, explicitly or otherwise, to quoting two-way bid-and-offer prices at all times in a particular market.
Market risk Risks related to changes in prices of tradable macroeconomic variables, such as exchange-rate risks.
Mark-to-market The act of revaluing securities to current market values. Such revaluations should include both coupon accrued on the securities outstanding and interest accrued on the cash.
Matched book This refers to the matching by a repo trader of securities repoed in and out. It carries no implications that the trader's position is "matched" in terms of exposure, for example, to short-term interest rates.
Maturity date Date on which stock is redeemed.
Mean Average.
Minmax option One of the strategies for reducing the cost of options by forgoing some of the potential for gain. The buyer of a currency option, for example, simultaneously sells an option on the same amount of currency but at a different strike price.
MLV Maximum likely potential increase in value.
Modified duration A measure of the proportional change in the price of a bond or other series of cash flows, relative to a change in yield. Mathematically:

$$\frac{dP}{di} / \textbf{dirty price}$$

See **duration**.
Modified following The convention that if a value date in the future falls on a nonbusiness day, the value date will be moved to the next following business day, unless this moves the value date to the next month; in which case, the value date is moved back to the last previous business day.
Momentum The strength behind an upward or downward movement in price.
Money market Short-term market (generally up to one year) for financial instruments. See **capital market**.
Money-market basis An interest rate quoted on an ACT/360 basis is said to be on a money-market basis. See **bond basis**.
Monte Carlo simulation Technique used to determine the likely value of a derivative or other contract by simulating the evolution of the underlying variables many times. The discounted **average** outcome of the simulation gives an approximation of the derivative's value. Monte Carlo simulation can be used to estimate the **value-at-risk** (**VaR**) of a portfolio. Here, it generates a simulation of many correlated market movements for the markets to which the portfolio is exposed, and the positions in the portfolio are revalued repeatedly in accordance with the simulated scenarios. This gives a probability distribution of portfolio gains and losses from which the **VaR** can be determined.
Moosmüller A method for calculating the yield of a bond.
Mortgage-backed security (MBS) Security guaranteed by pool of mortgages. MBS markets include the United States, UK, Japan, and Denmark.
Moving average convergence/divergence (MACD) The crossing of two exponentially smoothed moving averages that oscillate above and below an equilibrium line.
Multi-currency curve A yield curve that plots rates for more than one currency, after equalising for the forward FX basis differential.
Multi-index option **Option** which gives the holder the right to buy the asset that performs best out of a number of assets (usually two). The investor would typically buy a call allowing him to buy the **equity**.

N

Naked A naked **option** position is one not protected by an offsetting position in the **underlying**. See **covered call/put**.

Negative divergence When at least two indicators, indexes, or averages show conflicting or contradictory trends.
Negotiable A security that can be bought and sold in a **secondary market** is negotiable.
Net present value The net present value of a series of cash flows is the sum of the present values of each cash flow (some or all of which may be negative).
NLF National Loans Fund, the account which brings together all UK government lending and borrowing.
Noise Fluctuations in the market that can confuse or impede interpretation of market direction.
Nominal amount Same as **face value** of a security.
Nominal rate The quoted interest rate, rather than the **effective rate** to which it is equivalent.
Nondeliverable forward A forward FX contract that does not result in exchange of actual cash currency amounts on maturity, but instead has a single net payment representing the change between the traded forward rate and the spot rate on maturity.
Normal A normal **probability distribution** is a particular distribution assumed to prevail in a wide variety of circumstances, including the financial markets. Mathematically, it corresponds to the probability density function:

$$\frac{1}{\sqrt{2p}} e^{-\frac{1^{\sigma^2}}{2}}$$

Notional In a bond futures contract, the bond bought or sold is a standardised nonexistent notional bond, as opposed to the actual bonds which are **deliverable** at maturity. **Contracts for differences** also require a notional principal amount on which settlement can be calculated.
Novation Replacement of a contract or, more usually, a series of contracts with one new contract.
NPV See **net present value**.

O

O/N See **overnight**.
Odd date See **broken date**.
Off-balance-sheet A transaction whose nominal value is not entered on the balance sheet, because the principal amount is not traded. The standard accounting treatment for "contracts for differences".
Offer The price at which a market-maker will sell bonds. Also called **ask**.
Off-market A rate that is not the current market rate.
Off-market coupon swap Tax-driven swap strategy, in which the fixed-rate payments differ significantly from current market rates. There are high-and low-coupon swaps.
One-touch all-or-nothing Digital option. The option's put pays out a predetermined amount (the "all") if the index goes below (above) the strike price at any time during the option's life. How far below (above) the strike price the strike price the index moves is irrelevant; the payout will be the "all" or nothing.
Opening leg The first half of a repo transaction (see **closing leg**).
Open interest The quantity of **futures** contracts (of a particular specification) that have not yet been closed out by reversing. Either all **long** positions or all **short** positions are counted but not both.
Operational Market Notice Sets out the DMO's (previously the bank's) operations and procedures in the gilt market.
Operational risk Risk of loss occurring due to inadequate systems and control, human error, or management failure.
Opportunity cost Value of an action that could have been taken if the current action had not been chosen.
Option The right (but not the obligation) to buy or sell securities at a fixed price within a specified period.

Option forward See **time option**.

Ornstein-Uhlenbeck equation A standard equation that describes mean reversion. It can be used to characterise and measure commodity price behaviour.

OTC Over the counter. Strictly speaking, any transaction not conducted on a registered stock exchange. Trades conducted via the telephone between banks, and contracts such as FRAs and (non-exchange-traded) options are said to be over-the-counter instruments. OTC also refers to nonstandard instruments or contracts traded between two parties; for example, a client with a requirement for a specific risk to be hedged with a tailor-made instrument may enter into an OTC structured-option trade with a bank that makes markets in such products.

Out-of-the-money A **call** (**put**) **option** is out-of-the-money if the **underlying** is currently less (more) valuable than the strike price. See **at-the-money**, **in-the-money**.

Outright An outright (or **forward** outright) is the sale or purchase of one foreign currency against another value on any date other than spot. See **spot**, **swap**, **forward**, **short date**.

Overborrowed A position in which a dealer's liabilities (borrowings taken in) are of longer maturity than the assets (loans out).

Over the counter An OTC transaction is one dealt privately between any two parties, with all details agreed between them, as opposed to one dealt on an exchange—for example, a **forward** deal as opposed to a **futures contract**.

Overlent A position in which a dealer's assets (loans out) are of longer maturity than the liabilities (borrowings taken in).

Overnight A deal from today until the next working day ("tomorrow").

P

Paper Another term for a bond or debt issue.

Par The face value, or 100 percent. In foreign exchange, when the **outright** and **spot** exchange rates are equal, the **forward swap** is zero or par. When the price of a security is equal to the face value, usually expressed as 100, it is said to be trading at par. A par-swap rate is the current market rate for a fixed **interest-rate swap** against **Libor**.

Parity The official rate of exchange for one currency in terms of another, which a government is obliged to maintain by means of intervention.

Participation forward A product equivalent to a straightforward **option** plus a **forward** deal, but structured as a forward deal at an **off-market** rate plus the opportunity to benefit partially if the market rate improves.

Par yield curve A curve plotting maturity against **yield** for bonds priced at par.

Path-dependent A path-dependent **option** is one that depends on what happens to the **underlying** throughout the option's life (such as the **American** or **barrier** option) rather than only at expiry (a **European** option).

Peak exposure If the worst-case or the expected credit-risk exposures of an instrument are calculated over time, the resulting graph reveals a credit-risk exposure profile. The highest exposure marked out by the profile is the peak exposure generated by the instrument.

Periodic-resetting swap Swap where the floating-rate payment is an average of floating rates that have prevailed since the last payment, rather than the interest rate prevailing at the end of the period. For example, the average of six one-month **Libor** rates rather than one six-month Libor rate.

Pips See **points**.

Plain vanilla See **vanilla**.

Points The last two decimal places in an exchange rate. For example, when EUR/USD is 1.1910/1.1920, the points are 10/20. See **bid figure**.

Portfolio variance The square of the **standard deviation** of a portfolio's return from the mean.

Positive cash-flow collar Collar other than a zero-cost collar.

Potential exposure Estimate of the future replacement cost, or positive market value, of a derivative transaction. Potential exposure should be calculated using probability analysis based on broad confidence intervals (for example, two standard deviations) over the remaining term of the transaction.

Preference shares These are a form of corporate financing. They are normally fixed-interest shares whose holders have the right to receive dividends ahead of ordinary shareholders. If a company were to go into liquidation, preference shareholders would rank above ordinary shareholders for the repayment of their investment in the company. Preference shares ("prefs") are normally traded within the fixed-interest division of a bank or securities house.

Premium For a bond, the amount by which the price is over par. In the FX market, the amount by which a currency is more expensive, in terms of another currency, for future delivery than for spot, is the forward premium (in general, a reflection of interest-rate differentials between two currencies). If an exchange rate is "at a premium" (without specifying to which of the two currencies this refers), this generally means that the **variable currency** is at a premium. See **discount**.

Present value The amount of money that needs to be invested now to achieve a given amount in the future when interest is added. See **time value of money**, **future value**.

Presettlement risk As distinct from credit risk arising from intraday settlement risk, this term describes the risk of loss that might be suffered during the life of the contract if a counterparty to a trade defaulted and if, at the time of default, the instrument had a positive economic value.

Price-earnings ratio A ratio giving the price of a stock relative to the earnings per share.

Price factor See **conversion factor**.

Primary market The market for new debt, into which new bonds are issued. The primary market is made up of borrowers, investors, and the investment banks that place new debt into the market, usually with their clients. Bonds that trade after they have been issued are said to be part of the secondary market.

Probability distribution The mathematical description of how probable it is that the value of something is less than or equal to a particular level.

Probability of default The probability that an obligor will default during the next 12 months. A default probability calculated over a longer period of time than one year is a cumulative probability of default.

Probit procedures Methods for analysing qualitative dependent methods where the dependent variable is binary, taking the values zero or one.

Put A put option is an option to sell the commodity or instrument **underlying** the option. See **call** option.

Put-call parity The theory that demonstrates the relationship between the call price and put price of an option with otherwise identical terms.

PVB Price value of a basis point, the change in value of a bond or derivative contract resulting from a one-basis-point change in its yield, or in the level of interest rates.

Q

Quanto An option that has its final payoff linked to two or more underlying assets or reference rates.

Quanto swap A **swap** where the payments in one or both legs are based on a measurement (such as the interest rate) in one currency but payable in another currency.

Quasi-coupon date The regular date for which a **coupon** payment would be scheduled if there was a coupon payable. Used for price/yield calculations for **zero-coupon** bonds.

R

Range forward A **zero-cost collar** where the customer is obliged to deal with the same bank at spot if neither limit of the collar is breached at **expiry**.

Rate of recovery Estimate of the percentage of the amount exposed to default—that is, the credit-risk exposure—that is likely to be recovered by an institution if a counterparty defaults. The recovery value of a deposited asset is dependent on its rate of recovery.

Record date A **coupon** or other payment due on a security is paid by the issuer to whoever is registered on the record date as being the owner. See **ex-dividend (xd) date, cum-dividend**.

Redeem A security is said to be redeemed when the principal is repaid.

Redemptions When a borrower redeems all or part of the loan it has taken out.

Redemption yield The rate of interest at which all future payments (coupons and redemption) on a bond are discounted so that their total equals the current price of the bond (inversely related to price).

Redenomination A change in the currency unit in which the nominal value of a security is expressed (in context, from sterling to euro).

Reduced-cost option Generic term for options for which there is a reduced premium, either because the buyer undertakes to forgo a percentage of any gain, or because he offsets the cost by writing other options (for example, minmax, **range forward**). See also **zero-cost option**.

Refer The practice whereby a trader instructs a broker to put "under reference" any prices or rates he has quoted to him, meaning that they are no longer "firm" and the broker must refer to the trader before he can trade on the price initially quoted.

Register Record of ownership of securities. For gilts, excluding bearer bonds, entry in an official register confers title.

Registered bond A bond for which the issuer keeps a record (register) of its owners. Transfer of ownership must be notified and recorded in the register. Interest payments are posted (more usually electronically transferred) to the bondholder.

Registrar's Department Department of the Bank of England that maintains the register of holdings of gilts.

Reinvestment rate The rate at which interest paid during the life of an investment is reinvested to earn interest-on-interest, which, in practice, will generally not be the same as the original yield quoted on the investment.

Relative-performance option Option whose value varies in line with the relative value of two assets.

Replacement cost The present value of the expected future net cash flows of a derivative instrument. Aside from various conventions dealing with the **bid/ask** spread, synonymous with the "market value" or "current exposure" of an instrument.

Repo Usually refers in particular to **classic repo**. Also used as a term to include classic repos, **buy/sell-backs**, and **securities lending**.

Repo rate The return earned on a repo transaction expressed as an interest rate on the cash side of the transaction.

Repurchase agreement See **repo**.

Return on assets The net earnings of a company divided by its assets.

Return on equity The net earnings, of a company divided by its equity.

Return on value-at-risk (ROVAR) An analysis conducted to determine the relative rates of return on different risks, allowing corporations to compare different risk-capital allocations and capital-structure decisions effectively.

Revaluation An official one-off increase in the value of a currency in terms of other currencies. See **devaluation**.

Revenue volatility Any earnings stream that is subject to volatile or non-stable fluctuations.

Reverse See **reverse repo**.

Reverse repo The opposite of a **repo**.

Rho (ρ) The change in an option's value relative to a change in interest rates.

Risk-free rate The interest rate payable on an investment that carries zero credit risk. Usually associated with the 90-day Treasury bill rate.

Risk reversal Changing a long (or short) position in a call option to the same position in a put option by selling (or buying) forward, and vice versa.

Risk-weighted asset Assets that carry an element of credit risk and so must be weighted in accordance with relative risk, for capital adequacy purposes under Basel regulations.

Risk weighting The risk-adjusted exposure of an asset, undertaken for regulatory capital calculation. A credit risk-free asset has a 0% risk weighting.

Rollover See **tom/next**. Also refers to a renewal of a loan.

Rump A gilt issue so designated because it is illiquid, generally because there is a very small nominal amount left in existence.

Running yield Same as **current yield**.

S

SAFE See **synthetic agreement for forward exchange.**

Sale and repurchase agreement The full name for repo.

Secondary market The market in instruments after they have been issued. Bonds are bought and sold after their initial issue by the borrower, and the marketplace for this buying and selling is referred to as the secondary market. The new-issues market is the primary market.

Securities and Exchange Commission The central regulatory authority in the United States, responsible for policing the financial markets, including the bond markets.

Securities lending When a specific security is lent against some form of collateral. Also known as stock lending.

Security A financial asset sold initially for cash by a borrowing organisation (the "issuer"). The security is often negotiable and usually has a maturity date when it is redeemed.

Sell/buy-back Simultaneous spot sale and forward purchase of a security, with the forward price calculated to achieve an effect equivalent to a classic repo.

Settlement The process of transferring stock from seller to buyer and arranging the corresponding movement of funds between the two parties.

Settlement bank Bank that agrees to receive and make assured payments for gilts bought and sold by a CGO member.

Settlement date Date on which transfer of gilts and payment occur, usually the next working date after the trade is conducted.

Settlement risk The risk that occurs when there is a nonsimultaneous exchange of value. Also known as delivery risk and Herstatt risk.

Sharpe ratio A measure of the attractiveness of the return on an asset by comparing how much risk premium the investor can expect it to receive in return for the incremental risk (volatility) the investment carries. It is the ratio of the risk premium to the volatility of the asset.

Short A short position is a surplus of sales over purchases of a given currency or asset, or a situation that naturally gives rise to an organisation benefiting from a weakening of that currency or asset. To a money-market dealer, however, a short position is a surplus of money lent out over borrowings taken in (which give rise to a benefit if that currency strengthens rather than weakens). See **long**.

Short date A deal for value on a date other than spot but less than one month after spot.

Simple interest When interest on an investment is paid all at maturity or not reinvested to earn interest on interest, the interest is said to be simple. See **compound interest**.

Simple yield to maturity Bond coupon plus principal gain/loss amortised over the time to maturity, as a proportion of the clean price per 100. Does not take time value of money into account. See **yield to maturity**, **current yield**.

SLA Service Level Agreement, in this context between the DMO and service suppliers in the Bank and HM Treasury.

S/N See **spot/next**.
Special A security that for any reason is sought after in the repo market, thereby enabling any holder of the security to earn incremental income (in excess of the **general collateral** rate) through lending them via a repo transaction. The repo rate for a special will be below the GC rate, as this is the rate the borrower of the cash is paying in returning for supplying the special bond as collateral. An individual security can be in high demand for a variety of reasons; for instance, if there is sudden heavy investor demand for it, or (if it is a benchmark issue) it is required as a hedge against a new issue of similar-maturity paper.
Speculation A deal undertaken because the dealer expects prices to move in his favour, as opposed to **hedging** or **arbitrage**.
Spot A deal to be settled on the customary value date for that particular market. In the foreign-exchange market, this is for value in two working days' time.
Spot yield curve The current zero-coupon yield curve.
Spot/next A transaction from **spot** until the next working day.
Spread The difference between the bid and offer prices in a quotation. Also a strategy involving the purchase of an instrument and the simultaneous sale of a similar related instrument, such as the purchase of a **call option** at one **strike** and the sale of a call option at a different strike.
Square A position in which sales exactly match purchases, or in which assets exactly match liabilities. See **long**, **short**.
Standard deviation (σ) A measure of how much the values of something fluctuate around its mean value. Defined as the square root of the **variance**.
Step-down swap Swap in which the fixed-rate payment decreases over the life of the swap.
Step-up swap Swap in which the fixed-rate payment increases over the life of the swap.
Stock-index future Future on a stock index, allowing a hedge against, or bet on, a broad equity-market movement.
Stock-index option Option on a stock-index future.
Stock lending See **securities lending**.
Stock option Option on an individual stock.
Straddle A position combining the purchase of both a call and put at the same strike for the same date. See **strangle**.
Strangle A position combining the purchase of both a call and a put at different strikes for the same date. See **straddle**.
Street The "street" is a term for the market, deriving originally from "Wall Street". A U.S. term for market convention; so, in the U.S. market is the convention for quoting the price or yield for a particular instrument.
Stress testing Analysis that gives the value of a portfolio under a range of **worst-case** scenarios.
Strike The strike price or strike rate of an option is the price or rate at which the holder can insist on the underlying transaction being fulfilled.
Strip A zero-coupon bond that is produced by separating a standard coupon-bearing bond into its constituent principal and interest components. To strip a bond is to separate its principal amount and its coupons, and then trade each individual cash flow as a separate instrument ("**s**eparately **t**raded and **r**egistered for **i**nterest and **p**rincipal"). Also, a strip of **futures** is a series of short-term futures contracts with consecutive delivery dates, which together create the effect of a longer-term instrument (for example, four consecutive three-month futures contracts as a **hedge** against a one-year swap). A strip of **FRA**s is similar.
Swap A foreign-exchange swap is the purchase of one currency against another for delivery on one date, with a simultaneous sale to reverse the transaction on another value date. See also **interest-rate swap**, **currency swap**.
Swaption An **option** on an **interest-rate swap**, **currency swap**.
Switch Exchanges of one gilt holding for another, sometimes entered into between the DMO and a GEMM as part of the DMO's secondary-market operations.

Synthetic A package of transactions that is economically equivalent to a different transaction (for example, the purchase of a **call option** and simultaneous sale of a **put** option at the same **strike** is a synthetic **forward** purchase).

Synthetic agreement for forward exchange A generic term for **ERA**s and **FXA**s.

T

Tail The difference in maturity between assets and liabilities (also known as "gap" or interest rate tail). In FX markets, the exposure to interest rates over a forward-forward period arising from a mismatched position (such as a two-month borrowing against a three-month loan). A forward foreign-exchange dealer's exposure to **spot** movements.

Tap The issue of a gilt for exceptional market-management reasons and not on a pre-announced schedule.

TED spread A term referring to the spread in a trade involving a long/short futures position against a short/long government bond position. Also the futures strip hedge page on Bloomberg. Originally referred to as "Treasury-Eurodollar spread".

Term The time between the beginning and end of a deal or investment.

Theta (θ) The change in an option's value relative to a change in the time left to expiry.

Tick The minimum change allowed in a futures price.

Tick value The change in value of a futures contract for a one-tick movement in price.

Time bucket The maturity group into which a loan or other exposure is placed. For instance, time buckets of o/n, o/n—1 week, 1 week —3 month, 3 month—6 month, and 6 month—12 month may be calculated, and assets and liabilities placed in buckets according to their maturity.

Time deposit A **non-negotiable** deposit for a specific term.

Time option A forward currency deal in which the value date is set to be within a period rather than on a particular day. The customer sets the exact date two working days before settlement.

Time value of money The concept that a future cash flow can be valued as the amount of money that it is necessary to invest now in order to achieve that cash flow in the future. See **present value**, **future value**.

T/N See **tom/next.**

Today/tomorrow See **overnight**.

Tom/next A transaction from the next working day ("tomorrow") until the day after ("next day"—that is, **spot** in the foreign-exchange market).

Total-return swap Swap agreement in which the total return of bank loans or credit-sensitive securities is exchanged for some other cash flow usually tied to **Libor**, or other loans, or credit-sensitive securities. It allows participants to effectively go **long** or **short** the credit risk of the **underlying** asset.

Traded option Option that is listed on and cleared by an exchange, with standard terms and delivery months.

Trading book A bank's investment, trading, and short-term activity, grouped into the trading book for regulatory capital purposes.

Tranche One of a series of two or more issues with the same coupon rate and maturity date. The tranches become fungible at a future date, usually just after the first coupon date.

Transaction risk Extent to which the value of transactions that have already been agreed is affected by market risk.

Transparent A term used to refer to how clear asset prices are in a market. A transparent market is one in which a majority of market participants are aware of what level a particular bond or instrument is trading.

Trigger option See **barrier option**.

Treasury bill A short-term security issued by a government, generally with a zero **coupon**.

True yield The yield on a bond that is equivalent to the quoted discount or zero-coupon rate.

Tunnel The same as **collar**.

Tunnel options Set of collars, typically zero-cost, covering a series of maturities from the current date. They might, for example, be for dates 6, 12, or 24 months ahead. The special feature of a tunnel is that strike price on both sets of options, not just on the options bought, is constant.

U

Uncovered option When the writer of the option does not own the underlying security. Also known as a **naked option**.

Undated gilts Gilts for which there is no final date by which the gilt must be redeemed.

Underlying The underlying of a futures or option contract is the commodity of financial instrument on which the contract depends. Thus, underlying for a bond option is the bond; the underlying for a short-term interest-rate futures contract is typically a three-month deposit.

Underwriting An arrangement by which a company is guaranteed that an issue of debt (bonds) will raise a given amount of cash. Underwriting is carried out by investment banks, who undertake to purchase any part of the debt issue not taken up by the public. A commission is charged for this service.

Unexpected-default rate The distribution of future default rates is often characterised in terms of an expected-default rate (for example, 0.05%) and a worst-case default rate (for example, 1.05%). The difference between the worst-case default rate and the expected-default rate is often termed the "unexpected default" (that is, 1% = 1.05 – 0.05%).

Unexpected loss The distribution of credit losses associated with a derivative instrument is often characterised in terms of an unexpected loss or a **worst-case loss**. The unexpected loss associated with an instrument is the difference between these two measures.

Up-and-away option See **up-and-out option**.

Up-and-in option Type of barrier option that is activated if the value of the underlying goes above a predetermined level. See also **down-and-in option**.

Up-and-out option Type of barrier option that is extinguished if the value of the underlying goes above a predetermined level. See also **down-and-out option**.

V

Value-at-risk (VaR) Formally, the probabilistic bound of market losses over a given period of time (known as the holding period) expressed in terms of a specified degree of certainty (known as the confidence interval). Put more simply, the VaR is the worst-case loss that would be expected over the holding period within the probability set out by the confidence interval. Larger losses are possible but with a low probability. For instance, a portfolio whose VaR is $20 million over a one-day holding period, with a 95 percent confidence interval, would have only a 5 percent chance of suffering an overnight loss greater than $20 million.

Value date The date on which a deal is to be consummated. In some bond markets, the value date for coupon accruals can sometimes differ from the settlement date.

Vanilla A vanilla transaction is a straightforward one.

VaR See **value-at-risk.**

Variable currency Exchange rates are quoted in terms of the number of units of one currency (the variable or counter currency) that corresponds to one unit of the other currency (the **base currency**).

Variance (σ^2) A measure of how much the values of something fluctuate around its mean value. Defined as the average of (value – mean)^2. See **standard deviation**.

Variance-covariance methodology Methodology for calculating the **value-at-risk** of a portfolio as a function of the **volatility** of each asset or liability position in the portfolio and the correlation between the positions.

Variation margin The band agreed between the parties to a repo transaction at the outset within which the value of the collateral may fluctuate before triggering a right to call for cash or securities to reinstate the initial margin on the repo transaction.

Vega The change in an option's value relative to a change in the **underlying's volatility**.

Volatility The **standard deviation** of the continuously compounded return on the **underlying**. Volatility is generally annualised. It measures the price fluctuation of an asset or derivative. See **historic volatility, implied volatility**.

W

Warrant A security giving the holder a right to subscribe to a share or bond at a given price and from a certain date. If this right is not exercised before the maturity date, the warrant will expire worthless.

Warrant-driven swap Swap with a warrant attached allowing the issuer of the fixed-rate bond to go on paying a floating rate in the event that she exercises another warrant allowing her to prolong the life of the bond.

When-issued trading Trading a bond before the issue date; no interest is accrued during this period. Also known as the "grey market".

Worst-case (credit-risk) exposure Estimate of the highest positive market value a derivative contract or portfolio is likely to attain at a given moment or period in the future, with a given level of confidence.

Worst-case (credit-risk) loss Estimate of the largest amount a derivative counterparty is likely to lose, with a given level of probability, as a result of default from a derivatives contract or portfolio.

Worst-case default rate The highest rates of default that are likely to occur at a given moment or period in the future, with a given level of confidence.

Write To sell an option is to write it. The person selling an option is known as the **writer**.

Writer The same as "seller" of an **option**.

Writing What a seller does when transacting in an option. Writing an option is equivalent to selling an option.

X

X Used to denote the strike price of an option; sometimes this is denoted using the term K.

Y

Yield The interest rate that can be earned on an investment, currently quoted by the market or implied by the current market price for the investment—as opposed to the **coupon** paid by an issuer on a security, which is based on the coupon rate and the face value. For a bond, generally the same as yield to maturity unless otherwise specified.

Yield curve Graphical representation of the maturity structure of interest rates, plotting yields of bonds that are all of the same class or credit quality against the maturity of the bonds.

Yield-curve option Option that allows purchasers to take a view on a yield curve without having to take a view about a market's direction.

Yield-curve swap Swap in which the index rates of the two interest streams are at different points on the yield curve. Both payments are refixed with the same frequency whatever the index rate.

Yield to equivalent life The same as **yield to maturity** for a bond with partial redemptions.

Yield to maturity The **internal rate of return** of a bond—the yield necessary to discount all the bond's cash flows to an **NPV** equal to its current price. See **simple yield to maturity, current yield.**

YTM See **yield to maturity**.

Z

Zero-cost collar A **collar** where the premiums paid and received are equal, giving a net zero cost.

Zero-cost option An option structure whose premiums cancel out and net to zero, thereby requiring the buyer to pay no premium.

Zero-coupon A zero-coupon security is one that does not pay a **coupon**. Its price is correspondingly less to compensate for this. A zero-coupon **yield** is the yield that a zero-coupon investment for that term would have if it were consistent with the **par yield curve**.

Zero-coupon bond Bond on which no coupon is paid. It is either issued at a discount or redeemed at a premium to face value.

Zero-coupon rate The interest rate on a zero-coupon bond, sometimes called the spot rate. The two terms are not strictly synonymous, however.

Zero-coupon swap Swap converting the payment pattern of a zero-coupon bond, either to that of a normal, coupon-paying **fixed-rate** bond or to a **floating rate**.

Zero-coupon yield The yield returned on a zero-coupon bond.

Zero-premium option Generic term for options for which there is no premium, either because the buyer undertakes to forgo a percentage of any gain or because he offsets the cost by **writing** other options.

Fixed Income Markets: Management, Trading, Hedging, Second Edition.
Moorad Choudhry, David Moskovic and Max Wong.

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